

Geometry at the ICM

155 Problems, Their Histories, and Open Frontiers

**A research survey of the Geometry collection at icmconjectures.com
Frozen dataset edition: 28 July 2026**

Research compilation prepared with Codex

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Preface

This volume organizes all 155 records classified as *Geometry* in the public ICM Conjectures database as it stood on 28 July 2026. Its aim is simultaneously historical and forward-looking: each record contains the problem statement, its originating ICM source, the literature and progress narrative supplied by the database, and every separately listed remaining open problem. The organization follows the database’s ten geometry subfields, while the audit appendices make the coverage independently checkable.

The International Mathematical Union’s proceedings archive spans ICM volumes from 1893 through 2022 and provides the primary historical backbone for the collection: <https://www.mathunion.org/icm/proceedings/1920>. The present survey is not an official publication of the IMU, and the ICM Conjectures database is a separate project. The distinction matters: this is a research compilation of a live, editable dataset, not a declaration of mathematical consensus.

How to read this volume

Each numbered section corresponds to exactly one website record. The blue box reproduces the mathematical statement retained by the dataset. It is followed by the database’s historical and bibliographic account, including the stated verdict and discussion of what has and has not been proved. Red-edged boxes contain descendant questions explicitly entered in the website’s “Some remaining open problems” field. The final primary-source pointer preserves the originating ICM URL.

The label *partially resolved* should be read literally. It may mean that one implication is known, that special dimensions or regularity classes are settled, that a formerly broad program now has a precise surviving core, or that strong estimates fall short of a sharp conjectural bound. It is not a uniform measure of distance from a solution.

Editorial and verification status

The source payload itself marks only five of the 155 Geometry records as reviewed and 150 as unreviewed. This volume exposes that field for every record rather than hiding it. URLs, DOIs, and arXiv pointers are retained verbatim in the scholarly narratives and also extracted into a machine-readable link inventory. Bibliographic presence is therefore auditable, but inclusion in this edition must not be confused with independent peer review of every status claim.

Scope at a glance

Measure	Count
Geometry records	155
Subfields	10
Source-year span	1893–2022
Partially resolved records	149
Open or open-ended records	6
Resolved records	0
Unclassified records	0
Dataset-reviewed records	5
Dataset-unreviewed records	150
Separately listed remaining open problems	202
Records with at least one separate open problem	120
Unique original ICM source records	91
Unique URLs retained in the audit inventory	1396

The ten subfields and their record counts are:

Subfield	Records
Global/Riemannian geometry	40
Metric and topological geometry	21
Classical and local differential geometry	20
Manifolds and global analysis	19
Symplectic and contact geometry	19
Discrete and polyhedral geometry	13
Applications of geometry / global analysis	8
Geometric flows and geometric PDE	7
Incidence, axiomatic, and classical geometry	5
Convex geometry	3
Total	155

A historical map of the collection

From classical configurations to global structures

The earliest records belong to a period in which geometry was being recast through topology, Lie theory, and the calculus of variations. Problems about curves, surfaces, rigidity, and incidence retain classical formulations, yet their later histories increasingly depend on global invariants. This is one reason the collection resists a simple chronology: a question that looks local may ultimately be governed by index theory, a moduli space, or the topology of a transformation group.

Curvature as topology, analysis, and obstruction

Across the Riemannian and global-analysis chapters, curvature conditions serve three roles. They are geometric hypotheses controlling comparison and compactness; analytic coefficients in nonlinear PDE; and topological obstructions detected through characteristic classes or operator indices. The Hopf-type sign questions, scalar-curvature rigidity problems, and positive-curvature classification programs illustrate how these roles interlock. Progress often consists of proving the predicted conclusion under spin, symmetry, dimension, or regularity hypotheses while the unrestricted statement persists.

The symplectic and categorical turn

Late twentieth-century ICM lectures introduced a cluster of questions in which the relevant geometric object is no longer only a manifold, but a category, field theory, or enumerative package attached to it. Floer theory, Gromov–Witten invariants, homological mirror symmetry, and gauge-theoretic constructions appear throughout the collection. In this setting “solution” can require equivalence between independently developed frameworks, together with foundations strong enough to compare all cases in the original scope.

Metric geometry and quantitative modernity

The metric chapter shows a complementary movement: remove linear or smooth structure, then ask how much of it can be recovered quantitatively. Embedding distortion, dimension reduction, coarse assembly, and the structure of metric limits replace exact isomorphism by controlled approximation. Here the unresolved part is often a sharp exponent, a dimension-free constant, an endpoint, or the passage from average to worst-case control.

Why old problems remain mathematically active

The records collectively demonstrate that longevity does not imply stagnation. A conjecture may generate techniques that settle entire neighboring theories without deciding the original endpoint. Conversely, a new conceptual framework may reveal that an old statement contains several logically distinct claims. The problem-by-problem format preserves these differences and makes the open frontier explicit rather than compressing it into a single binary status.

Method and conventions

Frozen source

The complete JSON response from <https://icmconjectures.com/api/records> was downloaded on 28 July 2026. The source reported dataset version 1785131205351379430-6555336 and an update timestamp of 2026-07-27T05:46:45+00:00. The full payload and the extracted Geometry-only snapshot are included in the project.

Completeness rule

A record is in scope if and only if its API field `category` equals `Geometry`. The resulting count is 155. Records are ordered first by the website’s subfield order and then by source year and record identifier. Each API identifier occurs once in the body and once in the master coverage table.

Status normalization

For tabulation only, an early status phrase in each progress narrative is normalized to *Partially resolved*, *Open/open-ended*, *Resolved*, or *Unclassified*. No mathematical inference is made from this normalization. The full wording of the underlying assessment is retained in the corresponding record.

Citation convention

The original ICM source URL is displayed in the metadata box and the primary-source pointer. Later literature is presented locally inside each problem’s research record because the evidence is problem-specific and often explains exactly which theorem proves which subcase. DOI and arXiv URLs are left clickable. The source catalogue appendix consolidates original ICM sources, while the machine-readable link inventory covers every extracted URL.

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Part I.

The 155 Geometry Problems

1. Global/Riemannian geometry

The forty records in this chapter form the largest group in the collection. They run from global questions about curvature, topology, and rigidity to Kähler geometry, comparison geometry, geometric group actions, and scalar curvature. A recurring historical pattern is the replacement of a qualitative geometric question by an analytic or variational criterion: curvature is linked to index theory, canonical metrics to stability, and geometric limits to quantitative regularity. Several problems have celebrated solutions in important special cases while their dimension-free or fully general forms remain open.

Coverage: 40 website records; **source years:** 1958–2022; **separately listed descendant open problems:** 43.

1.1. 1. DIFFERENTIAL GEOMETRY AND INTEGRAL GEOMETRY [1958-geom-7]

Record: 1958-geom-7

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1958: DIFFERENTIAL GEOMETRY AND INTEGRAL GEOMETRY

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=505>

1.1.1. Problem statement

Let M^n be a compact differentiable n -manifold. For an immersion $x : M^n \rightarrow E^{n+N}$ into a Euclidean space, let B_ν be the bundle of unit normal vectors to the immersed submanifold $x(M^n)$. Fix an origin in E^{n+N} , let S_0 be the unit sphere centered at that origin, and let $T : B_\nu \rightarrow S_0$ send each unit normal vector to the endpoint of the parallel unit vector based at the origin. Define the total curvature $T_x(M^n)$ of the immersion x to be the volume of the image $T(B_\nu)$, with each point counted with multiplicity equal to the number of its preimages under T , divided by the volume of S_0 . Define

$$T(M^n) = \min_x T_x(M^n),$$

where the minimum is taken over all immersions x of M^n into Euclidean spaces. Prove or disprove that, for every compact differentiable manifold M^n , the minimum total curvature $T(M^n)$ is equal to the minimum total number of cells in any subdivision of M^n

as a cell complex.

1.1.2. Historical background, literature, and current progress

1.1.3. Verdict: partially resolved

1.1.3.1. Precise problem

The statement is the Chern-Lashof total absolute curvature problem. For a compact smooth differentiable manifold M , one averages, over unit normal directions, the number of preimages of the spherical normal map of a Euclidean immersion. Equivalently, for generic height functions, this total absolute curvature is the mean number of critical points of the height functions. The question is whether the best possible value over Euclidean immersions of M is exactly the smallest possible number of cells in a cell decomposition of M , usually expressed in the later literature as the Morse number $\mu(M)$, the minimum number of critical points of a Morse function on M .

1.1.3.2. Verdict

Partially resolved. The literature proves major versions of the Chern-Lashof problem, especially the infimum/Morse-number formulation and high-dimensional fixed-ambient variants, but I did not find a published result that covers the exact statement as written for every compact differentiable manifold with an actual minimum over all immersions. In particular, the modern references generally formulate the invariant as an infimum, and attainment is a separate tightness/minimal-immersion question.

1.1.3.3. Literature checked

1. On the total curvature of immersed manifolds

Authors: Shiing-Shen Chern and Richard K. Lashof.

Venue: *American Journal of Mathematics* 79(2), 306-318.

Year: 1957.

DOI/URL: <https://doi.org/10.2307/2372684>.

Role: Original source for the Euclidean total curvature theory. It proves foundational Chern-Lashof lower bounds, including lower bounds by topology and the convex hypersurface characterization in the minimal sphere case, but it does not settle the full all-manifolds equality asserted in the problem statement.

2. On the total curvature of immersed manifolds, II

Authors: Shiing-Shen Chern and Richard K. Lashof.

Venue: *Michigan Mathematical Journal* 5(1), 5-12.

Year: 1958.

DOI/URL: <https://doi.org/10.1307/mmj/1028998005>.

Role: Continuation of the original work and the source context for the Chern-Lashof conjectural comparison between total curvature and the minimal cell/Morse count. It provides foundational inequalities and special cases, not a full general proof.

3. Minimal total absolute curvature for immersions

Author: Nicolaas H. Kuiper.

Venue: *Inventiones Mathematicae* 10, 209-238.

Year: 1970.

DOI/URL: <https://doi.org/10.1007/BF01403250>.

Role: Develops the tight/minimal total absolute curvature viewpoint. The abstract explicitly says that many manifolds have no tight smooth immersion in any Euclidean space, so attainment of the optimal lower bound is not automatic in the exact “minimum” sense.

4. Geometry in total absolute curvature theory

Author: Nicolaas H. Kuiper.

Venue: *Perspectives in Mathematics*, Oberwolfach 1984, Birkhäuser, Basel, pp. 377-392.

Year: 1984.

URL: https://repo-archives.ihes.fr/FONDS_IHES/I_Prepublications/KUIPER/1972-1991/M_84_08/M_84_08_web.pdf.

Role: States the Morse-number formulation: if $\gamma(M)$ is the minimum number of critical points of a smooth nondegenerate real function, then the infimum of total absolute curvature over Euclidean embeddings is known to be $\gamma(M)$. This supports the infimum version, not universal attainment of a minimum.

5. Total absolute curvature and embedded Morse numbers

Author: R. W. Sharpe.

Venue: *Journal of Differential Geometry* 28(1), 59-92.

Year: 1988.

DOI/URL: <https://doi.org/10.4310/jdg/1214442160>.

Role: Computes infima of total absolute curvature for smooth embeddings in fixed ambient situations under mild hypotheses, using embedded Morse theory. This is substantial progress but has hypotheses and does not by itself cover the unrestricted all-dimensional minimum statement.

6. A proof of the Chern-Lashof conjecture in dimensions greater than five

Author: R. W. Sharpe.

Venue: *Commentarii Mathematici Helvetici* 64, 221-235.

Year: 1989.

DOI/URL: <https://doi.org/10.1007/BF02564672>.

Role: Proves that, for smooth manifolds of dimension $m > 5$, the best possible lower bound for total absolute curvature is the Morse number $\mu(M)$. The dimension restriction means it is not a full all-dimensional resolution of the statement in the problem statement.

7. Taut immersions into complete Riemannian manifolds

Authors: Chuu-Lian Terng and Gudlaugur Thorbergsson.

Venue: *Tight and Taut Submanifolds*, Mathematical Sciences Research Institute Publications 32, Cambridge University Press, pp. 181-228.

Year: 1997.

URL: <https://library.slmath.org/books/Book32/files/terng.pdf>.

Role: Survey/context reference. It states that Chern-Lashof’s proof gives $\tau(M) \geq \mu(M)$ and that Sharpe proved $\tau(M) = \mu(M)$ under mild dimension assumptions, confirming that the broad literature treats the available theorem as conditional rather than as a fully general minimum-attainment theorem.

8. The circular Morse-Smale characteristic of closed surfaces

Authors: Dorin Andrica, Dana Mangra, and Cornel Pinteau.

Venue: *Bulletin Mathématique de la Société des Sciences Mathématiques de Roumanie* 57(105), no. 3, 235-242.

Year: 2014.

URL: https://ssmr.ro/bulletin/pdf/57-3/articol_1.pdf.

Role: Modern reference stating the Chern-Lashof conjecture as the assertion that the Morse-Smale characteristic is the infimum of total curvatures over Euclidean embeddings and noting that it holds for several manifolds. This supports the conclusion that related special cases are known, but not an exact full-resolution claim for the problem as written.

1.1.3.4. What remains unresolved for this problem

The sources above justify strong partial progress: the optimal lower bound/infimum is identified with the Morse number in important formulations, and Sharpe proves a high-dimensional version. However, the problem asks for a minimum over all immersions of every compact differentiable manifold and equality with the minimum cell count. The literature I found does not provide a theorem proving that exact minimum-attainment statement for all compact differentiable manifolds. Because the modern formulation is usually an infimum and tight/minimal immersions may fail to exist for many manifolds, the exact statement is not fully resolved by the cited literature.

1.1.4. Remaining open problems recorded by the database

1.1.4.1. Open problem 1

Prove the Chern–Lashof conjecture for closed smooth 5-manifolds: show that the infimum of the total absolute curvatures of their Euclidean immersions equals their Morse number (the minimum number of critical points of a Morse function). This is the first dimension not covered by Sharpe’s theorem for dimensions greater than five.

Reference supplied by the database. R. W. Sharpe, “A proof of the Chern–Lashof conjecture in dimensions greater than five,” *Commentarii Mathematici Helvetici* **64** (1989), no. 2, 221–235, especially Theorem A (which proves the conjecture only for $m > 5$), DOI: [10.1007/BF02564672](https://doi.org/10.1007/BF02564672). The conjecture continued to be stated without a general low-dimensional solution in Dorin Andrica, Dana Mangra, and Cornel Pinte, “The circular Morse–Smale characteristic of closed surfaces,” *Bulletin Mathématique de la Société des Sciences Mathématiques de Roumanie* **57**(105) (2014), no. 3, 235–242, introduction, [article PDF](#).

1.1.5. Primary-source pointer

The originating ICM record is *DIFFERENTIAL GEOMETRY AND INTEGRAL GEOMETRY* (1958). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=505>.

1.2. 2. The Hopf Product Conjecture [1970-geom-12]

Record: 1970-geom-12

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: reviewed (reviewed flag: yes)

Original source: ICM 1970: Differential geometry; its past and its future

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>

1.2.1. Problem statement

Does $S^2 \times S^2$, the product of two 2-dimensional spheres, admit a Riemannian metric of strictly positive sectional curvature?

1.2.2. Historical background, literature, and current progress

1.2.3. Verdict: open

1.2.3.1. Precise problem

The problem asks the classical Hopf product problem: whether the smooth manifold $S^2 \times S^2$ admits any Riemannian metric of strictly positive sectional curvature.

The expected conjectural answer in the literature is negative, but I found no published or otherwise citable theorem that proves nonexistence for all Riemannian metrics on $S^2 \times S^2$, and no construction of such a positive-curvature metric.

1.2.3.2. Current status

The exact unrestricted problem remains open. Recent survey literature still describes it as open in general, while noting that it is settled in special metric classes. The correct status is therefore **Open**.

1.2.3.3. Original source and status references

- Title: *Differential Geometry; Its Past and Its Future*
- Authors: Shiing-Shen Chern; records a question asked by H. Hopf
- Venue/year: Actes, Congrès international des mathématiciens, Nice 1970, Tome 1, pp. 41-53
- DOI/arXiv/URL: <https://www.hiroleetanaka.com/pdfs/2014-fall-1-230a-icm1970-chern-differential-geometry.pdf>
- Relevance: Problem I asks exactly whether the product of two 2-dimensional spheres admits a Riemannian metric of strictly positive curvature.

- Title: *Bounds on Sectional Curvature and Interactions with Topology*
- Author: Manuel Amann
- Venue/year: Jahresbericht der Deutschen Mathematiker-Vereinigung 123, 27-55, 2021; published online 2020
- DOI/arXiv/URL: <https://doi.org/10.1365/s13291-020-00225-x>
- Relevance: States the Hopf conjecture for $S^2 \times S^2$, says it is settled only in special metric classes, and says it is widely open in general.
- Title: *Harmonic 2-forms and positively curved 4-manifolds*
- Authors: Kefeng Liu and Jianming Wan
- Venue/year: Science China Mathematics 64, 1613-1620, 2021; published online 2020
- DOI/arXiv/URL: <https://doi.org/10.1007/s11425-019-1641-9>; arXiv:1909.00110
- Relevance: The introduction says the Hopf conjecture $S^2 \times S^2$ has no positive sectional curvature metric is still widely open except for special cases.

1.2.3.4. Special cases progress

- Title: *On the topology of positively curved 4-manifolds with symmetry*
- Authors: Wu-Yi Hsiang and Bruce Kleiner
- Venue/year: Journal of Differential Geometry 29(3), 615-621, 1989
- DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1214443064>
- Result and limitation: A compact simply connected positively curved 4-manifold with an effective isometric circle action is homeomorphic to S^4 or $\mathbb{CP}^2$. Since $S^2 \times S^2$ is neither, this rules out positive-curvature metrics on $S^2 \times S^2$ with nontrivial circle symmetry. It does not rule out metrics with finite or trivial isometry group.
- Title: *A Remark On Positively Curved 4-Manifolds*
- Author: Christian Bär
- Venue/year: Differential Geometry and its Applications 4(1), 71-75, 1994
- DOI/arXiv/URL: [https://doi.org/10.1016/0926-2245\(94\)90009-4](https://doi.org/10.1016/0926-2245(94)90009-4); PDF: https://www.math.uni-potsdam.de/fileadmin/user_upload/Prof-Geometrie/Dokumente/Publikationen/hopf.pdf
- Result and limitation: If a compact oriented connected 4-manifold has indefinite intersection form, then it admits no metric with $K \geq 1$ and $\|\nabla R\|_{L^\infty} \leq 2/\pi$. This applies to $S^2 \times S^2$, but only under the extra derivative bound.
- Title: *Geometrically formal 4-manifolds with nonnegative sectional curvature*
- Author: Christian Bär
- Venue/year: Communications in Analysis and Geometry 23, 479-497, 2015
- DOI/arXiv/URL: <https://doi.org/10.4310/CAG.2015.v23.n3.a2>; arXiv:1212.1325
- Result and limitation: Shows that the Hopf product conjecture - namely, the nonexistence

of a positive-sectional-curvature metric on $S^2 \times S^2$ - holds for formal metrics and related harmonic-form length conditions. This is a restriction on the metric class and does not cover arbitrary metrics on $S^2 \times S^2$.

- Title: *Positive biorthogonal curvature on $S^2 \times S^2$*
- Author: Renato G. Bettiol
- Venue/year: Proceedings of the American Mathematical Society 142(12), 4341-4353, 2014
- DOI/arXiv/URL: <https://doi.org/10.1090/S0002-9939-2014-12173-3>; arXiv:1210.0043
- Result and limitation: Constructs metrics on $S^2 \times S^2$ with positive biorthogonal curvature (and in fact positive Ricci curvature and positive distance curvature $\sec_\theta$ for all $\theta > 0$), a package Bettiol describes as intermediate between admitting $\text{Ric} > 0$ and admitting $\sec > 0$.
- Title: *Harmonic 2-forms and positively curved 4-manifolds*
- Authors: Kefeng Liu and Jianming Wan
- Venue/year: Science China Mathematics 64, 1613-1620, 2021; published online 2020
- DOI/arXiv/URL: <https://doi.org/10.1007/s11425-019-1641-9>; arXiv:1909.00110
- Result and limitation: If a compact oriented 4-manifold has positive sectional curvature and all harmonic 2-forms satisfy $|\nabla \phi|^2 \geq 2|d|\phi||^2$, a Kato-type inequality, then its intersection form is definite. Since $S^2 \times S^2$ is indefinite, this would rule it out under the assumption of the Kato-type inequality. The inequality is not known for arbitrary positive-curvature metrics.
- Title: *Conformal Metrics to a Product or Doubly Warped Product on $S^2 \times S^2$ and the Hopf Conjecture*
- Authors: Thierno Seck and Athoumane Niang
- Venue/year: Journal of Mathematics 2022, Article ID 6268017, 5 pp., 2022
- DOI/arXiv/URL: <https://doi.org/10.1155/2022/6268017>
- Result and limitation: Proves no positive sectional curvature within two metric classes: conformal metrics to the product metric and doubly warped product metrics. These classes do not include all Riemannian metrics on $S^2 \times S^2$.
- Title: *Metrics Induced by $\mathbb{R}^{6+k}$ on the Graph of k -Smooth Functions and the Hopf Conjecture*
- Authors: Thierno Seck, Athoumane Niang, and Adama Thiandoum
- Venue/year: In *Nonlinear Analysis, Geometry and Applications*, Trends in Mathematics, Birkhäuser, pp. 285-298, 2024
- DOI/arXiv/URL: https://doi.org/10.1007/978-3-031-52681-7_13
- Result and limitation: By Perdomo's Theorem A (Isr. J. Math. 156 (2006)), every Riemannian metric on $S^2 \times S^2$ is, up to homothety, isometric to a graph metric over the standard embedding $S^2 \times S^2 \subset \mathbb{R}^6$ in sufficiently high codimension; since homothety preserves the sign of sectional curvature, the preprint's claimed theorem — that no graph of an $\mathbb{R}^k$ -valued function over $S^2 \times S^2$ has positive sectional curvature — would, if correct, resolve the conjecture in full. The claim is equivalent in strength to the conjecture itself; it appears only in an unrefereed preprint, has not been validated, and the refereed literature

through 2026 continues to treat the conjecture as open. We therefore do not count it. (Perdomo's published Theorem B establishes the graph statement only for scalar-valued F over the product metric.)

1.2.3.5. What remains unresolved

It remains unknown whether an arbitrary Riemannian metric on $S^2 \times S^2$ can have strictly positive sectional curvature everywhere, or whether such metrics are impossible without extra hypotheses such as symmetry, pinching/derivative bounds, formality-type conditions, Kato inequalities, conformal/warped-product form, or graph-induced form.

1.2.4. Remaining open problems recorded by the database

1.2.4.1. Open problem 1

Let ϕ be a harmonic 2-form on a compact Riemannian 4-manifold of positive sectional curvature. Must the strengthened Kato inequality $|\nabla\phi|^2 \geq 2|d\phi|^2$ hold?

Reference supplied by the database. Kefeng Liu and Jianming Wan, "Harmonic 2-forms and positively curved 4-manifolds," *Science China Mathematics* **64** (2021), no. 7, 1613–1620, explicit question at the end of Section 4; Theorem 1.1 proves that an affirmative answer would imply definiteness and hence the Hopf product conjecture, DOI: [10.1007/s11425-019-1641-9](https://doi.org/10.1007/s11425-019-1641-9), [arXiv:1909.00110](https://arxiv.org/abs/1909.00110).

1.2.5. Primary-source pointer

The originating ICM record is *Differential geometry; its past and its future* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>.

1.3. 3. Differential geometry; its past and its future [1970-geom-13]

Record: 1970-geom-13

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1970: Differential geometry; its past and its future

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>

1.3.1. Problem statement

Prove that if M is a compact orientable Riemannian manifold of even dimension n , and $\chi(M)$ denotes its Euler-Poincaré characteristic, then:

- if all sectional curvatures of M are nonnegative, then $\chi(M) \geq 0$;
- if all sectional curvatures of M are nonpositive, then $\chi(M) \geq 0$ when $n \equiv 0 \pmod{4}$, and $\chi(M) \leq 0$ when $n \equiv 2 \pmod{4}$.

1.3.2. Historical background, literature, and current progress

1.3.3. Verdict: partially resolved

1.3.3.1. Precise problem

This is the Hopf/Chern-Hopf Euler characteristic sign problem for compact even-dimensional Riemannian manifolds whose sectional curvatures have a fixed sign:

- if $\sec \geq 0$, prove $\chi(M) \geq 0$;
- if $\sec \leq 0$, prove $(-1)^{n/2} \chi(M) \geq 0$, equivalently $\chi(M) \geq 0$ for $n \equiv 0 \pmod{4}$ and $\chi(M) \leq 0$ for $n \equiv 2 \pmod{4}$.

The problem asks for both halves, in all compact orientable even dimensions.

1.3.3.2. Status assessment

The problem is not fully resolved. The nonnegative sectional-curvature half remains open in general: the positive-curvature Hopf conjecture, a special case of $\sec \geq 0$, is still described in modern literature as open in dimensions 6 and higher. There is substantial partial progress in dimension 4 and under additional hypotheses such as symmetry, Kähler structure, higher-order curvature positivity, or special topological/geometric assumptions.

For the nonpositive half, there is a recent arXiv preprint by Fang and Wang that explicitly claims the full nonpositive case. Its Theorem 1.9, as stated, has hypotheses and conclusion matching the nonpositive half of this problem. However, even accepting that citable preprint

as progress, it does not address the $\sec \geq 0$ half, so it cannot fully resolve the stated combined statement.

1.3.3.3. Key literature and progress

Original/source formulation

- Title: Differential geometry: its past and its future
- Authors: Shiing-Shen Chern
- Venue: Actes du Congrès International des Mathématiciens 1970, Vol. 1, Gauthier-Villars, Paris, pp. 41-53
- Year: 1971
- DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf> ; alternate PDF <https://www.hiroleetanaka.com/pdfs/2014-fall-230a-icm1970-chern-differential-geometry.pdf>
- Result/explanation: Problem III and Conjecture is exactly this problem's question: for compact orientable even-dimensional M , Chern asks for the Euler characteristic sign under $\sec \geq 0$ and $\sec \leq 0$, and notes that it was known for $n = 4$ and constant sectional curvature.

Low-dimensional result

- Title: On curvature and characteristic classes of a Riemannian manifold
- Authors: Shiing-Shen Chern
- Venue: Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg, 20, 117-126
- Year: 1955
- DOI/arXiv/URL: <https://doi.org/10.1007/BF02960745>
- Result/explanation: Gives the Chern-Gauss-Bonnet curvature approach used in the 4-dimensional case; Chern's 1970 ICM article cites this as part of the known $n = 4$ resolution.
- Title: Some implications of the generalized Gauss-Bonnet theorem
- Authors: Richard L. Bishop and Samuel I. Goldberg
- Venue: Transactions of the American Mathematical Society, 112(3), 508-535
- Year: 1964
- DOI/arXiv/URL: <https://doi.org/10.1090/S0002-9947-1964-0163271-8>
- Result/explanation: Theorem 1.1 proves that a compact oriented 4-manifold with non-negative or nonpositive sectional curvature has nonnegative Euler characteristic, with strict positivity in the strictly positive or strictly negative cases. This resolves the problem statement in dimension 4, but not in higher even dimensions.

Obstruction to the naive Gauss-Bonnet-integrand approach

- Title: Positive sectional curvatures does not imply positive Gauss-Bonnet integrand

- Authors: Robert Geroch
- Venue: Proceedings of the American Mathematical Society, 54, 267-270
- Year: 1976
- DOI/arXiv/URL: <https://doi.org/10.1090/S0002-9939-1976-0390961-8>
- Result/explanation: Constructs a 6-dimensional algebraic curvature tensor with positive sectional curvatures but negative Gauss-Bonnet integrand. This is not a counterexample to Hopf's conjecture, but it shows why pointwise positivity of the Gauss-Bonnet integrand cannot prove the general result.

Higher-order/special curvature conditions

- Title: Euler-Poincare characteristic and higher order sectional curvature. I
- Authors: Chuan-Chih Hsiung and Kenneth Michael Shiskowski
- Venue: Transactions of the American Mathematical Society, 305(1), 113-128
- Year: 1988
- DOI/arXiv/URL: <https://doi.org/10.1090/S0002-9947-1988-0920149-3>
- Result/explanation: Treats Hopf's conjecture under higher-order sectional-curvature assumptions. The paper itself describes the stated Hopf conjecture as a long-standing open conjecture, so it is partial progress only.

Kahler and related nonpositive-curvature progress

- Title: Kahler hyperbolicity and L2-Hodge theory
- Authors: Mikhail Gromov
- Venue: Journal of Differential Geometry, 33(1), 263-292
- Year: 1991
- DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1214446039>
- Result/explanation: Proves Euler characteristic sign results for Kahler hyperbolic manifolds using L2-Hodge theory. This covers important Kahler/negatively curved situations but assumes additional Kahler-type structure and does not prove the full Riemannian statement.
- Title: Kahler parabolicity and the Euler number of compact manifolds of non-positive sectional curvature
- Authors: Jianguo Cao and Frederico Xavier
- Venue: Mathematische Annalen, 319, 483-491
- Year: 2001
- DOI/arXiv/URL: <https://doi.org/10.1007/PL00004444>
- Result/explanation: Proves that if a compact $2n$ -dimensional Riemannian manifold has nonpositive sectional curvature and is homeomorphic to a Kahler manifold, then $(-1)^n \chi(M) \geq 0$. The extra hypothesis "homeomorphic to a Kahler manifold" means this is not the full nonpositive Riemannian case.

Symmetry results for positive/nonnegative side

- Title: On the Hopf conjecture with symmetry
- Authors: Lee Kennard
- Venue: Geometry & Topology, 17(1), 563-593
- Year: 2013
- DOI/arXiv/URL: <https://doi.org/10.2140/gt.2013.17.563> ; arXiv:1203.3808
- Result/explanation: Proves the positive-curvature Hopf conjecture under additional torus-symmetry assumptions. The introduction states that the conjecture holds in dimensions 2 and 4 but remains open in higher dimensions. Thus it is partial progress, not a full resolution of the problem.

Modern survey evidence that the nonnegative case remains open

- Title: Riemannian manifolds with positive sectional curvature
- Authors: Wolfgang Ziller
- Venue: Geometry of Manifolds with Non-negative Sectional Curvature, Lecture Notes in Mathematics 2110, Springer, pp. 1-19
- Year: 2014
- DOI/arXiv/URL: <https://www2.math.upenn.edu/~wziller/papers/SurveyMexico.pdf>
- Result/explanation: Lists Hopf's conjectures, including "A compact manifold with $\sec \geq 0$ has non-negative Euler characteristic" and the positive-curvature even-dimensional version, as conjectures.
- Title: Bounds on Sectional Curvature and Interactions with Topology
- Authors: Manuel Amann
- Venue: Jahresbericht der Deutschen Mathematiker-Vereinigung, 123, 27-55
- Year: 2021
- DOI/arXiv/URL: <https://doi.org/10.1365/s13291-020-00225-x>
- Result/explanation: A modern survey explaining that the Hopf conjecture for nonnegative curvature would follow from the Bott-Grove-Halperin conjecture. This places the nonnegative case among still-conjectural consequences rather than resolved theorems.

Claimed full nonpositive-curvature preprint

- Title: On non-elliptic symplectic manifolds
- Authors: Shouwen Fang and Hongyu Wang
- Venue: arXiv preprint, arXiv:1807.00326v4
- Year: submitted 2018, last revised 2024
- DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.1807.00326> ; <https://arxiv.org/abs/1807.00326>
- Result/explanation: Theorem 1.9 states: if (M^{2n}, g) is a closed Riemannian manifold with strictly negative sectional curvature, then $(-1)^n \chi(M^{2n}) > 0$; if it has nonpositive sectional curvature, then $(-1)^n \chi(M^{2n}) \geq 0$. The hypotheses and conclusion, as stated, cover the nonpositive half of this problem. The preprint does not address the nonnegative half, so it does not fully resolve the combined problem.

1.3.3.4. What remains unresolved

The all-dimensional $\sec \geq 0 \Rightarrow \chi(M) \geq 0$ assertion remains unresolved. Since strictly positive sectional curvature is a special case of nonnegative sectional curvature, a complete proof of the stated nonnegative half would also settle the classical positive-curvature Hopf conjecture in dimensions 6 and higher; modern papers and surveys still treat that as open except under additional assumptions. Therefore the problem is partially resolved, not fully resolved.

1.3.4. Remaining open problems recorded by the database

1.3.4.1. Open problem 1

Prove the first unresolved dimensional case of the positive-curvature Hopf conjecture: every closed orientable Riemannian 6-manifold with strictly positive sectional curvature has strictly positive Euler characteristic.

Reference supplied by the database. Lee Kennard, “On the Hopf conjecture with symmetry,” *Geometry & Topology* **17** (2013), no. 1, 563–593, introduction (stating the positive-curvature Euler-characteristic conjecture and that it is known in dimensions two and four), DOI: [10.2140/gt.2013.17.563](https://doi.org/10.2140/gt.2013.17.563), [arXiv:1203.3808](https://arxiv.org/abs/1203.3808). Current-status confirmation: Manuel Amann, “Bounds on sectional curvature and interactions with topology,” *Jahresbericht der Deutschen Mathematiker-Vereinigung* **123** (2021), 27–55, DOI: [10.1365/s13291-020-00225-x](https://doi.org/10.1365/s13291-020-00225-x).

1.3.4.2. Open problem 2

Prove the six-dimensional nonnegative-curvature case: every closed orientable Riemannian 6-manifold with nonnegative sectional curvature has nonnegative Euler characteristic.

Reference supplied by the database. Shiing-Shen Chern, “Differential geometry: its past and its future,” in *Actes du Congrès International des Mathématiciens (Nice, 1970)*, Tome 1, Gauthier-Villars, Paris, 1971, pp. 41–53, Problem III (the nonnegative-curvature Euler-sign conjecture, noting the four-dimensional case), [ICM proceedings PDF](#). Current-status confirmation: Wolfgang Ziller, “Riemannian manifolds with positive sectional curvature,” in *Geometry of Manifolds with Non-negative Sectional Curvature*, Lecture Notes in Mathematics 2110, Springer, Cham, 2014, pp. 1–19 (list of Hopf conjectures), DOI: [10.1007/978-3-319-06373-7_1](https://doi.org/10.1007/978-3-319-06373-7_1).

1.3.4.3. Open problem 3

Prove the Bott–Grove–Halperin conjecture: every closed simply connected Riemannian manifold with nonnegative sectional curvature is rationally elliptic, i.e. $\dim_{\mathbb{Q}}(\pi_*(M) \otimes \mathbb{Q}) < \infty$. This would in particular imply the nonnegative-curvature Euler-characteristic

inequality.

Reference supplied by the database. Karsten Grove, Burkhard Wilking, and Joseph Yeager, with an appendix by Steve Halperin, “Almost non-negative curvature and rational ellipticity in cohomogeneity two,” *Annales de l’Institut Fourier* **69** (2019), no. 7, 2921–2939, conjecture in the introduction, DOI: [10.5802/aif.3340](https://doi.org/10.5802/aif.3340). Current-status confirmation: Manuel Amann, “Bounds on sectional curvature and interactions with topology,” *Jahresbericht der Deutschen Mathematiker-Vereinigung* **123** (2021), 27–55, Conjecture (Bott–Grove–Halperin), DOI: [10.1365/s13291-020-00225-x](https://doi.org/10.1365/s13291-020-00225-x).

1.3.4.4. Open problem 4

Prove the Singer L^2 -cohomology conjecture for closed nonpositively curved manifolds: the reduced L^2 -homology of the universal cover vanishes in every degree except possibly the middle degree. This would imply the conjectured sign of the Euler characteristic on the nonpositive-curvature side.

Reference supplied by the database. Michael W. Davis and Boris Okun, “Vanishing theorems and conjectures for the ℓ^2 -homology of right-angled Coxeter groups,” *Geometry & Topology* **5** (2001), 7–74, Singer Conjecture and its implication for the Hopf–Chern sign conjecture, DOI: [10.2140/gt.2001.5.7](https://doi.org/10.2140/gt.2001.5.7), [arXiv:math/0102104](https://arxiv.org/abs/math/0102104). Current-status confirmation: Laurențiu Maxim, “On singular generalizations of the Singer–Hopf conjecture,” *Mathematische Nachrichten* **296** (2023), no. 11, 5232–5241, Conjecture 1.1 and introduction, DOI: [10.1002/mana.202200322](https://doi.org/10.1002/mana.202200322).

1.3.5. Primary-source pointer

The originating ICM record is *Differential geometry; its past and its future* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>.

1.4. 4. Geometric Aspects of the Generalized Plateau Problem [1974-geom-20]

Record: 1974-geom-20

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Geometric Aspects of the Generalized Plateau Problem

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf#page=15>

1.4.1. Problem statement

Prove or disprove that if M is a compact simply connected Riemannian n -manifold whose sectional curvatures are strictly $1/4$ -pinched, in the sense that after a constant rescaling of the metric

$$\frac{1}{4} < K(\sigma) \leq 1$$

for every tangent 2-plane $\sigma \subset T_x M$, then for every p with $1 \leq p \leq n - 1$, there is no nonzero rectifiable p -current V with compact support in M that is closed and stable; equivalently, there is no such V satisfying $dV = 0$, having first variation of mass equal to zero, and having second variation of mass nonnegative under all smooth deformations preserving its boundary.

1.4.2. Historical background, literature, and current progress

1.4.3. Verdict: partially resolved

1.4.3.1. Precise problem

This is the Lawson-Simons stable-current conjecture: if M^n is compact, simply connected, and strictly $1/4$ -pinched after rescaling,

$$1/4 < K(\sigma) \leq 1,$$

then for every $1 \leq p \leq n - 1$ there should be no nonzero closed stable rectifiable p -current in M . Equivalently, no nonzero compactly supported rectifiable p -current with $dV = 0$, zero first variation of mass, and nonnegative second variation under all smooth boundary-preserving deformations should exist.

1.4.3.2. Current status

I found no published or otherwise citable theorem proving this full statement for all compact simply connected strictly $1/4$ -pinched Riemannian manifolds and all p . Recent literature still describes the Lawson-Simons conjecture in full generality as open. The problem is nevertheless

not “fully open” in the sense of having no progress: there are substantial special cases and results with extra hypotheses.

The differentiable sphere theorem of Brendle-Schoen shows that such an M is diffeomorphic to a space form, and hence to S^n under simple connectivity, but this does not prove the stable-current statement. Absence of stable null-homologous currents is a variational/geometric assertion, not a consequence of the topology alone.

1.4.3.3. Progress and relevant literature

- H. Blaine Lawson, Jr.; James Simons, “On stable currents and their application to global problems in real and complex geometry,” *Annals of Mathematics* (2) 98, no. 3, 427-450, 1973. DOI: 10.2307/1970913. URL: <https://annals.math.princeton.edu/1973/98-3/p03>.

This is the foundational paper. It proves nonexistence of stable integral currents, currents mod p , and stable varifolds on the round Euclidean sphere, and proves related extrinsic nonexistence theorems for submanifolds of spheres. It is also cited by later papers as the source of the conjecture. These results cover important model cases but not arbitrary strictly $1/4$ -pinched metrics.

- H. Blaine Lawson, Jr., “Geometric aspects of the generalized Plateau problem,” *Proceedings of the International Congress of Mathematicians, Vancouver 1974*, vol. 2, Canadian Mathematical Congress, 1975, pp. 7-13. URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf>.

This ICM article is the original source listed in the dataset. It formulates the conjectural nonexistence of stable closed currents on compact simply connected manifolds with strictly quarter-pinched sectional curvature after surveying stable currents on spheres and projective spaces.

- Ralph Howard, “The nonexistence of stable submanifolds, varifolds, and harmonic maps in sufficiently pinched simply connected Riemannian manifolds,” *Michigan Mathematical Journal* 32, no. 3, 321-334, 1985. DOI/URL: <https://doi.org/10.1307/mmj/1029003241>.

Howard proves nonexistence under sufficiently strong pinching constants depending on the dimension and current/submanifold dimension. Later summaries note these constants are strictly larger than $1/4$, so this is partial progress and does not reach the exact $1/4$ -pinched range.

- Mario J. Micallef; John Douglas Moore, “Minimal two-spheres and the topology of manifolds with positive curvature on totally isotropic two-planes,” *Annals of Mathematics* (2) 127, no. 1, 199-227, 1988. DOI: 10.2307/1971420. URL: <https://annals.math.princeton.edu/1988/127-1/p06>.

Their minimal-two-sphere instability and index estimates apply to the two-sphere case under the relevant positive isotropic curvature consequences of quarter pinching. This is a low-dimensional/special-domain result, not the full stable-current statement.

- Yi-Bing Shen; Hui-Qun Xu, “On the nonexistence of stable minimal submanifolds in positively pinched Riemannian manifolds,” in *Geometry and Topology of Submanifolds, X* (Beijing/Berlin, 1999), World Scientific, 2000, pp. 274-283. MR 1801920.

This is repeatedly cited as a partial result toward Lawson-Simons, especially for extra hypotheses such as ambient hypersurface settings. Modern discussions do not treat it as resolving the full conjecture.

- Yi-Bing Shen; Qun He, “On stable currents and positively curved hypersurfaces,”

Proceedings of the American Mathematical Society 129, no. 1, 237-246, 2001. MR 1784025.

This proves nonexistence of stable currents or stable varifolds for complete pinched hypersurfaces in real space forms with nonnegative constant curvature. The hypersurface/space-form assumptions are extra hypotheses not present in the problem statement.

- **Ze-Jun Hu; Guo-Xin Wei, “On the nonexistence of stable minimal submanifolds and the Lawson-Simons conjecture,”** *Colloquium Mathematicum* 96, no. 2, 213-223, 2003. DOI: 10.4064/cm96-2-6. URL: <https://eudml.org/doc/284869>.

This proves no stable compact minimal submanifolds when the ambient compact Riemannian manifold satisfies certain curvature bounds and isometrically immerses as a hypersurface in Euclidean space or the unit sphere. This is important evidence but has extrinsic immersion assumptions and concerns minimal submanifolds, so it does not cover arbitrary rectifiable currents in arbitrary strictly $1/4$ -pinched manifolds.

- **Xueshan Zhang, “On the nonexistence of stable currents in submanifolds of a Euclidean space,”** *Tohoku Mathematical Journal* 56, no. 4, 491-499, 2004. DOI: 10.2748/tmj/1113246746. URL: https://www.jstage.jst.go.jp/article/tmj/56/4/56_4_491/_article/-char/ja.

Zhang states that the original Lawson-Simons conjecture is still open, and proves stable-current nonexistence under additional extrinsic principal-curvature/sectional-curvature inequalities for submanifolds of Euclidean or nonnegative space forms. These hypotheses are not implied by the stated intrinsic pinching assumption.

- **Ralph Howard; S. Walter Wei, “On the existence and nonexistence of stable submanifolds and currents in positively curved manifolds and the topology of submanifolds in Euclidean spaces,”** in *Geometry and Topology of Submanifolds and Currents, Contemporary Mathematics* 646, American Mathematical Society, 2015, pp. 127-167. MR 3384978.

This later survey/article is cited by recent papers as collecting existence and nonexistence results for stable submanifolds and currents in positively curved settings. It provides context and partial results, not a full proof of the strict $1/4$ -pinched conjecture.

- **Giada Franz; Federico Trinca, “On the Stability of Minimal Submanifolds in Conformal Spheres,”** *Journal of Geometric Analysis* 33, article 335, 2023. DOI: 10.1007/s12220-023-01398-4. arXiv:2211.01106. URL: <https://link.springer.com/article/10.1007/s12220-023-01398-4>.

This recent paper explicitly states that the Lawson-Simons conjecture in full generality is still open. It proves that an n -dimensional Riemannian sphere conformal to the round sphere and δ -pinched contains no closed stable minimal k -submanifold for $2 \leq k \leq n - \delta^{-1}$. For $\delta > 1/4$, this verifies part of the conjecture only for conformal spheres and only in the indicated dimension range.

- **Hang Chen, “A conformal invariant and its application to the nonexistence of minimal submanifolds,”** arXiv:2310.09724, version 2, 2024. URL: <https://arxiv.org/abs/2310.09724>.

This preprint also states that the Lawson-Simons conjecture is still open and proves further nonexistence theorems using a conformal invariant under nonnegative sectional curvature assumptions. The added conformal-invariant smallness and dimension-range restrictions make it partial progress, not a full resolution.

1.4.3.4. What remains unresolved

What remains open is the exact intrinsic statement for every compact simply connected strictly $1/4$ -pinched Riemannian n -manifold and every p , $1 \leq p \leq n - 1$, with no conformal-to-round, stronger-pinching, hypersurface, extrinsic principal-curvature, low-dimensional, or smooth-submanifold restriction. The literature found gives partial positive evidence but does not prove the full absence of nonzero closed stable rectifiable p -currents in all cases asked by the problem statement.

1.4.4. Remaining open problems recorded by the database

1.4.4.1. Open problem 1

Let (S^n, g) have sectional curvatures $\frac{1}{4} < K \leq 1$. Can it contain a closed stable minimal surface of arbitrary topology (equivalently, can one prove instability when the two-dimensional minimal surface is not assumed to be a two-sphere)?

Reference supplied by the database. Daren Cheng, Chao Li, and Christos Mantoulidis, “Topics in Differential Geometry,” lecture notes for Richard Schoen’s Stanford Math 286 course, Spring 2015, Conjecture 5.4 and the paragraph following it (“otherwise open, even when Σ is a general two dimensional surface”), https://web.stanford.edu/~rchlch/notes/Schoen_286_spring_2015.pdf; current partial-result/status reference: Giada Franz and Federico Trinca, “On the Stability of Minimal Submanifolds in Conformal Spheres,” The Journal of Geometric Analysis 33 (2023), no. 10, Article 335, <https://doi.org/10.1007/s12220-023-01398-4>.

1.4.5. Primary-source pointer

The originating ICM record is *Geometric Aspects of the Generalized Plateau Problem* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf#page=15>.

1.5. 5. Some Recent Advances in the Multidimensional Parametric Calculus of Variations [1974-geom-23]

Record: 1974-geom-23

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Some Recent Advances in the Multidimensional Parametric Calculus of Variations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf#page=239>

1.5.1. Problem statement

For every compact Riemannian manifold N and every integer k with $0 < k < \dim N$, does N contain a closed k -dimensional submanifold $M \subset N$ whose mean curvature vector is identically zero at every point of M ?

1.5.2. Historical background, literature, and current progress

1.5.3. Verdict: partially resolved

1.5.3.1. Precise problem checked

The problem asks whether every compact Riemannian manifold N^n , for every integer k with $0 < k < n$, contains a closed smooth k -dimensional submanifold $M \subset N$ whose mean curvature vector vanishes identically. Equivalently, it asks for a closed smooth minimal submanifold of every positive dimension below the ambient dimension, in every compact ambient Riemannian manifold.

1.5.3.2. Literature status

The problem is not fully resolved in this smooth-submanifold form. The literature gives major partial results: closed geodesics for $k = 1$, generalized stationary integral varifolds in all dimensions, and smooth embedded minimal hypersurfaces in low ambient dimensions. However, the all- N , all- k , smooth closed submanifold statement is not covered by the known general theorems. The main unresolved gap is higher codimension and/or high-dimensional smooth regularity: stationary varifolds or singular hypersurfaces are not the same as closed smooth submanifolds.

1.5.3.3. Key sources and progress

1. Title: "Some Recent Advances in the Multidimensional Parametric Calculus of Variations"
Authors: William K. Allard

Venue: Proceedings of the International Congress of Mathematicians, Vancouver 1974, Canadian Mathematical Congress, 1975, pp. 231-236

DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf>

Relevance: This appears to be the original source for the question. Allard states the problem of whether every compact Riemannian manifold contains a closed submanifold of prescribed dimension with zero mean curvature, and says the answer is not known in general. The same source records Almgren's generalized existence theorem and Pitts' then-new 3-dimensional result, so it already distinguishes smooth submanifolds from rectifiable/generalized solutions.

2. Title: "Variational problems on closed manifolds"

Authors: L. A. Lyusternik and A. I. Fet

Venue: Doklady Akademii Nauk SSSR (N.S.) 81, 1951, pp. 17-18

DOI/arXiv/URL: no DOI/arXiv found in live search

Relevance: This is the classical closed-geodesic theorem. It gives the $k = 1$ analogue: every compact Riemannian manifold has a nonconstant closed geodesic, hence a one-dimensional minimal object. It does not address $k \geq 2$.

3. Title: "The theory of varifolds: a variational calculus in the large for the k -dimensional area integrand"

Authors: Frederick J. Almgren, Jr.

Venue: Institute for Advanced Study / Princeton mimeographed notes, 1964/1965

DOI/arXiv/URL: <https://albert.ias.edu/20.500.12111/8025>

Relevance: Almgren's min-max theory gives nonzero stationary integral k -varifolds in compact Riemannian manifolds. This is broad progress for all k , but a stationary integral varifold may be singular and therefore does not prove the existence of a closed smooth k -submanifold.

4. Title: "Existence and Regularity of Minimal Surfaces on Riemannian Manifolds"

Authors: Jon T. Pitts

Venue: Mathematical Notes, vol. 27, Princeton University Press / University of Tokyo Press, 1981

DOI/arXiv/URL: <https://www.jstor.org/stable/j.ctt7zv66w>

Relevance: Pitts proves a general existence theorem for nonzero stationary integral varifolds with an almost-minimizing property, and a regular embedded minimal hypersurface theorem in a restricted dimension range. This is not the full statement because the all-dimensional result is in the varifold category, while the smooth regular theorem is codimension one and dimension-restricted.

5. Title: "Regularity of stable minimal hypersurfaces"

Authors: Richard Schoen and Leon Simon

Venue: Communications on Pure and Applied Mathematics 34(6), 1981, pp. 741-797

DOI/arXiv/URL: <https://doi.org/10.1002/cpa.3160340603>

Relevance: Together with Almgren-Pitts min-max theory, this gives closed minimal hypersurfaces with optimal regularity: smooth outside a singular set of codimension at least 7. In ambient dimension at most 7 this yields a smooth embedded closed minimal hypersurface. In higher ambient dimensions it allows singularities, so it does not resolve the smooth-submanifold question as stated.

6. Title: "Existence of infinitely many minimal hypersurfaces in closed manifolds"

Authors: Antoine Song

Venue: Annals of Mathematics 197(3), 2023, pp. 859-895

DOI/arXiv/URL: <https://doi.org/10.4007/annals.2023.197.3.1> ;

arXiv:1806.08816

Relevance: Song proves that every closed Riemannian manifold of dimension 3 through 7 contains infinitely many smoothly embedded closed minimal hypersurfaces. This is a strong resolution of the hypersurface case in low ambient dimensions, but it covers only $k = n - 1$ and $3 \leq n \leq 7$.

7. Title: “A viscosity method in the min-max theory of minimal surfaces”

Authors: Tristan Riviere

Venue: Publications Mathematiques de l’IHES 126, 2017, pp. 177-246

DOI/arXiv/URL: <https://doi.org/10.1007/s10240-017-0094-z>

Relevance: This is important progress for two-dimensional minimal surfaces/min-max in arbitrary codimension. It does not prove the existence of a closed smooth k -dimensional minimal submanifold for every k in every compact Riemannian manifold.

8. Title: “A proof of the multiplicity one conjecture for min-max minimal surfaces in arbitrary codimension”

Authors: Alessandro Pigati and Tristan Riviere

Venue: Duke Mathematical Journal 169(11), 2020, pp. 2005-2044

DOI/arXiv/URL: <https://doi.org/10.1215/00127094-2020-0002>

Relevance: This advances the two-dimensional arbitrary-codimension min-max theory, but it is not an all-dimensions existence theorem for smooth closed k -submanifolds.

9. Title: “Minimal submanifolds from the abelian Higgs model”

Authors: Alessandro Pigati and Daniel Stern

Venue: Inventiones Mathematicae 223, 2021, pp. 1027-1095

DOI/arXiv/URL: <https://doi.org/10.1007/s00222-020-01000-6>

Relevance: This produces codimension-two stationary integral varifold limits from a variational gauge-theoretic model. It is significant higher-codimension progress, but the output is again a generalized minimal object and does not settle the smooth closed submanifold existence question for all k .

1.5.3.4. What remains unresolved

No citable result found in the live search proves that for every compact Riemannian N^n and every $1 < k < n - 1$ there is a closed smooth k -dimensional minimal submanifold $M \subset N$. Nor do the high-dimensional hypersurface results prove the stated smooth-submanifold conclusion in ambient dimension at least 8, because the general min-max object may have a singular set. Therefore the exact problem is only partially resolved.

1.5.4. Remaining open problems recorded by the database

1.5.4.1. Open problem 1

Prove that Almgren’s higher-codimension min–max varifolds have the same regularity as area-minimizing integral currents; in particular, prove the corresponding regularity statement when the min–max submanifolds are two-dimensional.

Reference supplied by the database. Da Rong Cheng and Xin Zhou, “Existence of classical minimal surfaces in 4 and 5-manifolds,” arXiv:2606.24754 [math.DG], 2026, Introduc-

tion, subsection “Context” (the explicitly stated higher-codimension regularity conjecture).
<https://arxiv.org/abs/2606.24754>

1.5.5. Primary-source pointer

The originating ICM record is *Some Recent Advances in the Multidimensional Parametric Calculus of Variations* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf#page=239>.

1.6. 6. The Generalized Cartan-Hadamard Conjecture [1978-geom-25]

Record: 1978-geom-25

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: reviewed (reviewed flag: yes)

Original source: ICM 1978: Isoperimetric inequalities and eigenvalues of the Laplacian

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>

1.6.1. Problem statement

In dimension 2, if D is a simply connected domain on a Riemannian surface, $A = \text{area}(D)$, $L = \text{length}(\partial D)$, K is the Gauss curvature, and $M = \sup_D K$, then

$$L^2 \geq 4\pi A - MA^2,$$

with equality holds only when $K \equiv M$ on D and D is a geodesic disk.

Determine whether the higher-dimensional isoperimetric analogue holds for domains in Riemannian manifolds of dimension $n \geq 3$.

Determine whether the corresponding sharp isoperimetric comparison remains valid in dimension $n \geq 3$, using n -dimensional volume for D , $(n-1)$ -dimensional measure for ∂D , and the appropriate upper sectional-curvature bound in place of M , when the ambient Riemannian curvature is not constant.

1.6.2. Historical background, literature, and current progress

1.6.3. Verdict: partially resolved

1.6.3.1. Precise problem checked

The problem asks for the higher-dimensional analogue of the Bol-Fiala-Weil surface inequality

$$L^2 \geq 4\pi A - (\sup_D K)A^2$$

for a simply connected surface domain. In modern terminology the natural nonpositive-curvature formulation is the generalized Cartan-Hadamard isoperimetric comparison: if M^n is complete and simply connected with sectional curvature $K \leq \kappa \leq 0$, whether every domain $\Omega \subset M$ satisfy

$$|\partial\Omega| \geq I_{n,\kappa}(|\Omega|),$$

where $I_{n,\kappa}$ is the isoperimetric profile of the simply connected space form of constant curvature κ .

1.6.3.2. Current literature status

The problem is not fully resolved in the generality asked. The generalized Cartan-Hadamard comparison is known in important cases, but no published or otherwise citable theorem I found covers all $n \geq 3$, all volumes, and the full variable sectional-curvature comparison. Moreover, the naive higher-dimensional analogue for arbitrary Riemannian balls/topological balls is false.

Key references and what they prove

1. Robert Osserman, "The isoperimetric inequality", *Bulletin of the American Mathematical Society* 84, no. 6 (1978), 1182-1238, DOI: 10.1090/S0002-9904-1978-14553-4.
 - Records the two-dimensional variable-curvature inequalities, including the form saying that a simply connected surface domain has boundary length at least that of the model constant-curvature disk with curvature $\sup_D K$. This is the surface result whose higher-dimensional analogue is at issue.
2. Robert Osserman, "Isoperimetric inequalities and eigenvalues of the Laplacian", *Proceedings of the International Congress of Mathematicians, Helsinki 1978*, Vol. I, pp. 435-442, Acad. Sci. Fennica, Helsinki, 1980.
 - Original source connected to the dataset entry; discusses these isoperimetric inequalities in relation to eigenvalue estimates.
3. Andre Weil, "Sur les surfaces a courbure negative", *Comptes Rendus de l'Academie des Sciences de Paris* 182 (1926), 1069-1071.
 - Proves the $n = 2$, $\kappa = 0$ Cartan-Hadamard/Weil surface case.
4. Gerrit Bol, "Isoperimetrische Ungleichungen fur Bereiche auf Flächen", *Jahresbericht der Deutschen Mathematiker-Vereinigung* 51 (1941), 219-257.
 - Gives the sharp two-dimensional comparison for $\kappa \neq 0$, i.e. the surface model-space comparison underlying the stated formula.
5. Christopher B. Croke, "A sharp four-dimensional isoperimetric inequality", *Commentarii Mathematici Helvetici* 59, no. 2 (1984), 187-192.
 - Proves the sharp Euclidean Cartan-Hadamard inequality in dimension $n = 4$ for $\kappa = 0$. This is a major special case, not the full variable-curvature comparison for all dimensions or all κ .
6. Bruce Kleiner, "An isoperimetric comparison theorem", *Inventiones Mathematicae* 108, no. 1 (1992), 37-47.
 - Proves the generalized Cartan-Hadamard comparison in dimension $n = 3$ for all $\kappa \leq 0$. This fully resolves the three-dimensional Cartan-Hadamard version but not $n \geq 4$ for $\kappa < 0$, nor all higher dimensions.
7. Frank Morgan and David L. Johnson, "Some sharp isoperimetric theorems for Riemannian manifolds", *Indiana University Mathematics Journal* 49, no. 3 (2000), 1017-1041.
 - Establishes sharp small-volume isoperimetric results under curvature hypotheses. These are local/small-domain results and do not resolve the global problem.

8. Benoit R. Kloeckner and Greg Kuperberg, “The Cartan-Hadamard conjecture and the Little Prince”, *Revista Matematica Iberoamericana* 35, no. 4 (2019), 1195-1258, DOI: 10.4171/RMI/1082, arXiv:1303.3115.
 - States the generalized Cartan-Hadamard conjecture explicitly as the comparison for complete simply connected n -manifolds with $K \leq \kappa \leq 0$. Their results give unified proofs in dimensions 2 and 4 for $\kappa = 0$, partial $n = 4$ results for $\kappa < 0$ with an additional chord/volume smallness condition, and related positive-curvature versions under unique-geodesic and volume restrictions.
 - They also prove counterexamples to a naive extension: for every $\varepsilon > 0$, there are Riemannian 3-balls with nearly pinched negative curvature, bounded boundary volume, and arbitrarily large volume. Thus topology plus an upper curvature bound alone cannot be the right higher-dimensional analogue.
9. Mohammad Ghomi and Joel Spruck, “Total Curvature and the Isoperimetric Inequality in Cartan-Hadamard Manifolds”, *Journal of Geometric Analysis* 32, article 50 (2022), DOI: 10.1007/s12220-021-00801-2, arXiv:1908.09814.
 - Develops total-curvature comparison machinery and applications to the isoperimetric problem in nonpositive curvature. The accepted abstract does not claim a full proof of the Cartan-Hadamard conjecture; later work still treats the all-dimensional general case as open.
10. Mohammad Ghomi and John Ioannis Stavroulakis, “A Local Isoperimetric Inequality for Balls with Nonpositive Curvature”, *International Mathematics Research Notices* 2026, no. 6, rnag043, DOI: 10.1093/imrn/rnag043, arXiv:2505.11971.
 - Proves a sharp local version: small perturbations of the Euclidean ball metric to nonpositive curvature do not reduce the isoperimetric ratio. This is local and does not settle the global comparison.
11. Mohammad Ghomi, “Isoperimetric and total curvature inequalities in Cartan-Hadamard manifolds with nullity”, arXiv:2605.24638 (2026 preprint).
 - Proves the Euclidean Cartan-Hadamard isoperimetric inequality for Cartan-Hadamard manifolds whose nullity index is at least $n - 3$. This gives new high-dimensional classes but still imposes extra structure and does not cover arbitrary variable sectional curvature.

1.6.3.3. What remains unresolved

No citable result found proves the full sharp higher-dimensional model-space comparison for all domains in all dimensions $n \geq 3$ under only the appropriate upper sectional-curvature bound. The exact Cartan-Hadamard/generalized Cartan-Hadamard problem remains open beyond the known low-dimensional and special-geometry cases, especially for the full $\kappa < 0$ comparison in higher dimensions and for unrestricted $n \geq 5$. The broader naive version without ambient Cartan-Hadamard or unique-geodesic hypotheses is known to be false.

1.6.4. Remaining open problems recorded by the database

1.6.4.1. Open problem 1

Let (M^5, g) be a complete simply connected Riemannian 5-manifold with nonpositive sectional curvature. Prove that every bounded sufficiently regular domain $\Omega \subset M$ satisfies the sharp Euclidean isoperimetric inequality

$$|\partial\Omega|^5 \geq 5^5 \omega_5 |\Omega|^4,$$

with equality only for a Euclidean ball in a flat region.

Reference supplied by the database. Felix Schulze, “Optimal Isoperimetric Inequalities for Surfaces in Any Codimension in Cartan–Hadamard Manifolds,” *Geometric and Functional Analysis* 30 (2020), 255–288, Introduction, Cartan–Hadamard Conjecture and the statement that the cases beyond dimensions (2,3) and the flat (4)-dimensional case remain open. DOI: 10.1007/s00039-020-00522-8. <https://doi.org/10.1007/s00039-020-00522-8>

1.6.4.2. Open problem 2

Let (M^4, g) be complete and simply connected with sectional curvature bounded above by $-\kappa < 0$. Prove that every bounded sufficiently regular domain $\Omega \subset M$ has boundary 3-volume at least that of the geodesic ball in the 4-dimensional space form of constant sectional curvature $-\kappa$ having the same 4-volume as Ω , with the corresponding rigidity in the equality case.

Reference supplied by the database. Felix Schulze, “Optimal Isoperimetric Inequalities for Surfaces in Any Codimension in Cartan–Hadamard Manifolds,” *Geometric and Functional Analysis* 30 (2020), 255–288, Introduction, strengthened Cartan–Hadamard conjecture and the statement that only a partial result is known when $(n=4)$ and $\kappa \neq 0$. DOI: 10.1007/s00039-020-00522-8. <https://doi.org/10.1007/s00039-020-00522-8>

1.6.5. Primary-source pointer

The originating ICM record is *Isoperimetric inequalities and eigenvalues of the Laplacian* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>.

1.7. 7. The role of partial differential equations in differential geometry [1978-geom-28]

Record: 1978-geom-28

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: The role of partial differential equations in differential geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=237>

1.7.1. Problem statement

Let (M^n, g) be a compact Riemannian manifold whose scalar curvature is nonnegative at every point. Suppose that M is topologically covered by Euclidean space, equivalently that its universal cover is homeomorphic to $\mathbb{R}^n$. Must (M, g) be flat, i.e. have identically zero Riemannian curvature tensor?

1.7.2. Historical background, literature, and current progress

1.7.3. Status check

Verdict: partially resolved.

1.7.3.1. Problem identified

The question is Yau's scalar-curvature rigidity problem: if a compact Riemannian manifold (M^n, g) has scalar curvature $Sc_g \geq 0$ and its universal cover is homeomorphic to $\mathbb{R}^n$, must the full Riemann curvature tensor vanish identically?

Equivalently, this asks for scalar-curvature rigidity of closed manifolds topologically covered by Euclidean space. Such manifolds are aspherical, so the question is closely tied to the Gromov-Lawson/Schoen-Yau conjecture that closed aspherical manifolds admit no positive scalar curvature metric.

1.7.3.2. Why the exact problem is not fully resolved

No published or otherwise citable theorem found covers all dimensions and all compact manifolds whose universal cover is homeomorphic to $\mathbb{R}^n$. Recent literature still states that the aspherical positive-scalar-curvature conjecture is widely open in dimensions greater than five. Since an all-dimensional solution of that conjecture, together with standard scalar-flat rigidity, would imply an affirmative answer to this problem, the problem remains open in general, especially in dimensions $n \geq 6$.

The problem is not fully open, however: substantial classes are known, including all dimensions for enlargeable manifolds and all closed aspherical manifolds in dimensions 3, 4, 5.

1.7.3.3. Relevant literature and progress

Original source Title: The Role of Partial Differential Equations in Differential Geometry

Authors: Shing-Tung Yau

Venue/year: Proceedings of the International Congress of Mathematicians, Helsinki 1978, Acad. Sci. Fennica, Helsinki, 1980, pp. 237-250

URL: https://archive.ymsc.tsinghua.edu.cn/pacm_download/59/11114-Shing-Tung_Yau_217.pdf

Relevance: On p. 242 of the proceedings version, Yau asks: "If a compact manifold with nonnegative scalar curvature is covered by the euclidean space topologically, is it a flat manifold?" This is the exact problem statement.

Scalar-flat rigidity reduction Title: Scalar curvature and conformal deformation of Riemannian structure

Authors: Jerry L. Kazdan and F. W. Warner

Venue/year: Journal of Differential Geometry 10 (1975), 113-134

DOI/URL: <https://doi.org/10.4310/jdg/1214432678>

Relevance: Provides the scalar-curvature deformation background used in the standard reduction: if a closed manifold has a metric with $Sc \geq 0$ and $Sc > 0$ somewhere, one obtains a positive scalar curvature metric.

Title: Une stratification de l'espace des structures riemanniennes

Author: Jean-Pierre Bourguignon

Venue/year: Compositio Mathematica 30 (1975), no. 1, 1-41

URL: <http://eudml.org/doc/89245>

Relevance: Contains the scalar-flat deformation rigidity often used in this context: on a closed manifold admitting no positive scalar curvature metric, a nonnegative-scalar-curvature metric must be Ricci-flat.

Title: The structure of compact Ricci-flat Riemannian manifolds

Authors: Arthur E. Fischer and Joseph A. Wolf

Venue/year: Journal of Differential Geometry 10 (1975), 277-288

DOI/URL: <https://doi.org/10.4310/jdg/1214432794>

Relevance: Gives the compact Ricci-flat structure theorem. Combined with the Cheeger-Gromoll splitting theorem, a compact Ricci-flat manifold whose universal cover is $\mathbb{R}^n$ is flat. Thus positive-scalar-curvature obstructions for the underlying topology yield the desired nonnegative-scalar-curvature rigidity.

Enlargeable cases Title: Spin and scalar curvature in the presence of a fundamental group.

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Authors: Mikhail Gromov and H. Blaine Lawson, Jr.

Venue/year: Annals of Mathematics (2) 111 (1980), no. 2, 209-230

DOI/URL: <https://doi.org/10.2307/1971198>

Relevance: Theorem A proves that compact enlargeable manifolds admit no positive scalar curvature metric and that any metric with nonnegative scalar curvature on such a manifold is flat. This covers tori and many standard aspherical examples, but enlargeability is an extra hypothesis and does not cover every compact manifold topologically covered by $\mathbb{R}^n$.

Dimension 3 Title: Existence of incompressible minimal surfaces and the topology of three-dimensional manifolds with nonnegative scalar curvature

Authors: Richard Schoen and Shing-Tung Yau

Venue/year: Annals of Mathematics (2) 110 (1979), no. 1, 127-142

DOI/URL: <https://doi.org/10.2307/1971247>

Relevance: Foundational minimal-surface obstruction results for 3-manifolds with nonnegative/positive scalar curvature.

Title: Positive scalar curvature and the Dirac operator on complete Riemannian manifolds

Authors: Mikhail Gromov and H. Blaine Lawson, Jr.

Venue/year: Publications Mathématiques de l’IHES 58 (1983), 83-196

DOI/URL: <https://doi.org/10.1007/BF02953774>

Relevance: Later accounts cite these results, together with Schoen-Yau’s 3-manifold work, as verifying the closed aspherical case in dimension 3. Hence the problem statement is settled in dimension 3.

Dimensions 4 and 5 Title: Generalized soap bubbles and the topology of manifolds with positive scalar curvature

Authors: Otis Chodosh and Chao Li

Venue/year: Annals of Mathematics (2) 199 (2024), no. 2, 707-740

DOI/arXiv/URL: <https://doi.org/10.4007/annals.2024.199.2.3> ; arXiv:2008.11888, <https://arxiv.org/abs/2008.11888>

Relevance: Proves that for $n \in \{4, 5\}$, a closed aspherical n -manifold admits no positive scalar curvature metric. The final arXiv version states the rigidity form as Theorem 2: any metric of nonnegative scalar curvature on such a manifold is flat. Since universal cover $\mathbb{R}^n$ implies aspherical, this resolves the question in dimensions 4 and 5.

Title: No metrics with Positive Scalar Curvatures on Aspherical 5-Manifolds

Author: Misha Gromov

Venue/year: arXiv preprint, 2020

arXiv/URL: arXiv:2009.05332, <https://arxiv.org/abs/2009.05332>

Relevance: Independently proves the no-positive-scalar-curvature statement for closed aspherical 5-manifolds via μ -bubble methods. This is consistent with, but not broader than, the later published Chodosh-Li result for the problem.

Recent status and remaining gap Title: Positive scalar curvature metrics and aspherical summands

Authors: Shuli Chen, Jianchun Chu and Jintian Zhu

Venue/year: Accepted by Duke Mathematical Journal; arXiv preprint, submitted 2023, final version 2025

arXiv/URL: arXiv:2312.04698, <https://arxiv.org/abs/2312.04698>

Relevance: The introduction summarizes the current status: dimension 3 was verified by Gromov-Lawson, dimensions 4 and 5 by Chodosh-Li and Gromov, but “the aspherical conjecture is still widely open in dimensions greater than five.” The paper proves further noncompact connected-sum results for dimensions 3, 4, 5, not the all-dimensional compact rigidity asked in this problem.

1.7.3.4. What remains unresolved

The remaining unresolved part is the full all-dimensional statement for arbitrary compact n -manifolds, $n \geq 6$, whose universal cover is homeomorphic to $\mathbb{R}^n$. Known positive results require dimension bounds or extra hypotheses such as enlargeability, spin/index-theoretic assumptions, nonpositive sectional curvature, or special group-theoretic conditions. No source found proves that every compact manifold topologically covered by $\mathbb{R}^n$ and carrying a metric with $Sc \geq 0$ must be flat.

1.7.4. Remaining open problems recorded by the database

1.7.4.1. Open problem 1

Prove that no closed aspherical (6)-manifold admits a Riemannian metric of positive scalar curvature.

Reference supplied by the database. Liam Mazurowski, Tongrui Wang, and Xuan Yao, “On the Topology of Manifolds with Positive Intermediate Curvature,” *Advances in Mathematics* 485 (2026), Article 110731, Introduction, $(K\pi, 1)$ conjecture in dimension (6) (equivalently the unresolved $(n=6, m=5)$ scalar-curvature case of the paper’s conjecture). DOI: 10.1016/j.aim.2025.110731; arXiv:2503.13815. <https://doi.org/10.1016/j.aim.2025.110731>

1.7.5. Primary-source pointer

The originating ICM record is *The role of partial differential equations in differential geometry* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=237>.

1.8. 8. The role of partial differential equations in differential geometry [1978-geom-29]

Record: 1978-geom-29

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: The role of partial differential equations in differential geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=237>

1.8.1. Problem statement

Does every four dimensional compact simply-connected Riemannian manifold with zero Ricci curvature admit a Kähler structure?

1.8.2. Historical background, literature, and current progress

1.8.3. Status check

Verdict: partially resolved.

1.8.3.1. Precise problem

The problem asks whether every compact simply-connected 4-dimensional Riemannian manifold (M, g) with $\text{Ric}(g) = 0$ must admit a Kähler structure compatible with the Ricci-flat metric. In dimension 4 this is essentially the question whether such a metric must be locally hyperkähler, equivalently have reduced/special holonomy $SU(2) = \text{Sp}(1)$, rather than generic holonomy $SO(4)$.

1.8.3.2. Original source

- Title: “The Role of Partial Differential Equations in Differential Geometry”
- Author: Shing-Tung Yau
- Venue/year: Proceedings of the International Congress of Mathematicians, Helsinki 1978; published by Acad. Sci. Fennica, Helsinki, 1980, pp. 237-250
- DOI/arXiv/URL: https://archive.ymsc.tsinghua.edu.cn/pacm_download/59/11114-Shing-Tung_Yau_217.pdf
- Relevance: Yau states exactly the question: “Does every four dimensional compact simply-connected Riemannian manifold with zero Ricci curvature admit a Kahler structure?” He immediately records only partial positive evidence: Hitchin’s observation covers K3 surfaces and, more generally, compact spin manifolds with nonzero index. This is not a full solution of all simply-connected compact Ricci-flat 4-manifolds.

1.8.3.3. Partial progress and related literature

- Title: “Harmonic spinors”
- Author: Nigel J. Hitchin
- Venue/year: Advances in Mathematics 14 (1974), 1-55
- DOI/arXiv/URL: DOI 10.1016/0001-8708(74)90021-8; author listing: <https://people.maths.ox.ac.uk/hitchin/files/pubs.html>
- Relevance: The Lichnerowicz formula implies that on a scalar-flat/Ricci-flat spin manifold, harmonic spinors forced by nonzero Dirac index are parallel. In the present 4-dimensional simply-connected setting, this forces special holonomy and hence a hyperkähler/Kähler metric. This proves the nonzero-index spin cases mentioned by Yau, but leaves spin zero-index and non-spin possibilities outside its scope.
- Title: “On compact four-dimensional Einstein manifolds”
- Author: Nigel J. Hitchin
- Venue/year: Journal of Differential Geometry 9 (1974), 435-442
- DOI/arXiv/URL: DOI 10.4310/jdg/1214432419
- Relevance: Hitchin’s 4-dimensional Einstein results, together with the spinorial argument above, give strong restrictions and the K3/equality cases, but they do not exclude all compact simply-connected Ricci-flat 4-manifolds with generic holonomy.
- Title: “On the Ricci curvature of a compact Kähler manifold and the complex Monge-Ampère equation, I”
- Author: Shing-Tung Yau
- Venue/year: Communications on Pure and Applied Mathematics 31 (1978), 339-411
- DOI/arXiv/URL: DOI 10.1002/cpa.3160310304
- Relevance: Yau’s solution of the Calabi conjecture constructs Ricci-flat Kähler metrics on compact Kähler manifolds with vanishing first Chern class, in particular on K3 surfaces. This supplies the main known simply-connected compact 4-dimensional examples, but it is an existence theorem under a pre-existing Kähler/complex hypothesis, not a converse for arbitrary Ricci-flat Riemannian 4-manifolds.
- Title: “Conformal geometry of Ricci flat 4-manifolds”
- Author: Mitsuhiro Itoh
- Venue/year: Kodai Mathematical Journal 17 (1994), 179-200
- DOI/arXiv/URL: DOI 10.2996/kmj/1138039959
- Relevance: For the underlying K3 smooth 4-manifold, Ricci-flat metrics are shown to be Ricci-flat Kähler/hyperkähler and the moduli is described by period data. This resolves the question on the K3 diffeomorphism type, but assumes the manifold is K3 and therefore does not settle the general question.
- Title: “On noncollapsed almost Ricci-flat 4-manifolds”
- Authors: Vitali Kapovitch and John Lott
- Venue/year: American Journal of Mathematics 141 (2019), 737-755
- DOI/arXiv/URL: arXiv:1606.09041, <https://arxiv.org/abs/1606.09041>

- Relevance: Under noncollapsing, diameter/volume normalization, spin, nonzero A -hat genus, and sufficiently small Ricci curvature, the manifold is diffeomorphic to $K3$; the paper also notes that known closed Ricci-flat 4-manifolds are flat or finitely covered by $K3$. These are strong stability/near-Ricci-flat results, but they impose extra hypotheses and do not classify all exact Ricci-flat simply-connected 4-manifolds.
- Title: “Gluing Eguchi-Hanson Metrics and a Question of Page”
- Authors: Simon Brendle and Nikolaos Kapouleas
- Venue/year: Communications on Pure and Applied Mathematics 70 (2017), 1366-1401
- DOI/arXiv/URL: DOI 10.1002/cpa.21678; arXiv:1405.0056, <https://arxiv.org/abs/1405.0056>
- Relevance: Page asked whether the Kummer construction could be modified by reversing orientations of some Eguchi-Hanson pieces; success would have produced compact Einstein/Ricci-flat metrics that are neither Kähler nor self-dual. Brendle and Kapouleas find a nonvanishing obstruction in a highly symmetric mixed-orientation configuration. This is negative evidence against one proposed route to a counterexample, not a proof of the general statement.
- Title: “Higher order obstructions to the desingularization of Einstein metrics”
- Author: Tristan Ozuch
- Venue/year: Cambridge Journal of Mathematics 9 (4) (2022), 901-976
- DOI/arXiv/URL: arXiv:2012.13316, <https://arxiv.org/abs/2012.13316>
- Relevance: Ozuch studies higher-order obstructions to desingularizing Einstein orbifolds and applies them to Page’s proposed Ricci-flat but non-Kähler Kummer-type construction, finding many obstructions. Again, this obstructs a major construction strategy but does not rule out every possible generic-holonomy Ricci-flat 4-manifold.
- Title: “Supersymmetry, Ricci Flat Manifolds and the String Landscape”
- Author: Bobby Samir Acharya
- Venue/year: Journal of High Energy Physics 2020, article 128; arXiv version 2020
- DOI/arXiv/URL: arXiv:1906.06886, <https://arxiv.org/abs/1906.06886>
- Relevance: This survey formulates the broader conjecture that compact simply-connected Ricci-flat manifolds have special holonomy, explicitly noting the open possibility of compact simply-connected Ricci-flat manifolds with generic holonomy. In dimension 4 it identifies the expected special-holonomy case as $K3$ and records remaining unsettled cases such as possible Ricci-flat metrics on $S^2 \times S^2$ or other zero-index/non-spin topologies.
- Title: “Geometry of Gravitational Instantons”
- Authors: Mingyang Li and Song Sun
- Venue/year: arXiv preprint, 2026
- DOI/arXiv/URL: arXiv:2606.10443, <https://arxiv.org/abs/2606.10443>
- Relevance: This current survey states a closely equivalent open problem: “Is there a compact simply-connected Ricci-flat 4-manifold which is not diffeomorphic to the $K3$ manifold?” It notes that the Page/Gibbons-Pope mixed-orientation Kummer approach has only produced negative obstruction results. This confirms that the

literature still treats the general compact simply-connected Ricci-flat 4-dimensional classification/Kähler question as unresolved.

1.8.3.4. Applicability check

The available positive theorems do not cover the full statement in the problem statement. The K3 and spin nonzero-index cases imply a Kähler/hyperkähler structure, and Yau's theorem constructs Ricci-flat Kähler examples once a suitable complex/Kähler structure is already present. However, none of the cited results proves that every compact simply-connected Ricci-flat 4-manifold has K3 topology or reduced holonomy. The remaining obstruction is exactly the possible existence of a compact simply-connected Ricci-flat 4-manifold with generic holonomy $SO(4)$ (or, equivalently in this setting, a Ricci-flat metric not forced to be Kähler/hyperkähler). Current literature treats this as an open possibility and supplies only partial restrictions and failed construction attempts.

Therefore the problem is partially resolved, not fully resolved.

1.8.4. Remaining open problems recorded by the database

1.8.4.1. Open problem 1

Does the smooth (4)-sphere S^4 admit a Ricci-flat Riemannian metric?

Reference supplied by the database. Matthias Wink (joint work with Jan Nienhaus), "Einstein Metrics on Spheres," in Aaron Naber, André Neves, Eleonora Di Nezza, and Burkhard Wilking (organizers), "Geometrie," *Oberwolfach Reports* 21 (2024), No. 28, pp. 1604–1606, explicit open question on p. 1604. DOI: 10.4171/OWR/2024/28. <https://doi.org/10.4171/OWR/2024/28>

1.8.5. Primary-source pointer

The originating ICM record is *The role of partial differential equations in differential geometry* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=237>.

1.9. 9. Some recent developments in complex differential geometry [1983-geom-32]

Record: 1983-geom-32

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1983: Some recent developments in complex differential geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=349>

1.9.1. Problem statement

Every complete non-compact Kähler manifold of complex dimension n with positive sectional curvature is biholomorphic to $\mathbb{C}^n$.

1.9.2. Historical background, literature, and current progress

1.9.3. Status: partially resolved

1.9.3.1. Precise problem

The problem asks the Greene-Wu/Yau weak uniformization problem: whether every complete noncompact Kähler manifold of arbitrary complex dimension n , with positive real sectional curvature, must be biholomorphic to $\mathbb{C}^n$.

This is not merely the better-known positive holomorphic bisectional curvature version. Results for bisectional curvature with extra hypotheses such as maximal volume growth do not settle the problem statement, because the statement assumes no volume-growth, curvature-decay, bounded-curvature, or finite-Chern-number condition.

1.9.3.2. Verdict

Partially resolved. The statement is now proved in complex dimension 2, and dimension 1 is classical, but I found no citable theorem proving the full arbitrary-dimensional statement under only positive sectional curvature.

1.9.3.3. Progress and key literature

1. Yum-Tong Siu, "Some Recent Developments in Complex Differential Geometry." Proceedings of the International Congress of Mathematicians, Warsaw 1983, Vol. 1, PWN/North-Holland, 1984, pp. 287-297. URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf>

This is the dataset's original source context. Later papers identify the positive-sectional-curvature statement as the Greene-Wu/Yau weak form of Yau's uniformization conjecture.

2. **Ved Datar, Vamsi Pritham Pingali, Harish Seshadri, "Uniformisation of complete Kähler surfaces with positive sectional curvature."** arXiv preprint, 2026. arXiv:2604.11220. DOI/URL: <https://doi.org/10.48550/arXiv.2604.11220>

Theorem 1 proves: if (X, ω) is a complete noncompact Kähler surface with positive sectional curvature, then X is biholomorphic to $\mathbb{C}^2$. This exactly resolves the problem statement in complex dimension 2, with no extra assumptions at infinity. It does not cover $n \geq 3$.

3. **Ved V. Datar, Vamsi Pritham Pingali, Harish Seshadri, "The top Yau-Yang conjecture for Kähler manifolds with positive sectional curvature."** arXiv preprint, 2026. arXiv:2606.19806. DOI/URL: <https://doi.org/10.48550/arXiv.2606.19806>

Theorem 1.1 proves finite top Ricci mass $\int_X \text{Ric}_\omega^n < \infty$ for complete noncompact Kähler n -manifolds with positive sectional curvature. Theorem 1.2 gives quasiprojectivity under bounded sectional curvature. This is strong all-dimensional progress, but quasiprojectivity is weaker than being biholomorphic to $\mathbb{C}^n$. The paper explicitly notes that a higher-dimensional Ramanujam-type rigidity result is a challenging open problem.

4. **Ved Datar, Vamsi Pritham Pingali, Harish Seshadri, "The complex Monge-Ampere equation and an application to uniformisation of surfaces."** arXiv preprint, 2025. arXiv:2511.06849. DOI/URL: <https://doi.org/10.48550/arXiv.2511.06849>

Proves the surface case under positive and bounded sectional curvature by showing finiteness of the square of the Ricci form and applying Chen-Zhu/Ramanujam. This was superseded for the unbounded surface case by arXiv:2604.11220.

5. **Bing-Long Chen, Xi-Ping Zhu, "Positively curved complete noncompact Kähler manifolds."** Acta Mathematica Scientia 29B(4), 829-845, 2009. DOI: 10.1016/S0252-9602(09)60073-4.

Gives earlier partial uniformization results using finite analytic Chern number/top Ricci integral hypotheses; in complex dimension 2, combined with Ramanujam-type criteria, these imply $\mathbb{C}^2$ under additional assumptions. These hypotheses are not part of the problem.

6. **Bing-Long Chen, Xi-Ping Zhu, "Yau's uniformization conjecture for manifolds with non-maximal volume growth."** Acta Mathematica Scientia 38B(5), 1468-1484, 2018. DOI: 10.1016/S0252-9602(18)30828-2.

Surveys progress and proves partial results under non-maximal/minimal volume-growth and finite-Chern-type conditions. These are extra hypotheses and do not settle the arbitrary-dimensional positive-sectional statement.

7. **Gang Liu, "On Yau's uniformization conjecture."** Cambridge Journal of Mathematics 7(1-2), 33-70, 2019. DOI: 10.4310/CJM.2019.v7.n1.a2. arXiv:1606.08958.

Theorem 1.1 proves that a complete noncompact Kähler manifold with nonnegative bi-sectional curvature and maximal volume growth is biholomorphic to $\mathbb{C}^n$. This is important progress for Yau's bi-sectional-curvature conjecture, but maximal volume growth is an extra assumption not present in the problem statement.

8. **Man-Chun Lee, Luen-Fai Tam, “Chern-Ricci flows on noncompact complex manifolds.”** Journal of Differential Geometry 115(3), 529-564, 2020. DOI: 10.4310/jdg/1594260018. arXiv:1708.00141.

Gives an independent Chern-Ricci-flow proof of the maximal-volume-growth case with nonnegative bisectional curvature. Again, maximal volume growth is an extra hypothesis.

1.9.3.4. What remains unresolved

For complex dimensions $n \geq 3$, I found no published or otherwise citable theorem proving that every complete noncompact Kähler n -manifold with positive sectional curvature is biholomorphic to $\mathbb{C}^n$ without additional hypotheses. Current all-dimensional results reach finite top Ricci mass and quasiprojectivity under boundedness assumptions, but they do not imply $\mathbb{C}^n$ in higher dimensions.

1.9.4. Remaining open problems recorded by the database

1.9.4.1. Open problem 1

Let (X^n, ω) be a complete noncompact Kähler manifold with positive sectional curvature, with no boundedness assumption on the sectional curvature. Prove that X is biholomorphic to a quasiprojective variety.

Reference supplied by the database. Ved V. Datar, Vamsi P. Pingali, and Harish Seshadri, “The top Yau–Yang conjecture for Kähler manifolds with positive sectional curvature,” arXiv preprint (2026), arXiv:2606.19806, discussion immediately following Theorem 1.2 (the authors state that removing the bounded-sectional-curvature assumption is work in progress). <https://arxiv.org/abs/2606.19806>

1.9.4.2. Open problem 2

Let $n > 2$, and let X^n be a quasiprojective complex manifold admitting a complete Kähler metric of positive sectional curvature. Prove that X is biholomorphic to $\mathbb{C}^n$.

Reference supplied by the database. Ved V. Datar, Vamsi P. Pingali, and Harish Seshadri, “The top Yau–Yang conjecture for Kähler manifolds with positive sectional curvature,” arXiv preprint (2026), arXiv:2606.19806, discussion immediately following Theorem 1.2 (identified as the open higher-dimensional Ramanujam-type problem under curvature hypotheses). <https://arxiv.org/abs/2606.19806>

1.9.5. Primary-source pointer

The originating ICM record is *Some recent developments in complex differential geometry* (1983). The source link retained by the dataset is <https://www.mathunion.org/fileadmin>

[n/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=349.](#)

1.10. 10. Some recent developments in complex differential geometry [1983-geom-33]

Record: 1983-geom-33

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1983: Some recent developments in complex differential geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=349>

1.10.1. Problem statement

Prove that every complete noncompact Kähler manifold whose holomorphic bisectional curvature is positive is a Stein manifold.

1.10.2. Historical background, literature, and current progress

1.10.3. Verdict: partially resolved

1.10.3.1. Precise problem

The problem asks whether every complete noncompact Kahler manifold with strictly positive holomorphic bisectional curvature is necessarily a Stein manifold. This is the Greene-Wu/Siu Stein-type problem for positive bisectional curvature. It is weaker than Yau's uniformization conjecture, which asks whether such a manifold is biholomorphic to $\mathbb{C}^n$.

1.10.3.2. Current status

I did not find a published or otherwise citable result proving the full statement without extra hypotheses. Recent literature continues to treat the corresponding Stein/strong-Stein step under only $BK > 0$ as unknown. The problem is therefore not fully resolved. It is not completely untouched, however: several papers prove Steinness or the stronger conclusion $M \cong \mathbb{C}^n$ after adding volume growth, curvature decay, a pole/Busemann-exhaustion condition, or stronger positive sectional curvature.

1.10.3.3. Original/source literature

- Title: Some Recent Developments in Complex Differential Geometry
- Author: Yum-Tong Siu
- Venue: Proceedings of the International Congress of Mathematicians, Warszawa, 1983, pp. 287-297
- Year: 1983
- DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf>

- Relevance: This is the source named by the dataset. Later papers identify the Stein question for complete noncompact Kahler manifolds with positive bisectional curvature as a conjecture of Greene-Wu and Siu.

1.10.3.4. Partial progress

- Title: C infinity convex functions and manifolds of positive curvature
- Authors: R. E. Greene and H. Wu
- Venue: Acta Mathematica 137, 209-245
- Year: 1976
- DOI/arXiv/URL: <https://doi.org/10.1007/BF02392418>
- Result: Greene and Wu proved Steinness for complete noncompact Kahler manifolds under the stronger hypothesis of positive Riemannian sectional curvature.
- Why partial: Positive sectional curvature is a stronger geometric assumption than positive holomorphic bisectional curvature. This does not cover all manifolds in the problem statement.
- Title: On Kahler Manifolds of positive bisectional curvature and a theorem of Hartogs
- Authors: R. E. Greene and H. Wu
- Venue: Abhandlungen aus dem Mathematischen Seminar der Universitat Hamburg 47, 171-185
- Year: 1978
- DOI/arXiv/URL: <https://doi.org/10.1007/BF02941360>
- Result: Foundational work on function theory under positive bisectional curvature; frequently cited in later discussions of the Stein conjecture.
- Why partial: It does not prove that every complete noncompact Kahler manifold with positive bisectional curvature is Stein.
- Title: An embedding theorem of complete Kahler manifolds with positive bisectional curvature onto affine algebraic varieties
- Author: Ngaiming Mok
- Venue: Bulletin de la Societe Mathematique de France 112, 197-258
- Year: 1984
- DOI/arXiv/URL: <https://doi.org/10.24033/bsmf.2005>; <https://www.numdam.org/articles/10.24033/bsmf.2005/>
- Result: Mok explicitly states the conjecture that a noncompact complete Kahler manifold of positive bisectional curvature is Stein. He proves Steinness and affine-algebraic embedding under added hypotheses, including Euclidean volume lower growth and quadratic scalar-curvature decay.
- Why partial: The statement has no volume-growth or curvature-decay assumptions.
- Title: On complete noncompact Kahler manifolds with positive bisectional curvature
- Authors: Bing-Long Chen and Xi-Ping Zhu

- Venue: Mathematische Annalen 327, 1-23
- Year: 2003
- DOI/arXiv/URL: <https://doi.org/10.1007/s00208-003-0385-1> ;
arXiv:math/0211370, <https://arxiv.org/abs/math/0211370>
- Result: The paper proves that complete noncompact Kahler manifolds with positive bisectional curvature satisfying suitable growth conditions are biholomorphic to pseudoconvex domains of $\mathbb{C}^n$, and obtains stronger surface uniformization results under additional geometric hypotheses.
- Why partial: The theorem requires extra growth/curvature assumptions not present in the problem statement.
- Title: Plurisubharmonic functions and the structure of complete Kahler manifolds with nonnegative curvature
- Authors: Lei Ni and Luen-Fai Tam
- Venue: Journal of Differential Geometry 64(3), 457-524
- Year: 2003
- DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1090427001> ;
arXiv:math/0304096, <https://arxiv.org/abs/math/0304096>
- Result: Corollary 4.1 states that a complete Kahler manifold with nonnegative holomorphic bisectional curvature is Stein if it has maximum volume growth or has a pole.
- Why partial: Positive holomorphic bisectional curvature implies nonnegative bisectional curvature, but the statement does not assume maximum volume growth or existence of a pole.
- Title: On Yau's uniformization conjecture
- Author: Gang Liu
- Venue: Cambridge Journal of Mathematics 7, no. 1-2, 33-70
- Year: 2019
- DOI/arXiv/URL: https://archive.intlpress.com/site/pub/files/_fulltext/journals/cjm/2019/0007/0001/CJM-2019-0007-0001-a002.pdf
- Result: Proves the stronger conclusion that a complete noncompact Kahler manifold with nonnegative bisectional curvature and maximal volume growth is biholomorphic to $\mathbb{C}^n$.
- Why partial: Maximal volume growth is an additional hypothesis.
- Title: Chern-Ricci flows on noncompact complex manifolds
- Authors: Man-Chun Lee and Luen-Fai Tam
- Venue: Journal of Differential Geometry 115(3), 529-564
- Year: 2020
- DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1594260018>
- Result: Gives another proof, by Chern-Ricci flow methods, that the maximal-volume-growth case is biholomorphic to $\mathbb{C}^n$.
- Why partial: Again, the maximal volume growth assumption is not part of the problem.

- Title: The complex Monge-Ampere equation and an application to uniformisation of surfaces
- Authors: V. Datar, V. P. Pingali, and H. Seshadri
- Venue: arXiv preprint
- Year: 2025
- DOI/arXiv/URL: arXiv:2511.06849, <https://arxiv.org/abs/2511.06849>
- Result: The introduction notes that positive sectional curvature gives Steinness by Greene-Wu, but that neither Steinness nor the related exhaustion-function conclusion is known under the weaker assumption $BK > 0$. It proves related Monge-Ampere/psh-weight results under added hypotheses.
- Why partial/status evidence: This is recent evidence that the full bisectional-curvature-only Stein problem remains open.
- Title: Uniformisation of complete Kahler surfaces with positive sectional curvature
- Authors: V. Datar, V. P. Pingali, and H. Seshadri
- Venue: arXiv preprint
- Year: 2026
- DOI/arXiv/URL: arXiv:2604.11220, <https://arxiv.org/abs/2604.11220>
- Result: The paper discusses Yau's noncompact positive bisectional-curvature conjecture and states that proving a strong Stein-type exhaustion under $BK > 0$ is presently open; it proves results for surfaces under positive sectional curvature or additional assumptions.
- Why partial/status evidence: Its hypotheses are stronger or include extra conditions, and it does not resolve the exact statement.

1.10.3.5. What remains unresolved

No cited result proves that positive holomorphic bisectional curvature alone forces the existence of a strictly plurisubharmonic exhaustion, or otherwise proves Steinness, for every complete noncompact Kahler manifold. The unresolved gap is precisely the removal of the extra hypotheses used in known results: maximal volume growth, curvature decay, a pole/Busemann exhaustion, positive sectional curvature, bounded curvature, or surface-specific assumptions.

1.10.4. Remaining open problems recorded by the database

1.10.4.1. Open problem 1

Let (X, ω) be a complete noncompact Kähler manifold with positive holomorphic bisectional curvature. Must the cohomology at infinity $H_{\infty}^1(X; \mathbb{R})$ vanish?

Reference supplied by the database. Ved Datar, Vamsi Pritham Pingali, and Harish Seshadri, "Uniformisation of complete Kähler surfaces with positive sectional curvature," arXiv preprint (2026), arXiv:2604.11220, Section 8.3, "Topological aspects," concluding question. <https://arxiv.org/abs/2604.11220>

1.10.5. Primary-source pointer

The originating ICM record is *Some recent developments in complex differential geometry* (1983). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=349>.

1.11. 11. The geometry of Teichmüller space [1983-geom-35]

Record: 1983-geom-35

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1983: The geometry of Teichmüller space

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=727>

1.11.1. Problem statement

For a fixed closed surface S of genus g , let T_g be its Teichmüller space, the space of conformal structures on S modulo conformal equivalence by maps isotopic to the identity. Let μ, ν be measured geodesic laminations on S that together fill S , meaning that no essential simple closed curve is disjoint from both. Define the line of minima

$$L_{\mu, \nu} = \{X_t : 0 < t < 1\},$$

where, for each $t \in (0, 1)$, X_t is the unique point of T_g minimizing

$$X \mapsto t l_\mu(X) + (1 - t) l_\nu(X),$$

and $l_\lambda(X)$ denotes the geodesic length of the measured lamination λ in the structure X . Prove or disprove that any two distinct points of T_g lie on exactly one such line of minima.

1.11.2. Historical background, literature, and current progress

1.11.3. Status check

Verdict: partially resolved.

1.11.3.1. Precise problem

For a fixed closed surface S , Kerckhoff's line of minima associated to a filling pair of measured laminations μ, ν is the set of minimizers of positive linear combinations of the length functions l_μ and l_ν . The problem asks whether every two distinct points of Teichmüller space T_g lie on exactly one such line of minima. Since rescaling μ and ν only changes the parametrization, the relevant uniqueness is uniqueness of the line as a subset of T_g , equivalently of the projective filling pair up to the standard interchange/reparametrization ambiguity.

1.11.3.2. Current status

The problem is not fully resolved. Kerckhoff proved the existence half: any two points of Teichmüller space lie on at least one line of minima. I found no published or citable result

proving the missing uniqueness assertion for all closed surfaces and all pairs of points, nor a counterexample disproving it. Later literature develops the geometry, boundary behavior, quasi-geodesic properties, and geodesic-current projection framework for lines of minima, but these results do not imply uniqueness of a line through arbitrary two points of T_g .

1.11.3.3. Key literature and progress

1. Steven P. Kerckhoff, "The geometry of Teichmüller space", Proceedings of the International Congress of Mathematicians, Warsaw 1983, Vol. 1, PWN, Warsaw, 1984, pp. 665-678. URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf>

This is the original ICM source context for the question about geometric analogues in Teichmüller space.

2. Steven P. Kerckhoff, "Lines of minima in Teichmüller space", Duke Mathematical Journal 65, no. 2 (1992), 187-213. DOI: 10.1215/S0012-7094-92-06507-0. URL: <https://doi.org/10.1215/S0012-7094-92-06507-0>

Kerckhoff proved that for filling measured laminations the positive linear combination of length functions has a unique minimizer for each parameter, giving a well-defined line of minima. His Theorem 3.6/Corollary 3.7 proves existence of a line of minima between every pair of points in $T(S)$. This is substantial partial progress, but it is not the full statement because it does not prove uniqueness of that line.

3. Caroline Series, "Kerckhoff's lines of minima in Teichmüller space", in Handbook of Teichmüller Theory, Volume III, IRMA Lectures in Mathematics and Theoretical Physics 17, EMS Press, 2012, pp. 123-153. DOI: 10.4171/103-1/4. URL: <https://ems.press/books/irma/113/contents>

Series surveys Kerckhoff's theory and explicitly records the status: lines of minima for a fixed lamination foliate $T(S)$, any two points lie on at least one line, and whether that line is unique is unknown. This confirms that the existence theorem did not settle the exact uniqueness problem.

4. Raquel Diaz and Caroline Series, "Limit points of lines of minima in Thurston's boundary of Teichmüller space", Algebraic & Geometric Topology 3 (2003), 207-234. DOI: 10.2140/agt.2003.3.207. arXiv: math/0303108. URL: <https://arxiv.org/abs/math/0303108>

This paper studies limiting behavior at Thurston's boundary. It proves convergence for maximal uniquely ergodic laminations and different behavior for rational laminations. These results clarify boundary behavior of a given line, but they do not address uniqueness of a line through two interior points of T_g .

5. Young-Eun Choi, Kasra Rafi, and Caroline Series, "Lines of Minima and Teichmüller Geodesics", Geometric and Functional Analysis 18 (2008), 698-754. DOI: 10.1007/s00039-008-0675-6. arXiv: math/0605135. URL: <https://arxiv.org/abs/math/0605135>

This paper compares a fixed line of minima with the Teichmüller geodesic determined by the same filling pair, characterizing short curves and estimating distances. It proves strong structure for individual lines but not uniqueness through arbitrary point pairs.

6. Young-Eun Choi, Kasra Rafi, and Caroline Series, "Lines of minima are uniformly quasi-geodesic", Pacific Journal of Mathematics 237, no. 1 (2008), 21-44. DOI:

10.2140/pjm.2008.237.21. arXiv: 0706.2044. URL: <https://arxiv.org/abs/0706.2044>

This proves that every line of minima is a Teichmüller quasi-geodesic with uniform constants depending only on the surface topology. This is further geometric progress, but it is compatible with either uniqueness or non-uniqueness through two points.

7. Yuki Iguchi, “Lines of minima with no end in Thurston’s boundary of Teichmüller space”, *Conformal Geometry and Dynamics* 16 (2012), 22-43. DOI: 10.1090/S1088-4173-2012-00240-8. URL: <https://www.ams.org/ecgd/2012-16-02/S1088-4173-2012-00240-8/>

This gives examples of lines of minima with no limit in the Thurston compactification and locates their limit sets in a simplex. It concerns asymptotic behavior, not the uniqueness of a line through two prescribed interior points.

8. Sebastian Hensel and Jenya Sapir, “A projection from filling currents to Teichmüller space”, *Proceedings of the American Mathematical Society* 151 (2023), 3621-3633. DOI: 10.1090/proc/16311. arXiv: 2109.14768. URL: <https://arxiv.org/abs/2109.14768>

Hensel and Sapir generalize Kerckhoff’s minimizer construction to a mapping-class-group-equivariant, continuous, proper projection from projective filling geodesic currents to $T(S)$. Their result recovers/generalizes the existence of unique minimizers for filling currents, but it does not prove the injectivity or two-point uniqueness needed to conclude that two arbitrary points determine exactly one Kerckhoff line of minima.

1.11.3.4. What remains unresolved

The unresolved part is the uniqueness clause: given two distinct points $X, Y \in T_g$, determine whether there is at most one line of minima $L_{\mu, \nu}$ containing both. The literature proves existence and many structural properties of lines of minima, but I found no theorem covering all genera and all point pairs that proves this uniqueness, and no published counterexample that would settle the question negatively.

1.11.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.11.5. Primary-source pointer

The originating ICM record is *The geometry of Teichmüller space* (1983). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=727>.

1.12. 12. An algorithm to recognize the 3 -sphere [1994-geom-51]

Record: 1994-geom-51

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: An algorithm to recognize the 3 -sphere

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=673>

1.12.1. Problem statement

Prove that, for each fixed genus, analytic minimal surfaces in a bumpy Riemannian metric have uniformly bounded index of instability, where a bumpy metric is one for which minimal surfaces have no Jacobi fields, equivalently are isolated critical points of the area functional.

1.12.2. Historical background, literature, and current progress

1.12.3. Verdict: partially resolved

1.12.3.1. Precise problem checked

The problem asks whether, for a fixed genus, closed analytic minimal surfaces in a bumpy Riemannian metric have a uniform upper bound on Morse index (index of instability). I interpret the intended setting, following the Pitts-Rubinstein/Hass-Norbury-Rubinstein literature, as closed embedded minimal surfaces in a closed Riemannian 3-manifold. A bumpy metric means that closed minimal surfaces have no nontrivial Jacobi fields.

I did not find a published or citable theorem that proves this exact statement for all bumpy analytic metrics, nor a citable counterexample satisfying both the bumpy and analytic hypotheses. The literature gives strong related results, but the exact bumpy/analytic fixed-genus question remains unresolved.

1.12.3.2. Progress and near-results

1. Original/source context

- **Title:** *An algorithm to recognize the 3-sphere*
- **Author:** J. Hyam Rubinstein
- **Venue/year:** Proceedings of the International Congress of Mathematicians, Zurich 1994, Birkhauser, 1995
- **DOI/URL:** DOI 10.1007/978-3-0348-9078-6_54; Springer URL: https://link.springer.com/chapter/10.1007/978-3-0348-9078-6_54
- **Relevance:** This is the source for this problem context for the Pitts-Rubinstein program relating bounded Morse index for fixed-genus minimal surfaces to 3-manifold

recognition/spherical space form questions. It does not itself provide a full resolution of the stated analytic bumpy-metric statement.

2. Fixed-genus counterexamples for smooth metrics, but not bumpy/analytic as stated

- **Title:** *Minimal spheres of arbitrarily high Morse index*
- **Authors:** Joel Hass, Paul Norbury, J. Hyam Rubinstein
- **Venue/year:** Communications in Analysis and Geometry 11 (2003), no. 3, 425-439
- **DOI/arXiv/URL:** DOI 10.4310/CAG.2003.v11.n3.a2; arXiv math/0206286; URL <https://arxiv.org/abs/math/0206286>
- **Result:** Constructs a smooth Riemannian metric on any 3-manifold with embedded minimal spheres (genus zero) of arbitrarily high Morse index.
- **Applicability check:** This disproves any unrestricted smooth-metric version of a fixed-genus index bound. It does **not** resolve the problem statement as stated, because the theorem does not assert that the constructed metric is bumpy or analytic. The paper also notes that in positive Ricci curvature fixed-genus compactness/index bounds reappear, so the high-index phenomenon is not automatically preserved by perturbation.

3. Genus-one high-index examples

- **Title:** *Metrics without Morse index bounds*
- **Authors:** Tobias H. Colding, Nancy Hingston
- **Venue/year:** Duke Mathematical Journal 119 (2003), no. 2, 345-365
- **DOI/arXiv/URL:** DOI 10.1215/S0012-7094-03-11925-0; arXiv math/0210290; URL <https://arxiv.org/abs/math/0210290>
- **Result:** Gives metrics with embedded minimal tori of arbitrarily high Morse index.
- **Applicability check:** This covers fixed genus one for specially constructed smooth metrics. The authors discuss bumpy metrics and genericity, but the theorem does not establish a bumpy analytic counterexample to this problem.

4. All fixed genera for smooth metrics, including positive scalar curvature cases

- **Title:** *Singular limit laminations, Morse index, and positive scalar curvature*
- **Authors:** Tobias H. Colding, Camillo De Lellis
- **Venue/year:** Topology 44 (2005), no. 1, 25-45
- **DOI/arXiv/URL:** DOI 10.1016/j.top.2004.01.007; arXiv math/0208100; URL <https://arxiv.org/abs/math/0208100>
- **Result:** For any 3-manifold and any nonnegative genus, constructs metrics with embedded minimal surfaces of that genus whose Morse indices go to infinity; on manifolds admitting positive scalar curvature, the metric can be chosen with positive scalar curvature.
- **Applicability check:** This is the closest negative result to this problem and shows that fixed genus alone cannot bound index. However, the theorem is for constructed smooth metrics and does not impose the stated bumpy or analytic hypotheses.

5. Area plus genus gives index bounds

- **Title:** *Comparison between second variation of area and second variation of energy of a minimal surface*
- **Authors:** Norio Ejiri, Mario Micalef
- **Venue/year:** Advances in Calculus of Variations 1 (2008), no. 3, 223-239
- **DOI/arXiv/URL:** DOI 10.1515/ACV.2008.009; arXiv 0708.2188; URL <https://arxiv.org/abs/0708.2188>

- **Result:** Bounds the Morse index of a minimal surface in terms of its area, genus, and ambient geometry.
- **Applicability check:** This proves the desired index bound under an additional area bound. The statement has no area bound, and known examples show area cannot generally be controlled by genus in arbitrary metrics.

6. Bumpy metrics with fixed genus and area bound

- **Title:** *Embedded minimal surfaces without area bounds in 3-manifolds*
- **Authors:** Tobias H. Colding, William P. Minicozzi II
- **Venue/year:** in *Geometry and Topology: Aarhus*, Contemporary Mathematics 258, American Mathematical Society, 2000
- **DOI/arXiv/URL:** AMS URL <https://www.ams.org/books/conm/258>
- **Result used in later literature:** For a closed orientable 3-manifold with a bumpy metric, fixed genus and fixed area bound give only finitely many closed embedded minimal surfaces.
- **Applicability check:** Together with Ejiri-Micallef this yields an index bound when area is bounded. It does not settle the problem statement because the problem asks for a genus-only bound.

7. Positive Ricci curvature bumpy metrics: fixed genus is controlled, but this is an extra hypothesis

- **Title:** *Existence of minimal surfaces of arbitrarily large Morse index*
- **Authors:** Haozhao Li, Xin Zhou
- **Venue/year:** *Calculus of Variations and Partial Differential Equations* 55 (2016), article 64
- **DOI/arXiv/URL:** DOI 10.1007/s00526-016-1007-6; arXiv 1504.00970; URL <https://arxiv.org/abs/1504.00970>
- **Result:** For a closed 3-manifold with a bumpy metric of positive Ricci curvature, there are embedded minimal surfaces of arbitrarily large Morse index, and for each index bound there are only finitely many such surfaces. The paper also recalls that positive Ricci curvature gives area bounds for fixed genus via Choi-Wang/Yang-Yau, so fixed-genus families are finite in the bumpy positive-Ricci setting.
- **Applicability check:** This is important partial progress. It proves unbounded index occurs in bumpy metrics only with genus going to infinity in this setting; it does not answer the general bumpy analytic fixed-genus problem.

8. Positive scalar curvature and bounded-index compactness/finiteness

- **Title:** *Minimal hypersurfaces with bounded index*
- **Authors:** Otis Chodosh, Daniel Ketover, Davi Maximo
- **Venue/year:** *Inventiones Mathematicae* 209 (2017), no. 3, 617-664
- **DOI/arXiv/URL:** DOI 10.1007/s00222-017-0717-5; arXiv 1509.06724; URL <https://arxiv.org/abs/1509.06724>
- **Result:** In closed 3-manifolds with positive scalar curvature, a uniform index bound implies uniform area and genus bounds; for bumpy positive-scalar-curvature metrics, bounded-index minimal surfaces are finite.
- **Applicability check:** This is the converse direction from the problem statement: it starts with an index bound and derives topology/area control. It does not prove that fixed genus forces an index bound.

9. Generic finiteness with bounded index under positive scalar curvature

- **Title:** *Generic finiteness of minimal surfaces with bounded Morse index*

- **Author:** Alessandro Carlotto
- **Venue/year:** Annali della Scuola Normale Superiore di Pisa, Classe di Scienze 17 (2017), no. 3, 1153–1171
- **DOI/arXiv/URL:** DOI 10.2422/2036-2145.201510_003; arXiv 1509.07101; URL <https://arxiv.org/abs/1509.07101>
- **Result:** For bumpy positive-scalar-curvature metrics, minimal surfaces with a uniform Morse-index bound are finite, up to the stated projective-plane obstruction; for strongly bumpy metrics the conclusion holds for any closed 3-manifold with positive scalar curvature.
- **Applicability check:** This is another bounded-index-to-finiteness result under extra curvature/genericity assumptions. It does not establish the stated genus-to-index implication.

10. Recent related bumpy-metric preprint

- **Title:** *Closed minimal surfaces of index one in Riemannian manifolds*
- **Authors:** Fernando C. Marques, Andre Neves
- **Venue/year:** arXiv preprint, 2026
- **DOI/arXiv/URL:** arXiv 2606.03583; URL <https://arxiv.org/abs/2606.03583>
- **Result:** Gives new existence results for closed minimal hypersurfaces of index one, including bumpy-metric contexts with unbounded volume under topological assumptions.
- **Applicability check:** This is current related evidence that bumpy metrics do not by themselves impose area compactness. It does not settle the problem statement because the problem is about bounding index from fixed genus.

1.12.3.3. What remains unresolved

The unresolved point is exactly the bumpy/analytic fixed-genus case. Existing smooth counterexamples show fixed genus alone is not enough, but the key counterexamples do not verify the bumpy and analytic hypotheses. Existing positive results prove the desired boundedness only after adding hypotheses such as an area bound, positive Ricci curvature, or other positive-scalar-curvature/bounded-index assumptions. I found no citable theorem proving that every bumpy analytic metric has a uniform Morse-index bound for each fixed genus, and no citable counterexample with a bumpy analytic metric and a fixed-genus sequence of closed minimal surfaces with Morse index tending to infinity.

1.12.4. Remaining open problems recorded by the database

1.12.4.1. Open problem 1

For a generic Riemannian metric on S^3 , prove that the Morse indices of all closed embedded minimal tori are uniformly bounded.

Reference supplied by the database. Tobias H. Colding and Nancy Hingston, “Metrics without Morse index bounds,” *Duke Mathematical Journal* **119** (2003), no. 2, 345–365, especially the introduction’s discussion of the Pitts–Rubinstein fixed-topology/generic-metric question for embedded minimal tori in S^3 , DOI: [10.1215/S0012-7094-03-11925-0](https://doi.org/10.1215/S0012-7094-03-11925-0), [arXiv:math/0210290](https://arxiv.org/abs/math/0210290).

1.12.5. Primary-source pointer

The originating ICM record is *An algorithm to recognize the 3 -sphere* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=673>.

1.13. 13. Positive scalar curvature metrics - Existence and classification questions [1994-geom-59]

Record: 1994-geom-59

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Positive scalar curvature metrics - Existence and classification questions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=697>

1.13.1. Problem statement

Prove that the following stable positive scalar curvature statement holds. Let M be an n -dimensional spin manifold with fundamental group π . Let $u : M \rightarrow B\pi$ be the classifying map of the universal covering of M , and let $\alpha(M) = \alpha(M, u) \in KO_n(C_r^*\pi)$ denote the index of the Dirac operator. Fix a Bott manifold B , meaning a simply connected 8-dimensional spin manifold with $\alpha(B) = b$, where b is the generator of $KO_8(\mathbb{R}) \cong \mathbb{Z}$. Then the product $M \times B^k$, for a sufficiently large integer k , admits a positive scalar curvature metric if and only if $\alpha(M) = 0$.

1.13.2. Historical background, literature, and current progress

1.13.3. Verdict: partially resolved

1.13.3.1. Precise problem

The statement is the stable Gromov-Lawson-Rosenberg conjecture for closed connected spin manifolds, usually stated in dimensions at least 5. For a closed spin manifold M with fundamental group π , classifying map $u : M \rightarrow B\pi$, and Rosenberg/Dirac index

$$\alpha(M, u) \in KO_n(C_r^*(\pi)),$$

the question asks whether

$$M \times B^k$$

admits a positive-scalar-curvature metric for some sufficiently large k , where B is an 8-dimensional Bott manifold, if and only if $\alpha(M, u) = 0$.

The necessity direction is standard: positive scalar curvature forces the Rosenberg index to vanish, and multiplication by the Bott manifold identifies the index under Bott periodicity. The difficult part is the converse for arbitrary π .

1.13.3.2. Progress in the literature

1. Stephan Stolz, "Positive Scalar Curvature Metrics – Existence and Classification Questions", Proceedings of the International Congress of Mathematicians, Zurich 1994,

- Birkhauser, 1995, pp. 625-636. DOI: https://doi.org/10.1007/978-3-0348-9078-6_56. This ICM article is the source context for the stable formulation of the positive scalar curvature existence problem using the Dirac/Rosenberg index and Bott periodicity. It does not prove the full arbitrary- π statement.
2. Jonathan Rosenberg and Stephan Stolz, “A ‘stable’ version of the Gromov-Lawson conjecture”, in *The Cech Centennial, Contemporary Mathematics* 181, American Mathematical Society, 1995, pp. 405-418. DOI: <https://doi.org/10.1090/conm/181/02046>; arXiv:dg-ga/9407002; PDF: <https://math.umd.edu/~jmr/StableGLConj4.pdf>. This paper formulates the Stable Conjecture explicitly: M stably admits a positive-scalar-curvature metric, meaning $M \times B^k$ has one for some k , if and only if $\alpha(M, u) = 0$. It proves important cases:
 - Corollary 4.4: the stable conjecture holds if the assembly map $KO_n(B\pi) \rightarrow KO_n(C^*(\pi))$ is injective.
 - Theorem 4.6: the stable conjecture holds for finite groups π . These are genuine partial resolutions, but the hypotheses do not cover arbitrary fundamental groups.
 3. Jonathan Rosenberg and Stephan Stolz, “Metrics of positive scalar curvature and connections with surgery”, in *Surveys on Surgery Theory, Vol. 2, Annals of Mathematics Studies* 149, Princeton University Press, 2001, pp. 353-386. DOI: <https://doi.org/10.1515/9781400865215-010>; PDF: <https://www.math.umd.edu/~jmr/psc.pdf>. Conjecture 4.17 is the same stable Gromov-Lawson-Rosenberg conjecture with a Bott manifold. Theorem 4.18 proves it under injectivity of the ordinary assembly map, Theorem 4.19 records the finite fundamental group case, and Theorem 4.20 states the broader conditional result: the stable conjecture holds for π if the Baum-Connes assembly map $KO_*^\pi(E\pi) \rightarrow KO_*(C^*(\pi))$ is injective. This still leaves groups for which this injectivity is not known.
 4. Matthias Kreck and Wolfgang Lueck, *The Novikov Conjecture: Geometry and Algebra*, Oberwolfach Seminar notes, 2004, URL: <https://webhomes.maths.ed.ac.uk/~v1ranick/papers/owsem.pdf>. Section 22.5 states the stable Gromov-Lawson-Rosenberg conjecture using the real reduced group C^* -algebra $C_r^*(G; \mathbb{R})$. Theorem 22.15 says that injectivity of the real Baum-Connes assembly map $KO_n(E_{\text{FIN}}G) \rightarrow KO_n(C_r^*(G; \mathbb{R}))$ implies the stable conjecture for all closed spin manifolds of dimension at least 5 with fundamental group G . This verifies that the conditional theorem matches the reduced-real C^* -algebra formulation in the problem statement, but again it is conditional.
 5. James F. Davis and Wolfgang Lueck, “The topological K-theory of certain crystallographic groups”, *Journal of Noncommutative Geometry* 7 (2013), 373-431. DOI: <https://doi.org/10.4171/JNCG/121>; arXiv:1004.2660. For certain crystallographic groups $\Gamma = \mathbb{Z}^n \rtimes \mathbb{Z}/p$, the paper computes real and complex K-theory and proves the unstable Gromov-Lawson-Rosenberg conjecture. This is stronger than the stable statement for those special groups, but it is far from the arbitrary- π statement.
 6. Rudolf Zeidler, “Width, Largeness and Index Theory”, *SIGMA* 16 (2020), 127, 15 pages. DOI: <https://doi.org/10.3842/SIGMA.2020.127>; arXiv:2008.13754. Section 2 restates the stable GLR conjecture in essentially the stated Bott-manifold form and summarizes the current general theorem: Stolz’s result proves the conjecture whenever the real Baum-Connes assembly map is injective. Zeidler notes that this gives many known cases and that any counterexample would also be a counterexample to Baum-Connes. This confirms that the unconditional arbitrary-group statement is still not a theorem in the literature.

7. Noe Barcenas and Mario Velasquez, “Positive scalar curvature and crystallographic fundamental groups”, *New York Journal of Mathematics* 32 (2026), 676-690; arXiv:2503.11931; URL: <https://nyjm.albany.edu/j/2026/32-27.html>. This recent paper studies positive and negative results for the unstable GLR conjecture in crystallographic group families and cites Davis-Pearson, Davis-Lueck, and Hughes as sources for positive stable GLR results under Baum-Connes and extra group-theoretic hypotheses. It supports the current picture: progress is substantial for many classes, but remains group-dependent.

1.13.3.3. Hypothesis check

The literature contains results that match the conclusion of the problem only after adding hypotheses such as:

- π finite;
- injectivity of the relevant real Baum-Connes assembly map;
- membership in particular group classes where that assembly result or stronger GLR statements are known.

The problem asks for the statement for arbitrary fundamental group π . None of the citable results found removes the assembly-injectivity or special-group assumptions. Therefore the exact all-cases statement is not fully resolved.

1.13.3.4. What remains unresolved

The remaining open part is the converse direction for arbitrary discrete fundamental groups:

$$\alpha(M, u) = 0 \implies M \times B^k \text{ has a positive-scalar-curvature metric for some } k.$$

It is known for many groups and would follow from the relevant real Baum-Connes injectivity in general, but no published or otherwise citable theorem proves it for all π .

1.13.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.13.5. Primary-source pointer

The originating ICM record is *Positive scalar curvature metrics - Existence and classification questions* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=697>.

1.14. 14. Curvature Contents of Geometric Spaces [1998-geom-63]

Record: 1998-geom-63

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: reviewed (reviewed flag: yes)

Original source: ICM 1998: Curvature Contents of Geometric Spaces

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=383>

1.14.1. Problem statement

For a given space X , such as a Riemannian manifold, an Alexandrov space, or a specified family of such spaces, fix a curvature notion $\mathcal{K}$, for example Gaussian curvature on surfaces, scalar curvature, Ricci curvature, sectional curvature, or Alexandrov curvature. Also fix an admissible geometric class $\mathcal{A}$, such as metrics with normalized volume, normalized diameter, fixed conformal class, prescribed topology, or other constraints removing trivial scaling.

The curvature content or curvature capacity of X relative to $\mathcal{K}$ and $\mathcal{A}$ may be defined by

$$\sup\{k \in \mathbb{R} : \exists g \in \mathcal{A} \text{ such that } \mathcal{K}_g \geq k\}.$$

Equivalently, it is the supremum, over all admissible metrics or structures, of the lower bound for the chosen curvature. The problem is to determine whether this supremum is finite, whether it is attained, and what rigidity occurs at equality.

1.14.2. Historical background, literature, and current progress

1.14.3. Status: open-ended program

1.14.3.1. Precise problem

The problem is not a single formal conjecture with a fixed invariant. In Lohkamp's original formulation it is a programmatic question: for a fixed geometric setting and a chosen curvature notion (Gaussian curvature on surfaces; scalar, Ricci, sectional, or Alexandrov curvature in higher-dimensional or metric-space settings), should one be able to define a natural "curvature content" or "capacity" of a fixed manifold, or specified family of spaces, measuring the maximal amount of positive/lower-bounded curvature the space can carry, while curvature-decreasing deformations should have no analogous lower obstruction.

Thus the exact full question would require a citable definition and theorem covering all the listed curvature notions and their relevant constraints, including Riemannian manifolds and Alexandrov spaces.

1.14.3.2. Literature checked and current progress

- **Title: Curvature Contents of Geometric Spaces**
Authors: Joachim Lohkamp
Venue: Proceedings of the International Congress of Mathematicians, Vol. II (Berlin, 1998), Documenta Mathematica Extra Vol. II, 381-388
Year: 1998
DOI/arXiv/URL: <https://emis.de/ft/10232>
Role: Original source. It explicitly presents “curvature content” as an intuitive notion rather than a sharp definition, and it surveys special evidence: Gauss-Bonnet/uniformization for surfaces, scalar curvature deformation and positive mass rigidity, Ricci comparison and rigidity results, hyperbolization, Alexandrov curvature, and Gromov’s Betti-number bound.
- **Title: Existence and conformal deformation of metrics with prescribed Gaussian and scalar curvatures**
Authors: Jerry L. Kazdan, Frank W. Warner
Venue: Annals of Mathematics 101(2), 317-331
Year: 1975
DOI/arXiv/URL: <https://doi.org/10.2307/1970993>
Role: Gives fundamental prescribed-curvature/flexibility results for Gaussian and scalar curvature. This supports parts of the “no lower bound/decreasing curvature is flexible” theme, but it does not define or resolve a general capacity theory.
- **Title: A direct approach to the determination of Gaussian and scalar curvature functions**
Authors: Jerry L. Kazdan, Frank W. Warner
Venue: Inventiones Mathematicae 28, 227-230
Year: 1975
DOI/arXiv/URL: <https://doi.org/10.1007/BF01425558>
Role: Further prescribed scalar/Gaussian curvature results. Partial progress only.
- **Title: Metrics of negative Ricci curvature**
Authors: Joachim Lohkamp
Venue: Annals of Mathematics 140(3), 655-683
Year: 1994
DOI/arXiv/URL: <https://doi.org/10.2307/2118620>
Role: Establishes strong flexibility for negative Ricci curvature, a key part of the curvature-decreasing side of the program. It does not settle the positive-capacity side in general.
- **Title: Curvature h-principles**
Authors: Joachim Lohkamp
Venue: Annals of Mathematics 142(3), 457-498
Year: 1995
DOI/arXiv/URL: <https://doi.org/10.2307/2118552>
Role: Gives broad flexibility/deformation results for curvature conditions. This is substantial partial progress toward the “no lower bound” direction, not a full capacity theorem.
- **Title: Scalar curvature and hammocks**
Authors: Joachim Lohkamp
Venue: Mathematische Annalen 313, 385-407
Year: 1999

DOI/arXiv/URL: <https://doi.org/10.1007/s002080050266>

Role: Develops scalar curvature deformation ideas used in Lohkamp's discussion. Partial progress for scalar curvature only.

- **Title: On the proof of the positive mass conjecture in general relativity**
 Authors: Richard Schoen, Shing-Tung Yau
 Venue: Communications in Mathematical Physics 65, 45-76
 Year: 1979
 DOI/arXiv/URL: <https://doi.org/10.1007/BF01940959>
 Role: Positive mass rigidity is one of the scalar-curvature obstructions behind Lohkamp's "maximal content" discussion. It applies in the asymptotically flat scalar-curvature setting, not to all curvature contents.
- **Title: Proof of the positive mass theorem. II**
 Authors: Richard Schoen, Shing-Tung Yau
 Venue: Communications in Mathematical Physics 79, 231-260
 Year: 1981
 DOI/arXiv/URL: <https://doi.org/10.1007/BF01942062>
 Role: Further positive mass theorem work; partial scalar-curvature rigidity input.
- **Title: Sharp estimates and the Dirac operator**
 Authors: Marcelo Llarull
 Venue: Mathematische Annalen 310, 55-71
 Year: 1998
 DOI/arXiv/URL: <https://doi.org/10.1007/s002080050136>
 Role: Proves scalar-curvature rigidity of the round sphere under metric domination hypotheses. This is a sharp capacity-type result for a special scalar-curvature comparison problem, not the full problem.
- **Title: Metric Inequalities with Scalar Curvature**
 Authors: Misha Gromov
 Venue: Geometric and Functional Analysis 28, 645-726
 Year: 2018
 DOI/arXiv/URL: <https://doi.org/10.1007/s00039-018-0453-z>
 Role: Provides modern scalar-curvature inequalities and constraints under lower scalar curvature bounds. It advances scalar-curvature capacity-type questions but does not close the general program.
- **Title: Four Lectures on Scalar Curvature**
 Authors: Misha Gromov
 Venue: arXiv preprint; later in *Perspectives in Scalar Curvature*
 Year: 2019 / 2023
 DOI/arXiv/URL: arXiv:1908.10612, <https://arxiv.org/abs/1908.10612>
 Role: Survey of known scalar-curvature constraints and open directions. It confirms active partial progress rather than a completed general theory.
- **Title: Lower bounds on Ricci curvature and the almost rigidity of warped products**
 Authors: Jeff Cheeger, Tobias H. Colding
 Venue: Annals of Mathematics 144(1), 189-237
 Year: 1996
 DOI/arXiv/URL: <https://doi.org/10.2307/2118589>
 Role: Ricci lower-bound rigidity/comparison result, relevant to the Ricci-capacity examples cited by Lohkamp. It has specific hypotheses and does not resolve the full capacity principle.

- **Title: Minimal entropy and Mostow's rigidity theorems**
 Authors: Gerard Besson, Gilles Courtois, Sylvestre Gallot
 Venue: Ergodic Theory and Dynamical Systems 16(4), 623-649
 Year: 1996
 DOI/arXiv/URL: <https://doi.org/10.1017/S0143385700009019>
 Role: Gives sharp entropy/volume rigidity results for locally symmetric negatively curved spaces; relevant to rigidity and curvature bounds in special settings, not a full curvature-content theory.
- **Title: Hyperbolization of polyhedra**
 Authors: Michael W. Davis, Tadeusz Januszkiewicz
 Venue: Journal of Differential Geometry 34(2), 347-388
 Year: 1991
 DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1214447212>
 Role: Provides hyperbolization constructions, a main source of curvature-decreasing/topology-changing examples in the sectional/Alexandrov discussion. Partial progress only.
- **Title: Strict hyperbolization**
 Authors: Ruth M. Charney, Michael W. Davis
 Venue: Topology 34(2), 329-350
 Year: 1995
 DOI/arXiv/URL: [https://doi.org/10.1016/0040-9383\(94\)00027-I](https://doi.org/10.1016/0040-9383(94)00027-I)
 Role: Produces negatively curved spaces in the geodesic/Alexandrov sense and supports the decreasing-curvature side of the sectional/Alexandrov part. It does not give a general positive-capacity theorem.
- **Title: Curvature, diameter and Betti numbers**
 Authors: Mikhail Gromov
 Venue: Commentarii Mathematici Helvetici 56, 179-195
 Year: 1981
 DOI/arXiv/URL: <https://doi.org/10.1007/BF02566208>
 Role: Gives an upper bound for Betti numbers under lower sectional curvature and diameter bounds. Lohkamp identifies this as one concrete sectional-curvature capacity notion, but it is one normalized invariant, not the full program.
- **Title: A. D. Alexandrov spaces with curvature bounded below**
 Authors: Yuri Burago, Mikhail Gromov, Grigori Perelman
 Venue: Russian Mathematical Surveys 47(2), 1-58
 Year: 1992
 DOI/arXiv/URL: <https://doi.org/10.1070/RM1992v047n02ABEH000877>
 Role: Foundational Alexandrov-space theory. It supplies the metric-space framework in which such capacity questions can be posed, but it does not settle the broad principle.
- **Title: Scalar curvature, isoperimetric collapse and General Relativity in the Constant Mean Curvature gauge**
 Authors: Martin Reiris
 Venue: arXiv preprint
 Year: 2010
 DOI/arXiv/URL: arXiv:1002.3302, <https://arxiv.org/abs/1002.3302>
 Role: Later work explicitly states that a suitable scalar curvature capacity is still "to be found" in that context. This is evidence that even the scalar-curvature part is not closed in full generality.

1.14.3.3. Why the full statement is not resolved

The searched literature proves many sharp and influential results for particular curvature notions, normalizations, dimensions, comparison maps, topology classes, or metric-space categories. However, no source found gives a single precise definition of “curvature content/capacity” and a theorem proving the asserted upper-bound/no-lower-bound principle simultaneously for Gaussian, scalar, Ricci, sectional, and Alexandrov curvature under the broad “chosen constraints.” The original article itself avoids a sharp definition, and later scalar-curvature surveys and preprints still treat related capacity-type questions as active.

Therefore the exact problem is not fully resolved. It is also not fully open, because substantial special cases and components of the principle are established.

1.14.3.4. Verdict

open-ended program

1.14.4. Remaining open problems recorded by the database

1.14.4.1. Open problem 1

Let (X, g) be a compact Riemannian manifold with boundary Y and scalar curvature $R_g \geq \sigma$. Is $\int_Y H_g d\mu_{g|_Y}$ bounded above by a constant depending only on the boundary metric $(Y, g|_Y)$ and σ ?

Reference supplied by the database. Yuguang Shi, Wenlong Wang, and Guodong Wei, “Total Mean Curvature of the Boundary and Nonnegative Scalar Curvature Fill-ins,” *Journal für die reine und angewandte Mathematik (Crelle’s Journal)* 2022 (2022), no. 784, 215–250, Conjecture 1.1 (attributed there to M. Gromov, “Four Lectures on Scalar Curvature,” p. 23 of the 2019 preprint; published version in *Perspectives in Scalar Curvature*, Vol. 1, World Scientific, 2023, p. 232). DOI: 10.1515/crelle-2021-0072, arXiv:2007.06756.

Current partial results: Christian Bär, “Upper Bound for the Total Mean Curvature of Spin Fill-ins,” preprint, arXiv:2601.06713 (2026), proves the conjectured bound for spin fill-ins with mean curvature bounded below, $H \geq -\eta$, with the constant depending additionally on η (in particular, the conjecture holds as stated for spin fill-ins with $H \geq 0$). Independently and concurrently, Georg Frenck, Bernhard Hanke, and Sven Hirsch, “Surgery and Total Mean Curvature,” preprint, arXiv:2601.10617 (2026), prove the same η -dependent bound by surgery methods, without the spin assumption when $\dim Y \leq 6$ and $\eta = 0$. Whether the dependence on a lower mean curvature bound can be removed — i.e., the conjecture in full generality — remains open (see the concluding remarks in Bär’s preprint).

1.14.5. Primary-source pointer

The originating ICM record is *Curvature Contents of Geometric Spaces* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=383>.

1.15. 15. Dimension Theory and Large Riemannian Manifolds [1998-geom-64]

Record: 1998-geom-64

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Dimension Theory and Large Riemannian Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=425>

1.15.1. Problem statement

Prove that a closed aspherical manifold, that is, a closed manifold whose universal cover is contractible, does not admit a Riemannian metric whose scalar curvature is everywhere positive.

1.15.2. Historical background, literature, and current progress

1.15.3. Verdict: partially resolved

1.15.3.1. Precise problem

The problem asks whether every closed aspherical smooth manifold, equivalently every closed manifold with contractible universal cover, admits no Riemannian metric whose scalar curvature is everywhere positive. This is the usual Gromov-Lawson / Schoen-Yau aspherical positive scalar curvature conjecture. The statement is all-dimensional and has no spin, orientability, curvature-model, finite asymptotic dimension, Novikov, or group-theoretic hypothesis.

1.15.3.2. Status check

The problem is not fully resolved. The literature proves the statement in low dimensions, most notably through dimension 5, and proves many higher-dimensional cases under additional hypotheses. However, recent papers still explicitly treat the conjecture as widely open in dimensions greater than 5. No citable theorem found in the search proves the full all-dimensional statement for arbitrary closed aspherical manifolds.

1.15.3.3. Key literature and progress

Source and formulation

- Title: *Positive scalar curvature and the Dirac operator on complete Riemannian manifolds*
- Authors: Mikhail Gromov and H. Blaine Lawson, Jr.
- Venue/year: Publications Mathématiques de l'IHES 58, 83-196, 1983

- DOI/URL: <https://doi.org/10.1007/BF02953774>; https://www.numdam.org/item/PMIHES_1983__58__83_0/
- Relevance: Foundational source for the Gromov-Lawson positive scalar curvature program. Later literature cites it for the three-dimensional closed aspherical case and for broad obstruction methods. It does not prove the all-dimensional conjecture.
- Title: *Manifolds of positive scalar curvature*
- Authors: Jonathan Rosenberg and Stephan Stolz
- Venue/year: In *Algebraic Topology and Its Applications*, Springer, 1994, pp. 241-267
- URL: <https://www.math.umd.edu/~jmr/RosenbergStolz1994.pdf>
- Relevance: Records the explicit conjecture as Conjecture 6.1: no closed aspherical manifold admits a metric of positive scalar curvature. It also explains that many cases follow from strong Novikov-type hypotheses or good geometric properties, so these are conditional/special-case results rather than a complete resolution.
- Title: *Dimension theory and large Riemannian manifolds*
- Author: Alexander N. Dranishnikov
- Venue/year: Proceedings of the International Congress of Mathematicians, Berlin 1998, Doc. Math. Extra Vol. II, 423-432
- URL: <https://www2.math.ethz.ch/EMIS/journals/DMJDMV/xvol-icm/06/Dranishnikov.MAN.ps.gz>
- Relevance: This matches the dataset source context and discusses large-scale dimension ideas around scalar curvature and aspherical manifolds. It is background/source material, not a complete proof of the conjecture.

Low-dimensional resolved cases

- Title: *Generalized soap bubbles and the topology of manifolds with positive scalar curvature*
- Authors: Otis Chodosh and Chao Li
- Venue/year: *Annals of Mathematics* 199(2), 707-740, 2024
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2024.199.2.3>; arXiv:2008.11888
- Result: Theorem 2 proves that for n in $\{4,5\}$, a smooth closed aspherical n -manifold does not admit a smooth Riemannian metric with positive scalar curvature; any nonnegative scalar curvature metric is flat.
- Scope check: This fully covers dimensions 4 and 5, but only those dimensions.
- Title: *No metrics with Positive Scalar Curvatures on Aspherical 5-Manifolds*
- Author: Misha Gromov
- Venue/year: arXiv preprint, 2020
- DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2009.05332>; arXiv:2009.05332
- Result: Gives an independent mu-bubble argument implying no positive scalar curvature metrics on compact aspherical manifolds in dimensions up to 5.

- Scope check: Citable preprint evidence for the low-dimensional cases, but still not an all-dimensional theorem.
- Title: *Classifying sufficiently connected PSC manifolds in 4 and 5 dimensions*
- Authors: Otis Chodosh, Chao Li, and Yevgeny Liokumovich
- Venue/year: Geometry & Topology 27, 1635-1655, 2023
- DOI/arXiv: <https://doi.org/10.2140/gt.2023.27.1635> ; arXiv:2105.07306
- Result: Proves a mapping version: if N is a closed 4- or 5-manifold admitting a nonzero degree map to a closed aspherical manifold, then N admits no Riemannian metric with positive scalar curvature.
- Scope check: Stronger than the problem statement in dimensions 4 and 5, but still dimension-limited.

Higher-dimensional partial results

- Title: *On hypersphericity of manifolds with finite asymptotic dimension*
- Author: Alexander N. Dranishnikov
- Venue/year: Transactions of the American Mathematical Society 355(1), 155-167, 2003
- DOI/URL: <https://doi.org/10.1090/S0002-9947-02-03115-X> ; <https://www.jstor.org/stable/3072948>
- Result: Gives the finite-asymptotic-dimension case: a closed aspherical manifold whose fundamental group has finite asymptotic dimension admits no positive scalar curvature metric.
- Scope check: This is all-dimensional only after adding the extra hypothesis on $\pi_1(M)$, so it does not settle arbitrary closed aspherical manifolds.
- Title: *Two-dimension vanishing, splitting and positive scalar curvature*
- Author: Xingyu Zhu
- Venue/year: Mathematische Annalen 392, 2635-2656, 2025
- DOI/URL: <https://doi.org/10.1007/s00208-025-03139-0>
- Result: Reiterates the finite-asymptotic-dimension consequence as Theorem 2.7 and places it among current partial progress.
- Scope check: Again, the theorem has an extra finite-asymptotic-dimension assumption and is therefore partial for this problem.
- Title: *Positive scalar curvature metrics and aspherical summands*
- Authors: Shuli Chen, Jianchun Chu, and Jintian Zhu
- Venue/year: arXiv:2312.04698, submitted 2023, revised 2025; accepted by Duke Mathematical Journal
- DOI/arXiv: <https://doi.org/10.48550/arXiv.2312.04698> ; arXiv:2312.04698
- Result: Proves noncompact/aspherical-summand extensions in dimensions 3, 4, and 5, and surveys the state of the aspherical conjecture.

- Scope check: The paper explicitly states that despite progress through dimension 5, the aspherical conjecture remains widely open in dimensions greater than 5.

1.15.3.4. What remains unresolved

The unresolved part is the full statement for arbitrary closed aspherical manifolds in dimensions $n \geq 6$, without added assumptions such as spin, finite asymptotic dimension, strong Novikov/Baum-Connes hypotheses, nonpositive sectional curvature, enlargeability, special domination maps, or other geometric/group-theoretic restrictions.

1.15.4. Remaining open problems recorded by the database

1.15.4.1. Open problem 1

Prove that no closed aspherical 6-manifold admits a Riemannian metric of positive scalar curvature.

Reference supplied by the database. Liam Mazurowski, Tongrui Wang, and Xuan Yao, “On the Topology of Manifolds with Positive Intermediate Curvature,” *Advances in Mathematics* **485** (2026), Article 110731, Introduction, the $K(\pi, 1)$ conjecture in dimension 6 (equivalently the unresolved $(n, m) = (6, 5)$ scalar-curvature case of the paper’s conjecture), DOI: 10.1016/j.aim.2025.110731, arXiv:2503.13815, <https://doi.org/10.1016/j.aim.2025.110731>.

1.15.5. Primary-source pointer

The originating ICM record is *Dimension Theory and Large Riemannian Manifolds* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=425>.

1.16. 16. S. Gallot Curvature-Decreasing Maps are Volume-Decreasing [1998-geom-74]

Record: 1998-geom-74

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: S. Gallot Curvature-Decreasing Maps are Volume-Decreasing

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=341>

1.16.1. Problem statement

Let X be a compact n -dimensional manifold, with $n \geq 4$, and suppose that X admits a negatively curved locally symmetric Riemannian metric g_0 . Is every Einstein metric on X homothetic to g_0 , i.e. equal to g_0 up to pullback by a diffeomorphism of X and multiplication by a positive constant?

1.16.2. Historical background, literature, and current progress

1.16.3. Status check

Verdict: partially resolved.

1.16.3.1. Precise problem

The problem asks for global Einstein rigidity of compact negatively curved locally symmetric manifolds in all dimensions $n \geq 4$: if X admits a negatively curved locally symmetric metric g_0 , must every smooth Riemannian Einstein metric on the same smooth manifold be $\lambda\phi^*g_0$ for some $\lambda > 0$ and diffeomorphism $\phi: X \rightarrow X$?

This includes, in particular, the subcase of closed real hyperbolic n -manifolds for every $n \geq 4$.

1.16.3.2. Progress and literature

1. Sylvestre Gallot, "Curvature-Decreasing Maps are Volume-Decreasing", Proceedings of the International Congress of Mathematicians, Berlin 1998, Vol. II, Documenta Mathematica Extra Volume II (1998), 339-348. URL: <https://ems.press/content/book-chapter-files/27163>.

Gallot states the general question as an open conjectural problem after explaining that in dimension four the real and complex hyperbolic cases can be handled by global topological/curvature identities. The source records Theorem 3.6 for real hyperbolic 4-manifolds and Theorem 3.8 for complex hyperbolic 4-manifolds, but it does not prove the all- n statement.

2. Gerard Besson, Gilles Courtois, and Sylvestre Gallot, “Entropies et rigidites des espaces localement symetriques de courbure strictement negative”, *Geometric and Functional Analysis* 5 (1995), 731-799. DOI: <https://doi.org/10.1007/BF01897050>.

Their entropy-rigidity theorem is the key input behind the real hyperbolic four-dimensional uniqueness result. In Gallot’s ICM account, it yields: if a compact 4-manifold admits a real hyperbolic metric, then that metric is the only Einstein metric up to homothety. This covers only the real hyperbolic $n = 4$ part of the problem.

3. Claude LeBrun, “Einstein Metrics and Mostow Rigidity”, *Mathematical Research Letters* 2 (1995), 1-8. URL: <https://www.math.stonybrook.edu/~claudel/lmo.pdf>.

Theorem 2 proves that if M^4 is a compact quotient of complex hyperbolic 2-space $\mathbb{C}H^2$, then every Einstein metric is $\lambda\phi^*g_0$. This covers the remaining negatively curved locally symmetric case in real dimension four, but not higher dimensions.

4. Michael T. Anderson, “A survey of Einstein metrics on 4-manifolds”, in *Advances in Lectures in Mathematics* 14, International Press, 2010, 1-39. arXiv:0810.4830, URL: <https://arxiv.org/abs/0810.4830>.

Corollary 4.4 restates LeBrun’s complex-hyperbolic 4-manifold uniqueness, and the survey also presents the Besson-Courtois-Gallot entropy rigidity theorem. This is a useful secondary source confirming the four-dimensional scope of the known global uniqueness results.

5. Norihito Koiso, “Rigidity and stability of Einstein metrics: the case of compact symmetric spaces”, *Osaka Journal of Mathematics* 17 (1980), no. 1, 51-73. URL: <https://projecteuclid.org/journals/osaka-journal-of-mathematics/volume-17/issue-1/Rigidity-and-stability-of-Einstein-metrics---the-case/ojm/1200772808.pdf>.

Koiso proves infinitesimal/local rigidity results for Einstein metrics on locally symmetric spaces of non-compact type without two-dimensional factors. This is important local progress, but it does not imply the global uniqueness asked in the problem statement.

6. Richard H. Bamler, “Stability of symmetric spaces of noncompact type under Ricci flow”, *Geometric and Functional Analysis* 25 (2015), 342-416. arXiv:1011.4267, URL: <https://arxiv.org/abs/1011.4267>.

Corollary 1.3 gives a nearby Einstein-metric uniqueness result under small C^0 perturbations for many locally symmetric spaces of noncompact type, with exclusions including hyperbolic and complex-hyperbolic factors. This is local/stability progress only, not a solution of the global question.

7. Ursula Hamenstadt and Frieder Jackel, “Negatively curved Einstein metrics on Gromov-Thurston manifolds”, arXiv:2411.12956, submitted 2024, revised 2025. DOI: <https://doi.org/10.48550/arXiv.2411.12956>.

The paper constructs, for every $n \geq 4$, closed manifolds with negatively curved Einstein metrics but no locally symmetric metric. These examples are not counterexamples to this problem because their manifolds do not admit a locally symmetric metric. The introduction explicitly asks whether an Einstein metric on any closed hyperbolic manifold has constant curvature, showing that the hyperbolic higher-dimensional subcase remains an active unresolved question.

8. Ursula Hamenstadt, “The geometry of branched coverings of hyperbolic manifolds”, arXiv:2605.01027, 2026. DOI: <https://doi.org/10.48550/arXiv.2605.01027>.

01027.

This 2026 survey-style preprint reviews branched covers of hyperbolic manifolds and again uses the known fact that closed hyperbolic 4-manifolds have a unique Einstein metric up to scaling and isometry. It does not claim a higher-dimensional uniqueness theorem for closed hyperbolic manifolds.

1.16.3.3. What remains unresolved

No published or citable theorem found in the live literature search covers the full statement for every compact negatively curved locally symmetric X of every dimension $n \geq 4$. The four-dimensional cases are resolved: real hyperbolic 4-manifolds by Besson-Courtois-Gallot methods, and complex hyperbolic surfaces by LeBrun. There are also local/infinitesimal rigidity and Ricci-flow stability results in several settings.

However, the problem statement includes closed real hyperbolic n -manifolds for $n \geq 5$, and recent literature still phrases the question whether every Einstein metric on a closed hyperbolic manifold has constant curvature as open. Therefore the exact problem is not fully resolved; it is partially resolved.

1.16.4. Remaining open problems recorded by the database

1.16.4.1. Open problem 1

Must every Einstein metric on a closed real hyperbolic manifold have constant sectional curvature?

Reference supplied by the database. Ursula Hamenstädt and Frieder Jäkel, “Negatively curved Einstein metrics on Gromov–Thurston manifolds,” arXiv preprint (2024; revised 2025), unnumbered Question in the introduction (first question); arXiv: [2411.12956](https://arxiv.org/abs/2411.12956), DOI: [10.48550/arXiv.2411.12956](https://doi.org/10.48550/arXiv.2411.12956).

1.16.5. Primary-source pointer

The originating ICM record is *S. Gallot Curvature-Decreasing Maps are Volume-Decreasing* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=341>.

1.17. 17. Weyl Manifolds and Gaussian Thermostats [2002-geom-76]

Record: 2002-geom-76

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Weyl Manifolds and Gaussian Thermostats

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=498>

1.17.1. Problem statement

Prove that, for every integer $n \geq 2$, no n -dimensional torus T^n admits a Weyl structure whose Weyl sectional curvature is negative on every tangent 2-plane. Equivalently, prove that there is no Weyl structure for which $\widehat{K}(\Pi) < 0$ for every tangent 2-plane Π , where, for any orthonormal basis $\{X, Y\}$ of Π ,

$$\widehat{K}(\Pi) = \langle \widehat{R}_a(X, Y)Y, X \rangle = K(\Pi) - |E_\perp|^2 - \operatorname{div}_\Pi E,$$

$K(\Pi)$ is the Riemannian sectional curvature, $E_\perp$ is the component of the vector field E orthogonal to Π , and $\operatorname{div}_\Pi E = \langle \nabla_X E, X \rangle + \langle \nabla_Y E, Y \rangle$.

1.17.2. Historical background, literature, and current progress

Verdict: partially resolved.

1.17.2.1. Precise problem

The problem asks whether, for every integer $n \geq 2$, the smooth torus T^n admits no Weyl structure with strictly negative Weyl sectional curvature on every tangent 2-plane. In the notation used by Wojtkowski, a Weyl structure represented by a Riemannian metric g and vector field E has

$$\widehat{K}(\Pi) = K(\Pi) - |E_\perp|^2 - \operatorname{div}_\Pi E,$$

and the question is whether $\widehat{K}(\Pi) < 0$ for all tangent 2-planes Π is impossible on all tori.

1.17.2.2. Original source and status of the problem

Title: Weyl Manifolds and Gaussian Thermostats

Author: Maciej P. Wojtkowski

Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. III, Higher Education Press, Beijing, 2002, pp. 511-523

arXiv/URL: arXiv:math/0304461, <https://arxiv.org/abs/math/0304461>

Relevant result: Section 5 states Conjecture 5.3: "There are no Weyl structures of negative sectional curvature on n -dimensional tori." The paper also records Proposition 5.2, a local

obstruction near the homogeneous Weyl structure on a torus, and notes that the $n = 2$ case follows because in dimension 2 the Weyl curvature in the Gauduchon gauge is the Gaussian curvature. On T^2 , Gauss-Bonnet rules out everywhere negative Gaussian curvature in that gauge. This resolves the two-dimensional case but explicitly leaves the higher-dimensional conjecture open.

Title: W-flows on Weyl manifolds and Gaussian thermostats

Author: Maciej P. Wojtkowski

Venue/year: Journal de Mathematiques Pures et Appliquees (9) 79(10), 953-974, 2000

DOI/URL: [https://doi.org/10.1016/S0021-7824\(00\)00176-8](https://doi.org/10.1016/S0021-7824(00)00176-8)

Relevant result: This is the earlier foundational paper relating Weyl connections and Gaussian thermostats and giving dynamical consequences of negative Weyl curvature. It supplies background and motivation, but it does not prove the torus nonexistence conjecture in all dimensions.

1.17.2.3. Partial progress

Title: Rigidity of some Weyl manifolds with nonpositive sectional curvature

Author: Maciej P. Wojtkowski

Venue/year: Proceedings of the American Mathematical Society 133(11), 3395-3402, 2005

DOI/URL: <https://doi.org/10.1090/S0002-9939-05-07809-3> ; author page/PDF: <https://archive.math.arizona.edu/maciejw/>

Relevant result: The paper classifies locally metric Weyl connections with nonpositive sectional curvatures on n -tori and on $S^1 \times S^{n-1}$ with the standard conformal structures. For $S^1 \times S^{n-1}$ it excludes other nonpositive Weyl connections; for T^n the analogous conclusion is proved in the narrower integrable class. Later summaries of this paper state that, on tori with the flat Gauduchon metric, the only nonpositive examples are the constant-vector-field ones. These examples are nonpositive but not strictly negative because planes containing the constant field have zero Weyl curvature. This is important partial progress, but it does not cover arbitrary conformal/Gauduchon metrics or arbitrary Weyl connections on T^n .

Title: Hopf type rigidity for thermostats

Authors: Yernat M. Assylbekov and Nurlan S. Dairbekov

Venue/year: Ergodic Theory and Dynamical Systems 34(6), 1761-1769, 2014

DOI/arXiv/URL: <https://doi.org/10.1017/etds.2013.35> ; arXiv:1204.4380, <https://arxiv.org/abs/1204.4380>

Relevant result: The paper proves a Hopf-type rigidity theorem for magnetic/Gaussian thermostats on T^2 without conjugate points. This is consistent with, and gives another route to, rigidity phenomena for the two-dimensional Gaussian thermostat/Weyl setting. It does not address T^n for $n \geq 3$.

Title: Homogeneous Weyl connections of non-positive curvature

Authors: Gabriela Tereszkievicz and Maciej P. Wojtkowski

Venue/year: Annals of Global Analysis and Geometry 51, 209-229, 2017

DOI/arXiv/URL: <https://doi.org/10.1007/s10455-016-9526-0> ; arXiv:1506.08176, <https://arxiv.org/abs/1506.08176>

Relevant result: This paper studies homogeneous Weyl connections with nonpositive sectional curvature. Its introduction explicitly says that in dimensions 3 and higher the authors do not know examples of Weyl connections with negative sectional curvatures beyond small perturbations of negatively curved Levi-Civita connections. It also summarizes the torus situation: constant vector fields on flat tori give nonpositive examples; previous work gives no other nonpositive Weyl connections with flat Gauduchon metric; and only partial results are

known toward arbitrary Gauduchon metrics. The main theorem proves that homogeneous stretched-nonpositive Weyl connections modeled on unimodular Lie groups are locally product/parallel. These hypotheses are substantially narrower than the problem.

Title: Stretched non-positive Weyl connections on solvable Lie groups

Authors: Maciej Bochenski, Piotr Jastrzebski, and Aleksy Tralle

Venue/year: Annali di Matematica Pura ed Applicata 203, 1463-1481, 2024

DOI/arXiv/URL: <https://doi.org/10.1007/s10231-023-01409-9> ; arXiv:2207.00318, <https://arxiv.org/abs/2207.00318>

Relevant result: This recent paper classifies invariant stretched-nonpositive Weyl connections in several solvable Lie group settings, including a full classification in dimension 4 for unimodular solvable Lie groups. It explicitly does not address the harder problem of finding/classifying general Weyl connections of nonpositive curvature with more interesting dynamics. It therefore supplies current partial progress in homogeneous/SNP cases but not a solution to the torus conjecture.

Title: Thermostats without conjugate points

Authors: Javier Echevarria Cuesta and James Marshall Reber

Venue/year: Ergodic Theory and Dynamical Systems 46(6), 1494-1520, 2026

DOI/arXiv/URL: <https://doi.org/10.1017/etds.2026.10285> ; arXiv:2501.01923, <https://arxiv.org/abs/2501.01923>

Relevant result: This is current related literature on thermostat flows and no-conjugate-points phenomena. It treats general thermostats on surfaces and shows that Hopf-type rigidity does not extend to all thermostat functions on T^2 . Because the problem concerns the specific Weyl/Gaussian thermostat curvature condition and all dimensions, this does not resolve the stated torus problem.

1.17.2.4. What remains unresolved

I did not find any published or otherwise citable theorem proving that every T^n , $n \geq 3$, admits no Weyl structure with $\hat{K}(\Pi) < 0$ for every tangent 2-plane, without additional assumptions such as standard/flat conformal structure, locally metric or integrable Weyl connection, homogeneous/invariant connection, flat Gauduchon metric, or stretched-nonpositive behavior.

Thus the full all-dimensional statement is not fully resolved. The current literature resolves $n = 2$ and several restricted higher-dimensional settings, but the arbitrary higher-dimensional torus case remains open.

1.17.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.17.4. Primary-source pointer

The originating ICM record is *Weyl Manifolds and Gaussian Thermostats* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=498>.

1.18. 18. Geometry and Nonlinear Analysis [2002-geom-81]

Record: 2002-geom-81

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>

1.18.1. Problem statement

Determine whether every compact differentiable manifold M of dimension at least 5 admits a Riemannian metric g such that $\text{Ric}(g) = \lambda g$ for some constant λ .

1.18.2. Historical background, literature, and current progress

1.18.3. Status check

Verdict: partially resolved.

1.18.3.1. Precise question

Does every compact smooth differentiable manifold M of dimension at least 5 admit a Riemannian Einstein metric, i.e. a metric g satisfying $\text{Ric}(g) = \lambda g$ for some constant λ ?

1.18.3.2. Literature checked

1. Shing-Tung Yau, "Review of geometry and analysis," Asian Journal of Mathematics 4 (2000), no. 1, 235-278; also in Mathematics: Frontiers and Perspectives, Amer. Math. Soc., 2000, 353-401. URL: https://www.intlpress.com/site/pub/files/_fulltext/journals/ajm/2000/0004/0001/AJM-2000-0004-0001-a016.pdf. Yau formulates the broad problem as: classify compact smooth manifolds admitting Einstein metrics, and asks whether M always admits one when $\dim M > 4$. This is the source formulation, not a resolution.
2. Arthur L. Besse, Einstein Manifolds, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 10, Springer, 1987; reprint DOI: <https://doi.org/10.1007/978-3-540-74311-8>. This standard reference records the lack of known topological restrictions in dimensions greater than 4. It gives background and examples, but no universal existence theorem or high-dimensional counterexample.
3. Andrea Sambusetti, "Einstein manifolds and obstructions to the existence of Einstein metrics," Rendiconti di Matematica e delle sue Applicazioni, Serie VII, 18 (1998), 131-149. URL: <https://www1.mat.uniroma1.it/ricerca/rendiconti/ARCHIVIO/1998%281%29/131-149.pdf>. Sambusetti explicitly states the question

“Does every compact manifold of dimension greater than 4 have an Einstein metric?” and notes that no example was known of a compact manifold of dimension greater than 4 admitting no Einstein metric. This supports that the full problem was unresolved at that time.

4. Jeffrey S. Case, “Global obstructions to conformally Einstein metrics in dimension six,” *New York Journal of Mathematics* 28 (2022), 1085-1098; arXiv:2207.00645. URL: <https://nyjm.albany.edu/j/2022/28-45p.pdf>. Case constructs global conformal invariants that obstruct conformally Einstein metrics in dimension six and defines a possible diffeomorphism invariant obstruction to Einstein metrics. However, the paper states that no closed six-manifold is currently known for which the resulting diffeomorphism invariant is finite, let alone nonpositive, and that it is unclear whether the results can produce a closed six-manifold without an Einstein metric. Thus this is partial obstruction technology, not a counterexample to the universal question.
5. Charles P. Boyer, Krzysztof Galicki, and Janos Kollar, “Einstein metrics on spheres,” *Annals of Mathematics* (2) 162 (2005), no. 1, 557-580; DOI: <https://doi.org/10.4007/annals.2005.162.557>; arXiv:math/0309408. The paper constructs many Sasaki-Einstein metrics on odd-dimensional standard and exotic spheres. This proves many positive special cases in dimensions at least 5, but only for restricted diffeomorphism types.
6. Janos Kollar, “Einstein metrics on connected sums of $S^2 \times S^3$,” *Journal of Differential Geometry* 75 (2007), no. 2, 259-272; DOI: <https://doi.org/10.4310/jdg/1175266266>; arXiv:math/0402141. Kollar constructs infinitely many families of Einstein metrics on connected sums of arbitrarily many copies of $S^2 \times S^3$. This gives a large class of compact 5-manifolds with Einstein metrics, but it does not cover all compact 5-manifolds.
7. Charles P. Boyer and Krzysztof Galicki, “Sasakian geometry, hypersurface singularities, and Einstein metrics,” *Supplemento ai Rendiconti del Circolo Matematico di Palermo* (2), Suppl. 75 (2005), 57-87; arXiv:math/0405256. URL: <https://math.unm.edu/~cboyer/papers/srni.pdf>. Corollary 59 states that every compact smooth simply connected 5-manifold which is spin and has torsion-free $H_2(M, \mathbb{Z})$ admits a Sasakian-Einstein metric. This is a strong 5-dimensional positive result, but it assumes simply connected, spin, and torsion-free second homology.
8. Yuchen Liu, Taro Sano, and Luca Tasin, “Infinitely many families of Sasaki-Einstein metrics on spheres,” *Journal of Differential Geometry* 130 (2025), no. 1, 1-26; DOI: <https://doi.org/10.4310/jdg/1747062009>; arXiv:2203.08468. The paper proves infinitely many families of Sasaki-Einstein metrics on every odd-dimensional standard sphere of dimension at least 5 and on all odd-dimensional exotic spheres that bound parallelizable manifolds. This is further positive progress for special manifolds, not a universal theorem.

1.18.3.3. Assessment

the exact question remains unresolved. I found no published or otherwise citable theorem proving that all compact smooth manifolds of dimension at least 5 admit Einstein metrics, and no published or otherwise citable compact manifold of dimension at least 5 proved to admit no Einstein metric. The known results give many positive special cases and some sign-restricted or conformal-class obstruction technology, but they do not settle the full all-manifolds question.

What remains open is the simultaneous treatment of all possible Einstein constants $\lambda > 0$, $\lambda = 0$, and especially $\lambda < 0$ for arbitrary compact smooth manifolds in dimension at least 5.

1.18.4. Remaining open problems recorded by the database

1.18.4.1. Open problem 1

Does every closed smooth six-dimensional manifold admit an Einstein metric?

Reference supplied by the database. Jeffrey S. Case, “Global obstructions to conformally Einstein metrics in dimension six,” *New York Journal of Mathematics* **28** (2022), 1085–1098, discussion following Theorem 1.6 (the author states that it is unclear whether the developed obstructions can produce a closed six-manifold admitting no Einstein metric); [journal article](#); arXiv: [2207.00645](#).

1.18.4.2. Open problem 2

For a closed six-manifold M , define

$$\mathcal{L}_1(M) = \sup_{g \in \text{Met}(M)} \int_M \left(|\nabla W_g|^2 + 12E_s^t W_{tijk} W^{sijk} - 64|C_g|^2 \right) d\text{vol}_g,$$

where $W, E = \text{Ric} - \frac{R}{6}g$, and C are the Weyl, trace-free Ricci, and Cotton tensors. Is $\mathcal{L}_1(M)$ finite for any closed six-manifold M ; in particular, can it be nonpositive?

Reference supplied by the database. Jeffrey S. Case, “Global obstructions to conformally Einstein metrics in dimension six,” *New York Journal of Mathematics* **28** (2022), 1085–1098, discussion following Theorem 1.3; [journal article](#); arXiv: [2207.00645](#).

1.18.5. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

1.19. 19. Geometry and Nonlinear Analysis [2002-ag-91]

Record: 2002-ag-91

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>

1.19.1. Problem statement

Let M be a compact Kähler manifold with positive first Chern class $c_1(M) > 0$, and let $\eta(M)$ denote the Lie algebra of holomorphic vector fields on M . Suppose that $\eta(M) = \{0\}$. Prove that M admits a Kähler-Einstein metric if and only if M is asymptotically Chow stable with respect to its anticanonical polarization.

1.19.2. Historical background, literature, and current progress

1.19.3. Verdict: partially resolved

1.19.3.1. Precise problem

The statement is Yau's proposed Chow-stability version of the Kähler-Einstein criterion for smooth Fano manifolds: if M is a compact Kähler/Fano manifold with $c_1(M) > 0$ and no nonzero holomorphic vector fields $\eta(M) = 0$, is

$$M \text{ Kähler-Einstein} \iff (M, -K_M) \text{ asymptotically Chow stable?}$$

Here $\eta(M) = 0$ makes the automorphism group discrete/finite in the Fano setting, so the relevant Chow notion is ordinary asymptotic Chow stability rather than relative stability.

1.19.3.2. Current literature status

1. Original formulation / source.

Title: *Geometry and nonlinear analysis*

Author: Gang Tian

Venue: Proceedings of the ICM, Beijing 2002, vol. 1, pp. 475-493

Year: 2002

arXiv/DOI/URL: arXiv:math/0212404, <https://doi.org/10.48550/arXiv.math/0212404>

Role: Tian explicitly records Yau's conjecture in the form: if $\eta(M) = 0$, then M has a Kähler-Einstein metric iff M is asymptotically Chow stable. Tian notes that Donaldson's theorem gives only a partial answer and that comparing Chow stability with CM-stability is a further problem.

2. Proved direction: Kähler-Einstein implies asymptotic Chow stability.

Title: *Scalar Curvature and Projective Embeddings, I*

Author: Simon K. Donaldson

Venue: Journal of Differential Geometry 59(3), pp. 479-522

Year: 2001

DOI/URL: <https://doi.org/10.4310/jdg/1090349449>

Result used: Donaldson proves that a polarized manifold with a constant scalar curvature Kähler metric and discrete automorphism group is asymptotically Chow stable. Taking $L = -K_M$, a Kähler-Einstein metric is cscK in $c_1(-K_M)$, and $\eta(M) = 0$ gives the discrete automorphism hypothesis. This proves the “only if” direction of the problem statement.

3. Why the converse is not obtained from known Chow-to-K implications.

Title: *A study of the Hilbert-Mumford criterion for the stability of projective varieties*

Authors: Julius Ross and Richard P. Thomas

Venue: Journal of Algebraic Geometry 16(2), pp. 201-255

Year: 2007

DOI/arXiv/URL: 10.1090/S1056-3911-06-00461-9; arXiv:math/0412519

Result used: Their Theorem 3.9 and ensuing discussion give the standard implication chain

asymptotically Chow stable $\Rightarrow$ asymptotically Chow semistable $\Rightarrow$ K -semistable,

but not K -stable or K -polystable. Thus ordinary asymptotic Chow stability, as currently related to K -stability in general, does not trigger the Kähler-Einstein existence theorem.

4. Modern replacement theorem: K -stability, not Chow stability.

Title: *Kähler-Einstein metrics on Fano manifolds. I-III*

Authors: Xiuxiong Chen, Simon Donaldson, Song Sun

Venue: Journal of the American Mathematical Society 28(1), pp. 183-197, 199-234, 235-278

Year: 2015

DOI/arXiv/URL: 10.1090/S0894-0347-2014-00799-2; 10.1090/S0894-0347-2014-00800-6; 10.1090/S0894-0347-2014-00801-8; arXiv:1211.4566, arXiv:1211.4571, arXiv:1302.0282

Related later proof: Robert Berman, Sébastien Boucksom, Mattias Jonsson, *A variational approach to the Yau-Tian-Donaldson conjecture*, Journal of the American Mathematical Society 34(3), pp. 605-652, 2021, DOI 10.1090/jams/964, arXiv:1509.04561.

Role: These works resolve the Fano Yau-Tian-Donaldson criterion using K -polystability (and, with finite automorphism group, K -stability/uniform K -stability). They do not prove that asymptotic Chow stability implies K -polystability, so they do not settle the stated Chow-stability statement.

5. Near miss: a different “asymptotic stability” notion.

Title: *Mahler measures, Stable Pairs, and the Global coercive estimate for the Mabuchi Functional*

Author: Sean Timothy Paul

Venue: arXiv preprint

Year: 2021

arXiv/DOI/URL: arXiv:2105.01240, <https://doi.org/10.48550/arXiv.2105.01240>

Role: Corollary 1.1 states that an anticanonically polarized manifold with finite automorphism group is Kähler-Einstein iff it is “asymptotically stable.” However, the paper defines this through stable pairs involving the resultant and hyperdiscriminant with a Mahler-metric distance, not through the ordinary Chow point. This does not resolve the exact statement.

6. Special/related toric progress, not covering this problem.

Title: *Asymptotic Chow Semistability Implies Ding Polystability for Gorenstein Toric Fano Varieties*

Author: Naoto Yotsutani

Venue: *Mathematics* 11(19), Article 4114

Year: 2023

DOI/arXiv/URL: 10.3390/math11194114; arXiv:1711.10113

Role: Gives a toric/Gorenstein Fano result from asymptotic Chow semistability to Ding polystability in the toric setting. Smooth positive-dimensional toric Fano manifolds have nontrivial holomorphic torus vector fields, and the theorem is not a general result for all $\eta(M) = 0$ Fano manifolds.

Title: *Asymptotic Chow stability of uniformly K-stable toric varieties*

Authors: King Leung Lee and Naoto Yotsutani

Venue: arXiv preprint

Year: 2024; revised 2026

arXiv/DOI/URL: arXiv:2405.06883, <https://doi.org/10.48550/arXiv.2405.06883>

Role: Proves toric sufficient criteria for uniformly K-stable polarized toric varieties to be asymptotically Chow polystable under vanishing Futaki-Ono obstruction. This is a toric/extra-hypothesis result and does not prove the stated all-Fano converse.

7. Cautionary counterexamples outside the stated hypotheses.

Title: *An example of an asymptotically Chow unstable manifold with constant scalar curvature*

Authors: Hajime Ono, Yuji Sano, Naoto Yotsutani

Venue: *Annales de l'Institut Fourier* 62(4), pp. 1265-1287

Year: 2012

DOI/arXiv/URL: 10.5802/aif.2722; arXiv:0906.3836

Role: Gives Kähler-Einstein/cscK toric examples that are not asymptotically Chow semistable when the automorphism group is not discrete. These examples do not refute the problem statement because $\eta(M) \neq 0$, but they show why the no-vector-field hypothesis is essential.

1.19.3.3. Verdict

The problem is **partially resolved**. The implication

$$\text{Kähler-Einstein} \Rightarrow \text{asymptotically Chow stable}$$

is proved by Donaldson under exactly the stated no-holomorphic-vector-field hypothesis. The converse

$$\text{asymptotically Chow stable} \Rightarrow \text{Kähler-Einstein}$$

is not fully resolved in the published/citable literature found: known general implications from Chow stability reach only K -semistability, while the Kähler-Einstein existence theorem requires K -polystability/ K -stability. Related CM/stable-pair or toric results use different stability notions or extra hypotheses and therefore do not cover all cases in the problem statement.

1.19.4. Remaining open problems recorded by the database

1.19.4.1. Open problem 1

For a smooth anticanonically polarized Fano manifold $(M, -K_M)$ with $\eta(M) = 0$, determine whether asymptotic Chow stability forces CM-stability (equivalently, in the modern formulation, K-stability rather than only K-semistability), by comparing the Chow and CM polarizations on the relevant Hilbert scheme.

Reference supplied by the database. Gang Tian, “Geometry and Nonlinear Analysis,” in *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. I, Higher Education Press, Beijing, 2002, pp. 475–493, discussion following Yau’s asymptotic-Chow-stability conjecture on p. 480 (“a very interesting problem ... to compare the Chow stability with the CM-stability”). arXiv: [math/0212404](https://arxiv.org/abs/math/0212404); identifier: [10.48550/arXiv.math/0212404](https://doi.org/10.48550/arXiv.math/0212404). See also Sean Timothy Paul and Gang Tian, “Analysis of Geometric Stability,” *International Mathematics Research Notices* **2004** (2004), no. 48, 2555–2591, which identifies the difference between the Chow and CM polarizations without proving the required stability implication. DOI: [10.1155/S1073792804131759](https://doi.org/10.1155/S1073792804131759); arXiv: [math/0404223](https://arxiv.org/abs/math/0404223).

1.19.5. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

1.20. 20. Geometry and Nonlinear Analysis [2002-ag-92]

Record: 2002-ag-92

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>

1.20.1. Problem statement

Let (M, g) be a complete noncompact Calabi-Yau manifold of complex dimension n with quadratic curvature decay, meaning that for some base point $p \in M$ there are constants $C, R_0 > 0$ such that $|\mathrm{Rm}_g|(x) \leq C d_g(p, x)^{-2}$ whenever $d_g(p, x) \geq R_0$, and with Euclidean volume growth, meaning that there are constants $c, C' > 0$ such that $cR^{2n} \leq \mathrm{Vol}_g(B_g(p, R)) \leq C'R^{2n}$ for all sufficiently large R . Prove that M admits a compactification by a divisor at infinity: there exist a compact complex manifold $\bar{M}$ and an effective divisor $D \subset \bar{M}$ such that $M = \bar{M} \setminus D$, the divisor D is ample near itself (equivalently, the line bundle $\mathcal{O}_{\bar{M}}(D)$ is positive on a neighborhood of D), and the anticanonical bundle of $\bar{M}$ is proportional to the divisor class of D with coefficient greater than one, namely $K_{\bar{M}}^{-1} = \alpha[D]$ in $\mathrm{Pic}(\bar{M}) \otimes_{\mathbb{Z}} \mathbb{R}$ for some $\alpha > 1$, where $[D]$ denotes the class of $\mathcal{O}_{\bar{M}}(D)$.

1.20.2. Historical background, literature, and current progress

1.20.3. Verdict: partially resolved

1.20.3.1. Problem identified

The problem is the refined Yau/Tian compactification problem for complete noncompact Calabi-Yau manifolds: if a complete Calabi-Yau manifold has Euclidean volume growth and quadratic curvature decay, must it be the complement of a positive divisor at infinity in a compact complex manifold, with the anticanonical class proportional to that divisor by a coefficient greater than one?

The original source I found is:

- **Title:** Geometry and nonlinear analysis
- **Author:** Gang Tian
- **Venue:** Proceedings of the International Congress of Mathematicians, Vol. I, Beijing 2002, Higher Education Press
- **Year:** 2002
- **Pages:** 475-493
- **arXiv/URL:** arXiv:math/0212404, <https://arxiv.org/abs/math/0212404>
- **Problem location:** Around pp. 480-481 Tian states the refined conjecture: a complete Calabi-Yau manifold with quadratic curvature decay and Euclidean volume growth

should be of the form $\bar{M} \setminus D$, with D ample near itself and $K_{\bar{M}}^{-1} = \alpha[D]$ for some $\alpha > 1$.

1.20.3.2. Main current progress

- **Title:** No semistability at infinity for Calabi-Yau metrics asymptotic to cones
- **Authors:** Song Sun and Junsheng Zhang
- **Venue:** Inventiones Mathematicae 233, no. 1
- **Year:** 2023
- **Pages:** 461-494
- **DOI/arXiv/URL:** DOI 10.1007/s00222-023-01187-4; arXiv:2208.05098, <https://arxiv.org/abs/2208.05098>
- **Relevant result:** Under exactly the metric hypotheses used here, namely Euclidean volume growth and quadratic curvature decay for a complete Calabi-Yau manifold, Theorem 1.2 proves polynomial asymptotic convergence to the unique Calabi-Yau cone at infinity. The paper also proves the “no semistability at infinity” theorem identifying the intermediate Donaldson-Sun/Liu cone with the metric cone.
- **Applicability:** This removes the main analytic gap between the stated assumptions and the polynomially asymptotically conical setting classified by Conlon-Hein.
- **Title:** Classification of asymptotically conical Calabi-Yau manifolds
- **Authors:** Ronan J. Conlon and Hans-Joachim Hein
- **Venue:** Duke Mathematical Journal 173, no. 5
- **Year:** 2024
- **Pages:** 947-1015
- **DOI/arXiv/URL:** DOI 10.1215/00127094-2023-0030; arXiv:2201.00870, <https://arxiv.org/abs/2201.00870>
- **Relevant result:** The main Existence and Classification theorems classify smooth complete Calabi-Yau manifolds asymptotic to a Calabi-Yau cone at a polynomial rate. In the proof of classification, for an AC Calabi-Yau manifold M , they construct compactifications $X_i = M \cup D_i$ by admissible divisors, prove $K_{X_i} + \alpha_i[D_i]$ is trivial with $\alpha_i > 1$, prove projectivity, and show the corresponding Remmert reduction has an ample compactifying divisor.
- **Applicability:** Combined with Sun-Zhang, this gives a full algebro-geometric classification under the stated metric assumptions and proves the desired compactification statement in the projective orbifold/admissible-divisor sense.
- **Title:** On sharp rates and analytic compactifications of asymptotically conical Kähler metrics
- **Author:** Chi Li
- **Venue:** Duke Mathematical Journal 169, no. 8
- **Year:** 2020
- **Pages:** 1397-1483

- **DOI/arXiv/URL:** DOI 10.1215/00127094-2019-0073; arXiv:1405.2433, <https://arxiv.org/abs/1405.2433>
- **Relevant result:** Provides analytic compactification tools for asymptotically conical complex/Kähler manifolds. Conlon-Hein use and extend this compactification technology, including to orbifold divisors.
- **Title:** Compactification of certain Kähler manifolds with nonnegative Ricci curvature
- **Author:** Gang Liu
- **Venue:** Advances in Mathematics 382
- **Year:** 2021
- **Article:** 107652
- **DOI/URL:** DOI 10.1016/j.aim.2021.107652
- **Relevant result:** Removes a technical hypothesis from Donaldson-Sun's two-step degeneration theory in the complete Calabi-Yau setting with Euclidean volume growth and quadratic curvature decay; used by Sun-Zhang to place the stated manifolds in the algebraic degeneration framework.

1.20.3.3. Why this is not marked fully resolved

The literature now essentially resolves the problem in the modern algebraic/orbifold formulation: under the stated metric assumptions the manifold is polynomially AC, classified by Conlon-Hein, and compactifies by an admissible positive divisor with $K + \alpha D = 0$, $\alpha > 1$.

However, the problem statement as written asks for a **compact complex manifold** $\bar{M}$. The Conlon-Hein compactification produced in the general case is explicitly a **compact complex orbifold** $X_i = M \cup D_i$ or a normal projective variety with an admissible/orbifold divisor. I did not find a published theorem proving that, in all dimensions and for all possible asymptotic cones allowed by the problem statement, this can always be upgraded to a smooth compact complex manifold while preserving the exact anticanonical proportionality and positivity statement.

Therefore the exact statement is not fully resolved as stated, but it has very strong partial resolution: complete classification and the desired compactification in the projective orbifold/admissible-divisor sense.

1.20.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.20.5. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

1.21. 21. Geometry and Nonlinear Analysis [2002-ag-93]

Record: 2002-ag-93

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>

1.21.1. Problem statement

Let (M_i, g_i) be a sequence of compact real n -dimensional Einstein manifolds with the same special holonomy and uniformly bounded diameters, converging in the Gromov-Hausdorff topology to a compact metric space M_∞ . Let $S \subset M_\infty$ be the singular set in the Cheeger-Tian compactness theorem: S is a rectifiable closed subset, $M_\infty \setminus S$ is a smooth manifold carrying an Einstein metric g_∞ with the same holonomy as the g_i , and M_∞ is the metric completion of $M_\infty \setminus S$ for the distance induced by g_∞ .

Prove that S admits a stratification $S = \coprod_{0 \leq i \leq n-4} S_i$ such that each stratum S_i is a smooth manifold of real dimension at most i . If the manifolds (M_i, g_i) are Kähler-Einstein, prove that the odd strata do not occur separately: $S_{2j+1} = S_{2j}$ for every j for which these strata are defined.

1.21.2. Historical background, literature, and current progress

1.21.3. Verdict: partially resolved

1.21.3.1. Precise problem

The question is the regularity/stratification problem for the singular set in noncollapsed Gromov-Hausdorff limits of compact Einstein manifolds with fixed special holonomy. Starting from the Cheeger-Tian compactness picture, where the limit is smooth Einstein with the same holonomy away from a closed rectifiable singular set S of codimension at least 4, the proposed strengthening asks for a decomposition

$$S = \coprod_{0 \leq i \leq n-4} S_i$$

with each S_i a smooth manifold of real dimension at most i . In the Kähler-Einstein case it also asks for the parity improvement $S_{2j+1} = S_{2j}$.

The distinctive statement appears in Gang Tian's ICM article, where after the Cheeger-Tian compactness theorem he writes that the fundamental remaining problem is the regularity of S , conjecturing the above smooth stratification and the Kähler-Einstein parity statement.

1.21.3.2. Original/source literature

- **Geometry and nonlinear analysis**; Gang Tian; *Proceedings of the International Congress of Mathematicians, Vol. I (Beijing, 2002)*, Higher Education Press, 2002, pp. 475-493; arXiv:math/0212404, <https://arxiv.org/abs/math/0212404>. This is the closest match to the wording and explicitly formulates the conjectural smooth stratification of S and the Kähler-Einstein parity $S_{2j+1} = S_{2j}$.
- **Degeneration of Einstein metrics and metrics with special holonomy**; Jeff Cheeger; *Surveys in Differential Geometry*, Vol. VIII, International Press, 2003, pp. 29-73; DOI: 10.4310/SDG.2003.v8.n1.a2. This is the related special-holonomy degeneration source cited in later work.
- **Anti-Self-Duality of Curvature and Degeneration of Metrics with Special Holonomy**; Jeff Cheeger and Gang Tian; *Communications in Mathematical Physics* 255 (2005), 391-417; DOI: 10.1007/s00220-004-1279-0, <https://link.springer.com/article/10.1007/s00220-004-1279-0>. Proves smoothness off a closed codimension ≥ 4 subset for noncollapsed special-holonomy limits and gives further structure supporting, but not proving, the full smooth-stratification conjecture.

1.21.3.3. Current progress

- **On the singularities of spaces with bounded Ricci curvature**; Jeff Cheeger, Tobias H. Colding, and Gang Tian; *Geometric and Functional Analysis* 12(5) (2002), 873-914; DOI: 10.1007/PL00012649. Establishes major structural restrictions on Ricci-limit singularities. In the Kähler-Ricci setting, tangent cones split Euclidean factors in even real dimensions, which gives the expected no-odd-strata behavior for the standard Cheeger-Colding stratification.
- **Regularity of Einstein manifolds and the codimension 4 conjecture**; Jeff Cheeger and Aaron Naber; *Annals of Mathematics* 182(3) (2015), 1093-1165; DOI: 10.4007/annals.2015.182.3.5, <https://annals.math.princeton.edu/wp-content/uploads/annals-v182-n3-p05-p.pdf>. Solves the codimension-4 conjecture for noncollapsed limits with two-sided Ricci bounds: the singular set has Hausdorff/Minkowski dimension at most $n - 4$, and the limit is smooth away from a closed codimension-4 set. This proves the optimal dimension bound but not smooth manifold strata.
- **L^2 Curvature Bounds on Manifolds with Bounded Ricci Curvature**; Wenshuai Jiang and Aaron Naber; *Annals of Mathematics* 193(1) (2021), 107-222; DOI: 10.4007/annals.2021.193.1.2; arXiv:1605.05583, <https://arxiv.org/abs/1605.05583>. Proves L^2 curvature bounds and that, for noncollapsed two-sided Ricci limits, the singular set is $(n - 4)$ -rectifiable with uniformly bounded $\mathcal{H}^{n-4}$ -measure; at $\mathcal{H}^{n-4}$ -a.e. singular point the tangent cone is unique and of orbifold-cone type $\mathbb{R}^{n-4} \times C(S^3/\Gamma)$.
- **Rectifiability of singular sets of noncollapsed limit spaces with Ricci curvature bounded below**; Jeff Cheeger, Wenshuai Jiang, and Aaron Naber; *Annals of Mathematics* 193(2) (2021), 407-538; DOI: 10.4007/annals.2021.193.2.2, <https://annals.math.princeton.edu/2021/193-2/p02>. Proves the Cheeger-Colding strata S^k are k -rectifiable for all k ; under two-sided Ricci bounds it recovers $(n - 4)$ -rectifiability and tangent-cone uniqueness almost everywhere. Rectifiability is substantially weaker than a stratification by smooth manifolds at every point.
- **Gromov-Hausdorff limits of Kähler manifolds and algebraic geometry**; Simon Donaldson and Song Sun; *Acta Mathematica* 213(1) (2014), 63-106; DOI: 10.1007/s11511-014-0116-3, <https://arxiv.org/abs/1206.2609>. For polarized noncollapsed Kähler limits with two-sided Ricci bounds, proves the metric limit is homeomorphic to a

normal projective variety; in the Kähler-Einstein case the metric singular set matches the algebraic singular set. This gives complex-analytic stratifications in this Kähler-Einstein setting, hence no odd real-dimensional strata there, but it is not a theorem for all special holonomy types.

- **Gromov-Hausdorff limits of Kähler manifolds and algebraic geometry, II**; Simon Donaldson and Song Sun; *Journal of Differential Geometry* 107(2) (2017), 327–371; DOI: 10.4310/jdg/1506650422; arXiv:1507.05082, <https://arxiv.org/abs/1507.05082>. Gives further algebro-geometric structure of tangent cones in Kähler-Einstein limits.
- **Gromov-Hausdorff limits of Kähler manifolds with Ricci curvature bounded below**; Gang Liu and Gábor Székelyhidi; *Geometric and Functional Analysis* 32(2) (2022), 236–279; DOI: 10.1007/s00039-022-00594-8, <https://arxiv.org/abs/1804.08567>. Extends the Donaldson-Sun algebraic picture to polarized Kähler manifolds with Ricci bounded below, showing normal projective limits and metric singularities given by analytic subvarieties. Again, this is Kähler-specific and does not cover arbitrary special holonomy.

1.21.3.4. What remains unresolved

The literature fully resolves several important consequences of the conjecture: optimal codimension 4, rectifiability and finite measure of the top singular set, almost-everywhere uniqueness and orbifold-cone structure of tangent cones, and strong complex-analytic structure in polarized Kähler/Kähler-Einstein cases. However, I did not find a published theorem proving that every singular set S arising from all compact Einstein limits with the same special holonomy admits a global decomposition into smooth manifold strata S_i of the requested dimensions.

The gap is not just terminological: k -rectifiability and almost-everywhere tangent cone structure do not imply that each stratum is a smooth manifold, and the Kähler algebraic/analytic results do not cover non-Kähler special holonomy such as G_2 , $\text{Spin}(7)$, or quaternionic-Kähler settings. Therefore the exact statement is not fully resolved.

1.21.4. Remaining open problems recorded by the database

1.21.4.1. Open problem 1

Let (M_i^n, g_i) be a noncollapsed sequence of compact manifolds with special holonomy as in the Cheeger–Tian compactness theorem, and suppose in addition that the characteristic numbers $C(M_i^n)$ defined by Cheeger and Tian are uniformly bounded. Prove that the singularities of the Gromov–Hausdorff limit are of orbifold type outside a closed subset of real codimension at least 6.

Reference supplied by the database. Jeff Cheeger and Gang Tian, “Anti-Self-Duality of Curvature and Degeneration of Metrics with Special Holonomy,” *Communications in Mathematical Physics* 255 (2005), no. 2, 391–417. Main conjecture stated in the abstract and developed in the paper. DOI: [10.1007/s00220-004-1279-0](https://doi.org/10.1007/s00220-004-1279-0).

1.21.4.2. Open problem 2

Let (X, d) be a noncollapsed Gromov–Hausdorff limit of Einstein manifolds whose Einstein constants are uniformly bounded. Prove that the metric tangent cone at every point of X is unique, without assuming that one tangent cone has a smooth cross-section.

Reference supplied by the database. Tobias Holck Colding and William P. Minicozzi II, “On Uniqueness of Tangent Cones for Einstein Manifolds,” *Inventiones Mathematicae* **196** (2014), no. 3, 515–588. The general uniqueness question in the Introduction and its local Einstein-limit form surrounding Theorem 0.3; the paper proves uniqueness under the additional hypothesis that one tangent cone has a smooth cross-section. DOI: [10.1007/s00222-013-0474-z](https://doi.org/10.1007/s00222-013-0474-z); arXiv: [1206.4929](https://arxiv.org/abs/1206.4929).

1.21.4.3. Open problem 3

Let (X, d) be a noncollapsed Gromov–Hausdorff limit of Einstein n -manifolds. Prove that there is a closed subset $\mathcal{S}_{\text{orb}} \subset X$ of Hausdorff codimension at least 5 such that $X \setminus \mathcal{S}_{\text{orb}}$ is a smooth Einstein orbifold, whose orbifold singular set is a countable union of totally geodesic codimension-4 submanifolds and is locally modeled at every such point on $\mathbb{R}^{n-4} \times \mathbb{R}^4/\Gamma$ for a finite group $\Gamma \subset O(4)$.

Reference supplied by the database. Yiqi Huang and Tristan Ozuch, “Regularity of Einstein 5-Manifolds via 4-Dimensional Gap Theorems,” preprint, 2025 (revised 2026), Question 1.29 (“Codimension-5 conjecture”; numbered Question 1.29 in the arXiv version’s §1.3.2). arXiv: [2512.21317](https://arxiv.org/abs/2512.21317).

1.21.5. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

1.22. 22. Applications of loop group factorization to geometric soliton equations [2006-geom-91]

Record: 2006-geom-91

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Applications of loop group factorization to geometric soliton equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=937>

1.22.1. Problem statement

Let $x_1, \dots, x_n$ be the standard coordinates on $\mathbb{R}^n$, and let $F = (f_{ij})$ be a global smooth symmetric solution of the $U(n)/O(n)$ -system on $\mathbb{R}^n$, with $f_{ii} = 0$ for all i . Determine whether there exist global smooth functions $b_1, \dots, b_n : \mathbb{R}^n \rightarrow (0, \infty)$ satisfying the first-order system

$$\partial_{x_j} b_i = f_{ij} b_j, \quad i \neq j,$$

such that the associated diagonal Riemannian metric

$$ds^2 = \sum_{i=1}^n b_i^2 dx_i^2$$

is complete.

1.22.2. Historical background, literature, and current progress

this problem is fully open

1.22.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.22.4. Primary-source pointer

The originating ICM record is *Applications of loop group factorization to geometric soliton equations* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=937>.

1.23. 23. Metaphors in Systolic Geometry [2010-geom-114]

Record: 2010-geom-114

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Metaphors in Systolic Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>

1.23.1. Problem statement

Fix $r > 0$. For a Riemannian metric g on the n -dimensional torus T^n and a point $p \in T^n$, let $\text{Sc}_r(p)$ denote the macroscopic scalar curvature at scale r : if $\tilde{V}(p, r)$ is the volume, in the universal cover $(\tilde{T}^n, \tilde{g})$, of the ball of radius r centered at a lift of p , then $\text{Sc}_r(p)$ is the number S such that $\tilde{V}(p, r)$ equals the volume of a ball of radius r in the simply connected n -dimensional space of constant scalar curvature S .

Prove or disprove that there is no Riemannian metric g on T^n for which $\text{Sc}_r(p) > 0$ at every point $p \in T^n$. Equivalently, prove or disprove that for every Riemannian metric g on T^n , the universal cover $(\tilde{T}^n, \tilde{g})$ contains a ball of radius r and volume at least $\omega_n r^n$, where ω_n is the Euclidean volume of the unit n -ball.

1.23.2. Historical background, literature, and current progress

1.23.3. Status check: Generalized Geroch conjecture

Verdict: partially resolved.

1.23.3.1. Precise problem checked

The problem asks for the sharp macroscopic-scalar-curvature analogue of the Geroch conjecture for tori. For every dimension n , every Riemannian metric g on T^n , and every fixed scale $r > 0$, must the Riemannian universal cover $(\tilde{T}^n, \tilde{g})$ contain some radius- r ball of volume at least $\omega_n r^n$? Equivalently, can T^n carry a metric whose macroscopic scalar curvature at scale r is positive at every point?

This is the sharp Euclidean-ball constant statement. A nonsharp lower bound $c(n)r^n$, even for all r , is not enough to resolve it.

1.23.3.2. Source and formulation

- Title: Large Riemannian manifolds Authors: Mikhail Gromov Venue: Curvature and Topology of Riemannian Manifolds, Katata 1985, Lecture Notes in Mathematics 1201, Springer Year: 1986 DOI/URL: <https://doi.org/10.1007/BFb0075649> Role: Original source for Gromov's large-manifold conjectures behind this problem.

- Title: Metaphors in systolic geometry Authors: Larry Guth Venue: Proceedings of the International Congress of Mathematicians 2010, World Scientific, pp. 745-768 Year: 2010 URL: <https://math.mit.edu/~lguth/Exposition/icmguth.pdf> Role: States the exact problem in the form used in this problem: “Fix $r > 0$. The n -dimensional torus does not admit a metric with $Sc_r > 0$.” Guth also states the equivalent formulation that the universal cover contains a radius- r ball of volume at least $\omega_n r^n$, and says the conjecture was wide open at that time.

1.23.3.3. Main partial progress

- Title: Systolic inequalities and minimal hypersurfaces Authors: Larry Guth Venue: Geometric and Functional Analysis 19, 1688-1692 Year: 2010 arXiv/URL: arXiv:0903.5299, <https://arxiv.org/abs/0903.5299> Result used for status: Guth proves that if $R < \frac{1}{2} \text{Sys}(T^n, g)$, then some radius- R ball in T^n has volume at least $c_n R^n$. Why it is only partial: the constant is c_n , not the sharp Euclidean ω_n , and the statement is tied to the systole scale rather than the exact universal-cover statement for every prescribed r .
- Title: Volumes of balls in large Riemannian manifolds Authors: Larry Guth Venue: Annals of Mathematics 173(1), 51-76 Year: 2011 DOI/arXiv/URL: 10.4007/annals.2011.173.1.2; arXiv:math/0610212; <https://arxiv.org/abs/math/0610212> Result used for status: Corollary 3 proves that if M is a closed aspherical Riemannian n -manifold, then the supremal volume $V(R)$ of radius- R balls in the universal cover satisfies $V(R) \geq \delta(n)R^n$ for all R . This applies to T^n . Why it is only partial: $\delta(n)$ is an unspecified positive dimensional constant and is not shown to equal ω_n . Thus the theorem gives a qualitative/nonsharp macroscopic lower bound but not the sharp Euclidean threshold.
- Title: Volume and macroscopic scalar curvature Authors: Sabine Braun and Roman Sauer Venue: Geometric and Functional Analysis 31, 1321-1376 Year: 2021 DOI/arXiv/URL: 10.1007/s00039-021-00588-y; arXiv:2012.08999; <https://arxiv.org/abs/2012.08999> Result used for status: Theorem 1.3 proves that for every rationally essential Riemannian d -manifold and every $R > 0$, the universal cover contains a radius- R ball of volume $> \epsilon(d)R^d$. Since T^n is rationally essential, this applies to the torus. Hypothesis check: the torus is closed, oriented, and rationally essential, so the theorem covers the topological class in the problem statement. Why it is only partial: Braun-Sauer explicitly frame the theorem as a nonsharp macroscopic cousin; the exact conjecture would follow only if $\epsilon(d)$ could be chosen to be the Euclidean d -ball volume ω_d , which their theorem does not prove.
- Title: Uryson width and volume Authors: Panos Papasoglu Venue: Geometric and Functional Analysis 30, 574-587 Year: 2020 DOI/URL: 10.1007/s00039-020-00533-5; <https://link.springer.com/article/10.1007/s00039-020-00533-5> Result used for status: Gives a short proof and related width-volume approach to Guth-type nonsharp lower bounds. Why it is only partial: it does not replace the dimensional constants by the sharp Euclidean volume ω_n .
- Title: Linear bounds for constants in Gromov’s systolic inequality and related results Authors: Alexander Nabutovsky Venue: Geometry & Topology 26(7), 3123-3142 Year: 2022 DOI/arXiv/URL: 10.2140/gt.2022.26.3123; arXiv:1909.12225; <https://arxiv.org/abs/1909.12225> Result used for status: Improves constants in Guth/Gromov-type systolic and ball-volume estimates. Why it is only partial: improving nonsharp constants still does not prove the Euclidean sharp constant in the generalized Geroch

conjecture.

- Title: Macroscopic scalar curvature and local collapsing Authors: Stephane Sabourau Venue: Annales Scientifiques de l'Ecole Normale Supérieure (4) 55(4), 919-936 Year: 2022 arXiv/URL: arXiv:2006.00663; <https://arxiv.org/abs/2006.00663> Result used for status: Proves strong lower bounds for balls in universal covers of negatively curved manifolds under small-volume hypotheses and interprets them via macroscopic scalar curvature. Why it is only partial: the hypotheses and conclusion concern negatively curved/aspherical settings with different comparison constants and do not prove the sharp torus statement.
- Title: Macroscopic scalar curvature and areas of cycles Authors: Hannah Alpert and Kei Funano Venue: Geometric and Functional Analysis 27(4), 727-743 Year: 2017 DOI/arXiv/URL: 10.1007/s00039-017-0414-3; arXiv:1705.02923; <https://arxiv.org/abs/1705.02923> Result used for status: Develops Guth's macroscopic-scalar-curvature methods for area lower bounds of homology classes in $\Sigma \times S^1$. Why it is only partial: it is about hyperbolic products and cycle areas, not the sharp ball-volume lower bound for all metrics on T^n .
- Title: Macroscopic scalar curvature and codimension 2 width Authors: Hannah Alpert, Alexey Balitskiy, and Larry Guth Venue: Journal of Topology and Analysis 16(6), 979-987 Year: 2024 DOI/arXiv/URL: 10.1142/S1793525323500024; arXiv:2112.04594; <https://arxiv.org/abs/2112.04594> Result used for status: Gives width conclusions in dimension 3 under strong positive macroscopic-curvature and local acyclicity assumptions. Why it is only partial: the added hypotheses and width conclusion do not imply the exact $\omega_n r^n$ torus ball-volume statement.
- Title: Urysohn width and macroscopic scalar curvature Authors: Aditya Kumar and Balarka Sen Venue: arXiv preprint Year: 2026 arXiv/URL: arXiv:2601.14669; <https://arxiv.org/abs/2601.14669> Result used for status: Constructs examples with positive macroscopic scalar curvature in the context of circle bundles and shows that a macroscopic Urysohn-width analogue is false in dimensions at least four. Why it is only partial: these examples are not metrics on T^n satisfying the stated condition, and the result does not settle the generalized Geroch conjecture for tori.

1.23.3.4. Applicability check and remaining gap

The closest theorems that apply directly to T^n are Guth's aspherical-manifold ball-volume theorem and Braun-Sauer's rationally essential manifold theorem. Their hypotheses do cover the torus, and their conclusions hold at every scale after scaling. However, both conclusions have an unspecified dimensional positive constant rather than the sharp Euclidean constant ω_n . Since the problem asks exactly whether some ball has volume at least $\omega_n r^n$, these theorems do not fully resolve the problem.

I found no published or otherwise citable result proving the full sharp statement for all n , all $r > 0$, and all Riemannian metrics on T^n . The current literature therefore gives substantial nonsharp and related progress, but the exact generalized Geroch conjecture in this problem remains unresolved.

1.23.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.23.5. Primary-source pointer

The originating ICM record is *Metaphors in Systolic Geometry* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>.

1.24. 24. Metaphors in Systolic Geometry [2010-geom-115]

Record: 2010-geom-115

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Metaphors in Systolic Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>

1.24.1. Problem statement

Prove that if M^n is a closed n -dimensional hyperbolic manifold with hyperbolic Riemannian metric hyp , then for any Riemannian metric g on the underlying manifold M such that

$$\text{Sc}(g)(x) \geq \text{Sc}(\text{hyp})(x)$$

for every point $x \in M$, the Riemannian volumes satisfy

$$\text{Vol}(M, g) \geq \text{Vol}(M, \text{hyp}).$$

1.24.2. Historical background, literature, and current progress

1.24.3. Verdict: partially resolved

1.24.3.1. Precise problem

The problem asks for Schoen's hyperbolic volume comparison conjecture: for a closed hyperbolic Riemannian manifold (M^n, g_{hyp}) , does every smooth Riemannian metric g on the same underlying manifold satisfying

$$\text{Scal}_g \geq \text{Scal}_{g_{\text{hyp}}} = -n(n-1)$$

pointwise necessarily satisfy

$$\text{Vol}(M, g) \geq \text{Vol}(M, g_{\text{hyp}})?$$

For $n = 2$ this follows from Gauss-Bonnet. The substantive conjecture is for $n \geq 3$, and the full all-dimensional statement remains unresolved.

1.24.3.2. Status check

I found no citable theorem that proves the full statement for all closed hyperbolic n -manifolds and all smooth metrics satisfying only the scalar-curvature lower bound. Recent papers still describe the higher-dimensional case as open or only partially confirmed. The literature gives important special cases: dimension 3, metrics close to the hyperbolic metric, stronger Ricci-curvature hypotheses, conformal/self-dual cases in dimension 4, and macroscopic analogues. These do not imply the full statement because they add hypotheses not present in the conjecture or replace scalar curvature by a different large-scale condition.

1.24.3.3. Key literature and progress

1. Richard M. Schoen, “Variational theory for the total scalar curvature functional for Riemannian metrics and related topics”, in *Topics in Calculus of Variations*, Lecture Notes in Mathematics 1365, Springer, 1989, pp. 120-154. DOI: 10.1007/BFb0089180. URL: <https://link.springer.com/chapter/10.1007/BFb0089180>. This is the original source for Schoen’s conjecture on the Yamabe invariant of closed hyperbolic manifolds. Later literature identifies the volume-comparison formulation in this problem as equivalent to the hyperbolic Yamabe-invariant conjecture.
2. G. Besson, G. Courtois, and S. Gallot, “Volume et entropie minimale des espaces localement symétriques”, *Inventiones Mathematicae* 103 (1991), 417-445. DOI: 10.1007/BF01239520. URL: <https://link.springer.com/article/10.1007/BF01239520>. This gives local/entropy rigidity results used as partial evidence for Schoen’s conjecture. As summarized in later sources, it verifies the scalar-curvature volume comparison for metrics C^2 -close to the hyperbolic metric, but this closeness assumption is not in the problem statement.
3. G. Besson, G. Courtois, and S. Gallot, “Entropies et rigidités des espaces localement symétriques de courbure strictement négative”, *Geometric and Functional Analysis* 5 (1995), 731-799. DOI: 10.1007/BF01897050. URL: <https://link.springer.com/article/10.1007/BF01897050>. This proves a global minimal-entropy/rigidity theorem. Together with Bishop-Gromov it yields the desired volume lower bound under the stronger assumption $\text{Ric}_g \geq -(n-1)g$, but Ricci lower bounds are much stronger than the scalar-curvature lower bound in the problem statement.
4. Michael T. Anderson, “Scalar curvature and geometrization conjectures for 3-manifolds”, in *Comparison Geometry*, MSRI Publications 30, Cambridge University Press, 1997, pp. 49-82. URL: <https://library.slmath.org/books/Book30/files/anderson.pdf>. Anderson formulates the three-dimensional scalar-curvature/Yamabe-invariant picture. In dimension 3, Perelman’s Ricci-flow and geometrization work implies the corresponding closed hyperbolic 3-manifold volume comparison: if $R_g \geq -6$, then $\text{Vol}(M, g) \geq \text{Vol}(M, g_{\text{hyp}})$.
5. Michael T. Anderson, “Canonical metrics on 3-manifolds and 4-manifolds”, *Asian Journal of Mathematics* 10(1) (2006), 127-163. DOI: 10.4310/AJM.2006.v10.n1.a8. arXiv: math/0511034. URL: <https://arxiv.org/abs/math/0511034>. This records the post-Perelman 3-dimensional sigma/Yamabe picture. It supports the conclusion that the conjecture is resolved in dimension 3, but it does not settle dimensions $n \geq 4$.
6. Ian Agol, Peter A. Storm, and William P. Thurston, with appendix by Nathan Dunfield, “Lower bounds on volumes of hyperbolic Haken 3-manifolds”, *Journal of the American Mathematical Society* 20(4) (2007), 1053-1077. DOI: 10.1090/S0894-0347-07-00564-4. arXiv: math/0506338. URL: <https://arxiv.org/abs/math/0506338>. This uses Perelman’s work to prove volume inequalities for 3-manifolds under scalar-curvature/minimal-boundary hypotheses. It is important three-dimensional progress but not a proof of the all-dimensional closed-manifold conjecture.
7. Larry Guth, “Metaphors in systolic geometry”, in *Proceedings of the International Congress of Mathematicians 2010*, Vol. II, Hindustan Book Agency, 2010, pp. 745-768. DOI: 10.1142/9789814324359_0072. URL: <https://math.mit.edu/~lguth/Exposition/icmguth.pdf>. Guth presents the Schoen conjecture as open and notes that in dimensions $n \geq 5$ no scalar-curvature-only lower volume bound was known at

that time. He proves a non-sharp macroscopic analogue involving volumes of balls in the universal cover, not the pointwise scalar-curvature hypothesis itself.

8. **Wei Yuan**, “Volume comparison with respect to scalar curvature”, *Analysis & PDE* 16(1) (2023), 1-34. DOI: 10.2140/apde.2023.16.1. arXiv: 1609.08849. URL: <https://arxiv.org/abs/1609.08849>. Yuan states Schoen’s hyperbolic volume comparison conjecture explicitly and proves that it is equivalent to the hyperbolic Yamabe-invariant conjecture. The paper proves local volume comparison near strictly stable Einstein metrics and recovers the Besson-Courtois-Gallot local result for closed hyperbolic manifolds. It also states that the conjecture is known in dimension 3, while in higher dimensions the known results require C^2 -closeness or stronger Ricci hypotheses.
9. **Florent Balacheff and Stephane Karam**, “Macroscopic Schoen conjecture for manifolds with nonzero simplicial volume”, *Transactions of the American Mathematical Society* 372(10) (2019), 7071-7086. DOI: 10.1090/tran/7765. arXiv: 1709.10049. URL: <https://www.ams.org/tran/2019-372-10/S0002-9947-2019-07765-6/>. This proves macroscopic/non-sharp versions related to Guth’s framework. It concerns large-scale ball-volume hypotheses and simplicial volume, so it is partial evidence rather than a proof of the scalar-curvature conjecture in the problem statement.
10. **Sabine Braun and Roman Sauer**, “Volume and macroscopic scalar curvature”, *Geometric and Functional Analysis* 31 (2021), 1321-1376. DOI: 10.1007/s00039-021-00588-y. URL: <https://link.springer.com/article/10.1007/s00039-021-00588-y>. This proves macroscopic cousins of scalar-curvature conjectures, replacing scalar-curvature lower bounds by upper bounds on volumes of 1-balls in the universal cover. The authors explicitly describe Guth’s volume theorem as a non-sharp macroscopic cousin of Schoen’s conjecture, not a resolution of the original conjecture.
11. **Demetre Kazaras, Antoine Song, and Kai Xu**, “Scalar curvature and volume entropy of hyperbolic 3-manifolds”, *Journal of the European Mathematical Society* (2025), published online first. DOI: 10.4171/JEMS/1710. arXiv: 2312.00138. URL: <https://ems.press/journals/jems/articles/14299127>. This constructs metrics on closed hyperbolic 3-manifolds with scalar curvature at least -6 but volume entropy greater than 2, disproving a proposed entropy route to the 3-dimensional volume comparison. It does not refute Schoen’s conjecture; rather, it shows that the Besson-Courtois-Gallot entropy method cannot be extended from Ricci lower bounds to scalar lower bounds in this way.
12. **Jialong Deng**, “Scalar Curvature in Dimension 4”, arXiv preprint, 2025. arXiv: 2512.13528. DOI: 10.48550/arXiv.2512.13528. URL: <https://arxiv.org/abs/2512.13528>. This recent preprint gives partial dimension-4 confirmations, for example for self-dual, anti-self-dual, or locally conformally flat metrics on closed oriented hyperbolic 4-manifolds with scalar curvature bounded below by -12 . These are extra geometric restrictions on the metric and therefore do not prove the arbitrary-metric statement in dimension 4.

1.24.3.4. What remains unresolved

The unresolved part is the exact scalar-curvature-only comparison for arbitrary smooth metrics on closed hyperbolic n -manifolds in dimensions $n \geq 4$:

$$\text{Scal}_g \geq -n(n-1) \implies \text{Vol}(M, g) \geq \text{Vol}(M, g_{\text{hyp}}).$$

No result found removes the extra assumptions appearing in the known higher-dimensional results. Therefore the problem should be classified as **partially resolved**, not fully resolved.

1.24.4. Remaining open problems recorded by the database

1.24.4.1. Open problem 1

Let (M^4, g_{hyp}) be a closed oriented hyperbolic four-manifold. Prove that every Riemannian metric g on M satisfying $\text{Scal}_g \geq -12$ has $\text{Vol}(M, g) \geq \text{Vol}(M, g_{\text{hyp}})$, without imposing local conformal flatness, self-duality, anti-self-duality, or closeness to g_{hyp} .

Reference supplied by the database. Jialong Deng, “Scalar Curvature in Dimension 4,” arXiv preprint, 2025, §1 and Theorem A (explicitly treats the unrestricted dimension-four Schoen conjecture as open and proves only additional-hypothesis cases; no separately numbered conjecture). arXiv:2512.13528. <https://arxiv.org/abs/2512.13528>

1.24.5. Primary-source pointer

The originating ICM record is *Metaphors in Systolic Geometry* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>.

1.25. 25. Metaphors in Systolic Geometry [2010-geom-116]

Record: 2010-geom-116

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Metaphors in Systolic Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>

1.25.1. Problem statement

Let $r > 0$. Let (M^n, hyp) be a closed hyperbolic n -manifold, where hyp denotes its hyperbolic Riemannian metric. For a Riemannian metric h on M and a point $p \in M$, define the macroscopic scalar curvature $\text{Sc}_r^h(p)$ at scale r as follows: let $\tilde{V}_h(p, r)$ be the volume of the ball of radius r centered at any lift of p in the universal cover of (M, h) , and let $\tilde{V}_S(r)$ be the volume of the ball of radius r in the simply connected n -dimensional space of constant curvature and scalar curvature S ; then $\text{Sc}_r^h(p)$ is the number S such that $\tilde{V}_h(p, r) = \tilde{V}_S(r)$.

Prove that if g is a Riemannian metric on M such that $\text{Sc}_r^g(p) \geq \text{Sc}_r^{\text{hyp}}(p)$ for every $p \in M$, equivalently, every radius- r ball in the universal cover of (M, g) has volume at most the volume of a radius- r ball in the universal cover of (M, hyp) , then

$$\text{Vol}(M, g) \geq \text{Vol}(M, \text{hyp}).$$

1.25.2. Historical background, literature, and current progress

1.25.3. Verdict: partially resolved

1.25.3.1. Precise problem checked

The problem asks for the sharp macroscopic Schoen/Guth conjecture for closed hyperbolic manifolds:

For every closed hyperbolic n -manifold (M, hyp) , every Riemannian metric g on M , and every fixed radius $r > 0$, if every radius- r ball in the universal cover of (M, g) has volume at most the radius- r ball in H^n , equivalently $\text{Sc}_r^g(p) \geq \text{Sc}_r^{\text{hyp}}(p)$ for all p , then

$$\text{Vol}(M, g) \geq \text{Vol}(M, \text{hyp}).$$

Equivalently, by contrapositive, if $\text{Vol}(M, g) < \text{Vol}(M, \text{hyp})$, then for every $r > 0$ there must be some radius- r ball in the universal cover of (M, g) whose volume is larger than the hyperbolic radius- r ball. This is Guth's sharp Conjecture 2 / generalized Schoen conjecture.

1.25.3.2. Status

No citable paper found proves this exact sharp statement. The literature gives substantial partial progress, but all available theorems found either use a non-sharp small constant in place of 1, restrict the radius range, add extra hypotheses such as negative sectional curvature, or prove only an asymptotic large-radius conclusion.

1.25.3.3. Key literature and what it proves

1. Larry Guth, “Metaphors in systolic geometry”, Proceedings of the International Congress of Mathematicians, Vol. II, Hindustan Book Agency, 2010, pp. 745-768. DOI: 10.1142/9789814324359_0072. URL: <https://math.mit.edu/~lguth/Exposition>

This is the source containing the stated “Generalized Schoen conjecture” formulation: for every $r > 0$, $Sc_r(g) \geq Sc_r(hyp)$ should imply $Vol(M, g) \geq Vol(M, hyp)$. Guth explicitly describes this conjecture as far out of reach. The same exposition records only a non-sharp theorem: for $r = 1$, the unit-ball upper bound in the universal cover implies $Vol(M, g) \geq c(n) Vol(M, hyp)$, with a dimensional constant $c(n)$, not the sharp constant 1.

2. Larry Guth, “Volumes of balls in large Riemannian manifolds”, Annals of Mathematics 173(1), 51-76, 2011. DOI: 10.4007/annals.2011.173.1.2. arXiv: math.DG/0610212. URL: <https://annals.math.princeton.edu/2011/173-1/p02>.

Guth states Conjecture 2: if (M, hyp) is closed hyperbolic and $Vol(M, g) < Vol(M, hyp)$, then $V_{\{\widetilde{M}, \widetilde{g}\}}(R) > V_{\{H^n\}}(R)$ for all R . This is the contrapositive form of the sharp statement. Theorem 2 proves only the non-sharp case: for each dimension n there is $\delta(n) > 0$ such that $Vol(M, g) < \delta(n) Vol(M, hyp)$ forces a unit ball in the universal cover to be larger than the unit ball in H^n . Guth also notes that variants for other fixed radii have constants depending on the radius and that the constants deteriorate; this does not resolve the sharp conjecture.

3. Gerard Besson, Gilles Courtois, and Sylvestre Gallot, “Minimal entropy and Mostow’s rigidity theorems”, Ergodic Theory and Dynamical Systems 16(4), 623-649, 1996. DOI: 10.1017/S0143385700009019. URL: <https://www.cambridge.org/core/journal-theory-and-dynamical-systems/article/minimal-entropy-and-mostows-rigidity-theorems/7B7F706676FAC80C728846B8A29AFF00>.

The minimal entropy theorem implies the required ball-volume conclusion only for all sufficiently large radii, with the threshold depending on the metric g , when $Vol(M, g) < Vol(M, hyp)$. This is a sharp asymptotic/large-scale result, but it does not imply the stated fixed arbitrary radius statement, especially not small radii.

4. Steve Karam, “Growth of balls in the universal cover of surfaces and graphs”, Transactions of the American Mathematical Society 367(8), 5355-5373, 2015. DOI: 10.1090/S0002-9947-2015-06189-3. arXiv: 1304.3567. URL: <https://arxiv.org/>

Karam proves a uniform non-sharp version in dimension 2: for closed hyperbolic surfaces, $Area(M, h) \leq \delta Area(M, hyp)$ implies $V_{\{\widetilde{M}, \widetilde{h}\}}(R) \geq V_{\{H^2\}}(R)$ for every $R \geq 1$. This is important partial progress, but it is only dimension 2, uses a small constant δ instead of 1, and covers $R \geq 1$ rather than all $R > 0$.

5. Florent Balacheff and Steve Karam, “Macroscopic Schoen conjecture for manifolds with nonzero simplicial volume”, *Transactions of the American Mathematical Society* 372(10), 7071-7086, 2019. DOI: 10.1090/tran/7765. arXiv: 1709.10049. URL: <https://arxiv.org/abs/1709.10049>.

Theorem 1.2 gives a non-sharp simplicial-volume inequality for ratios of universal-cover ball volumes for $R \geq 1$. Theorem 1.3 gives, for a closed hyperbolic manifold with an auxiliary metric of negative sectional curvature and sufficiently small volume $\text{Vol}(M, g) < \beta_n \text{Vol}(M, \text{hyp})$, the conclusion that for every $R \geq 1$ some universal-cover R -ball is larger than the hyperbolic one. This has both a non-sharp constant and an extra negative-curvature hypothesis, and it does not cover small radii.

6. Sabine Braun and Roman Sauer, “Volume and macroscopic scalar curvature”, *Geometric and Functional Analysis* 31, 1321-1376, 2021. DOI: 10.1007/s00039-021-00588-y. arXiv: 2012.08999. URL: <https://arxiv.org/abs/2012.08999>.

Theorem 1.1 proves that an upper bound on volumes of unit balls in the universal cover gives $||M|| \leq \text{const}(d, V_1) \text{Vol}(M)$. For a closed hyperbolic manifold this implies a Guth-type volume lower bound only up to a dimensional constant. The paper explicitly describes Guth’s theorem as the non-sharp macroscopic cousin of Schoen’s conjecture because of this dimensional constant.

7. Stephane Sabourau, “Macroscopic scalar curvature and local collapsing”, *Annales Scientifiques de l’Ecole Normale Supérieure* 55(4), 919-936, 2022. DOI: 10.24033/asens.2509. arXiv: 2006.00663. URL: <https://arxiv.org/abs/2006.00663>.

Theorem 1.4 proves that, for a closed hyperbolic n -manifold and any metric g , if $\text{Vol}(M, g)$ is smaller than a dimensional absolute constant δ'_n , then $V_{\widetilde{M}}(r) \geq V_{H^n}(r)$ for every $r \geq 1$. This gives uniform radius control and removes the negative-curvature hypothesis, but the volume bound is not the sharp relative bound $\text{Vol}(M, \text{hyp})$ and the result still covers only $r \geq 1$.

8. Hannah Alpert, “Growth in the universal cover under large simplicial volume”, *Journal of Topology and Analysis* 18(4), 1033-1042, 2026. DOI: 10.1142/S1793525325500025. arXiv: 2402.04932. URL: <https://arxiv.org/abs/2402.04932>.

The main theorem proves that if $(\text{Vol}(M, g))^2 / ||M||_{\Delta} < \delta'_n$, then $V_r(\widetilde{M}) \geq V_r(H^n)$ for all $r \geq 1$. The introduction explicitly calls Guth’s sharp statement, $\text{Vol}(M, g) < \text{Vol}(M, \text{hyp})$ implying the ball-volume conclusion for all r , the “most ambitious conjecture” and describes the result as partial progress. Thus this paper also does not resolve the problem statement.

1.25.3.4. What remains unresolved

The unresolved part is exactly the sharp threshold and full scale range:

- replace all small constants $c(n)$, $\delta(n)$, β_n , or absolute δ'_n by the sharp constant 1 relative to $\text{Vol}(M, \text{hyp})$;
- allow every Riemannian metric g , without extra curvature assumptions;
- prove the conclusion for every fixed $r > 0$, including small radii, not only $r = 1$, $r \geq 1$, or sufficiently large r .

Therefore the correct status for the problem is **partially resolved**, not fully resolved.

1.25.4. Remaining open problems recorded by the database

1.25.4.1. Open problem 1

For every integer $n \geq 3$, prove that there exists $\delta_n > 0$ with the following property. If M is a closed, oriented, connected n -manifold admitting a hyperbolic metric and g is another Riemannian metric on M such that

$$\frac{\text{Vol}(M, g)}{\|M\|_{\Delta}} < \delta_n,$$

where $\|M\|_{\Delta}$ is the simplicial volume, then for every $r \geq 1$ the universal cover of (M, g) contains an r -ball whose volume is at least the volume of an r -ball in $\mathbb{H}^n$.

Reference supplied by the database. Hannah Alpert, “Growth in the Universal Cover under Large Simplicial Volume,” *Journal of Topology and Analysis* 18, no. 4 (2026), 1033–1042, Conjecture 4 (attributed to a suggestion of Larry Guth; the paper notes that only $n = 2$ is known). DOI: 10.1142/S1793525325500025. arXiv:2402.04932. <https://doi.org/10.1142/S1793525325500025>

1.25.5. Primary-source pointer

The originating ICM record is *Metaphors in Systolic Geometry* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>.

1.26. 26. Riemannian Manifolds of Positive Curvature [2010-geom-118]

Record: 2010-geom-118

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: reviewed (reviewed flag: yes)

Original source: ICM 2010: Riemannian Manifolds of Positive Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=442>

1.26.1. Problem statement

Prove that for every even integer n , if (M^n, g) is a compact n -dimensional Riemannian manifold whose sectional curvature is positive at every point and in every two-dimensional tangent plane, then the Euler characteristic $\chi(M)$ is positive.

1.26.2. Historical background, literature, and current progress

1.26.3. Verdict: partially resolved

1.26.3.1. Precise problem

This problem asks for the positive-sectional-curvature Euler-characteristic Hopf conjecture:

For every even integer n , every compact n -dimensional Riemannian manifold with positive sectional curvature in every tangent 2-plane has positive Euler characteristic.

Equivalently, for every closed positively curved Riemannian manifold M of even dimension, prove $\chi(M) > 0$.

1.26.3.2. Status

The literature does not fully resolve this exact all-even-dimensional statement. The unrestricted conjecture is known in low dimensions and under additional hypotheses such as large symmetry, but no published or otherwise citable result found proves the statement for all compact positively curved even-dimensional manifolds.

The problem is therefore partially resolved, not fully resolved. It is also not best recorded as “fully open”, because dimensions 2 and 4 and many special classes are known.

What remains unresolved is the general case in even dimensions at least 6 without extra hypotheses such as homogeneity, cohomogeneity bounds, or torus-symmetry assumptions.

1.26.3.3. Original source and current-status references

1. Title: Differentialgeometrie und topologische Gestalt Authors: Heinz Hopf Venue: Jahresbericht der Deutschen Mathematiker-Vereinigung 41, 209-229 Year: 1932 DOI/arXiv/URL: <https://eudml.org/doc/145926> Role: Original source for Hopf's questions in global differential geometry; later literature cites this as the source of the positive-curvature Euler-characteristic conjecture.
2. Title: Riemannian Manifolds with Positive Sectional Curvature Authors: Wolfgang Ziller Venue: Survey article; available from the author's webpage Year: 2010/2014 survey version DOI/arXiv/URL: <https://www2.math.upenn.edu/~wziller/papers/SurveyMexico.pdf> Role: Lists Hopf's conjecture that an even-dimensional manifold with positive curvature has positive Euler characteristic, and surveys progress via positive curvature and symmetry. It does not report a general solution.
3. Title: On the Hopf conjecture with symmetry Authors: Lee Kennard Venue: Geometry & Topology 17, 563-593 Year: 2013 DOI/arXiv/URL: DOI 10.2140/gt.2013.17.563; arXiv:1203.3808; <https://arxiv.org/abs/1203.3808> Role: States that the Hopf conjecture holds in dimensions 2 and 4 but remains open in higher dimensions. Proves a partial result: if n is divisible by 4 and M has positive sectional curvature with an effective isometric torus action of rank at least $2 \log_2(n)$, then $\chi(M) > 0$. This has an extra symmetry hypothesis and does not resolve the full statement.

1.26.3.4. Partial progress

1. Title: A simple intrinsic proof of the Gauss-Bonnet formula for closed Riemannian manifolds Authors: Shiing-Shen Chern Venue: Annals of Mathematics 45, 747-752 Year: 1944 DOI/arXiv/URL: <https://doi.org/10.2307/1969302> Role: Provides the intrinsic Gauss-Bonnet formula. In dimension 2, positive sectional curvature is positive Gaussian curvature, so Gauss-Bonnet gives $\chi(M) > 0$.
2. Title: On the curvature and characteristic classes of a Riemannian manifold Authors: Shiing-Shen Chern Venue: Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg 20, 117-126 Year: 1955 DOI/arXiv/URL: DOI 10.1007/BF02960745 Role: Standard reference in the Hopf-conjecture literature for curvature and characteristic classes. Later papers report that the algebraic Hopf conjecture was settled in dimension 4 by Milnor, unpublished, as cited through Chern; this supports the known four-dimensional case but does not extend to all dimensions.
3. Title: On the connectivity of spaces of positive curvature Authors: John L. Synge Venue: The Quarterly Journal of Mathematics (Oxford) os-7, 316-320 Year: 1936 <https://doi.org/10.1093/qmath/os-7.1.316> Role: Proves that a compact even-dimensional manifold with positive sectional curvature has fundamental group 0 or $\mathbb{Z}_2$ (simply connected when orientable). With Poincaré duality this gives $b_1 = b_3 = 0$, hence $\chi = 2 + b_2 > 0$ in dimension 4. This is a topological route to the four-dimensional case, independent of positivity of the Gauss-Bonnet-Chern integrand, and therefore not affected by Geroch's dimension-6 counterexample to the algebraic approach.
4. Title: Positive sectional curvature does not imply positive Gauss-Bonnet integrand Authors: Robert Geroch Venue: Proceedings of the American Mathematical Society 54, 267-270 Year: 1976 DOI/arXiv/URL: DOI 10.1090/S0002-9939-1976-0390961-8 Role: Shows that the stronger algebraic strategy fails in dimension 6: positive sectional curvature of an algebraic curvature tensor need not make the Gauss-Bonnet integrand positive. This

is not a counterexample to Hopf's global conjecture, but explains why Gauss-Bonnet integrand positivity does not solve the higher-dimensional cases.

5. Title: The Hopf conjecture for manifolds with low cohomogeneity or high symmetry rank
Authors: Thomas Puttmann and Catherine Searle Venue: Proceedings of the American Mathematical Society 130, no. 1, 163-166 Year: 2002 DOI/arXiv/URL: DOI 10.1090/S0002-9939-01-06039-7; <https://arxiv.org/abs/1207.4086> Role: Proves positive Euler characteristic for compact even-dimensional positively curved manifolds under low cohomogeneity or high symmetry rank assumptions. In particular, their Theorem 1 assumes an isometric compact Lie group action with $k - (\text{rank } G - \text{rank } H) \leq 5$; their Theorem 2 assumes a torus action of specified rank. These extra hypotheses do not cover the full statement.
6. Title: Splitting of torus representations and applications in the Grove symmetry program
Authors: Lee Kennard, Michael Wiemeler, and Burkhard Wilking Venue: arXiv preprint Year: 2021 DOI/arXiv/URL: DOI 10.48550/arXiv.2106.14723; arXiv:2106.14723; <https://arxiv.org/abs/2106.14723> Role: Proves the Hopf conclusion under the extra assumption that the isometry group has rank at least 5. This is strong progress in the Grove symmetry program, but it is conditional on symmetry and does not imply the unrestricted conjecture.
7. Title: An improved four-periodicity theorem and a conjecture of Hopf with symmetry
Authors: Jan Nienhaus Venue: arXiv preprint / PhD thesis Year: 2022 DOI/arXiv/URL: DOI 10.48550/arXiv.2211.13151; arXiv:2211.13151; <https://arxiv.org/abs/2211.13151> Role: Improves the symmetry result to effective isometric T^4 -actions: every even-dimensional closed positively curved manifold with such an action has $\chi(M) > 0$. The T^4 -action hypothesis is an additional assumption, so this does not resolve the full statement.
8. Title: Positive curvature, torus symmetry, and matroids
Authors: Lee Kennard, Michael Wiemeler, and Burkhard Wilking Venue: Journal of the European Mathematical Society (JEMS) Year: 2025 DOI/arXiv/URL: DOI 10.4171/JEMS/1656; arXiv:2212.08152; <https://arxiv.org/abs/2212.08152> Role: Recent progress on positive sectional curvature with torus symmetry, using matroid methods and torus-representation structure. Its theorems retain torus-symmetry and isotropy hypotheses, so they do not prove the full Hopf Euler-characteristic conjecture.

1.26.3.5. Applicability check

None of the cited higher-dimensional results covers every compact even-dimensional positively sectional-curved Riemannian manifold. The main modern positive results require extra assumptions such as a large torus action, low cohomogeneity, rank bounds on the isometry group, or special isotropy structure. These hypotheses are not present in the problem statement.

Thus the exact statement remains unresolved in general, with known partial cases and conditional progress only.

1.26.4. Remaining open problems recorded by the database

1.26.4.1. Open problem 1

Let p be a prime, let M be a topological space, and suppose that a nonzero class $x \in H^k(M; \mathbb{Z}_p)$ induces cohomological periodicity up to degree pk and has minimal degree among all classes with this property. Prove that:

1. if $p = 2$, then $k \in \{1, 2, 4, 8\}$, and if x induces periodicity up to degree $3k$, then $k \neq 8$;
2. if $p > 2$, then $k = 2\lambda$ for some $\lambda \mid (p - 1)$.

Here, a class $x \in H^k(M; \mathbb{Z}_p)$ induces periodicity up to degree c if multiplication by x is surjective $H^i(M; \mathbb{Z}_p) \rightarrow H^{i+k}(M; \mathbb{Z}_p)$ for $0 \leq i < c - k$ and injective for $0 < i \leq c - k$.

Reference supplied by the database. Lee Kennard, "On the Hopf Conjecture with Symmetry," *Geometry & Topology* **17** (2013), 563–593, Conjecture 6.3 and the discussion immediately following it, DOI: [10.2140/gt.2013.17.563](https://doi.org/10.2140/gt.2013.17.563), arXiv: [1203.3808](https://arxiv.org/abs/1203.3808).

1.26.5. Primary-source pointer

The originating ICM record is *Riemannian Manifolds of Positive Curvature* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=442>.

1.27. 27. Riemannian Manifolds of Positive Curvature [2010-geom-119]

Record: 2010-geom-119

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Riemannian Manifolds of Positive Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=442>

1.27.1. Problem statement

Call a Riemannian manifold (M, g) strictly δ -pinched in the pointwise sense if, for every point $p \in M$ and every pair of two-dimensional planes $\pi_1, \pi_2 \subset T_p M$, the sectional curvatures satisfy $0 < \delta K(\pi_1) < K(\pi_2)$, where $K(\pi)$ denotes the sectional curvature of π .

Prove that the following two assertions hold:

1. For every even integer n , there exists a real number $\delta_1(n) \in (0, 1/4)$ such that if (M, g_0) is a compact, simply connected Riemannian manifold of dimension n that is strictly $\delta_1(n)$ -pinched in the pointwise sense, then M is homeomorphic to S^n or diffeomorphic to a compact symmetric space of rank one.
2. For each integer $n \geq 4$, there exists a real number $\delta_2(n) \in (0, 1/4)$ such that if (M, g_0) is a compact, simply connected Riemannian manifold of dimension n that is strictly $\delta_2(n)$ -pinched in the pointwise sense, then M is diffeomorphic to S^n or to a compact symmetric space of rank one.

1.27.2. Historical background, literature, and current progress

1.27.3. Verdict: partially resolved

1.27.3.1. Precise problem checked

The problem asks for sub-quarter *pointwise* pinching constants. In the notation of the problem, the hypothesis is that for each point p , all sectional curvatures of two-planes in $T_p M$ have ratio $K_{\min}(p)/K_{\max}(p) > \delta_i(n)$, with $\delta_i(n) < 1/4$. This is weaker than a global pinching bound of the form $\sec_g \in [1/4 - \varepsilon, 1]$ after one normalization.

The requested conclusions are:

- for every even n , such pointwise $\delta_1(n)$ -pinching implies M is homeomorphic to S^n or diffeomorphic to a compact rank-one symmetric space;
- for every $n \geq 4$, such pointwise $\delta_2(n)$ -pinching implies M is diffeomorphic to S^n or to a compact rank-one symmetric space.

1.27.3.2. Literature found

1. Classification of Almost Quarter-Pinched Manifolds

Peter Petersen and Terence Tao, *Proceedings of the American Mathematical Society* 137 (2009), no. 7, 2437-2440. DOI: 10.1090/S0002-9939-09-09802-5. arXiv: 0807.1900. URL: <https://arxiv.org/abs/0807.1900>

Theorem 1.1 proves that for each n there is $\varepsilon(n) > 0$ such that any simply connected n -manifold with sectional curvatures globally in $[1/4 - \varepsilon, 1]$ is diffeomorphic to a sphere or compact rank-one symmetric space. This resolves the analogous *global* almost quarter-pinching problem and strengthens the older even-dimensional Berger conclusion. It does not cover the problem statement exactly, because the statement assumes only pointwise pinching.

2. Manifolds with 1/4-pinched Curvature are Space Forms

Simon Brendle and Richard Schoen, *Journal of the American Mathematical Society* 22 (2009), no. 1, 287-307. DOI: 10.1090/S0894-0347-08-00613-9. arXiv: 0705.0766. URL: <https://arxiv.org/abs/0705.0766>

Theorem 1 proves the differentiable sphere theorem under strictly pointwise 1/4-pinched sectional curvatures. This is a pointwise theorem, but it is an endpoint theorem with pinching constant 1/4, not a sub-quarter gap theorem for some $\delta(n) < 1/4$.

3. Classification of manifolds with weakly 1/4-pinched curvatures

Simon Brendle and Richard Schoen, *Acta Mathematica* 200 (2008), 1-13. DOI: 10.1007/s11511-008-0022-7. arXiv: 0705.3963. URL: <https://arxiv.org/abs/0705.3963>

The main theorem classifies compact locally irreducible manifolds M for which $M \times \mathbb{R}^2$ has nonnegative isotropic curvature, and gives the weak pointwise 1/4-pinched classification. Again, this is endpoint pointwise progress, not the requested $\delta(n) < 1/4$ pointwise result.

4. Sur les variétés riemanniennes pincées juste au-dessous de 1/4

Marcel Berger, *Annales de l'Institut Fourier* 33 (1983), no. 2, 135-150. URL: https://www.numdam.org/item/AIF_1983__33_2_135_0/

Berger proved an even-dimensional almost quarter-pinching result with the weaker conclusion that the manifold is homeomorphic to a sphere or diffeomorphic to a compact rank-one symmetric space. Petersen-Tao explicitly cite this as the earlier even-dimensional result. This is part of the global almost quarter-pinching line, not a full pointwise sub-quarter solution.

5. A sphere theorem with a pinching constant below 1/4

Uwe Abresch and Wolfgang T. Meyer, *Journal of Differential Geometry* 44 (1996), no. 2, 214-261. URL: <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-44/issue-2/A-sphere-theorem-with-a-pinching-constant-below-1-4/10.4310/jdg/1214458996.full>

This gives related below-1/4 sphere-theorem progress in the global pinching tradition, but it does not supply the full pointwise classification asked in the problem statement.

1.27.3.3. Status assessment

The closest complete theorem is Petersen-Tao, but its hypothesis is a global sectional-curvature interval $[1/4 - \varepsilon, 1]$. A compact pointwise δ -pinched metric need not satisfy any such global δ -pinching after a single rescaling, because the curvature scale can vary from point to point.

Therefore Petersen-Tao does not formally imply the pointwise statement.

The Brendle-Schoen theorems are pointwise, but they handle the exact $1/4$ endpoint, not the existence of a dimension-dependent constant below $1/4$ whose weaker pointwise hypothesis still forces the same classification.

I did not find a published or otherwise citable theorem that proves the full statement exactly as written for pointwise $\delta_i(n) < 1/4$. The problem is therefore not fully resolved as stated. The current literature gives strong partial progress: full resolution for global almost quarter-pinching and endpoint pointwise $1/4$ -pinching, but not the sub-quarter pointwise gap demanded here.

1.27.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.27.5. Primary-source pointer

The originating ICM record is *Riemannian Manifolds of Positive Curvature* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=442>.

1.28. 28. Extremal Kähler metrics [2014-geom-122]

Record: 2014-geom-122

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Extremal Kähler metrics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1035>

1.28.1. Problem statement

Let M be a compact complex projective manifold and let $L \rightarrow M$ be an ample line bundle, so that (M, L) is a polarized manifold. Suppose that M has no non-zero holomorphic vector fields. Prove that M admits a constant scalar curvature Kähler metric whose Kähler class is the first Chern class $c_1(L)$ if and only if the polarized pair (M, L) is K -stable in Donaldson's sense: for every test-configuration χ for (M, L) , meaning a $\mathbb{C}^*$ -equivariant flat polarized family $(M, \mathcal{L})$ over $\mathbb{C}$ with generic fiber isomorphic to (M, L^r) for some exponent r , the Futaki invariant satisfies $\text{Fut}(\chi) > 0$ whenever the norm $\|\chi\|$ is positive.

1.28.2. Historical background, literature, and current progress

1.28.3. Verdict: partially resolved

1.28.3.1. Precise problem checked

The problem asks for the general constant-scalar-curvature Kähler form of the Yau-Tian-Donaldson conjecture:

For every smooth compact complex projective polarized manifold (M, L) with no non-zero holomorphic vector fields, does M admit a cscK metric in $c_1(L)$ if and only if (M, L) is K -stable in Donaldson's original test-configuration sense, i.e. $\text{Fut}(\chi) > 0$ for every test configuration of positive norm?

This is stronger, as stated, than the currently proved general equivalences with uniform K -stability/non-Archimedean coercivity.

1.28.3.2. Status

I did not find a published or otherwise citable result proving the exact statement in the problem statement for all smooth polarized manifolds with discrete automorphism group. The literature proves major pieces:

- cscK existence implies K -stability/ K -polystability under the stated discrete-automorphism hypothesis;
- several important special cases are resolved;

- the full general existence theorem is known under uniform K-stability or equivalent non-Archimedean coercivity-type hypotheses.

What remains unresolved for the problem as written is the converse from plain Donaldson K-stability, with only strict positivity on individual positive-norm test configurations, to existence of a cscK metric in $c_1(L)$. Current general theorems require a uniform lower bound/non-Archimedean coercivity condition, and the sources explicitly distinguish this from ordinary K-stability.

1.28.3.3. Key literature checked

1. **Title:** Scalar curvature and stability of toric varieties
Authors: S. K. Donaldson
Venue/year: Journal of Differential Geometry 62 (2002), 289-349
DOI/arXiv/URL: DOI 10.4310/jdg/1090950195; projecteuclid.org/euclid.jdg/1090950195
Relevance: Introduces Donaldson's test configurations and the K-stability framework for polarized varieties, and formulates the cscK/stability expectation in the toric setting. This is part of the source of the stated "Donaldson's sense" condition.
2. **Title:** Lower bounds on the Calabi functional
Authors: S. K. Donaldson
Venue/year: Journal of Differential Geometry 70 (2005), 453-472
DOI/arXiv/URL: DOI 10.4310/jdg/1143643312
Relevance: Gives the lower-bound mechanism relating the Calabi functional to Donaldson-Futaki invariants. It supports the necessary direction but does not prove the full converse.
3. **Title:** K-stability of constant scalar curvature Kähler manifolds
Authors: J. Stoppa
Venue/year: Advances in Mathematics 221 (2009), 1397-1408
DOI/arXiv/URL: DOI 10.1016/j.aim.2009.02.009; arXiv:0803.4095
Relevance: Proves that a polarized manifold admitting a cscK metric and having finite automorphism group is K-stable. This covers the "only if" direction of the problem under the no-holomorphic-vector-fields hypothesis, but not the converse.
4. **Title:** Kähler-Einstein metrics and stability
Authors: X. X. Chen, S. Donaldson, and S. Sun
Venue/year: International Mathematics Research Notices 2014, no. 8, 2119-2125, plus the detailed series in Journal of the American Mathematical Society 28 (2015)
DOI/arXiv/URL: DOI 10.1093/imrn/rns275; arXiv:1210.7494 and related preprints
Relevance: Resolves the Fano Kähler-Einstein version of Yau-Tian-Donaldson. This is a landmark special case, not the full cscK polarized-manifold statement in the problem statement.
5. **Title:** An Introduction to Extremal Kähler Metrics
Author: G. Székelyhidi
Venue/year: Graduate Studies in Mathematics 152, American Mathematical Society, 2014
DOI/arXiv/URL: AMS book page, ISBN 978-1-4704-1047-4
Relevance: Surveys the cscK/extremal Yau-Tian-Donaldson problem and records the general cscK converse as a central conjectural problem rather than a theorem.
6. **Title:** On the constant scalar curvature Kähler metrics, I: A priori estimates; II: Existence results

- Authors:** X. X. Chen and J. Cheng
Venue/year: Journal of the American Mathematical Society 34 (2021), 909-936 and 937-1009
DOI/arXiv/URL: DOI 10.1090/jams/967 and DOI 10.1090/jams/966
Relevance: Proves deep analytic existence results for cscK metrics in terms of properness/coercivity of the Mabuchi K-energy. This is a major analytic component of YTD, but it is not the plain Donaldson K-stability criterion.
7. **Title:** Uniform stability of twisted constant scalar curvature Kähler metrics
Author: R. Dervan
Venue/year: International Mathematics Research Notices 2016, no. 15, 4728-4783
DOI/arXiv/URL: DOI 10.1093/imrn/rnv291; arXiv:1412.0648
Relevance: Introduces the minimum norm and formulates uniform K-stability as the stability notion analogous to coercivity of the Mabuchi functional. This paper explicitly shows why the pointwise positivity is not the same as the quantitative condition used in later existence theorems.
8. **Title:** Uniform K-stability, Duistermaat-Heckman measures and singularities of pairs
Authors: S. Boucksom, T. Hisamoto, and M. Jonsson
Venue/year: Annales de l'Institut Fourier 67 (2017), no. 2, 743-841
DOI/arXiv/URL: DOI 10.5802/aif.3096; arXiv:1504.00682
Relevance: Develops non-Archimedean and Duistermaat-Heckman tools for uniform K-stability. Its role in the literature is precisely to bridge the gap between pointwise positivity of Donaldson-Futaki invariants and analytic coercivity.
9. **Title:** Uniform K-stability and asymptotics of energy functionals in Kähler geometry
Authors: S. Boucksom, T. Hisamoto, and M. Jonsson
Venue/year: Journal of the European Mathematical Society 21 (2019), 2905-2944
DOI/arXiv/URL: DOI 10.4171/JEMS/894; arXiv:1603.01026
Relevance: Relates uniform K-stability/non-Archimedean functionals to slopes of analytic energy functionals. This is a key bridge toward uniform YTD results, but again uses uniform stability rather than the pointwise Donaldson K-stability.
10. **Title:** Geodesic rays and stability in the cscK problem
Author: C. Li
Venue/year: Annales scientifiques de l'École normale supérieure, accepted version on arXiv, 2021
DOI/arXiv/URL: arXiv:2001.01366
Relevance: Reduces the uniform Yau-Tian-Donaldson conjecture for cscK metrics to a Boucksom-Jonsson regularization conjecture and proves sufficient uniform stability conditions for cscK existence. This is important progress but is explicitly uniform, not the stated ordinary Donaldson K-stability.
11. **Title:** On the Yau-Tian-Donaldson conjecture for weighted cscK metrics
Authors: S. Boucksom and M. Jonsson
Venue/year: arXiv preprint, 2025
DOI/arXiv/URL: arXiv:2509.15016; <https://arxiv.org/abs/2509.15016>
Relevance: Establishes a version of the Yau-Tian-Donaldson conjecture for cscK and weighted extremal metrics using non-Archimedean pluripotential theory. The formulation is not the plain Donaldson positivity criterion.
12. **Title:** From complex to non-Archimedean geometry: an approach to the YTD conjecture
Authors: S. Boucksom and M. Jonsson
Venue/year: arXiv preprint/SRI notes, 2026

DOI/arXiv/URL: arXiv:2602.10800; <https://arxiv.org/abs/2602.10800>

Relevance: Surveys and sketches the non-Archimedean approach to a version of YTD for cscK metrics. This confirms the current modern formulation through non-Archimedean/uniform conditions.

13. **Title:** A solution to the Yau-Tian-Donaldson Conjecture

Author: A. Trusiani

Venue/year: arXiv preprint, 2026

DOI/arXiv/URL: arXiv:2605.30063; <https://arxiv.org/abs/2605.30063>

Relevance: Gives a recent citable claimed solution of the general YTD correspondence in uniform/non-Archimedean terms. The abstract and introduction distinguish K-polystability from the stronger uniform conditions, and state that existence implies K-polystability while the converse is not expected in that non-uniform form. Therefore this does not resolve the exact statement with only Donaldson's pointwise K-stability.

1.28.3.4. Why this is not fully resolved

The stated stability hypothesis is:

$$\text{Fut}(\chi) > 0 \quad \text{for every positive-norm test configuration } \chi.$$

This is pointwise positivity. The general existence theorems I found require a quantitative estimate of the form

$$\text{Fut}(\chi) \geq \delta \|\chi\| \quad \text{or an equivalent non-Archimedean/coercivity condition}$$

for some uniform $\delta > 0$, possibly modulo automorphisms. Since the statement assumes no non-zero holomorphic vector fields, the automorphism issue is simplified, but the distinction between pointwise K-stability and uniform K-stability remains the decisive gap. I found no citable theorem proving that the plain Donaldson K-stability is equivalent to uniform K-stability for all smooth polarized manifolds with discrete automorphism group, nor a theorem deriving cscK existence directly from the pointwise condition in full generality.

Therefore the current status of the exact problem is **partially resolved**, not fully resolved.

1.28.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.28.5. Primary-source pointer

The originating ICM record is *Extremal Kähler metrics* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1035>.

1.29. 29. New applications of Min-max Theory [2014-geom-128]

Record: 2014-geom-128

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: New applications of Min-max Theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>

1.29.1. Problem statement

Prove or disprove that, for every integer $n \geq 3$, among all non-totally geodesic minimal hypersurfaces in the unit round sphere

$$S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\},$$

the least possible $(n-1)$ -dimensional volume is attained by the following standard product hypersurface:

if $n = 2m$, then the minimizer is

$$S^{m-1}\left(\sqrt{\frac{m-1}{2m-1}}\right) \times S^m\left(\sqrt{\frac{m}{2m-1}}\right) \subset S^{2m} = S^n;$$

if $n = 2m - 1$, then the minimizer is

$$S^{m-1}\left(\frac{1}{\sqrt{2}}\right) \times S^{m-1}\left(\frac{1}{\sqrt{2}}\right) \subset S^{2m-1} = S^n.$$

Here $S^k(r)$ denotes the k -sphere of radius r , and each product is included in the unit sphere by the standard product embedding.

1.29.2. Historical background, literature, and current progress

1.29.3. Verdict: partially resolved

1.29.3.1. Precise problem checked

The problem asks for the sharp second-volume gap for closed non-totally geodesic minimal hypersurfaces in the unit round sphere. In the notation more common in the literature, for a hypersurface $M^d \subset S^{d+1}$, the conjecture says that the least possible d -volume among non-totally geodesic closed minimal hypersurfaces is the volume of the balanced Clifford hypersurface

$$S^k(\sqrt{k/d}) \times S^{d-k}(\sqrt{(d-k)/d}), \quad k = \lfloor d/2 \rfloor.$$

With $d = n - 1$, this is exactly the pair of products stated in the problem statement for ambient S^n . This is usually called Yau's conjecture or the Solomon-Yau/Solomon conjecture on the second lowest volume of minimal hypersurfaces in spheres.

1.29.3.2. Current status

No published or otherwise citable result found proves the full statement for all $n \geq 3$. The ambient S^3 case is resolved, and several higher-dimensional restricted cases are resolved, but the unrestricted embedded higher-dimensional case remains open. Therefore the problem is partially resolved, not fully resolved.

Recent literature still treats the higher-dimensional problem as partial. For example, Ding-Ge-Li (2025) explicitly says that for hypersurface dimension $d \geq 3$ “there are many partial results,” then lists rotational, asymptotic/cone, antipodal-symmetric, and nonembedded cases rather than a complete theorem.

1.29.3.3. Progress and relevant literature

1. Problem/source statement

- Title: *New applications of min-max theory*
- Author: Andre Neves
- Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, 2014
- arXiv/URL: arXiv:1409.7537, <https://arxiv.org/abs/1409.7537>
- Relevance: Section “Old questions” states exactly Solomon’s conjecture in the stated ambient-sphere notation, including the even and odd Clifford products. It records the S^3 case as confirmed by Marques-Neves and mentions only partial higher-dimensional progress.

2. Full solution only in ambient S^3

- Title: *The Min-Max theory and the Willmore conjecture*
- Authors: Fernando C. Marques and Andre Neves
- Venue/year: Annals of Mathematics 179 (2014), no. 2, 683-782
- DOI/arXiv/URL: DOI 10.4007/annals.2014.179.2.6; arXiv:1202.6036; <https://arxiv.org/abs/1202.6036>
- Result: Theorem B proves that every embedded closed minimal surface in S^3 of genus at least one has area at least $2\pi^2$, with equality only for the Clifford torus. Together with Almgren’s theorem that a minimal two-sphere in S^3 is totally geodesic and the Li-Yau lower bound for nonembedded surfaces, this proves the stated $n = 3$ case.

3. Nonembedded case in all dimensions

- Title: *Volume gap for minimal submanifolds in spheres*
- Authors: Jianquan Ge and Fagui Li
- Venue/year: arXiv preprint, updated 2024
- arXiv/URL: arXiv:2210.04654, <https://arxiv.org/abs/2210.04654>
- Result: Theorem 1.2 and Corollary 1.3 give a multiplicity/volume lower bound implying that if the volume is at most the Clifford value then the hypersurface must be embedded. The authors state this as verifying the nonembedded case of Yau’s conjecture. This does not settle embedded hypersurfaces in dimensions $d \geq 3$.

4. Independent nonembedded-case method

- Title: *Weighted monotonicity theorems and applications to minimal surfaces of $\mathbb{H}^n$ and S^n*
- Author: Manh Tien Nguyen

- Venue/year: Transactions of the American Mathematical Society 376 (2023), no. 8, 5899-5921
- DOI/arXiv/URL: DOI 10.1090/tran/8949; arXiv:2105.12625; <https://arxiv.org/abs/2105.12625>
- Result: Corollary 4.8 and Remark 4.9 show that a nonembedded minimal hypersurface in S^n has volume at least twice the equator volume, which is enough to confirm Yau's/Solomon's conjectured lower bound for nonembedded hypersurfaces. Again this leaves the embedded higher-dimensional case.

5. Antipodal-symmetric embedded case

- Title: *Isoperimetry and volume preserving stability in real projective spaces*
- Author: Celso Viana
- Venue/year: Journal of Differential Geometry 125 (2023), no. 1, 187-205
- DOI/arXiv/URL: DOI 10.4310/jdg/1695236595; arXiv:1907.09445; <https://arxiv.org/abs/1907.09445>
- Result: Theorem 1.4 proves a Willmore-type inequality for compact embedded hypersurfaces separating S^{d+1} into two antipodal-invariant regions, with equality only for the balanced Clifford hypersurface. Remark 1.5 says this confirms Solomon's conjecture in the class of minimal hypersurfaces with antipodal symmetry. The antipodal-invariance hypothesis is an extra restriction, so this is not the full statement.

6. Rotational hypersurfaces

- Title: *n-dimensional area of minimal rotational hypersurfaces in spheres*
- Authors: Oscar Perdomo and Guoxin Wei
- Venue/year: Nonlinear Analysis 125 (2015), 241-250
- DOI/URL: DOI 10.1016/j.na.2015.05.017
- Result: Verifies the conjectured lower bound for many compact minimal rotational hypersurfaces, reported in later papers as covering dimensions $2 \leq d \leq 100$ in that restricted rotational class.
- Title: *Area of minimal hypersurfaces in the unit sphere*
- Authors: Qing-Ming Cheng, Guoxin Wei, and Yuting Zeng
- Venue/year: Asian Journal of Mathematics 25 (2021), no. 2, 183-194
- DOI/arXiv/URL: DOI 10.4310/AJM.2021.v25.n2.a2; arXiv:1907.07314; <https://arxiv.org/abs/1907.07314>
- Result: Theorem 1.1 proves the Yau/Solomon conjecture for compact minimal rotational hypersurfaces.
- Title: *Area of Minimal Hypersurfaces in the Unit Sphere (II)*
- Authors: Qing-Ming Cheng, Junqi Lai, and Guoxin Wei
- Venue/year: Journal of Geometric Analysis 34 (2024), article 369
- DOI/URL: DOI 10.1007/s12220-024-01808-1; <https://doi.org/10.1007/s12220-024-01808-1>
- Result: Gives sharper area estimates for compact minimal rotational hypersurfaces and confirms conjectures from the preceding rotational papers. These theorems have a rotational-symmetry hypothesis, so they do not imply the unrestricted result.

7. Asymptotic/cone-type partial progress

- Title: *Sharp lower bounds on density for area-minimizing cones*
- Authors: Tom Ilmanen and Brian White
- Venue/year: Cambridge Journal of Mathematics 3 (2015), no. 1-2, 1-18
- DOI/URL: DOI 10.4310/CJM.2015.v3.n1.a1

- Result: Later papers summarize this as proving an asymptotic form of the conjecture for topologically nontrivial embedded hypersurfaces whose cones are area-minimizing. This has strong extra hypotheses and gives an asymptotic/cone statement, not the exact all-hypersurface Clifford minimizer statement in the problem statement.

8. Recent volume-gap improvements, still not the conjecture

- Title: *Volume gap between the minimal submanifold and the unit sphere*
- Authors: Weiran Ding, Jianquan Ge, and Fagui Li
- Venue/year: International Journal of Mathematics, 2025; arXiv preprint 2024
- DOI/arXiv/URL: DOI 10.1142/S0129167X25500296; arXiv:2412.01288; <https://arxiv.org/abs/2412.01288>
- Result: Improves Cheng-Li-Yau type volume gaps above the equator. Its introduction still describes the Solomon-Yau/Yau conjecture for $d \geq 3$ as having “many partial results” and lists the restricted results above. The gap it proves is not the sharp Clifford-hypersurface gap required by this problem.

1.29.3.4. What remains unresolved

For ambient spheres S^N with $N \geq 4$ (equivalently hypersurface dimension $d = N - 1 \geq 3$), no cited theorem found proves that every closed embedded non-totally geodesic minimal hypersurface has volume at least that of the balanced Clifford hypersurface, with equality only for that Clifford hypersurface, without extra assumptions such as rotational symmetry, antipodal symmetry, nonembeddedness, curvature conditions, or area-minimizing-cone hypotheses.

1.29.4. Remaining open problems recorded by the database

1.29.4.1. Open problem 1

Prove that every closed non-totally-geodesic minimal hypersurface $M^3 \subset S^4$ satisfies

$$\text{Vol}_3(M) \geq \text{Vol}_3\left(S^1\left(\sqrt{\frac{1}{3}}\right) \times S^2\left(\sqrt{\frac{2}{3}}\right)\right),$$

with equality only for that Clifford hypersurface up to an ambient isometry.

Reference supplied by the database. Qing-Ming Cheng, Guoxin Wei, and Yuting Zeng, “Area of minimal hypersurfaces in the unit sphere,” *Asian Journal of Mathematics* **25**(2) (2021), 183–194, introduction (the Solomon–Yau least-area conjecture, explicitly treated there only for rotational hypersurfaces). DOI: [10.4310/AJM.2021.v25.n2.a2](https://doi.org/10.4310/AJM.2021.v25.n2.a2). The fact that the conjecture remains open in every hypersurface dimension $n \geq 3$, hence in the displayed $M^3 \subset S^4$ case, is explicitly recorded in Jingwen Chen, “Symmetric mean curvature flow on the n -sphere,” *Nonlinear Analysis* **239** (2024), Article 113437, DOI: [10.1016/j.na.2023.113437](https://doi.org/10.1016/j.na.2023.113437).

1.29.5. Primary-source pointer

The originating ICM record is *New applications of Min-max Theory* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>.

1.30. 30. New applications of Min-max Theory [2014-geom-129]

Record: 2014-geom-129

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: New applications of Min-max Theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>

1.30.1. Problem statement

Prove that for every Riemannian metric on the 3-sphere S^3 , there exist at least five pairwise distinct embedded minimal surfaces in (S^3, g) that are tori.

1.30.2. Historical background, literature, and current progress

1.30.3. Verdict: partially resolved

1.30.3.1. Precise problem

The problem asks whether every smooth Riemannian metric g on the 3-sphere S^3 admits at least five pairwise distinct closed embedded minimal surfaces of genus one, i.e. embedded minimal tori, in (S^3, g) . Distinct means distinct embedded surfaces, not just different parametrizations or multiplicities.

1.30.3.2. Literature checked

1. New applications of Min-max Theory

Author: Andre Neves

Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, pp. 939-957; arXiv preprint, 2014

arXiv/URL: arXiv:1409.7537, <https://arxiv.org/abs/1409.7537>

Relevant statement: In the open-problems section Neves records White's conjecture: any metric on a 3-sphere has five distinct minimal embedded tori, and notes that White proved it for small perturbations of the round metric. This identifies the problem statement as White's enumerative minimal-torus conjecture, not merely a positive-Ricci or near-round statement.

2. Every three-sphere of positive Ricci curvature contains a minimal embedded torus

Author: Brian White

Venue/year: Bulletin of the American Mathematical Society (N.S.) 21(1), 71-75, 1989

DOI/URL: 10.1090/S0273-0979-1989-15765-0; <https://projecteuclid.org/journals/bulletin-of-the-american-mathematical-society-new-series/volume-21/issue-1/Every-three-sphere-of-positive-Ricci-curvature-contains-a-minimal/bams/1183555126.f>

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Relevant result: White proves existence of at least one embedded minimal torus for metrics on S^3 with positive Ricci curvature. This is important progress but it gives one torus and assumes positive Ricci curvature, so it does not imply the stated five-torus statement for arbitrary metrics.

3. The Space of Minimal Submanifolds for Varying Riemannian Metrics

Author: Brian White

Venue/year: Indiana University Mathematics Journal 40(1), 161-200, 1991

DOI/URL: 10.1512/iumj.1991.40.40008; <https://iumj.org/article/3473/>

Relevant result: Later papers cite White's perturbation theorem from this work as proving that every sufficiently round metric on S^3 has at least five embedded minimal tori, and a generic sufficiently round metric has at least nine. This covers a neighborhood of the round metric, not all Riemannian metrics on S^3 .

4. Existence of 5 minimal tori in 3-spheres of positive Ricci curvature

Authors: Adrian Chun-Pong Chu and Yangyang Li

Venue/year: arXiv preprint, 2024; current arXiv version dated July 31, 2025

arXiv/DOI/URL: arXiv:2409.09315; 10.48550/arXiv.2409.09315; <https://arxiv.org/abs/2409.09315>

Relevant theorem: Theorem A states that every smooth Riemannian 3-sphere of positive Ricci curvature contains at least five embedded minimal tori. This proves White's conjectured lower bound in the positive-Ricci class. It still has the extra hypothesis $\text{Ric} > 0$, so it does not cover arbitrary metrics on S^3 .

5. Existence of embedded minimal tori in three-spheres with positive Ricci curvature

Authors: Xingzhe Li and Zhichao Wang

Venue/year: arXiv preprint, 2024

arXiv/DOI/URL: arXiv:2409.10391; 10.48550/arXiv.2409.10391; <https://arxiv.org/abs/2409.10391>

Relevant theorem: Theorem 1.1 proves that a positive-Ricci metric on S^3 has at least four embedded minimal tori, and a bumpy positive-Ricci metric has at least nine. This independently confirms substantial positive-Ricci progress but also does not imply the arbitrary-metric five-torus statement.

6. An enumerative min-max theorem for minimal surfaces

Authors: Adrian Chun-Pong Chu, Yangyang Li, and Zhihan Wang

Venue/year: arXiv preprint, 2026

arXiv/URL: arXiv:2601.01736, <https://arxiv.org/abs/2601.01736>

Relevance: The introduction summarizes the current landscape: Poincaré's 2-sphere geodesic conjecture is fully solved; Yau's four-sphere conjecture is solved for bumpy metrics and positive Ricci curvature; and White's five-torus conjecture is proved for positive-Ricci 3-spheres. The paper's own enumerative theorem is also stated under positive Ricci curvature, so it does not remove the extra curvature hypothesis.

1.30.3.3. Conclusion

the full statement is not fully resolved. The strongest directly relevant result found is Chu–Li's theorem proving the exact lower bound of five embedded minimal tori for every S^3 metric with positive Ricci curvature. White's earlier work gives one torus under positive Ricci curvature and five tori only for sufficiently round metrics. Li–Wang give related positive-Ricci and bumpy-positive-Ricci lower bounds.

What remains unresolved is the full arbitrary-metric case: no published or citable preprint found proves that every Riemannian metric on S^3 , without positive Ricci curvature, bumpiness, or closeness to the round metric, contains at least five pairwise distinct embedded minimal tori.

1.30.4. Remaining open problems recorded by the database

1.30.4.1. Open problem 1

Prove that every bumpy Riemannian metric on S^3 , without any curvature assumption, admits at least five pairwise distinct embedded minimal tori.

Reference supplied by the database. Brian White, “Every three-sphere of positive Ricci curvature contains a minimal embedded torus,” *Bulletin of the American Mathematical Society* **21**(1) (1989), 71–75, where White proposed the five-minimal-tori conjecture. DOI: [10.1090/S0273-0979-1989-15765-0](https://doi.org/10.1090/S0273-0979-1989-15765-0). Current status: Adrian Chun-Pong Chu and Yangyang Li, “Existence of 5 minimal tori in 3-spheres of positive Ricci curvature,” preprint, revised 2025, proves the conjecture under positive Ricci curvature but leaves the general case—including bumpy metrics with no curvature assumption—open, [arXiv:2409.09315](https://arxiv.org/abs/2409.09315).

1.30.5. Primary-source pointer

The originating ICM record is *New applications of Min-max Theory* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>.

1.31. 31. Non-negatively curved manifolds and Tits geometry [2014-geom-130]

Record: 2014-geom-130

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: reviewed (reviewed flag: yes)

Original source: ICM 2014: Non-negatively curved manifolds and Tits geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=885>

1.31.1. Problem statement

Prove that there is no Riemannian metric with positive sectional curvature on the product $\mathbb{S}^2 \times \mathbb{S}^2$, where $\mathbb{S}^2$ denotes the 2-sphere. More generally, prove that if M and N are simply-connected connected closed (compact without boundary) positive-dimensional manifolds, then the product manifold $M \times N$ admits no Riemannian metric with positive sectional curvature; likewise, prove that the underlying smooth manifold of a compact symmetric space of rank at least two admits no Riemannian metric with positive sectional curvature.

1.31.2. Historical background, literature, and current progress

1.31.3. Verdict: partially resolved

1.31.3.1. Precise problem

The problem asks whether any Riemannian metric with strictly positive sectional curvature can exist on the smooth manifold $S^2 \times S^2$. It also asks the broader Hopf product/symmetric-space question: no nontrivial product $M \times N$ of closed positive-dimensional manifolds should admit positive sectional curvature, and no compact symmetric space of rank at least two should admit such a metric on its underlying smooth manifold.

1.31.3.2. Status

The full statement is not resolved. In particular, the basic case $S^2 \times S^2$ remains open in the published literature, so the product and rank-at-least-two symmetric-space assertions cannot be counted as fully proved. The literature contains substantial partial results, and some recent papers describe the generalized product conjecture as open.

1.31.3.3. Key literature and progress

1. Heinz Hopf, "Differentialgeometrie und topologische Gestalt", Jahresbericht der Deutschen Mathematiker-Vereinigung 41, 209-229, 1932. <https://eudml.org/doc/145926>. This is the historical source for Hopf-type curvature/topology

- questions; later sources identify the $S^2 \times S^2$ positive-curvature problem as Hopf's product conjecture.
2. Wolfgang Ziller, "Riemannian manifolds with positive sectional curvature", in *Geometry of Manifolds with Non-negative Sectional Curvature*, Lecture Notes in Mathematics 2110, Springer, 2014, pp. 1-19. DOI: 10.1007/978-3-319-06373-7_1. Ziller states exactly the relevant conjectural package: no positive sectional curvature on $S^2 \times S^2$, more generally no positive curvature on products of compact manifolds or on symmetric spaces of rank at least two. This is a survey/status source, not a resolution.
 3. Jean-Pierre Bourguignon, "Some constructions related to H. Hopf's conjecture on product manifolds", in *Differential Geometry, Proceedings of Symposia in Pure Mathematics 27*, American Mathematical Society, 1975, pp. 33-37. MR 0380906. This gives a local obstruction near the product metric on $S^2 \times S^2$. It does not exclude arbitrary metrics on $S^2 \times S^2$.
 4. Alan Weinstein, "Positively curved n -manifolds in R^{n+2} ", *Journal of Differential Geometry* 4(1), 1-4, 1970. DOI: 10.4310/jdg/1214429270. This rules out a positive-curvature metric on $S^2 \times S^2$ induced by an immersion in R^6 . It is an extrinsic special case, not a theorem about all intrinsic metrics.
 5. Wu-Yi Hsiang and Bruce Kleiner, "On the topology of positively curved 4-manifolds with symmetry", *Journal of Differential Geometry* 29(3), 615-621, 1989. DOI: 10.4310/jdg/1214443064. Their theorem implies that a simply connected positively curved 4-manifold with an isometric circle action is homeomorphic to S^4 or $\mathbb{CP}^2$. Thus $S^2 \times S^2$ cannot carry a positive-curvature metric with an isometric circle action. This leaves metrics with no such symmetry open.
 6. Lee Kennard, "Positively curved Riemannian metrics with logarithmic symmetry rank bounds", *Commentarii Mathematici Helvetici* 89(4), 937-962, 2014. DOI: 10.4171/CMH/340. Kennard proves rational-cohomology obstructions for positively curved manifolds with large symmetry rank and gives evidence for the symmetric-space version of Hopf's conjecture. The theorem assumes large isometric torus symmetry and does not cover arbitrary metrics.
 7. Manuel Amann and Lee Kennard, "On a generalized conjecture of Hopf with symmetry", *Compositio Mathematica* 153(2), 313-322, 2017. DOI: 10.1112/S0010437X16008150; arXiv:1402.7255. The abstract states the generalized conjecture for nontrivial products and proves evidence in the presence of symmetry, specifically excluding positively curved metrics on $N \times N$ with a certain degree of torus symmetry. This is a partial result because of the symmetry hypothesis and because it does not cover arbitrary $M \times N$.
 8. Oscar Perdomo, "Metrics on products of surfaces with non-positive sectional curvature", *Israel Journal of Mathematics* 156, 65-71, 2006. DOI: 10.1007/BF02773824. Perdomo proves that certain graph metrics over products of compact surfaces do not have positive sectional curvature. This is an extrinsic/graph-metric result, not a proof for all Riemannian metrics on $S^2 \times S^2$.
 9. Thierno Seck and Athoumane Niang, "Conformal Metrics to a Product or Doubly Warped Product on $S^2 \times S^2$ and the Hopf Conjecture", *Journal of Mathematics* 2022, Article ID 6268017, 2022. DOI: 10.1155/2022/6268017. The paper proves the conjecture for conformal metrics to the product metric and doubly warped product metrics on $S^2 \times S^2$, and explicitly says the generalized nontrivial-product conjecture is open. These are metric-class restrictions, so the full problem remains open.
 10. Athoumane Niang, Thierno Seck, and Adama Thiandoum, "Proof of the Hopf Conjec-

ture”, ResearchGate preprint, posted 2024, URL: https://www.researchgate.net/publication/377694075_PROOF_OF_THE_HOPF_CONJECTURE. This item claims to prove Hopf’s conjecture, but the theorem stated there concerns metrics induced on graphs $\{(x, F(x)) : x \in S^2 \times S^2\} \subset R^{6+k}$. That conclusion does not by itself cover all Riemannian metrics on $S^2 \times S^2$; an arbitrary metric need not be represented as such a graph over the standard $S^2 \times S^2 \subset R^6$. I therefore do not count it as a citable resolution of the problem.

1.31.3.4. What remains unresolved

No published or otherwise reliable citable theorem found in the search proves that every Riemannian metric on $S^2 \times S^2$ has some nonpositive sectional 2-plane. Since this basic special case is still unresolved, the full assertion for all compact products $M \times N$ and for all compact symmetric spaces of rank at least two is also unresolved. The known results above are genuine partial progress but all impose extra hypotheses such as proximity to the product metric, extrinsic graph/immersion form, conformal or warped product form, or nontrivial isometric symmetry.

1.31.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.31.5. Primary-source pointer

The originating ICM record is *Non-negatively curved manifolds and Tits geometry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=885>.

1.32. 32. The Geometry of Ricci Curvature [2014-geom-135]

Record: 2014-geom-135

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: The Geometry of Ricci Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>

1.32.1. Problem statement

Let $n \in \mathbb{N}$ and $v > 0$. Suppose that (M_j^n, g_j, p_j) is a sequence of complete pointed n -dimensional Riemannian manifolds converging in the pointed Gromov-Hausdorff sense to a pointed metric space (X, d, p) , with $\text{Ric}_{g_j} \geq -(n-1)$ and $\text{Vol}_{g_j}(B_1(p_j)) > v$ for every j . For $x \in X$, a tangent cone at x means any pointed Gromov-Hausdorff limit of rescalings $(X, r_i^{-1}d, x)$ along a sequence $r_i \rightarrow 0$. Prove that the set of points $x \in X$ at which there is more than one tangent cone, up to pointed isometry, has Hausdorff dimension at most $n - 3$.

1.32.2. Historical background, literature, and current progress

1.32.3. Verdict: partially resolved

1.32.3.1. Precise problem

The problem asks whether, for every noncollapsed pointed Gromov-Hausdorff limit of complete Riemannian n -manifolds with $\text{Ric} \geq -(n-1)$, the locus of points with more than one tangent cone has Hausdorff dimension at most $n - 3$. Equivalently: are tangent cones unique outside a codimension-three subset?

1.32.3.2. Verdict

This is not fully resolved in the generality stated. It is the codimension-three tangent-cone uniqueness conjecture for noncollapsed lower-Ricci limit spaces. Published general results give substantial partial progress, especially uniqueness outside an $\mathcal{H}^{n-2}$ -measure-zero set, but that does not imply Hausdorff dimension at most $n - 3$.

1.32.3.3. Literature checked

1. The geometry of Ricci curvature

Author: Aaron Naber

Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 911-937

URL: https://kyungmoon.com/shop/item.php?it_id=1759215764

Relevance: This is the source family for the problem. Later papers cite this ICM article as one of the places where the codimension-three tangent-cone uniqueness problem is formulated.

2. Conjectures and Open Questions on the Structure and Regularity of Spaces with Lower Ricci Curvature Bounds

Author: Aaron Naber

Venue/year: SIGMA Symmetry Integrability Geom. Methods Appl. 16 (2020), Paper No. 104

DOI/arXiv/URL: DOI 10.3842/SIGMA.2020.104; arXiv:2010.10031; <https://arxiv.org/abs/2010.10031>

Relevance: Conjecture 2.7 states exactly that if $(M_i^n, g_i, p_i) \rightarrow (X, d, p)$, $\text{Ric} \geq -(n-1)$, and $\text{Vol}(B_1(p_i)) > v > 0$, then tangent cones of X are unique away from a set of codimension three. The paper explicitly says the known result at that time was only uniqueness away from a set of $\mathcal{H}^{n-2}$ -measure zero.

3. Rectifiability of singular sets of noncollapsed limit spaces with Ricci curvature bounded below

Authors: Jeff Cheeger, Wenshuai Jiang, Aaron Naber

Venue/year: Annals of Mathematics (2) 193 (2021), no. 2, 407-538

DOI/arXiv/URL: DOI 10.4007/annals.2021.193.2.2; arXiv:1805.07988; <https://annals.math.princeton.edu/2021/193-2/p02>

Progress: For the same noncollapsed lower-Ricci hypotheses as in the problem statement, they prove that each stratum S^k is k -rectifiable and that for $\mathcal{H}^k$ -a.e. $x \in S^k$, every tangent cone is k -symmetric. As an application, tangent cones are unique $\mathcal{H}^{n-2}$ -a.e.

Why this is only partial: An exceptional set of zero $\mathcal{H}^{n-2}$ -measure can still have Hausdorff dimension $n-2$. The problem asks for Hausdorff dimension at most $n-3$, so this theorem does not imply the full statement.

4. Characterization of Tangent Cones of Noncollapsed Limits with Lower Ricci Bounds and Applications

Authors: Tobias H. Colding, Aaron Naber

Venue/year: Geometric and Functional Analysis 23 (2013), 134-148

DOI/arXiv/URL: DOI 10.1007/s00039-012-0202-7; arXiv:1108.3244; <https://arxiv.org/abs/1108.3244>

Progress/evidence: Shows that tangent cones in the exact noncollapsed lower-Ricci setting need not be unique at a point, including examples with nonunique and even non-homeomorphic tangent cones. This explains why the conjecture concerns the size of the nonuniqueness locus rather than pointwise uniqueness.

5. Examples of tangent cones of non-collapsed Ricci limit spaces

Author: Philipp Reiser

Venue/year: arXiv preprint, 2024

DOI/arXiv/URL: arXiv:2409.11954; <https://arxiv.org/abs/2409.11954>

Progress/evidence: Extends Colding-Naber type examples by constructing additional possible cross sections of tangent cones at the same point in noncollapsed Ricci limit spaces. This is further evidence about possible nonuniqueness phenomena, but it is not a dimension bound for the full nonuniqueness set.

6. Gromov-Hausdorff limits of Kähler manifolds with Ricci curvature bounded below

Authors: Gang Liu, Gábor Székelyhidi

Venue/year: Geometric and Functional Analysis 32 (2022), no. 2, 236-279

arXiv/URL: arXiv:1804.08567; <https://arxiv.org/abs/1804.08567>

Progress: Gives stronger structure in the Kähler setting; Naber's 2020 survey notes that

the Kähler case resolves the analogous refined stratification conjectures.

Why this is only partial for this problem: The statement has no Kähler, complex-analytic, or polarization hypothesis, so these results do not cover the full statement.

7. Regularity of Einstein 5-manifolds via 4-dimensional gap theorems

Authors: Yiqi Huang, Tristan Ozuch

Venue/year: arXiv preprint, 2025

arXiv/URL: arXiv:2512.21317; <https://arxiv.org/abs/2512.21317>

Progress: Proves uniqueness of tangent cones on the top stratum for noncollapsed limits of Einstein 5-manifolds and related bounded-Ricci settings.

Why this is only partial for this problem: It assumes dimension five and Einstein/bounded Ricci hypotheses, while the problem asks for arbitrary dimension and only a lower Ricci bound.

1.32.3.4. What remains unresolved

No citable result found proves that the full nonuniqueness locus has Hausdorff dimension at most $n - 3$ for arbitrary noncollapsed lower-Ricci limit spaces. The remaining gap is precisely to upgrade the known $\mathcal{H}^{n-2}$ -a.e. uniqueness/rectifiability theory to a genuine codimension-three Hausdorff-dimension bound for all points with nonunique tangent cones.

1.32.4. Remaining open problems recorded by the database

1.32.4.1. Open problem 1

In a noncollapsed three-dimensional Ricci limit space, prove that the set of points admitting two nonisometric tangent cones has Hausdorff dimension zero.

Reference supplied by the database. Aaron Naber, “Conjectures and Open Questions on the Structure and Regularity of Spaces with Lower Ricci Curvature Bounds,” *Symmetry, Integrability and Geometry: Methods and Applications (SIGMA)* **16** (2020), Article 104, Conjecture 2.7, specialized to $n = 3$. DOI: [10.3842/SIGMA.2020.104](https://doi.org/10.3842/SIGMA.2020.104).

1.32.5. Primary-source pointer

The originating ICM record is *The Geometry of Ricci Curvature* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>.

1.33. 33. The Geometry of Ricci Curvature [2014-geom-136]

Record: 2014-geom-136

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: The Geometry of Ricci Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>

1.33.1. Problem statement

Let n be fixed, let $v > 0$, and let (X, d, p) be the pointed Gromov-Hausdorff limit of pointed n -dimensional Riemannian manifolds (M_j^n, g_j, p_j) such that $\text{Ric}_{g_j} \geq -(n-1)$ and $\text{Vol}_{g_j}(B_1(p_j)) > v$ for all j . Prove that there exists a subset $E \subset X$ of Hausdorff dimension at most $n-3$ such that for every $x \in X \setminus E$, the tangent cone of X at x is unique and isometric to $\mathbb{R}^{n-2} \times C(S^1(r_x))$ for some radius $r_x \leq 1$, where $S^1(r_x)$ is the circle of radius r_x and $C(S^1(r_x))$ denotes its metric cone.

1.33.2. Historical background, literature, and current progress

1.33.3. Verdict: partially resolved

1.33.3.1. Precise problem

The statement asks for the codimension-three uniqueness conjecture for noncollapsed Ricci-limit spaces: if X is a pointed Gromov-Hausdorff limit of n -manifolds with $\text{Ric} \geq -(n-1)$ and a uniform lower volume bound, then outside a subset of Hausdorff dimension at most $n-3$, every point should have a unique tangent cone, and that cone should be the two-dimensional flat cone model

$$\mathbb{R}^{n-2} \times C(S^1(r_x)), \quad r_x \leq 1.$$

Regular points are included in this form by taking $r_x = 1$.

1.33.3.2. Verdict

This problem is partially resolved, but I did not find a citable published or preprint result proving the exact statement in full generality.

The main gap is the size of the exceptional set. The best general theorem found gives the same tangent-cone model and uniqueness outside a set of zero $\mathcal{H}^{n-2}$ -measure. That does not imply Hausdorff dimension at most $n-3$: an $\mathcal{H}^{n-2}$ -null set can still have Hausdorff dimension strictly larger than $n-3$, or even equal to $n-2$.

1.33.3.3. Key literature and progress

1. The Geometry of Ricci Curvature

Author: Aaron Naber

Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, Kyung Moon Sa, 2014, pp. 911-937

URL: <https://dokumen.pub/proceedings-of-the-international-congress-of-mathematicians-seoul-august-13-21-2014-korea-vol-2-9788961058032-8961058037-9788961058049-8961058056-8961058061-9788961058070-896105807x.html>

Relevance: Conjecture 6.2 in this ICM article is the source matching the problem statement: after removing a set of Hausdorff dimension at most $n - 3$, tangent cones should be unique and equal to $\mathbb{R}^{n-2} \times C(S_r^1)$.

2. Rectifiability of singular sets of noncollapsed limit spaces with Ricci curvature bounded below

Authors: Jeff Cheeger, Wenshuai Jiang, Aaron Naber

Venue/year: Annals of Mathematics 193 (2021), no. 2, 407-538

DOI/arXiv: DOI 10.4007/annals.2021.193.2.2; arXiv:1805.07988

Relevance: Theorem 1.15 gives the strongest general partial result found. It proves that for a noncollapsed limit space with Ricci curvature bounded below there is a subset $\tilde{S}$ with $\mathcal{H}^{n-2}(\tilde{S}) = 0$ such that at every point outside $\tilde{S}$, tangent cones are unique and are of the form $\mathbb{R}^{n-2} \times C(S_r^1)$. This covers the desired tangent-cone classification and uniqueness, but with the weaker exceptional-set bound $\mathcal{H}^{n-2} = 0$, not $\dim_{\mathcal{H}} \leq n - 3$.

3. Conjectures and Open Questions on the Structure and Regularity of Spaces with Lower Ricci Curvature Bounds

Author: Aaron Naber

Venue/year: SIGMA 16 (2020), Paper No. 104

DOI/arXiv: DOI 10.3842/SIGMA.2020.104; arXiv:2010.10031

Relevance: This survey/open-problems article records the same codimension-three tangent-cone uniqueness assertion as a conjectural statement and explicitly distinguishes it from the known $\mathcal{H}^{n-2}$ -measure-zero result of Cheeger-Jiang-Naber. It is a citable status reference that the full Hausdorff-dimension- $\leq n - 3$ statement remained open after the Annals result.

4. Characterization of tangent cones of noncollapsed limits with lower Ricci bounds and applications

Authors: Tobias H. Colding, Aaron Naber

Venue/year: Geometric and Functional Analysis 23 (2013), 134-148

DOI/arXiv: DOI 10.1007/s00039-012-0201-2; arXiv:1108.3244

Relevance: Gives foundational structure and examples around tangent cones of noncollapsed lower-Ricci limits, including phenomena showing why uniqueness is delicate. It does not prove the codimension-three uniqueness statement.

5. Topological regularity and stability of noncollapsed spaces with Ricci curvature bounded below

Authors: Elia Brue, Alessandro Pigati, Daniele Semola

Venue/year: arXiv preprint, 2024

DOI/arXiv: arXiv:2405.03839

Relevance: Provides later progress on topological regularity/stability for noncollapsed Ricci-limit spaces, but it does not prove the metric tangent-cone uniqueness assertion outside a Hausdorff-dimension- $\leq n - 3$ set in the general Riemannian lower-Ricci

setting.

1.33.3.4. What remains unresolved

To fully resolve the problem statement, one still needs to improve the general exceptional set from $\mathcal{H}^{n-2}$ -measure zero to Hausdorff dimension at most $n - 3$, or otherwise prove that every possible bad point in the top-dimensional singular stratum is contained in such a codimension-three set. The available general literature does not establish that improvement.

1.33.4. Remaining open problems recorded by the database

1.33.4.1. Open problem 1

Let X^n be a noncollapsed Ricci limit space. Prove that there is a set $E \subset X$ with $\dim_{\mathcal{H}} E \leq n - 3$ such that, at every $x \in X \setminus E$, every tangent cone is $(n - 2)$ -symmetric; equivalently, every tangent cone splits isometrically as

$$\mathbb{R}^{n-2} \times C(Z)$$

for some compact metric space Z . No uniqueness of the tangent cone is required.

Reference supplied by the database. Aaron Naber, “Conjectures and Open Questions on the Structure and Regularity of Spaces with Lower Ricci Curvature Bounds,” *Symmetry, Integrability and Geometry: Methods and Applications (SIGMA)* **16** (2020), Article 104, Conjecture 2.16 with $k = n - 3$. DOI: [10.3842/SIGMA.2020.104](https://doi.org/10.3842/SIGMA.2020.104). Its open status is reaffirmed in Elia Bruè, Alessandro Pigati, and Daniele Semola, “Topological regularity and stability of noncollapsed spaces with Ricci curvature bounded below,” preprint (2024), Remark 1.5, [arXiv:2405.03839](https://arxiv.org/abs/2405.03839).

1.33.5. Primary-source pointer

The originating ICM record is *The Geometry of Ricci Curvature* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>.

1.34. 34. The Geometry of Ricci Curvature [2014-geom-137]

Record: 2014-geom-137

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: The Geometry of Ricci Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>

1.34.1. Problem statement

Prove that if $n \in \mathbb{N}$ and $v > 0$, and if (M_j^n, g_j, p_j) is a sequence of pointed n -dimensional Einstein Riemannian manifolds such that $|\text{Ric}_{g_j}| \leq n-1$ and $\text{Vol}_{g_j}(B_1^{g_j}(p_j)) > v$ for every j , which converges in the pointed Gromov-Hausdorff sense to the pointed metric space (X, d, p) , then X is homeomorphic to a real analytic variety.

1.34.2. Historical background, literature, and current progress

1.34.3. Verdict: partially resolved

1.34.3.1. Precise problem

The problem is Naber's structural conjecture for noncollapsed pointed Gromov-Hausdorff limits of Einstein manifolds: for every dimension n and noncollapsing constant $v > 0$, if (M_j^n, g_j, p_j) are Einstein with $|\text{Ric}_{g_j}| \leq n-1$ and $\text{Vol}(B_1(p_j)) > v$, and if they converge to (X, d, p) , must the metric space X be homeomorphic to a real analytic variety?

1.34.3.2. Verdict

Partially resolved. I did not find a theorem that covers the full statement in all dimensions for arbitrary Riemannian Einstein manifolds. The literature proves strong regularity and several special cases, but the general real-analytic-variety conclusion remains open.

1.34.3.3. Original source

- Title: "The Geometry of Ricci Curvature"
- Author: Aaron Naber
- Venue/year: Proceedings of the International Congress of Mathematicians - Seoul 2014, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 911-937
- DOI/arXiv/URL: no DOI found; proceedings text visible at <https://dokumen.pub/proceedings-of-the-international-congress-of-mathematicians-seoul-august-13-21-2014-korea-vol-2-9788961058032-8961058037-9788961058049-8961058045-9788961058056-89610>

[58053-9788961058063-8961058061-9788961058070-896105807x.html](https://doi.org/10.4007/annals.2015.182.3.5)

- Relevance: Conjecture 6.4 states exactly the problem: a noncollapsed Gromov-Hausdorff limit of Einstein manifolds with $|\text{Ric}| \leq n-1$ should be homeomorphic to a real analytic variety. Naber notes progress for compact Kahler manifolds with positive first Chern class, but does not prove the general conjecture.

1.34.3.4. Main partial progress

- Title: “Regularity of Einstein manifolds and the codimension 4 conjecture”
- Authors: Jeff Cheeger and Aaron Naber
- Venue/year: Annals of Mathematics 182 (2015), no. 3, 1093-1165
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2015.182.3.5> ; arXiv:1406.6534 ; <https://annals.math.princeton.edu/2015/182-3/p05>
- What it proves: For noncollapsed manifolds with bounded Ricci curvature, including Einstein manifolds, a Gromov-Hausdorff limit is smooth away from a closed subset of codimension 4; it also gives curvature estimates.
- Why it is not a full resolution: Smoothness off a codimension-4 set and curvature estimates do not identify the entire metric space topologically with a real analytic variety.
- Title: “ L^2 curvature bounds on manifolds with bounded Ricci curvature”
- Authors: Wenshuai Jiang and Aaron Naber
- Venue/year: Annals of Mathematics 193 (2021), no. 1, 107-222
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2021.193.1.2> ; arXiv:1605.05583 ; <https://annals.math.princeton.edu/2021/193-1/p02>
- What it proves: Under the same bounded-Ricci and noncollapsing hypotheses, the singular set of a limit is $(n-4)$ -rectifiable with uniform Hausdorff measure estimates, and for H^{n-4} -a.e. singular point the tangent cone is uniquely $R^{n-4} \times C(S^3/\Gamma_x)$.
- Why it is not a full resolution: Rectifiability and almost-everywhere tangent-cone structure are weaker than a global homeomorphism to a real analytic variety.
- Title: “Rectifiability of singular sets of noncollapsed limit spaces with Ricci curvature bounded below”
- Authors: Jeff Cheeger, Wenshuai Jiang, and Aaron Naber
- Venue/year: Annals of Mathematics 193 (2021), no. 2, 407-538
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2021.193.2.2> ; arXiv:1805.07988 ; <https://annals.math.princeton.edu/2021/193-2/p02>
- What it proves: For noncollapsed lower-Ricci limit spaces, the strata of the singular set are rectifiable and have strong measure-theoretic structure.
- Why it is not a full resolution: The theorem applies even more generally than Einstein limits, but its conclusion is geometric-measure-theoretic, not a real analytic variety struc-

ture.

- Title: “Regularity of Einstein 5-manifolds via 4-dimensional gap theorems”
- Authors: Yiqi Huang and Tristan Ozuch
- Venue/year: arXiv preprint, arXiv:2512.21317, v1 2025 and v2 2026
- DOI/arXiv/URL: <https://arxiv.org/abs/2512.21317>; <https://arxiv.org/html/2512.21317>
- What it proves: In dimension 5, for noncollapsed Einstein limits, the orbifold-regular set is a real-analytic Einstein 5-orbifold and is homeomorphic to a real-analytic variety. The paper explicitly describes this as confirming Naber’s Conjecture 6.4 away from extreme points in dimension 5.
- Why it is not a full resolution: The result is dimension 5 and excludes the orbifold-singular/extreme-point set. The paper formulates further open questions/conjectures about the remaining singular set in dimension 5 and about higher-dimensional orbifold regularity.

1.34.3.5. Special cases with stronger conclusions

- Title: “Ricci curvature bounds and Einstein metrics on compact manifolds”
- Author: Michael T. Anderson
- Venue/year: Journal of the American Mathematical Society 2 (1989), no. 3, 455-490
- DOI/arXiv/URL: <https://doi.org/10.1090/S0894-0347-1989-0999661-1>
- Relevance: Part of the four-dimensional orbifold compactness theory for Einstein metrics.
- Limitation: It is a four-dimensional compactness/orbifold result, not the all-dimensional Riemannian Einstein-limit statement.
- Title: “On a construction of coordinates at infinity on manifolds with fast curvature decay and maximal volume growth”
- Authors: Shigetoshi Bando, Atsushi Kasue, and Hiraku Nakajima
- Venue/year: Inventiones Mathematicae 97 (1989), 313-349
- DOI/arXiv/URL: <https://doi.org/10.1007/BF01389045>
- Relevance: Provides analytic tools used in four-dimensional Einstein/ALE/orbifold compactness and removable singularity theory.
- Limitation: It does not prove the all-dimensional real-analytic-variety conclusion.
- Title: “Gromov-Hausdorff limits of Kahler manifolds and algebraic geometry”
- Authors: Simon Donaldson and Song Sun
- Venue/year: Acta Mathematica 213 (2014), no. 1, 63-106
- DOI/arXiv/URL: <https://doi.org/10.1007/s11511-014-0116-3>; arXiv:1206.2609
- What it proves: In the polarized Kahler setting with suitable noncollapsing and two-sided Ricci bounds, limit spaces acquire normal projective algebraic variety structures.

- Why it is not a full resolution: The statement has no Kahler, complex, polarization, compactness, or diameter hypothesis; it asks about arbitrary Riemannian Einstein limits.
- Title: “Gromov-Hausdorff limits of Kahler manifolds and algebraic geometry, II”
- Authors: Simon Donaldson and Song Sun
- Venue/year: Journal of Differential Geometry 107 (2017), no. 2, 327-371
- DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1506650422> ; arXiv:1507.05082
- What it proves: Further algebraic structure results for Kahler-Einstein limits and their tangent cones.
- Why it is not a full resolution: It remains within the Kahler/algebraic framework and does not cover general Riemannian Einstein manifolds.
- Title: “Gromov-Hausdorff limits of Kahler manifolds with Ricci curvature bounded below”
- Authors: Gang Liu and Gabor Székelyhidi
- Venue/year: Geometric and Functional Analysis 32 (2022), no. 2, 236-279
- DOI/arXiv/URL: <https://doi.org/10.1007/s00039-022-00594-8> ; arXiv:1804.08567
- What it proves: Noncollapsed Gromov-Hausdorff limits of polarized Kahler manifolds with Ricci curvature bounded below are homeomorphic to normal projective varieties, extending Donaldson-Sun beyond two-sided Ricci bounds in the polarized Kahler setting.
- Why it is not a full resolution: Again, the polarized Kahler hypotheses are extra assumptions absent from the problem statement.

1.34.3.6. Current obstruction

The remaining unresolved part is the full global topology/analytic structure of arbitrary noncollapsed Einstein limit spaces in all dimensions. Known results control the regular set, codimension, rectifiability, curvature integrability, and some tangent cones; special cases are orbifolds or complex algebraic varieties. None of these results, including the 2026 Huang-Ozuch preprint, proves that every all-dimensional Riemannian Einstein limit satisfying only the stated hypotheses is homeomorphic to a real analytic variety.

1.34.4. Remaining open problems recorded by the database

1.34.4.1. Open problem 1

Let (X, d) be a Gromov-Hausdorff limit of noncollapsed Einstein 5-manifolds. Prove that the non-orbifold set $\mathcal{S}_{\text{orb}} = X \setminus \mathcal{R}_{\text{orb}}$ is countable. Equivalently, away from a countable set, (X, d) should be a smooth real-analytic Einstein 5-orbifold, with singularities locally of the form $\mathbb{R} \times \mathbb{R}^4/\Gamma$ along countably many geodesics.

Reference supplied by the database. Yiqi Huang and Tristan Ozuch, “Regularity of Einstein 5-manifolds via 4-dimensional gap theorems,” arXiv preprint (2025; revised 2026), Conjecture 1.24 (“Codimension-5 conjecture in dimension 5”), arXiv:2512.21317v2 [math.DG], DOI: 10.48550/arXiv.2512.21317, <https://arxiv.org/abs/2512.21317>.

1.34.5. Primary-source pointer

The originating ICM record is *The Geometry of Ricci Curvature* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>.

1.35. 35. Analytic tools for the study of flows and inverse problems [2018-geom-138]

Record: 2018-geom-138

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Analytic tools for the study of flows and inverse problems

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=972>

1.35.1. Problem statement

Let (M_1, g_1) and (M_2, g_2) be Riemannian surfaces with common boundary $\partial M_1 = \partial M_2$. Assume that each boundary is strictly convex, that neither surface has conjugate points, and that the trapped set of each geodesic flow is hyperbolic, where the trapped set is the set of unit tangent vectors whose corresponding geodesics remain in the manifold for all forward and backward time.

For $i = 1, 2$, let SM_i be the unit tangent bundle. For $y = (x, v) \in \partial SM_i$, let $\gamma_{x,v}(t) = \exp_x^{g_i}(tv)$, and let $\tau_{g_i}^+(x, v) \in [0, \infty]$ be the maximal non-negative time for which this geodesic remains defined in the manifold. Set

$$\Gamma_-^{(i)} = \{y \in \partial SM_i : \tau_{g_i}^+(y) = +\infty\}.$$

For $y = (x, v) \in \partial SM_i \setminus \Gamma_-^{(i)}$, define the scattering map

$$\sigma_{g_i}(x, v) = (\gamma_{x,v}(\tau_{g_i}^+(x, v)), \dot{\gamma}_{x,v}(\tau_{g_i}^+(x, v))) \in \partial SM_i.$$

The lens data of (M_i, g_i) is the pair $(\tau_{g_i}^+, \sigma_{g_i})$.

Prove or disprove that if the full lens data agree,

$$\tau_{g_1}^+ = \tau_{g_2}^+ \quad \text{and} \quad \sigma_{g_1} = \sigma_{g_2},$$

then there exists a diffeomorphism $\psi : M_1 \rightarrow M_2$ fixing the boundary such that

$$\psi^* g_2 = g_1.$$

1.35.2. Historical background, literature, and current progress

1.35.3. Verdict: partially resolved

1.35.3.1. Precise problem

The problem asks for smooth lens rigidity for compact Riemannian surfaces with strictly convex boundary, no conjugate points, and hyperbolic trapped set (often called surfaces of Anosov type): if two such metrics have the same ordinary/unmarked lens data, namely the same travel-time function and scattering map on the boundary unit bundle, must they be isometric by a

diffeomorphism fixing the boundary? Equivalently, after Guillarmou's scattering-rigidity theorem gives a boundary-fixing conformal diffeomorphism, does the travel-time part of the lens data force the remaining conformal factor to be trivial?

1.35.3.2. Status

I did not find a published or otherwise citable theorem proving this full smooth unmarked statement. The closest current formulation is Conjecture 1.7 in Cekic–Guillarmou–Lefeuvre, which is exactly the unmarked lens rigidity problem for manifolds of Anosov type. That paper explicitly lists only partial answers and says, in the smooth two-dimensional setting of Guillarmou's theorem, that recovering the conformal factor remains open. Therefore the full statement is not fully resolved.

1.35.3.3. Main partial progress

1. Lens rigidity for manifolds with hyperbolic trapped set

Colin Guillarmou. *Journal of the American Mathematical Society* 30 (2017), no. 2, 561–599. DOI: <https://doi.org/10.1090/jams/865>. arXiv: <https://arxiv.org/abs/1412.1760>.

This is the source theorem behind the problem. It proves deformation lens rigidity under the no-conjugate-points/hyperbolic-trapped-set assumptions and, for surfaces, proves that scattering data determine the topology and conformal class. It does not determine the conformal factor from the full lens data in the general smooth case.

2. Local lens rigidity for manifolds of Anosov type

Mihajlo Cekic, Colin Guillarmou, Thibault Lefeuvre. *Analysis & PDE* 17 (2024), no. 8, 2737–2795. DOI: <https://doi.org/10.2140/apde.2024.17.2737>. arXiv: <https://arxiv.org/abs/2204.02476>.

The paper formulates the exact unmarked problem as Conjecture 1.7 and proves local rigidity: for a fixed Anosov-type metric in dimension 2, any sufficiently close smooth metric with the same lens data is boundary-fixing isometric to it. This is only a local result near a reference metric, not the full global smooth statement.

3. Marked boundary rigidity for surfaces of Anosov type

Alena Erchenko, Thibault Lefeuvre. *Mathematische Zeitschrift* 306, article 36 (2024). DOI: <https://doi.org/10.1007/s00209-024-03433-8>. arXiv: <https://arxiv.org/abs/2305.06893>.

This proves that two smooth surfaces of Anosov type with the same marked boundary distance are isometric. Marked boundary/lens data records the homotopy class or equivalently data on the universal cover. That is stronger/different data than the ordinary lens data in this problem, so it does not fully resolve the question.

4. Riemannian Anosov extension and applications

Dong Chen, Alena Erchenko, Andrey Gogolev. *Journal de l'Ecole polytechnique - Mathématiques* 10 (2023), 945–987. DOI: <https://doi.org/10.5802/jep.237>. arXiv: <https://arxiv.org/abs/2009.13665>.

Corollary B proves marked lens rigidity in a conformal class under the same Anosov-type hypotheses. This directly addresses conformal factors, but assumes marked lens equivalence, not just the unmarked lens data in the problem statement.

5. Scattering rigidity for analytic metrics

Yannick Guedes Bonthonneau, Colin Guillarmou, Malo Jezequel. *Cambridge Journal of*

Mathematics 12 (2024), no. 1, 165–222. DOI: <https://doi.org/10.4310/CJM.2024.v12.n1.a2>. arXiv: <https://arxiv.org/abs/2201.02100>.

This proves that for real analytic manifolds of Anosov type with analytic strictly convex boundary, the scattering map alone determines the metric up to boundary-fixing isometry. The analyticity assumption is an extra hypothesis absent from the problem statement, so it is only partial progress for the smooth problem.

6. Lens rigidity with trapped geodesics in two dimensions

Christopher B. Croke, Pilar Herreros. *Asian Journal of Mathematics* 20 (2016), no. 1, 47–58. arXiv: <https://arxiv.org/abs/1108.4938>.

This proves lens rigidity for specific two-dimensional trapped examples, including negatively curved cylinders with convex boundary, but not for all smooth surfaces with no conjugate points and hyperbolic trapped set.

7. Boundary and lens rigidity for non-convex manifolds

Colin Guillarmou, Marco Mazzucchelli, Leo Tzou. *American Journal of Mathematics* 143 (2021), no. 2, 533–575. DOI: <https://doi.org/10.1353/ajm.2021.0012>. arXiv: <https://arxiv.org/abs/1711.10059>.

This proves lens rigidity for non-trapping compact Riemannian surfaces with no conjugate points. The problem statement specifically allows trapping with hyperbolic trapped set, so this does not cover the full statement.

1.35.3.4. What remains unresolved

The missing step is global smooth rigidity from ordinary/unmarked full lens data for all surfaces with strictly convex boundary, no conjugate points, and hyperbolic trapped set. Existing theorems resolve marked data, analytic metrics, local neighborhoods of a fixed metric, non-trapping cases, or special topologies/curvature examples, but no located result removes all of these extra hypotheses.

1.35.4. Remaining open problems recorded by the database

1.35.4.1. Open problem 1

Let M be a compact surface with strictly convex boundary, no conjugate points, and hyperbolic trapped set. Suppose g_1 and g_2 are metrics of this type in the same conformal class and have identical lens data. Prove that their conformal factors agree up to a diffeomorphism fixing ∂M (equivalently, prove global lens rigidity within a fixed conformal class).

Reference supplied by the database. Mihajlo Cekić, Colin Guillarmou, and Thibault Lefeuvre, “Local lens rigidity for manifolds of Anosov type,” *Analysis & PDE* 17 (2024), no. 8, 2737–2795, discussion after Conjecture 1.7 (item (2): recovering the conformal factor is still open), DOI: 10.2140/apde.2024.17.2737, <https://doi.org/10.2140/apde.2024.17.2737>; arXiv:2204.02476, <https://arxiv.org/abs/2204.02476>.

1.35.5. Primary-source pointer

The originating ICM record is *Analytic tools for the study of flows and inverse problems* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=972>.

1.36. 36. Emergent complex geometry [2022-geom-161]

Record: 2022-geom-161

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Emergent complex geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>

1.36.1. Problem statement

Let X be a Fano manifold, i.e. a compact complex manifold whose anticanonical line bundle $-K_X$ is positive (ample). For each positive integer k , set

$$N_k := \dim_{\mathbb{C}} H^0(X, -kK_X).$$

Choose a basis $s_1^{(k)}, \dots, s_{N_k}^{(k)}$ of $H^0(X, -kK_X)$ and define the Slater determinant

$$(\det S^{(k)})(x_1, \dots, x_{N_k}) := \det(s_i^{(k)}(x_j))_{i,j=1}^{N_k},$$

a holomorphic section of $-kK_{X^{N_k}}$. On the complement of its zero locus, $|\det S^{(k)}|^{-2/k}$ defines a measure on X^{N_k} , and put

$$Z_{N_k} := \int_{X^{N_k}} |\det S^{(k)}|^{-2/k}.$$

Say that X is Gibbs stable at level k if $Z_{N_k} < \infty$, and Gibbs stable if this holds for all sufficiently large k . When $Z_{N_k} < \infty$, the corresponding canonical N_k -point process on X is the symmetric probability measure

$$\mu^{(N_k)} := Z_{N_k}^{-1} |\det S^{(k)}|^{-2/k}$$

on X^{N_k} ; after normalization it is independent of the chosen basis. For a configuration $(x_1, \dots, x_{N_k})$ sampled from this process, its empirical measure is

$$\delta_{N_k} := \frac{1}{N_k} \sum_{j=1}^{N_k} \delta_{x_j}.$$

Prove that X admits a unique Kähler-Einstein metric ω_{KE} with positive Ricci curvature, $\text{Ric } \omega_{\text{KE}} = \omega_{\text{KE}}$, if and only if X is Gibbs stable. Furthermore, prove that if X is Gibbs stable, then, as $k \rightarrow \infty$ (equivalently $N_k \rightarrow \infty$), the empirical measures δ_{N_k} of the above canonical point processes converge in probability to the normalized volume form $dV_{\text{KE}} := \omega_{\text{KE}}^{\dim_{\mathbb{C}} X} / \int_X \omega_{\text{KE}}^{\dim_{\mathbb{C}} X}$ of ω_{KE} .

1.36.2. Historical background, literature, and current progress

1.36.3. Verdict: partially resolved

1.36.3.1. Precise problem

The problem asks for Berman's Gibbs-stability conjecture for smooth Fano manifolds: for the canonical Slater-determinant point processes associated to sections of $-kK_X$, prove that X admits a unique positive Ricci Kähler-Einstein metric if and only if the partition functions Z_{N_k} are finite for all sufficiently large k , and, under this Gibbs stability hypothesis, prove convergence in probability of the empirical measures to the normalized Kähler-Einstein volume form.

1.36.3.2. Literature checked

1. **Kähler-Einstein metrics, canonical random point processes and birational geometry**
Author: Robert J. Berman
Venue: *Algebraic Geometry: Salt Lake City 2015*, Proceedings of Symposia in Pure Mathematics 97.1, AMS, 2018, pp. 29-73
DOI/arXiv/URL: DOI 10.1090/pspum/097.1/01669; arXiv 1307.3634, <https://arxiv.org/abs/1307.3634>
Status contribution: This is the original source of the exact problem. Conjecture 6.1 states the equivalence between asymptotic Gibbs stability and a unique Kähler-Einstein metric, plus convergence of the empirical measures. Theorem 6.3 proves only the stronger hypothesis "strong Gibbs stability" implies existence of a unique Kähler-Einstein metric. Thus it is partial progress, not a full resolution.
2. **Emergent complex geometry**
Author: Robert J. Berman
Venue: *Proceedings of the International Congress of Mathematicians 2022*, Vol. 4, EMS Press, 2023, pp. 2456-2483
DOI/arXiv/URL: DOI 10.4171/ICM2022/59; <https://ems.press/content/book-chapter-files/33209>
Status contribution: Restates the problem as Conjecture 2.3. It also records that Gibbs stability implies finite automorphism group, introduces the uniform Gibbs version, and states Theorem 2.5: uniform Gibbs stability implies uniform K-stability. The article explicitly treats the equivalence and empirical-measure convergence as conjectural.
3. **On Berman-Gibbs stability and K-stability of $\mathbb{Q}$ -Fano varieties**
Author: Kento Fujita
Venue: *Compositio Mathematica* 152(2), 2016, pp. 288-298
DOI/arXiv/URL: DOI 10.1112/S0010437X1500768X; Cambridge Core URL <https://www.cambridge.org/core/journals/compositio-mathematica/article/on-bermangibbs-stability-and-kstability-of-mathbbqfano-varieties/15DF205CC7BF193578B9B4AC09686933>
Status contribution: Proves that Berman-Gibbs stability or semistability implies the corresponding K-stability or K-semistability. Together with Yau-Tian-Donaldson results this gives one direction toward existence of Kähler-Einstein metrics, but it does not prove the converse Kähler-Einstein/unique Kähler-Einstein implies Gibbs stability, and it does not prove empirical-measure convergence.
4. **On the K-stability of Fano varieties and anticanonical divisors**
Authors: Kento Fujita and Yuji Odaka

Venue: *Tohoku Mathematical Journal*, Second Series 70(4), 2018, pp. 511-521

DOI/arXiv/URL: DOI 10.2748/tmj/1546570823; arXiv 1602.01305, <https://arxiv.org/abs/1602.01305>

Status contribution: Develops anticanonical-divisor criteria and relates them to Berman-Gibbs stability; in the survey/source literature this is cited for the implication from uniform Gibbs stability to uniform K-stability. This is a stronger-hypothesis/one-direction result and does not settle ordinary Gibbs stability equivalence or the point-process convergence statement.

5. Measure Preserving Holomorphic Vector Fields, Invariant Anti-Canonical Divisors and Gibbs Stability

Author: Robert J. Berman

Venue: *Analysis Mathematica* 48, 2022, pp. 347-375

DOI/arXiv/URL: DOI 10.1007/s10476-022-0154-6; arXiv 2201.03325, <https://arxiv.org/abs/2201.03325>

Status contribution: Proves that Gibbs stability rules out non-trivial holomorphic vector fields. This explains why the conjecture is phrased with uniqueness/finite automorphism behavior, but it is not a proof of the full equivalence or convergence.

6. The probabilistic vs the quantization approach to Kähler-Einstein geometry

Author: Robert J. Berman

Venue: *Mathematische Annalen* 388, 2024, pp. 4383-4404

DOI/arXiv/URL: DOI 10.1007/s00208-023-02627-5; arXiv 2109.06575, <https://arxiv.org/abs/2109.06575>

Status contribution: Theorem 1.2 and Corollary 1.3 give a direct analytic proof that uniform Gibbs stability implies existence of a unique Kähler-Einstein metric. The paper explicitly identifies this as one direction of the conjectural picture and notes that ordinary Gibbs stability is weaker than uniform Gibbs stability. It does not prove the converse or the empirical-measure convergence under ordinary Gibbs stability.

7. Gibbs polystability of Fano manifolds, stability thresholds and symmetry breaking

Authors: Rolf Andreasson, Robert J. Berman, Ludvig Svensson

Venue: arXiv preprint, 2025

DOI/arXiv/URL: arXiv 2511.16173, <https://arxiv.org/abs/2511.16173>

Status contribution: Extends the framework to log Fano manifolds with non-discrete automorphism groups by introducing Gibbs polystability. The paper states the polystable analogue as conjectural, proves several results for log Fano curves and some examples, and proves a strong-uniform Gibbs-polystability implication. It is significant progress but not a full resolution of the all-dimensional smooth Fano statement.

8. Open problems in K-stability of Fano varieties

Authors: Chenyang Xu and Ziquan Zhuang

Venue: arXiv preprint, 2026

DOI/arXiv/URL: arXiv 2601.15576v2, <https://arxiv.org/abs/2601.15576>

Status contribution: Section 2.3, “Gibbs stability”, describes the Fano Gibbs-stability picture as largely conjectural. It states the expected equivalence with K-stability as Conjecture 15 and says only one direction is known. It also says the empirical-measure convergence to the Kähler-Einstein volume form is conjectural.

1.36.3.3. What is resolved

- Gibbs stability or uniform Gibbs stability has been shown to imply algebraic K-stability or uniform K-stability in the cited algebraic work.
- Strong or uniform Gibbs stability implies existence of a unique Kähler-Einstein metric by the cited Berman results, and also via K-stability plus the Yau-Tian-Donaldson theorem.
- Gibbs stability is known to force discreteness/absence of holomorphic vector fields.
- Various special or modified settings are understood, including general type cases, log Fano curves, and polystable/symmetry-broken variants.

1.36.3.4. What remains unresolved

the exact statement is not fully proved in the literature. In particular, the converse direction that every Fano manifold with a unique positive Ricci Kähler-Einstein metric is Gibbs stable in the ordinary sense remains open in general. The convergence in probability of the canonical empirical measures to the Kähler-Einstein volume form under ordinary Gibbs stability also remains conjectural. Current results either prove only one direction, assume stronger stability such as strong or uniform Gibbs stability, treat special cases such as curves, or formulate broader polystable versions that remain conjectural.

1.36.4. Remaining open problems recorded by the database

1.36.4.1. Open problem 1

For a smooth Fano variety X , let

$$\gamma(X) = \liminf_{m \rightarrow \infty} \text{lct}\left(X^{N_m}, \frac{1}{m}D_m\right), \quad N_m = h^0(X, -mK_X),$$

where D_m is the determinantal divisor on X^{N_m} defined by a basis of $H^0(X, -mK_X)$. Is $\gamma(X) > 1$ equivalent to K-stability, and is $\gamma(X) \geq 1$ equivalent to K-semistability?

Reference supplied by the database. Chenyang Xu and Ziquan Zhuang, “Open Problems in K-Stability of Fano Varieties,” arXiv preprint, 2026, §2.3, Conjecture 15 (attributed to Berman), arXiv: [2601.15576](https://arxiv.org/abs/2601.15576). The underlying Gibbs-stability conjecture originates in Robert J. Berman, “Kähler-Einstein Metrics, Canonical Random Point Processes and Birational Geometry,” in *Algebraic Geometry: Salt Lake City 2015*, Proceedings of Symposia in Pure Mathematics **97.1**, American Mathematical Society, 2018, pp. 29–73, Conjecture 6.1, DOI: [10.1090/pspum/097.1/01669](https://doi.org/10.1090/pspum/097.1/01669), arXiv: [1307.3634](https://arxiv.org/abs/1307.3634).

1.36.4.2. Open problem 2

For a smooth Fano variety X whose stability threshold satisfies $\delta(X) \leq 1$, is the Gibbs threshold equal to the stability threshold, $\gamma(X) = \delta(X)$?

Reference supplied by the database. Chenyang Xu and Ziquan Zhuang, “Open Problems in K-Stability of Fano Varieties,” arXiv preprint, 2026, §2.3, the explicit expectation immediately following Conjecture 15 that $\delta(X) = \gamma(X)$ when $\delta(X) \leq 1$, together with the statement that only the inequality $\delta(X) \geq \gamma(X)$ is known, arXiv: [2601.15576](#). The known one-sided inequality is due to Kento Fujita, “On Berman–Gibbs Stability and K-Stability of $\mathbb{Q}$ -Fano Varieties,” *Compositio Mathematica* **152** (2016), no. 2, 288–298, DOI: [10.1112/S0010437X1500768X](#), arXiv: [1501.00248](#).

1.36.5. Primary-source pointer

The originating ICM record is *Emergent complex geometry* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>.

1.37. 37. Emergent complex geometry [2022-geom-162]

Record: 2022-geom-162

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Emergent complex geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>

1.37.1. Problem statement

Prove that, for a polarized projective algebraic manifold (X, L) , where X is a projective algebraic manifold and L is a holomorphic line bundle over X whose first Chern class $c_1(L)$ contains a Kähler form, there exists a Kähler metric whose Kähler form represents $c_1(L)$ and whose scalar curvature is constant if and only if (X, L) is K -polystable: for every test configuration $\rho = (\mathcal{X}, \mathcal{L})$ of (X, L) , that is, a $\mathbb{C}^*$ -equivariant degeneration over $\mathbb{C}$ restricting over $\mathbb{C}^*$ to the trivial polarized family $(X \times \mathbb{C}^*, L)$, with normal total space $\mathcal{X}$ and relatively ample $\mathbb{Q}$ -line bundle $\mathcal{L}$, the Donaldson-Futaki invariant satisfies $DF(\rho) \geq 0$, and equality occurs only when $\mathcal{X}$ is biholomorphic to $X \times \mathbb{C}$.

1.37.2. Historical background, literature, and current progress

1.37.3. Verdict: partially resolved

1.37.3.1. Precise problem

The problem asks for the general cscK form of the Yau-Tian-Donaldson conjecture for every smooth polarized projective manifold (X, L) : existence of a constant scalar curvature Kähler metric in $c_1(L)$ should be equivalent to the classical Donaldson-Futaki K -polystability condition tested on normal relatively ample test configurations, with equality only for product test configurations.

1.37.3.2. Status check

This exact statement is not fully resolved. The forward implication, cscK implies classical K -polystability, is known. Several important special cases are known, including the Fano Kähler-Einstein case and toric surfaces. Very recent work gives a full correspondence for stronger uniform/equivariant/non-Archimedean stability notions, but those hypotheses are not the same as the classical K -polystability condition. In fact, current papers explicitly distinguish these stronger notions from classical K -polystability and say the converse from classical K -polystability to cscK is not expected in this generality.

1.37.3.3. Key literature and applicability

1. Simon K. Donaldson, “Scalar curvature and stability of toric varieties”, *Journal of Differential Geometry* 62 (2002), no. 2, 289-349. DOI: 10.4310/jdg/1090950195. URL: <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-62/issue-2/Scalar-Curvature-and-Stability-of-Toric-Varieties/10.4310/jdg/1090950195.full>
 - Introduced the algebro-geometric K -stability framework for polarized varieties in the cscK setting and formulated the expected link with constant scalar curvature Kahler metrics. This is a source for the problem, not a full proof in all cases.
2. Simon K. Donaldson, “Constant Scalar Curvature Metrics on Toric Surfaces”, *Geometric and Functional Analysis* 19 (2009), 83-136. DOI: 10.1007/s00039-009-0714-y. arXiv:0805.0128. URL: <https://arxiv.org/abs/0805.0128>
 - Proves the existence direction in the toric surface case, assuming the relevant toric K -stability. This is a major special case but does not cover arbitrary polarized manifolds.
3. Jacopo Stoppa, “ K -stability of constant scalar curvature Kahler manifolds”, *Advances in Mathematics* 221 (2009), no. 4, 1397-1408. DOI: 10.1016/j.aim.2009.02.013. arXiv:0803.4095. URL: <https://arxiv.org/abs/0803.4095>
 - Proves that a polarized manifold admitting a cscK metric and having discrete automorphism group is K -stable. This is the forward implication under a discreteness hypothesis, not the converse and not all automorphism cases.
4. Robert J. Berman, Tamas Darvas, Chinh H. Lu, “Regularity of weak minimizers of the K -energy and applications to properness and K -stability”, *Annales Scientifiques de l’Ecole Normale Supérieure* 53 (2020), no. 2, 267-289. DOI: 10.24033/asens.2422. arXiv:1602.03114. URL: <https://arxiv.org/abs/1602.03114>
 - Proves that an ample line bundle (X, L) admitting a cscK metric in $c_1(L)$ is K -polystable. This covers the necessary direction of the problem statement for polarized manifolds, but it does not prove that classical K -polystability implies existence of a cscK metric.
5. Xiuxiong Chen, Simon Donaldson, Song Sun, “Kahler-Einstein metrics on Fano manifolds I-III”, *Journal of the American Mathematical Society* 28 (2015), no. 1, 183-197, 199-234, 235-278. DOI for I: 10.1090/S0894-0347-2014-00799-2; arXiv:1211.4566, arXiv:1212.4714, arXiv:1302.0282. URL: <https://arxiv.org/abs/1302.0282>
 - Resolves the Yau-Tian-Donaldson correspondence for smooth Fano manifolds with the anticanonical polarization. This is a central special case of the problem, but it does not imply the cscK statement for arbitrary polarizations.
6. Xiuxiong Chen and Jingrui Cheng, “On the constant scalar curvature Kahler metrics (I)–A priori estimates”, *Journal of the American Mathematical Society* 34 (2021), no. 4, 909-936, and “On the constant scalar curvature Kahler metrics (II)–Existence results”, *Journal of the American Mathematical Society* 34 (2021), no. 4, 937-1009. DOI for II: 10.1090/jams/966. arXiv:1712.06697 and arXiv:1801.00656. URL: <https://arxiv.org/abs/1801.00656>
 - Gives fundamental analytic existence/properness results for cscK metrics, including that properness/coercivity of the Mabuchi K -energy implies existence. This is a major ingredient in modern YTD work, but properness/coercivity is not the same as the classical Donaldson-Futaki K -polystability condition.

7. **Sebastien Boucksom, Tomoyuki Hisamoto, Mattias Jonsson**, “Uniform K-stability and asymptotics of energy functionals in Kahler geometry”, *Journal of the European Mathematical Society* 21 (2019), no. 9, 2905-2944; erratum *JEMS* 24 (2022), no. 2, 735-736. DOI: 10.4171/JEMS/894; erratum DOI: 10.4171/JEMS/1215. arXiv:1603.01026. URL: <https://arxiv.org/abs/1603.01026>
 - Relates slopes of energy functionals to non-Archimedean invariants and shows, among other results, that coercivity implies uniform K-stability. This supports stronger uniform versions of YTD but does not identify ordinary K-polystability with cscK existence in all cases.
8. **Chi Li**, “Geodesic rays and stability in the cscK problem”, *Annales Scientifiques de l’Ecole Normale Supérieure* 55 (2022), no. 6, 1529-1574. DOI: 10.24033/asens.2523. arXiv:2001.01366. URL: <https://arxiv.org/abs/2001.01366>
 - Proves existence results from strengthened stability notions over models/non-Archimedean objects. These are partial progress toward general cscK YTD, but the assumptions are stronger than classical K-polystability over normal ample test configurations.
9. **Sebastien Boucksom and Mattias Jonsson**, “On the Yau-Tian-Donaldson conjecture for weighted cscK metrics”, arXiv:2509.15016 (2025). DOI: 10.48550/arXiv.2509.15016. URL: <https://arxiv.org/abs/2509.15016>
 - Establishes a YTD-type correspondence for cscK and weighted extremal metrics using non-Archimedean pluripotential theory and stronger $\hat{K}$ -polystability-type notions on the completed non-Archimedean space, not merely the classical Donaldson-Futaki condition on normal test configurations. The introduction explicitly says that $\hat{K}$ -polystability is a stronger version of K-polystability and that the converse from classical K-polystability is not expected in general.
10. **Antonio Trusiani**, “A solution to the Yau-Tian-Donaldson Conjecture through Special Fujita Approximations”, arXiv:2605.30063 (2026). DOI: 10.48550/arXiv.2605.30063. URL: <https://arxiv.org/abs/2605.30063>
 - Proves a uniform/equivariant version: a smooth polarized projective variety admits a cscK metric iff it is $\text{Aut}^\circ(X, L)$ -uniformly K-stable, and relates several strengthened stability notions. The paper itself notes that existence of cscK metrics implies K-polystability, while the converse from classical K-polystability is not expected; therefore it does not prove the exact classical K-polystability equivalence.
11. **Vestislav Apostolov, David M. J. Calderbank, Paul Gauduchon, Christina W. Tonnesen-Friedman**, “Hamiltonian 2-forms in Kahler geometry. III. Extremal metrics and stability”, *Inventiones Mathematicae* 173 (2008), no. 3, 547-601. DOI: 10.1007/s00222-008-0126-x. URL: <https://doi.org/10.1007/s00222-008-0126-x>
 - Gives examples and computations in the extremal/cscK stability landscape that later authors cite as evidence that ordinary K-polystability is too weak for the general cscK existence problem. These examples are diagnostic evidence for the gap between classical and uniform stability, not a full resolution of the stated smooth polarized-manifold statement.
12. **Masafumi Hattori**, “A decomposition formula for J-stability and its applications”, arXiv:2103.04603 (2021). DOI: 10.48550/arXiv.2103.04603. URL: <https://arxiv.org/abs/2103.04603>

- Constructs K -stable but not uniformly K -stable examples in singular/log settings and is cited by recent YTD papers as further evidence that classical K -polystability should not be the final general cscK criterion. The constructions do not by themselves settle the stated smooth polarized projective manifold case.

1.37.3.4. What remains unresolved for this problem

The unresolved point is the converse in exactly the formulation: whether ordinary Donaldson-Futaki K -polystability for all normal relatively ample test configurations, with equality only for products, is sufficient for a cscK metric on every polarized smooth projective manifold. Current complete correspondences use stronger uniform, equivariant, model, or completed non-Archimedean stability notions. Since those hypotheses are not identical to the classical K -polystability, the full statement is not fully resolved.

1.37.4. Remaining open problems recorded by the database

1.37.4.1. Open problem 1

Let P be a Delzant polytope of dimension at least 3, with vanishing extremal affine function (equivalently, vanishing Futaki character). If P is torically K -polystable in Donaldson's classical sense— $\mathcal{L}_P(f) \geq 0$ for every rational convex piecewise-linear f , with equality only for affine f —must the corresponding polarized toric manifold admit a constant-scalar-curvature Kähler metric, equivalently a smooth symplectic potential satisfying the constant-scalar-curvature Abreu equation with Guillemin boundary behaviour?

Reference supplied by the database. Simon K. Donaldson, “Scalar Curvature and Stability of Toric Varieties,” *Journal of Differential Geometry* **62** (2002), no. 2, 289–349; the toric stability/cscK conjecture formulated in the Introduction (and reduced to the Abreu equation in the body of the paper). DOI: [10.4310/jdg/1090950195](https://doi.org/10.4310/jdg/1090950195). The two-dimensional case was proved in Simon K. Donaldson, “Constant Scalar Curvature Metrics on Toric Surfaces,” *Geometric and Functional Analysis* **19** (2009), 83–136, DOI: [10.1007/s00039-009-0714-y](https://doi.org/10.1007/s00039-009-0714-y), arXiv: [0805.0128](https://arxiv.org/abs/0805.0128); later all-dimensional existence results assume the stronger uniform stability condition, as in Bohui Chen, An-Min Li, and Li Sheng, “Interior Estimates for the n -Dimensional Abreu's Equation,” *Advances in Mathematics* **251** (2014), 35–46, DOI: [10.1016/j.aim.2013.10.022](https://doi.org/10.1016/j.aim.2013.10.022), and Chi Li, “Geodesic Rays and Stability in the cscK Problem,” *Annales Scientifiques de l'École Normale Supérieure* (4) **55** (2022), no. 6, 1529–1574, DOI: [10.24033/asens.2523](https://doi.org/10.24033/asens.2523), arXiv: [2001.01366](https://arxiv.org/abs/2001.01366).

1.37.5. Primary-source pointer

The originating ICM record is *Emergent complex geometry* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>.

1.38. 38. Kähler manifolds with curvature bounded below [2022-geom-163]

Record: 2022-geom-163

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Kähler manifolds with curvature bounded below

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219>

1.38.1. Problem statement

Prove that if M^n is a complete noncompact Kähler manifold of complex dimension n with positive holomorphic bisectional curvature, meaning that at every point of M , for all nonzero vectors $X, Y \in T^{1,0}M$ at that point,

$$\frac{R(X, \bar{X}, Y, \bar{Y})}{\|X\|^2\|Y\|^2 + |\langle X, \bar{Y} \rangle|^2} > 0,$$

then M is biholomorphic to the complex Euclidean space $\mathbb{C}^n$.

1.38.2. Historical background, literature, and current progress

1.38.3. Verdict: partially resolved

1.38.3.1. Precise problem

This is Yau's uniformization conjecture for complete noncompact Kahler manifolds: if a complete noncompact complex n -dimensional Kahler manifold has positive holomorphic bisectional curvature, then its underlying complex manifold is biholomorphic to $\mathbb{C}^n$. The statement imposes no maximal volume growth, bounded curvature, curvature decay, finite Chern number, topological, or dimension restriction.

1.38.3.2. Status check

The literature does not fully resolve the statement as written. The strongest all-dimensional theorems prove the conclusion after adding maximal volume growth, and several other papers prove special cases under bounded curvature, curvature decay, finite Chern-number, surface, or sectional-curvature hypotheses. These hypotheses are not consequences of the stated assumptions in the published results checked here. Recent literature in 2026 still explicitly describes the full conjecture as open.

1.38.3.3. Key literature and hypothesis comparison

1. Shing-Tung Yau, "A review of complex differential geometry," in *Several Complex Variables and Complex Geometry, Part 2* (Santa Cruz, CA, 1989), Proceedings of Symposia in Pure Mathematics 52, Part 2, American Mathematical Society, 1991, pp. 619-625. DOI: 10.1090/pspum/052.2/1128577.
This is one standard cited source for the conjecture. Later surveys quote it as asking whether every complete noncompact Kahler manifold with positive bisectional curvature is biholomorphic to $\mathbb{C}^n$.
2. Ngaiming Mok, Yum-Tong Siu, and Shing-Tung Yau, "The Poincare-Lelong equation on complete Kahler manifolds," *Compositio Mathematica* 44 (1981), no. 1-3, 183-218. URL: https://www.numdam.org/item/CM_1981__44_1-3_183_0/.
Foundational analytic progress used in later uniformization work, but not a proof of the full conjecture. It supplies techniques under growth hypotheses rather than proving the unrestricted biholomorphism conclusion.
3. Ngaiming Mok, "An embedding theorem of complete Kahler manifolds of positive bisectional curvature onto affine algebraic varieties," *Bulletin de la Societe Mathematique de France* 112 (1984), 197-258. DOI: 10.24033/bsmf.2005. URL: <https://www.numdam.org/articles/10.24033/bsmf.2005/>.
Mok proved affine-algebraic embedding under extra assumptions such as quadratic scalar-curvature decay and Euclidean/maximal volume growth. This is substantial progress but does not imply that every manifold in the problem statement is biholomorphic to $\mathbb{C}^n$.
4. Bing-Long Chen, Siu-Hung Tang, and Xi-Ping Zhu, "A Uniformization Theorem Of Complete Noncompact Kahler Surfaces With Positive Bisectional Curvature," *Journal of Differential Geometry* 67 (2004), no. 3, 519-570. DOI: 10.4310/jdg/1102091357. arXiv: math/0211372, <https://arxiv.org/abs/math/0211372>.
This proves a two-dimensional result under positive and bounded holomorphic bisectional curvature, Euclidean volume growth, and average scalar-curvature decay. It is only a special case and has extra hypotheses.
5. Albert Chau and Luen-Fai Tam, "On the complex structure of Kahler manifolds with nonnegative curvature," *Journal of Differential Geometry* 73 (2006), no. 3, 491-530. DOI: 10.4310/jdg/1146169936. arXiv: math/0504422, <https://arxiv.org/abs/math/0504422>.
The paper proves that a complete noncompact Kahler manifold with nonnegative bounded holomorphic bisectional curvature and maximal volume growth is biholomorphic to $\mathbb{C}^n$. The bounded-curvature and maximal-volume-growth assumptions are extra.
6. Bing-Long Chen and Xi-Ping Zhu, "Yau's uniformization conjecture for manifolds with non-maximal volume growth," *Acta Mathematica Scientia* 38B (2018), no. 5, 1468-1484. DOI: 10.1016/S0252-9602(18)30828-2.
This studies the nonmaximal-volume-growth regime and gives partial results, including conditions involving bounded curvature and finite Chern-number-type hypotheses. It does not prove the unrestricted conjecture.
7. Gang Liu, "On Yau's uniformization conjecture," *Cambridge Journal of Mathematics* 7 (2019), no. 1-2, 33-70. DOI: 10.4310/CJM.2019.v7.n1.a2. arXiv:1606.08958, <https://arxiv.org/abs/1606.08958>.
The main theorem states that if M^n is a complete noncompact Kahler manifold with

nonnegative bisectional curvature and maximal volume growth, then M is biholomorphic to $\mathbb{C}^n$. This confirms Yau's conjecture only in the maximal-volume-growth case. the problem statement has no such assumption.

8. Man-Chun Lee and Luen-Fai Tam, "Chern-Ricci flows on noncompact complex manifolds," *Journal of Differential Geometry* 115 (2020), no. 3, 529-564. DOI: 10.4310/jdg/1594260018. arXiv:1708.00141, <https://arxiv.org/abs/1708.00141>.

This gives another proof that a complete noncompact Kahler manifold with nonnegative bisectional curvature and maximal volume growth is biholomorphic to $\mathbb{C}^n$, explicitly calling it a partial answer to Yau's conjecture. Again, maximal volume growth is extra.

9. Gang Liu, "Kahler manifolds with curvature bounded below," in *Proceedings of the International Congress of Mathematicians 2022*, Vol. 4, EMS Press, 2023, pp. 2576-2592. DOI: 10.4171/ICM2022/6. URL: <https://ems.press/content/book-chapter-files/33214>.

This survey states Conjecture 4 exactly as the problem and records the known progress. It presents the unrestricted statement as a conjecture, while proving and surveying related results such as the maximal-volume-growth case and finite generation questions.

10. Yang Shi, "Minimal Degrees, Volume Growth, and Curvature Decay on Complete Kahler Manifolds," *International Mathematics Research Notices* 2026, Issue 9, rnag080, published 23 April 2026. DOI: 10.1093/imrn/rnag080. arXiv:2511.01263, <https://arxiv.org/abs/2511.01263>.

This 2026 paper states Yau's uniformization conjecture and explicitly says that the full conjecture is still widely open. It describes Liu's result and the Lee-Tam proof as resolving the maximal-volume-growth case, not the full statement.

11. Ved Datar, Vamsi Pritham Pingali, and Harish Seshadri, "Uniformisation of complete Kahler surfaces with positive sectional curvature," arXiv:2604.11220, 2026. URL: <https://arxiv.org/abs/2604.11220>.

This preprint proves that a complete noncompact Kahler surface with positive sectional curvature is biholomorphic to $\mathbb{C}^2$, without assumptions at infinity. It is recent progress on a weaker two-dimensional variant, but it assumes positive sectional curvature rather than the all-dimensional positive bisectional-curvature hypothesis.

1.38.3.4. What remains unresolved

The missing case is the full unrestricted one: complete noncompact Kahler manifolds of arbitrary complex dimension with positive holomorphic bisectional curvature but without assuming maximal volume growth, bounded curvature, curvature decay, finite Chern-number conditions, positive sectional curvature, or auxiliary topology. Existing citable theorems do not cover all such manifolds, so the correct status is partially resolved rather than fully resolved.

1.38.4. Remaining open problems recorded by the database

1.38.4.1. Open problem 1

Prove the complex-surface case of Yau's uniformization conjecture: every complete non-compact Kähler surface with positive holomorphic bisectional curvature is biholomorphic to $\mathbb{C}^2$, without imposing bounded curvature, volume-growth, curvature-decay, topological, or positive-sectional-curvature hypotheses.

Reference supplied by the database. Ved Datar, Vamsi Pritham Pingali, and Harish Seshadri, "Uniformisation of Complete Kähler Surfaces with Positive Sectional Curvature," 2026, Introduction and discussion following Theorem 1 (explicitly contrasting the solved positive-sectional-curvature surface case with the still-unknown positive-holomorphic-bisectional-curvature case), arXiv: [2604.11220](https://arxiv.org/abs/2604.11220). The original conjecture is attributed there to Shing-Tung Yau, "Open Problems in Geometry," in *Differential Geometry: Partial Differential Equations on Manifolds* (Los Angeles, CA, 1990), Proceedings of Symposia in Pure Mathematics 54, Part 1, American Mathematical Society, 1993, pp. 1–28.

1.38.4.2. Open problem 2

Does every complete noncompact Kähler manifold with positive holomorphic bisectional curvature admit a smooth proper plurisubharmonic exhaustion function ρ with bounded gradient (that is, is it strongly Stein in the sense used by Greene and Wu)?

Reference supplied by the database. Ved Datar, Vamsi Pritham Pingali, and Harish Seshadri, "Uniformisation of Complete Kähler Surfaces with Positive Sectional Curvature," 2026, Introduction, discussion immediately after Theorem 1 (identifying strong Steinness as known under positive sectional curvature but unknown under the weaker hypothesis of positive holomorphic bisectional curvature), arXiv: [2604.11220](https://arxiv.org/abs/2604.11220). The positive-sectional-curvature result being contrasted is Robert E. Greene and H. Wu, "On Kähler Manifolds of Positive Bisectional Curvature and a Theorem of Hartogs," *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg* 47 (1978), 171–185, DOI: [10.1007/BF02941481](https://doi.org/10.1007/BF02941481).

1.38.4.3. Open problem 3

Does the complex surface $\mathbb{C} \times \mathbb{C}^*$ admit a complete Kähler metric with positive holomorphic bisectional curvature?

Reference supplied by the database. Ved Datar, Vamsi Pritham Pingali, and Harish Seshadri, "Uniformisation of Complete Kähler Surfaces with Positive Sectional Curvature," 2026, Introduction, discussion immediately after Theorem 1 (where this specific existence question is explicitly identified as unknown), arXiv: [2604.11220](https://arxiv.org/abs/2604.11220).

1.38.5. Primary-source pointer

The originating ICM record is *Kähler manifolds with curvature bounded below* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219>.

1.39. 39. Kähler manifolds with curvature bounded below [2022-geom-164]

Record: 2022-geom-164

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Kähler manifolds with curvature bounded below

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219>

1.39.1. Problem statement

Prove that every complete noncompact Kähler manifold with positive Ricci curvature is biholomorphic to a Zariski open subset of some compact Kähler manifold.

1.39.2. Historical background, literature, and current progress

1.39.3. Verdict: partially resolved

1.39.3.1. Precise problem

The problem asks Yau's compactification question: whether every complete noncompact Kahler manifold whose Ricci curvature is everywhere positive is biholomorphic to a Zariski open subset of some compact Kahler manifold. No assumptions on dimension, bisectional/sectional curvature, volume growth, curvature decay, bounded curvature, finite topology, finite Chern/Ricci integral, or tangent cones are included in the problem statement.

1.39.3.2. Status check

I did not find a published or citable theorem proving this full statement. The closest results prove quasi-projectivity or compactification only after adding substantial hypotheses. Since those hypotheses are not consequences of positive Ricci curvature alone in the cited literature, they do not resolve the exact question.

The problem is not fully open, because there is significant partial progress. The current verdict is therefore **partially resolved**.

1.39.3.3. Key literature and progress

1. S. T. Yau, "S. S. Chern: A great geometer of the Twentieth Century." International Press, 1992. URL not found.

This is cited by Gang Liu's ICM survey as the original source of Question 71, p. 304: complete noncompact Kahler manifolds with positive Ricci curvature should be biholomorphic to a Zariski open set of a compact Kahler manifold.

2. Ngaiming Mok, “An embedding theorem of complete Kähler manifolds of positive bisectional curvature onto affine algebraic varieties.” *Bulletin de la Société Mathématique de France* 112 (1984), 197-258. DOI: 10.24033/bsmf.2005. URL: <https://www.numdam.org/articles/10.24033/bsmf.2005/>
Mok proved affine algebraicity under stronger curvature and asymptotic hypotheses: positive holomorphic bisectional curvature, maximal/Euclidean volume growth, and scalar-curvature decay. This is a compactification-type result, but it does not cover arbitrary positive Ricci curvature.
3. Ngaiming Mok, “An embedding theorem of complete Kähler manifolds of positive Ricci curvature onto quasi-projective varieties.” *Mathematische Annalen* 286 (1990), 373-408. DOI: 10.1007/BF01453581. URL: <https://link.springer.com/article/10.1007/BF01453581>
As summarized explicitly in Liu’s 2021 paper, Mok proved quasi-projectivity for complete noncompact Kähler manifolds with positive Ricci curvature assuming maximal volume growth, quadratic curvature decay $|Rm| \leq C/r^2$, and finite integral $\int_M Ric^n < \infty$. These extra assumptions mean Mok’s theorem is partial progress, not a full solution.
4. Wing-Keung To, “Quasi-projective embeddings of noncompact complete Kähler manifolds of positive Ricci curvature and satisfying certain topological conditions.” *Duke Mathematical Journal* 63(3) (1991), 745-789. DOI: 10.1215/S0012-7094-91-06332-5. URL: <https://projecteuclid.org/journals/duke-mathematical-journal/volume-63/issue-3/Quasi-projective-embeddings-of-noncompact-complete-K%C3%A4hler-manifolds-of-positive/10.1215/S0012-7094-91-06332-5.full>
This gives quasi-projective embeddings for positive-Ricci complete noncompact Kähler manifolds subject to additional topological conditions. The title itself records that the theorem is conditional; it does not establish Yau’s question for all positive-Ricci cases.
5. Bing-Long Chen and Xi-Ping Zhu, “Positively curved complete noncompact Kähler manifolds.” *Acta Mathematica Scientia* 29B(4) (2009), 829-845. DOI: 10.1016/S0252-9602(09)60073-4. arXiv: math/0211373. URL: <https://arxiv.org/abs/math/0211373>
The main theorem proves quasi-projectivity for complete noncompact Kähler manifolds with positive and bounded sectional curvature plus $\int_M Ric^n < \infty$, and gives C^2 in complex dimension 2. This concerns stronger positive sectional curvature and an integral finiteness condition, so it is only special-case progress for the problem.
6. Gang Liu, “Compactification of certain noncompact Kähler manifolds with non-negative Ricci curvature.” *Advances in Mathematics* 382 (2021), article 107652. DOI: 10.1016/j.aim.2021.107652. arXiv:1706.06067. URL: <https://arxiv.org/abs/1706.06067>
Theorem 1.1/3.1 states: if (M^n, p) is complete noncompact Kähler with nonnegative Ricci curvature and maximal volume growth, then in particular M is quasiprojective if Ricci curvature is positive and $|Rm| \leq C/r^2$. This strengthens Mok by removing the finite $\int_M Ric^n$ assumption, but still assumes maximal volume growth and quadratic curvature decay. Liu explicitly says this “supports” Yau’s Question 71, not that it settles it.
7. Gang Liu, “Kähler manifolds with curvature bounded below.” *Proceedings of the International Congress of Mathematicians 2022*, Vol. 4, EMS Press, pp. 2576-2592, 2022/2023. DOI: 10.4171/ICM2022/6. URL: <https://ems.press/content/book-chapter-files/33214>

This survey restates Yau's Question 71 and describes Liu's compactification theorem as supporting it under extra hypotheses. The survey also distinguishes positive Ricci curvature from stronger bisectional/sectional curvature assumptions.

8. **Gang Liu**, "Complete Kähler manifolds with nonnegative Ricci curvature." arXiv preprint, 2024. arXiv:2404.08537. URL: <https://arxiv.org/abs/2404.08537>

This later preprint studies scalar-curvature integrals, volume growth, and rigidity of Ricci-flat Kähler manifolds with nonnegative Ricci curvature. Its abstracted main results do not claim the compactification statement for all positive-Ricci complete noncompact Kähler manifolds.

9. **Ved V. Datar, Vamsi P. Pingali, and Harish Seshadri**, "Uniformisation of complete Kähler surfaces with positive sectional curvature." arXiv preprint, 2026. arXiv:2604.11220. URL: <https://arxiv.org/abs/2604.11220>

This resolves the weak Yau uniformization problem for complex surfaces with positive sectional curvature, proving biholomorphism to $\mathbb{C}^2$. Positive sectional curvature in complex dimension 2 is much stronger and narrower than positive Ricci curvature in arbitrary dimension, so it does not resolve this problem.

10. **Ved V. Datar, Vamsi P. Pingali, and Harish Seshadri**, "The top Yau–Yang conjecture for Kähler manifolds with positive sectional curvature." arXiv preprint, 2026. arXiv:2606.19806. URL: <https://arxiv.org/abs/2606.19806>

The abstract proves finiteness of the top wedge power of the Ricci form for complete noncompact Kähler manifolds with positive sectional curvature and obtains quasiprojectivity under bounded sectional curvature via Chen-Zhu. Again, this is positive sectional curvature plus boundedness for quasiprojectivity, not the full positive-Ricci problem.

1.39.3.4. What remains unresolved

The missing case is exactly the general one in the problem statement: a complete noncompact Kähler manifold with positive Ricci curvature and no further global control. The literature above does not show that positive Ricci curvature alone implies maximal volume growth, quadratic curvature decay, $\int_M \text{Ric}^n$, bounded sectional curvature, positive bisectional/sectional curvature, finite topology, or the topological assumptions used in the known embedding theorems. Therefore the full compactification question remains unresolved.

1.39.4. Remaining open problems recorded by the database

1.39.4.1. Open problem 1

Let M^n , $n \geq 3$, be a complete noncompact Kähler manifold with nonnegative Ricci curvature and maximal (Euclidean) volume growth. If its holomorphic bisectional curvature satisfies $BK(x) \geq -C/r(x)^2$, must M be biholomorphic to a Zariski open subset of a Moishezon manifold, without assuming that M has a unique tangent cone at infinity?

Reference supplied by the database. Gang Liu, "Compactification of Certain Noncompact Kähler Manifolds with Nonnegative Ricci Curvature," *Advances in Mathematics* **382** (2021), Article 107652, discussion immediately following Theorem 1.1(II) (explicitly asking to remove

uniqueness of the tangent cone at infinity). DOI: [10.1016/j.aim.2021.107652](https://doi.org/10.1016/j.aim.2021.107652); arXiv: [1706.06067](https://arxiv.org/abs/1706.06067). The complex-surface case is subsequent partial progress: Gang Liu, “Complete Kähler Manifolds with Nonnegative Ricci Curvature II,” 2025, main theorem, arXiv: [2510.10600](https://arxiv.org/abs/2510.10600); no corresponding all-dimensional result was found.

1.39.5. Primary-source pointer

The originating ICM record is *Kähler manifolds with curvature bounded below* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219>.

1.40. 40. Mean curvature and variational theory [2022-geom-166]

Record: 2022-geom-166

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Mean curvature and variational theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=339>

1.40.1. Problem statement

Prove or disprove that every closed Riemannian manifold (M^{n+1}, g) with $3 \leq n+1 \leq 7$ contains, for every constant $c > 0$, at least two distinct closed hypersurfaces whose mean curvature is identically equal to c .

1.40.2. Historical background, literature, and current progress

1.40.3. Verdict: partially resolved

1.40.3.1. Precise problem

The problem asks for the Twin Bubble Conjecture: for every fixed closed Riemannian manifold (M^{n+1}, g) with $3 \leq n+1 \leq 7$ and every prescribed constant $c > 0$, there should be at least two distinct closed hypersurfaces whose mean curvature is identically c . In the recent min-max literature this is usually stated for smooth closed almost-embedded hypersurfaces.

The literature does not currently prove this full fixed-metric, all- c statement. There is substantial progress in special regimes and under extra hypotheses, so the problem is not fully open, but it remains unresolved for arbitrary metrics and arbitrary intermediate values of c .

1.40.3.2. Original source

- Title: Mean curvature and variational theory
- Author: Xin Zhou
- Venue: ICM–International Congress of Mathematicians, Vol. IV, Sections 5-8, EMS Press, Berlin, pages 2696-2717
- Year: 2023
- DOI/URL: DOI 10.4171/ICM2022/27; <https://ems.press/books/standalone/276/5487>
- Relevance: This ICM survey formulates the conjectural statement that every closed manifold (M^{n+1}, g) , $3 \leq n+1 \leq 7$, contains at least two distinct closed CMC hypersurfaces with mean curvature c for any $c > 0$.

1.40.3.3. Progress and non-resolving results

- Title: Min-max theory for constant mean curvature hypersurfaces
- Authors: Xin Zhou and Jonathan J. Zhu
- Venue: *Inventiones Mathematicae* 218, 441-490
- Year: 2019
- DOI/arXiv/URL: DOI 10.1007/s00222-019-00886-1; arXiv:1707.08012; <https://arxiv.org/abs/1707.08012>
- Result: Proves existence of a nontrivial smooth closed almost-embedded CMC hypersurface of any prescribed mean curvature c in an arbitrary closed manifold; for nonzero c the min-max solution has multiplicity one.
- Why not full resolution: This gives one hypersurface for each c , not two.
- Title: Existence of multiple closed CMC hypersurfaces with small mean curvature
- Author: Akashdeep Dey
- Venue: *Journal of Differential Geometry* 125(2), 379-403
- Year: 2023
- DOI/arXiv/URL: DOI 10.4310/jdg/1696432925; arXiv:1910.00989; <https://arxiv.org/abs/1910.00989>
- Result: For every m , there is $c^*(m) > 0$ depending on g such that if $0 < c < c^*(m)$, then (M, g) contains at least m closed c -CMC hypersurfaces with optimal regularity. In dimensions $3 \leq n+1 \leq 7$, this gives smooth hypersurfaces.
- Why not full resolution: Taking $m = 2$ proves the conclusion only for sufficiently small c , not for every $c > 0$.
- Title: Foliation by constant mean curvature spheres
- Author: Rugang Ye
- Venue: *Pacific Journal of Mathematics* 147(2), 381-396
- Year: 1991
- DOI/arXiv/URL: <https://projecteuclid.org/journals/pacific-journal-of-mathematics/volume-147/issue-2/Foliation-by-constant-mean-curvature-spheres/pjm/1102644918.pdf>
- Result: Constructs local foliations by small CMC spheres near nondegenerate critical points of the scalar curvature, giving large-mean-curvature examples under a nondegeneracy hypothesis.
- Why not full resolution: This applies only in the large- c regime and uses extra scalar-curvature hypotheses.
- Title: Constant mean curvature spheres in Riemannian manifolds
- Authors: Frank Pacard and Xingwang Xu
- Venue: *Manuscripta Mathematica* 128, 275-295
- Year: 2009
- DOI/arXiv/URL: DOI 10.1007/s00229-008-0230-7; <https://link.springer.co>

[m/article/10.1007/s00229-008-0230-7](https://doi.org/10.1007/s00229-008-0230-7)

- Result: Proves existence of embedded spheres with large constant mean curvature in any compact Riemannian manifold, partially generalizing Ye's result and giving large- c multiplicity information via Lusternik-Schnirelmann category.
- Why not full resolution: This covers sufficiently large c , not all positive c .
- Title: The half-volume spectrum of a manifold
- Authors: Liam Mazurowski and Xin Zhou
- Venue: Calculus of Variations and Partial Differential Equations 64, article 155
- Year: 2025
- DOI/arXiv/URL: DOI 10.1007/s00526-025-02949-z; <https://link.springer.com/article/10.1007/s00526-025-02949-z>
- Result: Develops the half-volume spectrum and proves, in dimensions $3 \leq n+1 \leq 7$, existence of half-volume CMC hypersurfaces realizing phase-transition half-volume widths, possibly with additional minimal hypersurface components.
- Why not full resolution: The result controls enclosed volume rather than prescribing the exact mean curvature c .
- Title: Infinitely Many Half-Volume Constant Mean Curvature Hypersurfaces via Min-Max Theory
- Authors: Liam Mazurowski and Xin Zhou
- Venue: arXiv preprint
- Year: 2024
- DOI/arXiv/URL: DOI 10.48550/arXiv.2405.00595; arXiv:2405.00595; <https://arxiv.org/abs/2405.00595>
- Result: For closed manifolds of dimension $3 \leq n+1 \leq 5$, if the metric is generic or has positive Ricci curvature, there are infinitely many geometrically distinct half-volume CMC hypersurfaces.
- Why not full resolution: This imposes generic or positive Ricci hypotheses, is limited to dimensions up to five, and does not prescribe the mean curvature value c .
- Title: An Alternative for Constant Mean Curvature Hypersurfaces
- Authors: Liam Mazurowski and Xin Zhou
- Venue: arXiv preprint
- Year: 2024
- DOI/arXiv/URL: DOI 10.48550/arXiv.2408.13864; arXiv:2408.13864; <https://arxiv.org/abs/2408.13864>
- Result: For a generic metric on a closed manifold of dimension $3 \leq n+1 \leq 7$ and every $c > 0$, either there are infinitely many distinct closed almost-embedded hypersurfaces with CMC exactly c , or there are infinitely many closed almost-embedded CMC hypersurfaces with mean curvature less than c enclosing half the volume.
- Why not full resolution: It assumes a generic metric and the second alternative does not produce even two hypersurfaces with mean curvature exactly c .

- Title: Pairs of Embedded Spheres with Pinched Prescribed Mean Curvature
- Authors: Liam Mazurowski and Xin Zhou
- Venue: arXiv preprint
- Year: 2025
- DOI/arXiv/URL: DOI 10.48550/arXiv.2511.08228; arXiv:2511.08228; <https://arxiv.org/abs/2511.08228>
- Result: Proves two embedded two-spheres in the unit three-sphere under a pinching condition on a prescribed mean-curvature function. The introduction explicitly states the Twin Bubble Conjecture and says the relevant two-solution conjectures are still open.
- Why not full resolution: This is a special result for S^3 with a pinching condition and does not cover arbitrary closed Riemannian manifolds or all constants c .
- Title: Existence of multiple constant mean curvature hypersurfaces for varying Riemannian metrics
- Authors: Xiaoxiang Jiao and Wenduo Zou
- Venue: arXiv preprint, version 3
- Year: 2026
- DOI/arXiv/URL: DOI 10.48550/arXiv.2511.19102; arXiv:2511.19102; <https://arxiv.org/abs/2511.19102>
- Result: For a closed Riemannian manifold of dimension $3 \leq n+1 \leq 7$, if the number of closed c -CMC hypersurfaces in (M, g) is finite, then one can vary the metric to a metric h with strictly more c -CMC hypersurfaces while preserving the old ones, with a quantitative norm bound.
- Why not full resolution: The metric is changed from g to h ; the problem asks for the original fixed metric.

1.40.3.4. What remains unresolved

No citable result found proves that every fixed closed Riemannian manifold (M^{n+1}, g) , $3 \leq n+1 \leq 7$, has at least two distinct closed CMC hypersurfaces of every prescribed constant $c > 0$. The small- c and large- c regimes are covered by existing work, and generic/perturbative alternatives are known, but the arbitrary fixed metric and arbitrary intermediate prescribed mean curvature case remains open.

1.40.4. Remaining open problems recorded by the database

1.40.4.1. Open problem 1

Let g be an arbitrary Riemannian metric on S^3 . For every $c > 0$, are there at least two distinct immersions $S^2 \rightarrow (S^3, g)$ whose mean curvature is identically c ?

Reference supplied by the database. Liam Mazurowski and Xin Zhou, “Pairs of Embedded Spheres with Pinched Prescribed Mean Curvature,” 2025, Conjecture 1.6 (the modified

two-solution CMC-sphere conjecture, stated after the discussion of the counterexample to the embedded version), arXiv: [2511.08228](https://arxiv.org/abs/2511.08228).

1.40.4.2. Open problem 2

Let (M^{n+1}, g) , $3 \leq n + 1 \leq 5$, have a generic metric, and let $\{\Sigma_p\}$ be the sequence of geometrically distinct half-volume constant-mean-curvature hypersurfaces supplied by the half-volume min-max theory, with $\text{Area}(\Sigma_p) \rightarrow \infty$. Must their constant mean curvatures satisfy $H(\Sigma_p) \rightarrow 0$ as $p \rightarrow \infty$?

Reference supplied by the database. Liam Mazurowski and Xin Zhou, “An Alternative for Constant Mean Curvature Hypersurfaces,” *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* (to appear; accepted version available in 2026), §1.1, final paragraph (explicitly asking whether the mean curvatures of the half-volume sequence tend to zero), arXiv: [2408.13864](https://arxiv.org/abs/2408.13864). The sequence in the question is constructed in Liam Mazurowski and Xin Zhou, “Infinitely Many Half-Volume Constant Mean Curvature Hypersurfaces via Min-Max Theory,” 2024, arXiv: [2405.00595](https://arxiv.org/abs/2405.00595).

1.40.5. Primary-source pointer

The originating ICM record is *Mean curvature and variational theory* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=339>.

2. Metric and topological geometry

These records trace a broad arc from classical embedding and transformation questions to coarse geometry, the Ribe program, nonlinear dimension reduction, and the geometry of singular metric spaces. The central objects range from finite metric spaces and Banach spaces to Gromov–Hausdorff limits and large-scale assembly maps. The common theme is the extraction of topological, linear, or algorithmic structure from metric information, often with a sharp dependence on dimension or distortion as the unresolved point.

Coverage: 21 website records; **source years:** 1958–2018; **separately listed descendant open problems:** 19.

2.1. 41. MODERN DEVELOPMENT OF SURFACE THEORY [1958-geom-8]

Record: 1958-geom-8

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1958: MODERN DEVELOPMENT OF SURFACE THEORY

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=67>

2.1.1. Problem statement

Determine whether every abstract M.B.C. (manifold of bounded curvature) admits a local isometric realization in ordinary Euclidean space E^3 as a surface of bounded positive external curvature.

Here an abstract M.B.C. is a two-dimensional metric manifold with an intrinsic metric such that each point has a neighbourhood U for which the sums of the triangle excesses

$$\delta(T) = \bar{\alpha} + \bar{\beta} + \bar{\gamma} - \pi,$$

taken over arbitrary finite families of pairwise non-overlapping geodesic triangles $T \subset U$, are bounded above by a constant depending only on U ; $\bar{\alpha}, \bar{\beta}, \bar{\gamma}$ are the upper angles at the vertices, defined as the limiting upper values of the corresponding Euclidean comparison angles determined by the metric side lengths. A surface $S \subset E^3$ has bounded positive external curvature if, for every subdivision of S into pieces S_i , $\sigma^+(S_i)$ denotes the area on the unit sphere of the normal directions to supporting planes at the non-boundary points of S_i , and the sums $\sum_i \sigma^+(S_i)$ are bounded by a constant independent

of the subdivision.

2.1.2. Historical background, literature, and current progress

2.1.3. Verdict: partially resolved

2.1.3.1. Precise problem checked

The problem asks whether every abstract two-dimensional M.B.C. (manifold of bounded curvature, in Alexandrov's intrinsic sense of locally bounded sums of positive triangle excesses) admits local isometric realization in ordinary Euclidean 3-space as a surface of bounded positive external/extrinsic curvature. The target class is the class Alexandrov describes via uniformly bounded sums of spherical areas of supporting-plane normal directions over arbitrary subdivisions.

This is stronger than merely producing a Nash-Kuiper-type C^1 isometric embedding: Alexandrov explicitly wanted a "Gaussian" realization class where intrinsic and external curvature remain connected.

2.1.3.2. Current status

I found substantial partial progress, but no published or otherwise citable theorem that covers the full statement for all abstract two-dimensional M.B.C.s. The full local realization in E^3 by surfaces of bounded positive external curvature therefore remains unresolved.

2.1.3.3. Key literature and progress

Original source

- Title: "Modern development of surface theory"
- Author: A. D. Alexandrov
- Venue: Proceedings of the International Congress of Mathematicians, 14-21 August 1958, edited by J. A. Todd, Cambridge University Press
- Year: 1960
- DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf>
- Relevant content: Alexandrov defines M.B.C.s intrinsically using bounded sums of triangle excesses, then asks which Euclidean surfaces are M.B.C.s intrinsically and whether any M.B.C. can be embedded locally in E^3 . He says the desired embeddings are "Gaussian embeddings." He identifies surfaces of bounded positive external curvature as the expected general class, but states that at that time their curvature/Gauss theorem theory was not yet available and that general embedding results were known only for convex surfaces.

Convex and nonnegative-curvature cases

- Title: "Intrinsic Geometry of Convex Surfaces" / related Alexandrov convex-surface embedding theorem
- Author: A. D. Alexandrov; uniqueness developed by A. V. Pogorelov

- Venue: Russian original, Gostekhizdat; later references and modern accounts in Alexandrov geometry texts
- Year: 1948 and later
- DOI/arXiv/URL: modern accessible account: <https://anton-petrinin.github.io/alexandrov/alexandrov.pdf>
- Progress: The Alexandrov embedding theorem realizes geodesic CBB(0) metrics on S^2 as closed convex surfaces in E^3 , with uniqueness due to Pogorelov. This fully resolves important positive/nonnegative convex cases, but it does not cover arbitrary local M.B.C.s of mixed or negative curvature.

Foundations of M.B.C. theory

- Title: “Two-dimensional manifolds of bounded curvature (Foundations of the intrinsic geometry of surfaces)”
- Authors: A. D. Aleksandrov and V. A. Zalgaller
- Venue: Trudy Mat. Inst. Steklov 63; English translation as Intrinsic Geometry of Surfaces, Translations of Mathematical Monographs 15, AMS
- Year: 1962; English translation 1967
- DOI/arXiv/URL: <https://www.mathnet.ru/eng/tm1580>
- Progress: This is the foundational treatment of two-dimensional M.B.C.s. It develops the intrinsic class but does not supply the missing general local realization theorem in E^3 by surfaces of bounded positive external curvature.

General C^1 realization, but not in the bounded-positive-external-curvature class

- Title: “Realization of a metrized two-dimensional manifold by a surface in E^3 ”
- Author: Yu. D. Burago
- Venue: Dokl. Akad. Nauk SSSR 135:6, 1301-1302; English translation: Soviet Math. Dokl. 1, 1364-1365
- Year: 1960
- DOI/arXiv/URL: <https://www.mathnet.ru/eng/dan24409>
- Progress: Burago proves that if an orientable two-dimensional metric manifold admits proportional approximation by polyhedral metrics, then it is realized by a non-self-intersecting surface in E^3 with a tangent plane at every point. A corollary covers orientable M.B.C.s under extra exclusions such as no point of curvature 2π and suitable boundary/no boundary assumptions.
- Why this is not full resolution: The surface class in the theorem is a C^1 /tangent-plane class, not the class of surfaces of bounded positive external curvature. The proof is in the Nash-Kuiper flexibility direction and does not establish the Gaussian curvature connection Alexandrov required.

Bounded extrinsic/positive extrinsic curvature theory and structural results

- Title: “Surfaces of Bounded Extrinsic Curvature”
- Author: A. V. Pogorelov
- Venue: Kharkov University Press; later related material in Extrinsic Geometry of Convex Surfaces, AMS Translations of Mathematical Monographs 35
- Year: 1956; English book translation 1973
- DOI/arXiv/URL: Google Books record: <https://books.google.com/books>

/about/Extrinsic_Geometry_of_Convex_Surfaces.html?id=PI4mo2yX8HoC

- Progress: Pogorelov developed the theory of surfaces of bounded extrinsic curvature and Gauss-type relations in that setting. This is central background, but it treats a more regular/specific extrinsic class and does not prove that every abstract M.B.C. has the desired local realization.
- Title: “ C^1 -smooth isometric imbeddings”
- Author: S. Z. Shefel’
- Venue: Sibirsk. Mat. Zh. 15:6, 1372-1393; English translation in Siberian Mathematical Journal
- Year: 1974
- DOI/arXiv/URL: <https://doi.org/10.1007/BF00966565>; <https://link.springer.com/article/10.1007/BF00966565>
- Progress: Gives structural theorems for C^1 isometric immersions under sign and affine-stability hypotheses. In particular, C^1, α immersions with $\alpha > 2/3$ of metrics in the classes $A+$, $A0$, $A-$ have corresponding normal nonnegative/developable/saddle structure and bounded extrinsic curvature.
- Why this is partial: It assumes regularity/sign/stability hypotheses and is not an existence theorem for arbitrary M.B.C.s of mixed curvature.
- Title: “ C^1 -smooth surfaces of bounded positive extrinsic curvature”
- Author: S. Z. Shefel’
- Venue: Sibirsk. Mat. Zh. 16:5, 1122-1123; English translation: Siberian Math. J. 16:5, 863-864
- Year: 1975
- DOI/arXiv/URL: <https://doi.org/10.1007/BF00967119>; <https://www.mathnet.ru/eng/smj4203>
- Progress: Studies C^1 surfaces of bounded positive extrinsic curvature and proves that, if such a surface is an isometric immersion of an M.B.C. of class $A+$, $A0$, or $A-$, then it has the corresponding normal nonnegative-curvature, developable, or saddle extrinsic structure.
- Why this is partial: The theorem assumes the desired bounded-positive-extrinsic-curvature surface already exists and assumes intrinsic sign classes. It does not construct such a realization for every abstract M.B.C.
- Title: “On the extrinsic curvature and the extrinsic structure of normal developable C^1 surfaces”
- Author: I. Kh. Sabitov
- Venue: Mathematical Notes 87, 874-879
- Year: 2010
- DOI/arXiv/URL: <https://doi.org/10.1134/S0001434610050275>; <https://link.springer.com/article/10.1134/S0001434610050275>
- Progress: Modern continuation of the low-regularity extrinsic-curvature structural theory, proving that normal C^1 developable surfaces in Shefel’s sense have zero extrinsic

curvature in Pogorelov's sense.

- Why this is partial: It addresses a structural special case, not the general local realization of M.B.C.s.

Later surveys/background

- Title: "The Geometry of Surfaces in Euclidean Spaces"
- Authors: Yu. D. Burago and S. Z. Shefel'
- Venue: Geometry III, Encyclopaedia of Mathematical Sciences 48, Springer, pp. 1-85; Russian original in Itogi Nauki i Tekhniki, Sovremennye Problemy Matematiki, Fundamental'nye Napravleniya 48
- Year: 1992 English version; 1989 Russian version
- DOI/arXiv/URL: https://link.springer.com/chapter/10.1007/978-3-662-02751-6_1
- Progress: Survey of the connection between metric classes and surface classes in Euclidean spaces, including bounded extrinsic curvature, C^1 isometric immersions, convex/saddle/developable surfaces, and open directions. It does not give a theorem resolving the full Alexandrov expectation for arbitrary M.B.C.s.
- Title: "Two-Dimensional Manifolds of Bounded Curvature"
- Author: Yu. G. Reshetnyak
- Venue: Geometry IV, Encyclopaedia of Mathematical Sciences 70, Springer, pp. 3-163
- Year: 1993
- DOI/arXiv/URL: https://doi.org/10.1007/978-3-662-02897-1_1; https://link.springer.com/chapter/10.1007/978-3-662-02897-1_1
- Progress: A major survey of the intrinsic M.B.C. theory. Its bibliography records Alexandrov-Zalgaller, Burago, Shefel, Pogorelov, and related work, but not a full local E^3 realization theorem in the bounded-positive-external-curvature class.

Smooth local embedding literature: evidence of only special-case progress

- Title: "The local isometric embedding in R^3 of two-dimensional Riemannian manifolds with Gaussian curvature changing sign to finite order on a curve"
- Author: Marcus A. Khuri
- Venue: Journal of Differential Geometry 76, 249-291
- Year: 2007
- DOI/arXiv/URL: https://www.math.stonybrook.edu/~khuri/Khuri_JDG.pdf
- Progress: Khuri states that the local realization problem for two-dimensional Riemannian manifolds in R^3 "has only been partially answered" and proves existence under a finite-order sign-changing-curve hypothesis.
- Why this matters: Smooth Riemannian metrics are a subclass of M.B.C.s. The smooth literature still treats nonzero curvature, sign, finite-order vanishing, or other hypotheses rather than a general all-cases theorem. This supports the conclusion that no stronger published resolution of Alexandrov's arbitrary M.B.C. realization problem is available.

2.1.3.4. What remains unresolved

No located theorem proves that every abstract two-dimensional M.B.C., with no sign, smoothness, orientability/boundary exclusions, or curvature-zero exclusions beyond the definition itself, is locally isometric to a surface in E^3 whose positive external curvature is bounded in Alexandrov's supporting-plane normal-image sense.

The closest positive results either:

- realize only convex/nonnegative-curvature global cases;
- produce a C^1 /tangent-plane realization without bounded positive external curvature;
- assume the target surface already has bounded positive extrinsic curvature and then prove structure;
- impose sign, Holder regularity, affine-stability, finite-order vanishing, or other additional hypotheses.

Therefore the problem should be classified as partially resolved, not fully resolved and not fully open.

2.1.4. Remaining open problems recorded by the database

2.1.4.1. Open problem 1

Does every smooth two-dimensional Riemannian manifold admit a smooth local isometric embedding into $\mathbb{R}^3$? Equivalently, are there any smooth (or sufficiently smooth) counterexamples to the local isometric-embedding problem for surfaces?

Reference supplied by the database. Qing Han and Marcus Khuri, "On the local isometric embedding in $\mathbb{R}^3$ of surfaces with Gaussian curvature of mixed sign," *Communications in Analysis and Geometry* **18** (2010), no. 4, 649–704, opening question and Remark 1.1 (explicitly stating that the existence of smooth counterexamples remained open), DOI: [10.4310/CAG.2010.v18.n4.a2](https://doi.org/10.4310/CAG.2010.v18.n4.a2). The paper proves only the finite-order, transverse-zero-set case; subsequent special-case progress includes Siyuan Lu, "On the local isometric embedding in $\mathbb{R}^3$ of surfaces with zero sets of Gaussian curvature forming cusp domains," *Communications in Partial Differential Equations* **41** (2016), no. 10, 1541–1570, DOI: [10.1080/03605302.2016.1227336](https://doi.org/10.1080/03605302.2016.1227336), and does not settle the general smooth question.

2.1.5. Primary-source pointer

The originating ICM record is *MODERN DEVELOPMENT OF SURFACE THEORY* (1958). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=67>.

2.2. 42. Spaces with Curvature Bounded Below [1994-geom-61]

Record: 1994-geom-61

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Spaces with Curvature Bounded Below

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=589>

2.2.1. Problem statement

Formulate an Alexandrov-space analogue of the rational ellipticity conjecture for complete, locally compact, finite-dimensional Alexandrov spaces M with curvature bounded below by 0. The formulation should generalize the conjecture that every compact simply connected Riemannian manifold with nonnegative sectional curvature is rationally elliptic, meaning that the rational Betti numbers of its loop space grow at most polynomially, and it should incorporate not only the underlying topology of M but also the extremal subsets of M .

Here an extremal subset of an Alexandrov space M is a closed subset $F \subset M$ such that, for every distance function $f = \text{dist}_p$ with $p \notin F$, if $q \in F$ realizes the distance from p to F , then

$$\sup_{\xi \in \Sigma_q} f'_q(\xi) \leq 0,$$

where Σ_q is the space of directions at q . The formulation should also account for conjectures about positively curved manifolds, since spherical suspensions and joins of spaces with curvature at least 1 are Alexandrov spaces with curvature at least 1.

2.2.2. Historical background, literature, and current progress

2.2.3. Verdict: partially resolved

2.2.3.1. Precise problem checked

The statement asks for Perelman's speculative Alexandrov-space analogue of the rational ellipticity, or Bott-Grove-Halperin, conjecture. In its classical manifold form, the conjecture says that every compact simply connected Riemannian manifold with nonnegative sectional curvature is rationally elliptic, equivalently that the rational Betti numbers of its based loop space grow at most polynomially. The problem asks for an extension to complete, locally compact, finite-dimensional Alexandrov spaces with curvature bounded below by 0, and further says that the correct formulation should incorporate extremal subsets, not just the topology of the underlying space.

Because every compact nonnegatively curved Riemannian manifold is an Alexandrov space with curvature bounded below by 0, any theorem resolving the full Alexandrov statement would in particular resolve the classical Bott-Grove-Halperin conjecture for manifolds. The literature I found does not do this.

2.2.3.2. Original and status-setting sources

- Title:** Spaces with Curvature Bounded Below
Author: G. Perelman
Venue: Proceedings of the International Congress of Mathematicians, Zurich 1994, Birkhauser, Basel, pp. 517-525
Year: 1995
DOI/URL: DOI 10.1007/978-3-0348-9078-6_45, https://link.springer.com/chapter/10.1007/978-3-0348-9078-6_45
Relevance: This is the Perelman ICM source for Alexandrov spaces with curvature bounded below, including the extremal-subset viewpoint. It is the source context for the stated speculative extension, not a proof of rational ellipticity.
- Title:** Open Problems in Non-negative Sectional Curvature
Author/Compiler: Martin Kerin, compiled from the 2007 AIM workshop on nonnegative sectional curvature
Venue: American Institute of Mathematics workshop problem list
Year: 2007
URL: <https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf>
Relevance: Problem 12 explicitly asks for an Alexandrov analogue of rational ellipticity and, in particular, asks whether almost non-negatively curved manifolds are rationally elliptic. This is direct evidence that the Alexandrov analogue was open after Perelman's proposal.

2.2.3.3. Partial progress

- Title:** Nilpotency, almost nonnegative curvature, and the gradient flow on Alexandrov spaces
Authors: Vitali Kapovitch, Anton Petrunin, Wilderich Tuschmann
Venue: Annals of Mathematics, 171(1), 343-373
Year: 2010
DOI/URL: DOI 10.4007/annals.2010.171.343, <https://annals.math.princeton.edu/2010/171-1/p06>
Progress: The paper proves strong structural results for almost nonnegatively curved manifolds, including virtual nilpotency of fundamental groups and a fibration theorem. It formulates a rational-ellipticity conjecture for closed almost nonnegatively curved manifolds and reduces part of the problem to the simply connected case. This is substantial progress around the conjecture, but it does not prove rational ellipticity for all nonnegatively curved manifolds or any full Alexandrov/extremal-subset analogue.
- Title:** Almost non-negative curvature and rational ellipticity in cohomogeneity two
Authors: Karsten Grove, Burkhard Wilking, Joseph Yeager; appendix by Steve Halperin
Venue: Annales de l'Institut Fourier, 69(7), 2921-2939
Year: 2019
DOI/URL: DOI 10.5802/aif.3304, https://www.numdam.org/article/AIF_2019__69_7_2921_0/
Progress: The main theorem proves that simply connected almost non-negatively curved manifolds with an isometric action of cohomogeneity at most two are rationally elliptic. This verifies an important symmetry-restricted special case of the manifold conjecture, not the full manifold conjecture and not the full Alexandrov/extremal-subset problem.

- **Title:** Nonnegatively curved quotient spaces with boundary
Author: Wolfgang Spindeler
Venue: Manuscripta Mathematica, 153, 55-67
Year: 2017
DOI/arXiv/URL: DOI 10.1007/s00229-016-0875-y, arXiv 1510.01908, <https://arxiv.org/abs/1510.01908>
Progress: The paper proves that simply connected nonnegatively curved torus manifolds with nonempty boundary quotient are rationally elliptic and obtains related classification results. This is another special case under a strong torus-action hypothesis.
- **Title:** Torus manifolds and non-negative curvature
Author: Michael Wiemeler
Venue: Journal of the London Mathematical Society, 91(3), 667-692
Year: 2015
DOI/arXiv/URL: DOI 10.1112/jlms/jdv017, arXiv 1401.0403, <https://arxiv.org/abs/1401.0403>
Progress: This gives structure and rational-homotopy information for nonnegatively curved torus manifolds. It supports rational ellipticity in a highly symmetric manifold setting, but does not cover general nonnegatively curved manifolds or Alexandrov spaces with extremal subsets.
- **Title:** Rational ellipticity of G-manifolds from their quotients
Authors: Elahe Khalili Samani, Marco Radeschi
Venue: Compositio Mathematica, 161(2), 244-256
Year: 2025
DOI/arXiv/URL: DOI 10.1112/S0010437X24007656, arXiv 2312.08202, <https://arxiv.org/abs/2312.08202>
Progress: The paper proves rational ellipticity criteria for certain compact simply connected smooth manifolds from the topology of their metric quotients under isometric group actions, including a cohomogeneity-three infinitesimally polar case. Although metric quotients are Alexandrov spaces when the original manifold has lower curvature bounds, this is a quotient/submetry theorem for special group actions, not a full rational-ellipticity theorem for arbitrary Alexandrov spaces or arbitrary extremal-subset data.

2.2.3.4. What remains unresolved

No citable theorem found proves that every complete, locally compact, finite-dimensional Alexandrov space with curvature bounded below by 0 satisfies the proposed rational-ellipticity analogue, especially not in a formulation incorporating all extremal subsets. In fact, the full Alexandrov statement would imply the classical Bott-Grove-Halperin conjecture for compact simply connected nonnegatively curved Riemannian manifolds, and the cited recent literature continues to treat that manifold conjecture as open except in special cases.

Therefore the exact problem considered here is not fully resolved. The correct status is **partially resolved**: there is substantial progress in special cases and related structural results, but the full Alexandrov/extremal-subset analogue remains open.

2.2.4. Remaining open problems recorded by the database

2.2.4.1. Open problem 1

Is every closed simply connected almost nonnegatively curved manifold rationally elliptic? Here “almost nonnegatively curved” means that the manifold admits metrics g_i with $\sec(M, g_i) > -1/i$ and $\text{diam}(M, g_i) \leq 1/i$.

Reference supplied by the database. Vitali Kapovitch, Anton Petrunin, and Wilderich Tuschmann, “Nilpotency, Almost Nonnegative Curvature, and the Gradient Flow on Alexandrov Spaces,” *Annals of Mathematics* (2) **171** (2010), no. 1, 343–373, §6.3, Conjecture 6.3.1 (reduced there to the simply connected case), DOI: 10.4007/annals.2010.171.343, arXiv:math/0506273, <https://doi.org/10.4007/annals.2010.171.343>. See also Martin Kerin (compiler), “Open Problems in Non-negative Sectional Curvature,” American Institute of Mathematics workshop problem list, 2007, Problem 12, <https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf>.

2.2.5. Primary-source pointer

The originating ICM record is *Spaces with Curvature Bounded Below* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=589>.

2.3. 43. Dimension Theory and Large Riemannian Manifolds [1998-geom-65]

Record: 1998-geom-65

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Dimension Theory and Large Riemannian Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=425>

2.3.1. Problem statement

Prove or disprove that for every metric space X that is uniformly contractible and has bounded geometry, the coarse Baum-Connes assembly map

$$A : K_*(X) \rightarrow K_*(C^*(X))$$

is a monomorphism. Determine whether, in the stronger isomorphism version of the conjecture, this assembly map is an isomorphism.

Here X is uniformly contractible if for every $R > 0$ there is $S > 0$ such that every R -ball $B_R(x)$ centered at x can be contracted to a point inside $B_S(x)$, for every $x \in X$. The space X has bounded geometry if for every $\epsilon > 0$ and every $R > 0$ there is an integer m such that every R -ball in X contains an ϵ -net with fewer than m points. The notation $C^*(X)$ denotes the C^* -algebra associated to X in the coarse Baum-Connes assembly construction, and $K_*(C^*(X))$ denotes its K -theory.

2.3.2. Historical background, literature, and current progress

2.3.3. Verdict: partially resolved

2.3.3.1. Precise problem checked

The problem asks whether, for every uniformly contractible bounded-geometry metric space X , the coarse Baum-Connes assembly map

$$A : K_*(X) \rightarrow K_*(C^*(X))$$

is injective, and whether the stronger isomorphism form is true. In modern terminology, the injective form is the coarse Novikov conjecture for this class, while the isomorphism form is the coarse Baum-Connes conjecture for this class. The uniform contractibility and bounded-geometry hypotheses are part of the exact question.

2.3.3.2. Current status

I found no published or otherwise citable theorem that proves injectivity for all uniformly contractible bounded-geometry metric spaces, nor a citable counterexample satisfying both uni-

form contractibility and bounded geometry that disproves the injective statement. Therefore the exact problem is not fully resolved.

There is substantial partial progress. The assembly map is an isomorphism for many important subclasses, including finite-asymptotic-dimension spaces, spaces uniformly/coarsely embeddable into Hilbert space, Busemann non-positively curved spaces, coarsely convex spaces, and certain relative expanders. There are also counterexamples to the broader coarse Baum-Connes isomorphism conjecture for general bounded-geometry spaces, especially expanders and related warped-cone/asymptotic-expander constructions, but those do not prove failure of the exact uniformly contractible bounded-geometry injectivity statement in this problem.

2.3.3.3. Key literature and relevance

- **A. N. Dranishnikov**, “Dimension theory and large Riemannian manifolds”, *Documenta Mathematica*, Extra Volume ICM II, 1998, pp. 423-432. URL: <https://eudml.org/doc/231645> and <https://www2.math.ethz.ch/EMIS/journals/DMJDMV/xvol-icm/06/Dranishnikov.MAN.ps.gz>.
This is the dataset source. It states the coarse Baum-Connes/Novikov problem for uniformly contractible bounded-geometry metric spaces and notes the role of the bounded-geometry hypothesis.
- **John Roe**, “Coarse cohomology and index theory on complete Riemannian manifolds”, *Memoirs of the American Mathematical Society* 104, no. 497, 1993. URL: <https://doi.org/10.1090/memo/0497>.
Foundational source for coarse cohomology, Roe algebras, and the coarse assembly framework on complete Riemannian manifolds.
- **Nigel Higson and John Roe**, “On the coarse Baum-Connes conjecture”, in *Novikov Conjectures, Index Theorems and Rigidity*, Vol. 2, London Mathematical Society Lecture Note Series 227, Cambridge University Press, 1995, pp. 227-254.
Foundational formulation and analysis of the coarse Baum-Connes assembly map.
- **Guoliang Yu**, “The Novikov conjecture for groups with finite asymptotic dimension”, *Annals of Mathematics* (2) 147, no. 2, 1998, pp. 325-355. DOI: 10.2307/121011.
Proves the relevant Novikov/coarse injectivity consequences for finite-asymptotic-dimension settings. This is a major positive special case, but finite asymptotic dimension is an extra hypothesis not present in the problem statement.
- **Guoliang Yu**, “The coarse Baum-Connes conjecture for spaces which admit a uniform embedding into Hilbert space”, *Inventiones Mathematicae* 139, 2000, pp. 201-240. DOI: 10.1007/s002229900032.
Proves the coarse Baum-Connes assembly map is an isomorphism for bounded-geometry spaces admitting a uniform embedding into Hilbert space. This covers many uniformly contractible bounded-geometry examples but not all such spaces.
- **Gennadi Kasparov and Guoliang Yu**, “The coarse geometric Novikov conjecture and uniform convexity”, *Advances in Mathematics* 206, no. 1, 2006, pp. 1-56; arXiv:math/0507599.
Proves injectivity for bounded-geometry spaces uniformly embeddable into uniformly convex Banach spaces. Again, this is a strong special case with an additional embedding hypothesis.
- **Georges Skandalis, Jean-Louis Tu, and Guoliang Yu**, “The coarse Baum-Connes conjecture and groupoids”, *Topology* 41, no. 4, 2002, pp. 807-834. DOI: 10.1016/S0040-

9383(01)00004-0.

Identifies the coarse assembly map for bounded-geometry discrete metric spaces with a groupoid Baum-Connes map and gives another route to the Hilbert-space embedding theorem. It does not settle the full uniformly contractible bounded-geometry class.

- **Nigel Higson, Vincent Lafforgue, and Georges Skandalis, “Counterexamples to the Baum-Connes conjecture”, *Geometric and Functional Analysis* 12, 2002, pp. 330-354. DOI: 10.1007/s00039-002-8249-5.**
Gives counterexamples related to expanders and shows failure of surjectivity for coarse Baum-Connes maps in bounded-geometry settings. These are counterexamples to broader isomorphism formulations, not a resolution of the stated uniform-contractibility injectivity problem.
- **Rufus Willett and Guoliang Yu, “Higher index theory for certain expanders and Gromov monster groups I”, *Advances in Mathematics* 229, no. 3, 2012, pp. 1380-1416. DOI: 10.1016/j.aim.2011.10.024.**
Shows that for expander sequences with girth tending to infinity, the coarse Baum-Connes assembly map is injective but not surjective. This is important negative progress for the isomorphism form in general bounded-geometry spaces, but the expander coarse disjoint unions are not the uniformly contractible spaces of the problem.
- **Tomohiro Fukaya and Shin-ichi Oguni, “The coarse Baum-Connes conjecture for Busemann non-positively curved spaces”, *Kyoto Journal of Mathematics* 56, no. 1, 2016, pp. 1-12. DOI: 10.1215/21562261-3445129; arXiv:1304.3224.**
Proves isomorphism for proper Busemann non-positively curved spaces. This is a substantial positive subclass, not the full class.
- **Tomohiro Fukaya and Shin-ichi Oguni, “A coarse Cartan-Hadamard theorem with application to the coarse Baum-Connes conjecture”, *Journal of Topology and Analysis* 12, no. 3, 2020, pp. 857-895. DOI: 10.1142/S1793525319500675; arXiv:1705.05588.**
Proves the conjecture for proper coarsely convex spaces, including further geometric examples such as systolic complexes. This is still an extra coarse convexity hypothesis.
- **Jintao Deng, Qin Wang, and Guoliang Yu, “The coarse Baum-Connes conjecture for certain relative expanders”, *Advances in Mathematics* 424, 2023, Article 109047; arXiv:2102.10617.**
Proves the conjecture for a class of relative expanders and special box spaces that need not coarsely embed into Hilbert space. This enlarges the positive known class but remains a special construction.
- **Christos Kitsios, Thomas Schick, and Federico Vigolo, “Coarse Baum-Connes and warped cones: failure of surjectivity in odd degree”, arXiv:2504.21811, 2025. URL: <https://arxiv.org/abs/2504.21811>.**
Gives recent warped-cone counterexamples to surjectivity of the coarse Baum-Connes map. This further shows that broad isomorphism formulations fail, but it does not supply a full resolution of the stated injectivity claim for all uniformly contractible bounded-geometry spaces.
- **Jianguo Zhang, “On the quantitative coarse Baum-Connes conjecture with coefficients”, *Science China Mathematics* 69, 2026, pp. 1537-1564. DOI: 10.1007/s11425-024-2484-5.**
Recent work showing active development of quantitative/coefficient variants and listing the current landscape of positive classes and counterexamples. It does not resolve the exact question.

2.3.3.4. What remains unresolved

The unresolved core is the all-cases injectivity statement:

For every uniformly contractible bounded-geometry metric space X , $A : K_*(X) \rightarrow K_*(C^*(X))$ is injective.

The stronger isomorphism statement is known in many large subclasses and false in broader bounded-geometry settings, but I did not find a theorem whose hypotheses exactly include all uniformly contractible bounded-geometry spaces and whose conclusion settles either injectivity or isomorphism in all cases. Thus the problem should be classified as **partially resolved**, not fully resolved and not fully open.

2.3.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

2.3.5. Primary-source pointer

The originating ICM record is *Dimension Theory and Large Riemannian Manifolds* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=425>.

2.4. 44. Basis Problems in Combinatorial Set Theory [1998-geom-72]

Record: 1998-geom-72

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Basis Problems in Combinatorial Set Theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=45>

2.4.1. Problem statement

Prove or disprove that for every compact topological space X that does not contain a subspace homeomorphic to the discrete topological space on the first uncountable ordinal ω_1 , there exist a compact metric space M and a continuous surjection $f : X \rightarrow M$ such that $|f^{-1}(m)| \leq 2$ for every $m \in M$.

2.4.2. Historical background, literature, and current progress

2.4.3. Verdict: partially resolved

2.4.3.1. Precise problem

The problem asks whether every compact space K with no subspace homeomorphic to the discrete space $D(\omega_1)$ admits a continuous surjection $f : K \rightarrow M$ onto a compact metric space such that every fiber has size at most 2. Equivalently, must every compactum of countable spread be a premetric compactum of degree at most 2?

The original ICM source phrases this as Todorćević's Conjecture 6 with a forcing-axiom tag: under PFA, every compact space not containing $D(\omega_1)$ admits an at most 2-to-1 map onto a compact metric space. the problem statement omits the PFA tag, but the literature treats the general compact-space problem as a Fremlin/Todorćević topological basis problem rather than as a settled ZFC theorem.

2.4.3.2. Current status

I did not find a published or otherwise citable result proving the full statement for all compact spaces. Later sources continue to state the full compact-space version as an open problem, while proving substantial special cases. Therefore the problem should be treated as partially resolved, not fully resolved.

2.4.3.3. Key literature and progress

1. Basis Problems in Combinatorial Set Theory

Authors: Stevo Todorćević.

Venue: Documenta Mathematica, Extra Volume ICM 1998, Vol. II, pp. 43-52, 1998.

DOI/arXiv/URL: <https://emis.de/ft/10334> ; also in ICM 1998 proceedings.

Result/status: Conjecture 6 states, with the tag [PFA], that every compact space not containing $D(\omega_1)$ admits an at most 2-to-1 continuous map onto a compact metric space. The article says this is related to Fremlin's problem and then explicitly restricts to test cases because the needed Ramsey-theoretic principles were not available. It records a special-case theorem for pointwise compact subsets of the first Baire class, not a proof for arbitrary compact spaces.

2. Compact subsets of the first Baire class

Author: Stevo Todorćević.

Venue: Journal of the American Mathematical Society 12, no. 4, pp. 1179-1212, 1999.

DOI/arXiv/URL: DOI 10.1090/S0894-0347-99-00312-4; <https://www.ams.org/jams/1999-12-04/S0894-0347-99-00312-4/>

Result/status: Proves the relevant 2-to-1 conclusion for compact subsets of Baire class 1 functions on a Polish space, often called Rosenthal compacta: if such a compactum contains no uncountable discrete subspace, then it admits an at most 2-to-1 map onto a compact metric space. This verifies the conjecture for an important concrete subclass, but it assumes the compact space is Rosenthal/Baire-1 and so does not cover arbitrary compact spaces.

3. The Proper Forcing Axiom

Author: Justin Tatch Moore.

Venue: Proceedings of the International Congress of Mathematicians 2010, Vol. II, pp. 3-29, 2010.

DOI/arXiv/URL: <https://pi.math.cornell.edu/~justin/Ftp/ICM.pdf>

Result/status: Theorem 8.7 summarizes Todorćević's Baire-1/Rosenthal compacta result, including the at most 2-to-1 conclusion when there is no uncountable discrete subspace. Immediately afterward, Problem 8.9 asks whether, assuming PFA, every compact K with no uncountable discrete subspace admits an at most 2-to-1 map onto a metric space. This confirms that the full PFA compact-space question was still open in this survey.

4. Perfect compacta and basis problems in topology

Authors: Gary Gruenhage and Justin Tatch Moore.

Venue: Open Problems in Topology II, Elsevier, 2007, pp. 151-157; preprint dated 2005.

DOI/arXiv/URL: <https://pi.math.cornell.edu/~justin/Ftp/basisprob.pdf>

Result/status: Discusses Fremlin's central question asking whether it is consistent that every perfect compactum admits a continuous at most two-to-one map onto a metric space, and calls such spaces premetric of order 2. It records partial results: under Martin's Maximum, Fremlin obtained a map with fibers of size at most 2 on a comeager subset of the range; under PFA/OCA there are positive results for cometrizable spaces; and Todorćević proved the Rosenthal compacta case. These results are explicitly presented as partial approaches, and the paper notes that reductions such as the zero-dimensional case were still unknown.

5. Analysis of a topological basis problem

Authors: Yinhe Peng and Stevo Todorćević.

Venue: Acta Mathematica Hungarica 167, pp. 419-475, 2022.

DOI/arXiv/URL: DOI 10.1007/s10474-022-01253-y; <https://link.springer.com/article/10.1007/s10474-022-01253-y>

Result/status: Gives recent PFA analysis of the associated topological basis problem via inner/outer topologies and finite-to-one perfect maps. In accompanying 2022 slides, the Fremlin question “Is every compact space in $D(\aleph_1)^\perp$ an at most 2-to-1 continuous preimage of a compact metric space?” is still displayed as a question, while the Baire-class-one theorem is listed as a special-case theorem. The 2022 results provide structural progress but do not prove the full compact-space statement.

6. One-dimensional definable topological spaces in o-minimal structures

Authors: Chris Miller and Sergei Starchenko.

Venue: arXiv preprint, 2023.

DOI/arXiv/URL: arXiv:2310.04510; <https://arxiv.org/abs/2310.04510>

Result/status: Section 9.1 reviews Fremlin’s conjecture for perfectly normal compact Hausdorff spaces and states that it asks for consistency of every such space admitting an at most 2-to-1 map onto a compact metric space. The paper proves definable/o-minimal one-dimensional analogues, not the unrestricted compact-space conjecture.

2.4.3.4. What remains unresolved

No cited theorem covers all compact spaces K with no copy of $D(\omega_1)$. The existing positive results require extra hypotheses such as being a Rosenthal compactum/compact subset of Baire class 1 functions, metrizable, monotonically normal, definable in an o-minimal setting, or they provide only weaker fiber conclusions on a comeager subset. Thus the full statement in the problem statement remains unresolved, with substantial partial progress.

2.4.4. Remaining open problems recorded by the database

2.4.4.1. Open problem 1

Does every perfectly normal compact Hausdorff space admit a continuous map into a metric space all of whose fibers are metrizable?

Reference supplied by the database. Gary Gruenhage and Justin Tatch Moore, “Perfect Compacta and Basis Problems in Topology,” in Elliott Pearl (ed.), *Open Problems in Topology II*, Elsevier, 2007, pp. 151–157, Question 7, preprint dated June 16, 2006, <https://pi.math.cornell.edu/~justin/Ftp/basisprob.pdf>.

2.4.4.2. Open problem 2

If a perfectly normal compact Hausdorff space admits a continuous map into a metric space with metrizable fibers, must it admit an at-most-two-to-one continuous map into a metric space?

Reference supplied by the database. Gary Gruenhage and Justin Tatch Moore, “Perfect Compacta and Basis Problems in Topology,” in Elliott Pearl (ed.), *Open Problems in Topology II*, Elsevier, 2007, pp. 151–157, Question 8, preprint dated June 16, 2006, <https://pi.math.cornell.edu/~justin/Ftp/basisprob.pdf>.

2.4.5. Primary-source pointer

The originating ICM record is *Basis Problems in Combinatorial Set Theory* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=45>.

2.5. 45. Finite Metric Spaces - Combinatorics, Geometry and Algorithms [2002-geom-79]

Record: 2002-geom-79

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Finite Metric Spaces - Combinatorics, Geometry and Algorithms

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>

2.5.1. Problem statement

Prove that there exists a positive constant C such that for every finite graph G , equipped with its shortest-path graph metric, with girth g and minimum degree at least 3, the least distortion $c_2(G)$ of an embedding of this metric into a Euclidean ℓ_2 space satisfies

$$c_2(G) \geq Cg.$$

2.5.2. Historical background, literature, and current progress

2.5.3. Verdict: partially resolved

2.5.3.1. Precise problem

The problem asks whether there is a universal constant $C > 0$ such that every finite graph G , equipped with its shortest-path metric, whose vertices all have degree at least 3 and whose girth is g , satisfies

$$c_2(G) \geq Cg,$$

where $c_2(G)$ is the least bilipschitz distortion of an embedding of the graph metric into Hilbert/Euclidean ℓ_2 space.

2.5.3.2. Original source

Title: The geometry of graphs and some of its algorithmic applications

Authors: Nathan Linial, Eran London, Yuri Rabinovich

Venue/year: Combinatorica 15(2), 215-245, 1995; preliminary FOCS 1994 version

DOI/URL: <https://doi.org/10.1007/BF01200757>; <https://www.cs.huji.ac.il/~nati/PAPERS/llr.pdf>

Status contribution: Problem 8.7 asks exactly whether, if all vertices in G have degree at least 3, every embedding of G into Euclidean space has distortion $\Omega(\text{girth}(G))$.

2.5.3.3. Known partial progress

Title: Girth and Euclidean distortion

Authors: Nathan Linial, Avner Magen, Assaf Naor

Venue/year: Geometric and Functional Analysis 12(2), 380-394, 2002; STOC 2002 version

DOI/arXiv/URL: <https://doi.org/10.1007/s00039-002-8251-y>; STOC DOI <https://doi.org/10.1145/509907.510009>; https://www.cs.huji.ac.il/~nati/PAPERS/euc_girth.pdf

Progress: Theorem 1.3 proves that for every k -regular graph G , $k > 2$, with girth g , one has $c_2(G) \geq C\sqrt{g}$. The paper explicitly describes this as only a partial proof of the Linial-London-Rabinovich conjecture and states that the original $\Omega(g)$ conjecture remains open. This covers the regular subcase only and gives $\sqrt{g}$, not g .

Title: On metric Ramsey-type phenomena

Authors: Yair Bartal, Nathan Linial, Manor Mendel, Assaf Naor

Venue/year: Annals of Mathematics 162(2), 643-709, 2005

DOI/arXiv/URL: <https://doi.org/10.4007/annals.2005.162.643>; arXiv:math/0406353; <https://annals.math.princeton.edu/2005/162-2/p02>

Progress: Theorem 6.1 strengthens the $\sqrt{g}$ lower bound beyond regular graphs: if G has girth g and average degree δ , then for the full metric $c_2(G) \geq ((\delta - 2)/\delta)\sqrt{\lfloor g/2 \rfloor} - 1$. Since minimum degree at least 3 implies average degree $\delta \geq 3$, this gives $c_2(G) = \Omega(\sqrt{g})$ for the full class. It still falls short of the conjectured linear $\Omega(g)$ bound.

Title: Finite metric spaces - combinatorics, geometry and algorithms

Author: Nathan Linial

Venue/year: Proceedings of the International Congress of Mathematicians, Vol. III, Beijing 2002, Higher Ed. Press, 573-586, 2002

arXiv/URL: arXiv:math/0304466; <https://arxiv.org/abs/math/0304466>

Status contribution: Section 3.3 restates the problem for graphs of girth g with all degrees at least 3, records the Linial-Magen-Naor $\Omega(\sqrt{g})$ theorem, and lists the state of knowledge as between $\Omega(\sqrt{g})$ and $O(g)$.

Title: An introduction to the Ribe program

Author: Assaf Naor

Venue/year: Japanese Journal of Mathematics 7(2), 167-233, 2012

DOI/arXiv/URL: <https://doi.org/10.1007/s11537-012-1222-7>; arXiv:1205.5993; <https://arxiv.org/abs/1205.5993>

Status contribution: Section 9.4 records the average-degree $\Omega(\sqrt{g})$ lemma and then asks, even for finite 3-regular connected graphs of girth g , what the growth rate of $c_2(g)$ is and in particular whether it grows asymptotically faster than $\sqrt{g}$. Since 3-regular graphs are a subclass of the stated graphs, this later survey confirms that the Euclidean problem is not known even beyond the $\sqrt{g}$ scale, let alone at the conjectured linear scale.

2.5.3.4. Nearby but non-resolving work

Title: Low-distortion embeddings of graphs with large girth

Author: Mikhail I. Ostrovskii

Venue/year: Journal of Functional Analysis 262(8), 3548-3555, 2012

DOI/arXiv/URL: <https://doi.org/10.1016/j.jfa.2012.01.020> ;
arXiv:1108.2542 ; <https://arxiv.org/abs/1108.2542>

Relevance: This constructs bounded-degree graphs with growing girth and uniformly bounded ℓ_1 distortion, resolving a related ℓ_1 question. It does not prove or disprove the stated bilipschitz ℓ_2 lower bound $c_2(G) \geq \Omega(g)$; coarse embeddability or ℓ_1 -embeddability is not the same as uniformly bounded Euclidean distortion.

2.5.3.5. What remains unresolved

I found no published or citable theorem proving $c_2(G) \geq Cg$ for all finite graphs of girth g and minimum degree at least 3, nor a citable counterexample settling the exact conjecture negatively. The best directly applicable lower bound located is $\Omega(\sqrt{g})$, including the full minimum-degree-at-least-3 class via the average-degree theorem of Bartal-Linial-Mendel-Naor. Thus the problem is partially resolved, not fully resolved.

2.5.4. Remaining open problems recorded by the database

2.5.4.1. Open problem 1

For each g , let $c_2^{(3)}(g)$ be the infimum of the least Euclidean distortions $c_2(G)$ over all finite connected 3-regular graphs G of girth g . Does

$$\frac{c_2^{(3)}(g)}{\sqrt{g}} \rightarrow \infty \quad \text{as } g \rightarrow \infty?$$

Reference supplied by the database. Assaf Naor, “An introduction to the Ribe program,” *Japanese Journal of Mathematics* 7 (2012), no. 2, 167–233, Question 7; DOI: [10.1007/s11537-012-1222-7](https://doi.org/10.1007/s11537-012-1222-7); arXiv: [1205.5993](https://arxiv.org/abs/1205.5993).

2.5.5. Primary-source pointer

The originating ICM record is *Finite Metric Spaces - Combinatorics, Geometry and Algorithms* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>.

2.6. 46. Finite Metric Spaces - Combinatorics, Geometry and Algorithms [2002-geom-89]

Record: 2002-geom-89

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Finite Metric Spaces - Combinatorics, Geometry and Algorithms

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>

2.6.1. Problem statement

For given integers $n \geq 1$ and real numbers $\epsilon > 0$, what is the smallest integer $k = k(n, \epsilon)$ such that every finite metric space (X, d) with $|X| = n$ that embeds isometrically into some ℓ_1 space admits an embedding $f : X \rightarrow \ell_1^k$ with distortion less than $1 + \epsilon$, i.e. after rescaling, $d(x, y) \leq \|f(x) - f(y)\|_1 < (1 + \epsilon)d(x, y)$ for all distinct $x, y \in X$?

2.6.2. Historical background, literature, and current progress

2.6.3. Status: partially resolved

2.6.3.1. Precise problem

For each integer $n \geq 1$ and each $\epsilon > 0$, determine the least target dimension $k(n, \epsilon)$ such that every n -point metric space that is an ℓ_1 metric, equivalently every finite subset of some L_1 or ℓ_1 space, has a bi-Lipschitz embedding into ℓ_1^k with distortion less than $1 + \epsilon$. In cut-metric language this asks how many weighted cut coordinates are always enough, and sometimes necessary, to multiplicatively approximate every n -point ℓ_1 metric within factor $1 + \epsilon$.

2.6.3.2. Verdict

The problem is not fully resolved. The literature proves strong general upper and lower bounds, including a general upper bound $k(n, \epsilon) = O(n/\epsilon^2)$ and near-linear lower bounds for small ϵ , but these do not match and do not identify the smallest integer $k(n, \epsilon)$ for all n, ϵ .

2.6.3.3. Original source

- Title: Finite metric spaces–combinatorics, geometry and algorithms
- Author: Nathan Linial
- Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002, vol. 3, pp. 573-586; arXiv version submitted 2003
- DOI/arXiv/URL: arXiv:math/0304466, <https://arxiv.org/abs/math/0304466>

- Relevance: This ICM survey states Open Problem 5 in essentially the same form: determine the smallest $k = k(n, \epsilon)$ such that every n -point metric in ℓ_1 embeds into ℓ_1^k with distortion $< 1 + \epsilon$. At that time the recorded bounds were only roughly $\Omega(\log n) \leq k \leq O(n \log n)$.

2.6.3.4. Progress in the literature

- Title: More on embedding subspaces of L_p in l_r^n
- Author: Gideon Schechtman
- Venue/year: *Compositio Mathematica* 61(2), pp. 159-169, 1987
- DOI/arXiv/URL: <https://eudml.org/doc/89819>
- Result and relevance: Provides finite-dimensional embedding results for subspaces of L_p , including the historical L_1 subspace upper-bound line. For an n -point subset of L_1 , the span of the point differences has dimension at most $n - 1$, so these subspace results gave general finite-dimensional ℓ_1 upper bounds, but not the optimal $k(n, \epsilon)$.
- Title: Embedding subspaces of L_1 into l_1^N
- Author: Michel Talagrand
- Venue/year: *Proceedings of the American Mathematical Society* 108(2), pp. 363-369, 1990
- DOI/arXiv/URL: DOI 10.1090/S0002-9939-1990-0994792-4
- Result and relevance: Improved the finite-dimensional L_1 subspace embedding estimates. This is part of the upper-bound background cited in later dimension-reduction papers, but it does not settle the finite metric problem exactly.
- Title: Dimension Reduction in the ℓ_1 Norm
- Authors: Moses Charikar and Amit Sahai
- Venue/year: 43rd IEEE Symposium on Foundations of Computer Science (FOCS 2002), pp. 551-560, IEEE, 2002
- DOI/arXiv/URL: DOI 10.1109/SFCS.2002.1181979
- Result and relevance: Shows that linear dimension reduction is impossible for general ℓ_1 point sets and gives positive results for special ℓ_1 -embeddable metric classes. This is important partial progress but does not answer the nonlinear finite-metric question in the problem statement.
- Title: On the impossibility of dimension reduction in ℓ_1
- Authors: Bo Brinkman and Moses Charikar
- Venue/year: *Journal of the ACM* 52(5), pp. 766-788, 2005; preliminary version FOCS 2003
- DOI/arXiv/URL: DOI 10.1145/1089023.1089026; preliminary DOI 10.1109/SFCS.2003.1238224
- Result and relevance: Constructs n -point subsets of ℓ_1 for which every embedding into ℓ_1^d with distortion D requires $d \geq n^{\Omega(1/D^2)}$. For distortion $1 + \epsilon$, it also gives a lower bound $d \geq n^{1/2 - O(\epsilon \log(1/\epsilon))}$. This disproves a Johnson-Lindenstrauss-style logarithmic dimension theorem for ℓ_1 , but it is still only a lower bound.
- Title: Embedding the diamond graph in L_p and dimension reduction in L_1

- Authors: James R. Lee and Assaf Naor
- Venue/year: Geometric and Functional Analysis 14(4), pp. 745-747, 2004
- DOI/arXiv/URL: DOI 10.1007/s00039-004-0473-8; arXiv:math/0407520
- Result and relevance: Gives a short geometric proof using diamond graphs that there are n -point subsets of L_1 whose distortion- D embeddings into ℓ_1^d need $d \geq n^{\Omega(1/D^2)}$. This supports the lower-bound side but does not determine the optimal $k(n, \epsilon)$.
- Title: On Multiplicative λ -Approximations and Some Geometric Applications
- Authors: Ilan Newman and Yuri Rabinovich
- Venue/year: SIAM Journal on Computing 42(3), pp. 855-883, 2013; arXiv preprint "Finite Volume Spaces and Sparsification", 2010
- DOI/arXiv/URL: DOI 10.1137/100801809; arXiv:1002.3541, <https://arxiv.org/abs/1002.3541>
- Result and relevance: Proves that every ℓ_1 metric on n points can be embedded into $\ell_1^{O(n/\epsilon^2)}$ with distortion $(1 + \epsilon)/(1 - \epsilon)$. Reparameterizing ϵ gives a $1 + \epsilon$ distortion upper bound $k(n, \epsilon) = O(n/\epsilon^2)$ for exactly the full class in the problem statement. This is the best general upper-bound type result found, but it is not matched by known lower bounds.
- Title: Near Linear Lower Bound for Dimension Reduction in L_1
- Authors: Alexandr Andoni, Moses S. Charikar, Ofer Neiman, and Huy L. Nguyen
- Venue/year: 52nd IEEE Symposium on Foundations of Computer Science (FOCS 2011), pp. 315-323, IEEE, 2011
- DOI/arXiv/URL: DOI 10.1109/FOCS.2011.87; author PDF <https://www.cs.columbia.edu/~andoni/papers/dim2-focs.pdf>
- Result and relevance: Improves the $1 + \epsilon$ lower bound substantially: for small ϵ , some n -point subsets of ℓ_1 require dimension at least $n^{1-O(1/\log(1/\epsilon))}$. This is near-linear as $\epsilon \rightarrow 0$, but still falls short of matching the $O(n/\epsilon^2)$ upper bound and does not yield the exact smallest $k(n, \epsilon)$.
- Title: Entropy-based Bounds on Dimension Reduction in L_1
- Author: Oded Regev
- Venue/year: Israel Journal of Mathematics 195, pp. 825-832, 2013; arXiv version 2011
- DOI/arXiv/URL: DOI 10.1007/s11856-012-0137-6; arXiv:1108.1283, <https://arxiv.org/abs/1108.1283>
- Result and relevance: Gives an alternative entropy proof of the Brinkman-Charikar and Andoni-Charikar-Neiman-Nguyen lower bounds: $d \geq N^{\Omega(1/D^2)}$ for distortion $D > 1$, and $d \geq N^{1-O(1/\log(1/\epsilon))}$ for distortion $1 + \epsilon$. This clarifies the lower-bound mechanism but does not close the gap.
- Title: Dimension Reduction for Finite Trees in ℓ_1
- Authors: James R. Lee, Arnaud de Mesmay, and Mohammad Moharrami
- Venue/year: Discrete & Computational Geometry 50, pp. 977-1032, 2013
- DOI/arXiv/URL: DOI 10.1007/s00454-013-9536-7

- Result and relevance: Shows that every n -point tree metric embeds into $\ell_1^{C(\epsilon) \log n}$ with distortion $1 + \epsilon$, with related lower bounds for star metrics. This is significant for a special subclass of ℓ_1 metrics, but the problem asks for all n -point ℓ_1 metrics.
- Title: Metric dimension reduction: A snapshot of the Ribe program
- Author: Assaf Naor
- Venue/year: Proceedings of the International Congress of Mathematicians, Rio de Janeiro 2018, vol. I, pp. 759-837; arXiv:1809.02376
- DOI/arXiv/URL: arXiv:1809.02376, <https://arxiv.org/abs/1809.02376>
- Result and relevance: Survey context for metric dimension reduction and its open challenges. Later papers continue to cite the Brinkman-Charikar and Andoni-Charikar-Neiman-Nguyen bounds as the relevant ℓ_1 lower-bound state rather than a complete determination of $k(n, \epsilon)$.

2.6.3.5. What remains unresolved

The exact function $k(n, \epsilon)$, even up to matching asymptotic order in the main parameter ranges, remains unknown. The current general results imply

$$n^{1-O(1/\log(1/\epsilon))} \lesssim k(n, \epsilon) \lesssim O(n/\epsilon^2)$$

for $1 + \epsilon$ distortion, with earlier and complementary bounds for fixed larger distortion. Because no cited theorem gives matching upper and lower bounds, or an exact smallest integer, for all n and ϵ , the problem should be classified as partially resolved rather than fully resolved.

2.6.4. Remaining open problems recorded by the database

2.6.4.1. Open problem 1

Determine the asymptotic growth of $c_{10}(n)$, the least integer d such that every n -point subset of L_1 embeds into ℓ_1^d with distortion at most 10; in particular, close the gap between the known polynomial lower bound and linear upper bound.

Reference supplied by the database. James R. Lee, “Open problem: Dimension reduction in L_1 ,” *tcs math* (research blog), February 19, 2010, explicit fixed-distortion instance $c_{10}(n)$. [Post](#).

2.6.5. Primary-source pointer

The originating ICM record is *Finite Metric Spaces - Combinatorics, Geometry and Algorithms* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>.

2.7. 47. Finite Metric Spaces - Combinatorics, Geometry and Algorithms [2002-geom-90]

Record: 2002-geom-90

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Finite Metric Spaces - Combinatorics, Geometry and Algorithms

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>

2.7.1. Problem statement

For integers $n \geq 1$ and real numbers $t \geq 1$, what is the largest value of $f(n, t)$ such that every metric space (X, d) with $|X| = n$ contains a subset $Y \subseteq X$ with $|Y| \geq f(n, t)$ whose restricted metric $d|_{Y \times Y}$ has Euclidean distortion at most t ?

Equivalently, if $c_2(Y)$ denotes the least distortion of an embedding of the metric space $(Y, d|_{Y \times Y})$ into some finite-dimensional ℓ_2 space, where distortion at most t means that for some embedding $\varphi : Y \rightarrow \ell_2^m$ and constants $\alpha \geq \beta > 0$ one has

$$\beta d(y, y') \leq \|\varphi(y) - \varphi(y')\|_2 \leq \alpha d(y, y')$$

for all $y, y' \in Y$ and $\alpha/\beta \leq t$, determine the largest $f(n, t)$ for which every n -point metric space (X, d) has such a subset Y with $c_2(Y) \leq t$.

2.7.2. Historical background, literature, and current progress

2.7.3. Verdict: partially resolved

2.7.3.1. Problem identified

The problem asks for the finite metric Ramsey function for Hilbert/Euclidean targets: for each integer $n \geq 1$ and distortion $t \geq 1$, determine the largest integer $f(n, t)$ such that every n -point metric space contains a subset Y of size at least $f(n, t)$ with Euclidean distortion $c_2(Y) \leq t$. In the literature this is usually written as the Euclidean metric Ramsey function $R_2(t, n)$.

The original source found is:

- Title: *Finite Metric Spaces - Combinatorics, Geometry and Algorithms*
- Author: Nathan Linial
- Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002; arXiv version, 2003
- DOI/arXiv/URL: arXiv:math/0304466, <https://arxiv.org/abs/math/0304466>
- Relevance: Open Problem 8 asks exactly for the largest $f(\cdot, \cdot)$ such that every n -point

metric (X, d) has a subset Y with $|Y| \geq f(n, t)$ and $c_2(Y) \leq t$. It notes that for t close to 1 the answer is $\Theta(\log n)$, while larger t behaves differently.

2.7.3.2. Main progress

- Title: *On Hilbertian subsets of finite metric spaces*
- Authors: Jean Bourgain, Tadeusz Figiel, Vitali D. Milman
- Venue/year: Israel Journal of Mathematics 55(2), 147-152, 1986
- DOI/arXiv/URL: DOI 10.1007/BF02801990
- Result: For every $\epsilon > 0$, every finite metric space has a subset of size at least $C(\epsilon) \log |X|$ whose restricted metric embeds into Hilbert space with distortion $1+\epsilon$. They also prove logarithmic sharpness for some distortion $\alpha_0 > 1$. This establishes the original small-distortion nonlinear Dvoretzky/Ramsey phenomenon, but it does not determine $f(n, t)$ for all t .
- Title: *On metric Ramsey-type phenomena*
- Authors: Yair Bartal, Nathan Linial, Manor Mendel, Assaf Naor
- Venue/year: Annals of Mathematics 162(2), 643-709, 2005
- DOI/arXiv/URL: DOI 10.4007/annals.2005.162.643; arXiv:math/0406353
- Result: This paper explicitly studies $R_2(\alpha, n)$. It proves nearly tight upper and lower bounds, including a phase transition: for every fixed $1 < \alpha < 2$, $R_2(\alpha, n) = \Theta_\alpha(\log n)$, while for every fixed $\alpha > 2$, $R_2(\alpha, n)$ is polynomial in n with exponent strictly between 0 and 1. For large α , their bounds include $R_2(\alpha, n) \geq n^{\{1-C \log(2\alpha)/\alpha\}}$ and $R_2(\alpha, n) \leq C n^{\{1-c/\alpha\}}$. This is major partial progress, but not an exact determination of $f(n, t)$, and it leaves a logarithmic gap in the large-distortion lower bound.
- Title: *Some low distortion metric Ramsey problems*
- Authors: Yair Bartal, Nathan Linial, Manor Mendel, Assaf Naor
- Venue/year: Discrete & Computational Geometry 33(1), 27-41, 2005
- DOI/arXiv/URL: DOI 10.1007/s00454-004-1100-z; arXiv:math/0406358
- Result: This follow-up records and uses the ℓ_2 phase transition at distortion 2, and extends related low-distortion threshold questions to ℓ_p spaces. For this problem it is supporting evidence about the ℓ_2 low-distortion regime rather than a full determination of $f(n, t)$.
- Title: *Ramsey partitions and proximity data structures*
- Authors: Manor Mendel, Assaf Naor
- Venue/year: Journal of the European Mathematical Society 9(2), 253-275, 2007
- DOI/arXiv/URL: DOI 10.4171/JEMS/79; arXiv:cs/0511084
- Result: For every ϵ in $(0, 1)$, every n -point metric space has a subset of size at least $n^{\{1-\epsilon\}}$ that embeds into Hilbert space, indeed into an ultrametric and hence into Hilbert space, with distortion $O(1/\epsilon)$. The paper states this result is best possible, matching the known lower examples up to universal constants.

Equivalently, for large distortion t , this gives $R_2(t, n) \geq n^{1-O(1/t)}$ and the known upper examples give $R_2(t, n) \leq n^{1-\Omega(1/t)}$. This resolves the large-distortion order of the exponent up to constants, but still not the exact largest value requested in the problem statement.

- Title: *Scale-oblivious metric fragmentation and the nonlinear Dvoretzky theorem*
- Authors: Assaf Naor, Terence Tao
- Venue/year: Israel Journal of Mathematics 192, 489-504, 2012
- DOI/arXiv/URL: DOI 10.1007/s11856-012-0039-7; arXiv:1003.4013
- Result: Gives a simpler randomized proof for every distortion $D > 2$, producing a polynomial-size subset that embeds into an ultrametric with distortion D , with an explicit exponent from an optimization equation. The paper explicitly says the behavior at distortion $D = 2$ is unknown and asks related sharp-exponent questions near 2.
- Title: *Ultrametric subsets with large Hausdorff dimension*
- Authors: Manor Mendel, Assaf Naor
- Venue/year: Inventiones Mathematicae 192(1), 1-54, 2013
- DOI/arXiv/URL: DOI 10.1007/s00222-012-0402-7; arXiv:1106.0879
- Result: Extends the nonlinear Dvoretzky/metric Ramsey phenomenon to compact metric spaces and summarizes the finite results: for $D \in (1, 2)$ the finite Ramsey function is logarithmic in n , while for $D \in (2, \infty)$ it is polynomial in n ; for large D , the $O(1/\epsilon)$ distortion theorem and the matching expander-based obstruction are sharp up to constants.
- Title: *Advances in Metric Ramsey Theory and its Applications*
- Author: Yair Bartal
- Venue/year: arXiv preprint, 2021
- DOI/arXiv/URL: arXiv:2104.03484, <https://arxiv.org/abs/2104.03484>
- Result: Gives deterministic constructions and strengthened algorithmic versions of metric Ramsey theorems with tight asymptotic tradeoffs. This is later progress but does not claim an exact formula for $R_2(t, n)$ in all cases.

2.7.3.3. Why this is not fully resolved

The literature does not give a citable theorem determining the exact largest value of $f(n, t)$ for all integers $n \geq 1$ and all real $t \geq 1$. What is known is a strong collection of asymptotic and regime results:

- For fixed $1 < t < 2$, $f(n, t) = \Theta_t(\log n)$.
- For fixed $t > 2$, $f(n, t)$ is polynomial in n .
- For large t , $f(n, t)$ has exponent $1 - \Theta(1/t)$ in the sense that $n^{1-C/t} \leq f(n, t) \leq n^{1-c/t}$ up to universal constants.

This does not cover the exact question. In particular, the exact threshold value $t = 2$ is not determined by these theorems, and the exact constants/exponents for general t are not known. Therefore the correct status is partially resolved, not fully resolved.

2.7.4. Remaining open problems recorded by the database

2.7.4.1. Open problem 1

Determine the asymptotic order of $R_2(2, n)$, the largest integer m such that every n -point metric space contains an m -point subspace that embeds into Hilbert space with distortion at most 2.

Reference supplied by the database. Yair Bartal, Nathan Linial, Manor Mendel, and Asaf Naor, “On metric Ramsey-type phenomena,” *Annals of Mathematics* (2) 162, no. 2 (2005), 643–709, §7 (“Open Problems”), explicit question of determining $R_2(2, n)$. DOI: [10.4007/annals.2005.162.643](https://doi.org/10.4007/annals.2005.162.643). Preprint: [arXiv:math/0406353](https://arxiv.org/abs/math/0406353).

2.7.5. Primary-source pointer

The originating ICM record is *Finite Metric Spaces - Combinatorics, Geometry and Algorithms* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>.

2.8. 48. Higher index theory of elliptic operators and geometry of groups [2006-geom-95]

Record: 2006-geom-95

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Higher index theory of elliptic operators and geometry of groups

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1633>

2.8.1. Problem statement

Prove that for every discrete metric space Γ with bounded geometry, meaning that for every $r > 0$ there is a number $N(r) > 0$ such that every ball of radius r in Γ has at most $N(r)$ elements, the coarse Baum-Connes assembly map

$$\mu : \lim_{d \rightarrow \infty} K_*(P_d(\Gamma)) \longrightarrow K_*(C^*(\Gamma))$$

is injective. Here $P_d(\Gamma)$ is the Rips complex with vertex set Γ , in which a finite subset spans a simplex exactly when all pairwise distances are at most d , and $C^*(\Gamma)$ is the Roe algebra of Γ , the operator-norm closure of the locally compact finite-propagation operators on $\ell^2(\Gamma) \otimes H$ for a separable infinite-dimensional Hilbert space H .

2.8.2. Historical background, literature, and current progress

2.8.3. Status check: Coarse Strong Novikov Conjecture

Verdict: partially resolved.

2.8.3.1. Precise problem

The problem asks whether, for every discrete metric space X of bounded geometry, the reduced coarse Baum-Connes assembly map

$$\mu : \lim_{d \rightarrow \infty} K_*(P_d(X)) \longrightarrow K_*(C^*(X))$$

is injective. This is the injectivity part of the coarse Baum-Connes conjecture, usually called the coarse Novikov conjecture or coarse strong Novikov conjecture.

The problem is not the maximal coarse assembly map, not an equivariant/twisted/rational variant, and not the full isomorphism statement. A result resolves this problem only if it proves the above injectivity for all bounded-geometry discrete metric spaces, or gives a counterexample to that exact injectivity statement.

2.8.3.2. Current status

I found no published or citable theorem proving injectivity for all bounded-geometry discrete metric spaces, and no counterexample to this exact bounded-geometry injectivity statement. The literature gives many important special cases. Counterexamples to the full coarse Baum-Connes conjecture for bounded-geometry spaces are known, but the standard ones obstruct surjectivity, not the injectivity asked here.

Thus the full statement in the problem statement remains unresolved, while substantial special cases are known.

2.8.3.3. Original and background sources

1. Higher index theory of elliptic operators and geometry of groups

Authors: Guoliang Yu.

Venue/year: Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. II, European Mathematical Society, Zurich, 2006, pp. 1623-1639.

DOI/URL: DOI 10.4171/022-2/77; EMS page <https://ems.press/books/standalone>
IMU proceedings PDF <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM>

Relevance: This is the dataset's source context and discusses higher index theory, the coarse Baum-Connes framework, and the coarse Novikov/injectivity problem.

2. On the Coarse Baum-Connes Conjecture

Authors: Nigel Higson and John Roe.

Venue/year: In *Novikov Conjectures, Index Theorems and Rigidity, Vol. 2* (Oberwolfach, 1993), London Mathematical Society Lecture Note Series 227, Cambridge University Press, 1995, pp. 227-254.

DOI/arXiv/URL: no DOI found in the search results; bibliographic reference appears in later papers such as <https://ems.press/journals/jncg/articles/9871882>.

Relevance: Standard early formulation/reference for the coarse Baum-Connes assembly map and its injectivity/surjectivity problems.

2.8.3.4. Positive partial results

1. The Novikov conjecture for groups with finite asymptotic dimension

Authors: Guoliang Yu.

Venue/year: *Annals of Mathematics* 147(2), 325-355, 1998.

DOI/URL: DOI 10.2307/121011; Annals page <https://annals.math.princeton.edu/>

Result: Theorem 7.1 proves the coarse Baum-Connes conjecture for proper metric spaces with finite asymptotic dimension.

Applicability check: This implies the stated injectivity for bounded-geometry spaces of finite asymptotic dimension, but finite asymptotic dimension is an extra hypothesis and is not automatic for all bounded-geometry spaces.

2. The coarse Baum-Connes conjecture for spaces which admit a uniform embedding into Hilbert space

Authors: Guoliang Yu.

Venue/year: *Inventiones Mathematicae* 139, 201-240, 2000.

DOI/URL: DOI 10.1007/s002229900032; Springer page <https://link.springer.com/>

Result: Theorem 1.1 proves the coarse Baum-Connes conjecture for discrete bounded-geometry spaces admitting a uniform/coarse embedding into Hilbert space.

Applicability check: This is stronger than injectivity on that subclass, but the embedding hypothesis is additional and fails for important bounded-geometry examples such as many expanders.

3. The coarse Baum-Connes conjecture and groupoids

Authors: Georges Skandalis, Jean-Louis Tu, Guoliang Yu.

Venue/year: *Topology* 41(4), 807-834, 2002.

DOI/URL: DOI 10.1016/S0040-9383(01)00004-0; author PDF <https://webusers.imj-prg.fr/~georges.skandalis/Publications/Publi32.pdf>.

Result: Gives a groupoid approach and a new proof that bounded-geometry spaces uniformly embeddable into Hilbert space satisfy the coarse assembly isomorphism.

Applicability check: Again resolves the problem statement only for the Hilbert-embeddable subclass.

4. The coarse geometric Novikov conjecture and uniform convexity

Authors: Gennadi Kasparov and Guoliang Yu.

Venue/year: *Advances in Mathematics* 206(1), 1-56, 2006.

DOI/arXiv/URL: DOI 10.1016/j.aim.2005.08.004; arXiv math/0507599, <https://arxiv.org/abs/math/0507599>.

Result: Proves the coarse geometric Novikov conjecture for bounded-geometry spaces uniformly/coarsely embeddable into uniformly convex Banach spaces.

Applicability check: This proves injectivity under a Banach-space embeddability hypothesis, not for arbitrary bounded-geometry spaces.

5. The coarse Novikov conjecture and Banach spaces with Property (H)

Authors: Xiaoman Chen, Qin Wang, Guoliang Yu.

Venue/year: *Journal of Functional Analysis* 268(9), 2754-2786, 2015.

DOI/URL: DOI 10.1016/j.jfa.2015.02.001; bibliographic page <https://www.researchgate.net/publication/27542786>

Result: Proves the coarse Novikov conjecture for bounded-geometry metric spaces coarsely embeddable into Banach spaces with Property (H).

Applicability check: This includes spaces coarsely embeddable into many classical Banach spaces, but still assumes a coarse embedding into a target with Property (H).

6. Higher index theory for certain expanders and Gromov monster groups, I

Authors: Rufus Willett and Guoliang Yu.

Venue/year: *Advances in Mathematics* 229(3), 1380-1416, 2012.

DOI/arXiv/URL: DOI 10.1016/j.aim.2011.10.024; arXiv 1012.4150, <https://arxiv.org/abs/1012.4150>.

Result: For sequences of expander graphs with girth tending to infinity, the coarse assembly map is injective but not surjective.

Applicability check: This is important because it proves injectivity for some non-Hilbert-embeddable bounded-geometry spaces, but it is still a special family.

7. A Bott periodicity theorem for ℓ^p -spaces and the coarse Novikov conjecture at infinity

Authors: Liang Guo, Zheng Luo, Qin Wang, Yazhou Zhang.

Venue/year: *Journal of Functional Analysis* 286(2), article 110215, 2024.

DOI/arXiv/URL: DOI 10.1016/j.jfa.2023.110215; arXiv 2207.04193, <https://arxiv.org/abs/2207.04193>.

Result: Proves the coarse Novikov conjecture for bounded-geometry metric spaces admitting a fibred coarse embedding into an ℓ^p -space.

Applicability check: The fibred coarse embedding hypothesis is substantial and does not cover all bounded-geometry metric spaces.

8. ℓ^p -coarse Baum-Connes conjecture for ℓ^q -coarse embeddable spaces

Authors: Jinmin Wang, Zhizhang Xie, Guoliang Yu, Bo Zhu.

Venue/year: arXiv preprint, 2024; version 2 dated 2025.

DOI/arXiv/URL: DOI 10.48550/arXiv.2411.15070; arXiv 2411.15070, <https://arxiv.org/abs/2411.15070>.

Result: Theorem 1.1 proves the ordinary coarse Baum-Connes conjecture for discrete bounded-geometry spaces coarsely embedding into ℓ^q for some finite q .

Applicability check: This implies injectivity for that subclass and strengthens earlier Hilbert-space results, but still assumes coarse embeddability into ℓ^q . It does not prove the universal bounded-geometry statement.

2.8.3.5. Negative results and why they do not settle this problem

1. Counterexamples to the Baum-Connes conjecture

Authors: Nigel Higson, Vincent Lafforgue, Georges Skandalis.

Venue/year: *Geometric and Functional Analysis* 12(2), 330-354, 2002.

DOI/URL: DOI 10.1007/s00039-002-8249-5; Penn State bibliographic page

<https://pure.psu.edu/en/publications/counterexamples-to-the-baum-connes-conjecture/>; author PDF <https://vlafforg.perso.math.cnrs.fr/>

Result: Gives counterexamples to several Baum-Connes-type statements, including surjectivity failures for coarse geometric spaces.

Applicability check: the problem asks only for injectivity of the coarse assembly map. These bounded-geometry expander-type counterexamples do not give a failure of injectivity of the stated map.

2. Expanders are counterexamples to the ℓ^p coarse Baum-Connes conjecture

Authors: Yeong Chyuan Chung and Piotr W. Nowak.

Venue/year: *Journal of Noncommutative Geometry* 17(1), 305-331, 2023.

DOI/arXiv/URL: DOI 10.4171/JNCG/498; arXiv 1811.10457, <https://arxiv.org/abs/1811.10457>, EMS page <https://ems.press/journals/jncg/articles/9871882>.

Result: Shows non-surjectivity for an ℓ^p analogue of the coarse Baum-Connes assembly map for certain expanders; the paper's introduction also summarizes that classical expander counterexamples are surjectivity obstructions and that injectivity is known for certain classes of expanders.

Applicability check: This is not a counterexample to the original C^* -Roe-algebra injectivity statement in the problem statement.

3. Non-bounded-geometry counterexamples

Several sources note that Yu constructed a counterexample to coarse Novikov-type injectivity when the bounded-geometry hypothesis is omitted. For example, Chung-Nowak's 2023 paper explicitly states that such a counterexample exists without bounded geometry.

Applicability check: The statement includes bounded geometry as a central hypothesis, so these examples do not resolve the problem statement.

2.8.3.6. Status indicators from recent literature

1. The strongly quasi-local coarse Novikov conjecture and Banach spaces with Property (H)

Authors: Xiaoman Chen, Kun Gao, Jiawen Zhang.

Venue/year: *Kyoto Journal of Mathematics* 64(1), 31-74, 2024.

DOI/arXiv/URL: DOI 10.1215/21562261-2023-0010; arXiv 2109.11118, <https://arxiv.org/abs/2109.11118>; Project Euclid page <https://projecteuclid.org/journal-of-mathematics/volume-64/issue-1/The-strongly-quasi-local-coarse-Novikov-conjecture-and-Banach-spaces/10.1215/21562261-2023-0010.full>.

Relevance: Its introduction says that, despite many positive results, no counterexample to the coarse Novikov conjecture had been discovered. The paper proves a strongly quasi-local variant under Property (H) embeddability, not the universal statement in this problem.

2.8.3.7. What remains unresolved

It remains unknown whether the reduced coarse assembly map

$$\mu : \lim_{d \rightarrow \infty} K_*(P_d(X)) \rightarrow K_*(C^*(X))$$

is injective for every discrete bounded-geometry metric space X . Existing theorems prove injectivity, or even full isomorphism, for large classes such as finite asymptotic dimension spaces, Hilbert-embeddable spaces, uniformly convex Banach-embeddable spaces, Property (H)-embeddable spaces, ℓ^q -embeddable spaces, and some fibred-coarse-embeddable or expander families. None of these hypotheses is part of the problem statement, and no theorem found removes them for all bounded-geometry spaces.

2.8.4. Remaining open problems recorded by the database

2.8.4.1. Open problem 1

Does there exist a sequence $(p_n)_{n \in \mathbb{N}}$ with $p_n > 1$ and $p_n \rightarrow \infty$ such that the Hilbertian direct sum $(\bigoplus_{n=1}^{\infty} \ell_{p_n})_{\ell_2}$ has Kasparov–Yu Property (H)?

Reference supplied by the database. Qingjin Cheng, “Sphere equivalence, Property H, and Banach expanders,” *Studia Mathematica* **233** (2016), no. 1, 67–83, Question 3.7. DOI: [10.4064/sm8396-4-2016](https://doi.org/10.4064/sm8396-4-2016).

2.8.4.2. Open problem 2

Does the Banach sequence space c_0 have Kasparov–Yu Property (H)?

Reference supplied by the database. Qingjin Cheng, “Sphere equivalence, Property H, and Banach expanders,” *Studia Mathematica* **233** (2016), no. 1, 67–83, Remark 3.8(ii). DOI: [10.4064/sm8396-4-2016](https://doi.org/10.4064/sm8396-4-2016).

2.8.5. Primary-source pointer

The originating ICM record is *Higher index theory of elliptic operators and geometry of groups* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1633>.

2.9. 49. Quasiconformal geometry of fractals [2006-geom-101]

Record: 2006-geom-101

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Quasiconformal geometry of fractals

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1359>

2.9.1. Problem statement

Let X be the metric two-sphere obtained from a subdivision rule on $\mathbb{S}^2$ by the associated tile metric d_λ , in which level- n tiles have diameter comparable to λ^n for some $\lambda < 1$ sufficiently close to 1, and let $f : X \rightarrow X$ be the postcritically finite topologically holomorphic expanding map associated with the subdivision rule. Let P_f be the finite postcritical set of f . The Ahlfors regular conformal dimension of X is

$\dim_c X = \inf\{q : \text{there exists an Ahlfors } q\text{-regular metric space quasisymmetrically equivalent to } X\}.$

Here Ahlfors q -regular means that, with respect to q -dimensional Hausdorff measure, the measure of each ball of radius r not exceeding the diameter of the space is bounded above and below by constant multiples of r^q , and quasisymmetrically equivalent means related by a quasimetric homeomorphism.

For an f -stable multicurve $\Gamma = \{\gamma_1, \dots, \gamma_n\}$, meaning a finite collection of pairwise disjoint, pairwise non-homotopic, nonperipheral Jordan curves in $\mathbb{S}^2 \setminus P_f$ such that every component of $f^{-1}(\gamma_j)$ is either peripheral or homotopic in $\mathbb{S}^2 \setminus P_f$ to one of the curves γ_i , where nonperipheral means that each complementary component of the curve contains at least two points of P_f , define for each $Q \geq 2$ the modified Thurston matrix $A(\Gamma, Q) = (m_{ij}^Q)$ by

$$m_{ij}^Q = \sum_{\alpha} \frac{1}{d_{i,j,\alpha}^{Q-1}},$$

where the sum is over the components $\gamma_{i,j,\alpha}$ of $f^{-1}(\gamma_j)$ homotopic to γ_i in $\mathbb{S}^2 \setminus P_f$, and $d_{i,j,\alpha}$ is the degree of $f : \gamma_{i,j,\alpha} \rightarrow \gamma_j$. Let $\lambda(f, \Gamma, Q)$ be the largest nonnegative eigenvalue of $A(\Gamma, Q)$.

Prove that

$$\dim_c X = \inf\{Q \geq 2 : \lambda(f, \Gamma, Q) < 1 \text{ for every } f\text{-stable multicurve } \Gamma\}.$$

2.9.2. Historical background, literature, and current progress

2.9.3. Verdict: partially resolved

2.9.3.1. Precise problem checked

The problem asks for the Bonk-Geyer-Pilgrim conformal-dimension formula for visual/tile-metric two-spheres coming from finite subdivision rules, equivalently expanding postcritically finite Thurston-type maps on the sphere: whether

$$\text{ARCdim}(X) = \inf\{Q \geq 2 : \lambda(f, \Gamma, Q) < 1 \text{ for every } f\text{-stable multicurve } \Gamma\}.$$

This is an exact equality statement. It is not just asking for one inequality, for the rational/unobstructed case, for examples, or for a different critical exponent defined by all curve families or graph energies.

2.9.3.2. Literature found

1. Mario Bonk, “Quasiconformal geometry of fractals”, *Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. II, European Mathematical Society, 2007*, pp. 1349-1373. DOI: 10.4171/022-2/64. URL: <https://ems.press/books/standalone/21/448>

This is the source context for the conjectural formula. Bonk describes metric two-spheres associated with subdivision rules and, citing discussions with Lukas Geyer and Kevin Pilgrim, states the conjecture that the Ahlfors regular conformal dimension is the infimum of the parameters $Q \geq 2$ for which all stable multicurve modified Thurston matrices have spectral radius < 1 . This is the exact problem statement.

2. Peter Haissinsky and Kevin M. Pilgrim, “Thurston obstructions and Ahlfors regular conformal dimension”, *Journal de Mathématiques Pures et Appliquées* (9) 90 (2008), no. 3, 229-241. DOI: 10.1016/j.matpur.2008.04.006. arXiv:0706.1123. URL: <https://arxiv.org/abs/0706.1123>

This is the main partial resolution. For an expanding branched covering $f : S^2 \rightarrow S^2$ with finite postcritical set, the paper associates a canonical quasisymmetry class of Ahlfors regular metrics and proves a lower bound for the Ahlfors regular conformal dimension in terms of the same multicurve spectral data. In the notation of the problem, it proves the obstruction direction: if some stable multicurve has $\lambda(f, \Gamma, Q) \geq 1$, then the conformal dimension cannot be below the corresponding Q . This gives $\text{ARCdim}(X) \geq$ the multicurve critical exponent. It does not prove the reverse inequality, i.e. it does not construct Ahlfors regular metrics of every dimension above the multicurve threshold.

3. John M. Mackay and Jeremy T. Tyson, “Conformal Dimension: Theory and Application”, *University Lecture Series 54, American Mathematical Society, 2010*. DOI: 10.1090/ulect/054.

Section 7.4 formulates the same statement as Conjecture 7.4.6, attributed to Bonk-Geyer-Pilgrim, in the form $\text{ARCdim}(\mathcal{G}_f) = p_f$, where p_f is the supremum of the critical exponents of irreducible stable multicurves. The book explicitly records that one direction is known and cites Haissinsky-Pilgrim for $\text{ARCdim} \mathcal{G}_f \geq p_f$. This confirms that, as of that survey, the equality was not known.

4. **Peter Haissinsky and Kevin M. Pilgrim, “Coarse expanding conformal dynamics”, *Asterisque* 325 (2009), viii+139 pp. arXiv:math/0612617. URL: <https://arxiv.org/abs/math/0612617>**

This monograph develops the canonical conformal gauge/coarse expanding conformal framework that includes finite subdivision rule examples, and records how finite subdivision rules with mesh going to zero and bounded valence fit into topologically coarse expanding conformal dynamics. It supplies the ambient setting for the Haissinsky-Pilgrim lower-bound theorem, but it does not prove the full Bonk-Geyer-Pilgrim equality.

5. **Peter Haissinsky and Kevin M. Pilgrim, “Examples of coarse expanding conformal maps”, *Discrete and Continuous Dynamical Systems* 32 (2012), no. 7, 2403-2416. DOI: 10.3934/dcds.2012.32.2403. URL: <https://www.aims sciences.org/article/doi/10.3934/dcds.2012.32.2403>**

This paper extends the class of examples and discusses implications for computing conformal dimension. It gives useful examples and applications of the coarse expanding conformal framework, but not a theorem covering all subdivision-rule spheres and all stable multicurves in the stated formula.

6. **Peter Haissinsky and Kevin M. Pilgrim, “Minimal Ahlfors regular conformal dimension of coarse expanding conformal dynamics on the sphere”, *Duke Mathematical Journal* 163 (2014), no. 13, 2517-2559. DOI: 10.1215/00127094-2819408. arXiv:1103.4019. URL: <https://arxiv.org/abs/1103.4019>**

This proves strong results about when the Ahlfors regular conformal dimension of a topologically coarse expanding conformal sphere is attained. It is relevant to the “moreover” part often stated with the conjecture, but it does not identify the conformal dimension with the multicurve critical exponent in all cases.

7. **Mario Bonk and Daniel Meyer, “Expanding Thurston Maps”, *Mathematical Surveys and Monographs* 225, American Mathematical Society, 2017. DOI: 10.1090/surv/225. arXiv:1009.3647. URL: <https://arxiv.org/abs/1009.3647>**

This monograph gives the modern theory of expanding Thurston maps and visual metrics, including that expanding Thurston maps produce visual spheres and that rationality corresponds to quasisymmetric equivalence with the Riemann sphere in the relevant setting. It supports the interpretation of the stated d_λ tile metrics as visual metrics, but it does not resolve the Bonk-Geyer-Pilgrim conformal-dimension formula.

8. **Matias Carrasco Piaggio, “On the conformal gauge of a compact metric space”, *Annales scientifiques de l’Ecole Normale Supérieure, Serie 4*, 46 (2013), no. 3, 495-548. DOI: 10.24033/asens.2195. URL: <https://www.numdam.org/articles/10.24033/asens.2195/>**

Carrasco gives a general combinatorial-modulus description of Ahlfors regular conformal gauges and conformal dimension for compact metric spaces. This is important background and provides a broad critical-exponent framework, but it does not specialize the critical exponent for all subdivision-rule visual spheres to the stable-multicurve Thurston-matrix quantity appearing in the problem statement.

9. **Kevin M. Pilgrim and Dylan P. Thurston, “Ahlfors-regular conformal dimension and energies of graph maps”, *Groups, Geometry, and Dynamics* 20 (2026), no. 3, 971-1050. DOI: 10.4171/GGD/914. arXiv:2112.09041. URL: <https://ems.press/journals/ggd/articles/14299346>**

This recent work gives upper and lower bounds for the Ahlfors regular conformal di-

mension of connected Julia sets of hyperbolic rational maps using asymptotic conformal energies of graph maps. It is substantial later progress in the same area, but its hypotheses and conclusion differ from the problem statement: it concerns hyperbolic rational-map Julia sets and graph-energy bounds, not the exact all-stable-multicurve formula for arbitrary subdivision-rule visual two-spheres.

10. Insung Park, “Julia sets with Ahlfors-regular conformal dimension one”, arXiv:2209.13384, version 2, 2026. URL: <https://arxiv.org/abs/2209.13384>

Park characterizes postcritically finite hyperbolic rational maps whose Julia sets have Ahlfors regular conformal dimension one and uses finite subdivision rules/graph virtual endomorphisms to estimate conformal dimensions. This is later progress for a different extremal problem involving Julia sets, not a proof of the stated dimension formula for metric two-spheres.

2.9.3.3. Why the problem is not fully resolved

The exact formula requires both inequalities. The literature proves the obstruction/lower-bound direction via Haissinsky-Pilgrim: multicurves with large modified Thurston spectral radius force the Ahlfors regular conformal dimension to be at least the associated critical exponent.

I found no published or otherwise citable result proving the converse in the full stated generality: for every subdivision-rule visual/tile sphere and every Q above the multicurve threshold, there exists an Ahlfors Q -regular metric quasisymmetrically equivalent to X . Later work supplies general combinatorial-modulus frameworks, attainment results, examples, and graph-energy estimates in related rational-map settings, but none of these hypotheses and conclusions imply the exact Bonk-Geyer-Pilgrim equality for all stable multicurves and all subdivision-rule visual spheres.

Therefore the current status is **partially resolved**.

2.9.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

2.9.5. Primary-source pointer

The originating ICM record is *Quasiconformal geometry of fractals* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1359>.

2.10. 50. Volume Comparison via Boundary Distances [2010-geom-121]

Record: 2010-geom-121

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Volume Comparison via Boundary Distances

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=775>

2.10.1. Problem statement

Let S^1 be the circle with its intrinsic length metric and total length 2π . Let M be a compact orientable Riemannian surface whose boundary is identified with S^1 , and suppose that for every $x, y \in \partial M = S^1$, the distance $d_M(x, y)$ in M is at least the intrinsic circle distance between x and y . Prove that $\text{area}(M) \geq 2\pi$. Equivalently, prove that the filling volume of S^1 , defined as the infimum of $\text{area}(M)$ over all such M , is 2π , with equality realized by the standard round hemisphere of radius 1.

2.10.2. Historical background, literature, and current progress

2.10.3. Verdict: partially resolved

2.10.3.1. Precise problem

The problem asks for Gromov's circle filling conjecture for the Riemannian circle of length 2π : if M is any compact orientable Riemannian surface with ∂M identified with that circle and with no boundary shortcuts, then $\text{area}(M) \geq 2\pi$, with the round unit hemisphere realizing equality. Since the boundary itself contains arcs of length equal to the intrinsic circle distance, the hypothesis $d_M(x, y) \geq d_{S^1}(x, y)$ is equivalent here to equality of the boundary distance function $d_M|_{\partial M \times \partial M} = d_{S^1}$.

The literature I found does not prove this for all compact orientable filling surfaces. It proves important special cases and lower bounds, but the higher-genus case remains open.

2.10.3.2. Main sources and progress

1. M. Gromov, "Filling Riemannian manifolds", *Journal of Differential Geometry* **18** (1983), 1-147. DOI/URL: <https://doi.org/10.4310/jdg/1214509283>. This is the original source for filling volume and the circle filling problem. Later papers consistently cite it as the source of the conjecture. The disk/simply connected filling case is known from the Pu inequality argument: identify antipodal boundary points to get a projective-plane metric with systole at least π , giving area at least 2π . This does not cover fillings of arbitrary genus.

2. Victor Bangert, Christopher Croke, Sergei V. Ivanov, and Mikhail G. Katz, “Filling area conjecture and ovalless real hyperelliptic surfaces”, *Geometric and Functional Analysis* 15 (2005), 577-597. arXiv:math/0405583, URL: <https://arxiv.org/abs/math/0405583>. Theorem 1.7 proves a relative Pu-type inequality for ovalless real hyperelliptic surfaces, and Corollary 1.8 states that all orientable genus-one fillings of the circle satisfy the filling area conjecture. The hypotheses/conclusion are not enough for the full statement, because the statement quantifies over all compact orientable surfaces, including arbitrary higher genus.
3. Neil N. Katz, “A modulus for curves from distance”, *Annales Academiae Scientiarum Fennicae Mathematica* 35 (2010), 3-21. DOI: <https://doi.org/10.5186/aasfm.2010.3501>. The introduction summarizes the status: Gromov proved the disk case, Bangert-Croke-Ivanov-M. Katz extended it to a torus with a disk removed, and the higher-genus question remains open. This is useful because it directly addresses the same filling-volume formulation and confirms that the known genus-one result is only partial.
4. Sergei Ivanov, “Volume comparison via boundary distances”, *Proceedings of the International Congress of Mathematicians 2010*, Vol. II, 769-784; arXiv:1004.2505, URL: <https://arxiv.org/abs/1004.2505>. Section 2.2 states Gromov’s circle filling conjecture in the same form and says the disk case is well known via Pu’s inequality, the genus-one case follows from Bangert-Croke-Ivanov-Katz, and higher genus remains open. This is a survey/status source, not a full solution.
5. Gregory R. Chambers, “On minimal higher genus fillings”, *Journal of Topology and Analysis*, online ready, DOI: <https://doi.org/10.1142/S1793525326500184>; arXiv:2202.01342, URL: <https://arxiv.org/abs/2202.01342>. The main theorem proves that if M has genus G and is compared with a disk D with unique geodesics and dominating boundary distances, then

$$\left(1 + \frac{2G}{\pi}\right) \text{Area}(M) \geq \text{Area}(D).$$

Taking D to be the hemisphere would only give $\text{Area}(M) \geq 2\pi/(1 + 2G/\pi)$, which is weaker than 2π for $G > 0$. Thus it is genuine higher-genus progress but not a resolution of Gromov’s conjecture.

6. Roger Züst, “The Riemannian hemisphere is almost calibrated in the injective hull of its boundary”, *Calculus of Variations and Partial Differential Equations* 64, Article 297 (2025). DOI: <https://doi.org/10.1007/s00526-025-03133-z>; arXiv:2104.04498, URL: <https://arxiv.org/abs/2104.04498>. This paper explicitly states that the higher-genus question is widely open. It proves an almost-calibration/stability lower bound for metric-current or oriented Lipschitz-surface fillings near the hemisphere in the injective hull. The result supplies positive evidence and local/near-hemisphere control, but not the global 2π area lower bound for every orientable Riemannian filling.
7. Joseph Briggs and Chris Wells, “A discrete view of Gromov’s filling area conjecture”, arXiv:2602.17859v1 (2026), URL: <https://arxiv.org/abs/2602.17859>. This recent preprint states that the full conjecture remains unresolved and proves Theorem 1.2: any compact Riemannian surface isometrically filling the Riemannian circle of length 2π has area at least $(\sqrt{3}/4)\pi^2$, approximately 1.36π . This is an all-topology quantitative lower bound, even without orientability, but it is strictly below the conjectured 2π .

2.10.3.3. What remains unresolved

the exact statement remains open for compact orientable Riemannian fillings of genus at least 2. No located published or citable preprint proves that every such filling has area at least 2π . The known results cover genus 0, genus 1, certain stability/near-hemisphere settings, and weaker universal lower bounds, so the correct status is **partially resolved**.

2.10.4. Remaining open problems recorded by the database

2.10.4.1. Open problem 1

Prove that every compact orientable Riemannian surface (M) of genus (2) with one boundary component identified with the Riemannian circle S^1 of length 2π , and satisfying

$$d_M(x, y) \geq d_{S^1}(x, y) \quad \text{for all } x, y \in \partial M,$$

has area at least 2π .

Reference supplied by the database. Mikhail G. Katz, “Filling Area Conjecture,” *Website for Systolic Geometry and Topology*, section explicitly offering a prize for the genus-(2) case, accessed July 21, 2026, <https://u.cs.biu.ac.il/~katzmik/sgtdirectory/filling.html>. Current open-status verification: Joseph Briggs and Chris Wells, “A Discrete View of Gromov’s Filling Area Conjecture,” arXiv preprint (2026), Conjecture 1.1 and the discussion following it, arXiv: [2602.17859](https://arxiv.org/abs/2602.17859).

2.10.5. Primary-source pointer

The originating ICM record is *Volume Comparison via Boundary Distances* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=775>.

2.11. 51. Isoperimetric inequalities and asymptotic geometry [2014-geom-126]

Record: 2014-geom-126

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Isoperimetric inequalities and asymptotic geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1067>

2.11.1. Problem statement

Let X be a proper cocompact $\text{CAT}(0)$ -space, and let its Euclidean rank be k , meaning that k is the maximal dimension of an isometrically embedded copy of Euclidean space in X . For a metric space X and $r \geq 0$, define $\text{FV}_{m+1}(X, r)$ to be the supremum of $\text{Fillvol}_X(z)$ over all Lipschitz m -cycles z in X with mass $\mathbf{M}(z) \leq r$, where $\text{Fillvol}_X(z)$ is the least mass of a Lipschitz $(m+1)$ -chain c in X satisfying $\partial c = z$, and is $+\infty$ if no such c exists. Prove that for every integer $m \geq k$, there exists a constant C such that

$$\text{FV}_{m+1}(X, r) \leq Cr$$

for all $r \geq 0$.

2.11.2. Historical background, literature, and current progress

2.11.3. Verdict: partially resolved

2.11.3.1. Precise problem

The problem asks for Gromov's isoperimetric gap/rank conjecture for proper cocompact $\text{CAT}(0)$ spaces: if X has Euclidean rank k , then every Lipschitz m -cycle in X , for every $m \geq k$, should admit a Lipschitz $(m+1)$ -chain filling with mass bounded linearly by the mass of the cycle. Equivalently, since the asymptotic rank equals the Euclidean rank for proper cocompact $\text{CAT}(0)$ spaces, it asks whether $\text{CAT}(0)$ spaces of asymptotic rank at most m satisfy a linear m -dimensional filling inequality for all m -cycles.

2.11.3.2. Status

I did not find a published or citable theorem proving the full statement for all proper cocompact $\text{CAT}(0)$ spaces and all $m \geq k$. The literature gives substantial progress and several important special cases, but recent papers still state the general $\text{CAT}(0)$ rank ≥ 2 case as incomplete. Therefore the correct verdict is **partially resolved**.

2.11.3.3. Original source and formulation

- **Title:** *Asymptotic invariants of infinite groups*
- **Author:** Mikhail Gromov
- **Venue/year:** In *Geometric Group Theory, Vol. 2 (Sussex, 1991)*, London Mathematical Society Lecture Note Series 182, Cambridge University Press, 1993, pp. 1–295
- **URL:** <https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/492.pdf>
- **Role:** The conjectural linear filling statement is traced in later papers to Gromov, section 6.B2, p. 128(b).
- **Title:** *Guido's Book of Conjectures*
- **Collector/editor:** Indira Chatterji
- **Venue/year:** L'Enseignement Mathématique 54 (2008)
- **URL:** <https://chatterji.perso.math.cnrs.fr/papers/GMFinal.pdf>
- **Role:** Conjecture 45.1 records the statement in essentially the same form: every proper cocompact CAT(0) space of Euclidean rank r should admit a linear isoperimetric inequality for k -cycles for every $k \geq r$.

2.11.3.4. Progress and related results

- **Title:** *Higher-dimensional linear isoperimetric inequalities in hyperbolic groups*
- **Author:** Urs Lang
- **Venue/year:** International Mathematics Research Notices 2000, no. 13, 709–717
- **DOI:** 10.1155/S1073792800000384
- **Progress:** Proves higher-dimensional linear filling inequalities in the hyperbolic setting. This covers the rank-one/hyperbolic part of the conjectural picture under the standard geometric hypotheses, but not general higher-rank CAT(0) spaces.
- **Title:** *The asymptotic rank of metric spaces*
- **Author:** Stefan Wenger
- **Venue/year:** Commentarii Mathematici Helvetici 86 (2011), 247–275
- **DOI:** 10.4171/CMH/223
- **arXiv:** math/0701212
- **Progress:** Shows that for complete metric spaces with cone-type inequalities, including CAT(0) spaces, cycles in dimensions at least the asymptotic rank satisfy a sub-Euclidean filling inequality: $FV_{m+1}(r) = o(r^{(m+1)/m})$. It also records that for proper cocompact CAT(0) spaces the asymptotic rank equals Euclidean rank. This is strong evidence/progress but is weaker than the required linear bound $O(r)$.
- **Title:** *Characterizations of higher rank hyperbolicity*
- **Authors:** Tommaso Goldhirsch and Urs Lang
- **Venue/year:** Mathematische Zeitschrift 305 (2023), Article 13

- **DOI:** 10.1007/s00209-023-03339-x
- **arXiv:** 2112.15565
- **Progress:** Establishes a linear filling inequality for n -cycles with controlled density in spaces of asymptotic rank at most n . This is a restricted class of cycles, so it does not imply the stated supremum over all Lipschitz m -cycles.
- **Title:** *Linear isoperimetric inequality for homogeneous Hadamard manifolds*
- **Author:** Hjalti Isleifsson
- **Venue/year:** Journal of Topology and Analysis, online 2023 / issue 2025, 8 pp.
- **DOI:** 10.1142/S1793525323500334
- **arXiv:** 2203.09166
- **Progress:** Proves the expected linear inequality for homogeneous Hadamard manifolds, extending the known symmetric-space case. This is a significant special class, not all proper cocompact CAT(0) spaces.
- **Title:** *Minimal tetrahedra and an isoperimetric gap theorem in non-positive curvature*
- **Authors:** Cornelia Drutu, Urs Lang, Panos Papasoglu, and Stephan Stadler
- **Venue/year:** arXiv preprint, 2025
- **arXiv/DOI:** arXiv:2502.03389, <https://doi.org/10.48550/arXiv.2502.03389>
- **Progress:** For proper CAT(0) spaces of asymptotic rank at most 2, proves a homotopical δ -isoperimetric inequality for Lipschitz 2-spheres, with exponent $1 + \delta$, and related results for bounded-genus surfaces. The paper explicitly notes that this is weaker than the conjectured linear bound and does not handle homological fillings of general 2-cycles.
- **Title:** *Isoperimetric inequalities in Hadamard spaces of asymptotic rank two*
- **Authors:** Urs Lang, Stephan Stadler, and David Urech
- **Venue/year:** arXiv preprint, 2025
- **arXiv/DOI:** arXiv:2506.04882, <https://doi.org/10.48550/arXiv.2506.04882>
- **Progress:** Proves a homological δ -isoperimetric inequality $M(V) \leq CM(T)^{1+\delta}$ for general k -cycles, $k \geq 2$, in CAT(0) spaces of asymptotic rank at most 2 and finite asymptotic Nagata dimension; it covers examples such as Hadamard 3-manifolds and finite-dimensional CAT(0) cube complexes. The authors explicitly leave open whether a linear filling inequality holds even under their assumptions and whether the finite asymptotic Nagata dimension assumption can be removed.

2.11.3.5. What remains unresolved

The full problem asks for a uniform linear filling bound for **all** Lipschitz m -cycles in **every** proper cocompact CAT(0) space, for **every** $m \geq$ Euclidean rank. Current literature proves the conjectured linear bound only in special settings such as hyperbolic/rank-one cases, symmetric or homogeneous Hadamard manifolds, and restricted cycle classes such as controlled-density cycles. For general proper cocompact CAT(0) spaces of rank at least 2, the best general theorem I found is sub-Euclidean or $1 + \delta$ -type, not linear. Thus the exact statement is not fully resolved.

2.11.4. Remaining open problems recorded by the database

2.11.4.1. Open problem 1

Let X be a CAT(0) space of asymptotic rank at most 2 and finite asymptotic Nagata dimension. Prove that, for every integer $k \geq 2$, there is a constant $C = C(X, k)$ such that every compactly supported integral k -cycle T in X has a compactly supported integral filling V satisfying

$$\partial V = T, \quad \mathbf{M}(V) \leq C \mathbf{M}(T).$$

Reference supplied by the database. Urs Lang, Stephan Stadler, and David Urech, “Isoperimetric inequalities in Hadamard spaces of asymptotic rank two,” *Advances in Mathematics* **502** (2026), Article 111123. Open question immediately following Theorem 1.1 (whether the theorem’s $\mathbf{M}(T)^{1+\delta}$ estimate can be replaced by a linear isoperimetric filling inequality). DOI: [10.1016/j.aim.2026.111123](https://doi.org/10.1016/j.aim.2026.111123). Preprint: [arXiv:2506.04882](https://arxiv.org/abs/2506.04882).

2.11.5. Primary-source pointer

The originating ICM record is *Isoperimetric inequalities and asymptotic geometry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1067>.

2.12. 52. Metric dimension reduction A snapshot of the Ribe program [2018-geom-144]

Record: 2018-geom-144

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.12.1. Problem statement

Prove or disprove that for every sequence $(X_n)_{n=1}^{\infty}$ such that each X_n is a linear subspace of $\ell_{\infty}^n = (\mathbb{R}^n, \|\cdot\|_{\infty})$ and

$$\limsup_{n \rightarrow \infty} \frac{\dim(X_n)}{\log n} = \infty,$$

the Pythagorean direct sum

$$X = \left(\bigoplus_{n=1}^{\infty} X_n \right)_2 = \left\{ (x_n)_{n=1}^{\infty} : x_n \in X_n, \sum_{n=1}^{\infty} \|x_n\|_{\infty}^2 < \infty \right\}, \quad \|(x_n)\|_X = \left(\sum_{n=1}^{\infty} \|x_n\|_{\infty}^2 \right)^{1/2},$$

is finitarily universal: there exists a constant $\alpha \geq 1$ such that every finite metric space $(\mathcal{F}, d_{\mathcal{F}})$ admits a bi-Lipschitz embedding into X with distortion at most α , i.e., there are a map $f : \mathcal{F} \rightarrow X$ and a scaling factor $\tau > 0$ such that for all $u, v \in \mathcal{F}$,

$$\tau d_{\mathcal{F}}(u, v) \leq \|f(u) - f(v)\|_X \leq \alpha \tau d_{\mathcal{F}}(u, v).$$

2.12.2. Historical background, literature, and current progress

2.12.3. Verdict: partially resolved

2.12.3.1. Precise problem

The problem asks whether every sequence of linear subspaces $X_n \subseteq \ell_{\infty}^n$ with

$$\limsup_{n \rightarrow \infty} \frac{\dim X_n}{\log n} = \infty$$

has the property that the Hilbert/Pythagorean sum $(\oplus_n X_n)_2$ is finitarily universal, i.e. embeds every finite metric space with a distortion bound independent of the finite metric space.

Equivalently in the formulation used in the literature, this is the universality version of Pisier's dichotomy problem. A standard sufficient route is to prove that, along such a sequence, the subspaces X_n contain uniformly isomorphic copies of $\ell_{\infty}^{m_n}$ with $m_n \rightarrow \infty$. That would imply finitary universality, since every finite metric space embeds isometrically into some finite-dimensional ℓ_{∞}^m . The unresolved point is whether the hypothesis $\dim X_n / \log n \rightarrow \infty$ along a subsequence is already enough for this.

2.12.3.2. Literature checked

Original source

- **Title:** *Remarques sur un résultat non publié de B. Maurey*
- **Author:** Gilles Pisier
- **Venue/year:** Séminaire d'analyse fonctionnelle 1980-1981, Exposé V, École Polytechnique, Palaiseau, 1981
- **DOI/arXiv/URL:** no DOI found; Numdam URL: https://www.numdam.org/item/SAF_1980-1981____A5_0/
- **Status relevance:** Pisier formulates the subspace analogue of Maurey's quotient theorem and the associated threshold question for subspaces of ℓ_∞^m . The note identifies the logarithmic threshold as the natural boundary: if $\dim E$ is only of order $\log m$, no conclusion of this kind can be forced, while the conjectural statement asks for ℓ_∞^k -subspaces when $\dim E$ is asymptotically larger than $\log m$. It is the source of the problem, not a resolution.

Survey formulation of the exact problem

- **Title:** *Metric dimension reduction: A snapshot of the Ribe program*
- **Author:** Assaf Naor
- **Venue/year:** Proceedings of the International Congress of Mathematicians, Rio de Janeiro 2018, Vol. I, pp. 759-837; World Scientific, 2018/2019
- **DOI/arXiv/URL:** DOI 10.1142/9789813272880_0029; arXiv:1809.02376
- **Status relevance:** Naor states exactly the stated metric-universality question as Question 6, "Pisier's dichotomy problem": for arbitrary $X_n \subseteq \ell_\infty^n$ with $\limsup \dim(X_n)/\log n = \infty$, does $(X_1 \oplus X_2 \oplus \dots)_2$ become universal? The survey explicitly lists this among concrete universality questions whose answer is unknown, and records only stronger-growth cases as known.

Partial progress: proportional/large-dimensional cases

- **Title:** *Large subspaces of ℓ_∞^n and estimates of the Gordon-Lewis constant*
- **Authors:** Tadeusz Figiel and William B. Johnson
- **Venue/year:** Israel Journal of Mathematics 37 (1980), no. 1-2, pp. 92-112
- **DOI/arXiv/URL:** DOI 10.1007/BF02762871
- **Status relevance:** This gives sharp information for very large/proportional-dimensional subspaces of ℓ_∞^N . It is far from the logarithmic-threshold regime in the problem statement.
- **Title:** *Subspaces of L_∞^N , arithmetical diameter and Sidon sets*
- **Author:** Jean Bourgain
- **Venue/year:** In *Probability in Banach Spaces V*, Lecture Notes in Mathematics 1153, Springer, 1985, pp. 96-127
- **DOI/arXiv/URL:** DOI 10.1007/BFb0074947; IHES archive PDF: https://repositaires.ihs.fr/FONDS_IHES/I_Prepublications/BOURGAIN/1984-1989/M_85_11/M_85_11_web.pdf
- **Status relevance:** Bourgain proves a partial form of the desired conclusion for substantially larger subspaces. As summarized by Naor and by Schechtman-Tomczak-Jaegermann, Bourgain's method yields the desired uniform ℓ_∞^k -subspaces, and hence

the stated finitary universality conclusion, under a stronger hypothesis such as $\dim X_N$ exceeding about $(\log N)^4$. This does not cover arbitrary sequences with $\dim X_N / \log N \rightarrow \infty$.

Partial progress: current best general quantitative bound found

- **Title:** *Polylog Dimensional Subspaces of ℓ_∞^N*
- **Authors:** Gideon Schechtman and Nicole Tomczak-Jaegermann
- **Venue/year:** In *Geometric Aspects of Functional Analysis: Israel Seminar (GAFA) 2017-2019, Volume II*, Lecture Notes in Mathematics 2266, Springer, 2020, pp. 325-334
- **DOI/arXiv/URL:** DOI 10.1007/978-3-030-46762-3_15; arXiv:1812.03678
- **Status relevance:** The paper states Pisier's dichotomy problem in the equivalent Banach-space form: whether for every sequence $n_N \leq N$ with $n_N / \log N \rightarrow \infty$, every n_N -dimensional subspace of ℓ_∞^N contains uniformly isomorphic $\ell_\infty^{m_N}$ with $m_N \rightarrow \infty$. Its Theorem 1 proves the stronger-hypothesis result: if $n > (\log N \log \log N)^2$, then every n -dimensional subspace of ℓ_∞^N contains a subspace of dimension

$$m = c(\eta) \frac{\sqrt{n}}{\log N \log \log N}$$

whose distance from ℓ_∞^m is at most $1 + \eta$. Thus the conclusion follows when

$$\limsup_{N \rightarrow \infty} \frac{\dim X_N}{(\log N)^2 (\log \log N)^2} = \infty,$$

but this is much stronger than the stated assumption $\limsup \dim X_N / \log N = \infty$.

Generic-case progress

- **Title:** *Random Banach spaces: The limitations of the method*
- **Authors:** Piotr Mankiewicz and Stanislaw J. Szarek
- **Venue/year:** *Mathematika* 41 (1994), no. 2, pp. 239-250; corrigendum *Mathematika* 42 (1995), pp. 220-221
- **DOI/arXiv/URL:** DOI 10.1112/S002557930000735X; arXiv:math/9305203
- **Status relevance:** The authors prove a version of "Pisier's dichotomy conjecture" for generic subspaces of ℓ_∞^N in the Haar-measure sense. This is not a resolution of the problem because the statement quantifies over every possible sequence of subspaces.

2.12.3.3. Why the full statement is not resolved

The strongest general result found is Schechtman-Tomczak-Jaegermann's 2020 theorem, which requires roughly $(\log N)^2 (\log \log N)^2$ -scale dimension growth. Bourgain's earlier result requires an even stronger polylogarithmic condition. Generic/random-subspace results do not address arbitrary subspaces. I found no published or citable theorem proving the exact hypothesis, and no published or citable counterexample disproving it.

What remains open is the full logarithmic-threshold case: arbitrary $X_N \subseteq \ell_\infty^N$ with $\dim X_N / \log N$ unbounded but possibly much smaller than $(\log N)^2 (\log \log N)^2$. The current literature therefore gives partial progress, not a full resolution.

2.12.4. Remaining open problems recorded by the database

2.12.4.1. Open problem 1

Construct sequences of finite-dimensional normed spaces E_n, F_n with $\dim E_n = \dim F_n = n$ and subspaces $Z_n \subseteq \mathcal{L}(E_n, F_n)$ such that $\dim Z_n/n \rightarrow \infty$, while there is a universal integer k_0 for which no Z_n contains a subspace that is 2-isomorphic to ℓ_∞^k for any $k > k_0$.

Reference supplied by the database. Gideon Schechtman and Nicole Tomczak-Jaegermann, “Polylog Dimensional Subspaces of ℓ_∞^N ,” in Bo’az Klartag and Emanuel Milman (eds.), *Geometric Aspects of Functional Analysis: Israel Seminar (GAFA) 2017–2019, Volume II*, Lecture Notes in Mathematics **2266**, Springer, 2020, pp. 325–334, final Remark 1 (the proposed operator-space route to a negative solution of Pisier’s dichotomy problem), https://doi.org/10.1007/978-3-030-46762-3_15; arXiv:1812.03678.

2.12.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

2.13. 53. Metric dimension reduction A snapshot of the Ribe program [2018-geom-145]

Record: 2018-geom-145

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.13.1. Problem statement

For each positive integer k , let $P_1(\mathbb{R}^k)$ denote the set of Borel probability measures μ on Euclidean $\mathbb{R}^k$ with finite first moment, meaning that $\int_{\mathbb{R}^k} \|x - y\|_2 d\mu(y) < \infty$ for every $x \in \mathbb{R}^k$, equipped with the Wasserstein-1 metric

$$W_1(\mu, \nu) = \inf_{\pi} \int_{\mathbb{R}^k \times \mathbb{R}^k} \|x - y\|_2 d\pi(x, y),$$

where the infimum is over all Borel probability measures π on $\mathbb{R}^k \times \mathbb{R}^k$ whose marginals are μ and ν . Is $(P_1(\mathbb{R}^2), W_1)$ universal in the finitary bi-Lipschitz sense: does there exist a constant $\alpha \geq 1$ such that every finite metric space $(\mathcal{F}, d_{\mathcal{F}})$ admits a map $f : \mathcal{F} \rightarrow P_1(\mathbb{R}^2)$ and a scaling factor $\tau > 0$ satisfying

$$\tau d_{\mathcal{F}}(x, y) \leq W_1(f(x), f(y)) \leq \alpha \tau d_{\mathcal{F}}(x, y)$$

for all $x, y \in \mathcal{F}$? More generally, does there exist an integer $k \geq 2$ for which $(P_1(\mathbb{R}^k), W_1)$ has this universality property?

2.13.2. Historical background, literature, and current progress

this problem is fully open

2.13.3. Remaining open problems recorded by the database

2.13.3.1. Open problem 1

Is the Wasserstein space $(P_1(\ell_2), W_1)$ universal in the finitary bi-Lipschitz sense?

Reference supplied by the database. Assaf Naor, “Metric Dimension Reduction: A Snapshot of the Ribe Program,” in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018*, Vol. I: Plenary Lectures, eds. Boyan Sirakov, Paulo Ney de Souza, and Marcelo Viana, World Scientific, 2018, pp. 759–837, §1.2.2, immediately following “Bourgain’s

universality problem” (explicitly stating that it is unknown whether $P_1(\ell_2)$ is universal). arXiv:1809.02376, <https://arxiv.org/abs/1809.02376>.

2.13.4. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

2.14. 54. Metric dimension reduction A snapshot of the Ribe program [2018-geom-146]

Record: 2018-geom-146

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.14.1. Problem statement

Let $p \in (1, 2)$. Let $\ell_p \widehat{\otimes} \ell_2$ denote the projective tensor product of the Banach spaces ℓ_p and ℓ_2 , i.e. the completion of their algebraic tensor product under the norm whose unit ball is the convex hull of $\{x \otimes y : \|x\|_{\ell_p} \leq 1, \|y\|_{\ell_2} \leq 1\}$. Determine whether $\ell_p \widehat{\otimes} \ell_2$, viewed as a metric space with its norm metric, is universal in the following finitary sense: does there exist a constant $\alpha \geq 1$ such that every finite metric space $(\mathcal{F}, d_{\mathcal{F}})$ admits a map $f : \mathcal{F} \rightarrow \ell_p \widehat{\otimes} \ell_2$ and a scaling factor $\tau > 0$ satisfying

$$\tau d_{\mathcal{F}}(x, y) \leq \|f(x) - f(y)\|_{\ell_p \widehat{\otimes} \ell_2} \leq \alpha \tau d_{\mathcal{F}}(x, y)$$

for all $x, y \in \mathcal{F}$.

2.14.2. Historical background, literature, and current progress

2.14.3. Verdict: partially resolved

2.14.3.1. Precise problem

For each fixed $p \in (1, 2)$, decide whether the Banach space

$$X_p = \ell_p \widehat{\otimes} \pi \ell_2$$

is finitarily universal as a metric space, i.e. whether there is a constant α independent of the finite metric space such that every finite metric embeds into X_p with distortion at most α , after rescaling.

For Banach targets this is equivalent to asking whether X_p fails to have finite cotype. Indeed, failure of finite cotype is equivalent, by Maurey-Pisier, to uniform finite representability of ℓ_∞^n , and every n -point metric embeds isometrically into ℓ_∞^n . Conversely, finite cotype gives the standard metric-cotype obstruction to finite-metric universality.

2.14.3.2. Literature and progress

1. Gilles Pisier, “Random series of trace class operators”, in *Proceedings Cuarto CLAPEM Mexico 1990. Contribuciones en probabilidad y estadística matemática*,

- pp. 29-42, 1992. arXiv:1103.2090, URL: <https://arxiv.org/abs/1103.2090>. Pisier formulates the relevant open finite-cotype problem for projective tensor products. In the final remarks he states that $L_p \widehat{\otimes}_\pi L_q$ has cotype $\max(p, q)$ when $2 \leq p, q < \infty$, but that it remains open whether $L_p \widehat{\otimes}_\pi L_q$ has finite cotype for $1 < p < 2$ and $1 < q \leq 2$, explicitly including $L_p \widehat{\otimes}_\pi L_2$. Since ℓ_p is an L_p -space, this is the original source-level open problem covering the stated case.
2. Gilles Pisier, “Factorization of operator valued analytic functions”, *Advances in Mathematics* 93(1):61-125, 1992. DOI: 10.1016/0001-8708(92)90025-G. This gives the positive finite-cotype/nonuniversality result for $L_p \widehat{\otimes}_\pi L_q$ in the range $p, q \in [2, \infty)$. It verifies nonuniversality at and beyond the Hilbert endpoint, but it does not cover the stated range $1 < p < 2, q = 2$.
 3. Nicole Tomczak-Jaegermann, “The moduli of smoothness and convexity and the Rademacher averages of the trace classes S_p ($1 \leq p < \infty$)”, *Studia Mathematica* 50(2):163-182, 1974. URL: <https://eudml.org/doc/217886>. This includes the finite-cotype behavior of trace classes; in particular $\ell_2 \widehat{\otimes}_\pi \ell_2 \cong S_1$ has cotype 2, so the endpoint $p = 2$ is not universal. This is an endpoint result, not a resolution for $1 < p < 2$.
 4. Jop Briët, Assaf Naor, and Oded Regev, “Locally decodable codes and the failure of cotype for projective tensor products”, *Electronic Research Announcements in Mathematical Sciences* 19:120-130, 2012. DOI: 10.3934/era.2012.19.120; arXiv:1208.0539, URL: <https://arxiv.org/abs/1208.0539>. Theorem 1.1 proves that if $p_1, p_2, p_3 \in (1, \infty)$ and $1/p_1 + 1/p_2 + 1/p_3 \leq 1$, then $\ell_{p_1} \widehat{\otimes}_\pi \ell_{p_2} \widehat{\otimes}_\pi \ell_{p_3}$ does not have finite cotype. It also implies that for every $p \in (1, \infty)$ there exists a Banach space X of finite cotype such that $\ell_p \widehat{\otimes}_\pi X$ fails finite cotype. This is substantial partial progress, but X is not ℓ_2 , and the theorem is three-fold rather than the two-fold tensor product in the problem statement.
 5. Assaf Naor, “Metric dimension reduction: A snapshot of the Ribe program”, in *Proceedings of the International Congress of Mathematicians - Rio de Janeiro 2018, Vol. I. Plenary Lectures*, pp. 759-837, World Scientific, 2018/2019. DOI: 10.1142/9789813272880_0029; arXiv:1809.02376, URL: <https://arxiv.org/abs/1809.02376>. Question 8 in this survey is exactly the question: for $p \in (1, 2)$, is $\ell_p \widehat{\otimes}_\pi \ell_2$ universal? The survey records the known endpoint results and identifies Briët-Naor-Regev’s three-fold theorem as the best available result in this direction, not as a solution of the two-fold problem.
 6. Qingying Bu and Patrick N. Dowling, “On type and cotype of tensor products of Banach spaces”, *Journal of Mathematical Analysis and Applications* 553(2), Article 129913, 2026. DOI: 10.1016/j.jmaa.2025.129913, URL: <https://www.sciencedirect.com/science/article/pii/S0022247X25006948>. This recent adjacent paper studies type/cotype phenomena for tensor products, but the available article metadata and abstract do not state a theorem resolving whether $\ell_p \widehat{\otimes}_\pi \ell_2$ has finite cotype for $1 < p < 2$. Searches for the exact phrase, the Pisier/Naor question, and post-2018 finite-cotype tensor-product results found no citable full resolution.

2.14.3.3. What remains unresolved

No published or otherwise citable theorem found in the literature proves either:

- $\ell_p \widehat{\otimes}_\pi \ell_2$ fails finite cotype for every $1 < p < 2$, which would imply metric universality; or
- $\ell_p \widehat{\otimes}_\pi \ell_2$ has finite cotype for every $1 < p < 2$, which would imply nonuniversality.

Thus the exact problem remains open, while related endpoint cases and higher-fold/projective-

tensor-product variants constitute partial progress.

2.14.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

2.14.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

2.15. 55. Metric dimension reduction A snapshot of the Ribe program [2018-geom-147]

Record: 2018-geom-147

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.15.1. Problem statement

For metric spaces (F, d_F) and (X, d_X) , let $C_X(F)$ denote the infimum of all $\alpha \geq 1$ for which there exist a map $f : F \rightarrow X$ and a scaling factor $\tau > 0$ such that

$$\tau d_F(u, v) \leq d_X(f(u), f(v)) \leq \alpha \tau d_F(u, v)$$

for every $u, v \in F$. Say that a metric space X is finitarily universal if there is a constant $\alpha < \infty$ such that every finite metric space F satisfies $C_X(F) \leq \alpha$.

Is there a universal constant $\theta_0 > 0$ such that every metric space X that is not finitarily universal admits an exponent $\theta_X > \theta_0$ and a constant $c_X > 0$ with the following property: for arbitrarily large integers n , there is an n -point metric space G_n such that

$$C_X(G_n) \geq c_X (\log n)^{\theta_X}?$$

2.15.2. Historical background, literature, and current progress

2.15.3. Verdict: partially resolved

2.15.3.1. Precise problem

The problem asks for the uniform-exponent version of the metric cotype dichotomy. Mendel and Naor's metric cotype dichotomy says that if a metric space X is not finitarily universal, then X admits arbitrarily large finite metric spaces whose least distortion into X grows at least like a positive power of $\log n$. The open part in this problem is whether that positive exponent can always be chosen larger than one universal constant $\theta_0 > 0$, independent of X .

2.15.3.2. Current literature

1. Metric cotype

- Title: *Metric Cotype*
- Authors: Manor Mendel and Assaf Naor
- Venue: *Annals of Mathematics* 168(1), 247-298
- Year: 2008

- DOI/arXiv/URL: DOI 10.4007/annals.2008.168.247; arXiv math/0506201; <https://arxiv.org/abs/math/0506201>
- Status relevance: The paper proves the metric cotype dichotomy in an X -dependent form. In the notation of the problem, it gives: if X is not universal/finitarily universal, then there is some exponent $\alpha(X) > 0$ and arbitrarily large finite metric spaces M_n with $C_X(M_n) \gtrsim_X (\log |M_n|)^{\alpha(X)}$. This is substantial partial progress, but it does not give a lower bound on $\alpha(X)$ that is uniform over all non-finitarily universal metric spaces.

2. Metric dimension reduction: A snapshot of the Ribe program

- Title: *Metric dimension reduction: A snapshot of the Ribe program*
- Author: Assaf Naor
- Venue: *Proceedings of the International Congress of Mathematicians - Rio de Janeiro 2018*, Vol. I, World Scientific, 759-837
- Year: 2018
- DOI/arXiv/URL: arXiv 1809.02376; <https://arxiv.org/abs/1809.02376>
- Status relevance: The survey restates the Mendel-Naor theorem as Theorem 9 and then asks the exact uniform-exponent problem as Question 10. It explicitly identifies the unknown point as whether the logarithmic power $\theta(X)$ in the dichotomy can be bounded below by a universal positive constant. It also notes that known computed examples at that time were consistent even with exponent at least 1, while a proposed Wasserstein scenario could give a negative answer.

3. Snowflake universality of Wasserstein spaces

- Title: *Snowflake universality of Wasserstein spaces*
- Authors: Alexandr Andoni, Assaf Naor, and Ofer Neiman
- Venue: *Annales Scientifiques de l'Ecole Normale Supérieure* 51(3), 657-700
- Year: 2018
- DOI/arXiv/URL: DOI 10.24033/asens.2363; arXiv 1509.08677; <http://numdam.org/articles/10.24033/asens.2363/>
- Status relevance: Section 1.5.1 formulates the same metric cotype dichotomy problem as Question 13. The paper gives Wasserstein snowflake universality results and asks Question 14 about whether every n -point metric space embeds into $P_p(\mathbb{R}^3)$ with distortion $\lesssim_p (\log n)^{1-1/p}$. The authors explain that a positive answer to that Wasserstein question would imply exponents tending to 0 as $p \rightarrow 1^+$, hence would disprove the uniform-exponent problem. However, Question 14 is left open and therefore does not resolve the problem.

2.15.3.3. What remains unresolved

I found no published or citable preprint result proving either of the two conclusions needed for a full resolution:

- a positive theorem giving a universal $\theta_0 > 0$ for every non-finitarily universal metric space X ; or
- a negative construction of non-finitarily universal spaces X_j for which every admissible logarithmic obstruction exponent tends to 0.

Thus the exact problem is not fully resolved. The correct status is **partially resolved**: the X -dependent metric cotype dichotomy is known, and Wasserstein-space work identifies a plau-

sible route to a negative answer, but the universal-exponent question remains open.

2.15.4. Remaining open problems recorded by the database

2.15.4.1. Open problem 1

For $p \in (1, 2]$, is there a constant $C_p < \infty$ such that every n -point metric space X embeds into $(\mathcal{D}_p(\mathbb{R}^3), W_p)$ with distortion at most $C_p(\log n)^{1-1/p}$?

Reference supplied by the database. Alexandr Andoni, Assaf Naor, and Ofer Neiman, “Snowflake Universality of Wasserstein Spaces,” *Annales Scientifiques de l’École Normale Supérieure*, 4e série, 51 (2018), no. 3, 657–700, Question 14 (§1.5.1). DOI: 10.24033/asens.2363. arXiv:1509.08677. <https://doi.org/10.24033/asens.2363>.

2.15.4.2. Open problem 2

Let G be an n -vertex 3-regular graph whose normalized adjacency matrix has second-largest eigenvalue $\lambda_2(G) \leq 99/100$, and let $\Sigma_{\lceil \log n \rceil}(G)$ be obtained by replacing every edge of G by a path of $\lceil \log n \rceil$ edges. What is the asymptotic growth rate of the least distortion $c_{\mathcal{D}_2(\mathbb{R}^3)}(\Sigma_{\lceil \log n \rceil}(G))$? In particular, is it $O(\sqrt{\log n})$, or even $o(\log n)$?

Reference supplied by the database. Alexandr Andoni, Assaf Naor, and Ofer Neiman, “Snowflake Universality of Wasserstein Spaces,” *Annales Scientifiques de l’École Normale Supérieure*, 4e série, 51 (2018), no. 3, 657–700, Question 36 (§6.3, p. 694). DOI: 10.24033/asens.2363. arXiv:1509.08677. <https://doi.org/10.24033/asens.2363>.

2.15.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

2.16. 56. Metric dimension reduction A snapshot of the Ribe program [2018-geom-148]

Record: 2018-geom-148

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.16.1. Problem statement

Suppose that $p \in [1, \infty) \setminus \{2\}$. Do there exist constants $\alpha = \alpha(p)$ and $\beta = \beta(p)$ in $[1, \infty)$, depending only on p , such that for every $n \in \mathbb{N}$ and every subset $C \subseteq \ell_p$ with $|C| = n$, there is a finite-dimensional normed space $(Y, \|\cdot\|_Y)$ with $\dim Y \leq \beta \log n$ and a map $f : C \rightarrow Y$ for which there exists a scaling factor $\tau > 0$ satisfying

$$\tau \|x - y\|_{\ell_p} \leq \|f(x) - f(y)\|_Y \leq \alpha \tau \|x - y\|_{\ell_p}$$

for all $x, y \in C$?

2.16.2. Historical background, literature, and current progress

2.16.3. Verdict: partially resolved

2.16.3.1. Precise problem

For a fixed $p \in [1, \infty) \setminus \{2\}$, decide whether there are constants $\alpha(p), \beta(p) < \infty$ such that every n -point subset $C \subseteq \ell_p$ embeds with bi-Lipschitz distortion at most $\alpha(p)$ into some normed space Y of dimension at most $\beta(p) \log n$. The target space is allowed to depend on C and is not required to be an ℓ_p space or a subspace of ℓ_p .

2.16.3.2. Status

I found no published or citable result that proves either the asserted $O(\log n)$ constant-distortion reduction for all finite subsets of ℓ_p , $p \neq 2$, or a counterexample to it when the target is an arbitrary finite-dimensional normed space. The problem is therefore not fully resolved.

It is not best described as fully open, because there is substantial partial progress on nearby formulations: growing-distortion Hilbert embeddings for subsets of L_p , strong lower bounds when the target is restricted to ℓ_1^k or ℓ_p^k , and recent arbitrary-target lower bounds for Hamming metrics. None of these results has the hypotheses and conclusion needed to settle the exact question above.

2.16.3.3. Key literature and applicability checks

1. Metric dimension reduction: A snapshot of the Ribe program

Assaf Naor. Proceedings of the International Congress of Mathematicians–Rio de Janeiro 2018, Vol. I, World Scientific, 2018, pp. 759–837. DOI: 10.1142/9789813272880_0029; arXiv: 1809.02376; URL: <https://arxiv.org/abs/1809.02376>.

This is the source of the question. It explicitly asks whether every n -point subset of ℓ_p , $p \neq 2$, embeds with constant distortion into some $O(\log n)$ -dimensional normed space, and notes that even an $o(n)$ dimensional bound was open at the time. This source does not resolve the problem.

2. Euclidean distortion and the sparsest cut

Sanjeev Arora, James R. Lee, Assaf Naor. Journal of the American Mathematical Society 21(1):1–21, 2008. DOI: 10.1090/S0894-0347-07-00573-5; arXiv: math/0508154; URL: <https://arxiv.org/abs/math/0508154>.

This proves that every n -point metric of negative type, in particular every n -point subset of L_1 , embeds into Hilbert space with distortion $O(\sqrt{\log n \log \log n})$. As summarized by Naor's ICM article, this gives nonconstant Hilbert distortion for $p \in [1, 2]$. After Johnson-Lindenstrauss, the target can be made $O(\log n)$ -dimensional, but the distortion still grows with n . Hence it is only partial progress, not a solution to the constant distortion question.

3. Euclidean embedding, randomized clustering, and Lipschitz extension for finite and doubling subsets of L_p when $p > 2$

Assaf Naor, Kevin Ren. Ars Inveniendi Analytica, 2026, Paper No. 2, 60 pp. DOI: 10.15781/d72xfr59; arXiv: 2502.10543; URL: <https://arxiv.org/abs/2502.10543>.

This proves that for fixed $p > 2$, every n -point subset of L_p embeds into Hilbert space with distortion $p^3(\log n)^{1/2+o(1)}$, improving Bourgain's general $O(\log n)$ guarantee. Since $\ell_p \subset L_p$, this applies to the stated spaces for $p > 2$. Applying Johnson-Lindenstrauss gives an $O(\log n)$ -dimensional Hilbert target, but the distortion is still unbounded in n , so it does not imply the requested constants $\alpha(p), \beta(p)$.

4. The Power of Recursive Embeddings for ℓ_p Metrics

Robert Krauthgamer, Nir Petruschka, Shay Sapir. arXiv preprint, 2025. arXiv: 2503.18508; URL: <https://arxiv.org/abs/2503.18508>.

This gives additional state-of-the-art recursive embedding and decomposition results for ℓ_p metrics with $p > 2$, including improved Hilbert distortion bounds in some regimes. These are again growing-distortion results and therefore do not settle the constant-distortion $O(\log n)$ -dimension problem.

5. On the impossibility of dimension reduction in ℓ_1

Bo Brinkman, Moses Charikar. Journal of the ACM 52(5):766–788, 2005. DOI: 10.1145/1089023.1089026; URL: <https://dl.acm.org/doi/10.1145/1089023.1089026>.

This gives n -point subsets of ℓ_1 requiring polynomial dimension for constant-distortion embeddings into ℓ_1^k . It is strong evidence against a Johnson-Lindenstrauss analogue with the same target norm, but it does not rule out embeddings into an arbitrary k -dimensional normed space as allowed in the question.

6. Dimensionality of Hamming Metrics and Rademacher Type

Alexandros Eskenazis. 40th International Symposium on Computational Geometry (SoCG 2024), LIPIcs 293, Art. 55, 2024. DOI: 10.4230/LIPIcs.SocG.2024.55;

URL: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.SoCG.2024.55>.

This proves arbitrary-target dimension lower bounds for embeddings of Hamming cubes in terms of Rademacher type constants. Since Hamming cubes are subsets of ℓ_1 , this is relevant to the $p = 1$ case, but the lower bounds are compatible with dimension proportional to $\log|C|$ for arbitrary targets and hence do not disprove the desired statement.

2.16.3.4. What remains unresolved

The remaining gap is exactly the question: for any fixed $p \in [1, \infty) \setminus \{2\}$, no cited theorem gives a constant $\alpha(p)$ and logarithmic dimension constant $\beta(p)$ for embeddings of all finite subsets of ℓ_p into arbitrary finite-dimensional normed spaces, and no cited theorem proves that such constants cannot exist.

2.16.4. Remaining open problems recorded by the database

2.16.4.1. Open problem 1

Fix $p \in [1, \infty) \setminus \{2\}$. Do there exist a constant $\alpha_p < \infty$ and a function $k_p(n) = o(n)$ such that every n -point subset of ℓ_p embeds with distortion at most α_p into some normed space of dimension at most $k_p(n)$?

Reference supplied by the database. Assaf Naor, “Metric Dimension Reduction: A Snapshot of the Ribe Program,” in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018*, Vol. I: Plenary Lectures, eds. Boyan Sirakov, Paulo Ney de Souza, and Marcelo Viana, World Scientific, 2018, pp. 759–837, immediately after Question 13 (§1.3, p. 767), explicitly stating that even the bound $k = o(n)$ is open for every fixed $p \neq 2$. arXiv:1809.02376. <https://arxiv.org/abs/1809.02376>.

2.16.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

2.17. 57. Metric dimension reduction A snapshot of the Ribe program [2018-geom-150]

Record: 2018-geom-150

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.17.1. Problem statement

Does every subset S of a Hilbert space H , equipped with the metric d_S induced by H , that is doubling admit a bi-Lipschitz embedding into $\mathbb{R}^k$ for some $k \in \mathbb{N}$? Here “doubling” means that there exists $K \in [2, \infty)$ such that every closed ball $B_S(x, r)$ in S can be covered by at most K closed balls in S of radius $r/2$, and “bi-Lipschitz embedding into $\mathbb{R}^k$ ” means that there are $k \in \mathbb{N}$, a map $f : S \rightarrow \mathbb{R}^k$, and constants $0 < c \leq C < \infty$ such that

$$c d_S(x, y) \leq \|f(x) - f(y)\| \leq C d_S(x, y)$$

for all $x, y \in S$.

2.17.2. Historical background, literature, and current progress

2.17.3. Verdict: partially resolved

2.17.3.1. Precise problem

The problem asks the Lang-Plaut problem: if S is an arbitrary doubling subset of a Hilbert space H , with the metric inherited from H , must S admit a bi-Lipschitz embedding into some finite-dimensional Euclidean space $\mathbb{R}^k$?

The exact problem remains open in full generality. I found no published or citable preprint result proving either that every doubling Hilbert subset embeds bi-Lipschitzly into some $\mathbb{R}^k$, or constructing a doubling subset of Hilbert space that fails to do so.

2.17.3.2. Original source

- Title: “Bilipschitz Embeddings of Metric Spaces into Space Forms”
- Authors: Urs Lang and Conrad Plaut
- Venue: *Geometriae Dedicata* 87, 285-307
- Year: 2001
- DOI/URL: <https://doi.org/10.1023/A:1012093209450>
- Relevance: Question 2.4 asks whether every doubling subset of a Hilbert space admits a bilipschitz embedding into some Euclidean space. The paper proves positive embedding

results under additional geodesic/convex-bicombing hypotheses, but those hypotheses do not cover arbitrary subsets of Hilbert space.

2.17.3.3. Main partial progress and status evidence

- Title: “Assouad’s theorem with dimension independent of the snowflaking”
- Authors: Assaf Naor and Ofer Neiman
- Venue: Revista Matematica Iberoamericana 28(4), 1123-1142
- Year: 2012
- DOI/arXiv/URL: <https://doi.org/10.4171/RMI/706> ; arXiv:1012.2307
- Result/progress: For every doubling metric space (X, d) , each snowflake $(X, d^{1-\epsilon})$, $0 < \epsilon < 1/2$, embeds bi-Lipschitzly into $\mathbb{R}^{N(K)}$, where the dimension is independent of ϵ . Section 1.1 states the Lang-Plaut problem and explains that it remains open; the paper also proposes possible counterexample routes via snowflakes of the Heisenberg group. This is partial because it embeds snowflaked metrics, not the original Hilbert-subset metric.
- Title: “A doubling subset of L_p for $p > 2$ that is inherently infinite dimensional”
- Authors: Vincent Lafforgue and Assaf Naor
- Venue: Geometriae Dedicata 172, 387-398
- Year: 2014
- DOI/arXiv/URL: <https://doi.org/10.1007/s10711-013-9924-4> ; arXiv:1308.4554
- Result/progress: Theorem 1.1 constructs, for every $p > 2$, a doubling subset of L_p that does not admit a bi-Lipschitz embedding into any $\mathbb{R}^k$. The introduction explicitly says the Hilbert case and the Lang-Plaut question remain open. This is strong evidence against naive positive approaches, but it does not resolve the Hilbert case $p = 2$.
- Title: “On the Impossibility of Dimension Reduction for Doubling Subsets of ℓ_p ”
- Authors: Yair Bartal, Lee-Ad Gottlieb, and Ofer Neiman
- Venue: SIAM Journal on Discrete Mathematics 29(3), 1207-1222
- Year: 2015
- DOI/arXiv/URL: <https://doi.org/10.1137/140977655> ; arXiv:1308.4996
- Result/progress: Gives finite doubling subsets of ℓ_p , $p > 2$, for which embeddings into low-dimensional ℓ_p^d require large distortion, yielding another negative answer to the $p > 2$ analogue. It does not apply to ℓ_2 /Hilbert.
- Title: “No dimension reduction for doubling subsets of ℓ_q when $q > 2$ revisited”
- Authors: Florent P. Baudier, Krzysztof Swiecicki, and Andrew Swift
- Venue: Journal of Mathematical Analysis and Applications 504(2), Paper 125407
- Year: 2021
- DOI/arXiv/URL: <https://doi.org/10.1016/j.jmaa.2021.125407> ; arXiv:2103.05080

- Result/progress: Reproves and extends the $q > 2$ impossibility results, including obstructions for broader target classes. The paper formulates the Hilbert/ell_2 version as the original open problem and treats the $q > 2$ results as analogues, not as a solution.
- Title: “Bi-Lipschitz embeddings revisited”
- Authors: H. Movahedi-Lankarani and R. Wells
- Venue: arXiv preprint
- Year: 2025
- DOI/arXiv/URL: arXiv:2501.07648, <https://arxiv.org/abs/2501.07648>
- Result/progress: Gives sufficient conditions for compact metric-measure spaces, including compact subsets of Hilbert space satisfying additional hypotheses involving the canonical distance map and a measure. The paper explicitly states that the problem of whether every doubling subset of Hilbert space embeds into Euclidean space is still open. These hypotheses are not implied by doubling alone.
- Title: “Coarse and pointwise tangent fields”
- Authors: Guy C. David, Sylvester Eriksson-Bique, and Raanan Schul
- Venue: arXiv preprint
- Year: 2025; version checked: arXiv v2 dated 17 Apr 2026
- DOI/arXiv/URL: arXiv:2508.04653, <https://arxiv.org/abs/2508.04653>
- Result/progress: Shows that doubling subsets of Hilbert space admit pointwise and coarse tangent fields with dimension bounded by Nagata/Assouad dimension. This gives direct structural information for the exact class in the Lang-Plaut problem, but it is not a bi-Lipschitz embedding theorem into finite-dimensional Euclidean space.

2.17.3.4. What remains unresolved

The full arbitrary-Hilbert-subset case remains unresolved. The available positive results either require snowflaking, geodesic/convex structure, compact metric-measure hypotheses, or give structural tangent-field information rather than an embedding. The available negative results solve analogues in L_p or ℓ_p for $p > 2$, not in Hilbert space $p = 2$. Therefore no cited theorem’s hypotheses and conclusion cover all doubling subsets of Hilbert space with the original induced metric.

2.17.4. Remaining open problems recorded by the database

2.17.4.1. Open problem 1

For every $K > 1$, do there exist constants $a(K), b(K) > 0$ such that, for every metric space (X, d) with doubling constant at most K and every $\varepsilon \in (0, 1/2)$, there is a map $f : X \rightarrow \ell_2$ satisfying

$$a(K) \sqrt{\varepsilon} d(x, y)^{1-\varepsilon} \leq \|f(x) - f(y)\|_2 \leq d(x, y)^{1-\varepsilon} \quad (x, y \in X),$$

whose image $f(X)$ has doubling constant at most $b(K)$?

Reference supplied by the database. Assaf Naor and Ofer Neiman, “Assouad’s Theorem with Dimension Independent of the Snowflaking,” *Revista Matemática Iberoamericana* 28 (2012), no. 4, 1123–1142, Question 2 (§1.1). DOI: 10.4171/RMI/706. arXiv:1012.2307. <https://doi.org/10.4171/RMI/706>.

2.17.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

2.18. 58. Metric dimension reduction A snapshot of the Ribe program [2018-geom-151]

Record: 2018-geom-151

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.18.1. Problem statement

Does ℓ_1 admit average metric dimension reduction, in the following precise sense: is there a constant $\alpha \in [1, \infty)$ such that for every $n \in \mathbb{N}$ there is an integer k_n with

$$\lim_{n \rightarrow \infty} \frac{\log k_n}{\log n} = 0$$

and such that, for every n -point subset $C \subseteq \ell_1$, there are a linear subspace $F \subseteq \ell_1$ with $\dim(F) = k_n$ and a map $f : C \rightarrow F$ satisfying

$$\|f(x) - f(y)\|_1 \leq \alpha \|x - y\|_1 \quad \text{for all } x, y \in C$$

and

$$\frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|f(x) - f(y)\|_1 \geq \frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|x - y\|_1?$$

More generally, for $p \in [1, 2)$, does ℓ_p have the same property, with the ℓ_p norm in place of the ℓ_1 norm and with the constant α and the sequence $k_n = n^{o(1)}$ allowed to depend on p ?

2.18.2. Historical background, literature, and current progress

this problem is fully open

2.18.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

2.18.4. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

[n/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758.](#)

2.19. 59. Metric dimension reduction A snapshot of the Ribe program [2018-geom-152]

Record: 2018-geom-152

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.19.1. Problem statement

Let $X = \ell_3 \widehat{\otimes} \ell_3$, where ℓ_3 is the Banach space of scalar sequences $a = (a_i)$ with $\sum_i |a_i|^3 < \infty$, equipped with the norm $\|a\|_3 = (\sum_i |a_i|^3)^{1/3}$, and $\widehat{\otimes}$ denotes the projective tensor product, i.e. the completion of $\ell_3 \otimes \ell_3$ under the norm whose unit ball is the convex hull of $\{u \otimes v : \|u\|_3 \leq 1, \|v\|_3 \leq 1\}$. Does there exist a constant $\alpha \in [1, \infty)$ and a sequence of integers k_n with $\lim_{n \rightarrow \infty} (\log k_n) / \log n = 0$ such that, for every $n \in \mathbb{N}$ and every n -point subset $C \subseteq X$, there are a linear subspace $F \subseteq X$ with $\dim(F) = k_n$ and a map $f : C \rightarrow F$ satisfying

$$\|f(x) - f(y)\|_X \leq \alpha \|x - y\|_X \quad \text{for all } x, y \in C,$$

and

$$\frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|f(x) - f(y)\|_X \geq \frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|x - y\|_X?$$

2.19.2. Historical background, literature, and current progress

this problem is fully open

2.19.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

2.19.4. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

2.20. 60. Metric dimension reduction A snapshot of the Ribe program [2018-geom-153]

Record: 2018-geom-153

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.20.1. Problem statement

Determine whether there exists an infinite-dimensional Banach space Y such that Y admits average metric dimension reduction, but the projective tensor product $\ell_2 \widehat{\otimes} Y$ fails to admit average metric dimension reduction.

Here, for Banach spaces X and Z , $X \widehat{\otimes} Z$ is the completion of $X \otimes Z$ under the projective tensor norm whose unit ball is the convex hull of

$$\{x \otimes z \in X \otimes Z : \|x\|_X \leq 1, \|z\|_Z \leq 1\}.$$

An infinite-dimensional Banach space X admits average metric dimension reduction if there is a constant $\alpha_X \in [1, \infty)$ and a sequence of integers k_n with

$$\lim_{n \rightarrow \infty} \frac{\log k_n}{\log n} = 0$$

such that for every $n \in \mathbb{N}$ and every n -point subset $C \subseteq X$, there is a linear subspace $F \subseteq X$ with $\dim(F) = k_n$ and a map $f : C \rightarrow F$ satisfying

$$\|f(x) - f(y)\|_X \leq \alpha_X \|x - y\|_X \quad \text{for all } x, y \in C,$$

and

$$\frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|f(x) - f(y)\|_X \geq \frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|x - y\|_X.$$

2.20.2. Historical background, literature, and current progress

2.20.3. Verdict: partially resolved

2.20.3.1. Precise problem

The problem asks whether there is an infinite-dimensional Banach space Y such that Y admits average metric dimension reduction, but the projective tensor product $\ell_2 \widehat{\otimes} Y$ fails to admit average metric dimension reduction.

2.20.3.2. Current literature status

I did not find a published or citable result that resolves the exact ℓ_2 -endpoint question. The closest available theorem proves the analogous non-stability phenomenon when the first tensor factor is ℓ_p for every $p > 2$, and the same source explicitly formulates the $p = 2$ endpoint as an open question.

2.20.3.3. Relevant literature and progress

1. Metric dimension reduction: A snapshot of the Ribe program

- Authors: Assaf Naor
- Venue: Proceedings of the International Congress of Mathematicians–Rio de Janeiro 2018, Vol. I, World Scientific, 2018, pp. 759–837
- arXiv: 1809.02376
- URL: <https://arxiv.org/abs/1809.02376>
- Relevant result: Theorem 29 proves that there exist Banach spaces X, Y that admit average metric dimension reduction while $X \widehat{\otimes} Y$ does not; moreover, for every $p \in (2, \infty)$, one can take $X = \ell_p$.
- Why this is only partial: Immediately after Theorem 29, Naor asks Question 31: whether the same conclusion holds for $p = 2$, namely whether there is Y admitting average metric dimension reduction such that $\ell_2 \widehat{\otimes} Y$ does not. Thus Theorem 29 does not cover the exact statement.

2. Locally decodable codes and the failure of cotype for projective tensor products

- Authors: Jop Briët, Assaf Naor, Oded Regev
- Venue: Electronic Research Announcements in Mathematical Sciences, 19, 2012, pp. 120–138
- DOI: 10.3934/era.2012.19.120
- arXiv: 1208.0539
- URL: <https://arxiv.org/abs/1208.0539>
- Relevant result: Theorem 1.1 proves that if $p_1, p_2, p_3 \in (1, \infty)$ and $1/p_1 + 1/p_2 + 1/p_3 \leq 1$, then $\ell_{p_1} \widehat{\otimes} \ell_{p_2} \widehat{\otimes} \ell_{p_3}$ does not have finite cotype. This is the tensor-product obstruction used in Naor's proof of Theorem 29.
- Why this is only partial: It gives finite-cotype/universality input for triple tensor products, but it does not itself prove failure of average metric dimension reduction for any space of the form $\ell_2 \widehat{\otimes} Y$ with Y admitting average metric dimension reduction.

3. Impossibility of dimension reduction in the nuclear norm

- Authors: Assaf Naor, Gilles Pisier, Gideon Schechtman
- Venue: Discrete & Computational Geometry, 63(2), 2020, pp. 319–345
- DOI: 10.1007/s00454-019-00162-2
- arXiv: 1710.08896
- URL: <https://arxiv.org/abs/1710.08896>
- Relevant result: Theorem 1 proves strong lower bounds for ordinary bi-Lipschitz metric dimension reduction in S_1 , where $S_1 \cong \ell_2 \widehat{\otimes} \ell_2$.
- Why this is only partial: the problem concerns average metric dimension reduction. Naor's ICM survey explicitly notes that it is unknown whether S_1 admits average metric dimension reduction, so this result does not resolve the problem statement by taking $Y = \ell_2$.

2.20.3.4. What remains unresolved

The exact endpoint question remains unresolved: no located result proves either the existence or nonexistence of an infinite-dimensional Banach space Y that admits average metric dimension reduction while $\ell_2 \widehat{\otimes} Y$ fails to admit average metric dimension reduction. The $p > 2$ projective tensor result and the non-average S_1 dimension-reduction obstruction are genuine partial progress, but they do not cover all hypotheses and conclusions of the problem statement.

2.20.4. Remaining open problems recorded by the database

2.20.4.1. Open problem 1

Does the Schatten–von Neumann trace class $S_1 \cong \ell_2 \widehat{\otimes} \ell_2$ admit average metric dimension reduction? Explicitly, is there a constant $D < \infty$ and a sequence $k_n = n^{o(1)}$ such that every n -point subset $C \subseteq S_1$ has a D -Lipschitz map into a k_n -dimensional subspace of S_1 whose average pairwise distance is at least that of C ?

Reference supplied by the database. Assaf Naor, “Metric Dimension Reduction: A Snapshot of the Ribe Program,” in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018*, Vol. I: Plenary Lectures, eds. Boyan Sirakov, Paulo Ney de Souza, and Marcelo Viana, World Scientific, 2018, pp. 759–837, §1.4.1, discussion between Corollary 28 and Theorem 29 (explicitly stating that it is unknown whether S_1 admits average metric dimension reduction). arXiv:1809.02376. <https://arxiv.org/abs/1809.02376>.

2.20.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

2.21. 61. Metric dimension reduction A snapshot of the Ribe program [2018-geom-154]

Record: 2018-geom-154

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.21.1. Problem statement

Let (W, d_W) be a metric space and let $p \in [1, \infty)$. For a metric space (X, d_X) , let $P_p(X)$ denote the set of Borel probability measures μ on X with finite p -th moment, and equip $P_p(X)$ with the Wasserstein distance

$$W_p(\mu, \nu) = \inf_{\pi} \left(\int_{X \times X} d_X(x, y)^p d\pi(x, y) \right)^{1/p},$$

where the infimum is over all couplings π of μ and ν . Fix $\alpha > 1$, $n \in \mathbb{N}$, and $\mu_1, \dots, \mu_n \in P_p(W)$. What is the asymptotic behavior of the smallest $k \in \mathbb{N}$ for which there exists a subset $S \subseteq W$, equipped with the metric inherited from W , with $|S| \leq k$, and a map $f : \{\mu_1, \dots, \mu_n\} \rightarrow P_p(S)$ such that, for some scaling factor $\tau > 0$,

$$\tau W_p(\mu_i, \mu_j) \leq W_p(f(\mu_i), f(\mu_j)) \leq \alpha \tau W_p(\mu_i, \mu_j)$$

for all $i, j \in \{1, \dots, n\}$?

2.21.2. Historical background, literature, and current progress

2.21.3. Verdict: partially resolved

2.21.3.1. Precise problem

The problem is Naor's Wasserstein version of metric dimension reduction: given an n -point subset $\{\mu_1, \dots, \mu_n\}$ of a Wasserstein space $P_p(W)$, determine the asymptotic size of the smallest subset S of the original ground metric space W such that the finite metric space induced by Wasserstein distances on the μ_i embeds with distortion α into $P_p(S)$. Equivalently, it asks how well one can replace arbitrary measures by measures supported on a small subset of the same ambient metric space while preserving all pairwise Wasserstein distances up to a fixed bi-Lipschitz distortion.

2.21.3.2. Status

I found no published or otherwise citable theorem that determines this minimal support size in the full generality of the problem: arbitrary metric ground space W , all $p \in [1, \infty)$, all $\alpha > 1$, and all finite subsets of $P_p(W)$. The literature gives important obstructions and special/adjacent results, but not the full asymptotic answer.

2.21.3.3. Partial progress and related literature

1. Title: Metric dimension reduction: A snapshot of the Ribe program Authors: Assaf Naor Venue/year: Proceedings of the International Congress of Mathematicians–Rio de Janeiro 2018, Vol. I, World Scientific, 2018, pp. 759–837 DOI/arXiv/URL: DOI 10.1142/9789813272880_0029; arXiv:1809.02376; <https://arxiv.org/abs/1809.02376> Relevance: This is the original source. The problem appears as Question 32 in the Wasserstein-spaces subsection. Naor records that the bi-Lipschitz structure of Wasserstein spaces is poorly understood in this dimension-reduction setting. He also gives obstructions: for unrestricted W , one cannot hope for $k = n^{o(1)}$ with constant distortion, since point masses already contain arbitrary hard finite metrics; and even for $W = \ell_1$, universality of $P_1(\ell_1)$ gives hard examples obstructing subpolynomial support-size reduction.
2. Title: Similarity estimation techniques from rounding algorithms Authors: Moses S. Charikar Venue/year: Proceedings of the 34th Annual ACM Symposium on Theory of Computing (STOC 2002), pp. 380–388 DOI/arXiv/URL: DOI 10.1145/509907.509965; <https://dl.acm.org/doi/10.1145/509907.509965> Relevance: Together with Indyk-Thaper below, this gives logarithmic-distortion embeddings of finite earthmover/Wasserstein-1 metrics into ℓ_1 , used in Naor’s obstruction discussion. It does not solve the support-size question.
3. Title: Fast Image Retrieval via Embeddings Authors: Piotr Indyk and Nitin Thaper Venue/year: 3rd International Workshop on Statistical and Computational Theories of Vision, 2003 DOI/arXiv/URL: <https://people.csail.mit.edu/indyk/emd.pdf> Relevance: Provides the standard embedding approach for earthmover distance used in the same obstruction framework. It concerns embeddings into ℓ_1 , not embeddings into $P_p(S)$ with $S \subset W$.
4. Title: The metrical interpretation of superreflexivity in Banach spaces Authors: Jean Bourgain Venue/year: Israel Journal of Mathematics 56, 222–230, 1986 DOI/arXiv/URL: DOI 10.1007/BF02766125; <https://link.springer.com/article/10.1007/BF02766125> Relevance: Cited in the Wasserstein literature for the universality of $P_1(\ell_1)$. This supplies negative evidence against small-support reduction even when the ground space itself is ℓ_1 , but it is not a full answer to the question.
5. Title: Snowflake universality of Wasserstein spaces Authors: Alexandr Andoni, Assaf Naor, Ofer Neiman Venue/year: Annales scientifiques de l’Ecole Normale Supérieure, Serie 4, 51(3), 657–700, 2018 DOI/arXiv/URL: DOI 10.24033/asens.2363; arXiv:1509.08677; <https://www.numdam.org/articles/10.24033/asens.2363/> Relevance: Proves that for $p > 1$, every finite metric space has its $1/p$ -snowflake embedded almost isometrically into $P_p(\mathbb{R}^3)$, with sharpness results for $p \in (1, 2]$. Naor uses this as partial information for Question 32 when $W = \mathbb{R}^3$ and $p > 1$. It still does not answer the stated support-size problem because it does not show how small $S \subset \mathbb{R}^3$ can be, nor does it determine whether $P_p(S)$ for small S has the necessary low-dimensional normed embedding behavior.

6. Title: Impossibility of Sketching of the 3D Transportation Metric with Quadratic Cost
Authors: Alexandr Andoni, Assaf Naor, Ofer Neiman Venue/year: 43rd International Colloquium on Automata, Languages, and Programming (ICALP 2016), LIPIcs 55, article 83, pp. 83:1-83:14 DOI/arXiv/URL: DOI 10.4230/LIPIcs.ICALP.2016.83; <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ICALP.2016.83> Relevance: Conference/CS version of the 3D transportation-metric obstruction. It supports the same partial picture for $P_p(\mathbb{R}^3)$ but is not a full support-size theorem.
7. Title: Wasserstein distance and metric trees Authors: Maxime Mathey-Prevot and Alain Valette Venue/year: L'Enseignement Mathématique 69(3/4), 315-333, 2023 DOI/arXiv/URL: DOI 10.4171/LEM/1052; arXiv:2110.02115; <https://ems.press/journals/lem/articles/11033098> Relevance: Shows that if a finite metric space X stochastically embeds with distortion D into finite metric trees, then $P_1(X)$ embeds bi-Lipschitzly into ell_1 with distortion D , and gives tree formulas for Wasserstein distance. This is useful structural progress for tree-like ground spaces, but it does not determine the minimal $|S|$ in Naor's question.
8. Title: Coarse Embeddability of Wasserstein Space and the Space of Persistence Diagrams Authors: Neil Pritchard and Thomas Weighill Venue/year: Discrete & Computational Geometry 74, 358-373, 2025 DOI/arXiv/URL: DOI 10.1007/s00454-024-00674-6; arXiv:2307.12884; <https://link.springer.com/article/10.1007/s00454-024-00674-6> Relevance: Recent work showing equivalences between embeddability questions for Wasserstein space on $\mathbb{R}^2$ and persistence diagrams, and explicitly noting that important embeddability cases for Wasserstein space on the plane remain open. This reinforces that the low-dimensional Wasserstein embedding landscape is still unresolved, but it does not answer Question 32.

2.21.3.4. What remains unresolved

The full asymptotic behavior of the smallest support size k remains unknown. In particular, the literature does not give a theorem covering all metric spaces W and all p , nor does it settle the main “nice ground space” cases highlighted by Naor, such as $p=1$ over low-dimensional Euclidean spaces. Existing results rule out some overly optimistic answers and give related embedding theorems, but they do not determine the support-size reduction modulus asked for in the problem statement.

2.21.4. Remaining open problems recorded by the database

2.21.4.1. Open problem 1

Fix $\alpha > 1$. Given arbitrary measures $\mu_1, \dots, \mu_n \in P_1(\mathbb{R}^2)$, determine the asymptotic order of the smallest $k = k(n, \alpha)$ for which there is a set $S \subseteq \mathbb{R}^2$ with $|S| \leq k$ such that the finite metric space $(\{\mu_1, \dots, \mu_n\}, W_1)$ embeds with distortion at most α into $(P_1(S), W_1)$.

Reference supplied by the database. Assaf Naor, “Metric Dimension Reduction: A Snapshot of the Ribe Program,” in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018*, Vol. I: Plenary Lectures, eds. Boyan Sirakov, Paulo Ney de Souza, and Marcelo Viana, World Scientific, 2018, pp. 759–837, Question 32 and the ensuing discussion in §1.4.2

(explicitly singling out low-dimensional ground spaces and the image-retrieval case $p = 1$, $\mathbb{M} \subset \mathbb{R}^2$). arXiv:1809.02376. <https://arxiv.org/abs/1809.02376>.

2.21.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

3. Classical and local differential geometry

This chapter collects problems about submanifolds, minimal and constant-mean-curvature surfaces, local invariants, special coordinate systems, and classical rigidity. The historical record repeatedly shows local differential equations interacting with global completeness, embeddedness, or classification. Modern techniques—geometric measure theory, integrable systems, min–max constructions, and elliptic PDE—have resolved major subcases without eliminating the original classification problems.

Coverage: 20 website records; **source years:** 1912–2022; **separately listed descendant open problems:** 22.

3.1. 62. Conformal geometry [1912-geom-4]

Record: 1912-geom-4

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1912: Conformal geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1912.2/ICM1912.2.ocr.pdf#page=81>

3.1.1. Problem statement

Let a and b be regular real analytic arcs through a common vertex V forming a horn angle, i.e. a curvilinear angle of magnitude 0. Suppose that the contact of a and b at V is exactly of order $h - 1$, equivalently that the horn angle has type H_h and the two arcs have h consecutive points in common. Prove or disprove that, whenever h is odd, there exists an analytic curve c through V which is the external bisector of the horn angle, in the sense that a and b are symmetric with respect to c , analogous to the external angle bisector for two straight lines.

3.1.2. Historical background, literature, and current progress

3.1.3. Verdict: partially resolved

3.1.3.1. Precise problem

Kasner's problem is the conformal/Schwarz-reflection bisection problem for a horn angle. Given two regular real analytic arcs through a common vertex V , tangent with exact contact order $h - 1$ (type H_h), decide whether there is a real analytic arc c through V such that Schwarz reflection in c interchanges the two given arcs. After conformal normalization, this is equivalent to asking whether a germ

$$g(u) = u + b_h u^h + \cdots, \quad b_h \neq 0,$$

has an analytic iterative square root

$$f(u) = -u + \cdots, \quad f(f(u)) = g(u),$$

the sign $f'(0) = -1$ corresponding to the external bisector. the assertion is the proposed parity criterion: an external analytic bisector should exist whenever h is odd.

3.1.3.2. Literature and progress

Original source

- **Title:** Conformal geometry
Author: Edward Kasner
Venue: Proceedings of the Fifth International Congress of Mathematicians, Cambridge 1912, vol. 2, Cambridge University Press, pp. 81-90
Year: 1912/1913
URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1912.2/ICM1912.2.ocr.pdf>
Relevant result/status: Kasner defines horn angles of type H_h by exact contact order $h - 1$, proves the finite-order invariant statement for horn angles, and states the bisection issue explicitly: type H_2 has an internal but no external bisector, type H_3 has both, and in general if the type is even no external bisector exists while if the type is odd one "probably" exists. Thus the odd case was posed conjecturally, not proved.

Formal solution and uniqueness results

- **Title:** On the Conformal Geometry of Analytic Arcs
Author: G. A. Pfeiffer
Venue: American Journal of Mathematics, vol. 37, no. 4, pp. 395-430
Year: 1915
DOI/URL: <https://doi.org/10.2307/2370214>; <https://archive.org/download/jstor-2370214/2370214.pdf>
Relevant result/status: Pfeiffer reduces the bisection problem to $f(f(u)) = g(u)$. In the tangent horn case $g(u) = u + b_r u^r + \cdots$, his coefficient calculation gives one formal external solution with $f'(0) = -1$ when the first nonzero term is of odd degree $r = 2p + 1$, and no such formal solution when it is of even degree $r = 2p$. He also proves uniqueness-type statements. This is strong partial progress, but it is formal; the convergence needed for an analytic curve is not proved in the horn/parabolic multiplier-one case.

- **Title:** The Functional Equation $f[f(x)] = g(x)$
Author: G. A. Pfeiffer
Venue: Annals of Mathematics, Second Series, vol. 20, no. 1, pp. 13-22
Year: 1918
DOI/URL: <https://doi.org/10.2307/1967374> ; <https://archive.org/download/jstor-1967374/1967374.pdf>
Relevant result/status: This follow-up studies convergence and divergence for $f(f(x)) = g(x)$, but under the irrational neutral condition $|a_1| = 1, a_1^n \neq 1$. That excludes the horn case $g'(0) = 1$. It therefore does not resolve Kasner's odd-type external-bisector problem.

Modern classification and obstruction results

- **Title:** The classification of curvilinear angles in the complex plane and the groups of $\pm$ holomorphic diffeomorphisms
Author: Isao Nakai
Venue: Annales de la Faculté des Sciences de Toulouse, vol. 7, no. 2, pp. 313-334
Year: 1998
URL: https://www.numdam.org/item/AFST_1998_6_7_2_313_0.pdf
Relevant result/status: Nakai reformulates tangent curvilinear angles through the group generated by Schwarz reflections and uses Écalle-Voronin theory to describe moduli. The paper makes clear that formal equivalence of tangent curvilinear angles need not converge in general. This supports the conclusion that Pfeiffer's formal odd-degree solution is not by itself an analytic existence theorem.
- **Title:** A Complete Classification for Pairs of Real Analytic Curves in the Complex Plane with Tangential Intersection
Authors: Patrick Ahern and Xianghong Gong
Venue: Journal of Dynamical and Control Systems, vol. 11, pp. 1-71
Year: 2005; erratum 2007
DOI/URL: <https://doi.org/10.1007/s10883-005-0001-7> ; erratum <https://doi.org/10.1007/s10883-007-9015-7>
Relevant result/status: Ahern and Gong give a complete Écalle-Voronin classification for pairs of real analytic curves with tangential intersection and classify related involution/centralizer structures. Their abstract explicitly notes examples that are formally reversible by an antiholomorphic mapping but not by a convergent one. This is modern evidence of analytic obstructions beyond formal parity, but it is not a theorem in the simple form "all odd H_h horn angles have analytic external bisectors."
- **Title:** The root extraction problem
Author: Christiane Rousseau
Venue: Journal of Differential Equations, vol. 234, no. 1, pp. 110-141
Year: 2007
DOI/URL: <https://doi.org/10.1016/j.jde.2006.12.003>
Relevant result/status: Rousseau studies the N -th iterative root problem for analytic germs and states that, depending on the germ and multiplier, both formal and analytic obstructions can occur. The abstract explicitly says the theory is applied to sectioning a curvilinear angle in N equal parts in conformal geometry. This is directly relevant to bisection as $N = 2$, and again indicates that formal solvability is not the same as analytic bisection.

Special cases and survey-level status

- **Title:** Bisecting Horn Angles

Author: Sergiy Koshkin

Venue: The College Mathematics Journal, vol. 51, no. 2, pp. 124-131

Year: 2020

DOI/arXiv/URL: <https://doi.org/10.1080/07468342.2020.1700745> ; arXiv:2206.09499, <https://arxiv.org/abs/2206.09499>

Relevant result/status: Koshkin gives explicit metric and conformal bisectors for line-circle and circle-circle horn angles. For the general conformal bisection problem he cites Pfeiffer and states that a conformal bisector may not exist at all, while when it does it is unique. These special cases do not cover arbitrary analytic arcs of odd type H_h .

3.1.3.3. What remains unresolved

The literature fully settles only parts of the question: even type has no external formal/analytic candidate in Kasner-Pfeiffer's sense; odd type has a unique formal external bisector; and special line/circle horn angles have explicit analytic conformal bisectors. Modern Écalle-Voronin/root-extraction work gives analytic moduli and obstruction theory, showing that convergence is a genuine issue.

I did not find a published theorem proving the full statement that every odd-type real analytic horn angle H_h has an analytic external bisector. Nor did I find a citable theorem stated exactly as a complete odd-type horn-bisector classification. Therefore the current status of the exact problem is **partially resolved**, not fully resolved and not fully open.

3.1.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.1.5. Primary-source pointer

The originating ICM record is *Conformal geometry* (1912). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1912.2/ICM1912.2.ocr.pdf#page=81>.

3.2. 63. MODERN DEVELOPMENT OF SURFACE THEORY [1958-geom-9]

Record: 1958-geom-9

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1958: MODERN DEVELOPMENT OF SURFACE THEORY

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=67>

3.2.1. Problem statement

Let ds^2 be a prescribed two-dimensional line element defining an intrinsic metric on a surface M , and let S be a convex surface in a three-dimensional space of constant curvature that realizes this metric isometrically (in Euclidean space, this means that some parametrization $\mathbf{x}(u, v)$ of S satisfies $d\mathbf{x}^2 = ds^2$). Determine the minimal conditions on the prescribed line element ds^2 that ensure that the realizing convex surface S is twice differentiable, respectively three times differentiable.

3.2.2. Historical background, literature, and current progress

3.2.3. Verdict: partially resolved

3.2.3.1. Precise problem

The problem asks for a regularity criterion for convex isometric realizations of a prescribed two-dimensional line element. In modern terms: given a two-dimensional intrinsic metric ds^2 on a surface and a convex isometric realization in a three-dimensional space form, determine the weakest conditions on ds^2 which force the realizing surface/embedding to be C^2 , respectively C^3 .

The important point is that this is a minimal-conditions problem, not just an existence theorem for smooth metrics. In a space form of sectional curvature K_0 , the strictly elliptic case corresponds to intrinsic Gauss curvature $K > K_0$; the degenerate convex case allows $K \geq K_0$ and includes flat/developable behavior.

3.2.3.2. Original source

- Title: Modern development of surface theory
- Author: A. D. Alexandrov
- Venue: Proceedings of the International Congress of Mathematicians, Edinburgh 1958, Cambridge University Press, 1960, pp. 3-18
- DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf>

- Relevance: Section 6.4 states that known results gave differentiability of the convex surface from stronger differentiability assumptions on the line element, but that finding the minimal conditions ensuring two or three differentiability remained unsolved.

3.2.3.3. Main progress

- Title: On the regularity of a convex surface with a regular metric in Euclidean space
- Author: A. V. Pogorelov
- Venue: Dokl. Akad. Nauk SSSR 139:5, 1056-1058; English translation in Soviet Math. Dokl. 2, 1961
- DOI/arXiv/URL: MathNet listing: <https://www.mathnet.ru/eng/person18697>
- Result/progress: This filled a major gap left in the 1958 discussion. Later summaries state Pogorelov's theorem as: a convex surface with a C^n regular metric, $n \geq 2$, of positive Gauss curvature is $C^{n-1, \alpha}$ for every $0 < \alpha < 1$, and an analytic metric gives an analytic surface. This gives strong sufficient regularity in the strictly positive curvature case, but it has a derivative loss in plain C^n classes and is not a necessary-and-sufficient minimal criterion for all convex realizations.
- Title: Extrinsic Geometry of Convex Surfaces
- Author: A. V. Pogorelov
- Venue: Translations of Mathematical Monographs, vol. 35, American Mathematical Society, 1973; translated from the Russian edition, Nauka, 1969
- DOI/arXiv/URL: AMS book record; also cited in later literature as the standard source
- Result/progress: Gives the general theory of convex surfaces and regularity/rigidity results used in subsequent Weyl problem literature. It supports the positive-curvature regularity theory but does not settle the full minimal-conditions question in the problem statement.
- Title: The Weyl and Minkowski problems in differential geometry in the large
- Author: Louis Nirenberg
- Venue: Communications on Pure and Applied Mathematics 6, 337-394, 1953
- DOI/arXiv/URL: <https://doi.org/10.1002/cpa.3160060303>
- Result/progress: Solved the classical smooth Weyl embedding problem for positively curved metrics on S^2 under high regularity hypotheses, classically C^4 . This is foundational but gives sufficient hypotheses, not the minimal differentiability threshold asked in the problem statement.
- Title: On Weyl's embedding problem
- Author: Erhard Heinz
- Venue: Journal of Mathematics and Mechanics 11, 421-454, 1962
- DOI/arXiv/URL: JSTOR stable URL: <https://www.jstor.org/stable/24900718>
- Result/progress: Settled the C^3 positive-curvature Weyl case for obtaining a C^2 convex embedding in Euclidean space. Later papers describe this as a reduction from Niren-

berg's C^4 requirement to C^3 . This is major progress for the twice differentiable part in the strictly elliptic closed-sphere case, but it is not the full all-cases minimal answer.

- Title: The regularity of convex regions with a metric that is regular in the Holder classes
- Author: I. Kh. Sabitov
- Venue: Sibirsk. Mat. Zh. 17, 907-915, 1976; English translation, Siberian Mathematical Journal 17, 681-687, 1976
- DOI/arXiv/URL: <https://doi.org/10.1007/BF00971678>
- Result/progress: In Holder classes there is no derivative loss: a convex surface with $C^{\{n, \alpha\}}$ metric of positive curvature is $C^{\{n, \alpha\}}$. Thus $C^{\{2, \alpha\}}$ metrics give $C^{\{2, \alpha\}}$ convex surfaces and $C^{\{3, \alpha\}}$ metrics give $C^{\{3, \alpha\}}$ convex surfaces in the positive-curvature setting. This is close to the desired regularity statement but assumes Holder control and positive curvature.
- Title: The connections between the order of smoothness of a surface and its metric
- Authors: I. Kh. Sabitov and S. Z. Shefel'
- Venue: Sibirsk. Mat. Zh. 17, 916-925, 1976; English translation, Siberian Mathematical Journal 17, 687-694, 1976
- DOI/arXiv/URL: <https://doi.org/10.1007/BF00971679>
- Result/progress: Gives converse-type information relating the smoothness of a submanifold and of its induced metric. This helps clarify sharpness within Holder categories, but it is not a complete minimal criterion for arbitrary convex isometric realizations in space forms.
- Title: Regularity of locally convex surfaces
- Author: Friedmar Schulz
- Venue: Bulletin of the Australian Mathematical Society 42:3, 487-497, 1990
- DOI/arXiv/URL: <https://doi.org/10.1017/S0004972700028653>
- Result/progress: Derived interior estimates for locally convex surfaces and relaxed positive-curvature Weyl regularity toward $C^{\{2, \alpha\}}$ metric assumptions for C^2 regularity. This is partial progress on the twice differentiable threshold, not a full minimal classification.
- Title: On Weyl's Embedding Problem in Riemannian Manifolds
- Author: Siyuan Lu
- Venue: International Mathematics Research Notices 2020:11, 3229-3259; originally arXiv:1608.07539
- DOI/arXiv/URL: <https://doi.org/10.1093/imrn/rny109>; <https://arxiv.org/abs/1608.07539>
- Result/progress: Theorem 1.3 proves that if g is a C^2 metric on S^2 , D^2 g is Dini continuous, and the scalar curvature satisfies $R > 2K$, then (S^2, g) embeds in the three-dimensional space form of curvature K as a C^2 surface. This is one of the strongest modern sufficient criteria for the C^2 part in the strictly elliptic space-form case. It still assumes a Dini modulus and strict curvature and does not prove a necessary-and-sufficient minimal condition.
- Title: The Weyl problem of isometric immersions revisited

- Author: Siran Li
- Venue: Bulletin of the London Mathematical Society 53:1, 220-230, 2021; arXiv:2004.05532
- DOI/arXiv/URL: <https://doi.org/10.1112/blms.12413>; <https://arxiv.org/abs/2004.05532>
- Result/progress: Revisits the Weyl problem and its generalizations, gives sufficient conditions in degenerate settings, and records the strict-curvature consequence that C^3 metrics in suitable ambient settings yield C^3 surfaces. The introduction also notes obstructions in degenerate/nonnegative curvature cases, including metrics real-analytic and positive except at one point that do not admit global C^3 embeddings into R^3 . This reinforces that the full minimal-conditions question is not settled merely by the strict positive-curvature theory.

3.2.3.4. Why this is not fully resolved

The literature fully resolves many important special regimes: smooth or analytic metrics, strictly positive curvature on S^2 , Holder classes, and modern Dini-type sufficient conditions for C^2 embeddings in space forms. These results are too strong or too specialized to answer the problem statement exactly.

What remains unresolved is the complete minimal criterion on the prescribed line element that is necessary and sufficient for C^2 and for C^3 regularity of the realizing convex surface in the full space-form convex-realization problem, including degenerate convex metrics and the transition from strict curvature to nonnegative extrinsic curvature. Current results should therefore be classified as partial progress, not a full resolution.

3.2.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.2.5. Primary-source pointer

The originating ICM record is *MODERN DEVELOPMENT OF SURFACE THEORY* (1958). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=67>.

3.3. 64. Differential geometry; its past and its future [1970-geom-14]

Record: 1970-geom-14

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1970: Differential geometry; its past and its future

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>

3.3.1. Problem statement

Let M be a compact differentiable surface, let $I = (-1, 1)$, and let E^3 denote three-dimensional Euclidean space. Let

$$f : M \times I \rightarrow E^3$$

be a differentiable mapping such that, for each $t \in I$, the map

$$f_t : M \rightarrow E^3, \quad f_t(x) = f(x, t),$$

is an immersion. Suppose that the Riemannian metric ds_t^2 on M induced by the immersion f_t is independent of t . Does there exist, for each $t \in I$, a rigid motion $g(t)$ of E^3 such that

$$f_t(x) = g(t)(f_0(x))$$

for every $x \in M$?

3.3.2. Historical background, literature, and current progress

3.3.3. Verdict: partially resolved

3.3.3.1. Precise problem

This is Chern's Problem VIII from S.-S. Chern, "Differential Geometry; Its Past and Its Future" (ICM Nice 1970): whether every differentiable one-parameter isometric deformation of a compact surface immersed in Euclidean 3-space is trivial, i.e. obtained from the initial immersion by ambient rigid motions.

Equivalently, in the usual smooth/ C^2 interpretation of the classical surface-theory problem, it asks whether closed compact smooth immersed surfaces in $\mathbb{R}^3$ are isometrically flexible.

3.3.3.2. Why the full problem is not resolved

I found no published or otherwise citable theorem that covers arbitrary compact smooth immersed surfaces in $\mathbb{R}^3$ and proves that every one-parameter isometric deformation is induced

by rigid motions, nor a smooth compact counterexample giving such a one-parameter deformation. Modern open-problem accounts still list the existence of a closed C^2 flexible surface in $\mathbb{R}^3$ as open.

the exact question is therefore not fully resolved. The literature contains substantial special-case results and adjacent non-uniqueness results, but they do not imply the full statement/all cases asked by Chern's problem.

3.3.3.3. Key literature and status

Original source

- Title: "Differential Geometry; Its Past and Its Future"
- Author: Shiing-Shen Chern
- Venue: Actes du Congrès International des Mathématiciens, Nice 1970, Tome 1, pp. 41-53
- Year: 1970
- DOI/arXiv/URL: <https://www.hiroleetanaka.com/pdfs/2014-fall-1-230a-icml1970-chern-differential-geometry.pdf>
- Relevant result/status: Chern states Problem VIII exactly in the form of the problem. He notes classical rigidity for compact surfaces under curvature hypotheses, but explicitly says the problem is not known once the curvature condition is dropped.

Convex and nonnegative-curvature special cases

- Title: "Zwei Satze uber die Starrheit der Eiflachen"
- Author: Stephan Cohn-Vossen
- Venue: Nachrichten von der Gesellschaft der Wissenschaften zu Gottingen, Mathematisch-Physikalische Klasse
- Year: 1927
- DOI/arXiv/URL: no DOI found; cited in Chern 1970 and later surveys
- Relevant result/status: Rigidity of closed convex/positively curved surfaces. This supports an affirmative answer only for convex or positive-curvature cases, not arbitrary compact immersed surfaces.
- Title: "Differentialgeometrie geschlossener Flachen im euklidischen Raum"
- Author: K. Voss
- Venue: Jahresbericht der Deutschen Mathematiker-Vereinigung, Vol. 63, pp. 117-136
- Year: 1960-1961
- DOI/arXiv/URL: cited as reference [35] in Chern 1970, URL above
- Relevant result/status: Covers an additional curvature-restricted class only. It does not settle surfaces of mixed curvature or arbitrary immersed compact surfaces.

Open-problem surveys

- Title: "Open Problems in Geometry of Curves and Surfaces"
- Author: Mohammad Ghomi
- Venue: unpublished/open-problems manuscript

- Year: 2017-2019 versions circulated online
- DOI/arXiv/URL: <https://ghomi.math.gatech.edu/Papers/op.pdf>
- Relevant result/status: Section 1.1 states that no closed C^2 , or even $C^{1,1}$, flexible surface in $\mathbb{R}^3$ is known, and asks whether a closed C^2 surface in $\mathbb{R}^3$ can be flexible. This is essentially the smooth regularity form of Chern's one-parameter deformation problem.
- Title: "Review of Geometry and Analysis"
- Author: Shing-Tung Yau
- Venue: Mathematics: Frontiers and Perspectives, American Mathematical Society
- Year: 2000
- DOI/arXiv/URL: https://www.intlpress.com/site/pub/files/_fulltext/journals/ajm/2000/0004/0001/AJM-2000-0004-0001-a016.pdf
- Relevant result/status: Discusses the global rigidity problem for closed surfaces in E^3 as an open classical question and asks for projective/affine formulations. This does not resolve the full problem.

Low-regularity flexibility, not the classical smooth problem

- Title: " C^1 Isometric Imbeddings"
- Author: John Nash
- Venue: Annals of Mathematics, Second Series, Vol. 60, No. 3, pp. 383-396
- Year: 1954
- DOI/arXiv/URL: <https://doi.org/10.2307/1969840>
- Relevant result/status: Nash's C^1 theorem begins the low-regularity flexibility theory. In the C^1 category, isometric embeddings are flexible in ways impossible for C^2 /smooth surfaces, but this does not resolve the classical C^2/C^∞ compact-surface problem.
- Title: "On C^1 -isometric imbeddings. I, II"
- Author: Nicolaas H. Kuiper
- Venue: Indagationes Mathematicae, Vol. 17, pp. 545-556 and 683-689
- Year: 1955
- DOI/arXiv/URL: MR0075640; MathSciNet/Zbl references
- Relevant result/status: Extends Nash's C^1 flexibility. Again, this is a low-regularity counterpoint, not a solution to the smooth/ C^2 version of Chern's problem.

Adjacent non-uniqueness results, not one-parameter flexing

- Title: "Compact Bonnet pairs: isometric tori with the same curvatures"
- Authors: Alexander I. Bobenko, Tim Hoffmann, Andrew O. Sageman-Furnas
- Venue: Publications Mathématiques de l'IHES, Vol. 142, pp. 241-293
- Year: 2025
- DOI/arXiv/URL: <https://doi.org/10.1007/s10240-025-00159-z> ;

arXiv:2110.06335

- Relevant result/status: Constructs two non-congruent smooth/real-analytic immersed tori in $\mathbb{R}^3$ related by a mean-curvature-preserving isometry. This resolves a Bonnet-pair/non-uniqueness problem, but a pair of non-congruent isometric compact immersions is weaker than Chern's requested differentiable one-parameter isometric deformation. The paper itself distinguishes Bonnet pairs from Bonnet families and notes that compact Bonnet families cannot occur in the nonconstant-mean-curvature classification.

Claimed embedded local rigidity result not counted as a resolution

- Title: "Local rigidity of surfaces in space forms"
- Author: Michael T. Anderson
- Venue: arXiv preprint
- Year: 2007, withdrawn 2009
- DOI/arXiv/URL: <https://arxiv.org/abs/0709.3202>
- Relevant result/status: Earlier versions claimed local rigidity of closed embedded surfaces in $\mathbb{R}^3$, which would be only an embedded special case anyway. The arXiv record is withdrawn, so I did not count it as citable resolving literature.

3.3.3.4. What remains unresolved

What remains open is the full smooth/ C^2 compact immersed case with no convexity, nonnegative-curvature, tightness, embeddedness, constant-mean-curvature, or other extra hypotheses: given a differentiable/smooth one-parameter family of compact surface immersions in $\mathbb{R}^3$ with the induced metric independent of the parameter, must the family be induced by ambient rigid motions?

Current verdict for the problem: partially resolved.

3.3.4. Remaining open problems recorded by the database

3.3.4.1. Open problem 1

Does there exist a closed C^2 surface in $\mathbb{R}^3$ which is flexible, i.e. which admits a nontrivial continuous isometric deformation?

Reference supplied by the database. Mohammad Ghomi, "Open Problems in Geometry of Curves and Surfaces," unpublished open-problem manuscript, revised September 2, 2019 (online version current in 2026), Problem 1.1 (attributed there to Leonhard Euler, 1776, and James Clerk Maxwell, 1819), <https://ghomi.math.gatech.edu/Papers/op.pdf>.

3.3.4.2. Open problem 2

Are all smooth tight surfaces in $\mathbb{R}^3$ rigid?

Reference supplied by the database. Mohammad Ghomi, “Open Problems in Geometry of Curves and Surfaces,” unpublished open-problem manuscript, revised September 2, 2019 (online version current in 2026), Problem 1.2 (attributed there to A. D. Alexandrov, 1938, and Louis Nirenberg, 1962), <https://ghomi.math.gatech.edu/Papers/op.pdf>.

3.3.4.3. Open problem 3

Are negatively curved annuli in $\mathbb{R}^3$ bounded by a pair of fixed convex planar curves rigid?

Reference supplied by the database. Mohammad Ghomi, “Open Problems in Geometry of Curves and Surfaces,” unpublished open-problem manuscript, revised September 2, 2019 (online version current in 2026), Problem 1.3, <https://ghomi.math.gatech.edu/Papers/op.pdf>.

3.3.4.4. Open problem 4

Can a negatively curved annulus in $\mathbb{R}^3$, bounded by a pair of convex planar curves, have a closed asymptotic curve?

Reference supplied by the database. Mohammad Ghomi, “Open Problems in Geometry of Curves and Surfaces,” unpublished open-problem manuscript, revised September 2, 2019 (online version current in 2026), unnumbered question immediately following Problem 1.3, <https://ghomi.math.gatech.edu/Papers/op.pdf>; current partial progress: Mohammad Ghomi and Matteo Raffaelli, “Topology of Closed Asymptotic Curves on Negatively Curved Surfaces,” arXiv:2412.19266 [math.DG], 2024 (revised 2025), <https://doi.org/10.48550/arXiv.2412.19266>.

3.3.4.5. Open problem 5

Let Γ be a smooth closed curve immersed in $\mathbb{R}^3$ with a continuous one-to-one binormal vector field B . Must the ribbon (Γ, B) be twisted—that is, must the linking number of Γ and $\Gamma + \varepsilon B$ be nonzero?

Reference supplied by the database. Mohammad Ghomi, “Open Problems in Geometry of Curves and Surfaces,” unpublished open-problem manuscript, revised September 2, 2019 (online version current in 2026), Problem 1.4, <https://ghomi.math.gatech.edu/Papers/op.pdf>; current partial progress: Mohammad Ghomi and Matteo Raffaelli, “Topology of Closed Asymptotic Curves on Negatively Curved Surfaces,” arXiv:2412.19266 [math.DG], 2024 (revised 2025), <https://doi.org/10.48550/arXiv.2412.19266>.

3.3.5. Primary-source pointer

The originating ICM record is *Differential geometry; its past and its future* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>.

3.4. 65. Isoperimetric inequalities and eigenvalues of the Laplacian [1978-geom-24]

Record: 1978-geom-24

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Isoperimetric inequalities and eigenvalues of the Laplacian

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>

3.4.1. Problem statement

Prove or disprove that if a minimal surface in Euclidean space $\mathbb{R}^n$ has boundary consisting of several curves, and if A denotes the area of the surface while L denotes the total length of its boundary curves, then

$$L^2 \geq 4\pi A.$$

3.4.2. Historical background, literature, and current progress

3.4.3. Status check

Verdict: partially resolved.

3.4.3.1. Precise problem

The problem asks for the sharp Euclidean isoperimetric inequality for a compact two-dimensional minimal surface in Euclidean space, with boundary a finite union of closed curves. If A is the area of the surface and L is the total length of all boundary components, the question is whether

$$L^2 \geq 4\pi A$$

holds without restrictions on the number of boundary components, the topology of the surface, or the ambient dimension R^n .

3.4.3.2. Source and current progress

1. Title: "Isoperimetric inequalities and eigenvalues of the Laplacian" Authors: Robert Osserman Venue: Proceedings of the International Congress of Mathematicians, Helsinki 1978, pp. 435-442, Acad. Sci. Fennica, Helsinki Year: 1980 DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf> Relevance: This is the apparent source of the problem. Osserman states that the classical inequality holds for minimal surfaces in R^n bounded by a

single curve, and then formulates the several-boundary-curves case as a conjecture. At that time it was known only for doubly connected surfaces, by Osserman-Schiffer in R^3 and Feinberg in arbitrary R^n .

2. Title: "Doubly-connected minimal surfaces" Authors: Robert Osserman and Menahem Schiffer Venue: Archive for Rational Mechanics and Analysis 58, pp. 285-307 Year: 1975 DOI/arXiv/URL: not located in the live search Relevance: Proves the sharp isoperimetric inequality for doubly connected minimal surfaces in R^3 . This is a two-boundary-component/topological special case, not the full several-curve problem.
3. Title: "The isoperimetric inequality for doubly-connected minimal surfaces in R^n " Authors: Jerry M. Feinberg Venue: Journal d'Analyse Mathématique 32, pp. 249-278 Year: 1977 DOI/arXiv/URL: not located in the live search Relevance: Extends the doubly connected case to arbitrary ambient R^n . This still assumes the surface is doubly connected and therefore does not cover arbitrary several boundary curves.
4. Title: "On the isoperimetric inequality for minimal surfaces" Authors: Peter Li, Richard Schoen, Shing-Tung Yau Venue: Annali della Scuola Normale Superiore di Pisa, Classe di Scienze (4) 11(2), pp. 237-244 Year: 1984 DOI/arXiv/URL: https://www.numdam.org/item/ASNSP_1984_4_11_2_237_0/ Relevance: Theorem 1 proves the sharp inequality for compact minimal surfaces in R^n whose boundary is "weakly connected". Theorem 2 proves the sharp inequality in R^3 when the boundary has at most two connected components, without assuming the surface is doubly connected. This is major progress, but the weak-connectedness or at-most-two-components hypotheses do not cover every finite union of boundary curves.
5. Title: "Optimal isoperimetric inequalities" Authors: Frederick J. Almgren, Jr. Venue: Indiana University Mathematics Journal 35, pp. 451-547 Year: 1986 DOI/arXiv/URL: not located in the live search Relevance: Proves sharp isoperimetric inequalities for absolutely area-minimizing integral currents. This covers area-minimizing minimal surfaces, but the problem asks about minimal surfaces in general, not only absolutely area-minimizing ones.
6. Title: "The isoperimetric inequality for a minimal surface with radially connected boundary" Authors: Jaigyoung Choe Venue: Annali della Scuola Normale Superiore di Pisa, Classe di Scienze (4) 17(4), pp. 583-593 Year: 1990 DOI/arXiv/URL: https://www.numdam.org/item/ASNSP_1990_4_17_4_583_0/ Relevance: Theorem 2 proves the sharp inequality for minimal surfaces in R^n whose boundary is radially connected from a point of the surface. Corollary 2 proves that every minimal surface in R^n whose boundary has two connected components satisfies the sharp inequality. Choe also records remaining issues for three or four boundary components and examples not covered by weak or radial connectedness.
7. Title: "On the isoperimetric inequality on a minimal surface" Authors: Andrew Stone Venue: Calculus of Variations and Partial Differential Equations 17, pp. 369-391 Year: 2003 DOI/arXiv/URL: DOI 10.1007/s00526-002-0174-9 Relevance: Gives a formula for the isoperimetric defect of arbitrary minimal surfaces in R^n , and improves the universal non-sharp estimate to $L^2 \geq 2\sqrt{2}\pi A$. Since the constant is below 4π , this is not a resolution of the sharp conjecture.
8. Title: "The isoperimetric inequality for a minimal submanifold in Euclidean space" Authors: Simon Brendle Venue: Journal of the American Mathematical Society 34(2),

pp. 595-603 Year: 2021 DOI/arXiv/URL: DOI 10.1090/jams/969; arXiv:1907.09446
 Relevance: Proves a sharp isoperimetric inequality for compact minimal submanifolds of codimension at most 2. For two-dimensional minimal surfaces this covers arbitrary boundary topology in ambient R^3 and R^4 , and surfaces contained in an affine subspace of dimension at most 4. It does not cover the full arbitrary R^n statement when the ambient codimension is at least 3.

9. Title: “Minimal hypersurfaces and geometric inequalities” Authors: Simon Brendle
 Venue: Annales de la Faculte des sciences de Toulouse: Mathematiques, Serie 6, 32(1), pp. 179-201 Year: 2023 DOI/arXiv/URL: DOI 10.5802/afst.1734; arXiv:2010.03425
 Relevance: A recent survey/exposition. It states the sharp codimension at most 2 result and notes that in codimension 3 or higher only a non-sharp version of the inequality is available.

3.4.3.3. Why this is not fully resolved

Several strong theorems prove the desired $L^2 \geq 4\pi A$ in important subclasses: one boundary curve, two boundary components in arbitrary R^n , weakly connected or radially connected boundary, absolutely area-minimizing surfaces, and arbitrary boundary components in codimension at most 2.

However, I found no citable theorem proving the exact statement for every compact minimal surface in arbitrary Euclidean R^n with an arbitrary finite number of boundary curves. In particular, for two-dimensional minimal surfaces in ambient R^N with $N \geq 5$, the codimension is at least 3; Brendle’s published sharp theorem does not cover this range, and recent exposition describes only non-sharp constants there.

What remains unresolved is the full sharp inequality for arbitrary compact minimal surfaces of high codimension with several boundary components, without weak/radial connectedness, area-minimizing, or codimension-at-most-2 hypotheses.

3.4.4. Remaining open problems recorded by the database

3.4.4.1. Open problem 1

Let $S \subset \mathbb{R}^n$ be a compact minimal surface with boundary

$$\partial S = \bigcup_{i=1}^k C_i,$$

where no single component C_i can be separated from all the other boundary components by a hyperplane. If $A = \text{Area}(S)$ and $L_i = \text{Length}(C_i)$, prove that

$$4\pi A \leq \sum_{i=1}^k L_i^2.$$

Reference supplied by the database. Jaigyoung Choe, “Isoperimetric Inequalities of Minimal Submanifolds,” in David Hoffman (ed.), *Global Theory of Minimal Surfaces*, Clay Mathematics Proceedings, Vol. 2, American Mathematical Society, Providence, RI, 2005, pp. 325–369,

Open Problem 2. MR 2167266. <https://www.ams.org/books/cmip/002/>

3.4.5. Primary-source pointer

The originating ICM record is *Isoperimetric inequalities and eigenvalues of the Laplacian* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>.

3.5. 66. Isoperimetric inequalities and eigenvalues of the Laplacian [1978-geom-26]

Record: 1978-geom-26

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Isoperimetric inequalities and eigenvalues of the Laplacian

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>

3.5.1. Problem statement

For any integers n and N with $N > n$, does the Euclidean n -dimensional isoperimetric inequality

$$S^n \geq n^n \omega_n V^{n-1}$$

hold for every domain D on an arbitrary n -dimensional minimal variety in $\mathbb{R}^N$, where V is the n -dimensional volume of D , S is the $(n-1)$ -dimensional measure of the boundary ∂D , and ω_n is the volume of the unit ball in $\mathbb{R}^n$?

3.5.2. Historical background, literature, and current progress

3.5.3. Verdict: partially resolved

3.5.3.1. Precise problem

The problem asks whether the sharp Euclidean n -dimensional isoperimetric inequality

$$S^n \geq n^n \omega_n V^{n-1}$$

holds for every domain D in every n -dimensional minimal variety in $\mathbb{R}^N$, for all integers $N > n$. Equivalently, in the smooth compact-with-boundary formulation, it asks whether every compact n -dimensional minimal submanifold/domain $\Sigma^n \subset \mathbb{R}^{n+m}$, in every codimension $m = N - n \geq 1$, satisfies

$$|\partial \Sigma| \geq n \omega_n^{1/n} |\Sigma|^{(n-1)/n}.$$

3.5.3.2. Current literature status

The literature does not appear to fully resolve this exact all-codimension, arbitrary-minimal-variety statement. There is major partial progress:

1. Simon Brendle, "The isoperimetric inequality for a minimal submanifold in Euclidean space", *Journal of the American Mathematical Society* 34(2), 595-603, 2021. DOI: <https://doi.org/10.1090/jams/969>. arXiv:1907.09446, <https://arxiv.org/abs/1907.09446>.

Brendle proves a sharp Sobolev/isoperimetric inequality for compact smooth minimal submanifolds in Euclidean space of codimension at most 2. In particular, his Theorem 3 states that if Σ is a compact n -dimensional minimal submanifold of $\mathbb{R}^{n+2}$, then

$$|\partial\Sigma| \geq n|B^n|^{1/n}|\Sigma|^{(n-1)/n},$$

with equality only for a flat round ball; codimension 1 follows by embedding $\mathbb{R}^{n+1}$ into $\mathbb{R}^{n+2}$. Brendle's general inequality in higher codimension has a codimension-dependent constant and is stated to be sharp only when the codimension is at most 2, so it does not prove the full assertion for all $N > n$.

2. **Frederick J. Almgren Jr., "Optimal isoperimetric inequalities", *Indiana University Mathematics Journal* 35(3), 451-547, 1986. DOI: <https://doi.org/10.1512/iumj.1986.35.35028>.**

Almgren's optimal filling/isoperimetric inequalities imply the sharp Euclidean inequality for area-minimizing submanifolds/currents in all dimensions and codimensions. This is an important full-codimension special case, but "area-minimizing" is a stronger assumption than "minimal" or "stationary"; it therefore does not cover arbitrary minimal varieties/domains in the problem statement.

3. **Jaigyoung Choe, "The isoperimetric inequality for a minimal surface with radially connected boundary", *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* (4) 17(4), 583-593, 1990. URL: https://www.numdam.org/article/ASNSP_1990_4_17_4_583_0.pdf.**

Choe surveys earlier two-dimensional progress and proves the sharp inequality for minimal surfaces satisfying a radial-connectedness condition on the boundary. The paper also records earlier surface cases due to Carleman, Osserman-Schiffer, Feinberg, and Li-Schoen-Yau. These results are restricted to dimension 2 and/or extra topological/boundary hypotheses, so they are not a full solution to this problem.

4. **Wenkui Du, Yuchao Yi, and Ziyi Zhao, "Wulff inequality for minimal submanifolds in Euclidean space", arXiv:2412.19063, 2024. URL: <https://arxiv.org/html/2412.19063v1>.**

This recent preprint still states the sharp isoperimetric inequality for smooth minimal submanifolds as a conjecture, and summarizes the known status as: Carleman for disk-type minimal surfaces, Almgren for arbitrary-codimension area-minimizing submanifolds, and Brendle for codimension one and two. This supports that the general arbitrary-codimension case is not considered settled in the current literature.

3.5.3.3. What remains unresolved

The full statement asks for every n , every $N > n$, and every domain on an arbitrary n -dimensional minimal variety in $\mathbb{R}^N$. The available resolving theorem of Brendle covers smooth minimal submanifolds only in codimension 1 and 2; Almgren covers all codimensions only with the additional area-minimizing hypothesis; and the older surface literature covers dimension 2 under special assumptions. I found no published or otherwise citable theorem proving the sharp Euclidean inequality for arbitrary minimal varieties/submanifolds in codimension $m \geq 3$, let alone in the full singular "minimal variety" generality.

Therefore the exact problem is **partially resolved**, not fully resolved.

3.5.4. Remaining open problems recorded by the database

3.5.4.1. Open problem 1

Let $S \subset \mathbb{R}^N$ be a compact two-dimensional minimal submanifold with boundary

$$\partial S = \bigcup_{i=1}^k C_i,$$

where no single component C_i can be separated from all the other boundary components by a hyperplane. If $A = \text{Area}(S)$ and $L_i = \text{Length}(C_i)$, prove the strengthened isoperimetric inequality

$$4\pi A \leq \sum_{i=1}^k L_i^2.$$

Reference supplied by the database. Jaigyoung Choe, “Isoperimetric Inequalities of Minimal Submanifolds,” in David Hoffman (ed.), *Global Theory of Minimal Surfaces*, Clay Mathematics Proceedings, Vol. 2, American Mathematical Society, Providence, RI, 2005, pp. 325–369, Open Problem 2. MR 2167266. <https://www.ams.org/books/cmip/002/>

3.5.5. Primary-source pointer

The originating ICM record is *Isoperimetric inequalities and eigenvalues of the Laplacian* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>.

3.6. 67. W. H., Recent progress on the geometry of surfaces in $\mathbb{R}^3$ and on the use of computer... [1986-geom-41]

Record: 1986-geom-41

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: W. H., Recent progress on the geometry of surfaces in $\mathbb{R}^3$ and on the use of computer graphics as a research tool

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>

3.6.1. Problem statement

Prove or disprove that for every properly embedded minimal surface $M \subset \mathbb{R}^3$, every intrinsic isometry $f : M \rightarrow M$ of M with its induced Riemannian metric extends to a Euclidean isometry $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$; that is, $F(M) = M$ and $F|_M = f$.

3.6.2. Historical background, literature, and current progress

3.6.3. Verdict: partially resolved

3.6.3.1. Precise problem

The statement is the Choi–Meeks–White/Meeks Isometry Conjecture for properly embedded minimal surfaces in Euclidean 3-space: if $M \subset \mathbb{R}^3$ is a properly embedded minimal surface and $f : M \rightarrow M$ is an intrinsic isometry for the induced metric, must f be the restriction of an ambient Euclidean isometry F with $F(M) = M$?

This is weaker than the often-cited minimal rigidity formulation, which asks uniqueness up to congruence of isometric minimal immersions, but that stronger formulation implies the extension statement.

3.6.3.2. Status

The exact all-cases statement is not fully resolved in the literature I found. It is proved in several large classes, especially surfaces with more than one end and surfaces of finite genus, but current survey literature still lists the unrestricted statement as a conjecture. The remaining case is essentially one-ended, infinite-genus properly embedded minimal surfaces; more refined reductions point to zero-flux/bounded-curvature/quasi-dilation-periodic scenarios.

3.6.3.3. Key literature and progress

1. Recent progress on the geometry of surfaces in $\mathbb{R}^3$ and on the use of computer graphics as a research tool

Author: William H. Meeks III

Venue/year: Proceedings of the International Congress of Mathematicians, Berkeley 1986, 1987, pp. 551–556.

URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf>

Role: This is the dataset's indicated source area for Meeks' problem. Later surveys identify the question as Meeks' isometry/rigidity conjecture for properly embedded minimal surfaces.

2. Some recent developments in the theory of properly embedded minimal surfaces in $\mathbb{R}^3$

Author: Harold Rosenberg

Venue/year: Séminaire Bourbaki, Vol. 34 (1991–1992), pp. 463–535.

URL: <https://eudml.org/doc/110163>

Role: Records the conjectural status and early partial results: Rosenberg states that Meeks conjectures every symmetry of a properly embedded minimal surface extends to an isometry of $\mathbb{R}^3$, and notes Choi–Meeks–White's proof for surfaces with more than one end and Meeks–Rosenberg's result for doubly periodic surfaces.

3. A rigidity theorem for properly embedded minimal surfaces in $\mathbb{R}^3$

Authors: Hyeon In Choi, William H. Meeks III, Brian White

Venue/year: Journal of Differential Geometry 32 (1990), no. 1, 65–76.

DOI/URL: 10.4310/jdg/1214445037, <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-32/issue-1/A-rigidity-theorem-for-properly-embedded-minimal-surfaces-in-bf/10.4310/jdg/1214445037.pdf>

Result: Proves the extension/rigidity result when the properly embedded minimal surface has more than one end. This is a major partial resolution but does not cover one-ended surfaces.

4. A rigidity theorem for periodic minimal surfaces

Author: Joaquín Pérez

Venue/year: Communications in Analysis and Geometry 7 (1999), no. 1, 95–104.

URL: https://www.intlpress.com/site/pub/files/_fulltext/journals/cag/1999/0007/0001/CAG-1999-0007-0001-a003.pdf

Result: Proves rigidity for periodic properly embedded minimal surfaces with finite-topology quotient, apart from the helicoid as the non-rigid associate-family exception. This is a special periodic case, not the unrestricted problem.

5. The rigidity of embedded constant mean curvature surfaces

Authors: William H. Meeks III, Giuseppe Tinaglia

Venue/year: Journal für die reine und angewandte Mathematik 660 (2011), 181–190.

DOI/arXiv/URL: 10.1515/CRELLE.2011.073; arXiv:0801.3409, <https://arxiv.org/abs/0801.3409>

Result: Theorem 1.2, the Finite Genus Isometry Extension Theorem, states that if $M \subset \mathbb{R}^3$ is complete embedded CMC of finite genus and is minimal, then every intrinsic isometry $\sigma : M \rightarrow M$ extends to an isometry of $\mathbb{R}^3$. This covers all finite-genus minimal cases but not infinite-genus one-ended surfaces.

6. The classical theory of minimal surfaces

Authors: William H. Meeks III, Joaquín Pérez

Venue/year: Bulletin of the American Mathematical Society 48 (2011), no. 3, 325–407.

DOI/URL: 10.1090/S0273-0979-2011-01334-9, <https://www.ams.org/bull/2011-48-03/>

Result/status summary: Conjecture 11.10 is exactly the Isometry Conjecture: if $M \in \mathcal{C}$, every intrinsic isometry of M extends to an ambient isometry of $\mathbb{R}^3$. The survey says it is known for more-than-one-end surfaces, doubly periodic surfaces, periodic finite-topology quotients, and finite genus. It also states the reduction to checking one-ended infinite-genus properly embedded examples with a plane intersection containing a simple closed curve. This confirms the full statement was not then resolved.

7. A Survey on Classical Minimal Surface Theory

Authors: William H. Meeks III, Joaquín Pérez

Venue/year: American Mathematical Society, University Lecture Series 60, 2012.

DOI/URL: 10.1090/ulect/060, <https://www.ams.org/books/ulect/060/>

Result/status summary: Repeats the Isometry Conjecture as an outstanding problem and lists the same known cases. It further notes reductions using the Dynamics Theorem and Local Picture Theorem, but does not claim a full proof.

8. The local picture theorem on the scale of topology

Authors: William H. Meeks III, Joaquín Pérez, Antonio Ros

Venue/year: Journal of Differential Geometry 109 (2018), no. 3, 509–565.

DOI/arXiv/URL: 10.4310/jdg/1531188195; arXiv:1505.06761, <https://arxiv.org/abs/1505.06761>

Result/status summary: Gives structural restrictions on potential counterexamples. Remark 5.6 says a complete embedded minimal surface with an intrinsic isometry not extending to an ambient isometry would have to have zero flux and constrained local pictures; the authors describe this as potentially useful for proving the classical conjecture. This is diagnostic progress, not a proof of the full conjecture.

3.6.3.4. What remains unresolved

The published theorems cover properly embedded minimal surfaces with more than one end and all finite-genus cases, among other periodic/special cases. I found no citable theorem proving the full statement for every properly embedded minimal surface. The unresolved part is the possible one-ended, infinite-genus properly embedded minimal surface case, especially the zero-flux configurations singled out by the later reduction and local-picture work. Therefore the correct verdict for the problem is **partially resolved**.

3.6.4. Remaining open problems recorded by the database

3.6.4.1. Open problem 1

Let $M \subset \mathbb{R}^3$ be a properly embedded minimal surface with exactly one end and infinite genus. Prove that every intrinsic isometry of M extends to an ambient Euclidean isometry of $\mathbb{R}^3$.

Reference supplied by the database. William H. Meeks III and Giuseppe Tinaglia, “The rigidity of embedded constant mean curvature surfaces,” *Journal für die reine und angewandte Mathematik* **660** (2011), 181–190, especially the outstanding isometry conjecture in the introduction and Theorem 1.2 (the finite-genus case), DOI: [10.1515/CRELLE.2011.073](https://doi.org/10.1515/CRELLE.2011.073). Together with Hyeong In Choi, William H. Meeks III, and Brian White, “A rigidity theorem for properly embedded minimal surfaces in $\mathbb{R}^3$,” *Journal of Differential Geometry* **32** (1990), no. 1, 65–76

(the case of at least two ends), DOI: [10.4310/jdg/1214445037](https://doi.org/10.4310/jdg/1214445037), these results isolate the stated one-ended infinite-genus case as the remaining open case.

3.6.5. Primary-source pointer

The originating ICM record is *W. H., Recent progress on the geometry of surfaces in R and on the use of computer graphics as a research tool* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>.

3.7. 68. W. H., Recent progress on the geometry of surfaces in $\mathbb{R}$ and on the use of computer... [1986-geom-42]

Record: 1986-geom-42

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: W. H., Recent progress on the geometry of surfaces in $\mathbb{R}$ and on the use of computer graphics as a research tool

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>

3.7.1. Problem statement

Prove that for every surface M that is homeomorphic to a closed surface with at least three points removed, there exists a proper embedding $f : M \rightarrow \mathbb{R}^3$ such that the embedded surface $f(M)$ has constant mean curvature 1.

3.7.2. Historical background, literature, and current progress

3.7.3. Verdict: partially resolved

3.7.3.1. Precise problem

The problem asks for an existence theorem for complete finite-topology constant-mean-curvature surfaces in Euclidean three-space: for every orientable surface of type (g, k) , i.e. a closed genus g surface with $k \geq 3$ punctures, does there exist a proper embedding in $\mathbb{R}^3$ with constant mean curvature $H = 1$?

The normalization $H = 1$ is harmless up to Euclidean dilation. The word “embedding” is essential here: results for properly immersed or merely Alexandrov/almost embedded CMC surfaces do not by themselves settle the problem statement.

3.7.3.2. Current progress

1. Complete constant mean curvature surfaces in Euclidean three-space

- Authors: Nicolaos Kapouleas
- Venue: Annals of Mathematics (2), 131(2), 239-330
- Year: 1990
- DOI/URL: DOI 10.2307/1971494; JSTOR <https://www.jstor.org/stable/1971494>
- Relevance: Kapouleas constructed many complete finite-topology CMC surfaces by gluing Delaunay pieces and spheres. The 1987 announcement states the construction gives properly immersed complete CMC surfaces of genus g with m ends in the

relevant ranges, and later CMC moduli papers cite Kapouleas for nonemptiness of the Alexandrov-embedded moduli spaces $M_{g,k}$ for $k > 2$.

- Limitation: this is not a direct proof of the exact problem because the problem asks for proper embeddings. The citable statements I found distinguish properly immersed/Alexandrov embedded surfaces from strict embedded surfaces.

2. Triunduloids: Embedded constant mean curvature surfaces with three ends and genus zero

- Authors: Karsten Grosse-Brauckmann, Robert B. Kusner, John M. Sullivan
- Venue: Journal für die reine und angewandte Mathematik, 564, 35-61
- Year: 2003
- DOI/arXiv/URL: DOI 10.1515/crll.2003.093; arXiv:math/0102183
- Relevance: this constructs/classifies the complete embedded CMC surfaces with exactly three ends and genus zero. Thus the problem is resolved for the sphere minus three points.
- Limitation: it does not cover three-ended surfaces of positive genus.

3. Embedded constant mean curvature surfaces in Euclidean three-space

- Authors: Christine Breiner, Nikolaos Kapouleas
- Venue: Mathematische Annalen, 360(3-4), 1041-1108
- Year: 2014
- DOI/arXiv/URL: arXiv:1210.3394
- Relevance: this refines Kapouleas' construction and proves embeddedness for a large class of finite-topology CMC examples arising from flexible pre-embedded graphs.
- Limitation: the theorem is graph-hypothesis based and is presented as producing a large class of embedded examples, not every pair (g, k) with $k \geq 3$. It therefore does not by itself imply the full problem.

4. Constant Mean Curvature surfaces with prescribed finite topologies

- Author: Stephen J. Kleene
- Venue: arXiv preprint
- Year: 2023
- DOI/arXiv/URL: arXiv:2309.08344
- Relevance: Theorem 1.1 states that for every $g \geq 0$ and every $k \geq 4$, there is a family of complete embedded CMC surfaces in $\mathbb{R}^3$ with genus g and k ends. This fully resolves the problem statement for all surfaces with at least four punctures.
- Limitation: the theorem assumes $k \geq 4$. The same preprint lists "Complete embedded CMC surfaces with three ends and any genus" as a work "in preparation", not as a published or arXiv theorem.

5. The structure of complete embedded surfaces with constant mean curvature

- Authors: Nicholas J. Korevaar, Robert Kusner, Bruce Solomon
- Venue: Journal of Differential Geometry, 30(2), 465-503
- Year: 1989
- DOI/URL: DOI 10.4310/jdg/1214443598
- Relevance: this proves the Delaunay asymptotic structure of annular ends and the classification of the two-ended embedded finite-topology case as Delaunay. It explains why the original question starts at three punctures.
- Limitation: it is structural/classificatory, not an existence theorem for all $k \geq 3$.

3.7.3.3. What remains unresolved

The citable literature found fully covers all genera with $k \geq 4$ ends and covers the genus-zero case with $k = 3$ ends. I did not find a published or arXiv result proving the missing case: proper embedded CMC $H = 1$ surfaces of arbitrary positive genus with exactly three ends.

Therefore the exact statement is not fully resolved. The correct status is **partially resolved**.

3.7.4. Remaining open problems recorded by the database

3.7.4.1. Open problem 1

For every integer $g \geq 1$, construct a complete properly embedded constant-mean-curvature-one surface in $\mathbb{R}^3$ having genus g and exactly three ends.

Reference supplied by the database. Stephen J. Kleene, “Constant Mean Curvature surfaces with prescribed finite topologies,” arXiv preprint (2023), Theorem 1.1 and bibliography item “Complete embedded CMC surfaces with three ends and any genus, in preparation,” [arXiv:2309.08344](https://arxiv.org/abs/2309.08344).

3.7.5. Primary-source pointer

The originating ICM record is W. H., *Recent progress on the geometry of surfaces in R and on the use of computer graphics as a research tool* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>.

3.8. 69. W. H., Recent progress on the geometry of surfaces in $\mathbb{R}^3$ and on the use of computer... [1986-geom-43]

Record: 1986-geom-43

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: W. H., Recent progress on the geometry of surfaces in $\mathbb{R}^3$ and on the use of computer graphics as a research tool

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>

3.8.1. Problem statement

Prove that every properly embedded surface $M \subset \mathbb{R}^3$ with constant mean curvature that is contained in a half-space $H \subset \mathbb{R}^3$ is invariant under reflection across some plane parallel to the boundary plane ∂H of the half-space.

3.8.2. Historical background, literature, and current progress

3.8.3. Verdict: partially resolved

3.8.3.1. Precise problem

The problem asks the half-space Alexandrov-symmetry problem for constant-mean-curvature surfaces in Euclidean 3-space: if a properly embedded CMC surface $M \subset \mathbb{R}^3$ is contained in a half-space H , must M be invariant under reflection in some plane parallel to ∂H ?

This is stronger than the known finite-topology/finite-ends statements. If such a reflecting plane exists for a surface lying on one side of ∂H , then the surface is also trapped in a slab parallel to ∂H . Thus unresolved slab-symmetry questions are directly relevant to this problem.

3.8.3.2. Literature found

1. Recent Progress on the Geometry of Surfaces in $\mathbb{R}^3$ and on the Use of Computer Graphics as a Research Tool

Authors: William H. Meeks III.

Venue/year: Proceedings of the International Congress of Mathematicians, Berkeley, 1986, published 1987.

DOI/arXiv/URL: no DOI found; IMU proceedings OCR: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf>; author bibliography: <https://profmeeks.wordpress.com/my-research-papers-2/>

Relevance: this is the source family for the problem. It places the question in the Alexandrov reflection theory of properly embedded CMC surfaces.

2. The strong halfspace theorem for minimal surfaces

Authors: David Hoffman and William H. Meeks III.

Venue/year: *Inventiones Mathematicae* 101, 373-377, 1990.

DOI/arXiv/URL: DOI 10.1007/BF01231507; bibliographic listing at https://www.bimsa.cn/research_detail/mincmcr3.html

Result: a connected proper nonplanar minimal surface in $\mathbb{R}^3$ is not contained in a half-space.

Applicability: this resolves the zero-mean-curvature subcase of the problem. A minimal properly embedded surface in a half-space is therefore a plane parallel to the boundary plane, hence has the required reflection symmetry. This is only one subcase.

3. The topology and geometry of embedded surfaces of constant mean curvature

Authors: William H. Meeks III.

Venue/year: *Journal of Differential Geometry* 27, 539-552, 1988.

DOI/arXiv/URL: DOI 10.4310/jdg/1214442008; Project Euclid page <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-27/issue-3/The-topology-and-geometry-of-embedded-surfaces-of-constant-mean/10.4310/jdg/1214442008.full>

Result/progress: proves major topology restrictions for properly embedded nonzero CMC surfaces, including the no-one-end phenomenon cited in later surveys.

Applicability: important background, but it does not prove the full half-space reflection conclusion.

4. The structure of complete embedded surfaces with constant mean curvature

Authors: Nicholas J. Korevaar, Rob Kusner, and Bruce Solomon.

Venue/year: *Journal of Differential Geometry* 30, 465-503, 1989.

DOI/arXiv/URL: DOI 10.4310/jdg/1214443598; bibliographic listing at https://www.bimsa.cn/research_detail/mincmcr3.html

Result: the finite-ended version is resolved. Expositions of the KKS theorem state that if a properly embedded CMC surface has finitely many ends and is contained in a half-space bounded by a plane π , then it has a plane of symmetry parallel to π , and hence lies in a slab. See also Jesse Ratzkin, *An End-to-End Gluing Construction for Surfaces of Constant Mean Curvature*, thesis, 2001, Appendix C, Theorem 17, which explicitly quotes this KKS consequence: <https://sites.math.washington.edu/~pollack/students/Ratzkin-thesis-2001.pdf>

Applicability: this is the closest positive result, but the extra hypothesis “finitely many ends” or finite topology is not present in the problem statement.

5. Properly embedded surfaces with constant mean curvature

Authors: Antonio Ros and Harold Rosenberg.

Venue/year: *American Journal of Mathematics* 132(6), 1429-1443, 2010.

DOI/arXiv/URL: DOI 10.1353/ajm.2010.0015; JSTOR stable page <https://www.jstor.org/stable/40931045>; author PDF via ResearchGate https://www.researchgate.net/publication/228395641_Properly_embedded_surfaces_with_constant_mean_curvature

Result/progress: proves a maximum principle at infinity for properly embedded H -surfaces. The paper explicitly identifies as unsolved whether a properly embedded CMC surface in a horizontal slab admits a horizontal plane of symmetry.

Applicability: because the stated half-space statement would imply the corresponding slab symmetry statement, this is evidence that the unrestricted problem was not resolved by the Ros-Rosenberg maximum principle.

6. The Dynamics Theorem for CMC surfaces in $\mathbb{R}^3$

Authors: William H. Meeks III and Giuseppe Tinaglia.

Venue/year: Journal of Differential Geometry 85(1), 141-173, 2010.

DOI/arXiv/URL: DOI 10.4310/jdg/1284557928; arXiv:0805.1427, <https://arxiv.org/abs/0805.1427>

Result/progress: for a properly embedded nonzero CMC surface with bounded second fundamental form, the paper studies translational limits $T(M)$. Its abstract states that if M has infinitely many ends, there is a nonempty minimal T -invariant set in $T(M)$ consisting entirely of surfaces with planes of Alexandrov symmetry. Searchable text of the paper also states that if M lies in a half-space, a minimal element of $T(M)$ has a horizontal Alexandrov symmetry plane.

Applicability: this is substantial asymptotic progress, but it proves symmetry for certain translational limit surfaces, not for the original M . It also assumes bounded second fundamental form, which is not in the problem statement.

7. Geometry of constant mean curvature surfaces in $\mathbb{R}^3$

Authors: William H. Meeks III and Giuseppe Tinaglia.

Venue/year: Journal of the European Mathematical Society 27(6), 2467-2476, 2025.

DOI/arXiv/URL: DOI 10.4171/JEMS/1416; EMS Press PDF <https://ems.press/content/serial-article-files/50378>

Result/progress: proves major global structure results for nonzero CMC surfaces, including that complete simply connected embedded nonzero CMC surfaces in $\mathbb{R}^3$ are round spheres and further finite-topology consequences.

Applicability: these results do not remove the finite-topology or asymptotic restrictions needed to prove the exact half-space reflection statement for every properly embedded CMC surface.

3.8.3.3. Conclusion

The problem is not fully resolved as stated. The minimal case is settled by Hoffman-Meeks, compact embedded CMC surfaces are classical Alexandrov spheres, and the nonzero CMC finite-ended/finite-topology half-space case is settled by Korevaar-Kusner-Solomon. Later Ros-Rosenberg and Meeks-Tinaglia papers provide maximum-principle and dynamical/asymptotic progress, but they do not prove that an arbitrary properly embedded CMC surface contained in a half-space is itself symmetric about a plane parallel to the boundary plane.

What remains unresolved is the unrestricted case with no finite-topology, finite-end, bounded-curvature, or asymptotic-limit replacement hypothesis.

3.8.4. Remaining open problems recorded by the database

3.8.4.1. Open problem 1

Prove that every properly embedded constant-mean-curvature surface in $\mathbb{R}^3$ that is contained between two parallel planes has a plane of reflection symmetry parallel to those planes.

Reference supplied by the database. Antonio Ros and Harold Rosenberg, "Properly embedded surfaces with constant mean curvature," *American Journal of Mathematics* **132**

(2010), no. 6, 1429–1443, explicit unsolved slab-symmetry problem in the introduction, DOI: [10.1353/ajm.2010.0015](https://doi.org/10.1353/ajm.2010.0015).

3.8.5. Primary-source pointer

The originating ICM record is W. H., *Recent progress on the geometry of surfaces in R and on the use of computer graphics as a research tool* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>.

3.9. 70. W. H., Recent progress on the geometry of surfaces in $\mathbb{R}^3$ and on the use of computer... [1986-geom-44]

Record: 1986-geom-44

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: W. H., Recent progress on the geometry of surfaces in $\mathbb{R}^3$ and on the use of computer graphics as a research tool

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>

3.9.1. Problem statement

Prove that every smooth embedded closed surface $M \subset \mathbb{R}^3$ of genus 2 whose total Gaussian curvature

$$K(M) = \int_M |K_G| dA$$

is less than or equal to the total Gaussian curvature of every smooth embedded closed surface ambiently isotopic to M in $\mathbb{R}^3$ satisfies

$$K(M) = 4\pi(2 + 1) = 12\pi.$$

3.9.2. Historical background, literature, and current progress

3.9.3. Verdict: partially resolved

3.9.3.1. Precise problem

Let $M \subset \mathbb{R}^3$ be a smooth embedded closed orientable surface of genus 2, and let

$$\mathcal{K}(M) = \int_M |K_G| dA.$$

The problem asks whether every genus-2 embedding that minimizes $\mathcal{K}$ among all smooth embedded closed surfaces ambiently isotopic to it must be absolutely tight, i.e. must satisfy

$$\mathcal{K}(M) = 4\pi(2 + 1) = 12\pi.$$

Equivalently, in the normalized convention $\tau(M) = \mathcal{K}(M)/(2\pi)$, the question is whether every genus-2 isotopy-tight embedding has $\tau = 6$, and whether a non-tight genus-2 isotopy-tight embedding can exist.

3.9.3.2. Current status

I did not find a published or otherwise citable result that fully resolves this exact genus-2 question. The literature gives strong partial results: the Chern-Lashof lower bound gives $\mathcal{K}(M) \geq 12\pi$; surfaces of genus 2 below the next threshold $\mathcal{K} < 20\pi$ are unknotted; Kuiper

and Meeks rule out the first possible non-tight genus-2 isotopy-tight level $\mathcal{K} = 20\pi$; the torus analogue is fully solved; and knotted isotopy-tight examples exist in every genus $g \geq 3$. What remains unresolved is whether a genus-2 isotopy-tight embedding with higher total absolute curvature, necessarily beyond the excluded 20π level, exists.

3.9.3.3. Key literature and applicability

1. On the total curvature of immersed manifolds

- Authors: S.-S. Chern and R. K. Lashof
- Venue/year: American Journal of Mathematics 79 (1957), 306-318
- DOI/arXiv/URL: no arXiv; bibliographic listing at <https://www.ams.org/tran/1965-115-00/S0002-9947-1965-0202094-9/>
- Contribution: Establishes the total absolute curvature lower bound in terms of Betti numbers. For a closed orientable genus-2 surface, this gives $\mathcal{K}(M) \geq 2\pi(2 + 2g) = 12\pi$. This proves the absolute lower bound but not that every isotopy-class minimizer attains it.

2. On the total curvature of immersed manifolds. II

- Authors: S.-S. Chern and R. K. Lashof
- Venue/year: Michigan Mathematical Journal 5 (1958), 5-12
- DOI/arXiv/URL: DOI 10.1307/mmj/1028998005; <https://projecteuclid.org/journals/michigan-mathematical-journal/volume-5/issue-1/On-the-total-curvature-of-immersed-manifolds-II/10.1307/mmj/1028998005.full>
- Contribution: Continuation of the Chern-Lashof total curvature theory. It supports the tightness framework but does not address ambient isotopy classes of knotted genus-2 surfaces.

3. A criterion for an embedded surface in R^3 to be unknotted

- Author: H. R. Morton
- Venue/year: Lecture Notes in Mathematics 722, Springer, 1979, pp. 92-98
- DOI/arXiv/URL: cited in Springer reference list at <https://link.springer.com/article/10.1007/BF01389134>
- Contribution: One of the sources, together with work of Meeks and Langevin-Rosenberg, for the unknottedness threshold for embedded surfaces. In the later Banchoff-Kuhnel survey this appears as: if $(1/2\pi) \int_M |K| < 8 - \chi(M)$, then M is unknotted. For genus 2, $8 - \chi(M) = 10$, so this covers $\mathcal{K} < 20\pi$, not the full isotopy-tight question.

4. On curvature integrals and knots

- Authors: R. Langevin and H. Rosenberg
- Venue/year: Topology 15 (1976), 405-416
- DOI/arXiv/URL: no arXiv found; cited at <https://link.springer.com/article/10.1007/BF01871945>
- Contribution: Early curvature-integral work relating total curvature and knotting, including the torus direction. It is part of the background for the unknottedness threshold but does not settle genus-2 isotopy tightness.

5. Total curvature for knotted surfaces

- Authors: N. H. Kuiper and W. H. Meeks III

- Venue/year: *Inventiones Mathematicae* 77:1 (1984), 25-69
- DOI/arXiv/URL: DOI 10.1007/BF01389134; <https://link.springer.com/article/10.1007/BF01389134>; IHES PDF https://repo-archives.ihes.fr/FONDS_IHES/I_Prepublications/KUIPER/1972-1991/M_83_37/M_83_37_web.pdf
- Contribution: Defines isotopy-tight embedded surfaces and proves the key structural theorem: a relatively/isotopy-tight genus g surface has normalized curvature in the discrete form $2 + 2g + 4(k - 1)$. It proves that if a genus-2 surface had the first non-tight value $\tau = 10$, equivalently $\mathcal{K} = 20\pi$, then it is unknotted. It also constructs knotted isotopy-tight examples for every $g > 2$ with $\tau = 2g + 6$. This is major partial progress but does not rule out genus-2 isotopy-tight embeddings with $\tau \geq 14$.

6. Minimal total curvature, a look at old and new results

- Author: N. H. Kuiper
- Venue/year: *Russian Mathematical Surveys* 40:4 (1985), 49-55
- DOI/arXiv/URL: https://www.mathnet.ru/php/getFT.phtml?jrn_id=rm&option_lang=rus&paperid=2703&what=fullteng
- Contribution: Survey of the total-absolute-curvature theory. It summarizes the unknottedness threshold, Kuiper-Meeks examples, the nonattainment of the $\beta + 4$ level for genera 1 and 2, and the fundamental lemma for isotopy-tight surfaces. It explicitly leaves many related problems open and does not provide a full genus-2 resolution.

7. The total curvature of a knotted torus

- Authors: N. H. Kuiper and W. H. Meeks III
- Venue/year: *Journal of Differential Geometry* 26:3 (1987), 371-384
- DOI/arXiv/URL: DOI 10.4310/jdg/1214441483; <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-26/issue-3/The-total-curvature-of-a-knotted-torus/10.4310/jdg/1214441483.pdf>
- Contribution: Fully resolves the genus-1 analogue by proving there is no isotopy-tight knotted torus. The introduction states that the corresponding genus-2 non-tight isotopy-tight existence question was still unknown. This confirms that the 1984 results had not resolved the exact problem.

8. Recent progress on the geometry of surfaces in R^3 and on the use of computer graphics as a research tool

- Author: W. H. Meeks III
- Venue/year: *Proceedings of the International Congress of Mathematicians, Berkeley 1986*, Vol. 1, 551-560, published 1987
- DOI/arXiv/URL: no arXiv found; IMU proceedings OCR PDF <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf>
- Contribution: Original source context for the problem. It points to the total-curvature/isotopy-tight surface problem area but does not supply a theorem resolving the genus-2 case.

9. Tight Submanifolds, Smooth and Polyhedral

- Authors: Thomas F. Banchoff and Wolfgang Kuhnel

- Venue/year: in *Tight and Taut Submanifolds*, MSRI Publications 32, Cambridge University Press, 1997, pp. 51-118
- DOI/arXiv/URL: DOI 10.1017/9781009701815.003; <https://www.cambridge.org/core/books/abs/tight-and-taut-submanifolds/tight-submanifolds-smooth-and-polyhedral/D0929532D954165E5470391EF8A5EF42>
- Contribution: Later survey restating the relevant known theorem: a knotted torus is never isotopy tight, and knotted isotopy-tight surfaces exist for genus $g \geq 3$ with $(1/2\pi) \int |K| = 8 - \chi(M)$. It does not state a solution for genus 2, which remains the missing case between the torus theorem and the genus $g \geq 3$ examples.

10. Regular homotopy and total curvature II: sphere immersions into 3-space

- Author: Tobias Ekholm
- Venue/year: Algebraic & Geometric Topology 6 (2006), 493-512
- DOI/arXiv/URL: DOI 10.2140/agt.2006.6.493; <https://msp.org/agt/2006/6-1/p17.xhtml>; arXiv version <https://arxiv.org/pdf/math/0310266>
- Contribution: Later work on relatively isotopy-tight immersions and local minima of the total curvature functional. It cites the Kuiper-Meeks structure theorem and proves non-injectivity results for certain RIT sphere immersions. Its hypotheses concern sphere immersions/local minima, not embedded genus-2 ambient isotopy-class minimizers, so it does not resolve this problem.

3.9.3.4. What remains unresolved

The exact missing theorem would have to prove that no non-tight genus-2 isotopy-tight embedding exists, or equivalently that every genus-2 ambient-isotopy-class minimizer has $\mathcal{K} = 12\pi$. Existing results cover the absolute lower bound, the unknotted region below 20π , the first non-tight level 20π , the torus analogue, and the existence of higher-genus counterexamples. They do not eliminate possible genus-2 isotopy-tight embeddings at higher normalized curvature levels such as $\tau \geq 14$. Therefore the literature only partially resolves the problem.

3.9.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.9.5. Primary-source pointer

The originating ICM record is W. H., *Recent progress on the geometry of surfaces in R and on the use of computer graphics as a research tool* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>.

3.10. 71. Properly embedded minimal surfaces with finite topology [2006-ag-115]

Record: 2006-ag-115

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Properly embedded minimal surfaces with finite topology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=917>

3.10.1. Problem statement

For integers $k \geq 0$ and $r \geq 1$, let $\mathcal{M}(k, r)$ denote the moduli space of properly embedded minimal surfaces $\Sigma \subset \mathbb{R}^3$ of finite topology, genus k , and exactly r horizontal ends. Prove that $\mathcal{M}(k, r)$ is empty whenever $r > k + 2$.

3.10.2. Historical background, literature, and current progress

3.10.3. Status: partially resolved

3.10.3.1. Precise problem

For integers $k \geq 0$ and $r \geq 1$, let $\mathcal{M}(k, r)$ be the moduli space of properly embedded minimal surfaces in $\mathbb{R}^3$ with finite topology, genus k , and exactly r horizontal ends. The problem asks for the nonexistence statement

$$\mathcal{M}(k, r) = \emptyset \quad \text{whenever } r > k + 2.$$

This is the nonexistence half of the Hoffman-Meeks finite topology conjecture, in the formulation for finite-topology properly embedded minimal surfaces with horizontal ends.

3.10.3.2. Literature checked

1. Embedded minimal surfaces of finite topology

Authors: David Hoffman and William H. Meeks III

Venue/year: Annals of Mathematics (2) 131, no. 1, 1-34, 1990

DOI/URL: <https://doi.org/10.2307/1971506>; <https://annals.math.princeton.edu/1990/131-1/p01>

Relevance: Original Hoffman-Meeks finite-topology context and examples. Later papers and surveys attribute the finite-topology genus/ends conjecture to Hoffman and Meeks.

2. Properly embedded minimal surfaces with finite topology

Author: Antonio Ros

Venue/year: Proceedings of the International Congress of Mathematicians, Madrid 2006,

Vol. II, 907-926, European Mathematical Society, published 2007

DOI/URL: <https://doi.org/10.4171/022-2/43>; <https://ems.press/books/standalone/21>

Relevance: States the exact $M(k, r)$ formulation: $M(k, r)$ should be empty for $r > k + 2$. It records that $k = 0$ is known and that for higher genus only a sufficiently-large- r nonexistence result was known at that time.

3. Topologie et courbure des surfaces minimales proprement plongées de $\mathbb{R}^3$

Author: Pascal Collin

Venue/year: Annals of Mathematics (2) 145, no. 1, 1-31, 1997

DOI/URL: <https://doi.org/10.2307/2951822>; <https://annals.math.princeton.edu/articles/13049>

Relevance: Proves that a properly embedded minimal surface in $\mathbb{R}^3$ with finite topology and at least two ends has finite total curvature. Since $r > k + 2$ implies $r > 1$, this reduces the stated setting to the finite-total-curvature/plane-or-catenoid-end setting used in the later literature, but does not prove the $r \leq k + 2$ bound.

4. On embedded complete minimal surfaces of genus zero

Authors: Francisco J. López and Antonio Ros

Venue/year: Journal of Differential Geometry 33, no. 1, 293-300, 1991

DOI/URL: <https://doi.org/10.4310/jdg/1214446040>; <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-33/issue-1/On-embedded-complete-minimal-surfaces-of-genus-zero/10.4310/jdg/1214446040.full>

Relevance: Proves the genus-zero case: the only embedded complete genus-zero finite-total-curvature examples are the plane and catenoid. Therefore $M(0, r) = \emptyset$ for $r > 2$. This is only the $k = 0$ case of the problem.

5. Bounds on the topology and index of minimal surfaces

Authors: William H. Meeks III, Joaquín Pérez, Antonio Ros

Venue/year: Acta Mathematica 223, no. 1, 113-149, 2019

DOI/URL: <https://doi.org/10.4310/ACTA.2019.v223.n1.a2>; <https://www.intlpress.com/api/bgcloud-front/resource/pdf/volume/1805558284297904130-1805558284297904130-fb9a5257cf584f50b9d843f7768fe5a9.pdf>

Relevance: Theorem 1.4 proves that, for fixed genus, a properly embedded finite-topology minimal surface has a bound on the number of ends depending only on the genus. This gives $M(k, r) = \emptyset$ for all sufficiently large r depending on k , but it does not identify the sharp bound $k + 2$.

6. Minimal Surfaces of Finite Genus: Classification, Dynamics and Laminations

Author: Joaquín Pérez

Venue/year: arXiv preprint/survey, 2025

arXiv/URL: arXiv:2510.12319; <https://arxiv.org/abs/2510.12319>; <https://arxiv.org/html/2510.12319v1>

Relevance: Section 5 restates the Hoffman-Meeks finite topology conjecture and says that even the genus $g = 1$ case remains open. It identifies the Meeks-Pérez-Ros genus-dependent end bound as the best partial answer up to date.

3.10.3.3. Conclusion

The literature does **not** fully resolve the exact statement in the problem statement. Known results prove the genus-zero case and prove that, for each fixed genus k , there is some finite

upper bound on the number of ends. However, the sharp bound $r \leq k + 2$ remains unproved in positive genus; in particular the first case $k = 1, r = 4$ is still not covered by a published or citable theorem. Therefore the correct verdict is **partially resolved**.

3.10.4. Remaining open problems recorded by the database

3.10.4.1. Open problem 1

Prove the genus-one case of the Hoffman–Meeks end bound: a complete properly embedded minimal surface in $\mathbb{R}^3$ with finite total curvature and genus one has at most three ends; equivalently, $\mathcal{M}(1, r) = \emptyset$ for every $r \geq 4$.

Reference supplied by the database. Matthias Weber, “Recent Progress on Minimal Surfaces with Low Genus,” in Olivier Biquard, Bernhard Leeb, and Gang Tian (organizers), “Differentialgeometrie im Großen,” *Oberwolfach Reports* 6 (2009), 1943–1972, contribution at pp. 1966–1969, especially Conjectures 4.1 and 4.3 and the intervening genus-one classification discussion; DOI: [10.4171/OWR/2009/35](https://doi.org/10.4171/OWR/2009/35).

3.10.5. Primary-source pointer

The originating ICM record is *Properly embedded minimal surfaces with finite topology* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=917>.

3.11. 72. Properly embedded minimal surfaces with finite topology [2006-ag-116]

Record: 2006-ag-116

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Properly embedded minimal surfaces with finite topology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=917>

3.11.1. Problem statement

For integers $k \geq 0$ and $r \geq 1$, let $M(k, r)$ denote the moduli space of properly embedded minimal surfaces $\Sigma \subset \mathbb{R}^3$ of genus k with r horizontal ends. Prove that $M(k, r)$ is either empty or noncompact; equivalently, if it is nonempty, it is not compact in the sense that not every sequence in $M(k, r)$ has a subsequence converging to a single limit surface still lying in $M(k, r)$.

3.11.2. Historical background, literature, and current progress

Verdict: partially resolved.

Precise problem checked: for each genus $k \geq 0$ and number $r \geq 1$ of horizontal ends, decide whether the moduli space $M(k, r)$ of properly embedded minimal surfaces in $\mathbb{R}^3$ is either empty or noncompact. In the source literature this is the Ros compactness/noncompactness question for finite-topology embedded minimal surfaces, most explicitly cited for $r \geq 4$; the low-end cases $r < 4$ are treated separately.

Key source:

- Title: Properly embedded minimal surfaces with finite topology Authors: Antonio Ros Venue: Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. II, pp. 907-926, European Mathematical Society Year: 2006/2007 DOI/URL: DOI 10.4171/022-2/43, <https://ems.press/books/standalone/21/427> Role: This is the source of the moduli-space question. Ros defines $M(k, r)$ for properly embedded finite-topology minimal surfaces with horizontal ends and records the state of compactness and classification questions. Later authors cite Ros's conjecture for $r \geq 4$: if $M(g, r)$ is nonempty, then it should be noncompact.

Global partial progress:

- Title: Bounds on the topology and index of minimal surfaces Authors: William H. Meeks III, Joaquin Perez, Antonio Ros Venue: Acta Mathematica 223(1), 113-149 Year: 2019 DOI/URL: <https://intlpress.com/api/bgcloud-front/resource/pdf/volume/1805558284297904130-1805558284297904130-fb9a5257cf584f50b9d843f7768fe5a9.pdf> Result used: Theorem 1.4 proves that a properly embedded minimal surface in $\mathbb{R}^3$ with finite topology has a bound on

the number of ends depending only on its genus. Why only partial: this gives a major finiteness/topological obstruction toward the Hoffman-Meeks finite topology conjecture, but it does not prove that every nonempty $\mathcal{M}(k, r)$ is noncompact, nor that the compact cases predicted by Ros are empty.

- Title: Minimal Surfaces of Finite Genus: Classification, Dynamics and Laminations Author: Joaquin Perez Venue: arXiv preprint / survey Year: 2025 arXiv/URL: arXiv:2510.12319, <https://arxiv.org/html/2510.12319v1> Result/status used: The survey states that the Hoffman-Meeks finite topology conjecture remains open, even in the simplest positive-genus finite-topology cases, and lists open problems including sharper bounds on the number of ends. Why only partial: it documents the current state after the Meeks-Perez-Ros program; it does not report a theorem resolving Ros's full noncompactness question.

Special cases and related progress:

- Title: On embedded complete minimal surfaces of genus zero Authors: Francisco J. Lopez, Antonio Ros Venue: Journal of Differential Geometry 33(1), 293-300 Year: 1991 DOI/arXiv/URL: bibliographic reference in the current surveys; no arXiv found Result used: the only complete embedded genus-zero finite-total-curvature examples are the plane and catenoid. Why only partial: this resolves the genus-zero finite-total-curvature cases but not positive genus and $r \geq 4$.
- Title: Uniqueness, symmetry, and embeddedness of minimal surfaces Author: Richard M. Schoen Venue: Journal of Differential Geometry 18, 791-809 Year: 1983 DOI/arXiv/URL: bibliographic reference in the current surveys Result used: the complete embedded finite-total-curvature two-ended case is the catenoid. Why only partial: this resolves $r = 2$, but not higher-ended moduli spaces.
- Title: Properly embedded minimal planar domains Authors: William H. Meeks III, Joaquin Perez, Antonio Ros Venue: Annals of Mathematics 181(2), 473-546 Year: 2015 DOI/arXiv/URL: cited in the 2025 survey, <https://arxiv.org/html/2510.12319v1> Result used: classification of properly embedded minimal planar domains, including the Riemann minimal examples in the infinite-topology genus-zero setting. Why only partial: it is a genus-zero classification theorem and does not settle the finite-topology positive-genus compactness question.
- Title: Non-compact families of complete, properly immersed minimal surfaces with finite topology Authors: Stephen J. Kleene, Niels Martin Moller Venue: arXiv preprint Year: 2014, revised 2024 arXiv/URL: arXiv:1409.8381, <https://arxiv.org/pdf/1409.8381> Result used: Theorem 1.1 constructs continuous families of complete minimal surfaces of genus g with four catenoidal ends for large g , giving evidence for noncompactness phenomena. Why only partial: the authors explicitly state that their constructed surfaces are a priori non-embedded and explain that the embedded analogue is expected but not proved there; hence this does not resolve $\mathcal{M}(k, r)$ for properly embedded surfaces.

Conclusion: I found no published or otherwise citable theorem that proves the full statement for all $k \geq 0$ and $r \geq 1$. The literature resolves several low-topology or genus-zero cases and proves strong global bounds on the number of ends in fixed genus, but the exact Ros noncompactness/emptiness statement for the embedded moduli spaces remains open in the remaining higher-genus, higher-end cases.

3.11.3. Remaining open problems recorded by the database

3.11.3.1. Open problem 1

For every integer $G \geq 2$, prove that every complete, properly embedded minimal surface in $\mathbb{R}^3$ with finite total curvature, genus G , and exactly three horizontal ends is a Costa–Hoffman–Meeks surface of genus G ; equivalently, prove that $\mathcal{M}(G, 3)$ consists precisely of the Costa–Hoffman–Meeks surfaces.

Reference supplied by the database. Martin Traizet, “A balancing condition for weak limits of families of minimal surfaces,” *Commentarii Mathematici Helvetici* **79** (2004), no. 4, 798–825, Conjecture 2, DOI: [10.1007/s00014-004-0805-1](https://doi.org/10.1007/s00014-004-0805-1).

3.11.4. Primary-source pointer

The originating ICM record is *Properly embedded minimal surfaces with finite topology* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=917>.

3.12. 73. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-104]

Record: 2010-geom-104

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.12.1. Problem statement

Prove or disprove that every compact immersed surface S in the product space $\mathbb{H}^2 \times \mathbb{R}$ with constant mean curvature and nonnegative second variation for all compactly supported volume-preserving variations, equivalently for all smooth variational functions f on S with $\int_S f = 0$, is a rotational constant-mean-curvature sphere, meaning a sphere invariant under rotations about a vertical axis in $\mathbb{H}^2 \times \mathbb{R}$.

3.12.2. Historical background, literature, and current progress

Verdict: partially resolved.

Precise problem checked: classify compact immersed constant-mean-curvature surfaces in $\mathbb{H}^2 \times \mathbb{R}$ that are stable for all compactly supported volume-preserving variations, equivalently whose CMC stability quadratic form is nonnegative on all smooth functions with zero integral. The proposed conclusion is that every such surface must be a rotational CMC sphere.

Current progress:

1. Title: Global properties of constant mean curvature surfaces in $\mathbb{H}^2 \times \mathbb{R}$ Authors: Barbara Nelli and Harold Rosenberg Venue: Pacific Journal of Mathematics 226(1), 137-152 Year: 2006 DOI/URL: <https://doi.org/10.2140/pjm.2006.226.137> ; <https://msp.org/pjm/2006/226-1/p07.xhtml> Relevance: Establishes foundational global restrictions for CMC surfaces in $\mathbb{H}^2 \times \mathbb{R}$. In particular, later papers cite it for the fact that compact immersed CMC surfaces in $\mathbb{H}^2 \times \mathbb{R}$ must have $H > 1/2$. This narrows the possible range but does not classify stable compact immersed examples.
2. Title: A Hopf differential for constant mean curvature surfaces in $S^2 \times \mathbb{R}$ and $\mathbb{H}^2 \times \mathbb{R}$ Authors: Uwe Abresch and Harold Rosenberg Venue: Acta Mathematica 193, 141-174 Year: 2004 DOI/URL: <https://doi.org/10.1007/BF02392562> Relevance: Introduces the Abresch-Rosenberg holomorphic quadratic differential and gives the Hopf-type classification of immersed CMC spheres in these product spaces. Thus, if stability could be used to prove that the surface is topologically a sphere, this result would identify it as rotational. It does not by itself prove that every stable compact immersed CMC surface is a sphere.
3. Title: On stable constant mean curvature surfaces in $S^2 \times \mathbb{R}$ and $\mathbb{H}^2 \times \mathbb{R}$ Author: Rabah

Souam Venue: Transactions of the American Mathematical Society 362(6), 2845-2857
 Year: 2010 DOI/arXiv/URL: <https://doi.org/10.1090/S0002-9947-10-04826-9> ; arXiv:0709.4204 ; <https://arxiv.org/abs/0709.4204> Relevance: This is the main partial result for the exact stability question. It proves that all rotational CMC spheres in $\mathbb{H}^2 \times \mathbb{R}$ are stable, and proves the desired converse only under the additional hypothesis $H \geq 1/\sqrt{2}$: a closed stable immersed CMC surface with $H \geq 1/\sqrt{2}$ is a rotational sphere. For lower H , it gives genus bounds, for example genus at most one when $1/\sqrt{3} < H < 1/\sqrt{2}$ and genus at most two at $H = 1/\sqrt{3}$, but does not rule out stable nonspherical compact immersed surfaces throughout the full remaining range.

4. Title: On the uniqueness of isoperimetric solutions and imbedded soap bubbles in non-compact symmetric spaces, I Authors: Wu-Teh Hsiang and Wu-Yi Hsiang Venue: Inventiones Mathematicae 98, 39-58 Year: 1989 DOI/URL: <https://doi.org/10.1007/BF01388843> Relevance: Gives uniqueness/isoperimetric and embedded soap-bubble results for the rotational CMC spheres in the relevant product spaces. This supports the rotational-sphere picture for embedded or isoperimetric hypersurfaces, but the problem asks about compact immersed weakly stable CMC surfaces, so this does not cover the full statement.
5. Title: Constant mean curvature surfaces in 3-dimensional Thurston geometries Authors: Isabel Fernandez and Pablo Mira Venue: Invited contribution to the Proceedings of the International Congress of Mathematicians 2010; Proceedings of the ICM, Volume II, Hindustan Book Agency, pp. 830-861 Year: 2010 arXiv/URL: arXiv:1004.4752 ; <https://arxiv.org/abs/1004.4752> Relevance: Survey of compact CMC surfaces in Thurston geometries. It explicitly records the Nelli-Rosenberg conjectural status: compact weakly stable CMC surfaces in $\mathbb{H}^2 \times \mathbb{R}$ are expected to be rotational CMC spheres. This survey source treats the statement as conjectural/open rather than settled.
6. Title: On Stability and Isoperimetry of Constant Mean Curvature Spheres of $\mathbb{H}^n \times \mathbb{R}$ and $\mathbb{S}^n \times \mathbb{R}$ Authors: Ronaldo F. de Lima, Maria F. Elbert, and Barbara Nelli Venue: To appear in Mathematische Zeitschrift; preprint Year: 2023 preprint, revised 2024 arXiv/URL: arXiv:2301.11038 ; <https://arxiv.org/abs/2301.11038> Relevance: Extends and refines stability/isoperimetry properties of the known rotational CMC spheres, proving in particular that all rotational CMC spheres in $\mathbb{H}^n \times \mathbb{R}$ are stable. It does not prove the converse classification of all compact immersed weakly stable CMC surfaces in $\mathbb{H}^2 \times \mathbb{R}$.

What remains unresolved: the exact statement still lacks a citable theorem covering every compact immersed weakly stable CMC surface in $\mathbb{H}^2 \times \mathbb{R}$. The literature proves the desired conclusion for $H \geq 1/\sqrt{2}$, proves the known rotational spheres are stable, and proves related embedded/isoperimetric uniqueness results. I did not find a published or citable result ruling out all possible stable nonspherical compact immersed CMC surfaces in the remaining mean-curvature regime $1/2 < H < 1/\sqrt{2}$. Therefore the problem is partially resolved, not fully resolved and not fully open.

3.12.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.12.4. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.13. 74. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-105]

Record: 2010-geom-105

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.13.1. Problem statement

Prove or disprove that, in the Heisenberg space $\text{Nil}_3 = \mathbb{E}^3(0, \tau)$ with $\tau \neq 0$, the simply connected three-dimensional Heisenberg group endowed with its standard left-invariant homogeneous metric, the isoperimetric regions, namely regions of prescribed volume minimizing boundary area, are precisely the regions bounded by the canonical rotational constant-mean-curvature spheres; that is, the embedded CMC spheres invariant under rotations about the vertical axis of the fibration $\mathbb{E}^3(0, \tau) \rightarrow \mathbb{R}^2$, unique up to ambient isometry for their mean curvature.

3.13.2. Historical background, literature, and current progress

3.13.3. Verdict: partially resolved

3.13.3.1. Precise problem

The problem asks for the full isoperimetric classification in the Riemannian Heisenberg space $\text{Nil}_3 = \mathbb{E}^3(0, \tau)$, $\tau \neq 0$: for every prescribed volume, are the perimeter-minimizing regions exactly the regions enclosed by the canonical rotational constant-mean-curvature spheres, up to ambient isometry?

3.13.3.2. Current literature status

No citable published result found proves the full statement. The strongest current results classify the candidate CMC spheres and prove several necessary, conditional, asymptotic, or small-volume facts, but they do not prove that every isoperimetric region in Nil_3 has spherical topology, nor that every canonical rotational CMC sphere is globally isoperimetric for its enclosed volume.

Key literature:

- **Constant mean curvature surfaces in 3-dimensional Thurston geometries;** Isabel Fernández and Pablo Mira; *Proceedings of the International Congress of Mathematicians, Volume II (Invited Lectures)*, Hindustan Book Agency, 2010, pp. 830-861; arXiv:1004.4752,

<https://arxiv.org/abs/1004.4752>. This is the source survey. It states that for the remaining $E^3(\kappa, \tau)$ spaces, including Nil_3 , the isoperimetric problem is open; it records only the small-volume consequence that isoperimetric solutions are canonical rotational CMC spheres with large mean curvature.

- **CMC spheres in the Heisenberg group**; Valentina Franceschi, Francescopaolo Montefalcone, Roberto Monti; *Analysis and Geometry in Metric Spaces* 7(1), 109-129, 2019; DOI: 10.1515/agms-2019-0006; arXiv:1611.09215, <https://arxiv.org/abs/1611.09215>. This paper studies the explicit Riemannian Heisenberg CMC spheres and says they are conjectured to be the isoperimetric sets. It proves supporting results, including classification of topological CMC spheres as translations of the model spheres and a restricted quantitative stability inequality, but explicitly does not prove the isoperimetric property.
- **Isoperimetric domains of large volume in homogeneous three-manifolds**; William H. Meeks III, Pablo Mira, Joaquín Pérez, Antonio Ros; *Advances in Mathematics* 264, 546-592, 2014; DOI: 10.1016/j.aim.2014.07.025; arXiv:1303.4222, <https://arxiv.org/abs/1303.4222>. This gives asymptotic information for large-volume isoperimetric domains in homogeneous three-manifolds. It does not classify Nil_3 isoperimetric domains as canonical rotational CMC spheres.
- **Constant mean curvature spheres in homogeneous three-manifolds**; William H. Meeks III, Pablo Mira, Joaquín Pérez, Antonio Ros; *Inventiones Mathematicae* 224, 147-244, 2021; DOI: 10.1007/s00222-020-01008-y; <https://link.springer.com/article/10.1007/s00222-020-01008-y>. This fully classifies immersed CMC spheres in homogeneous three-manifolds. In particular, combined with the Abresch-Rosenberg theory, CMC spheres in Nil_3 are the rotational ones and are unique up to isometry for their mean curvature. This is not enough to solve the problem statement because an isoperimetric boundary is only known a priori to be a stable embedded CMC surface, not necessarily a sphere.
- **Uniqueness of Constant Mean Curvature Spheres**; Pablo Mira and Joaquín Pérez; in *New Trends in Geometric Analysis*, RSME Springer Series, vol. 10, Springer, 2023, pp. 365-393; DOI: 10.1007/978-3-031-39916-9_13. This recent survey explicitly says that in Nil_3 , although all CMC spheres are rotational and weakly stable, it is not known whether isoperimetric regions are topological balls; the obstruction is the lack of reflections/Alexandrov reflection methods in Nil_3 .

Important non-resolution:

- **Isoperimetric domains in homogeneous three-manifolds and the isoperimetric constant of the Heisenberg group H^1** ; Jih-Hsin Cheng, Andrea Malchiodi, Paul Yang; arXiv preprint, 2015; arXiv:1501.07357, <https://arxiv.org/abs/1501.07357>. This preprint claimed a broad topological-ball theorem that would be relevant, but arXiv marks it as withdrawn due to a problem in Section 2. It should not be treated as resolving literature.

3.13.3.3. What remains unresolved

The missing full step is to exclude non-spherical isoperimetric boundaries in Nil_3 , or equivalently to prove a theorem strong enough to show that every isoperimetric region is a topological ball/sphere-bounded region. Once that topology is known, the CMC-sphere classification would identify the boundary with a canonical rotational CMC sphere. Conversely, the global

minimizing property of the entire canonical family is still only conjectural outside the known small-volume/conditional regimes.

3.13.4. Remaining open problems recorded by the database

3.13.4.1. Open problem 1

Prove or disprove that every isoperimetric region in the Riemannian Heisenberg space Nil_3 is a topological three-ball (equivalently, that its boundary is a topological two-sphere).

Reference supplied by the database. Pablo Mira and Joaquín Pérez, “Uniqueness of Constant Mean Curvature Spheres,” in *New Trends in Geometric Analysis*, RSME Springer Series, vol. 10, Springer, Cham, 2023, pp. 365–393; discussion of the open isoperimetric-topology step for Nil_3 (no numbered problem). DOI: 10.1007/978-3-031-39916-9_13. https://doi.org/10.1007/978-3-031-39916-9_13

3.13.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.14. 75. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-106]

Record: 2010-geom-106

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.14.1. Problem statement

Let $\mathbb{E}^3(\kappa, \tau)$ be a fixed homogeneous space with $\kappa < 0$ and $\tau \neq 0$, equivalently the universal cover of $\mathrm{PSL}(2, \mathbb{R})$ endowed with one of its standard left-invariant homogeneous metrics. Prove or disprove that the isoperimetric regions in $\mathbb{E}^3(\kappa, \tau)$ are exactly the compact regions bounded by the canonical rotational constant-mean-curvature spheres. Here an isoperimetric region means a region whose boundary has least area among regions with the same enclosed volume. The canonical rotational constant-mean-curvature spheres are the embedded CMC H -spheres in $\mathbb{E}^3(\kappa, \tau)$ that are invariant under rotations about a vertical fiber; for $\kappa - 4\tau^2 < 0$, such a sphere exists uniquely up to ambient isometry precisely when $H^2 > -\kappa/4$.

3.14.2. Historical background, literature, and current progress

3.14.3. Verdict: partially resolved

3.14.3.1. Precise problem checked

For a fixed simply connected homogeneous space $\mathbb{E}^3(\kappa, \tau)$ with $\kappa < 0$ and $\tau \neq 0$, i.e. the universal cover of $\mathrm{PSL}(2, \mathbb{R})$ with one of its standard 4-dimensional-isometry-group metrics, determine whether every isoperimetric region of every prescribed volume is exactly the compact region bounded by the canonical rotational embedded CMC sphere, and conversely whether every such spherical region is isoperimetric.

3.14.3.2. Literature and current progress

1. Constant mean curvature surfaces in 3-dimensional Thurston geometries

Authors: Isabel Fernández and Pablo Mira.

Venue/year: Proceedings of the International Congress of Mathematicians 2010; arXiv preprint, 2010.

DOI/arXiv/URL: arXiv:1004.4752, <https://arxiv.org/abs/1004.4752>.

Contribution: This survey is the apparent source of the problem. In Section 4.4–4.5 it explicitly says that the isoperimetric problem is open in Nil_3 , the universal cover of

$\mathrm{PSL}(2, \mathbb{R})$, and Berger spheres outside the Torralbo–Urbano range. It also records the partial fact that for small volumes the general isoperimetric theory plus the Abresch–Rosenberg CMC-sphere uniqueness theorem gives canonical rotational CMC spheres with large H as the isoperimetric solutions.

2. Rotationally invariant constant mean curvature surfaces in homogeneous 3-manifolds

Author: Francisco Torralbo.

Venue/year: *Differential Geometry and its Applications* 28 (2010), no. 5, 593–607.

DOI/arXiv/URL: DOI 10.1016/j.difgeo.2010.04.007; arXiv:0911.5128, <https://arxiv.org/abs/0911.5128>.

Contribution: Classifies rotational CMC surfaces in Berger spheres and in $\mathrm{SL}(2, \mathbb{R})$ -type homogeneous spaces, including the canonical rotational CMC spheres used in the conjecture. This supplies the candidate spheres and their ODE geometry, but it does not prove that their enclosed domains minimize area for all prescribed volumes.

3. Compact stable constant mean curvature surfaces in homogeneous 3-manifolds

Authors: Francisco Torralbo and Francisco Urbano.

Venue/year: *Indiana University Mathematics Journal* 61 (2012), no. 3, 1129–1156.

DOI/arXiv/URL: DOI 10.1512/iumj.2012.61.4667; <https://iumj.org/article/5606/>.

Contribution: Proves that the constant-mean-curvature spheres in the special linear group case and in Nil_3 are stable, and gives partial classification results for compact stable CMC surfaces under additional hypotheses. Stability is necessary for isoperimetry, but not sufficient; the paper does not prove the full isoperimetric classification in $\mathrm{PSL}(2, \mathbb{R})$.

4. Isoperimetric domains of large volume in homogeneous three-manifolds

Authors: William H. Meeks III, Pablo Mira, Joaquín Pérez, and Antonio Ros.

Venue/year: *Advances in Mathematics* 264 (2014), 546–592.

DOI/arXiv/URL: DOI 10.1016/j.aim.2014.07.025; arXiv:1303.4222, <https://arxiv.org/abs/1303.4222>.

Contribution: Gives large-volume asymptotics for isoperimetric domains in noncompact simply connected homogeneous three-manifolds: radii diverge, area/volume ratios converge to the Cheeger constant, boundary mean curvatures converge to $\mathrm{Ch}(X)/2$, and large boundaries are approximated by a CMC foliation. This is substantial asymptotic progress, but it does not identify the exact minimizers for every finite volume as the canonical rotational CMC spheres.

5. Constant mean curvature spheres in homogeneous three-manifolds

Authors: William H. Meeks III, Pablo Mira, Joaquín Pérez, and Antonio Ros.

Venue/year: *Inventiones Mathematicae* 224 (2021), 147–244.

DOI/arXiv/URL: DOI 10.1007/s00222-020-01008-y; arXiv:1706.09394, <https://arxiv.org/abs/1706.09394>.

Contribution: Completely classifies immersed CMC spheres in homogeneous three-manifolds. For X diffeomorphic to $\mathbb{R}^3$, CMC spheres exist exactly for $|H| > \mathrm{Ch}(X)/2$, are unique up to isometry for fixed H , are maximally symmetric, and in rotationally symmetric $\mathbb{E}^3(\kappa, \tau)$ spaces are rotational. This proves the uniqueness/classification of the candidate boundaries, but it does not prove that every isoperimetric boundary has genus zero, nor that each candidate sphere is globally area-minimizing for its enclosed volume.

6. Isoperimetric domains in homogeneous three-manifolds and the isoperimetric constant of the Heisenberg group H^1

Authors: Jih-Hsin Cheng, Andrea Malchiodi, and Paul Yang.

Venue/year: arXiv preprint, 2015.

DOI/arXiv/URL: arXiv:1501.07357, <https://arxiv.org/abs/1501.07357>.

Contribution: The preprint claimed that isoperimetric sets in homogeneous three-spaces diffeomorphic to $\mathbb{R}^3$ are topological balls, which together with the CMC-sphere classification would have been close to resolving this problem. However, the arXiv record states that the paper was withdrawn due to a problem in Section 2, so it cannot be counted as a valid resolution.

3.14.3.3. What remains unresolved

No valid published or otherwise citable result found proves the full biconditional in the problem statement: for every volume in every $\mathbb{E}^3(\kappa, \tau)$ with $\kappa < 0$, $\tau \neq 0$, the isoperimetric regions are exactly the compact regions bounded by canonical rotational CMC spheres. The remaining gap is global isoperimetric minimality and exclusion of all other possible isoperimetric boundaries for all volumes. The literature gives strong partial progress: small-volume solutions, stability of the candidate spheres, large-volume asymptotics, and complete uniqueness/classification of CMC spheres.

3.14.4. Remaining open problems recorded by the database

3.14.4.1. Open problem 1

Prove or disprove that every isoperimetric region in a fixed homogeneous space $\mathbb{E}^3(\kappa, \tau)$ with $\kappa < 0$ and $\tau \neq 0$ is a topological three-ball (equivalently, that every isoperimetric boundary is a topological two-sphere).

Reference supplied by the database. William H. Meeks III, Joaquín Pérez, and Giuseppe Tinaglia, “Constant Mean Curvature Surfaces,” *Surveys in Differential Geometry* 21 (2016), 179–287, Conjecture 13.8(1), “Isoperimetric Domains Conjecture” (numbered Conjecture 15.18(1) in the author-manuscript pagination). DOI: 10.4310/SDG.2016.v21.n1.a6. arXiv:1605.02512. <https://doi.org/10.4310/SDG.2016.v21.n1.a6>

3.14.4.2. Open problem 2

Prove or disprove that every canonical rotational constant-mean-curvature sphere in a fixed homogeneous space $\mathbb{E}^3(\kappa, \tau)$ with $\kappa < 0$ and $\tau \neq 0$ bounds an isoperimetric region.

Reference supplied by the database. William H. Meeks III, Joaquín Pérez, and Giuseppe Tinaglia, “Constant Mean Curvature Surfaces,” *Surveys in Differential Geometry* 21 (2016), 179–287, Conjecture 13.8(2), “Isoperimetric Domains Conjecture” (numbered Conjecture 15.18(2) in the author-manuscript pagination). DOI: 10.4310/SDG.2016.v21.n1.a6. arXiv:1605.02512. <https://doi.org/10.4310/SDG.2016.v21.n1.a6>

3.14.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.15. 76. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-107]

Record: 2010-geom-107

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.15.1. Problem statement

Prove or disprove that the following Alexandrov classification holds in the non-product homogeneous spaces $\mathbb{E}^3(\kappa, \tau)$ with $\tau \neq 0$: in the Heisenberg space Nil_3 , in the universal covering of $\text{PSL}(2, \mathbb{R})$ with its left-invariant homogeneous metric, and in hemispheres of Berger 3-spheres, every compact embedded surface with constant mean curvature is, up to an ambient isometry, the canonical rotational constant-mean-curvature sphere with the same mean curvature; that is, the rotational CMC 2-sphere invariant under rotations about a fiber (the vertical axis) of the canonical fibration of the corresponding $\mathbb{E}^3(\kappa, \tau)$ space over a 2-dimensional space form, unique among rotational CMC spheres for that mean curvature whenever such a sphere exists.

3.15.2. Historical background, literature, and current progress

3.15.3. Verdict: partially resolved

3.15.3.1. Precise problem

The problem asks for the Alexandrov classification of compact embedded constant-mean-curvature surfaces in the non-product homogeneous spaces $\mathbb{E}^3(\kappa, \tau)$, $\tau \neq 0$: Nil_3 , $\widetilde{\text{PSL}}(2, \mathbb{R})$ / $\widetilde{\text{SL}}_2(\mathbb{R})$, and Berger hemispheres. The desired conclusion is that every such surface is congruent to the canonical rotational CMC sphere with the same mean curvature, whenever that sphere exists. This is stronger than the Hopf problem, because no genus-zero hypothesis is allowed.

3.15.3.2. Literature checked and current progress

1. Isabel Fernández and Pablo Mira, “Constant mean curvature surfaces in 3-dimensional Thurston geometries”, *Proceedings of the International Congress of Mathematicians, Hyderabad 2010*, Vol. II, Hindustan Book Agency, 2010, pp. 830-861. arXiv:1004.4752. URL: <https://arxiv.org/abs/1004.4752>.

This is the source of the problem. Section 4.5 states that the Alexandrov problem for $\tau \neq 0$, namely in Nil_3 , $\widetilde{\text{PSL}}(2, \mathbb{R})$, and Berger hemispheres, is a major open problem and

conjectures that the canonical rotational spheres are the only compact embedded CMC surfaces. The same survey records the known partial results: existence/uniqueness of rotational CMC spheres, Hopf uniqueness for immersed CMC spheres, and some isoperimetric/stability results.

2. Uwe Abresch and Harold Rosenberg, "A Hopf differential for constant mean curvature surfaces in $S^2 \times \mathbb{R}$ and $H^2 \times \mathbb{R}$ ", *Acta Mathematica* 193(2), 141-174, 2004. DOI: 10.1007/BF02392562. URL: <https://doi.org/10.1007/BF02392562>. Uwe Abresch and Harold Rosenberg, "Generalized Hopf differentials", *Matemática Contemporânea* 28, 1-28, 2005. MR 2195187.

These papers give the Abresch-Rosenberg holomorphic differential and solve the Hopf-type genus-zero problem in the four-dimensional-isometry homogeneous spaces: any immersed CMC sphere is rotational. This is important partial progress but does not imply the Alexandrov statement, because an arbitrary compact embedded CMC surface need not be known to be a sphere.

3. Francisco Torralbo, "Rotationally invariant constant mean curvature surfaces in homogeneous 3-manifolds", *Differential Geometry and its Applications* 28(5), 593-607, 2010. DOI: 10.1016/j.difgeo.2010.04.007. arXiv:0911.5128. URL: <https://arxiv.org/abs/0911.5128>.

This classifies rotational CMC surfaces in Berger spheres and in the special linear group setting, including the rotational CMC spheres and Delaunay-type examples. It verifies the rotational model family, but only within the rotational class, so it does not rule out non-rotational compact embedded CMC surfaces.

4. Francisco Torralbo and Francisco Urbano, "Compact stable constant mean curvature surfaces in homogeneous 3-manifolds", *Indiana University Mathematics Journal* 61(3), 1129-1156, 2012. DOI: 10.1512/iumj.2012.61.4667. arXiv:0906.1439. URL: <https://arxiv.org/abs/0906.1439>.

This classifies stable CMC spheres in Berger spheres, Nil_3 , and $\widetilde{\text{SL}}_2(\mathbb{R})$, and classifies stable compact orientable CMC surfaces in certain Berger-sphere parameter ranges, with isoperimetric consequences. The hypotheses are stability and/or restricted Berger parameters, so this is not the unrestricted Alexandrov classification.

5. William H. Meeks III, Pablo Mira, Joaquín Pérez and Antonio Ros, "Constant mean curvature spheres in homogeneous three-manifolds", *Inventiones Mathematicae* 224, 147-244, 2021. DOI: 10.1007/s00222-020-01008-y. URL: <https://doi.org/10.1007/s00222-020-01008-y>.

This gives a complete classification of immersed CMC spheres in arbitrary homogeneous three-manifolds. It subsumes and extends Hopf-type uniqueness, but its theorem is explicitly about genus-zero immersed spheres. It therefore does not prove that every compact embedded CMC surface in Nil_3 or $\widetilde{\text{SL}}_2(\mathbb{R})$ has genus zero.

6. William H. Meeks III, Pablo Mira, Joaquín Pérez and Antonio Ros, "Constant mean curvature spheres in homogeneous three-spheres", *Journal of Differential Geometry* 120(2), 307-343, 2022. DOI: 10.4310/jdg/1645207520. arXiv:1308.2612. URL: <https://doi.org/10.4310/jdg/1645207520>.

This resolves uniqueness/existence of CMC spheres in homogeneous three-spheres, including Berger-sphere settings. Again, it is a CMC-sphere theorem rather than a classification of all compact embedded CMC surfaces in the Berger-hemisphere Alexandrov sense.

7. Antonio Bueno, "Delaunay surfaces of prescribed mean curvature in Nil_3 and $\widetilde{\text{SL}}_2(\mathbb{R})$ ", arXiv preprint, submitted 2020, revised 2021. arXiv:2003.13018. DOI:

10.48550/arXiv.2003.13018. URL: <https://arxiv.org/abs/2003.13018>.

The introduction explicitly states that, for CMC surfaces, the Alexandrov problem in Nil_3 and $\widetilde{\text{SL}}_2(\mathbb{R})$ is still an outstanding major open problem. The paper studies rotational prescribed-mean-curvature analogues and does not resolve the CMC Alexandrov classification.

3.15.3.3. What remains unresolved

No published or otherwise citable theorem found covers the full statement. The literature proves uniqueness of CMC spheres and gives strong rotational, stable, and isoperimetric special cases, but the decisive missing step for the full Alexandrov classification is to show that an arbitrary compact embedded CMC surface in Nil_3 and in $\widetilde{\text{PSL}}(2, \mathbb{R})$ must be genus zero/rotational. Because those two cases are still described in recent literature as open, the problem is not fully resolved.

3.15.4. Remaining open problems recorded by the database

3.15.4.1. Open problem 1

Prove or disprove that every compact embedded constant-mean-curvature surface in the Heisenberg space Nil_3 is a canonical rotational constant-mean-curvature sphere.

Reference supplied by the database. Pablo Mira and Joaquín Pérez, “Uniqueness of Constant Mean Curvature Spheres,” in *New Trends in Geometric Analysis*, RSME Springer Series, vol. 10, Springer, Cham, 2023, pp. 365–393, Conjecture 9.2. DOI: 10.1007/978-3-031-39916-9_13. https://doi.org/10.1007/978-3-031-39916-9_13

3.15.4.2. Open problem 2

Prove or disprove that every compact embedded constant-mean-curvature surface in the universal cover of $\text{PSL}(2, \mathbb{R})$, equipped with a standard left-invariant homogeneous metric $\mathbb{E}^3(\kappa, \tau)$ with $\kappa < 0$ and $\tau \neq 0$, is a canonical rotational constant-mean-curvature sphere.

Reference supplied by the database. Isabel Fernández and Pablo Mira, “Constant Mean Curvature Surfaces in 3-Dimensional Thurston Geometries,” in *Proceedings of the International Congress of Mathematicians, Hyderabad 2010*, vol. II, Hindustan Book Agency, New Delhi, 2010, pp. 830–861, §4.5 “Open problems” (the $\widetilde{\text{PSL}}(2, \mathbb{R})$ special case of the Alexandrov problem; no numbered conjecture). arXiv:1004.4752. <https://arxiv.org/abs/1004.4752>

3.15.4.3. Open problem 3

Prove or disprove that every compact embedded constant-mean-curvature surface contained in a hemisphere of a Berger three-sphere is a canonical rotational constant-mean-curvature sphere.

Reference supplied by the database. Isabel Fernández and Pablo Mira, “Constant Mean Curvature Surfaces in 3-Dimensional Thurston Geometries,” in *Proceedings of the International Congress of Mathematicians, Hyderabad 2010*, vol. II, Hindustan Book Agency, New Delhi, 2010, pp. 830–861, §4.5 “Open problems” (the Berger-hemisphere special case of the Alexandrov problem; no numbered conjecture). arXiv:1004.4752. <https://arxiv.org/abs/1004.4752>

3.15.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.16. 77. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-108]

Record: 2010-geom-108

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.16.1. Problem statement

Prove that every immersed two-sphere S of constant mean curvature in the Thurston geometry Sol_3 , realized as the Lie group $\mathbb{R}^3$ with left-invariant metric

$$ds^2 = e^{2x_3} dx_1^2 + e^{-2x_3} dx_2^2 + dx_3^2,$$

is weakly stable: if η is a unit normal to S , $\mathcal{B}$ is its second fundamental form, and $\mathcal{L} = \Delta + \|\mathcal{B}\|^2 + \text{Ric}(\eta)$ is its Jacobi operator, then

$$-\int_S f \mathcal{L}(f) dA \geq 0$$

for every smooth function f on S satisfying $\int_S f dA = 0$.

3.16.2. Historical background, literature, and current progress

3.16.3. Verdict: partially resolved

3.16.3.1. Problem checked

The problem asks whether every immersed constant-mean-curvature two-sphere in the standard Thurston geometry Sol_3 ,

$$ds^2 = e^{2x_3} dx_1^2 + e^{-2x_3} dx_2^2 + dx_3^2,$$

is weakly stable for volume-preserving variations, i.e. whether

$$-\int_S f (\Delta f + \|\mathcal{B}\|^2 f + \text{Ric}(\eta)f) dA \geq 0$$

for every smooth zero-mean function f on S .

3.16.3.2. Status

I did not find a published or otherwise citable theorem proving this full weak-stability statement for all CMC spheres in Sol_3 . Recent citable literature still treats the exact assertion as

open. The problem is not fully open, however, because the CMC spheres in Sol3 have been classified and are known to be embedded and of index one.

3.16.3.3. Progress in the literature

1. Existence and uniqueness of constant mean curvature spheres in Sol3

Authors: Benoit Daniel and Pablo Mira.

Venue: Journal für die reine und angewandte Mathematik 685 (2013), 1-32.

DOI/arXiv/URL: DOI 10.1515/crelle-2012-0016; arXiv 0812.3059; <http://arxiv.org/abs/0812.3059>.

Result: Theorems 1.1 and 1.2 prove existence/uniqueness and embeddedness for $H > 1/\sqrt{3}$, with index one, and prove a conditional uniqueness theorem when a CMC H -sphere satisfies one of several hypotheses including weak stability, index one, or Gauss-map diffeomorphism. This gives major partial progress but does not prove weak stability for every CMC sphere.

2. Constant mean curvature spheres in Sol3

Author: William H. Meeks III.

Venue: American Journal of Mathematics 135(3) (2013), 763-775.

DOI/arXiv/URL: DOI 10.1353/ajm.2013.0025; JSTOR page <https://www.jstor.org/stable/23525906>.

Result: Completes the Sol3 CMC-sphere classification for every $H > 0$: existence, uniqueness up to left translation, embeddedness, and index one. This still does not imply the volume-preserving weak-stability inequality, since index one is weaker than weak stability.

3. Constant mean curvature spheres in homogeneous three-manifolds

Authors: William H. Meeks III, Pablo Mira, Joaquin Perez, Antonio Ros.

Venue: Inventiones Mathematicae 224 (2021), 147-244.

DOI/arXiv/URL: DOI 10.1007/s00222-020-01008-y; arXiv 1706.09394; <https://link.springer.com/article/10.1007/s00222-020-01008-y>.

Result: Gives the general homogeneous-three-manifold classification. For Sol3 it confirms the known picture: unique embedded CMC spheres with index one and nullity three. The theorem's conclusions do not include weak stability.

4. Uniqueness of Constant Mean Curvature Spheres

Authors: Pablo Mira and Joaquin Perez.

Venue: New Trends in Geometric Analysis, RSME Springer Series 10 (2023), 365-393.

DOI/URL: DOI 10.1007/978-3-031-39916-9_13; https://link.springer.com/chapter/10.1007/978-3-031-39916-9_13.

Result: This recent survey explicitly states that the Sol3 CMC spheres classified by Daniel-Mira and Meeks are embedded and have index one, but that it is not known whether they are weakly stable; it records the conjecture "Any CMC sphere in Sol3 is weakly stable."

3.16.3.4. What remains unresolved

The missing step is a citable proof that the known index-one CMC spheres in Sol3 satisfy the full zero-mean Jacobi quadratic-form inequality. The available classification and index-one theorems are necessary background, but they do not by themselves prove weak stability.

I also checked the arXiv preprint **Isoperimetric domains in homogeneous three-manifolds**

and the isoperimetric constant of the Heisenberg group H_1 by Jih-Hsin Cheng, Andrea Malchiodi, and Paul Yang, arXiv 1501.07357; arXiv marks it withdrawn because of a problem in Section 2, so I did not count it as resolving literature.

3.16.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.16.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.17. 78. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-109]

Record: 2010-geom-109

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.17.1. Problem statement

Prove that every constant mean curvature sphere in the Thurston geometry Sol_3 , realized as $\mathbb{R}^3$ with the standard left-invariant metric $ds^2 = e^{2x_3} dx_1^2 + e^{-2x_3} dx_2^2 + dx_3^2$, is the boundary of an isoperimetric region in Sol_3 , that is, of a region whose boundary has least area among regions with the same enclosed volume.

3.17.2. Historical background, literature, and current progress

3.17.3. Verdict: partially resolved

3.17.3.1. Precise problem

The problem asks whether, in the standard Thurston geometry $\text{Sol}_3 = (\mathbb{R}^3, e^{2x_3} dx_1^2 + e^{-2x_3} dx_2^2 + dx_3^2)$, every compact constant mean curvature two-sphere S is the boundary of an isoperimetric region: equivalently, whether the domain enclosed by S minimizes boundary area among all regions in Sol_3 with the same enclosed volume.

3.17.3.2. Current status

No published or otherwise reliable citable result found proves this for all CMC spheres in Sol_3 . The literature fully classifies the CMC spheres in Sol_3 , and proves the desired isoperimetric conclusion in the small-volume/large-mean-curvature regime, but the all- $H > 0$ isoperimetric assertion remains open. In particular, a 2023 survey by Mira and Perez still records that it is not known whether the Sol_3 CMC spheres are weakly stable, and says that a positive answer would imply that all CMC spheres in Sol_3 bound isoperimetric regions.

3.17.3.3. Key literature

1. “Constant mean curvature surfaces in 3-dimensional Thurston geometries”

Authors: Isabel Fernandez and Pablo Mira.

Venue/year: Proceedings of the International Congress of Mathematicians, Volume II, Hindustan Book Agency, 2010, pp. 830-861.

arXiv/DOI/URL: arXiv:1004.4752; DOI 10.1142/9789814324359_0076; <https://arxiv.org/abs/1004.4752>.

Role: This is the original survey source matching the problem statement. It states the Sol_3 classification theorem of Daniel-Mira and Meeks, but then lists as open: whether CMC spheres in Sol_3 are weakly stable and whether they all bound isoperimetric regions.

2. **“Existence and uniqueness of constant mean curvature spheres in Sol_3 ”**

Authors: Benoit Daniel and Pablo Mira.

Venue/year: Journal fuer die reine und angewandte Mathematik (Crelle’s Journal) 685 (2013), 1-32.

arXiv/DOI/URL: arXiv:0812.3059; DOI 10.1515/crelle-2012-0016; <https://arxiv.org/abs/0812.3059>.

Role: Proves existence and uniqueness of embedded CMC H -spheres for $H > 1/\sqrt{3}$, and proves a conditional uniqueness theorem for spheres satisfying properties such as being isoperimetric, weakly stable, index one, or having diffeomorphic Gauss map. This is major progress but does not prove that every CMC sphere is isoperimetric.

3. **“Constant mean curvature spheres in Sol_3 ”**

Author: William H. Meeks III.

Venue/year: American Journal of Mathematics 135 no. 3 (2013), 763-775.

DOI/URL: DOI 10.1353/ajm.2013.0025; JSTOR <https://www.jstor.org/stable/23525906>.

Role: Completes the Sol_3 classification by proving existence and uniqueness of an immersed CMC H -sphere for each $H > 0$. The result gives uniqueness/embeddedness/index-one information, not the global area-minimizing isoperimetric property.

4. **“Constant mean curvature spheres in homogeneous three-manifolds”**

Authors: William H. Meeks III, Pablo Mira, Joaquin Perez, Antonio Ros.

Venue/year: Inventiones Mathematicae 224 (2021), 147-244.

arXiv/DOI/URL: arXiv:1706.09394; DOI 10.1007/s00222-020-01008-y; <https://arxiv.org/abs/1706.09394>.

Role: Gives a broad classification of CMC spheres in homogeneous three-manifolds. It records, in the proof discussion around Proposition 3.4, that for H sufficiently large the spheres S_H bound small isoperimetric regions. This is only a large- H special case and does not cover all CMC spheres in Sol_3 .

5. **“Uniqueness of Constant Mean Curvature Spheres”**

Authors: Pablo Mira and Joaquin Perez.

Venue/year: RSME Springer Series chapter, Springer, 2023.

DOI/URL: DOI 10.1007/978-3-031-39916-9_13; author PDF <https://wpd.ugr.es/~jperez/wordpress/wp-content/uploads/authorsample-MiraPerez.pdf>.

Role: A recent survey of the classification theory. In its open-problems section it states that, although the Sol_3 CMC spheres have index one, it is not known whether they are weakly stable; it records Daniel-Mira’s conjecture that every Sol_3 CMC sphere is weakly stable and notes that this would imply that all such spheres bound isoperimetric regions. This is strong evidence that the exact question is not fully resolved as of the current literature.

6. **“Isoperimetric domains in homogeneous three-manifolds and the isoperimetric constant of the Heisenberg group H^1 ”**

Authors: Jih-Hsin Cheng, Andrea Malchiodi, Paul Yang.

Venue/year: arXiv preprint, 2015.

arXiv/URL: arXiv:1501.07357; <https://arxiv.org/abs/1501.07357>.

Role: This appeared in searches because it claimed structural results for isoperimetric sets in homogeneous spaces diffeomorphic to $\mathbb{R}^3$, but arXiv marks it withdrawn due to a problem in Section 2. I therefore did not treat it as resolving citable evidence.

3.17.3.4. What remains unresolved

The missing step is to prove, for every $H > 0$, that the unique CMC H -sphere $S_H \subset \text{Sol}_3$ is the boundary of a globally area-minimizing region for its enclosed volume. Known index-one and uniqueness results do not imply this, and the available isoperimetric result only covers sufficiently large H (small enclosed volumes). Thus the exact all-cases statement is not fully resolved.

3.17.4. Remaining open problems recorded by the database

3.17.4.1. Open problem 1

Prove or disprove that every constant-mean-curvature sphere in Sol_3 is weakly stable under volume-preserving variations.

Reference supplied by the database. Pablo Mira and Joaquín Pérez, “Uniqueness of Constant Mean Curvature Spheres,” in *New Trends in Geometric Analysis*, RSME Springer Series, vol. 10, Springer, Cham, 2023, pp. 365–393, Conjecture 9.3 (attributed there to Benoît Daniel and Pablo Mira). DOI: 10.1007/978-3-031-39916-9_13. https://doi.org/10.1007/978-3-031-39916-9_13

3.17.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.18. 79. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-110]

Record: 2010-geom-110

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.18.1. Problem statement

Prove that there is no complete stable immersed surface in $\mathbb{H}^2 \times \mathbb{R}$ with constant mean curvature H satisfying $H > 1/2$.

3.18.2. Historical background, literature, and current progress

3.18.3. Verdict: partially resolved

3.18.3.1. Precise problem

The problem asks for the nonexistence of complete stable immersed constant-mean-curvature surfaces in the product space $\mathbb{H}^2 \times \mathbb{R}$ when the constant mean curvature satisfies $H > 1/2$. In the original CMC-surface context, “stable” is the strong/two-sided stability condition for compactly supported variations, not merely weak volume-preserving stability of closed CMC spheres.

3.18.3.2. Current progress

Original/open-problem source Title: Constant mean curvature surfaces in 3-dimensional Thurston geometries

Authors: Isabel Fernández and Pablo Mira

Venue/year: Proceedings of the International Congress of Mathematicians 2010, Vol. II, Hindustan Book Agency/World Scientific, pp. 830-861, 2010/2011

DOI/arXiv/URL: arXiv:1004.4752; DOI: 10.1142/9789814324359_0076; <https://arxiv.org/abs/1004.4752>

Status contribution: This survey is the source matching the dataset entry. It records that there are no complete stable CMC surfaces in $\mathbb{H}^2 \times \mathbb{R}$ for $H > 1/\sqrt{3}$, and says the expected optimal statement is for $H > 1/2$. It also records the parabolic-conformal-type case as known. Thus the survey itself presents the exact $H > 1/2$ problem as not fully proved.

Stronger mean-curvature range Title: Global properties of constant mean curvature surfaces in $\mathbb{H}^2 \times \mathbb{R}$

Authors: Barbara Nelli and Harold Rosenberg

Venue/year: Pacific Journal of Mathematics 226(1), 137-152, 2006

DOI/arXiv/URL: DOI: 10.2140/pjm.2006.226.137; <https://msp.org/pjm/2006/226-1/pjm-v226-n1-p07-p.pdf>

Key result: Theorem B proves that in $\mathbb{H}^2 \times \mathbb{R}$ there is no complete noncompact stable H -surface with $H > 1/\sqrt{3}$, with compact boundary or without boundary.

Why partial: Since $1/\sqrt{3} \approx 0.577 > 1/2$, this leaves the interval $1/2 < H \leq 1/\sqrt{3}$ open for arbitrary complete stable immersed surfaces.

Parabolic conformal type Title: Parabolic stable surfaces with constant mean curvature

Authors: José M. Manzano, Joaquín Pérez, and M. Magdalena Rodríguez

Venue/year: Calculus of Variations and Partial Differential Equations 42(1-2), 137-152, 2011

DOI/arXiv/URL: arXiv:0910.5373; DOI: 10.1007/s00526-010-0383-6; <https://arxiv.org/abs/0910.5373>

Key result: For an orientable, parabolic, complete immersed stable surface with constant mean curvature H in $\mathbb{H}^2 \times \mathbb{R}$, one has $|H| \leq 1/2$; if equality holds, the surface is an entire graph or a vertical horocylinder.

Why partial: This proves the desired nonexistence only under the additional assumption that the surface is parabolic as a Riemannian/conformal surface. It does not rule out possible hyperbolic conformal type examples with $1/2 < H \leq 1/\sqrt{3}$.

Curvature decay/integrability hypotheses Title: Finite Index Operators on Surfaces

Authors: José M. Espinar

Venue/year: Journal of Geometric Analysis 23, 415-437, 2013

DOI/arXiv/URL: DOI: 10.1007/s12220-011-9259-z; <https://link.springer.com/article/10.1007/s12220-011-9259-z>

Key result: Among other applications, the paper proves that there are no complete stable H -surfaces $\Sigma \subset \mathbb{H}^2 \times \mathbb{R}$ with $H > 1/2$ when either $K_e^+ = \max\{0, K_e\} \in L^1(\Sigma)$ or the extrinsic curvature satisfies a decay bound $|K_e(q)| \leq M/d(p_0, q)$.

Why partial: The threshold is exactly $H > 1/2$, but the theorem imposes extra integrability/decay assumptions on the extrinsic curvature. These hypotheses are not present in the problem statement.

Related compact/weak-stability result Title: On stable constant mean curvature surfaces in $\mathbb{S}^2 \times \mathbb{R}$ and $\mathbb{H}^2 \times \mathbb{R}$

Authors: Rabah Souam

Venue/year: Transactions of the American Mathematical Society 362(6), 2845-2857, 2010

DOI/arXiv/URL: arXiv:0709.4204; DOI: 10.48550/arXiv.0709.4204; <https://arxiv.org/abs/0709.4204>

Status contribution: This paper studies compact weakly stable CMC surfaces, proving for example stability/classification results for rotational CMC spheres and genus bounds. This is useful context but does not resolve the strong complete-stability nonexistence question in the problem statement.

3.18.3.3. What remains unresolved

I did not find published or otherwise citable literature proving the unrestricted statement that every complete strongly stable immersed CMC surface in $\mathbb{H}^2 \times \mathbb{R}$ must satisfy $|H| \leq 1/2$. The unresolved part is the general case, especially possible complete stable surfaces of hyperbolic conformal type with $1/2 < H \leq 1/\sqrt{3}$, without the parabolicity or extrinsic-curvature hypotheses used in the known partial results.

3.18.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.18.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.19. 80. New applications of Min-max Theory [2014-geom-127]

Record: 2014-geom-127

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: New applications of Min-max Theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>

3.19.1. Problem statement

Prove or disprove that, for every integer $g \geq 2$, the Lawson embedded minimal surface $\xi_{1,g} \subset S^3$ of genus g , in the round unit 3-sphere, minimizes the Willmore energy among all compact genus- g surfaces in S^3 ; equivalently, every compact surface $\Sigma \subset S^3$ of genus g satisfies

$$\mathcal{W}(\Sigma) \geq \mathcal{W}(\xi_{1,g}),$$

where

$$\mathcal{W}(\Sigma) = \int_{\Sigma} \left(1 + \left(\frac{k_1 + k_2}{2} \right)^2 \right) d\mu$$

and k_1, k_2 are the principal curvatures of Σ with respect to the standard metric on S^3 .

3.19.2. Historical background, literature, and current progress

3.19.3. Verdict: partially resolved

3.19.3.1. Precise problem

For every integer $g \geq 2$, decide whether the Lawson embedded minimal surface $\xi_{1,g} \subset S^3$ of genus g minimizes the Willmore energy

$$\mathcal{W}(\Sigma) = \int_{\Sigma} \left(1 + \left(\frac{k_1 + k_2}{2} \right)^2 \right) d\mu$$

among all compact genus- g surfaces in the round unit S^3 . The literature often writes the same Lawson genus- g family as $\xi_{g,1}$; the Lawson indices are interchangeable up to the usual symmetry of the construction.

3.19.3.2. Verdict

This is not fully resolved. The best available literature gives important special cases, existence/asymptotic results, numerical evidence, and a recent symmetry-restricted theorem, but no published or citable result proves

$$\mathcal{W}(\Sigma) \geq \mathcal{W}(\xi_{1,g})$$

for every compact genus- g surface in S^3 for every $g \geq 2$.

3.19.3.3. Key literature and current progress

1. Complete minimal surfaces in S^3

Authors: H. Blaine Lawson, Jr.

Venue/year: Annals of Mathematics (2) 92, 335-374, 1970.

DOI/URL: [doi:10.2307/1970625](https://doi.org/10.2307/1970625); JSTOR stable URL <http://www.jstor.org/stable/1970625>.

Role: Constructs the Lawson minimal surfaces $\xi_{m,k}$ in S^3 . This supplies the candidate surfaces but does not address Willmore minimization.

2. Comparison surfaces for the Willmore problem

Author: Robert Kusner.

Venue/year: Pacific Journal of Mathematics 138(2), 317-345, 1989.

DOI/URL: [doi:10.2140/pjm.1989.138.317](https://doi.org/10.2140/pjm.1989.138.317); <https://msp.org/pjm/1989/138-2/p08.xhtml>.

Role: Original source of Kusner's higher-genus Lawson minimizer conjecture (Conjecture 8.4 in later citations) and comparison-surface estimates such as $\mathcal{W}(\xi_{g,1}) < 8\pi$. It proposes the problem; it does not prove the full conjecture.

3. Minimizing the squared mean curvature integral for surfaces in space forms

Authors: Lucas Hsu, Rob Kusner, John Sullivan.

Venue/year: Experimental Mathematics 1(3), 191-207, 1992.

DOI/URL: [doi:10.1080/10586458.1992.10504258](https://doi.org/10.1080/10586458.1992.10504258); Project Euclid record <https://projecteuclid.org/euclid.em/1048622023>.

Role: Numerical and experimental evidence for Lawson-type minimizers in low genus. This is supporting evidence, not a proof of the full inequality.

4. Existence of surfaces minimizing the Willmore functional and Existence of minimizing Willmore surfaces of prescribed genus

Authors: Leon Simon; Matthias Bauer and Ernst Kuwert.

Venues/years: Communications in Analysis and Geometry 1(2), 281-326, 1993, [doi:10.4310/CAG.1993.v1.n2.a4](https://doi.org/10.4310/CAG.1993.v1.n2.a4); International Mathematics Research Notices 2003(10), 553-576, 2003, [doi:10.1155/S1073792803208072](https://doi.org/10.1155/S1073792803208072).

Role: These works establish existence of Willmore minimizers for prescribed genus under the necessary compactness/Douglas-type framework. They do not identify the minimizer as $\xi_{1,g}$.

5. The large genus limit of the infimum of the Willmore energy

Authors: Ernst Kuwert, Yuxiang Li, Reiner Schaetzle.

Venue/year: American Journal of Mathematics 132(1), 37-51, 2010.

DOI/URL: [doi:10.1353/ajm.0.0100](https://doi.org/10.1353/ajm.0.0100).

Role: Shows the large-genus limit of the infimum is 8π , consistent with the Lawson-surface picture. This is asymptotic information about the infimum, not identification of the minimizers for each g .

6. Min-Max theory and the Willmore conjecture

Authors: Fernando C. Marques, Andre Neves.

Venue/year: Annals of Mathematics 179(2), 683-782, 2014.

DOI/URL: [doi:10.4007/annals.2014.179.2.6](https://doi.org/10.4007/annals.2014.179.2.6).

Role: Fully resolves the genus-1 Willmore conjecture, identifying the Clifford torus. It gives a general lower bound $\mathcal{W} \geq 2\pi^2$ for positive genus, but this is weaker than the

genus $g \geq 2$ Lawson-minimizer claim and does not prove the problem statement.

7. Area estimates for high genus Lawson surfaces via DPW

Authors: Lynn Heller, Sebastian Heller, Martin Traizet.

Venue/year: Journal of Differential Geometry 124(1), 1-35, 2023; preprint arXiv:1907.07139.

DOI/URL: [doi:10.4310/jdg/1685121318](https://doi.org/10.4310/jdg/1685121318); <https://arxiv.org/abs/1907.07139>.

Role: Gives high-genus asymptotic area estimates for Lawson surfaces, for example $\text{Area}(\zeta_{1,g}) = 8\pi(1 - \frac{\log 2}{2g} + O(g^{-2}))$ in the relevant case. Since a minimal surface in S^3 has Willmore energy equal to area, this estimates the conjectured minimum, but it does not prove minimality among all genus- g surfaces.

8. The Lawson surfaces are determined by their symmetries and topology

Authors: Nikolaos Kapouleas, David Wiygul.

Venue/year: Journal fuer die reine und angewandte Mathematik 786, 155-173, 2022.

DOI/URL: [doi:10.1515/crelle-2021-0087](https://doi.org/10.1515/crelle-2021-0087).

Role: Proves that a closed embedded minimal surface in the round S^3 with the symmetries and genus of a Lawson surface is congruent to that Lawson surface. This is a minimal-surface characterization with symmetry assumptions, not a global Willmore minimization theorem.

9. The Willmore problem for surfaces with symmetry

Authors: Rob Kusner, Ying Lu, Peng Wang.

Venue/year: arXiv preprint, submitted 2024, revised 2025.

DOI/arXiv/URL: arXiv:2410.12582v2; [doi:10.48550/arXiv.2410.12582](https://doi.org/10.48550/arXiv.2410.12582); <https://arxiv.org/abs/2410.12582>.

Key result: Theorem 1.1 proves that if $M \subset S^3$ is a closed embedded genus mk surface invariant under specified finite ambient symmetry groups associated to $\zeta_{m,k}$, then

$$\mathcal{W}(M) \geq \mathcal{W}(\zeta_{m,k}) = \text{Area}(\zeta_{m,k}) \geq \mathcal{W}(\zeta_{mk,1}),$$

with equality conditions identifying the Lawson surface, and equality in the second inequality only for $k = 1$.

Applicability check: Taking $k = 1$ gives the desired Lawson lower bound for the symmetric embedded competitors of genus m . the problem has no symmetry hypothesis and asks for all compact genus- g competitors, so this is only partial progress. The paper's abstract explicitly states that the general genus $g > 1$ Lawson minimizer conjecture remains open.

10. Large topology asymptotics for spectrally extremal minimal surfaces in $\mathbb{B}^3$ and $\mathbb{S}^3$

Authors: Mikhail Karpukhin, Peter McGrath, Daniel Stern.

Venue/year: Calculus of Variations and Partial Differential Equations 65, article 119, 2026; preprint arXiv:2502.10225.

DOI/arXiv/URL: [doi:10.1007/s00526-026-03288-3](https://doi.org/10.1007/s00526-026-03288-3); <https://arxiv.org/abs/2502.10225>.

Role: Confirms, in a class of spectrally extremal low-area minimal examples, the special role of Lawson $\zeta_{\gamma,1}$ surfaces and compares large-topology area behavior. This is strong contextual evidence but does not cover all Willmore competitors.

11. The Analysis of Willmore Surfaces and its Generalizations in Higher Dimensions

Authors: Tian Lan, Dorian Martino, Tristan Riviere.

Venue/year: arXiv survey/preprint, 2025, revised 2026.

DOI/arXiv/URL: arXiv:2511.01777v3; [doi:10.48550/arXiv.2511.01777](https://doi.org/10.48550/arXiv.2511.01777); <https://arxiv.org/abs/2511.01777>.

Role: A current survey of Willmore-surface analysis. It states that after Marques-Neves the genus-1 case is solved, but the higher-genus Lawson candidate problem remains largely open. This agrees with the theorem-by-theorem applicability check above.

3.19.3.4. What remains unresolved

The open part is the full, unrestricted global minimization statement for every $g \geq 2$: proving that every compact genus- g surface in round S^3 , without imposed Lawson symmetries, fixed conformal structure, minimality, or membership in a special constructed family, satisfies $\mathcal{W}(\Sigma) \geq \mathcal{W}(\xi_{1,g})$. Existing results do not remove these restrictions.

3.19.4. Remaining open problems recorded by the database

3.19.4.1. Open problem 1

Prove that the genus-two Lawson surface $\xi_{1,2} \subset S^3$ minimizes the Willmore energy among all compact genus-two surfaces in S^3 , with equality only for Möbius images of $\xi_{1,2}$.

Reference supplied by the database. Robert Kusner, “Comparison surfaces for the Willmore problem,” *Pacific Journal of Mathematics* **138**(2) (1989), 317–345; the genus-two case of Kusner’s proposed higher-genus Lawson-surface minimization conjecture. DOI: [10.2140/pjm.1989.138.317](https://doi.org/10.2140/pjm.1989.138.317). Its open status, including for $g = 2$, is reaffirmed in Rob Kusner, Ying Lü, and Peng Wang, “The Willmore problem for surfaces with symmetry,” preprint (2024), introduction, [arXiv:2410.12582](https://arxiv.org/abs/2410.12582).

3.19.5. Primary-source pointer

The originating ICM record is *New applications of Min-max Theory* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>.

3.20. 81. Evolution of form and shape [2022-ag-217]

Record: 2022-ag-217

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Evolution of form and shape

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=267>

3.20.1. Problem statement

Prove or disprove the Calabi-Yau conjectures for embedded minimal surfaces: let $\Sigma \subset \mathbf{R}^3$ be a complete embedded minimal surface, meaning a smooth embedded surface that is complete for the Riemannian metric induced from Euclidean space and whose mean curvature vector is identically zero. Prove or disprove that Σ is unbounded in $\mathbf{R}^3$. More ambitiously, prove or disprove that if Σ is non-flat (equivalently, not a plane), then for every line $L \subset \mathbf{R}^3$, the orthogonal projection of Σ onto L is an unbounded subset of L .

3.20.2. Historical background, literature, and current progress

3.20.3. Verdict: partially resolved

3.20.3.1. Precise problem

The problem asks whether every complete embedded minimal surface Σ in $\mathbf{R}^3$ is unbounded, and whether the stronger Calabi-Yau projection statement holds: if Σ is not a plane, then for every line L in $\mathbf{R}^3$, the orthogonal projection of Σ to L is unbounded.

Equivalently, the stronger part asks for a halfspace/slab exclusion with no topology, genus, curvature, injectivity-radius, or end-space assumptions: a nonflat complete embedded minimal surface should not be contained in any halfspace perpendicular to a chosen line.

3.20.3.2. Status

This is not fully resolved as stated. The literature proves the statement in important embedded cases, especially finite topology and finite genus with countably many ends. However, the unrestricted embedded case remains open in the infinite-genus or uncountably-many-ends regime. Recent literature explicitly notes that it is still open whether a complete bounded embedded minimal surface exists in $\mathbf{R}^3$; such a surface, if it exists, would necessarily lie outside the finite-topology and finite-genus/countable-end cases.

The correct verdict for this problem is therefore `partially_resolved`.

3.20.3.3. Progress and key literature

1. Original source

- Title: “Problems in Differential Geometry”
- Author: E. Calabi
- Venue: Proceedings of the United States-Japan Seminar in Differential Geometry, Kyoto, Japan, 1965, ed. S. Kobayashi and J. Eells, Jr.; Nippon Hyoronsha, Tokyo
- Year: 1966
- DOI/arXiv/URL: no DOI found; cited in Colding-Minicozzi as p. 170
- Relevance: This is the source of the Calabi conjectures that a complete minimal hypersurface in Euclidean space is unbounded, and that a complete nonflat minimal hypersurface has unbounded projection in every relevant linear direction.

2. Immersed counterexamples, not embedded

- Title: “A complete minimal surface in R^3 between two parallel planes”
- Authors: Luquesio P. de M. Jorge, Frederico Xavier
- Venue: Annals of Mathematics (2), 112(1), 203-206
- Year: 1980
- DOI/URL: <https://doi.org/10.2307/1971325> ; Annals page <https://annals.math.princeton.edu/1980/112-1/p08>
- Relevance: Gives a complete nonflat minimal immersion contained between two parallel planes, so the projection conjecture is false for immersed surfaces. It does not resolve the problem statement because the statement requires embeddedness.
- Title: “Hadamard’s and Calabi-Yau’s conjectures on negatively curved and minimal surfaces”
- Author: Nikolai Nadirashvili
- Venue: Inventiones Mathematicae, 126, 457-465
- Year: 1996
- DOI/URL: <https://doi.org/10.1007/s002220050106>
- Relevance: Constructs a complete minimal immersion in the unit ball, disproving the unboundedness conjecture for immersed surfaces. It does not resolve the embedded problem.

3. Proper surfaces imply the projection statement

- Title: “The strong halfspace theorem for minimal surfaces”
- Authors: David Hoffman, William H. Meeks III
- Venue: Inventiones Mathematicae, 101, 373-377
- Year: 1990
- DOI/URL: <https://doi.org/10.1007/BF01231506>
- Relevance: A connected proper minimal surface in R^3 contained in a halfspace must be a plane. Therefore, once properness is known for a class of complete embedded minimal surfaces, the stated stronger projection claim follows for that class: bounded projection to a line would put the surface in a halfspace perpendicular to that line.

4. Finite topology: resolved

- Title: “The Calabi-Yau conjectures for embedded surfaces”
- Authors: Tobias H. Colding, William P. Minicozzi II
- Venue: *Annals of Mathematics* (2), 167(1), 211-243
- Year: 2008
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2008.167.211> ; arXiv:math/0404197
- Key result: Corollary 0.12 says the plane is the only complete embedded minimal surface with finite topology in a halfspace of $\mathbb{R}^3$; Corollary 0.13 says a complete embedded minimal surface with finite topology in $\mathbb{R}^3$ is proper.
- Relevance: This proves both parts of the problem for complete embedded minimal surfaces of finite topology, but it has an extra finite-topology hypothesis and therefore does not cover the full statement.

5. Positive injectivity radius: resolved

- Title: “The minimal lamination closure theorem”
- Authors: William H. Meeks III, Harold Rosenberg
- Venue: *Duke Mathematical Journal*, 133(3), 467-497
- Year: 2006
- DOI/URL: <https://doi.org/10.1215/S0012-7094-06-13332-X>
- Relevance: The closure theorem implies properness in $\mathbb{R}^3$ for complete embedded minimal surfaces with positive injectivity radius. This is another major partial result, but the statement does not assume positive injectivity radius.

6. Finite genus with countably many ends: resolved

- Title: “The embedded Calabi-Yau conjecture for finite genus”
- Authors: William H. Meeks III, Joaquin Perez, Antonio Ros
- Venue: *Duke Mathematical Journal*, 170(13), 2891-2956
- Year: 2021
- DOI/arXiv/URL: <https://doi.org/10.1215/00127094-2020-0087> ; arXiv:1806.03104
- Key result: For a complete embedded minimal surface in $\mathbb{R}^3$ of finite genus, properness is obtained under countability of the end space; the arXiv abstract states the finite-genus equivalences involving properness, at most two limit ends, and countability conditions.
- Relevance: This extends the finite-topology progress substantially. It still has extra hypotheses and does not prove the unrestricted statement.

7. Recent survey/current status

- Title: “Complete minimal surfaces with Cantor ends in minimally convex domains”
- Author: Antonio Alarcon
- Venue: arXiv preprint
- Year: 2024
- DOI/arXiv/URL: arXiv:2410.13687, <https://arxiv.org/abs/2410.13687>
- Relevance: This recent survey/preprint explicitly records the current boundary: finite topology and finite genus with countably many ends are covered by Colding-Minicozzi and Meeks-Perez-Ros, while it remains open whether there exists a complete bounded embedded minimal surface in $\mathbb{R}^3$ of infinite genus or with uncount-

ably many ends.

- Title: “Minimal Surfaces of Finite Genus: Classification, Dynamics and Laminations”
- Author: Joaquín Pérez
- Venue: arXiv/preprint survey
- Year: 2025
- DOI/arXiv/URL: arXiv:2510.12319, <https://arxiv.org/html/2510.12319v1>
- Relevance: Surveys the finite-genus program and describes the embedded Calabi-Yau problem as resolved for finite genus with countably many ends, while emphasizing remaining properness questions beyond the known hypotheses.

3.20.3.4. What remains unresolved

No citable theorem found proves the full unrestricted statement for every complete embedded minimal surface in $\mathbb{R}^3$. The unresolved cases are complete embedded minimal surfaces without finite topology, without finite genus/countable-end hypotheses, and without positive injectivity-radius or curvature hypotheses. In particular, the possible infinite-genus or uncountably-many-ends cases are not covered by the resolving theorems above.

3.20.4. Remaining open problems recorded by the database

3.20.4.1. Open problem 1

Prove that every complete embedded minimal surface in $\mathbb{R}^3$ of finite genus is properly embedded in $\mathbb{R}^3$.

Reference supplied by the database. William H. Meeks III, Joaquín Pérez, and Antonio Ros, “The embedded Calabi–Yau conjecture for finite genus,” *Duke Mathematical Journal* 170(13) (2021), 2891–2956, Conjecture 1.1. DOI: [10.1215/00127094-2020-0087](https://doi.org/10.1215/00127094-2020-0087); arXiv: [1806.03104](https://arxiv.org/abs/1806.03104). Its still-open status and the best known countable-ends case are recorded in Joaquín Pérez, “Minimal Surfaces of Finite Genus: Classification, Dynamics and Laminations,” 2025, Section 6 and Theorem 6.1, arXiv: [2510.12319](https://arxiv.org/abs/2510.12319).

3.20.4.2. Open problem 2

Let M be an orientable open surface every end of which has infinite genus. Prove that M admits a complete proper minimal embedding into every smooth bounded domain $D \subset \mathbb{R}^3$.

Reference supplied by the database. Leonor Ferrer, Francisco Martín, and William H. Meeks III, “Existence of proper minimal surfaces of arbitrary topological type,” *Advances*

in *Mathematics* 231(1) (2012), 378–413, Embedded Calabi–Yau Conjecture 1(1) (attributed there to Martín, Meeks, Nadirashvili, Pérez, and Ros). DOI: [10.1016/j.aim.2012.05.007](https://doi.org/10.1016/j.aim.2012.05.007); arXiv: [0903.4194](https://arxiv.org/abs/0903.4194). The underlying existence of any nonproper complete embedded minimal surface in $\mathbb{R}^3$ is still listed as open in Joaquín Pérez, “Minimal Surfaces of Finite Genus: Classification, Dynamics and Laminations,” 2025, Section 7, arXiv: [2510.12319](https://arxiv.org/abs/2510.12319).

3.20.5. Primary-source pointer

The originating ICM record is *Evolution of form and shape* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=267>.

4. Manifolds and global analysis

The nineteen problems here connect the topology of manifolds with elliptic operators, index theory, moduli spaces, gauge theory, and global nonlinear analysis. Their histories often pass through a succession of increasingly flexible categories: smooth, topological, coarse, equivariant, or derived. Partial results typically isolate hypotheses under which compactness, transversality, or an index obstruction can be controlled.

Coverage: 19 website records; **source years:** 1978–2022; **separately listed descendant open problems:** 34.

4.1. 82. Algebraic Curves of Infinite Genus Arising in the Theory of Nonlinear Waves [1978-ag-18]

Record: 1978-ag-18

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Algebraic Curves of Infinite Genus Arising in the Theory of Nonlinear Waves

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=273>

4.1.1. Problem statement

Let $Q = -D^2 + q(x)$ be Hill's operator with periodic potential, and let $\Delta(\lambda)$ be one-half the trace of the monodromy matrix of $Qf = \lambda f$. Assume the exceptional double spectra are absent, so the multiplier curve

$$\Omega = \{(\lambda, m) : m^2 - 2\Delta(\lambda)m + 1 = 0\}$$

is a nonsingular hyperelliptic curve with infinitely many branch points accumulating at the ramified point ∞ . Its genus is $g = \infty$. The roots of $\Delta = +1$ and $\Delta = -1$ give the periodic and antiperiodic spectra: the ground state is λ_0 , and the remaining spectral points occur in pairs $\lambda_{2j-1} \leq \lambda_{2j}$ tending to $+\infty$. The j -th lacuna is the interval $[\lambda_{2j-1}, \lambda_{2j}]$. Let DFK be the class of differentials of the first kind consisting of differentials

$$\omega = \frac{\varphi(\lambda) d\lambda}{\sqrt{\Delta(\lambda)^2 - 1}},$$

where φ is entire of order $1/2$ and type at most 1 , satisfies $\varphi(\lambda^*) = \varphi(\lambda)^*$, satisfies

$$\int_0^\infty |\varphi(\lambda)|^2 \lambda^{3/2} d\lambda < \infty,$$

and has values on the lacunae that vanish rapidly as $j \rightarrow \infty$.

Formulate and prove a Riemann-Roch theorem for Ω and the class DFK that gives a useful replacement for the classical dimension and codimension terms in this infinite-genus setting, where such dimensions and codimensions may be infinite, and that is compatible with the following zero-counting behavior: every differential of the first kind on Ω with a zero of finite multiplicity $2m$ at ∞ has $2g - 2 - 2m$ additional finite zeros on Ω , counted in the special Hill-surface sense used for these curves; equivalently, it has $2g - 2$ zeros in all. Account also for the fact that a differential ω in DFK can have a zero of infinite multiplicity at ∞ without vanishing identically, but this cannot occur if, for example, the lacuna lengths decay exponentially: there exist constants $a, b > 0$ such that

$$\lambda_{2j} - \lambda_{2j-1} \leq a \exp(-bj)$$

as $j \rightarrow \infty$.

4.1.2. Historical background, literature, and current progress

4.1.3. Verdict: partially resolved

4.1.3.1. Precise problem checked

The problem asks for the status of a Riemann-Roch-type theorem for the infinite-genus hyperelliptic multiplier curve of a periodic Hill operator in the KdV case, using the McKean-Trubowitz/DFK class of holomorphic differentials

$$\omega = \frac{\varphi(\lambda)d\lambda}{\sqrt{\Delta(\lambda)^2 - 1}},$$

where φ has the stated entire-function, L^2 , reality, and lacuna-decay conditions. The unresolved part is not the construction of the Hill surface itself, but whether there is a useful Riemann-Roch theorem for this infinite-genus curve and this natural class of differentials/divisors: a theorem determining the relevant dimensions/codimensions and replacing the finite-genus formula without imposing extra hypotheses not present in the problem statement.

4.1.3.2. Literature found

1. H. P. McKean, "Algebraic curves of infinite genus arising in the theory of nonlinear waves," in *Proceedings of the International Congress of Mathematicians (Helsinki, 1978)*, vol. 2, Acad. Sci. Fennica, 1980, pp. 777-783. URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf>. This is the apparent source for the problem. Live searches of the proceedings locate the explicit statement that the Riemann-Roch theorem is open for these Hill surfaces, while the zero-counting aspect and the possibility of infinite multiplicity at infinity are known.

2. H. P. McKean and E. Trubowitz, "Hill's surfaces and their theta functions," *Bulletin of the American Mathematical Society* 84, no. 6 (1978), 1042-1085. DOI: 10.1090/S0002-9904-1978-14542-X. URL: <https://projecteuclid.org/journals/bulletin-of-the-american-mathematical-society/volume-84/issue-6/Hills-surfaces-and-their-theta-functions/bams/1183541462.pdf>. This is the main Hill-surface/theta-function reference. It develops the infinite-genus Hill surface, DFK-type differentials, Abel sums, and theta functions, but does not give a full Riemann-Roch theorem for the DFK class.
3. H. P. McKean and E. Trubowitz, "Hill's operator and hyperelliptic function theory in the presence of infinitely many branch points," *Communications on Pure and Applied Mathematics* 29 (1976), 143-226. DOI: 10.1002/cpa.3160290203. This gives the foundational hyperelliptic function theory for Hill operators with infinitely many branch points. It is essential background but not a full Riemann-Roch resolution.
4. M. U. Schmidt, "Integrable systems and Riemann surfaces of infinite genus," *Memoirs of the American Mathematical Society* 122, no. 581 (1996), viii+111 pp.; preprint arXiv:solv-int/9412006. URL: <https://arxiv.org/pdf/solv-int/9412006>. Schmidt proves a Riemann-Roch theorem for divisors of "Riemann-Roch type" using modified cohomology. The theorem is important partial progress, but its own formulation restricts to divisors satisfying special asymptotic/trivialization conditions; the paper notes that identifying enough such divisors is a separate main problem. Thus it does not cover all DFK differentials/divisors on arbitrary Hill curves in the problem statement.
5. Franz Merkl, "A Riemann Roch Theorem for infinite genus Riemann surfaces," *Inventiones Mathematicae* 139, no. 2 (2000), 391-437. DOI: 10.1007/s002229900031. URL: <http://research.tue.nl/en/publications/a-riemann-roch-theorem-for-infinite-genus-riemann-surfaces>. Merkl proves a Riemann-Roch theorem for a class of holomorphic line bundles over infinite-genus Riemann surfaces, with pointwise asymptotic growth conditions and asymptotic bounds on gluing and transition functions. This is a genuine infinite-genus Riemann-Roch theorem, but the hypotheses and section spaces are not shown to coincide with the McKean-Trubowitz DFK class for every nonsingular Hill multiplier curve in the problem statement.
6. Joel Feldman, Horst Knorrer, and Eugene Trubowitz, *Riemann Surfaces of Infinite Genus*, CRM Monograph Series 20, American Mathematical Society, Providence, 2003. DOI: 10.1090/crmm/020. URL: <https://bookstore.ams.org/CRMM/20> and author PDF <https://www.math.ubc.ca/~feldman/papers/allriem.pdf>. The monograph develops a broad theory for marked infinite-genus surfaces satisfying explicit geometric hypotheses, including Royden-Riemann-Roch for finite divisors on parabolic surfaces and theta-function results. This is not the requested full Hill/DFK Riemann-Roch theorem: it works with different hypotheses and does not settle the dimensions/codimensions for the DFK class in the problem statement.
7. Ilya Zakharevich, "Curves of infinite genus I. Riemann-Roch theorem for small degree," arXiv:math/0202057, 2002. URL: <https://arxiv.org/abs/math/0202057>; and "Quasi-algebraic geometry of curves I. Riemann-Roch theorem and Jacobian," arXiv:alg-geom/9710013, 1997. URL: <https://arxiv.org/abs/alg-geom/9710013>. These preprints develop Riemann-Roch analogues for infinite-genus curves with growth-at-infinity conditions and handle-length restrictions. They are related partial progress, but they are not a published theorem covering the full Hill/DFK problem as stated.
8. Thomas Kappeler and Peter Topalov, "On normalized differentials on hyperelliptic

curves of infinite genus,” *Journal of Differential Geometry* 105, no. 2 (2017), 209-248. DOI: 10.4310/jdg/1486522814. arXiv:1408.7107, URL: <https://arxiv.org/abs/1408.7107>. This constructs normalized differentials on infinite-genus hyperelliptic curves and gives zero-distribution estimates. It is strong later progress on differentials but not a Riemann-Roch theorem for the DFK class.

9. Kenneth T-R. McLaughlin and Patrik V. Nabelek, “A Riemann-Hilbert problem approach to infinite gap Hill’s operators and the Korteweg-de Vries equation,” *International Mathematics Research Notices* 2021, no. 2, 1288-1348. DOI: 10.1093/imrn/rnz156. arXiv:1810.07818, URL: <https://arxiv.org/abs/1810.07818>. This gives a Riemann-Hilbert formulation and uniqueness results for periodic infinite-gap Hill operators. It is current Hill/KdV infinite-gap literature, but it does not prove the missing Riemann-Roch theorem.

4.1.3.3. Why this is not fully resolved

The later literature does contain several Riemann-Roch-type theorems for infinite-genus surfaces, but none found in the search proves a theorem covering exactly the full setting: arbitrary nonsingular KdV/Hill multiplier curves with the McKean-Trubowitz DFK class and no extra exponential/asymptotic/trivialization hypotheses beyond those in the statement. The available results either:

- treat finite divisors or different function spaces;
- impose explicit geometric, gluing, handle-length, or growth conditions;
- restrict to divisors of “Riemann-Roch type” or admissible line bundles;
- prove normalized differential or inverse spectral/Riemann-Hilbert results rather than Riemann-Roch dimensions for DFK.

Therefore the correct current status is **partially resolved**: the original zero-counting and special-hypothesis results remain established, and substantial related infinite-genus Riemann-Roch theory exists, but the full Hill/DFK Riemann-Roch problem is not fully resolved by citable literature.

4.1.4. Remaining open problems recorded by the database

4.1.4.1. Open problem 1

For a completed infinite-genus spectral curve Y in Schmidt’s framework, find and geometrically characterize sufficiently many divisors of Riemann–Roch type, beyond the classes covered by the known generic separation and admissibility criteria, so that the infinite-genus Riemann–Roch theorem applies to the spectral line bundles of interest.

Reference supplied by the database. Martin U. Schmidt, “Integrable Systems and Riemann Surfaces of Infinite Genus,” *Memoirs of the American Mathematical Society* 122, no. 581 (1996), viii+111 pp., §5, immediately after Serre Duality Theorem 5.3 (explicitly identified as “the main problem”). ISBN: 978-0-8218-0445-2. DOI: 10.1090/memo/0581. arXiv: solv-int/9412006. <https://arxiv.org/abs/solv-int/9412006>

4.1.5. Primary-source pointer

The originating ICM record is *Algebraic Curves of Infinite Genus Arising in the Theory of Nonlinear Waves* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=273>.

4.2. 83. Some Problems in Complex Analytic Geometry with Growth Conditions [1978-geom-27]

Record: 1978-geom-27

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Some Problems in Complex Analytic Geometry with Growth Conditions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=141>

4.2.1. Problem statement

Let $X \subset \mathbb{C}^n$ be a pure d -dimensional entire analytic set, let

$$X[r] = X \cap \{z : \|z\| < r\},$$

let φ be the standard Kahler form on $\mathbb{C}^n$, and let Ω_X be the curvature matrix of X at its smooth points. Define the Chern forms $c_i(\Omega_X)$ by

$$\det \left(tI + \frac{\sqrt{-1}}{2\pi} \Omega_X \right) = \sum_{i=0}^d (-1)^i t^{d-i} c_i(\Omega_X),$$

and, for $0 \leq i \leq d$, define

$$\mu_i(X, r) = \frac{1}{(d-i)! r^{2(d-i)}} \int_{X[r]} c_i(\Omega_X) \wedge \varphi^{d-i}.$$

Set

$$\mu(X, r) = \sum_{i=0}^d \mu_i(X, r).$$

Determine whether there is a Bezout-type estimate bounding the refined growth indicator $\mu(V \cap W, r)$ of the intersection of entire analytic subvarieties $V, W \subset \mathbb{C}^n$ in terms of the refined growth indicators $\mu(V, r)$ and $\mu(W, r)$.

4.2.2. Historical background, literature, and current progress

4.2.3. Verdict: partially resolved

4.2.3.1. Precise problem

The problem asks whether Griffiths's refined curvature-integral growth invariant for a pure-dimensional entire analytic set $X \subset \mathbb{C}^n$,

$$\mu(X, r) = \sum_i \mu_i(X, r), \quad \mu_i(X, r) = \frac{1}{(d-i)! r^{2(d-i)}} \int_{X[r]} c_i(\Omega_X) \wedge \varphi^{d-i},$$

satisfies a Bezout-type inequality for intersections: namely, whether the refined growth indicator of $V \cap W$ can be bounded in terms of the refined growth indicators of the entire analytic subvarieties V and W themselves, without restricting to the ordinary Euclidean volume growth term.

4.2.3.2. Literature checked

1. Phillip A. Griffiths, "Some Problems in Complex Analytic Geometry with Growth Conditions," Proceedings of the International Congress of Mathematicians, Helsinki 1978, pp. 645-651, Acad. Sci. Fennica, 1980. URL: <https://publications.ias.edu/sites/default/files/someproblems.pdf>. This is the original source of the question. Griffiths states the ordinary transcendental Bezout problem, recalls that the ordinary estimate holds in some hypersurface/linear-section settings but fails in higher codimension, then proposes the curvature-integral refinement and asks whether there is a Bezout estimate for $\mu(V, r) = \sum_i \mu_i(V, r)$.
2. Phillip A. Griffiths, "Complex differential and integral geometry and curvature integrals associated to singularities of complex analytic varieties," Duke Mathematical Journal 45 no. 3 (1978), 427-512. DOI: 10.1215/S0012-7094-78-04522-2. URL: <https://publications.ias.edu/sites/default/files/complexdiff.pdf>. This paper develops the Chern-form curvature integrals, monotonicity, tube-volume interpretation, and complex Crofton/kinematic formulas underlying the refined indicator. It supplies the geometric framework for the question, but it does not prove a global Bezout inequality for $\mu(V \cap W, r)$ in terms of $\mu(V, r)$ and $\mu(W, r)$.
3. Wilhelm Stoll, "A Bezout estimate for complete intersections," Annals of Mathematics (2) 96 no. 2 (1972), 361-401. DOI: 10.2307/1970793. URL: <https://annals.math.princeton.edu/1972/96-2/p07>. Stoll proves a Bezout estimate in the complete-intersection/value-distribution setting. This is important partial progress on the classical transcendental Bezout problem, but it is for the ordinary growth framework and special complete-intersection hypotheses, not the full curvature-integral refined indicator for arbitrary analytic subvariety intersections.
4. Maurizio Cornalba and Bernard Shiffman, "A counterexample to the 'Transcendental Bezout problem'," Annals of Mathematics (2) 96 no. 2 (1972), 402-406. DOI: 10.2307/1970794. URL: <https://annals.math.princeton.edu/1972/96-2/p08>. This disproves the naive ordinary maximum-modulus/volume version in dimension at least two. It motivates Griffiths's refined curvature question but does not settle the refined question.
5. Phillip A. Griffiths, "On the Bezout problem for entire analytic sets," Annals of Mathematics (2) 100 no. 3 (1974), 533-552. DOI: 10.2307/1970957. URL: <https://annals.math.princeton.edu/1974/100-3/p02>. This gives partial positive results for the ordinary entire-analytic-set Bezout problem, especially around curves and linear sections. It does not prove the later 1978 refined Chern-form growth estimate for all V, W .
6. James A. Carlson, "A moving lemma for the transcendental Bezout problem," Annals of Mathematics (2) 103 no. 2 (1976), 305-330. DOI: 10.2307/1971008. URL: <https://annals.math.princeton.edu/1976/103-2/p06>. Carlson proves moving/generic-intersection results for the transcendental Bezout problem. These are partial results under movement/genericity and ordinary growth notions; they do not imply a bound for the refined curvature indicator of arbitrary intersections.

7. James R. King, “Refined residues, Chern forms, and intersections,” in *Value Distribution Theory, Part A, Pure and Applied Mathematics 25*, Marcel Dekker, New York, 1974, pp. 169-190. MR50 #7565; Zbl 0307.32002. King’s work is relevant to Chern forms and intersection theory, but the available citations and later use concern residue/intersection formulas rather than a full transcendental Bezout growth estimate for Griffiths’s $\sum_i \mu_i$.
8. Shanyu Ji, “Bezout estimate for entire holomorphic maps and their Jacobians,” *American Journal of Mathematics* 117 no. 2 (1995), 395-403. DOI: 10.2307/2374920. URL: <https://www.jstor.org/stable/2374920>. Ji obtains Bezout-type estimates for entire maps under hypotheses involving the Jacobian. This is a substantial special/conditional result for maps and zeros, but the hypotheses and conclusion do not cover arbitrary analytic subvarieties V, W with Griffiths’s refined curvature growth.
9. B. Q. Li and B. A. Taylor, “On the Bezout problem and area of interpolating varieties in $\mathbb{C}^n$,” *American Journal of Mathematics* 118 no. 5 (1996), 989-1010; and B. Q. Li, “On the Bezout problem and area of interpolating varieties in $\mathbb{C}^n$, II,” *American Journal of Mathematics* 120 no. 6 (1998), 1191-1198. These papers give further partial results for area/interpolation formulations of the Bezout problem. They do not address the full refined Chern-form growth indicator for arbitrary intersections.
10. Lev Buhovsky, Iosif Polterovich, Leonid Polterovich, Egor Shelukhin, and Vukasin Stojisavljevic, “Persistent transcendental Bezout theorems,” *Forum of Mathematics, Sigma* 12 (2024), e72, 1-28. DOI: 10.1017/fms.2024.49. arXiv:2307.02937. URL: <https://www.cambridge.org/core/journals/forum-of-mathematics-sigma/article/persistent-transcendental-bezout-theorems/C973B8F28E9FAD19237DEE5DD2AE0426>. This recent paper proves bounds for modified coarse/persistent zero counts of entire maps. It explicitly changes the counting invariant to a persistence-based coarse count and uses maximum-modulus growth, so it is not a resolution of Griffiths’s curvature-integral refined Bezout problem.

4.2.3.3. What remains unresolved

I found no published or otherwise citable theorem whose hypotheses and conclusion imply the full statement in the problem statement: a Bezout-type estimate bounding the refined curvature-integral growth indicator $\mu(V \cap W, r)$ of arbitrary intersections of pure-dimensional entire analytic subvarieties in $\mathbb{C}^n$ in terms of the corresponding refined indicators of V and W .

The literature gives important partial progress and related variants, including complete-intersection estimates, moving/generic-intersection results, estimates with Jacobian or interpolation hypotheses, average/area estimates, and recent persistent/coarse zero-count bounds. Each changes the invariant, adds extra hypotheses, or treats a special setting. Thus the exact Griffiths refined curvature-integral Bezout question remains unresolved in full.

4.2.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

4.2.5. Primary-source pointer

The originating ICM record is *Some Problems in Complex Analytic Geometry with Growth Conditions* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=141>.

4.3. 84. Spectral geometry and non-compact Riemannian manifolds [1983-geom-34]

Record: 1983-geom-34

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1983: Spectral geometry and non-compact Riemannian manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=627>

4.3.1. Problem statement

Prove that for every discrete torsion-free subgroup $\Gamma \subset \mathrm{SL}(2, \mathbb{R})$ for which the hyperbolic quotient $\Gamma \backslash \mathbb{H}$ has finite volume, where $\mathbb{H}$ is the upper half-plane with the Poincaré metric, the self-adjoint Laplace-Beltrami operator on L^2 functions on $\Gamma \backslash \mathbb{H}$ has infinitely many eigenvalues in its point spectrum.

4.3.2. Historical background, literature, and current progress

4.3.3. Verdict: partially resolved

4.3.3.1. Precise problem

The statement asks whether every torsion-free cofinite Fuchsian group $\Gamma \subset \mathrm{SL}(2, \mathbb{R})$ has the property that the Laplace-Beltrami operator on $L^2(\Gamma \backslash \mathbb{H})$ has infinitely many eigenvalues in its point spectrum.

For cocompact Γ , this is a standard consequence of compactness and Weyl's law. The real Roelcke-Selberg problem is the noncompact finite-area case, where the continuous spectrum is $[1/4, \infty)$ and only the discrete part of the spectrum is counted. Since the residual spectrum and the eigenvalues below $1/4$ are finite in the usual finite-area setting, the question is essentially whether every torsion-free nonuniform lattice has infinitely many cuspidal L^2 eigenfunctions.

4.3.3.2. Verdict

Partially resolved. The full statement for all torsion-free cofinite noncompact Fuchsian groups is not resolved by citable literature found in the search. It is true in important special cases, especially congruence/arithmetic groups, but no valid theorem found covers all cofinite torsion-free groups. There is also strong contrary/generic evidence, from Phillips-Sarnak type deformation theory and later work, that the full universal assertion may fail.

4.3.3.3. Original and background sources

- Title: Über die Wellengleichung bei Grenzkreisgruppen erster Art Authors: Walter Roelcke Venue: Sitzungsberichte der Heidelberger Akademie der Wissenschaften, Mathematisch-naturwissenschaftliche Klasse 1953/1955, 159-267 Year: 1956 DOI/arXiv/URL: MR18,476d Relevance: One of Roelcke's original spectral-analysis sources for finite area Fuchsian quotients. Later literature identifies the infinite-discrete-spectrum expectation as the Roelcke-Selberg conjecture or hypothesis, though Petridis-Risager note that it is unclear how formally Roelcke and Selberg stated it as a conjecture.
- Title: Das Eigenwertproblem der automorphen Formen in der hyperbolischen Ebene. I, II Authors: Walter Roelcke Venue: Mathematische Annalen Year: 1966-1967 DOI/arXiv/URL: Part I, Math. Ann. 167 (1966), 292-337; Part II, Math. Ann. 168 (1967), 261-324 Relevance: Foundational source on the automorphic Laplace eigenvalue problem. It does not settle the modern all-cofinite-groups question.
- Title: Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet series Authors: Atle Selberg Venue: Journal of the Indian Mathematical Society (N.S.) 20, 47-87 Year: 1956 DOI/arXiv/URL: MR0088511 Relevance: Selberg's trace formula proves abundance/Weyl-law results for congruence groups. This gives infinitely many cuspidal eigenvalues for the relevant torsion-free congruence subgroups, but it is an arithmetic special case, not the full statement.

4.3.3.4. Later progress and limitations

- Title: Artin-Takagi formula for the Selberg zeta function and the Roelcke hypothesis Authors: Alexei B. Venkov Venue: Doklady Akademii Nauk SSSR 247, no. 3, 540-543 Year: 1979 DOI/arXiv/URL: Soviet Math. Dokl. translation 20, no. 4, 745-748 Relevance: Gives further arithmetic/commensurability-type progress around the Roelcke hypothesis. It is not a proof for arbitrary torsion-free cofinite Fuchsian groups.
- Title: On cusp forms for co-finite subgroups of $\mathrm{PSL}(2, \mathbb{R})$ Authors: Ralph Phillips; Peter Sarnak Venue: Inventiones Mathematicae 80, no. 2, 339-364 Year: 1985 DOI/arXiv/URL: <https://doi.org/10.1007/BF01388610> Relevance: Developed the deformation/instability theory of cusp forms for cofinite groups. This work is central to why the original universal Roelcke-Selberg expectation is no longer generally believed, but it does not itself give a full counterexample to the torsion-free quotient statement.
- Title: The Weyl theorem and the deformation of discrete groups Authors: Ralph Phillips; Peter Sarnak Venue: Communications on Pure and Applied Mathematics 38, 853-866 Year: 1985 DOI/arXiv/URL: MR87f:11035 Relevance: Shows how Weyl-law behavior for discrete spectrum can be destroyed under deformation. This is strong partial/generic evidence, not a complete resolution of all cases.
- Title: Spectra of hyperbolic surfaces Authors: Peter Sarnak Venue: Bulletin of the American Mathematical Society (N.S.) 40, no. 4, 441-478 Year: 2003 DOI/arXiv/URL: <https://doi.org/10.1090/S0273-0979-03-00991-1> Relevance: Survey source. It records that modular/congruence surfaces have abundant Maass cusp forms, while for general finite-area hyperbolic surfaces the existence problem remains subtle. It states Phillips-Sarnak type conjectures that generic finite-area surfaces are not essentially cuspidal and, except for the once-punctured torus Teichmüller space, generically have only finitely many discrete eigenvalues. These conjectures are incompatible with a naive universal Weyl-law version and show the full statement is not settled by the known arith-

metric results.

- Title: Hyperbolic triangles without embedded eigenvalues Authors: Luc Hillairet; Chris Judge Venue: *Annals of Mathematics* (2) 187, no. 2, 301-377 Year: 2018 DOI/arXiv/URL: <https://doi.org/10.4007/annals.2018.187.2.1> ; arXiv:1402.4533 Relevance: Proves that generic one-cusp hyperbolic triangles with Neumann boundary have no nonconstant square-integrable eigenfunctions. The paper also explicitly presents the corresponding finite-area hyperbolic-surface question as a basic unresolved problem. This is important negative model evidence but not a counterexample to the problem statement, because the result concerns triangle Neumann problems/model orbifold-type settings, not every torsion-free quotient $\Gamma \backslash \mathbb{H}$.
- Title: Higher Order Deformations of Hyperbolic Spectra Authors: Yiannis N. Petridis; Morten S. Røisager Venue: *Schrödinger Operators, Spectral Analysis and Number Theory*, Springer Proceedings in Mathematics & Statistics 348, Springer, Cham, 223-239 Year: 2021 DOI/arXiv/URL: https://doi.org/10.1007/978-3-030-68490-7_11 ; arXiv:1907.00800 Relevance: Modern review of the Roelcke-Selberg conjecture and deformation theory. It states that for a finite-volume hyperbolic quotient the discrete spectrum may be finite or infinite, that Selberg proved Weyl law for congruence groups, and that the belief $N_d(\lambda) \rightarrow \infty$ for general cofinite groups is called the Roelcke-Selberg conjecture. It also explains that the difficulty is counting embedded eigenvalues in the continuous spectrum and that Phillips-Sarnak's work weakened general belief in the conjecture.

4.3.3.5. Claimed full resolution not accepted

- Title: Selberg Formula for Cofinite Groups and the Roelke Conjecture Authors: Dmitry A. Popov Venue: arXiv preprints Year: 2015/2017, withdrawn 2019 DOI/arXiv/URL: arXiv:1506.08886 and arXiv:1705.05612 Relevance: These preprints claimed to prove the Roelke conjecture for arbitrary cofinite Fuchsian groups, but arXiv records the later version arXiv:1705.05612 as withdrawn and unavailable. The earlier arXiv record states that a reviewer pointed out an unproved assertion and that the preprint was withdrawn. Therefore this cannot be treated as published or citable resolving literature.

4.3.3.6. What remains unresolved

No located published or otherwise reliable citable theorem proves that every torsion-free non-compact cofinite $\Gamma \subset \mathrm{SL}(2, \mathbb{R})$ has infinitely many L^2 point-spectrum eigenvalues. The known positive theorems cover compact quotients and arithmetic/congruence families; the general nonarithmetic finite-area torsion-free case remains unresolved.

4.3.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

4.3.5. Primary-source pointer

The originating ICM record is *Spectral geometry and non-compact Riemannian manifolds* (1983). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=627>.

4.4. 85. Nishio) Trace Formulae in Spectral Geometry [1990-geom-47]

Record: 1990-geom-47

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1990: Nishio) Trace Formulae in Spectral Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=665>

4.4.1. Problem statement

Let M be a closed nonpositively curved Riemannian manifold, let $X = \widetilde{M}$ be its universal covering with deck transformation group $\Gamma = \pi_1(M)$, and let Δ_X be the Laplacian on X . For the identity conjugacy class $[1] \subset \Gamma$, equivalently the null-homotopic free homotopy class of closed curves in M , define the even wave-trace distribution $\Theta_{[1]}$ on $\mathbb{R}$ by

$$\langle \Theta_{[1]}, f \rangle = \int_{\mathcal{D}} U_f(x, x) dx, \quad f \in C_c^\infty(\mathbb{R}),$$

where $\mathcal{D}$ is a fundamental domain for the Γ -action on X , and $U_f(x, y)$ is the kernel of

$$\hat{f}(\sqrt{\Delta_X}) = \int_{-\infty}^{\infty} f(t) \cos(t\sqrt{\Delta_X}) dt, \quad \hat{f}(s) = \int_{-\infty}^{\infty} f(t) \cos(st) dt.$$

Prove that $\Theta_{[1]}$, regarded as a function away from its possible singularity at $t = 0$, satisfies

$$\Theta_{[1]}(t) = O(|t|^{-\alpha}) \quad \text{as } t \rightarrow +\infty \text{ and as } t \rightarrow -\infty$$

for some $\alpha > 0$.

4.4.2. Historical background, literature, and current progress

4.4.3. Status: partially resolved

4.4.3.1. Precise problem checked

The problem asks whether, for every closed nonpositively curved Riemannian manifold M , the identity-conjugacy-class term in the Nishio-Sunada wave trace formula

$$\Theta_{[1]}(t) = \text{Tr}_{\Gamma} (\cos(t\sqrt{\Delta_{\widetilde{M}}}))$$

is a smooth function for $t \neq 0$ whose long-time size satisfies $\Theta_{[1]}(t) = O(|t|^{-\alpha})$ as $t \rightarrow \pm\infty$ for some $\alpha > 0$.

4.4.3.2. Verdict

I did not find a published or citable result proving this full statement for all closed nonpositively curved Riemannian manifolds. The literature gives the trace-formula framework, temperedness of the identity term, related heat-kernel/density-of-states results, and special homogeneous cases, but not the stated pointwise polynomial decay of the wave trace in the full variable-curvature nonpositive-curvature setting.

4.4.3.3. Relevant literature and what it proves

1. Trace formulae in spectral geometry

Authors: Toshikazu Sunada and Manabu Nishio.

Venue/year: Proceedings of the International Congress of Mathematicians, Kyoto 1990, Vol. I, Springer-Verlag, Tokyo, 1991, pp. 577-585.

DOI/arXiv/URL: no DOI found; full text/source page: https://www.researchgate.net/publication/236210645_Trace_formulae_in_spectral_geometry; ICM proceedings OCR: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.2/ICM1990.2.ocr.pdf>

Relevance: This is the original source for the conjugacy-class distributions $\Theta_{[\sigma]}$. It proves the generalized trace formula, support/singular-support statements, and that the identity-class distribution is tempered. Temperedness is much weaker than $\Theta_{[1]}(t) = O(|t|^{-\alpha})$ as a function at infinity.

2. An overview of Sunada's work up to age 60

Authors: Atsushi Katsuda and Polly W. Sy.

Venue/year: Contemporary Mathematics 484, *Spectral Analysis in Geometry and Number Theory*, 2009, pp. 7-42.

DOI/arXiv/URL: DOI 10.1090/conm/484/09462; PDF: <https://www.mims.meiji.ac.jp/publications/2008/abst00013.pdf>

Relevance: The survey restates the Nishio-Sunada generalized trace formula and the explicit formula for $\Theta_{[\sigma]}$. It confirms the framework and known properties but does not state a resolution of the polynomial decay question.

3. On the wave equation on a compact Riemannian manifold without conjugate points

Author: Pierre H. Bérard.

Venue/year: Mathematische Zeitschrift 155, 1977, pp. 249-276.

DOI/arXiv/URL: DOI 10.1007/BF02028444.

Relevance: Bérard's no-conjugate-points wave parametrix is foundational for later spectral estimates on such manifolds. It gives strong control useful for logarithmic Weyl remainders, but it does not prove long-time polynomial decay of the von Neumann wave trace on the universal cover.

4. Density of states in spectral geometry

Authors: Toshiaki Adachi and Toshikazu Sunada.

Venue/year: Commentarii Mathematici Helvetici 68, 1993, pp. 480-493.

DOI/arXiv/URL: DOI 10.1007/BF02565831; Springer page: <https://link.springer.com/article/10.1007/BF02565831>

Relevance: Develops density-of-states methods in spectral geometry related to the von Neumann trace. This is relevant background for $\Theta_{[1]}$, but it concerns density-of-states/heat-type spectral information rather than the required pointwise large- $|t|$ decay of the wave trace.

5. Von Neumann spectra near the spectral gap

Authors: Alan L. Carey, Thierry Coulhon, Varghese Mathai, and John Phillips.

Venue/year: Bulletin des Sciences Mathématiques 122, 1998, pp. 203-242.

DOI/arXiv/URL: DOI 10.1016/S0007-4497(98)80088-2; arXiv:dg-ga/9610019, <https://arxiv.org/abs/dg-ga/9610019>

Relevance: Studies von Neumann heat-theta functions on universal covers and introduces invariants $\beta_j(M, g)$. The paper notes that Sunada's positivity conjecture for the function case $\beta_0(M, g) > 0$ was proved by Terry Lyons. This gives related polynomial heat-trace information, but a heat-trace estimate does not by itself imply the stated oscillatory wave-trace decay.

6. Asymptotic behaviour of spectra of compact quotients of certain symmetric spaces

Author: Ramesh Gangolli.

Venue/year: Acta Mathematica 121, 1968, pp. 151-192.

DOI/arXiv/URL: DOI 10.1007/BF02391912.

Relevance: In locally symmetric/nonpositive-curvature settings, the identity contribution can be analyzed through the Selberg trace formula and Harish-Chandra Plancherel theory. These homogeneous cases are important supporting examples, but they impose much stronger hypotheses than an arbitrary closed nonpositively curved manifold.

4.4.3.4. What remains unresolved

The missing result is a theorem covering every closed nonpositively curved Riemannian manifold, with no local-symmetry, rank-one, pinching, analyticity, or other extra assumptions, proving that the identity-class wave trace is represented away from 0 by a function with some polynomial decay at both infinities. Existing results establish temperedness and related heat/density-of-states estimates, but I found no citable theorem converting those into the exact wave-trace bound required in the problem statement.

4.4.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

4.4.5. Primary-source pointer

The originating ICM record is *Nishio*) *Trace Formulae in Spectral Geometry* (1990). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.occr.pdf#page=665>.

4.5. 86. Positive scalar curvature metrics - Existence and classification questions [1994-geom-60]

Record: 1994-geom-60

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Positive scalar curvature metrics - Existence and classification questions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=697>

4.5.1. Problem statement

For every discrete group π and integer n , let $B\pi$ denote the classifying space of π , and let $C_r^*\pi$ be the reduced real C^* -algebra of π . Let $R_n(B\pi)$ be Stolz's relative spin bordism group: its elements are represented by triples (N, f, h) , where N is a compact n -dimensional spin manifold, possibly with boundary, $f : N \rightarrow B\pi$ is a map, and h is a positive scalar curvature metric on ∂N ; two such triples are bordant if their boundary triples are joined by a positive-scalar-curvature bordism and the closed spin manifold obtained by gluing the two representatives to that boundary bordism bounds over $B\pi$. The index homomorphism

$$\theta : R_n(B\pi) \longrightarrow KO_n(C_r^*\pi)$$

sends the class of (N, f, h) to the real K -theoretic index of the Dirac operator on N determined by f and the boundary metric h . Fix a Bott manifold $\mathcal{B}$, meaning a simply connected 8-dimensional spin manifold with $\hat{A}(\mathcal{B}) = 1$. Taking products with $\mathcal{B}$ gives homomorphisms $R_{n+8k}(B\pi) \rightarrow R_{n+8(k+1)}(B\pi)$, and

$$R_n(B\pi)[\mathcal{B}^{-1}] := \varinjlim_{k \geq 0} R_{n+8k}(B\pi).$$

Prove that, under the Bott-periodicity identifications $KO_{n+8k}(C_r^*\pi) \cong KO_n(C_r^*\pi)$, the homomorphisms θ induce an isomorphism

$$R_n(B\pi)[\mathcal{B}^{-1}] \xrightarrow{\cong} KO_n(C_r^*\pi).$$

4.5.2. Historical background, literature, and current progress

4.5.3. Verdict: partially resolved

4.5.3.1. Precise problem

The statement is the Bott-stabilized Stolz relative index conjecture for positive scalar curvature. In spin notation, it asks whether, for every discrete group Γ and every integer n , the relative

higher index map

$$\theta : R_n^{\text{spin}}(B\Gamma)[B^{-1}] \longrightarrow KO_n(C_r^*(\Gamma; \mathbb{R}))$$

is an isomorphism, where $R_n^{\text{spin}}(B\Gamma)[B^{-1}]$ is obtained from Stolz's R -group by taking the direct limit under products with an 8-dimensional Bott manifold.

4.5.3.2. Verdict

This problem is not fully resolved. It has substantial partial progress, mainly on surjectivity/rational surjectivity under Baum-Connes-type hypotheses, but the full integral isomorphism for all discrete groups is not known. In particular, recent citable literature explicitly formulates exactly this map as the Stolz conjecture and states that injectivity of the Stolz map is still wide open, even for the trivial group.

4.5.3.3. Literature and progress

1. **Stephan Stolz, "Positive Scalar Curvature Metrics – Existence and Classification Questions",** *Proceedings of the International Congress of Mathematicians, Zurich 1994, Vol. 1, Birkhauser, 1995, pp. 625–636.* DOI/URL: https://doi.org/10.1007/978-3-0348-9078-6_56. This is the original ICM source for the existence/classification framework around positive scalar curvature metrics and Stolz's R -groups.
2. **Jonathan Rosenberg and Stephan Stolz, "Metrics of positive scalar curvature and connections with surgery",** *Surveys on Surgery Theory, Vol. 2, Annals of Mathematics Studies 149, Princeton University Press, 2001, pp. 353–386.* URL: <https://www.math.umd.edu/~jmr/psc.pdf>. Section 5 defines the index homomorphism $\theta : R_n(\gamma) \rightarrow KO_n(C_\gamma^*)$, observes that it is not generally injective before stabilization, defines $R_n(\gamma)[Bt^{-1}]$, and states Conjecture 5.7: the stable homomorphism $\theta^{st} : R_n(\gamma)[Bt^{-1}] \rightarrow KO_n(C_\gamma^*)$ is an isomorphism. The spin case is the problem statement. The same discussion notes that surjectivity is related to Baum-Connes and that injectivity appears hard.
3. **Jonathan Rosenberg, "Manifolds of Positive Scalar Curvature: A Progress Report",** *Surveys in Differential Geometry 11, 2006, pp. 259–294.* URL: <https://www.math.umd.edu/~jmr/psc2006.pdf>. Conjecture 2.4 restates Stolz's problem: the index map $R_{n+1}(\pi) \rightarrow KO_{n+1}(C_R^*(\pi))$ should be stably an isomorphism, with stability given by the direct limit under product with the Bott manifold. This matches the problem.
4. **Paolo Piazza and Thomas Schick, "Rho-classes, index theory and Stolz' positive scalar curvature sequence",** *Journal of Topology 7 (2014), no. 4, 965–1004.* DOI/arXiv: <https://doi.org/10.1112/jtopol/jtt048>, arXiv:1210.6892. They construct a map from Stolz's positive scalar curvature exact sequence to the Higson-Roe analytic surgery exact sequence and identify the analytic targets of the R -group index and rho classes. This is important structural progress, but it does not prove the Bott-stabilized index map is an isomorphism.
5. **Zhizhang Xie and Guoliang Yu, "Positive scalar curvature, higher rho invariants and localization algebras",** *Advances in Mathematics 262 (2014), 823–866.* DOI/arXiv: <https://doi.org/10.1016/j.aim.2014.06.001>, arXiv:1302.4418. They give localization-algebra constructions relating higher rho invariants and relative higher

indices, including a map from Stolz's sequence to analytic surgery. Again, this supplies comparison machinery but not the full isomorphism in the conjecture.

6. Zhizhang Xie, Guoliang Yu and Rudolf Zeidler, “On the range of the relative higher index and the higher rho-invariant for positive scalar curvature”, *Advances in Mathematics* 390 (2021), Article 107897. DOI/arXiv: <https://doi.org/10.1016/j.aim.2021.107897>, arXiv:1712.03722. Corollary 1.6/3.16 proves that if the real Baum-Connes assembly map for Γ is rationally surjective, then the Bott-stabilized relative index map

$$R_n^{\text{spin}}(B\Gamma)[Bt^{-1}] \rightarrow KO_n(C_r^*\Gamma)$$

is rationally surjective. This is strong partial progress on the range of the map, but it is only rational surjectivity under an additional hypothesis and does not address full integral injectivity.

7. Jinmin Wang, Zhizhang Xie and Guoliang Yu, “Approximations of delocalized eta invariants by their finite analogues”, *L'Enseignement Mathématique* 69 (2023), 51–91. DOI/URL: <https://doi.org/10.4171/LEM/1048>, <https://ems.press/content/serial-article-files/44474>. Appendix A formulates exactly the assertion as Conjecture A.3, “Stolz conjecture”: the map

$$\theta : R_n^{\text{spin}}(B\Gamma)[B^{-1}] \rightarrow KO_n(C_r^*(\Gamma; \mathbb{R}))$$

is an isomorphism. The authors then state that the injectivity of the Stolz map, and of the generalized Stolz map, is “wide open” and not even known for Γ trivial. This directly rules out a fully resolved verdict.

4.5.3.4. What remains unresolved

The unresolved part is the full isomorphism assertion in all cases. Known results do not prove integral surjectivity and injectivity for every discrete group. The principal missing point is injectivity of the Bott-stabilized Stolz map; current literature records it as open even in the simplest case of the trivial group. Therefore the correct status is **partially resolved**, not fully resolved.

4.5.4. Remaining open problems recorded by the database

4.5.4.1. Open problem 1

Prove, already for the trivial group $\Gamma = \{e\}$, that the Bott-stabilized Stolz relative index map $\theta : R_n^{\text{spin}}(\text{pt})[B^{-1}] \rightarrow KO_n(\mathbb{R})$ is injective.

Reference supplied by the database. Jinmin Wang, Zhizhang Xie, and Guoliang Yu, “Approximations of Delocalized Eta Invariants by Their Finite Analogues,” *L'Enseignement Mathématique* 69 (2023), no. 1–2, 51–91, Appendix A, Conjecture A.3 and the paragraph following Conjecture A.4 (explicitly stating that injectivity is not known even for the trivial group), DOI: 10.4171/LEM/1048, arXiv:2003.03401, <https://doi.org/10.4171/LEM/1048>.

4.5.5. Primary-source pointer

The originating ICM record is *Positive scalar curvature metrics - Existence and classification questions* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=697>.

4.6. 87. Finite Dimensional Approximations in Geometry [2002-ag-88]

Record: 2002-ag-88

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Finite Dimensional Approximations in Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=380>

4.6.1. Problem statement

Let $\overline{M}$ be a compactified moduli space, as in symplectic geometry or Donaldson theory and in the Kotschick-Morgan conjecture, whose local structure is encoded by Fukaya-Ono Kuranishi data or Ruan virtual-neighborhood data. Determine whether there exists a finite-dimensional approximation A of the ambient infinite-dimensional configuration space such that $\overline{M} \subset A$, the Kuranishi or virtual-neighborhood structure is retained sufficiently that changes of auxiliary choices are represented by cobordisms, and the resulting cobordism class can be interpreted by a stable Pontrjagin-Thom construction.

4.6.2. Historical background, literature, and current progress

4.6.3. Verdict: partially resolved

4.6.3.1. Precise problem

The problem is not just the classical Kotschick-Morgan wall-crossing conjecture. The source problem is Furuta's broader "global approximation" question: given a compactified moduli space described locally by Fukaya-Ono Kuranishi data or Ruan virtual-neighborhood data, construct a finite-dimensional ambient approximation of the original infinite-dimensional configuration space that contains the compactified moduli space, keeps enough of the local Kuranishi/virtual-neighborhood data to make changes of choices into cobordisms, and places the resulting cobordism class in a stable Pontrjagin-Thom/stable homotopy framework.

4.6.3.2. Original source

- Title: *Finite dimensional approximations in geometry*
- Author: Mikio Furuta
- Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002, vol. 2, pp. 395-404; arXiv version submitted 2003
- DOI/arXiv/URL: arXiv:math/0304295; <https://doi.org/10.48550/arXiv.math/0304295>; <https://arxiv.org/abs/math/0304295>
- Relevance: Section 4 states the problem in essentially this form. Furuta observes that for symplectic geometry or Donaldson theory this stable Pontrjagin-Thom/global finite-

dimensional approximation construction had not been done, and that the same difficulty occurs for the Kotschick-Morgan conjecture. The paper contrasts this with the Seiberg-Witten case, where compactness allows a global finite-dimensional approximation leading to the Bauer-Furuta stable cohomotopy invariant.

4.6.3.3. Progress found

- Title: *Global Kuranishi charts and a product formula in symplectic Gromov-Witten theory*
- Authors: Amanda Hirschi and Mohan Swaminathan
- Venue/year: *Selecta Mathematica*, New Series, 30, article 87, 2024
- DOI/arXiv/URL: <https://doi.org/10.1007/s00029-024-00982-y>; arXiv:2212.11797; <https://arxiv.org/abs/2212.11797>
- Result/progress: Constructs global Kuranishi charts for moduli spaces of stable pseudo-holomorphic maps to closed symplectic manifolds in all genera and proves a product formula for symplectic Gromov-Witten invariants. This is strong progress on the symplectic Gromov-Witten part of Furuta's question, including global finite-dimensional chart technology and cobordism/equivalence control for choices, but it is not a full solution for arbitrary Fukaya-Ono/Ruan moduli problems or Donaldson/Kotschick-Morgan theory.
- Title: *Gromov-Witten Invariants in Complex and Morava-Local K-Theories*
- Authors: Mohammed Abouzaid, Mark McLean, and Ivan Smith
- Venue/year: *Geometric and Functional Analysis* 34, pp. 1647-1733, 2024
- DOI/arXiv/URL: <https://doi.org/10.1007/s00039-024-00697-4>; arXiv:2307.01883; <https://arxiv.org/abs/2307.01883>
- Result/progress: Constructs Gromov-Witten-type invariants valued in complex K-theory and Morava-local complex-oriented cohomology theories. The key geometric input is global Kuranishi charts for stable maps of arbitrary genus. This advances the generalized-cohomology/stable-homotopy direction for symplectic Gromov-Witten theory, but it remains a special symplectic GW framework rather than the full ambient approximation problem for all Kuranishi/virtual-neighborhood compactifications.
- Title: *Complex cobordism, Hamiltonian loops and global Kuranishi charts*
- Authors: Mohammed Abouzaid, Mark McLean, and Ivan Smith
- Venue/year: Preprint, 2021
- DOI/arXiv/URL: arXiv:2110.14320; <https://doi.org/10.48550/arXiv.2110.14320>; <https://arxiv.org/abs/2110.14320>
- Result/progress: Builds global Kuranishi charts for moduli spaces of pseudoholomorphic spheres and uses them in generalized cohomology arguments. This is an important genus-zero symplectic case, but it does not address Donaldson theory or the full generality of Furuta's problem.
- Title: *An integral Euler cycle in normally complex orbifolds and $\mathbb{Z}$ -valued Gromov-Witten type invariants*
- Authors: Shaoyun Bai and Guangbo Xu
- Venue/year: Preprint, 2022

- DOI/arXiv/URL: arXiv:2201.02688; <https://arxiv.org/abs/2201.02688>
- Result/progress: Provides derived orbifold chart/bordism technology for genus-zero Gromov-Witten-type invariants, including invariance under choices in a bordism group. This is relevant partial progress toward Pontrjagin-Thom-type refinements, but it is not a complete solution of the general ambient finite-dimensional approximation problem.
- Title: *An $SO(3)$ -Monopole Cobordism Formula Relating Donaldson and Seiberg-Witten Invariants*
- Authors: Paul M. N. Feehan and Thomas G. Leness
- Venue/year: *Memoirs of the American Mathematical Society* 256, no. 1226, 2018
- DOI/arXiv/URL: <https://doi.org/10.1090/memo/1226>; arXiv:math/0203047; <https://arxiv.org/abs/math/0203047>
- Result/progress: Proves an $SO(3)$ -monopole analogue of the Kotschick-Morgan wall-crossing phenomenon and reduces the classical Kotschick-Morgan conjecture to further anti-self-dual gluing input. This is related gauge-theoretic progress but does not give the finite-dimensional ambient approximation and stable Pontrjagin-Thom construction asked for in the problem statement.

4.6.3.4. Claimed resolution checked

- Title: *A Smooth Compactification of the Moduli Space of Instantons and Its Application*
- Author: Bohui Chen
- Venue/year: Preprint, 2002
- DOI/arXiv/URL: arXiv:math/0204287; <https://arxiv.org/abs/math/0204287>
- Claim and assessment: The abstract claims a proof of the Kotschick-Morgan conjecture via a smooth compactification of Donaldson moduli spaces. Even if read as addressing the classical Donaldson wall-crossing conjecture, this is narrower than Furuta's global finite-dimensional ambient approximation/stable Pontrjagin-Thom problem. Later Feehan-Leness discussions describe Chen's approach as an attempt requiring difficult transition-map/gluing analysis, and their published monograph treats related results as analogues/reductions rather than a full resolution of Furuta's stated problem. This item therefore does not fully resolve the problem.

4.6.3.5. What remains unresolved

No published or otherwise citable theorem found gives a construction covering the full statement of the problem: a general finite-dimensional ambient approximation for compactified moduli spaces with Fukaya-Ono Kuranishi or Ruan virtual-neighborhood structures, including Donaldson theory and the Kotschick-Morgan setting, with changes of auxiliary choices represented by cobordisms and with the resulting class interpreted by a stable Pontrjagin-Thom construction.

The symplectic Gromov-Witten case has substantial modern solutions via global Kuranishi charts and generalized cohomology-valued invariants, but these are special cases with extra structure and do not settle the full cross-context problem Furuta asked.

4.6.4. Remaining open problems recorded by the database

4.6.4.1. Open problem 1

Let $X = M/\mathbb{Z}_\alpha$ be a four-orbifold of the type considered by Qiao, with $b_2^+(X) = 1$, and let $D(e)$ be the link of the reducible pseudo-connections associated with an orbifold wall W_e . Prove the orbifold Kotschick–Morgan conjecture: if q is the intersection form and $r = (e^2 - p_1(P))/4$, then

$$\langle (2\mu)^d, D(e) \rangle = \sum_{i=0}^r a_i(r, O, X) q^{r-i} e^{d-2r-2i},$$

where the coefficients $a_i(r, O, X)$ depend only on r , the equivariant bundle data O , and the homotopy type of X .

Reference supplied by the database. Shuaige Qiao, *Bubble Tree Compactification of Instanton Moduli Spaces*, Ph.D. thesis, Australian National University, 2021, Chapter 7, Section 7.1, “The Kotschick–Morgan conjecture on orbifolds,” displayed “Conjecture (Kotschick–Morgan conjecture for orbifolds),” pp. 133–135. DOI: [10.25911/B3M9-WX61](https://doi.org/10.25911/B3M9-WX61); repository record: <http://hdl.handle.net/1885/233020>.

4.6.4.2. Open problem 2

Construct the proposed bubble-tree compactification of the universal instanton bundle over a general smooth four-manifold, with compactified total space $\mathcal{D}_{k,1}^0$, compactified marked-point base $\mathcal{D}_{k,1}$, and maps $\mathcal{D}_{k,1}^0 \rightarrow \mathcal{D}_{k,1} \rightarrow \overline{M}_k$ and $\mathcal{D}_{k,1} \rightarrow X$, so that the family index class extends over the lower strata and gives a gauge-theoretic definition of K-theoretic Donaldson invariants beyond nonsingular projective surfaces.

Reference supplied by the database. Shuaige Qiao, *Bubble Tree Compactification of Instanton Moduli Spaces*, Ph.D. thesis, Australian National University, 2021, Chapter 7, Section 7.2, “K-theoretic Donaldson invariant,” especially the proposed compactification diagram and the statement “Details of this construction need to be clarified,” pp. 135–137. DOI: [10.25911/B3M9-WX61](https://doi.org/10.25911/B3M9-WX61); repository record: <http://hdl.handle.net/1885/233020>.

4.6.4.3. Open problem 3

Construct a bubble-tree compactification of the $U(2)$ -monopole moduli space that incorporates virtual-neighborhood data around the nonregular reducible monopoles in every bubbling stratum, and use it to complete the $PU(2)$ -monopole cobordism approach to Witten’s conjectural formula relating Donaldson and Seiberg–Witten invariants.

Reference supplied by the database. Shuaige Qiao, *Bubble Tree Compactification of Instanton Moduli Spaces*, Ph.D. thesis, Australian National University, 2021, Chapter 7, Section 7.3, “Witten’s conjecture and bubble tree for non-abelian monopole moduli space,” pp. 137–138. DOI: [10.25911/B3M9-WX61](https://doi.org/10.25911/B3M9-WX61); repository record: <http://hdl.handle.net/1885/233020>.

4.6.4.4. Open problem 4

Associate to general Floer data a well-defined stable homotopy type whose ordinary homology recovers the corresponding Floer homology; in particular, construct a true model in the case of nontrivial polarization and disc bubbling by combining the required twisted-spectrum structure with a compatible Kuranishi- or polyfold-type compactification of the flow category.

Reference supplied by the database. Alice Hedenlund and Tasos Moulinos, “Twisted Spectra Revisited,” *Algebraic & Geometric Topology* **25** (2025), no. 9, 5693–5754, Question 0.1, Question 0.2, and Remark 0.3, pp. 5694–5695. DOI: [10.2140/agt.2025.25.5693](https://doi.org/10.2140/agt.2025.25.5693); arXiv: [2312.07403](https://arxiv.org/abs/2312.07403).

4.6.5. Primary-source pointer

The originating ICM record is *Finite Dimensional Approximations in Geometry* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=380>.

4.7. 88. Kähler manifolds and transcendental techniques in algebraic geometry [2006-geom-97]

Record: 2006-geom-97

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Kähler manifolds and transcendental techniques in algebraic geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>

4.7.1. Problem statement

Prove or disprove that if $\pi : \mathcal{X} \rightarrow S$ is a deformation of compact complex manifolds over an irreducible analytic base space S , that is, π is a proper holomorphic map whose fibers are smooth compact complex manifolds, and locally near every point of every fiber $\mathcal{X}$ is isomorphic to a product over S , with π identified with the projection to the base, and if X_{t_0} is Kähler for some $t_0 \in S$, where $X_t = \pi^{-1}(t)$ for $t \in S$, then there exists a finite, or possibly countable, union $\bigcup_v S_v$ of proper analytic strata $S_v \subsetneq S$ such that X_t is Kähler for every $t \in S \setminus \bigcup_v S_v$, and X_t is bimeromorphic to a Kähler manifold for every $t \in \bigcup_v S_v$.

4.7.2. Historical background, literature, and current progress

4.7.3. Verdict: partially resolved

4.7.3.1. Precise problem

The problem is Demailly's/Demailly–Paun's deformation-stability question for the Kahler property. In modern terms it asks:

Given a locally trivial proper holomorphic family of compact complex manifolds over an irreducible analytic base, with at least one Kahler fiber, is the Kahler locus the complement of a finite or countable union of proper analytic strata, and are all remaining fibers at least in Fujiki class C, i.e. bimeromorphic to compact Kahler manifolds?

Equivalently, it combines two assertions:

1. global/countable-Zariski openness of the Kahler property from one Kahler fiber;
2. closedness of the Fujiki-class property for the exceptional/non-Kahler fibers, already including the disk-limit problem: if all punctured fibers are Kahler, is the central fiber in Fujiki class C?

4.7.3.2. Current literature status

The exact statement is not fully resolved. The most difficult remaining point is the Fujiki-class conclusion for limits of Kahler manifolds; even the one-parameter disk version is still treated

in recent papers as a conjecture, with only equivalent criteria or extra hypotheses known. The generic/countable-Zariski openness part from a single Kahler fiber is also not known in the full generality of the problem.

4.7.3.3. Key literature and progress

Original source and foundational results

- Title: Kähler manifolds and transcendental techniques in algebraic geometry
 Authors: Jean-Pierre Demailly
 Venue: Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. I, EMS, pp. 153-186
 Year: 2007
 DOI/URL: <https://doi.org/10.4171/022-1/8>; https://www-fourier.univ-grenoble-alpes.fr/~demailly/manuscripts/madrid_icm2006_print.pdf
 Relevance: This is the source context for the problem. It states the conjectural countable analytic stratification and the expected Fujiki-class conclusion for the exceptional fibers.
- Title: Numerical characterization of the Kähler cone of a compact Kähler manifold
 Authors: Jean-Pierre Demailly and Mihai Păun
 Venue: Annals of Mathematics (2) 159, no. 3, 1247-1274
 Year: 2004
 DOI/arXiv/URL: <https://doi.org/10.4007/annals.2004.159.1247>; [arXiv:math/0105176](https://arxiv.org/abs/math/0105176); <https://annals.math.princeton.edu/2004/159-3/p05>
 Relevance: Theorem 0.9 proves invariance of the Kahler cone for very general fibers in a deformation whose fibers are already compact Kahler manifolds. Section 5 formulates Conjecture 5.1: if one fiber is Kahler, then outside a countable union of proper analytic subsets the fibers should be Kahler. The paper explicitly presents this as a conjecture, so it does not prove the full hypothesis.
- Title: On deformations of complex analytic structures. III. Stability theorems for complex structures
 Authors: Kunihiko Kodaira and Donald C. Spencer
 Venue: Annals of Mathematics (2) 71, 43-76
 Year: 1960
 DOI/URL: <https://doi.org/10.2307/1969879>
 Relevance: Proves local openness of the Kahler property: nearby fibers of a Kahler fiber are Kahler. This is much weaker than the global countable-stratification assertion in the problem statement.
- Title: An example of a non-Kählerian complex-analytic deformation of Kählerian complex structures
 Authors: Heisuke Hironaka
 Venue: Annals of Mathematics (2) 75, 190-208
 Year: 1962
 URL: <https://www.jstor.org/stable/1970426>
 Relevance: Shows that a limit of Kahler/projective manifolds need not itself be Kahler in dimension 3. Hironaka's central fiber is Moishezon, hence in Fujiki class C, so it supports the expected weaker conclusion but does not prove it generally.

Partial results for Moishezon/projective analogues

- Title: Deformation limits of projective manifolds: Hodge numbers and strongly Gauduchon metrics
 Authors: Dan Popovici
 Venue: *Inventiones Mathematicae* 194, 515-534
 Year: 2013
 DOI/URL: <https://doi.org/10.1007/s00222-013-0449-0>
 Relevance: Proves that limits of projective fibers are Moishezon under additional hypotheses such as local constancy of certain Hodge numbers or the existence of a strongly Gauduchon metric on the limit fiber. This is a projective/Moishezon analogue with extra assumptions, not the Kahler-to-Fujiki statement.
- Title: Deformation limit and bimeromorphic embedding of Moishezon manifolds
 Authors: Sheng Rao and I-Hsun Tsai
 Venue: *Communications in Contemporary Mathematics* 23, no. 8, Paper No. 2050087
 Year: 2021
 DOI/arXiv/URL: <https://doi.org/10.1142/S021919972050087X> ; arXiv:1901.10627
 Relevance: If uncountably many fibers are Moishezon and the reference fiber satisfies deformation invariance of the Hodge number of type (0,1) or admits a strongly Gauduchon metric, then the reference fiber is Moishezon. Still conditional and about Moishezon fibers.
- Title: Smooth deformation limit of Moishezon manifolds is Moishezon
 Authors: Mu-Lin Li, Sheng Rao, Kai Wang, and Meng-jiao Wang
 Venue: arXiv preprint
 Year: 2024/2025
 DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2407.02022> ; arXiv:2407.02022
 Relevance: Proves the Moishezon disk-limit conjecture and even that uncountably many Moishezon fibers force all fibers to be Moishezon. This is important progress for the algebraic/bimeromorphic-projective analogue, but it does not apply to general Kahler fibers, which need not be Moishezon.

Partial results for Kahler/Fujiki limits

- Title: Compact Complex Manifolds with Small Gauduchon Cone
 Authors: Dan Popovici and Luis Ugarte
 Venue: *Proceedings of the London Mathematical Society*
 Year: 2018
 DOI/arXiv/URL: <https://doi.org/10.1112/plms.12110> ; arXiv:1407.5070
 Relevance: Surveys and develops a program toward closedness of Fujiki class C under holomorphic deformations. It states the class-C limit conjecture and gives stability results for related Gauduchon-type conditions. It is explicitly programmatic, not a proof of the problem.
- Title: The limits of Kahler manifolds under holomorphic deformations
 Authors: Mu-Lin Li and Wanmin Liu
 Venue: arXiv preprint
 Year: 2024, revised 2026
 DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2406.14076> ; arXiv:2406.14076

Relevance: Treats the disk conjecture as longstanding. It proves that a Kahler limit is actually Kahler under extra metric and cohomological assumptions on the central fiber, and recovers the surface case. These hypotheses are not present in the problem statement.

- Title: The holomorphic limit of Kahler manifolds

Authors: Mu-Lin Li

Venue: arXiv preprint

Year: 2024

DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2409.19957> ; arXiv:2409.19957

Relevance: For a smooth family over a disk with all punctured fibers Kahler, Theorem 1.2 gives an equivalence: the central fiber is in Fujiki class C if and only if it has finite upper volume. The same paper says it is an important open problem whether this upper volume is always finite. Therefore this is a strong criterion, not a full resolution of the problem.

- Title: Volumes of Bott-Chern Classes

Authors: Sébastien Boucksom, Vincent Guedj, and Chinh H. Lu

Venue: Peking Mathematical Journal

Year: 2025

DOI/URL: <https://doi.org/10.1007/s42543-025-00105-2>

Relevance: Gives a characterization of Fujiki class C via the bounded mass property plus existence of a closed Bott-Chern class with positive volume. This supplies major analytic context for recent work on Kahler limits, but it still leaves the needed bounded-mass/positive-volume conditions to be proved for arbitrary Kahler deformation limits.

4.7.3.4. Why the full problem is not resolved

No cited theorem found proves the full implication:

if one fiber in an arbitrary locally trivial proper family over an irreducible analytic base is Kahler, then all very general fibers are Kahler and every exceptional fiber is in Fujiki class C.

The closest general disk statement,

if all punctured fibers are Kahler, then the central fiber is in Fujiki class C,

is still presented as a conjecture in 2024-2026 work and is only known under additional hypotheses or through equivalent criteria such as finite upper volume/bounded mass. Since the statement requires all bases, all dimensions, no projectivity/Moishezon assumption, and no extra metric or cohomological condition on exceptional fibers, the literature gives partial progress only.

4.7.4. Remaining open problems recorded by the database

4.7.4.1. Open problem 1

Let $\pi : \mathcal{X} \rightarrow \Delta$ be a proper holomorphic submersion over the unit disc, with compact complex fibers X_t . If X_t is Kähler for every $t \neq 0$, must the central fiber X_0 belong to Fujiki class C, i.e. be bimeromorphic to a compact Kähler manifold?

Reference supplied by the database. Mu-Lin Li and Wanmin Liu, “The limits of Kähler manifolds under holomorphic deformations,” arXiv preprint (2024; revised 2026), Conjecture 1.1, [arXiv:2406.14076](#).

4.7.4.2. Open problem 2

If a compact complex manifold X carries a closed positive $(1, 1)$ -current T whose Bott–Chern class has positive volume $\text{vol}_n(\{T\}) > 0$, must X belong to Fujiki class C ?

Reference supplied by the database. Mu-Lin Li, “The holomorphic limit of Kähler manifolds,” arXiv preprint (2024; revised 2026), Conjecture 1.3 (the Boucksom–Demailly–Păun conjecture), [arXiv:2409.19957](#).

4.7.5. Primary-source pointer

The originating ICM record is *Kähler manifolds and transcendental techniques in algebraic geometry* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>.

4.8. 89. Kähler manifolds and transcendental techniques in algebraic geometry [2006-geom-98]

Record: 2006-geom-98

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Kähler manifolds and transcendental techniques in algebraic geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>

4.8.1. Problem statement

Prove or disprove that for every compact Kähler manifold X of complex dimension n , the Kodaira dimension $\kappa(X)$ equals the numerical dimension of the canonical class:

$$\kappa(X) = \nu(X) := \nu(c_1(K_X)),$$

where $K_X = \Lambda^n T_X^*$ is the canonical bundle of X , $c_1(K_X) \in H^{1,1}(X, \mathbb{R})$ is its first Chern class, and the numerical dimension of a class $\alpha \in H^{1,1}(X, \mathbb{R})$ is defined by

$$\nu(\alpha) = -\infty \quad \text{if } \alpha \text{ is not pseudo-effective,}$$

while, if α is pseudo-effective, meaning that it is represented by a closed positive $(1,1)$ -current,

$$\nu(\alpha) = \max\{p \in \{0, 1, \dots, n\} : \langle \alpha^p \rangle \neq 0\},$$

with $\langle \alpha^p \rangle$ denoting the p -fold movable intersection product of α .

4.8.2. Historical background, literature, and current progress

4.8.3. Verdict: partially resolved

4.8.3.1. Precise problem

The problem asks for the generalized abundance conjecture for compact Kähler manifolds: for every compact Kähler manifold X , in every dimension, the Kodaira dimension of X should equal the numerical dimension of the canonical class,

$$\kappa(X) = \nu(c_1(K_X)),$$

where ν is defined using the movable/non-pluripolar intersection product and is $-\infty$ for non-pseudo-effective classes.

This is an all-dimensional compact Kähler statement, not just the projective case, not just minimal models, and not just the case where K_X is nef.

4.8.3.2. Status check

I found no citable literature proving this full all-dimensional compact Kahler statement. Recent literature still treats the corresponding Kahler abundance problem as open in higher dimension. Since projective manifolds are compact Kahler manifolds, the problem statement would in particular cover the projective generalized abundance problem in all dimensions; the projective/Kahler abundance literature still has open cases in dimension at least 4.

The correct status is therefore partially resolved: substantial cases are known, especially the extreme numerical-dimension cases and compact Kahler threefolds, but the full statement remains open.

4.8.3.3. Key literature and progress

1. Original/source formulation

- **Title:** “Kahler manifolds and transcendental techniques in algebraic geometry”
- **Author:** Jean-Pierre Demailly
- **Venue:** Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. I, EMS Press, pp. 153-186
- **Year:** 2007
- **DOI/URL:** DOI 10.4171/022-1/8; <https://ems.press/books/standalone/20/363>
- **Relevant result:** Conjecture 5.9 states the generalized abundance conjecture for arbitrary compact Kahler manifolds: $\kappa(X) = \nu(c_1(K_X))$. Theorem 5.10 records known extreme cases $\nu(X) = -\infty, 0, n$.
- **Scope:** This is the source/problem statement and partial progress only; it does not resolve the intermediate cases in arbitrary dimension.

2. Movable intersection/numerical dimension framework and early partial cases

- **Title:** “The pseudo-effective cone of a compact Kahler manifold and varieties of negative Kodaira dimension”
- **Authors:** Sebastien Boucksom, Jean-Pierre Demailly, Mihai Paun, Thomas Peternell
- **Venue:** Journal of Algebraic Geometry 22, no. 2, pp. 201-248
- **Year:** 2013
- **DOI/arXiv/URL:** DOI 10.1090/S1056-3911-2012-00574-8; arXiv math/0405285; <https://arxiv.org/abs/math/0405285>
- **Relevant result:** Section 3.8 formulates the generalized abundance conjecture using the numerical dimension; remarks record $\kappa(X) \leq \nu(X)$, the easy cases $\nu = -\infty, n$, and partial information for $\nu = 0$, including projective dimension ≤ 4 under the hypotheses stated there.
- **Scope:** Gives foundational tools and partial cases, not the full conjecture.

3. Relation with the usual projective abundance conjecture

- **Title:** “Numerical analogues of the Kodaira dimension and the Abundance Conjecture”
- **Author:** Thomas Eckl
- **Venue:** Manuscripta Mathematica 150, pp. 337-356
- **Year:** 2016
- **DOI/arXiv/URL:** DOI 10.1007/s00229-015-0815-x; arXiv 1505.01262; <https://arxiv.org/abs/1505.01262>

- **Relevant result:** Shows that, for non-uniruled smooth complex projective varieties, the usual MMP abundance conjecture and generalized abundance formulated with numerical analogues of Kodaira dimension are equivalent.
- **Scope:** This confirms that the conjecture contains deep unresolved projective abundance cases; it is not a full compact Kahler proof.

4. Compact Kahler threefold minimal models

- **Title:** “Minimal models for Kahler threefolds”
- **Authors:** Andreas Hoering, Thomas Peternell
- **Venue:** *Inventiones Mathematicae* 203, no. 1, pp. 217-264
- **Year:** 2016
- **DOI/arXiv/URL:** DOI 10.1007/s00222-015-0592-x; arXiv 1304.4013; <https://arxiv.org/abs/1304.4013>
- **Relevant result:** Proves that a non-uniruled compact Kahler threefold has a minimal model.
- **Scope:** Supplies the threefold MMP input; it is dimension 3 only.

5. Abundance for compact Kahler threefolds

- **Title:** “Abundance for Kahler threefolds”
- **Authors:** Frederic Campana, Andreas Hoering, Thomas Peternell
- **Venue:** *Annales scientifiques de l’Ecole Normale Supérieure*, Serie 4, 49, no. 4, pp. 971-1025
- **Year:** 2016
- **DOI/URL:** DOI 10.24033/asens.2301; <https://www.numdam.org/articles/10.24033/asens.2301/>
- **Relevant result:** Theorem 1.1 proves that if X is a normal $\mathbb{Q}$ -factorial compact Kahler threefold with terminal singularities and K_X nef, then K_X is semi-ample. The paper explains that for nef K_X , equality $\kappa(X) = \nu(X)$ is sufficient for abundance in this setting.
- **Scope:** Resolves the minimal threefold case, but not arbitrary dimension.

6. Log abundance for compact Kahler threefolds, first part

- **Title:** “On the log abundance for compact Kahler 3-folds”
- **Authors:** Omprokash Das, Wenhao Ou
- **Venue:** *Manuscripta Mathematica* 173, no. 1-2, pp. 341-404
- **Year:** 2024
- **DOI/arXiv/URL:** DOI 10.1007/s00229-023-01467-6; arXiv 2201.01202; <https://arxiv.org/abs/2201.01202>
- **Relevant result:** Proves log abundance for log canonical compact Kahler threefold pairs when $K_X + \Delta$ is nef and $\nu(K_X + \Delta) \neq 2$, including non-vanishing and semi-ampleness in the stated cases.
- **Scope:** Major dimension-3/log progress; excludes $\nu = 2$, later handled separately.

7. Log abundance for compact Kahler threefolds, completion of the $\nu = 2$ case

- **Title:** “On the Log Abundance for Compact Kahler threefolds II”
- **Authors:** Omprokash Das, Wenhao Ou
- **Venue:** *Proceedings of the London Mathematical Society* 132, no. 3, article e70141
- **Year:** 2026
- **DOI/arXiv/URL:** DOI 10.1112/plms.70141; arXiv 2306.00671; <https://arxiv.org/abs/2306.00671>

- **Relevant result:** Proves that for a log canonical compact Kahler threefold pair with $K_X + \Delta$ nef and $\nu(X, K_X + \Delta) = 2$, $K_X + \Delta$ is semi-ample. Combined with the previous Das-Ou paper, this gives log abundance for log canonical compact Kahler threefold pairs.
- **Scope:** Completes the threefold log case, but still only in dimension 3.

8. Current higher-dimensional status

- **Title:** “An approach to the abundance conjecture for Kahler varieties via algebraic reduction”
- **Author:** Zhiyuan Jiang
- **Venue:** arXiv preprint
- **Year:** 2026
- **arXiv/URL:** arXiv 2604.03879; <https://arxiv.org/abs/2604.03879>
- **Relevant result/status statement:** The introduction states that abundance is open in dimension ≥ 4 , that there are major open cases even in dimension 4, and that abundance for Kahler varieties of arbitrary dimension is a major open problem. The paper proves an inductive criterion and some special Kahler fourfold cases, e.g. $a(X) = 3$, under its hypotheses.
- **Scope:** Confirms that the full statement is not resolved; it gives only conditional or special higher-dimensional progress.

4.8.3.4. What remains unresolved

The full conjecture remains open for compact Kahler manifolds in arbitrary dimension, especially intermediate numerical dimensions $0 < \nu(c_1(K_X)) < \dim X$ in dimension at least 4. Known threefold and selected fourfold/projective results do not cover all compact Kahler manifolds and do not imply the full equality $\kappa(X) = \nu(c_1(K_X))$ in every dimension.

4.8.4. Remaining open problems recorded by the database

4.8.4.1. Open problem 1

Let X be a compact Kähler fourfold whose canonical bundle K_X is nef. Must K_X be semiample, i.e. must some positive tensor power $K_X^{\otimes m}$ be generated by global sections?

Reference supplied by the database. Zhiyuan Jiang, “An approach to the abundance conjecture for Kähler varieties via algebraic reduction,” arXiv preprint (2026), Conjecture 1.1 and the discussion identifying dimension four as open, [arXiv:2604.03879](https://arxiv.org/abs/2604.03879).

4.8.4.2. Open problem 2

If the canonical class K_X of a compact Kähler manifold X is pseudo-effective, must X have nonnegative Kodaira dimension, $\kappa(X) \geq 0$?

Reference supplied by the database. Jean-Pierre Demailly, “Kähler manifolds and transcendental techniques in algebraic geometry,” in *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Vol. I, European Mathematical Society, Zürich, 2007, pp. 153–186, §5.5 (the non-vanishing consequence of Conjecture 5.9). DOI: [10.4171/022-1/8](https://doi.org/10.4171/022-1/8).

4.8.4.3. Open problem 3

Is the algebraic reduction map of every compact Kähler variety almost holomorphic?

Reference supplied by the database. Zhiyuan Jiang, “An approach to the abundance conjecture for Kähler varieties via algebraic reduction,” arXiv preprint (2026), Conjecture 1.2, [arXiv:2604.03879](https://arxiv.org/abs/2604.03879).

4.8.5. Primary-source pointer

The originating ICM record is *Kähler manifolds and transcendental techniques in algebraic geometry* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>.

4.9. 90. Kähler manifolds and transcendental techniques in algebraic geometry [2006-geom-99]

Record: 2006-geom-99

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Kähler manifolds and transcendental techniques in algebraic geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>

4.9.1. Problem statement

Prove that for every compact Kähler manifold X of complex dimension n , if $\mathcal{E} \subset H^{1,1}(X, \mathbb{R})$ is the pseudo-effective cone, namely the closed convex cone of cohomology classes $\{T\}$ of closed positive currents T of type $(1, 1)$ on X , and $\mathcal{M} \subset H_{\mathbb{R}}^{n-1, n-1}(X)$ is the cone of movable classes, namely the closure of the convex cone generated by cohomology classes of currents of the form $\mu_*(\tilde{\omega}_1 \wedge \cdots \wedge \tilde{\omega}_{n-1})$, where $\mu : \tilde{X} \rightarrow X$ ranges over modifications and $\tilde{\omega}_1, \dots, \tilde{\omega}_{n-1}$ range over Kähler forms on $\tilde{X}$, then $\mathcal{E}$ and $\mathcal{M}$ are dual under the Poincaré duality pairing $H^{1,1}(X, \mathbb{R}) \times H_{\mathbb{R}}^{n-1, n-1}(X) \rightarrow \mathbb{R}$, $(\alpha, \beta) \mapsto \int_X \alpha \wedge \beta$; equivalently, $\mathcal{E} = \mathcal{M}^\vee$ and $\mathcal{M} = \mathcal{E}^\vee$ with respect to this pairing.

4.9.2. Historical background, literature, and current progress

4.9.3. Verdict: partially resolved

4.9.3.1. Precise problem

The problem asks for the Boucksom-Demailly-Paun-Peternell cone-duality conjecture in its full compact Kähler form: for every compact Kähler manifold X , the pseudo-effective cone $\mathcal{E} \subset H^{1,1}(X, \mathbb{R})$ of closed positive $(1, 1)$ -current classes should be dual, under Poincaré duality, to the movable cone $\mathcal{M} \subset H_{\mathbb{R}}^{n-1, n-1}(X, \mathbb{R})$ generated by pushforwards $\mu_*(\tilde{\omega}_1 \wedge \cdots \wedge \tilde{\omega}_{n-1})$ from modifications.

4.9.3.2. Literature status

1. Jean-Pierre Demailly, "Kähler manifolds and transcendental techniques in algebraic geometry", Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. I, European Mathematical Society, 2007, pp. 153-186. DOI: 10.4171/022-1/8. URL: <https://ems.press/books/standalone/20> This ICM source states the Kähler analogue as a conjecture: for arbitrary compact Kähler X , $\mathcal{E}$ and $\mathcal{M}$ should be dual cones.
2. Sébastien Boucksom, Jean-Pierre Demailly, Mihai Păun, Thomas Peternell, "The pseudo-effective cone of a compact Kähler manifold and varieties of negative Ko-

- daira dimension”, *Journal of Algebraic Geometry* 22 (2013), no. 2, 201-248. DOI: 10.1090/S1056-3911-2012-00574-8; arXiv: math/0405285; URL: <https://arxiv.org/abs/math/0405285> This is the original BDPP source for the cone-duality conjecture. It formulates the arbitrary compact Kähler statement as Conjecture 2.3, and proves projective/algebraic consequences, including the projective movable-curve duality in the Néron-Severi setting. This does not prove the full problem because the statement allows all compact Kähler manifolds and all real $(1, 1)$ -classes.
3. David Witt Nyström, “Duality between the pseudoeffective and the movable cone on a projective manifold”, with an appendix by Sébastien Boucksom, *Journal of the American Mathematical Society* 32 (2019), no. 3, 675-689. DOI: 10.1090/jams/922; arXiv: 1602.03778; URL: <https://arxiv.org/abs/1602.03778> The main theorem proves exactly the $\mathcal{E}$ - $\mathcal{M}$ duality in full real Hodge cohomology when X is projective. This is substantial progress, but projectivity is an extra hypothesis not present in the problem statement.
 4. Jian Xiao, “Weak transcendental holomorphic Morse inequalities on compact Kähler manifolds”, *Annales de l’Institut Fourier* 65 (2015), no. 3, 1367-1379. DOI: 10.5802/aif.2959; arXiv: 1308.2878; URL: <https://aif.centre-mersenne.org/articles/10.5802/aif.2959/> This proves weak transcendental Morse inequalities and is explicitly described as a partial improvement of BDPP. It supplies related partial progress, not the full cone duality.
 5. Nicholas McCleerey, “Plurisupported Currents on Compact Kähler Manifolds”, arXiv preprint, 2021. DOI: 10.48550/arXiv.2106.12017; arXiv: 2106.12017; URL: <https://arxiv.org/abs/2106.12017> Theorem 1.6 / Theorem 3.12 proves the BDPP cone-duality conjecture for compact Kähler manifolds Y admitting a proper modification $\mu : X \rightarrow Y$ where X carries a plurisupported current whose class is Kähler. This is conditional and does not cover arbitrary compact Kähler manifolds.
 6. Hsueh-Yung Lin, “On the dual positive cones and the algebraicity of a compact Kähler manifold”, *Épjournal de Géométrie Algébrique*, special volume in honour of Claire Voisin, 2024, epiga:10771. DOI: 10.46298/epiga.2024.10771; arXiv: 2001.10654; URL: <https://arxiv.org/abs/2001.10654> This recent work still refers to the equality of the movable cone with the dual of the pseudo-effective cone as conjectural in the compact Kähler setting, and studies related algebraicity consequences instead of proving the cone duality.
 7. Junyu Cao and Valentino Tosatti, “Regularity of the volume function”, *Bulletin of the London Mathematical Society*, 2026. DOI: 10.1112/blms.70212; arXiv: 2505.14495; URL: <https://arxiv.org/abs/2505.14495> The results are stated for projective manifolds, or more generally compact Kähler manifolds satisfying the BDPP conjecture, and the paper notes that proving the corresponding result for all compact Kähler manifolds without BDPP remains interesting. This is further evidence that the full compact Kähler cone-duality statement is not presently settled.

4.9.3.3. Hypothesis check

The strongest citable theorem matching the conclusion is Witt Nyström’s theorem, but it assumes X is projective. McCleerey’s theorem covers additional compact Kähler manifolds only under the existence of a suitable plurisupported Kähler class after modification. These hypotheses are not automatic for an arbitrary compact Kähler manifold.

Therefore the exact problem is not fully resolved. It is also not fully open, since the projective

case and some conditional compact Kähler cases are proved. The correct status is partially resolved.

4.9.4. Remaining open problems recorded by the database

4.9.4.1. Open problem 1

Let X be a compact Kähler n -fold and let α be a pseudo-effective real $(1,1)$ -class. Does the orthogonality identity

$$\langle \alpha^{n-1} \rangle \cdot \alpha = \text{Vol}(\alpha)$$

always hold, where $\langle \alpha^{n-1} \rangle$ is the movable positive product?

Reference supplied by the database. Valentino Tosatti, “Orthogonality of divisorial Zariski decompositions for classes with volume zero,” *Tohoku Mathematical Journal* (2) **71** (2019), no. 1, 1–8, Conjecture 1.1. DOI: [10.2748/tmj/1552100439](https://doi.org/10.2748/tmj/1552100439).

4.9.4.2. Open problem 2

Let X be a compact Kähler n -fold and let $\alpha, \beta \in H^{1,1}(X, \mathbb{R})$ be nef classes. Is the sharp transcendental Morse inequality

$$\text{Vol}(\alpha - \beta) \geq \alpha^n - n \alpha^{n-1} \cdot \beta$$

always valid?

Reference supplied by the database. Jian Xiao, “Weak transcendental holomorphic Morse inequalities on compact Kähler manifolds,” *Annales de l’Institut Fourier* **65** (2015), no. 3, 1367–1379, Conjecture 1.1(ii) (originally Boucksom–Demailly–Păun–Peternell, Conjecture 10.1(ii)). DOI: [10.5802/aif.2959](https://doi.org/10.5802/aif.2959).

4.9.5. Primary-source pointer

The originating ICM record is *Kähler manifolds and transcendental techniques in algebraic geometry* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>.

4.10. 91. A Hyperbolic Dispersion Estimate, with Applications to the Linear Schrödinger Equation [2010-geom-103]

Record: 2010-geom-103

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: A Hyperbolic Dispersion Estimate, with Applications to the Linear Schrödinger Equation

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.3/ICM2010.3.pdf#page=663>

4.10.1. Problem statement

Let X be a compact d -dimensional Riemannian manifold with negative sectional curvatures, and let Φ^t be the geodesic flow on the unit cotangent bundle

$$S^*X = \{(x, \xi) \in T^*X : \|\xi\|_x = 1\}.$$

For any family of L^2 -normalized Laplace-Beltrami eigenfunctions ϕ_λ satisfying $-\Delta\phi_\lambda = \lambda\phi_\lambda$ with $\lambda \rightarrow +\infty$, let μ be any associated semiclassical measure, i.e. any weak limit of the microlocal distributions

$$\langle \mu_\lambda, a \rangle = \langle \phi_\lambda, \text{Op}_{\lambda^{-1/2}}(a)\phi_\lambda \rangle_{L^2(X)}, \quad a \in C_c^\infty(T^*X).$$

Prove that

$$h_{KS}(\mu; \Phi^t) \geq \frac{1}{2} \int_{S^*X} \sum_{j=1}^{d-1} \lambda_j^+(\rho) d\mu(\rho),$$

where $h_{KS}(\mu; \Phi^t)$ is the Kolmogorov-Sinai entropy of the Φ^t -invariant probability measure μ on S^*X , and $\lambda_j^+(\rho)$ are the nonnegative Lyapunov exponents of the geodesic flow at $\rho = (x, \xi) \in S^*X$.

4.10.2. Historical background, literature, and current progress

4.10.3. Status verdict: partially resolved

4.10.3.1. Precise problem

The problem asks whether every semiclassical measure of Laplace-Beltrami eigenfunctions on an arbitrary compact negatively curved d -dimensional Riemannian manifold satisfies the sharp lower bound

$$h_{KS}(\mu; \Phi^t) \geq \frac{1}{2} \int_{S^*X} \sum_{j=1}^{d-1} \lambda_j^+(\rho) d\mu(\rho).$$

Equivalently, this is the conjectural lower bound by one half of the Ruelle-Pesin upper bound, for all compact negatively curved manifolds and all dimensions.

4.10.3.2. Literature checked

Original source / formulation

- Title: A Hyperbolic Dispersion Estimate, with Applications to the Linear Schrödinger Equation
- Author: Nalini Anantharaman
- Venue: Proceedings of the International Congress of Mathematicians 2010, Volume III, Hindustan Book Agency, pp. 1839-1861
- Year: 2010
- DOI/URL: https://doi.org/10.1142/9789814324359_0124 ; PDF copy: https://www.math.u-bordeaux.fr/~pthieull/Recherche/KamFaible/Publications/Anantharaman2010_01.pdf
- Relevance: Section 3.2 states the conjectured bound exactly for variable negative curvature. The same discussion says the existing method proves positivity of entropy in variable negative curvature and the half-maximal entropy bound in constant curvature, but not the stated variable-curvature formula. It also records that the inequality had been proved in dimension $d = 2$ by Rivière and that, at that time, it was not known how to prove it when the stable/unstable bundles have dimension $d - 1 > 1$.

General positive entropy and constant-curvature progress

- Title: Entropy and the localization of eigenfunctions
- Author: Nalini Anantharaman
- Venue: Annals of Mathematics, Second Series 168(2), 435-475
- Year: 2008
- DOI/URL: <https://doi.org/10.4007/annals.2008.168.435> ; <https://annals.math.princeton.edu/2008/168-2/p02>
- Relevance: Proves that semiclassical measures for Anosov geodesic flows have positive metric entropy. This rules out complete concentration on closed geodesics, but it does not prove the explicit half-Ruelle lower bound in the problem statement.
- Title: Half-delocalization of eigenfunctions for the Laplacian on an Anosov manifold
- Authors: Nalini Anantharaman and Stéphane Nonnenmacher
- Venue: Annales de l'Institut Fourier 57(7), 2465-2523
- Year: 2007
- DOI/arXiv/URL: <https://doi.org/10.5802/aif.2340> ; arXiv:math-ph/0610019 ; <https://numdam.org/articles/10.5802/aif.2340/>
- Relevance: Gives an entropy lower bound for Anosov manifolds that equals half the maximal entropy in constant sectional curvature. For variable curvature, the bound is weaker and is not the integral of half the sum of positive Lyapunov exponents.
- Title: Entropy of eigenfunctions
- Authors: Nalini Anantharaman, Herbert Koch, and Stéphane Nonnenmacher
- Venue: New Trends in Mathematical Physics, Proceedings of the XVth International Congress on Mathematical Physics, Springer, pp. 1-22
- Year: 2009

- DOI/arXiv/URL: https://doi.org/10.1007/978-90-481-2810-5_1 ; arXiv:0704.1564
- Relevance: Improves the explicit lower bound in variable negative curvature, but the theorem still has a maximal-expansion error term and does not imply the sharp half-Ruelle bound in general.

Cases where the requested bound is proved

- Title: Entropy of semiclassical measures in dimension 2
- Author: Gabriel Rivière
- Venue: Duke Mathematical Journal 155(2), 271-335
- Year: 2010
- DOI/arXiv/URL: <https://doi.org/10.1215/00127094-2010-056> ; arXiv:0809.0230 ; <https://projecteuclid.org/journals/duke-mathematical-journal/volume-155/issue-2/Entropy-of-semiclassical-measures-in-dimension-2/10.1215/00127094-2010-056.full>
- Relevance: Theorem 1.2 proves that on compact Riemannian surfaces with Anosov geodesic flow, every semiclassical measure satisfies the lower bound by one half of the Ruelle upper bound. This resolves the problem statement when $d = 2$, but not all dimensions.
- Title: Entropy of Semiclassical Measures for Nonpositively Curved Surfaces
- Author: Gabriel Rivière
- Venue: Annales Henri Poincaré 11(6), 1085-1116
- Year: 2010
- DOI/arXiv/URL: <https://doi.org/10.1007/s00023-010-0055-2> ; arXiv:0911.1840 ; <https://link.springer.com/article/10.1007/s00023-010-0055-2>
- Relevance: Extends the surface entropy lower bound to compact nonpositively curved surfaces with the appropriate Lyapunov exponent. This is stronger in curvature assumptions for surfaces, but still only two-dimensional.

Later surveys and newer related progress

- Title: Semiclassical behaviour of quantum eigenstates
- Author: Gabriel Rivière
- Venue: arXiv preprint / lecture notes
- Year: 2019
- DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.1905.12303> ; arXiv:1905.12303
- Relevance: States the general Anosov conjecture as Conjecture 6.17 and then says that the theorem answers the question in dimension 2, while in higher dimensions the conjecture remains open. It lists only partial explicit bounds for locally symmetric spaces and

multidimensional torus maps beyond the surface case.

- Title: Quantum ergodicity and semiclassical measures: mathematical results
- Author: Stéphane Nonnenmacher
- Venue: arXiv preprint, chapter for Comprehensive Quantum Mechanics
- Year: 2026
- DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2606.12098> ; arXiv:2606.12098
- Relevance: A current survey of entropy and delocalization constraints. It cites the classical entropy-bound works and describes recent higher-dimensional results as delocalization/support results in locally symmetric settings, not as a proof of the full sharp half-Ruelle entropy lower bound for arbitrary compact variable negatively curved manifolds.
- Title: Semiclassical Measures on Hyperbolic Manifolds
- Authors: Elena Kim and Nicholas Miller
- Venue: arXiv preprint
- Year: 2025
- DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2503.01528> ; arXiv:2503.01528
- Relevance: Proves support restrictions for semiclassical measures on compact real hyperbolic manifolds. These are constant-curvature locally symmetric manifolds and the result is about support/delocalization, not the full variable-curvature entropy bound.
- Title: Semiclassical measures for complex hyperbolic quotients
- Authors: Jayadev Athreya, Semyon Dyatlov, and Nicholas Miller
- Venue: Geometric and Functional Analysis 35(4), 979-1050
- Year: 2025
- DOI/arXiv/URL: arXiv:2402.06477 ; <https://link.springer.com/journal/139/articles>
- Relevance: Gives support restrictions for compact complex hyperbolic quotients, a locally symmetric negative-curvature class with different Lyapunov exponents. This is important related progress but neither covers arbitrary negatively curved manifolds nor proves the requested entropy inequality.

4.10.3.3. Hypothesis/conclusion check

The statement quantifies over every compact d -dimensional Riemannian manifold with negative sectional curvatures, including variable curvature and dimensions $d \geq 3$. The published theorems above cover:

- positive entropy in the general Anosov/negative-curvature setting;
- the sharp half-Ruelle bound in dimension 2;
- the sharp half-maximal entropy bound in constant real hyperbolic curvature and isotropic-expansion cases;
- weaker explicit entropy bounds in variable curvature;
- newer support/delocalization results for surfaces and locally symmetric higher-

dimensional spaces.

None of these hypotheses and conclusions imply the full statement for arbitrary variable negatively curved manifolds in dimensions $d \geq 3$. The obstruction repeatedly identified in the literature is the presence of several distinct Lyapunov exponents and higher-dimensional stable/unstable directions.

4.10.3.4. What remains unresolved

The exact inequality

$$h_{KS}(\mu; \Phi^t) \geq \frac{1}{2} \int_{S^*X} \sum_{j=1}^{d-1} \lambda_j^+(\rho) d\mu(\rho)$$

remains open for general compact variable negatively curved manifolds in dimensions $d \geq 3$. Therefore the problem is not fully resolved, but it is not fully open because it is proved for surfaces and for constant-curvature/isotropic-expansion special cases, with additional weaker general entropy bounds.

4.10.4. Remaining open problems recorded by the database

4.10.4.1. Open problem 1

Let X be a compact three-dimensional Riemannian manifold of variable negative sectional curvature. Does every semiclassical measure μ of Laplace eigenfunctions satisfy

$$h_{KS}(\mu; \Phi^t) \geq \frac{1}{2} \int_{S^*X} (\lambda_1^+(\rho) + \lambda_2^+(\rho)) d\mu(\rho)?$$

Reference supplied by the database. Gabriel Rivi re, “Semiclassical behaviour of quantum eigenstates,” lecture notes, 2019, Conjecture 6.17 and the subsequent discussion identifying only the two-dimensional case as proved, [arXiv:1905.12303](https://arxiv.org/abs/1905.12303).

4.10.5. Primary-source pointer

The originating ICM record is *A Hyperbolic Dispersion Estimate, with Applications to the Linear Schr dinger Equation* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.3/ICM2010.3.pdf#page=663>.

4.11. 92. Equidistribution in Homogeneous Spaces and Number Theory [2010-geom-111]

Record: 2010-geom-111

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Equidistribution in Homogeneous Spaces and Number Theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=522>

4.11.1. Problem statement

Let M be a compact Riemannian manifold with negative sectional curvature, let $d \text{ vol}$ be its Riemannian volume measure, and let

$$\text{vol}(M) = \int_M d \text{ vol}.$$

Let $(\phi_i)_{i \geq 1}$ be an $L^2(M, d \text{ vol})$ -orthonormal sequence of eigenfunctions of the Laplacian on M , so that $\Delta \phi_i = \lambda_i \phi_i$ for some eigenvalue λ_i . Prove that, for every $f \in C^\infty(M)$,

$$\lim_{i \rightarrow \infty} \int_M f(x) |\phi_i(x)|^2 d \text{ vol}(x) = \frac{1}{\text{vol}(M)} \int_M f(x) d \text{ vol}(x).$$

4.11.2. Historical background, literature, and current progress

4.11.3. Verdict: partially resolved

4.11.3.1. Precise problem

The statement is the physical-space form of the Rudnick-Sarnak Quantum Unique Ergodicity (QUE) conjecture. It asks whether, for every compact Riemannian manifold M of negative sectional curvature and every orthonormal sequence of Laplace eigenfunctions ϕ_i with eigenvalues tending to infinity, the probability measures

$$|\phi_i(x)|^2 d \text{ vol}(x)$$

converge weakly to the normalized Riemannian volume measure $d \text{ vol} / \text{vol}(M)$. Equivalently, it asks that there be no exceptional subsequence of eigenfunctions whose mass has a different macroscopic limit.

4.11.3.2. Verdict

Partially resolved. The exact full statement for arbitrary compact negatively curved manifolds and arbitrary orthonormal eigenfunction sequences is not known. The literature proves

density-one equidistribution and several strong restrictions on possible exceptional limits, and it proves QUE in special arithmetic/Hecke settings, but all of these have extra hypotheses or weaker conclusions.

4.11.3.3. Literature and status

1. Original source and formulation:

- Title: The behaviour of eigenstates of arithmetic hyperbolic manifolds
- Authors: Zeev Rudnick and Peter Sarnak
- Venue: Communications in Mathematical Physics 161, no. 1, 195-213
- Year: 1994
- DOI/URL: <https://doi.org/10.1007/BF02099418>; <https://projecteuclid.org/journals/communications-in-mathematical-physics/volume-161/issue-1/The-behaviour-of-eigenstates-of-arithmetic-hyperbolic-manifolds/cmp/1104269797.full>
- Relevance: This is the Rudnick-Sarnak source associated with QUE. It studies arithmetic hyperbolic manifolds and proves no strong localization (“scarring”) onto totally geodesic submanifolds in that arithmetic setting. It does not prove the full conjecture for all compact negatively curved manifolds.

2. problem source / exact Conjecture 8:

- Title: Equidistribution in homogeneous spaces and number theory
- Author: Elon Lindenstrauss
- Venue: Proceedings of the International Congress of Mathematicians 2010, Vol. I, Hindustan Book Agency, pp. 531-557
- Year: 2010
- URL: <https://math.huji.ac.il/~elon/Publications/ICM2010.pdf>
- Relevance: Section 4.8 states Conjecture 8 in the same form as this problem and notes that both the physical-space and stronger phase-space versions are open. It also records the arithmetic Hecke-eigenfunction theorem as a special case, not a solution of the general conjecture.

3. Quantum ergodicity, a density-one result:

- Title: Ergodic properties of eigenfunctions
- Author: A. I. Shnirelman
- Venue: Uspekhi Mat. Nauk 29, no. 6(180), 181-182
- Year: 1974
- URL: <https://www.mathnet.ru/eng/rm4463>
- Title: Ergodicite et fonctions propres du laplacien
- Author: Yves Colin de Verdiere
- Venue: Communications in Mathematical Physics 102, 497-502
- Year: 1985
- DOI/URL: <https://doi.org/10.1007/BF01209296>
- Title: Uniform distribution of eigenfunctions on compact hyperbolic surfaces

- Author: Steven Zelditch
- Venue: Duke Mathematical Journal 55, no. 4, 919-941
- Year: 1987
- DOI/URL: <https://doi.org/10.1215/S0012-7094-87-05546-3>
- Relevance: These works prove the Quantum Ergodicity theorem: if the geodesic flow is ergodic, then a density-one subsequence of any eigenbasis equidistributes. Negative sectional curvature implies ergodicity/Anosov dynamics, so this applies here. It is only a density-one statement and does not exclude exceptional subsequences, so it does not resolve QUE.

4. Arithmetic QUE special cases:

- Title: Invariant measures and arithmetic quantum unique ergodicity
- Author: Elon Lindenstrauss
- Venue: Annals of Mathematics 163, no. 1, 165-219
- Year: 2006
- DOI/URL: <https://doi.org/10.4007/annals.2006.163.165>
- Relevance: Proves arithmetic QUE for compact arithmetic hyperbolic surfaces, using Hecke symmetry and measure rigidity. The hypotheses require arithmetic structure and Hecke-Laplace eigenfunctions, so this does not cover arbitrary compact negatively curved manifolds or arbitrary eigenbases.
- Title: Quantum unique ergodicity for $SL_2(\mathbb{Z}) \backslash \mathbb{H}$
- Author: Kannan Soundararajan
- Venue: Annals of Mathematics 172, no. 2, 1529-1538
- Year: 2010
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2010.172.1529> ; arXiv:0901.4060
- Relevance: Eliminates escape of mass for Hecke-Maass forms on the modular surface; combined with Lindenstrauss this establishes QUE for the noncompact modular surface. This is not the compact general statement and still uses Hecke eigenfunctions.
- Title: Joint quasimodes, positive entropy, and quantum unique ergodicity
- Authors: Shimon Brooks and Elon Lindenstrauss
- Venue: Inventiones Mathematicae 198, no. 1, 219-259
- Year: 2014
- DOI/arXiv/URL: <https://doi.org/10.1007/s00222-014-0502-7> ; arXiv:1112.5311
- Relevance: Gives QUE for certain joint quasimodes/eigenfunctions involving Hecke operators on compact congruence surfaces. This again depends on arithmetic/Hecke structure and is not the unrestricted Rudnick-Sarnak conjecture.

5. Restrictions on exceptional quantum limits:

- Title: Entropy and the localization of eigenfunctions

- Author: Nalini Anantharaman
 - Venue: Annals of Mathematics 168, no. 2, 435-475
 - Year: 2008
 - DOI/URL: <https://doi.org/10.4007/annals.2008.168.435>
 - Relevance: For compact negatively curved, more generally Anosov, manifolds, Wigner measures of eigenfunctions have positive metric entropy. This rules out total concentration on a closed geodesic but does not force the Liouville measure.
 - Title: Half-delocalization of eigenfunctions for the Laplacian on an Anosov manifold
 - Authors: Nalini Anantharaman and Stephane Nonnenmacher
 - Venue: Annales de l'Institut Fourier 57, no. 7, 2465-2523
 - Year: 2007
 - DOI/arXiv/URL: <https://doi.org/10.5802/aif.2340> ; arXiv:math-ph/0610019
 - Relevance: Gives stronger entropy lower bounds, equal to half the maximal entropy in constant negative curvature. This is a major partial result but still permits many non-Liouville invariant measures.
 - Title: Semiclassical measures on hyperbolic surfaces have full support
 - Authors: Semyon Dyatlov and Long Jin
 - Venue: Acta Mathematica 220, no. 2, 297-339
 - Year: 2018
 - DOI/arXiv/URL: <https://doi.org/10.4310/ACTA.2018.v220.n2.a3> ; arXiv:1705.05019
 - Relevance: Shows every semiclassical measure on a compact hyperbolic surface has full support. Full support is weaker than being Liouville, so this is partial progress only.
 - Title: Control of eigenfunctions on surfaces of variable curvature
 - Authors: Semyon Dyatlov, Long Jin, and Stephane Nonnenmacher
 - Venue: Journal of the American Mathematical Society 35, no. 2, 361-465
 - Year: 2022
 - DOI/arXiv/URL: <https://doi.org/10.1090/jams/979> ; arXiv:1906.08923
 - Relevance: Extends the full-support theorem to compact negatively curved surfaces and more generally surfaces with Anosov geodesic flow. Again, full support does not imply equidistribution to normalized volume.
6. Recent surveys confirming the current status:
- Title: Recent progress on the quantum unique ergodicity conjecture
 - Author: Peter Sarnak
 - Venue: Bulletin of the American Mathematical Society 48, no. 2, 211-228
 - Year: 2011
 - DOI/URL: <https://doi.org/10.1090/S0273-0979-2011-01323-4>

- Relevance: Surveys arithmetic QUE breakthroughs and general microlocal progress; it treats the general Rudnick-Sarnak conjecture as unresolved.
- Title: Macroscopic limits of chaotic eigenfunctions
- Author: Semyon Dyatlov
- Venue: Proceedings of the International Congress of Mathematicians 2022, Vol. 5, EMS Press, pp. 3704-3723
- Year: 2023
- DOI/arXiv/URL: <https://doi.org/10.4171/ICM2022/22> ; arXiv:2109.09053
- Relevance: States the QUE conjecture for Anosov systems and says it is known in arithmetic hyperbolic surface cases with Hecke symmetry, but open in general despite entropy and full-support progress.
- Title: Quantum ergodicity and semiclassical measures: mathematical results
- Author: Stephane Nonnenmacher
- Venue: arXiv preprint
- Year: 2026
- arXiv/URL: arXiv:2606.12098 ; <https://arxiv.org/abs/2606.12098>
- Relevance: A current survey that formulates QUE for compact manifolds of negative curvature and states that the conjecture remains open, even for constant negative curvature, outside the arithmetic Hecke-eigenfunction setting.

4.11.3.4. What remains unresolved

No published or citable result found proves that every orthonormal sequence of Laplace eigenfunctions on every compact negatively curved Riemannian manifold satisfies the stated equidistribution limit. The unresolved part is exactly the removal of the density-one restriction and the removal of all extra arithmetic/Hecke/special-surface hypotheses. In particular, the literature still does not prove that the normalized Liouville/volume measure is the only possible quantum limit for arbitrary compact negatively curved manifolds.

4.11.4. Remaining open problems recorded by the database

4.11.4.1. Open problem 1

Let M be a compact hyperbolic surface. Prove that every sequence of L^2 -normalized Laplace eigenfunctions whose eigenvalues tend to infinity equidistributes in physical space, without assuming that the eigenfunctions are simultaneous Hecke eigenfunctions.

Reference supplied by the database. Ankit Bisain, Peter Humphries, Andrei Mandelsham, Noah Walsh, and Xun Wang, “Subconvexity Implies Effective Quantum Unique Ergodicity for Hecke–Maaß Cusp Forms on $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{R})$,” *Essential Number Theory* 3, no. 1 (2024), 101–148, §1A (explicitly identifies QUE for compact hyperbolic surfaces, without the arith-

metic Hecke restriction, as the Rudnick–Sarnak conjecture and as far from proved; no separate conjecture number). DOI: 10.2140/ent.2024.3.101. <https://doi.org/10.2140/ent.2024.3.101>

4.11.5. Primary-source pointer

The originating ICM record is *Equidistribution in Homogeneous Spaces and Number Theory* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=522>.

4.12. 93. Moduli spaces of manifolds [2014-ag-161]

Record: 2014-ag-161

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Moduli spaces of manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1217>

4.12.1. Problem statement

For $n = 1$, let $W_g = \#^g(S^1 \times S^1)$ be the closed oriented surface of genus g , and fix an embedded disk $D^2 \subset W_g$. For a smooth oriented W_g -bundle $\pi : E \rightarrow B$, with vertical tangent bundle $T_\pi E$ and Euler class $e(T_\pi E)$, define

$$\kappa_{e^k}(\pi) = \pi_! (e(T_\pi E)^k) \in H^{2k-2}(B; \mathbb{Q}), \quad k \geq 2.$$

Determine, for each genus g , all universal relations among the Miller-Morita-Mumford classes κ_{e^k} in the tautological subrings, equivalently determine the kernels of the natural homomorphisms

$$\mathbb{Q}[\kappa_{e^k} \mid k \geq 2] \rightarrow H^*(B \operatorname{Diff}(W_g, D^2); \mathbb{Q})$$

and

$$\mathbb{Q}[\kappa_{e^k} \mid k \geq 2] \rightarrow H^*(B \operatorname{Diff}^+(W_g); \mathbb{Q}),$$

where $\operatorname{Diff}(W_g, D^2)$ denotes diffeomorphisms restricting to the identity near D^2 .

4.12.2. Historical background, literature, and current progress

4.12.3. Status check

Verdict: partially resolved.

4.12.3.1. Precise problem

For the genus g oriented surface $W_g = \#^g(S^1 \times S^1)$, determine the full kernel of the natural map

$$\mathbb{Q}[\kappa_{e^k} \mid k \geq 2] \rightarrow H^*(B \operatorname{Diff}(W_g, D^2); \mathbb{Q})$$

and similarly for

$$\mathbb{Q}[\kappa_{e^k} \mid k \geq 2] \rightarrow H^*(B \operatorname{Diff}^+(W_g); \mathbb{Q}).$$

Equivalently, after identifying $B \operatorname{Diff}^+(W_g)$ with the orbifold moduli space $\mathcal{M}_g$ and κ_{e^k} with the usual surface κ_{k-1} classes, the question asks for all universal relations among the κ -classes in the tautological κ -subring, in every genus, including the disk-fixed/marked-disk version.

4.12.3.2. Literature and progress

1. Carel Faber, “A conjectural description of the tautological ring of the moduli space of curves”, in *Moduli of Curves and Abelian Varieties, Aspects of Mathematics E33, Vieweg, 1999*, pp. 109-129. arXiv:math/9711218. URL: <https://arxiv.org/abs/math/9711218> This is the original source of the all-genus conjectural description of $R^*(M_g)$. The arXiv abstract states that the paper formulates conjectures giving a rather complete description of the tautological ring of M_g . This is the source of the conjectural all-genus answer, not a proof of the full all-genus relations.
2. Eduard Looijenga, “On the tautological ring of M_g ”, *Inventiones Mathematicae* 121 (1995), 411-419. arXiv:alg-geom/9501010. URL: <https://arxiv.org/abs/alg-geom/9501010> Looijenga proved the key vanishing/socle result: tautological products in degree $d > g - 2$ vanish and degree $g - 2$ is at most one-dimensional. This is major structural progress, but it does not determine all relations among the κ -classes.
3. Carel Faber, “Tautological algebras of moduli spaces of curves”, in *Moduli Spaces of Riemann Surfaces, IAS/Park City Mathematics Series 20, AMS/IAS, 2013*, pp. 197-219. URL: <https://www.staff.science.uu.nl/~faber010/faber.jc pB.pdf> Faber’s survey records the state of the conjecture: large parts are proved, the complete conjecture was proved computationally for $g \leq 23$, and the relations used for $g \leq 23$ appear insufficient to prove the Gorenstein prediction in genus 24. This supports the stated “known up to $g \leq 23$ ” statement but also confirms that the all-genus problem was not settled there.
4. Qizheng Yin, “On the tautological rings of $M_{g,1}$ and its universal Jacobian”, arXiv:1206.3783, 2012. URL: <https://arxiv.org/abs/1206.3783> Yin gives additional relations for the one-pointed case and proves Faber-type conjectures for $M_{g,1}$ up to $g = 19$, and by pushing forward obtains another proof for M_g up to $g = 23$. This is partial evidence for the disk-fixed/marked version, not an all-genus solution.
5. R. Pandharipande and A. Pixton, “Relations in the tautological ring of the moduli space of curves”, *Pure and Applied Mathematics Quarterly* 17 (2021), no. 2, 717-771. arXiv:1301.4561. URL: <https://arxiv.org/abs/1301.4561> DOI: <https://doi.org/10.48550/arXiv.1301.4561> This paper proves the Faber-Zagier relations in the Chow tautological ring of M_g . It explicitly defines the kernel problem for $\mathbb{Q}[\kappa_1, \kappa_2, \dots] \rightarrow R^*(M_g)$, proves that the Faber-Zagier expressions are genuine relations, and states as Conjecture 2 that the Faber-Zagier ideal is the entire kernel for all g . The same paper says that whether additional cohomological relations exist is not settled and that no non-Faber-Zagier relation has been found. Therefore it proves a large, central family of relations but not completeness.
6. R. Pandharipande, A. Pixton, and D. Zvonkine, “Relations on $\overline{M}_{g,n}$ via 3-spin structures”, *Journal of the American Mathematical Society* 28 (2015), 279-309. arXiv:1303.1043. URL: <https://arxiv.org/abs/1303.1043> This proves Pixton’s generalized relations in cohomology for stable pointed curves. It includes the Faber-Zagier relations on M_g as a special/open-interior shadow, but it does not prove that these relations are complete.
7. Felix Janda, “Relations on $\overline{M}_{g,n}$ via equivariant Gromov-Witten theory of $\mathbb{P}^1$ ”, *Algebraic Geometry* 4 (2017), 311-336. URL: <https://algebraicgeometry.nl/2017-4/2017-4-006.pdf> DOI: 10.14231/AG-2017-006 Janda proves Pixton’s relations in Chow using stable maps to $\mathbb{P}^1$. This strengthens validity of the known relations but still does not establish that they generate all relations.

8. Samir Canning, “The tautological ring of $\overline{M}_{g,n}$ is rarely Gorenstein”, *Geometry & Topology* 29 (2025), no. 7, 3905–3919; arXiv:2406.10516. URL: <https://arxiv.org/abs/2406.10516> DOI: 10.2140/gt.2025.29.3905 This recent work concerns the compactified spaces $\overline{M}_{g,n}$, proving non-Gorenstein results when $g \geq 2$ and $2g + n \geq 24$, and explicitly states that finding a complete set of relations among tautological generators remains a fundamental open problem. Its hypotheses and conclusion are not the same as the smooth κ -subring problem considered here.

4.12.3.3. Why this is not fully resolved

The strongest directly relevant all-genus theorem proves that the Faber-Zagier relations are true relations. The missing step is completeness: no published or otherwise citable result found proves that the Faber-Zagier ideal, or any other explicit ideal, is exactly the full kernel of the κ -class map for all genera in rational cohomology of $B\text{Diff}^+(W_g)$ and $B\text{Diff}(W_g, D^2)$.

The literature repeatedly formulates completeness as conjectural: Pandharipande-Pixton state the all-genus kernel equality as a conjecture, and later work on Pixton relations proves validity rather than completeness. The known complete computations for small genus, especially $g \leq 23$ for M_g , and the proven Faber-Zagier/Pixton relations are substantial progress. The all-genus universal relation problem remains unresolved.

4.12.4. Remaining open problems recorded by the database

4.12.4.1. Open problem 1

For every genus $g \geq 2$ and every $0 \leq d \leq g - 2$, is the multiplication pairing

$$R^d(M_g) \times R^{g-2-d}(M_g) \longrightarrow R^{g-2}(M_g) \cong \mathbb{Q}$$

perfect (equivalently, is $R^*(M_g)$ Gorenstein with socle in degree $g - 2$)?

Reference supplied by the database. Carel Faber, “A Conjectural Description of the Tautological Ring of the Moduli Space of Curves,” in Carel Faber and Eduard Looijenga (eds.), *Moduli of Curves and Abelian Varieties*, Aspects of Mathematics E33, Vieweg, 1999, pp. 109–129, Conjecture 1 (perfect-pairing/Gorenstein part). DOI: 10.1007/978-3-322-90172-9_6. https://doi.org/10.1007/978-3-322-90172-9_6; arXiv:math/9711218, <https://arxiv.org/abs/math/9711218>

4.12.4.2. Open problem 2

For every genus $g \geq 2$ and every $0 \leq d \leq g - 1$, is the multiplication pairing

$$R^d(M_{g,1}) \times R^{g-1-d}(M_{g,1}) \longrightarrow R^{g-1}(M_{g,1}) \cong \mathbb{Q}$$

perfect (equivalently, is $R^*(M_{g,1})$ Gorenstein with socle in degree $g - 1$)?

Reference supplied by the database. Qizheng Yin, “On the Tautological Rings of $\mathcal{M}_{g,1}$ and Its Universal Jacobian,” preprint, first posted 2012 and revised 2026, Conjecture 3.1(i) (Faber’s conjecture for $\mathcal{M}_{g,1}$); arXiv:1206.3783. <https://arxiv.org/abs/1206.3783>

4.12.4.3. Open problem 3

Is the tautological ring $R^*(\mathcal{M}_{24})$ not Gorenstein?

Reference supplied by the database. Aaron Pixton, *The Tautological Ring of the Moduli Space of Curves*, Ph.D. dissertation, Princeton University, 2013, Appendix A, Conjecture A.1 (the $\mathcal{M}_{24}$ case), ProQuest LLC, MR 3153424. <https://math.mit.edu/~apixton/papers/tautological.pdf>

4.12.5. Primary-source pointer

The originating ICM record is *Moduli spaces of manifolds* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1217>.

4.13. 94. Algebraic surfaces with minimal Betti numbers [2018-ag-177]

Record: 2018-ag-177

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Algebraic surfaces with minimal Betti numbers

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=784>

4.13.1. Problem statement

Every faithful smooth action of the circle group S^1 on the 5-sphere S^5 that is pseudofree, meaning that it is free except for finitely many orbits whose isotropy groups are nontrivial finite cyclic groups, has at most three exceptional orbits, where the exceptional orbits are precisely those non-free orbits with finite cyclic isotropy.

4.13.2. Historical background, literature, and current progress

4.13.3. Verdict: partially resolved

4.13.3.1. Precise problem checked

The problem asks whether every faithful smooth pseudofree circle action

$$S^1 \times S^5 \rightarrow S^5$$

has at most three exceptional/non-free orbits, where pseudofree means free away from finitely many orbits with nontrivial finite cyclic isotropy. Equivalently, in the standard orbit-space formulation, it asks whether the associated simply connected smooth compact 4-manifold with $H_2 \cong \mathbb{Z}$ can have boundary a connected sum of more than three lens spaces with pairwise coprime isotropy orders.

4.13.3.2. Status

I did not find any published or otherwise citable theorem proving the full unrestricted smooth statement. The literature contains substantial partial progress, especially for algebraic orbit spaces and for additional geometric/topological hypotheses, but the full smooth pseudofree S^1 -action problem on S^5 remains unresolved.

The verdict is therefore **partially resolved**, not fully resolved: there are citable special-case and obstruction results, but they do not cover all smooth pseudofree circle actions on S^5 .

4.13.3.3. Key literature and what it proves

1. Deane Montgomery and C. T. Yang, "Differentiable pseudo-free circle actions on homotopy seven spheres," *Proceedings of the Second Conference on Compact Transformation Groups, Lecture Notes in Mathematics 298*, Springer, 1972, pp. 41-101. DOI: 10.1007/BFb0070033.

This is one of the sources motivating the problem: in dimension 7 and higher odd dimensions there are pseudofree circle actions with arbitrarily many exceptional orbits. It does not resolve the S^5 case.

2. Ronald Fintushel and Ronald J. Stern, "Pseudofree orbifolds," *Annals of Mathematics* (2) 122(2), 1985, pp. 335-364. DOI: 10.2307/1971306.

Develops the pseudofree orbifold/4-manifold viewpoint used in later formulations. It supplies background and tools, not a full proof of the Montgomery-Yang bound.

3. Ronald Fintushel and Ronald J. Stern, "O(2) actions on the 5-sphere," *Inventiones Mathematicae* 87(3), 1987, pp. 457-476. DOI: 10.1007/BF01389237.

This is the standard cited source for the Montgomery-Yang problem in this form. Kollár later notes that the proposed partial solution in this paper is incorrect, citing Math Reviews MR0874031; therefore it cannot be counted as a full resolution.

4. János Kollár, "Is there a topological Bogomolov-Miyaoka-Yau inequality?," *Pure and Applied Mathematics Quarterly* 4(2), 2008, pp. 203-236. DOI: 10.4310/PAMQ.2008.v4.n2.a1. arXiv: math/0602562.

Kollár states the Montgomery-Yang problem for circle actions on S^5 , gives the 4-manifold/lens-space correspondence, and formulates algebraic analogues. He explicitly records that the Fintushel-Stern proposed partial solution is incorrect. This paper frames the problem and related conjectures but does not prove the full statement.

5. Jin Hong Kim, "The number of exceptional orbits of a pseudofree circle action on S^5 ," *Journal of Fixed Point Theory and Applications* 9, 2011, pp. 55-62. DOI: 10.1007/s11784-010-0039-3.

Proves bounds under additional Riemannian hypotheses: for pseudofree isometric S^1 -actions on nonnegatively curved 5-manifolds, and in particular on S^5 , the number of exceptional circle orbits is at most five under the paper's hypotheses. This is related progress but weaker than "at most three" and assumes extra geometry.

6. DongSeon Hwang and JongHae Keum, "The maximum number of singular points on rational homology projective planes," *Journal of Algebraic Geometry* 20(3), 2011, pp. 495-523. DOI: 10.1090/S1056-3911-10-00532-1.

Gives algebraic bounds for quotient singularities on rational homology projective planes. This supports the algebraic Montgomery-Yang program but does not cover arbitrary smooth orbit spaces.

7. DongSeon Hwang and JongHae Keum, "Algebraic Montgomery-Yang problem: the non-cyclic case," *Mathematische Annalen* 350(3), 2011, pp. 721-754. DOI: 10.1007/s00208-010-0565-8. arXiv: 0904.2975.

Proves the algebraic Montgomery-Yang conjecture when the rational homology projective plane has at least one non-cyclic singularity. This is a genuine special case of the algebraic analogue, not a solution of the full smooth S^5 action problem.

8. DongSeon Hwang and JongHae Keum, "Algebraic Montgomery-Yang problem: the

nonrational surface case,” *Michigan Mathematical Journal* 62(1), 2013, pp. 3-37. DOI: 10.1307/mmj/1363958239.

Solves another algebraic special case, namely nonrational surfaces. It leaves the rational cyclic ample-canonical case outside its scope.

9. **DongSeon Hwang and JongHae Keum, “Algebraic Montgomery-Yang problem: the log del Pezzo surface case,”** *Journal of the Mathematical Society of Japan* 66(4), 2014, pp. 1073-1089. DOI: 10.2969/jmsj/06641073.

Proves the algebraic Montgomery-Yang statement for log del Pezzo surfaces of Picard number one. Again, this is a special algebraic setting.

10. **DongSeon Hwang, “Algebraic Montgomery-Yang problem and cascade conjecture,”** *arXiv preprint*, 2020/2021. arXiv: 2012.13355.

States that the algebraic Montgomery-Yang problem remains open for rational $\mathbb{Q}$ -homology projective planes with cyclic quotient singularities and ample canonical divisor. It proves the algebraic conjecture assuming the additional cascade conjecture, so the result is conditional.

11. **Woohyeok Jo, Jongil Park, and Kyungbae Park, “Algebraic Montgomery-Yang problem and smooth obstructions,”** *Transactions of the American Mathematical Society* 378(4), 2025, pp. 2969-3003. DOI: 10.1090/tran/9364. arXiv: 2402.04569.

This is the strongest recent progress found. It states that the algebraic Montgomery-Yang problem has not been completely resolved, summarizes known partial results, formulates a smooth 4-manifold obstruction problem, and rules out many four-singularity types using Donaldson theory, Heegaard Floer correction terms, and related conditions. The paper explicitly says that infinitely many singularity types are still not ruled out by its conditions, so it does not resolve the full problem.

12. **Jihao Liu and Wenfei Liu, “The minimal volume of stable surfaces of rank one,”** *arXiv preprint*, 2026. arXiv: 2605.05641.

This very recent preprint applies a plurigenus filter to Kollár’s algebraic Montgomery-Yang problem after the Jo-Park-Park reductions. It reports that Jo-Park-Park leave 16 values of $p \leq 50000$ to be excluded in a remaining algebraic family and that the new filter excludes 8 of those 16. This is additional partial progress in the algebraic analogue, not a full resolution of either the algebraic problem or the original unrestricted smooth S^5 problem.

4.13.3.4. What remains unresolved

No cited result proves that every faithful smooth pseudofree S^1 -action on the standard S^5 has at most three exceptional orbits. Existing results either:

- treat algebraic/projective orbit spaces rather than arbitrary smooth orbit spaces;
- impose extra assumptions such as non-cyclic singularities, nonrationality, log del Pezzo structure, cascade existence, bounded/computational singularity families, curvature, or isometric action;
- prove a weaker numerical bound such as at most five exceptional orbits; or
- rule out many candidate boundary/singularity types without excluding all possible cases.

Thus the exact statement is not fully resolved by the current literature.

4.13.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

4.13.5. Primary-source pointer

The originating ICM record is *Algebraic surfaces with minimal Betti numbers* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=784>.

4.14. 95. Lattice subgroups acting on manifolds [2022-geom-165]

Record: 2022-geom-165

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Lattice subgroups acting on manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=33>

4.14.1. Problem statement

Let G be a connected noncompact simple Lie group of real rank at least 2, meaning that the dimension of a maximal $\mathbb{R}$ -split abelian subalgebra of the Lie algebra of G is at least 2. Let $\Gamma \subset G$ be a lattice, meaning a discrete subgroup such that G/Γ has finite invariant volume. Let M be a compact manifold, and let vol be a smooth volume form on M . Fix a differentiability regularity class, write $\text{Diff}(M)$ for the diffeomorphism group of M in that regularity class, and write $\text{Diff}_{\text{vol}}(M)$ for the subgroup of diffeomorphisms preserving vol .

Define

$$v(G) = \min\{\dim(G/H) : H \subset G \text{ is a proper closed subgroup}\},$$

let $n(G)$ be the minimal dimension of a nontrivial real linear representation of the Lie algebra of G , and define

$$d(G) = \min\{\dim(G_{\text{cpt}}/H) : H \subset G_{\text{cpt}} \text{ is a proper closed subgroup}\},$$

where G_{cpt} denotes the connected compact Lie group whose Lie algebra is a compact real form of the complexification of the Lie algebra of G .

Prove that the following assertions hold:

1. If $\dim(M) < \min\{n(G), d(G), v(G)\}$, then every homomorphism $\alpha : \Gamma \rightarrow \text{Diff}(M)$ has finite image.
2. If $\dim(M) < \min\{n(G), d(G)\}$, then every homomorphism $\alpha : \Gamma \rightarrow \text{Diff}_{\text{vol}}(M)$ has finite image.
3. If $\dim(M) < \min\{v(G), n(G)\}$, then for every homomorphism $\alpha : \Gamma \rightarrow \text{Diff}(M)$, there exists an $\alpha(\Gamma)$ -invariant Riemannian metric on M , at least a C^0 Riemannian metric.
4. If $\dim(M) < n(G)$, then for every homomorphism $\alpha : \Gamma \rightarrow \text{Diff}_{\text{vol}}(M)$, there exists an $\alpha(\Gamma)$ -invariant Riemannian metric on M , at least a C^0 Riemannian metric.

4.14.2. Historical background, literature, and current progress

4.14.3. Verdict: partially resolved

4.14.3.1. Problem identified

The problem asks for the general Zimmer/Farb-Shalen low-dimensional action conjecture for a lattice Γ in a connected noncompact simple Lie group G of real rank at least 2. In the notation of the problem, it asks whether actions of Γ on compact manifolds below the critical dimensions defined by $n(G)$, $d(G)$, and $v(G)$ must either have finite image or preserve a Riemannian metric, with separate volume-preserving and non-volume-preserving versions.

4.14.3.2. Status summary

This exact general statement is not fully resolved. There is major progress, including sharp results for real-split simple groups and many non-split families, but I found no published or standard arXiv-citable theorem covering all connected noncompact simple real-rank-at-least-2 Lie groups and all four clauses at the stated bounds. Recent non-split work still states results for specified families and weaker bounds in some cases.

4.14.3.3. Key literature and progress

1. **Robert J. Zimmer, “Actions of semisimple groups and discrete subgroups.”**
 Venue/year: Proceedings of the International Congress of Mathematicians, Berkeley, 1986, American Mathematical Society, 1987, pp. 1247-1258.
 DOI/arXiv/URL: no DOI found; bibliographic URL: <https://mathscinet.ams.org/mathscinet-getitem?mr=934298>
 Role: original Zimmer-program source for rigidity questions about actions of higher-rank lattices on manifolds.
2. **Aaron Brown, “Lattice subgroups acting on manifolds.”**
 Venue/year: Proceedings of the International Congress of Mathematicians 2022, Vol. 5, EMS Press, 2023, pp. 3388-3411.
 DOI/arXiv/URL: DOI 10.4171/ICM2022/163; <https://ems.press/books/standalone/277/5519>
 Progress: survey/source for the formulation and current Zimmer-program landscape. It discusses low-dimensional existence questions under the name Zimmer’s conjecture, but it is not itself a full proof of the general statement.
3. **Aaron Brown, David Fisher, Sebastian Hurtado, “Zimmer’s conjecture: Subexponential growth, measure rigidity, and strong property (T).”**
 Venue/year: Annals of Mathematics (2) 196 (2022), no. 3, 891-940.
 DOI/arXiv/URL: DOI 10.4007/annals.2022.196.3.1; arXiv:1608.04995; <https://annals.math.princeton.edu/2022/196-3/p01>
 Progress: states the full Zimmer/Farb-Shalen conjecture and proves several major cocompact-lattice cases. It proves the non-volume-preserving case for cocompact lattices in higher-rank split simple Lie groups and certain volume-preserving cases, but explicitly gives only partial results for non-split groups.
4. **Aaron Brown, David Fisher, Sebastian Hurtado, “Zimmer’s conjecture for actions of $SL(m, \mathbb{Z})$.”**

Venue/year: *Inventiones Mathematicae* 221 (2020), no. 3, 1001-1060.

DOI/arXiv/URL: DOI 10.1007/s00222-020-00962-x; arXiv:1710.02735.

Progress: proves Zimmer's conjecture for C^2 actions by finite-index subgroups of $SL(m, \mathbb{Z})$ for $m > 3$. This is a sharp and important nonuniform arithmetic case, but it does not cover arbitrary simple G .

5. **Aaron Brown, David Fisher, Sebastian Hurtado, "Zimmer's conjecture for non-uniform lattices: escape of mass and growth of cocycles."**

Venue/year: arXiv preprint, 2021; current version v3, 2024.

DOI/arXiv/URL: arXiv:2105.14541; DOI 10.48550/arXiv.2105.14541; <https://arxiv.org/abs/2105.14541>

Progress: extends the BFH method to nonuniform lattices. Theorem A proves the expected bounds for lattices in connected $\mathbb{R}$ -split simple Lie groups of real rank at least 2. Theorem B gives broader bounds using the minimal resonant codimension $r(G)$. For non- $\mathbb{R}$ -split groups, $r(G)$ can be strictly smaller than $v(G)$, so this is partial for the full statement.

6. **Aaron Brown, Danijela Damjanović, Zhiyuan Zhang, " C^1 actions on manifolds by lattices in Lie groups."**

Venue/year: *Compositio Mathematica* 158 (2022), no. 3, 529-549.

DOI/arXiv/URL: DOI 10.1112/S0010437X22007278; arXiv:1801.04009; <https://www.cambridge.org/core/journals/compositio-mathematica/article/c1-actions-on-manifolds-by-lattices-in-lie-groups/05C8DE5533527E5AEE7764FDE4D40897>

Progress: obtains C^1 -regularity results when the rank of a uniform lattice is larger than the manifold dimension, with sharp bounds for lattices in $SL(n, \mathbb{R})$. These are important low-regularity results but not a full resolution of the general $n(G), d(G), v(G)$ statement.

7. **Jinpeng An, Aaron Brown, Zhiyuan Zhang, "Zimmer's conjecture for non-split semisimple Lie groups."**

Venue/year: arXiv preprint, 2024.

DOI/arXiv/URL: arXiv:2411.13858; DOI 10.48550/arXiv.2411.13858; <https://arxiv.org/abs/2411.13858>

Progress: proves many new non- $\mathbb{R}$ -split cases. Theorem 1.3 covers connected complex semisimple groups without rank-one simple factors and specified families including $SL(n, \mathbb{H})$, $SO^+(m, n)$, $SU(m, n)$, $SO^*(2n)$, and EII , with stated parameter ranges; it also gives weaker $v(G) - 2$ bounds for some $Sp(m, n)$. The paper's own formulation is "many new cases", not all cases, so it does not fully resolve the problem statement.

4.14.3.4. What remains unresolved

The unresolved part is the full non- $\mathbb{R}$ -split generality: all connected noncompact simple Lie groups of real rank at least 2, all lattices, and all four finite-image/invariant-metric conclusions at the exact critical dimensions involving $n(G)$, $d(G)$, and $v(G)$. Current literature covers the real-split case and many non-split families, but not every simple real form and not every volume-preserving/invariant-metric clause at the stated sharp bounds.

4.14.4. Remaining open problems recorded by the database

4.14.4.1. Open problem 1

Let Γ be a lattice in a connected Lie group whose Lie algebra is $\mathfrak{so}(n, n+1)$ for $n \geq 3$, or $\mathfrak{so}(n, n)$ for $n \geq 4$. Prove that every sufficiently smooth volume-preserving action of Γ on a compact manifold of dimension $n(\mathfrak{g}) - 1$ has finite image (thus, respectively, in dimensions $2n$ and $2n - 1$).

Reference supplied by the database. Aaron Brown, Federico Rodriguez Hertz, and Zhiren Wang, “Boundary Actions by Higher-Rank Lattices: Classification and Embedding in Low Dimensions, Local Rigidity, Smooth Factors,” 2024, §2.8, especially the paragraph following Theorem 2.17 (explicitly identifying this as the missing dimension in the volume-preserving Zimmer conjecture and reducing it to ruling out an embedded projective subaction), arXiv: [2405.16202](https://arxiv.org/abs/2405.16202).

4.14.4.2. Open problem 2

Let $G = \mathrm{Sp}(m, n)$ with $6 \leq n \leq m \leq (n^2 - 3n + 6)/4$, and let $\Gamma < G$ be a lattice. If M is a compact manifold of either of the two still-uncovered dimensions

$$\dim M = v(G) - 2 = 4m + 4n - 7 \quad \text{or} \quad \dim M = v(G) - 1 = 4m + 4n - 6,$$

must every $C^{1+\varepsilon}$ action $\Gamma \rightarrow \mathrm{Diff}(M)$ have finite image?

Reference supplied by the database. Jinpeng An, Aaron Brown, and Zhiyuan Zhang, “Zimmer’s Conjecture for Non-Split Semisimple Lie Groups,” 2024, Conjecture 1.1 (non-volume-preserving Zimmer conjecture), the paragraph preceding Conjecture 1.2, and Theorem 3.12 (which proves only the weaker range $\dim M < v(G) - 2$ for the displayed $\mathrm{Sp}(m, n)$ family). arXiv: [2411.13858](https://arxiv.org/abs/2411.13858).

4.14.5. Primary-source pointer

The originating ICM record is *Lattice subgroups acting on manifolds* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=33>.

4.15. 96. Rigidity, lattices, and invariant measures beyond homogeneous dynamics [2022-geom-168]

Record: 2022-geom-168

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Rigidity, lattices, and invariant measures beyond homogeneous dynamics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=129>

4.15.1. Problem statement

Let G be a simple Lie group of real rank at least 2, let $\Gamma < G$ be a lattice, let M be a compact manifold, and let $\rho : \Gamma \rightarrow \text{Diff}(M)$ be an action. Suppose that either $\rho(\Gamma)$ preserves a rigid geometric structure on M , or there exists $\gamma \in \Gamma$ such that the diffeomorphism $f = \rho(\gamma)$ admits a dominated splitting of TM , meaning a nontrivial continuous Df -invariant splitting $TM = E \oplus F$ for which there are constants $K > 0$ and $0 < \lambda < 1$ such that

$$\|Df^n|_{E_x}\| \|(Df^n|_{F_x})^{-1}\| \leq K\lambda^n$$

for every $x \in M$ and every $n \geq 0$. Prove that the action ρ is generalized quasiasfine. Here generalized quasiasfine means that, after identifying M equivariantly with a skew-product space, the Γ -action has the following form. There is a collection of data consisting of a topological group A , a closed subgroup $B < A$, a compact group C of affine diffeomorphisms of A/B , an affine action of Γ on A/B commuting with the action of C , a compact Riemannian manifold F , and a C^1 cocycle

$$\alpha : \Gamma \times C \backslash A/B \rightarrow \text{Isom}(F),$$

such that Γ acts on $C \backslash A/B \times F$ by

$$\gamma \cdot (C[a], y) = (C(\gamma \cdot [a]), \alpha(\gamma, C[a])y).$$

An affine diffeomorphism of A/B means that each acting element d sends $[a]$ to $[t_d L_d(a)]$, where $t_d \in A$ and L_d is a continuous automorphism of A .

4.15.2. Historical background, literature, and current progress

4.15.3. Verdict: partially resolved

4.15.3.1. Precise problem

The problem is Fisher's Conjecture 2.10 from the Zimmer-program literature. It asks whether every action $\rho : \Gamma \rightarrow \text{Diff}(M)$ of a lattice Γ in a simple Lie group G of real rank at least 2 on a compact manifold M is generalized quasiasfine, assuming either:

1. $\rho(\Gamma)$ preserves a rigid geometric structure on M , or
2. some element $\rho(\gamma)$ has a nontrivial dominated splitting of TM .

Here generalized quasiasfine is the skew-product/compact-quotient affine model described in the statement.

4.15.3.2. Status

I found no published or citable theorem proving this full statement for arbitrary higher-rank simple G , arbitrary lattice $\Gamma < G$, arbitrary compact M , and either of the two stated hypotheses. The problem is not fully open, because there is substantial special-case progress, but the exact conjecture remains unresolved.

4.15.3.3. Original source

Title: Rigidity, lattices, and invariant measures beyond homogeneous dynamics

Author: David Fisher

Venue: Proceedings of the International Congress of Mathematicians 2022, EMS Press, 2023; also arXiv preprint

Year: 2022/2023

arXiv/URL: arXiv:2111.14922, <https://arxiv.org/abs/2111.14922>; EMS URL: <https://ems.press/content/book-chapter-files/33258>

Relevant item: Conjecture 2.10 states exactly this rigid-structure/dominated-splitting-to-generalized-quasiasfine problem. Immediately after the conjecture, Fisher says the best evidence for the dynamical form is Homin Lee's work in dimension n , while the geometric form has more evidence going back to Zimmer but "the question remains mostly open." This source therefore does not resolve the problem statement; it identifies it as an open conjecture with partial evidence.

4.15.3.4. Partial progress

Title: Global rigidity of higher rank lattice actions with dominated splitting

Author: Homin Lee

Venue: Ergodic Theory and Dynamical Systems, 44(3), 799-828

Year: 2024

DOI/arXiv/URL: DOI 10.1017/etds.2023.31; arXiv:2010.11874, <https://arxiv.org/abs/2010.11874>

What it proves: For volume-preserving smooth actions of lattices in $SL(n, \mathbb{R})$, $n \geq 3$, on closed n -manifolds, if one element admits a dominated splitting, then the manifold is the flat n -torus and the action is smoothly conjugate to an affine action. Analogous results are given for lattices in $Sp(2n, \mathbb{R})$ and $SO(n, n)$ on closed $2n$ -manifolds.

Why it is only partial: It assumes volume preservation, restricts the ambient groups and dimensions, and concludes standard affine behavior in those cases. It does not cover arbitrary simple real-rank-at-least-2 G , arbitrary compact M , or the rigid-geometric-structure alternative.

Title: Positive entropy actions by higher-rank lattices

Authors: Aaron Brown and Homin Lee

Venue: arXiv preprint

Year: 2024, revised 2025

DOI/arXiv/URL: arXiv:2409.05991, <https://arxiv.org/abs/2409.05991>

What it proves: For smooth actions by lattices in higher-rank simple Lie groups, assuming one element has positive topological entropy, the paper proves rigidity results. In particular, for lattices in $\mathrm{SL}(n, \mathbb{R})$ acting on n -manifolds, positive entropy forces the lattice to be commensurable with $\mathrm{SL}(n, \mathbb{Z})$, and such actions admit an absolutely continuous probability measure with positive metric entropy and are measurably conjugate to affine actions on infra-tori.

Why it is only partial: Positive entropy is not the same as the stated dominated-splitting or rigid-structure hypotheses, the main stated n -manifold conclusion is for $\mathrm{SL}(n, \mathbb{R})$, and the conclusion is measurable/structural rather than the full generalized quasiaffine classification in all cases.

Title: Partially hyperbolic lattice actions on 2-step nilmanifolds

Authors: Homin Lee and Sven Sandfeldt

Venue: arXiv preprint

Year: 2024

DOI/arXiv/URL: arXiv:2410.00784, <https://arxiv.org/abs/2410.00784>

What it proves: Global rigidity for higher-rank lattice actions on tori or 2-step nilmanifolds containing a partially hyperbolic element with one-dimensional center, under a technical assumption on the partially hyperbolic element; the action is forced to be affine in that setting.

Why it is only partial: It is restricted to tori/2-step nilmanifolds, partially hyperbolic elements with one-dimensional center, and an extra technical assumption. It does not cover general compact manifolds or the rigid-geometric-structure alternative.

Title: Global smooth and topological rigidity of hyperbolic lattice actions

Authors: Aaron Brown, Federico Rodriguez Hertz, and Zhiren Wang

Venue: Annals of Mathematics, 186(3), 913-972

Year: 2017

DOI/arXiv/URL: DOI 10.4007/annals.2017.186.3.3; arXiv:1512.06720, <https://arxiv.org/abs/1512.06720>

What it proves: For smooth actions of higher-rank lattices on compact nilmanifolds, if the action lifts to the universal cover and its linear data contains a hyperbolic element, there is a semiconjugacy to the linear data; with an Anosov element, the semiconjugacy becomes a smooth conjugacy.

Why it is only partial: It assumes nilmanifold topology, lifting/linear-data hypotheses, and hyperbolicity/Anosov-type behavior; it is evidence for the dominated-splitting branch but not a proof of the general conjecture.

Title: Rigidity of weakly hyperbolic actions of higher real rank semisimple Lie groups and their lattices

Authors: Gregory A. Margulis and Nantian Qian

Venue: Ergodic Theory and Dynamical Systems, 21(1), 121-164

Year: 2001

DOI/URL: DOI 10.1017/S0143385701001109

What it proves: Local rigidity for standard algebraic actions under weak hyperbolicity conditions, and C^0 global rigidity for volume-preserving higher-rank lattice Anosov actions on nilmanifolds with a finite orbit.

Why it is only partial: It treats standard/nilmanifold/Anosov settings with additional hypotheses and does not classify arbitrary compact-manifold actions with a dominated splitting as generalized quasiaffine.

Title: Invariant Rigid Geometric Structures and Smooth Projective Factors

Authors: Amos Nevo and Robert J. Zimmer

Venue: Geometric and Functional Analysis, 19, 520-535

Year: 2009

DOI/arXiv/URL: DOI 10.1007/s00039-009-0005-7; arXiv:0708.0766, <https://arxiv.org/abs/0708.0766>

What it proves: For actions of noncompact simple Lie groups preserving analytic rigid geometric structures of algebraic type on compact manifolds, under a full-support stationary-measure hypothesis, either there is a smooth equivariant map to a homogeneous projective variety on an open dense conull set or a Gromov fundamental-group representation has Zariski closure containing the acting Lie algebra.

Why it is only partial: It concerns connected Lie group actions rather than the full lattice-action conjecture as stated, assumes analyticity/algebraic type and full-support stationary measures, and gives a projective-factor/fundamental-group alternative rather than generalized quasiaffine classification.

Title: Rigid Geometric Structures and Representations of Fundamental Groups

Author: David Fisher

Venue: Rigidity in Dynamics and Geometry, Springer, pp. 135-147

Year: 2002

DOI/URL: DOI 10.1007/978-3-662-04743-9_6; https://link.springer.com/chapter/10.1007/978-3-662-04743-9_6

What it proves: Under analytic, ergodic, volume-preserving, rigid-geometric-structure hypotheses, a higher-rank lattice action is either isometric with finite $\pi_1(M)$, or $\pi_1(M)$ has a large linear representation; related results describe zero-entropy extensions of standard arithmetic examples.

Why it is only partial: These are strong restrictions under additional hypotheses, not a classification of all rigid-structure-preserving lattice actions as generalized quasiaffine.

Title: Rigid geometric structures and actions of semisimple Lie groups

Author: Renato Feres

Venue: Rigidite, groupe fondamental et dynamique, Panoramas et Syntheses 13, Societe Mathematique de France, pp. 121-167

Year: 2002

URL: <https://www.math.wustl.edu/~feres/strassbourg.pdf>

What it provides: A survey and framework for rigid geometric structures, including results of Zimmer/Gromov giving algebraic-hull, local-symmetry, and fundamental-group restrictions for semisimple actions preserving rigid structures.

Why it is only partial: It supplies restrictions and background machinery, not the full generalized quasiaffine conclusion for all higher-rank lattice actions in the problem statement.

Title: Lattices in semisimple groups and distal geometric structures

Author: Robert J. Zimmer

Venue: Inventiones Mathematicae, 80, 123-137

Year: 1985

DOI/URL: DOI 10.1007/BF01388551

What it proves: Early Zimmer-program rigidity results for lattice actions preserving distal geometric structures.

Why it is only partial: It covers a special class of geometric structures/actions and does not establish the full generalized quasiaffine classification.

Title: On the automorphism group of a compact Lorentz manifold and other geometric manifolds

Author: Robert J. Zimmer

Venue: Inventiones Mathematicae, 83, 411-424

Year: 1986

DOI/URL: DOI 10.1007/BF01394415

What it proves: Restrictions on automorphism groups of compact Lorentz and related geometric manifolds.

Why it is only partial: It is important evidence for the rigid-geometric-structure branch, but it is not a full classification theorem for arbitrary higher-rank lattice actions as generalized quasilinear.

4.15.3.5. What remains unresolved

The unresolved part is exactly the broad statement in the problem statement: without dimension restrictions, volume preservation, nilmanifold/topological assumptions, Anosov/partial-hyperbolicity strengthening, analytic/full-support stationary-measure assumptions, or restrictions to special ambient groups, prove that every higher-rank lattice action preserving a rigid geometric structure or containing an element with a dominated splitting is generalized quasilinear.

I found no post-2022 paper claiming a theorem with hypotheses broad enough to cover both alternatives in this exact form. The correct current verdict is therefore partially resolved.

4.15.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

4.15.5. Primary-source pointer

The originating ICM record is *Rigidity, lattices, and invariant measures beyond homogeneous dynamics* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=129>.

4.16. 97. The regularity theory for the area functional (in geometric measure theory) [2022-geom-170]

Record: 2022-geom-170

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: The regularity theory for the area functional (in geometric measure theory)

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>

4.16.1. Problem statement

Prove that if Σ is an m -dimensional stationary integral varifold in an open set $U \subset \mathbb{R}^{m+n}$, and $\text{Sing}(\Sigma)$ denotes the set of points of the support of Σ at which Σ is not locally a smooth m -dimensional minimal submanifold, then

$$\dim_{\mathcal{H}} \text{Sing}(\Sigma) \leq m - 1.$$

4.16.2. Historical background, literature, and current progress

4.16.3. Status check

4.16.3.1. Verdict

Partially resolved. The exact statement remains open for arbitrary stationary integral varifolds, but there is substantial progress in special classes and on related structural problems.

4.16.3.2. Precise problem

The problem asks whether every stationary integral m -varifold V in an open set U subset $\mathbb{R}^{m+n}$ has interior singular set

$\text{Sing}(V) = \{\text{points of } \text{spt}(V) \text{ at which } \text{spt}(V) \text{ is not locally a smooth } m\text{-dimensional minimal submanifold}\}$

with Hausdorff dimension at most $m - 1$. Equivalently, the interior singular set should have codimension at least 1 in the support. The statement has no extra assumptions such as area minimizing, stability, codimension one, graphicality, multiplicity one, absence of classical singularities, or an epsilon-regularity hypothesis.

4.16.3.3. Why the full statement is not resolved

The best current general theory does not prove this all-cases bound. The obstruction is not the usual Almgren stratification S^k : for stationary integral varifolds, S^k has dimension at

most k and is rectifiable, but singular points with a flat tangent cone of multiplicity greater than one can lie in the top stratum $S^m \setminus S^{m-1}$. Controlling these flat higher-multiplicity or branch-type singular points is precisely the unresolved part.

Current literature even records the weaker question “is the singular set H^m -null?” as open for general stationary integral varifolds. Since the stated bound $\dim_H \text{Sing}(V) \leq m - 1$ would imply $H^m(\text{Sing}(V)) = 0$, these statements confirm that the problem is not fully resolved.

4.16.3.4. Main status sources

1. Title: The regularity theory for the area functional (in geometric measure theory) Authors: Camillo De Lellis Venue: Proceedings of the International Congress of Mathematicians 2022, Vol. 2, Plenary Lectures, EMS Press, Berlin, 2023, pp. 872-913 DOI/arXiv/URL: DOI 10.4171/ICM2022/122; arXiv:2110.11324; <https://ems.press/books/standalone/274/5418> Relevance: This ICM survey is the source context for the problem. It states Almgren’s theorem that the stratum S_k of a stationary integral varifold has Hausdorff dimension at most k and Naber-Valtorta’s rectifiability improvement. It then explains that Allard’s multiplicity-one epsilon-regularity does not control flat tangent cones of higher multiplicity, and says that one could reasonably conjecture that the singular set of a stationary integral varifold has dimension at most $m - 1$. The stated best general consequence is only that the singular set is closed with empty relative interior. Thus it does not resolve the problem statement.
2. Title: Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri Authors: Camillo Brena, Stefano Decio, Camillo De Lellis Venue: Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. 36 (2025), no. 3, pp. 409-443; published online by EMS Press in 2026 DOI/arXiv/URL: DOI 10.4171/RLM/1077; arXiv:2503.00651; <https://ems.press/journals/rlm/articles/14299484> Relevance: This paper states the exact codimension-one assertion as Conjecture 1.1: the interior singular set of a stationary integer rectifiable m -dimensional varifold has codimension not smaller than 1. It also states the weaker Conjecture 1.2 that the interior singular set is H^m -null and explicitly says this remains open even decades after Allard. The paper proves and reproves height, Lipschitz approximation, and excess-decay tools, but these are partial results and not a proof of the problem.
3. Title: A Branch Set Stratification for Stationary Varifolds with Epsilon-Regularity Authors: Brian Krummel, Paul Minter, Neshan Wickramasekera Venue: arXiv preprint, 2026 DOI/arXiv/URL: arXiv:2606.01511; <https://arxiv.org/abs/2606.01511> Relevance: This very recent preprint says that for general stationary integral n -varifolds it is still open whether the singular set, equivalently the branch set in the relevant context, can have positive H^n measure, and that in full generality essentially only density of the regular set is known. Its main theorem proves that branch points of density $\leq Q$ have Hausdorff dimension $\leq n - 2$ only for classes satisfying an epsilon-regularity property near planes of integer multiplicity $\leq Q$. The paper then obtains consequences for two-valued Lipschitz graphs and stable codimension-one classes. These extra hypotheses do not cover arbitrary stationary integral varifolds.

4.16.3.5. Important partial progress

1. Title: On the First Variation of a Varifold Authors: William K. Allard Venue: *Annals of Mathematics* (2) 95 (1972), no. 3, pp. 417-491 DOI/arXiv/URL: DOI 10.2307/1970868; <https://www.jstor.org/stable/1970868> Progress: Allard's interior regularity theorem gives smooth graphical regularity near multiplicity-one flat points and implies that the regular set of a stationary integral varifold is relatively open and dense. It does not bound the Hausdorff dimension of the entire singular set because higher-multiplicity flat tangent cones are outside the multiplicity-one hypothesis.
2. Title: The singular structure and regularity of stationary varifolds Authors: Aaron Naber, Daniele Valtorta Venue: *Journal of the European Mathematical Society* 22 (2020), no. 10, pp. 3305-3382 DOI/arXiv/URL: DOI 10.4171/JEMS/987; arXiv:1505.03428; <https://ems.press/content/serial-article-files/32493> Progress: For integral varifolds with bounded mean curvature, including stationary varifolds, the classical strata S^k are countably k -rectifiable, with quantitative Minkowski estimates for the quantitative strata. This is a major structural theorem, but it does not imply $\text{Sing}(V) \subset S^{m-1}$; flat higher-multiplicity singular points can be top-stratum points.
3. Title: C^∞ rectifiability of stationary varifolds Authors: Camillo Brena, Camillo De Lellis, Federico Franceschini Venue: arXiv preprint, 2025 DOI/arXiv/URL: arXiv:2503.00649; <https://arxiv.org/abs/2503.00649> Progress: Proves that the support of any stationary m -dimensional integer rectifiable varifold in $U \subset \mathbb{R}^{m+n}$ is C^∞ rectifiable up to an H^m -null set. This is general and important, but it is not the requested singular-set dimension bound: being covered a.e. by smooth graphs does not rule out an H^m -large set of points where the support is not locally a single smooth minimal submanifold.
4. Title: Dimension of the singular set for 2-valued stationary Lipschitz graphs Authors: Jonas Hirsch, Luca Spolaor Venue: *Duke Mathematical Journal* 175 (2026), no. 8, pp. 1453-1518 DOI/arXiv/URL: DOI 10.1215/00127094-2025-0060; arXiv:2401.00279; <https://projecteuclid.org/journals/duke-mathematical-journal/volume-175/issue-8/Dimension-of-the-singular-set-for-2-valued-stationary-Lipschitz/10.1215/00127094-2025-0060.full> Progress: Proves the optimal codimension-one bound for the singular set of a two-valued Lipschitz graph whose associated graph varifold is stationary. This is a genuine stationary result, but it assumes multiplicity two and Lipschitz graphicality, so it is a special case.
5. Title: On the Nature of Stationary Integral Varifolds near Multiplicity 2 Planes Authors: Spencer Becker-Kahn, Paul Minter, Neshan Wickramasekera Venue: arXiv preprint, 2025 DOI/arXiv/URL: arXiv:2507.13148; <https://arxiv.org/abs/2507.13148> Progress: Proves an epsilon-regularity theorem near multiplicity-two planes under a topological structural condition on the lower-density part of the support, and obtains consequences for stationary two-valued Lipschitz graphs. The theorem is local and conditional, not a general theorem for all stationary integral varifolds.
6. Title: A general regularity theory for stable codimension 1 integral varifolds Authors: Neshan Wickramasekera Venue: *Annals of Mathematics* (2) 179 (2014), no. 3, pp. 843-1007 DOI/arXiv/URL: DOI 10.4007/annals.2014.179.3.2; arXiv:0911.4883; <https://annals.math.princeton.edu/2014/179-3/p02> Progress: Gives optimal regularity for stable codimension-one integral varifolds satisfying the alpha-structural hypothesis. The stability, codimension-one, and structural assumptions are not present in the problem statement.

7. Title: A structure theory for stable codimension 1 integral varifolds with applications to area minimising hypersurfaces mod p Authors: Paul Minter, Neshan Wickramasekera Venue: Journal of the American Mathematical Society 37 (2024), pp. 861-927 DOI/arXiv/URL: DOI 10.1090/jams/1032; arXiv:2111.11202; <https://www.ams.org/journals/jams/2024-37-03/S0894-0347-2023-01032-X/> Progress: Develops a structure theory for stable codimension-one stationary integral varifolds admitting no low-density classical singularities, with applications to area-minimizing hypersurfaces mod p . Again this has extra stability and structural hypotheses.
8. Title: Regularity of area minimizing currents mod p Authors: Camillo De Lellis, Jonas Hirsch, Andrea Marchese, Salvatore Stuvard Venue: Geometric and Functional Analysis 30 (2020), no. 5, pp. 1224-1336 DOI/arXiv/URL: DOI 10.1007/s00039-020-00546-0; arXiv:1909.05172; <https://link.springer.com/article/10.1007/s00039-020-00546-0> Progress: Proves that the interior singular set of an m -dimensional area-minimizing current mod p has Hausdorff dimension at most $m - 1$. This matches the numerical bound but only for area-minimizing currents mod p , a much stronger variational assumption than stationarity of an arbitrary integral varifold.

4.16.3.6. What remains unresolved

What remains open is the flat higher-multiplicity/branch-point part of the singular set for a completely arbitrary stationary integral varifold. Existing theorems either handle multiplicity-one points, stratify non-flat tangent-cone points, prove rectifiability of the support up to H^m -null sets, or impose extra assumptions such as minimizing, stability, codimension one, graphicality, multiplicity two, no classical singularities, or an epsilon-regularity class condition. None of these hypotheses is available in the problem statement.

Therefore the literature does not fully resolve the problem as stated, but the problem is not “fully open” because the sources above give substantial partial progress and solve important special cases.

4.16.4. Remaining open problems recorded by the database

4.16.4.1. Open problem 1

Let V be a stationary integer rectifiable m -dimensional varifold in an open set $U \subset \mathbb{R}^{m+n}$. Is its interior singular set $\mathcal{H}^m$ -null?

Reference supplied by the database. Camillo Brena, Stefano Decio, and Camillo De Lellis, “Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri,” *Atti della Accademia Nazionale dei Lincei, Classe di Scienze Fisiche, Matematiche e Naturali. Rendiconti Lincei. Matematica e Applicazioni* **36** (2025), no. 3, 409–443 (published online 3 February 2026), Conjecture 1.2. DOI: [10.4171/RLM/1077](https://doi.org/10.4171/RLM/1077); arXiv: [2503.00651](https://arxiv.org/abs/2503.00651).

4.16.4.2. Open problem 2

Let V be a stationary integer rectifiable m -dimensional varifold in an open set $U \subset \mathbb{R}^{m+n}$, and let $x_0 \in \text{spt}(V) \cap U$. Suppose that V has a unique flat tangent cone at x_0 and

$$\lim_{r \downarrow 0} \frac{\|V\|(\{x : \Theta(V, x) \neq \Theta(V, x_0)\} \cap B_r(x_0))}{r^m} = 0.$$

Must x_0 be a regular point of V ?

Reference supplied by the database. Camillo Brena, Stefano Decio, and Camillo De Lellis, “Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri,” *Atti della Accademia Nazionale dei Lincei, Classe di Scienze Fisiche, Matematiche e Naturali. Rendiconti Lincei. Matematica e Applicazioni* **36** (2025), no. 3, 409–443 (published online 3 February 2026), Conjecture 1.3. DOI: [10.4171/RLM/1077](https://doi.org/10.4171/RLM/1077); arXiv: [2503.00651](https://arxiv.org/abs/2503.00651).

4.16.4.3. Open problem 3

Let V and x_0 satisfy the following conditions: V is a stationary integer rectifiable m -dimensional varifold in an open set $U \subset \mathbb{R}^{m+n}$, $x_0 \in \text{spt}(V) \cap U$, V has a unique flat tangent cone at x_0 , and

$$\lim_{r \downarrow 0} \frac{\|V\|(\{x : \Theta(V, x) \neq \Theta(V, x_0)\} \cap B_r(x_0))}{r^m} = 0.$$

Does there exist a smooth classical minimal m -dimensional graph M in a neighborhood of x_0 such that

$$\int_{B_r(x_0)} \text{dist}(x, M)^2 d\|V\|(x) = o(r^N)$$

for every $N \in \mathbb{N}$?

Reference supplied by the database. Camillo Brena, Stefano Decio, and Camillo De Lellis, “Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri,” *Atti della Accademia Nazionale dei Lincei, Classe di Scienze Fisiche, Matematiche e Naturali. Rendiconti Lincei. Matematica e Applicazioni* **36** (2025), no. 3, 409–443 (published online 3 February 2026), Conjecture 1.4. DOI: [10.4171/RLM/1077](https://doi.org/10.4171/RLM/1077); arXiv: [2503.00651](https://arxiv.org/abs/2503.00651).

4.16.4.4. Open problem 4

Let V be a stationary m -dimensional varifold in an open set $U \subset \mathbb{R}^{m+n}$, let $x_0 \in \text{spt}(V) \cap U$, and let M be a classical smooth minimal m -dimensional surface. Suppose that

$$\int_{B_r(x_0)} \text{dist}(x, M)^2 d\|V\|(x) = o(r^N)$$

for every $N \in \mathbb{N}$. Must $\text{spt}(V)$ be contained in M in some neighborhood of x_0 ?

Reference supplied by the database. Camillo Brena, Stefano Decio, and Camillo De Lellis, “Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri,” *Atti della Accademia Nazionale dei Lincei, Classe di Scienze Fisiche, Matematiche e Naturali. Rendiconti Lincei. Matematica e Applicazioni* **36** (2025), no. 3, 409–443 (published online 3 February 2026), Conjecture 1.5. DOI: [10.4171/RLM/1077](https://doi.org/10.4171/RLM/1077); arXiv: [2503.00651](https://arxiv.org/abs/2503.00651).

4.16.4.5. Open problem 5

Let V be a stationary m -dimensional varifold in an open set $U \subset \mathbb{R}^{m+n}$, and let $h \in C^1(U)$ be harmonic on V , meaning that

$$\int \nabla_{T_V} h \cdot \nabla_{T_V} \varphi \, d\|V\| = 0$$

for every $\varphi \in C_c^1(U)$. If $x_0 \in \text{spt}(V) \cap U$ and

$$\int_{B_r(x_0)} h^2 \, d\|V\| = o(r^N)$$

for every $N \in \mathbb{N}$, must h vanish identically on $\text{spt}(V) \cap B_\rho(x_0)$ for some $0 < \rho < \text{dist}(x_0, \partial U)$?

Reference supplied by the database. Camillo Brena, Stefano Decio, and Camillo De Lellis, “Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri,” *Atti della Accademia Nazionale dei Lincei, Classe di Scienze Fisiche, Matematiche e Naturali. Rendiconti Lincei. Matematica e Applicazioni* **36** (2025), no. 3, 409–443 (published online 3 February 2026), Conjecture 1.7. DOI: [10.4171/RLM/1077](https://doi.org/10.4171/RLM/1077); arXiv: [2503.00651](https://arxiv.org/abs/2503.00651).

4.16.5. Primary-source pointer

The originating ICM record is *The regularity theory for the area functional (in geometric measure theory)* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>.

4.17. 98. The regularity theory for the area functional (in geometric measure theory) [2022-geom-172]

Record: 2022-geom-172

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: The regularity theory for the area functional (in geometric measure theory)

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>

4.17.1. Problem statement

Let $m, n \geq 1$, let $\Gamma \subset \mathbb{R}^{m+n}$ be a smooth closed oriented $(m-1)$ -dimensional submanifold, and let Σ be an m -dimensional area-minimizing integral current in $\mathbb{R}^{m+n}$ whose boundary is Γ counted with multiplicity one. Call a point $p \in \Gamma$ a boundary regular point of Σ if, in some neighborhood U of p , either $\Sigma \cap U$ is a smooth minimal m -dimensional submanifold with boundary $\Gamma \cap U$, or there is a smooth minimal m -dimensional submanifold $\Lambda \subset U$ without boundary such that $\Gamma \cap U \subset \Lambda$, $\Gamma \cap U$ separates Λ into two regions Λ^+ and Λ^- , and $\Sigma \cap U = (k+1)\Lambda^+ + k\Lambda^-$ as currents for some positive integer k . Prove that the set of boundary singular points, namely the points of Γ that are not boundary regular points, has zero $(m-1)$ -dimensional Hausdorff measure.

4.17.2. Historical background, literature, and current progress

Verdict: partially resolved.

4.17.2.1. Problem identified

The statement asks whether, for every $m, n \geq 1$, every smooth closed oriented $(m-1)$ -submanifold $\Gamma \subset \mathbb{R}^{m+n}$, and every m -dimensional area-minimizing integral current Σ with $\partial\Sigma = [\Gamma]$, the boundary singular set

$$\Gamma \setminus \text{Reg}_b(\Sigma)$$

has zero $\mathcal{H}^{m-1}$ -measure, where boundary regularity includes both the ordinary one-sided smooth boundary case and the collapsed/two-sided local model $(k+1)\Lambda^+ + k\Lambda^-$.

4.17.2.2. Current status

The exact full statement is not known. The literature proves substantial special cases and density/rectifiability results, but I found no published or citable result proving $\mathcal{H}^{m-1}(\text{Sing}_b(\Sigma)) = 0$ for arbitrary codimension and general smooth multiplicity-one boundary. Recent work explicitly lists this as open, even for $Q = 1$.

4.17.2.3. Key literature and progress

1. Boundary regularity and embedded solutions for the oriented Plateau problem

Authors: Robert Hardt, Leon Simon.

Venue/year: Annals of Mathematics (2) 110, no. 3, 439-486, 1979.

DOI/URL: <https://doi.org/10.2307/1971233>

Relevance: Resolves the codimension-one hypersurface case by proving boundary regularity for oriented area-minimizing hypersurfaces. This gives an empty boundary singular set in that special case, but does not cover arbitrary codimension.

2. On the boundary behavior of mass-minimizing integral currents

Authors: Camillo De Lellis, Guido De Philippis, Jonas Hirsch, Annalisa Massaccesi.

Venue/year: Memoirs of the American Mathematical Society 291, no. 1446, 2023.

DOI/arXiv/URL: <https://doi.org/10.1090/memo/1446> ; arXiv:1809.09457, <https://arxiv.org/abs/1809.09457>

Relevance: Theorem 1.6 proves that the boundary regular set is dense in Γ under the general higher-codimension multiplicity-one hypotheses. The same work states Conjecture 2.8: $\mathcal{H}^{m-1}(\text{Sing}_b(T)) = 0$. Thus it is partial progress, not a resolution.

3. The regularity theory for the area functional (in geometric measure theory)

Author: Camillo De Lellis.

Venue/year: ICM 2022, Vol. 2 Plenary Lectures, EMS Press, 872-913, 2023.

DOI/arXiv/URL: <https://doi.org/10.4171/ICM2022/122> ; arXiv:2110.11324, <https://arxiv.org/abs/2110.11324>

Relevance: Theorem 13.2 states the dense relatively open boundary regular set result using exactly the two regularity alternatives in the problem statement. Section 16.5 asks exactly whether this can be improved to zero $(m - 1)$ -dimensional Hausdorff measure for the boundary singular set.

4. Nonclassical minimizing surfaces with smooth boundary

Authors: Camillo De Lellis, Guido De Philippis, Jonas Hirsch.

Venue/year: Journal of Differential Geometry 122, no. 2, 205-222, 2022.

DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1669998183> ; arXiv:1906.09488, <https://arxiv.org/abs/1906.09488>

Relevance: Constructs smooth boundary examples with large boundary singular sets; the Memoirs paper records an example with boundary singular set of Hausdorff dimension 1 but $\mathcal{H}^1$ -null. This rules out a naive dimension estimate in full smooth generality, but does not disprove or prove the measure-null conjecture.

5. An Allard-type boundary regularity theorem for $2d$ minimizing currents at smooth curves with arbitrary multiplicity

Authors: Camillo De Lellis, Stefano Nardulli, Simone Steinbrüchel.

Venue/year: Publications mathématiques de l'IHÉS 140, 37-154, 2024.

DOI/arXiv/URL: <https://doi.org/10.1007/s10240-024-00144-y> ; arXiv:2111.02991, <https://arxiv.org/abs/2111.02991>

Relevance: Gives a sharp Allard-type boundary regularity theorem for two-dimensional currents under density and convex-barrier hypotheses. This is important special-case progress, but its hypotheses do not cover the full statement.

6. Density of the boundary regular set of $2d$ area minimizing currents with arbitrary codimension and multiplicity

Authors: Stefano Nardulli, Reinaldo Resende.

Venue/year: Advances in Mathematics 455, article 109889, 2024.

DOI/arXiv/URL: <https://doi.org/10.1016/j.aim.2024.109889> ;

arXiv:2204.11947, <https://arxiv.org/abs/2204.11947>

Relevance: Extends density of the boundary regular set in dimension $m = 2$ and arbitrary codimension/multiplicity. It remains a density result, not a proof of $\mathcal{H}^{m-1}$ -nullity of all boundary singular points.

7. On the regularity of area minimizing currents at boundaries with arbitrary multiplicity

Authors: Ian Fleschler, Reinaldo Resende.

Venue/year: arXiv preprint, 2024; revised version 2025.

DOI/arXiv/URL: arXiv:2410.04566, <https://arxiv.org/abs/2410.04566>

Relevance: Proves $\mathcal{H}^{m-3}$ -rectifiability for density $Q/2$ boundary singular points and an open dense boundary regular set under broad arbitrary-multiplicity assumptions. It explicitly states as Open question 2 whether $\text{Sing}_b(T)$ is $\mathcal{H}^{m-1}$ -null, says this was originally Conjecture 2.8, and says it remains widely open even when $Q = 1$. This directly prevents a “fully resolved” verdict for the stated multiplicity-one problem.

8. An essential one sided boundary singularity for a 3-dimensional area minimizing current in $\mathbb{R}^5$

Author: Ian Fleschler.

Venue/year: arXiv preprint, 2025.

DOI/arXiv/URL: arXiv:2502.15047, <https://arxiv.org/abs/2502.15047>

Relevance: Constructs essential one-sided singularities in a higher-multiplicity setting and shows sharpness of related rectifiability theory. It is not a counterexample to the stated $Q = 1$ null-measure statement, but it clarifies why boundary regularity remains difficult.

4.17.2.4. What remains unresolved

The unresolved part is exactly the full statement: for arbitrary codimension n , arbitrary dimension m , and a general smooth multiplicity-one boundary Γ , no known theorem proves that all non-regular boundary points form an $\mathcal{H}^{m-1}$ -null subset of Γ . Existing results give full regularity in codimension one, open/dense regularity in the general multiplicity-one case, and stronger rectifiability under density, dimension-two, convex-barrier, or arbitrary-multiplicity subcase hypotheses.

4.17.3. Remaining open problems recorded by the database

4.17.3.1. Open problem 1

Let T be an m -dimensional area-minimizing integral current with smooth multiplicity-one boundary Γ in a smooth ambient manifold Σ . A boundary singular point is called *genuine* if it is not of crossing type, i.e. it cannot locally be separated as $T = T_1 + T_2$ with $\partial T_1 = 0$ and the point boundary-regular for T_2 . Is the Hausdorff dimension of the set of genuine boundary singularities at most $m - 2$?

Reference supplied by the database. Camillo De Lellis, Guido De Philippis, Jonas Hirsch, and Annalisa Massaccesi, *On the Boundary Behavior of Mass-Minimizing Integral Currents*, *Mem-oirs of the American Mathematical Society* **291** (2023), no. 1446, Conjecture 2.7 (also stated as Conjecture 1.7). DOI: [10.1090/memo/1446](https://doi.org/10.1090/memo/1446); arXiv: [1809.09457](https://arxiv.org/abs/1809.09457).

4.17.3.2. Open problem 2

Let (g^+, g^-) be a $(Q - \frac{1}{2})$ -valued function with a C^3 interface (γ, φ) over a domain $\Omega \subset \mathbb{R}^m$, and suppose that (g^+, g^-) minimizes the Dirichlet energy. Is the Hausdorff dimension of its set of genuine boundary singularities at most $m - 2$?

Reference supplied by the database. Camillo De Lellis, Guido De Philippis, Jonas Hirsch, and Annalisa Massaccesi, *On the Boundary Behavior of Mass-Minimizing Integral Currents*, *Memoirs of the American Mathematical Society* **291** (2023), no. 1446, Conjecture 2.11. DOI: [10.1090/memo/1446](https://doi.org/10.1090/memo/1446); arXiv: [1809.09457](https://arxiv.org/abs/1809.09457).

4.17.3.3. Open problem 3

Let (g^+, g^-) be a $(Q - \frac{1}{2})$ -valued function with a C^3 interface (γ, φ) over a domain $\Omega \subset \mathbb{R}^m$, and suppose that (g^+, g^-) minimizes the Dirichlet energy. Is its boundary singular set $\mathcal{H}^{m-1}$ -null?

Reference supplied by the database. Camillo De Lellis, Guido De Philippis, Jonas Hirsch, and Annalisa Massaccesi, *On the Boundary Behavior of Mass-Minimizing Integral Currents*, *Memoirs of the American Mathematical Society* **291** (2023), no. 1446, Conjecture 2.12. DOI: [10.1090/memo/1446](https://doi.org/10.1090/memo/1446); arXiv: [1809.09457](https://arxiv.org/abs/1809.09457).

4.17.3.4. Open problem 4

Let T be a two-dimensional area-minimizing integral current with smooth multiplicity-one boundary Γ in a smooth ambient manifold Σ , and assume that both Σ and Γ are real analytic. Is the union of the boundary singular set and the interior singular set of T discrete?

Reference supplied by the database. Camillo De Lellis, Guido De Philippis, Jonas Hirsch, and Annalisa Massaccesi, *On the Boundary Behavior of Mass-Minimizing Integral Currents*, *Memoirs of the American Mathematical Society* **291** (2023), no. 1446, Conjecture 2.14 (the White conjecture). DOI: [10.1090/memo/1446](https://doi.org/10.1090/memo/1446); arXiv: [1809.09457](https://arxiv.org/abs/1809.09457).

4.17.3.5. Open problem 5

Does there exist a two-dimensional area-minimizing integral current in Euclidean space, with smooth multiplicity-one boundary, having a boundary singular point that is an accumulation point of boundary branch points?

Reference supplied by the database. Camillo De Lellis, Guido De Philippis, Jonas Hirsch, and Annalisa Massaccesi, *On the Boundary Behavior of Mass-Minimizing Integral Currents*, *Memoirs of the American Mathematical Society* **291** (2023), no. 1446, Question 2.15. DOI: [10.1090/memo/1446](https://doi.org/10.1090/memo/1446); arXiv: [1809.09457](https://arxiv.org/abs/1809.09457).

4.17.4. Primary-source pointer

The originating ICM record is *The regularity theory for the area functional (in geometric measure theory)* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>.

4.18. 99. The regularity theory for the area functional (in geometric measure theory) [2022-geom-174]

Record: 2022-geom-174

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: The regularity theory for the area functional (in geometric measure theory)

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>

4.18.1. Problem statement

Let Σ be an m -dimensional area-minimizing integral current in $\mathbb{R}^{m+n}$, with $m \geq 3$ and $n \geq 2$, and let p be an interior singular point of Σ . Suppose that, for some sequence of scales $r_k \downarrow 0$, the rescaled currents $(\Sigma - p)/r_k$ converge to a tangent cone Σ_0 at p , and that Σ_0 is the union of two distinct m -dimensional linear planes in $\mathbb{R}^{m+n}$, each counted with multiplicity 1. Must Σ_0 be independent of the choice of blow-up sequence, equivalently the unique tangent cone to Σ at p ?

4.18.2. Historical background, literature, and current progress

4.18.3. Verdict: partially resolved

4.18.3.1. Precise problem

The problem asks whether an arbitrary interior singular point of an m -dimensional area-minimizing integral current $T \subset \mathbb{R}^{m+n}$, $m \geq 3$, $n \geq 2$, has a unique tangent cone once one blow-up cone is

$$T_0 = \llbracket P_1 \rrbracket + \llbracket P_2 \rrbracket,$$

where $P_1 \neq P_2$ are m -dimensional linear planes, each with multiplicity one. Equivalently, it asks whether every sequence of blow-up scales at that same point converges to the same two-plane cone.

4.18.3.2. Status

No citable theorem found resolves this exact pointwise question in all cases. The literature gives important partial results: conditional uniqueness for cylindrical cones under extra hypotheses such as Simon's no-hole condition, uniqueness in low-dimensional or smooth-link settings, and recent $\mathcal{H}^{m-2}$ -a.e. uniqueness and rectifiability theorems for higher-codimension area-minimizing currents. These do not imply uniqueness at the particular point p in the problem statement, because the statement has no no-hole assumption and is not restricted to almost every singular point.

4.18.3.3. Key literature and comparison

1. **The regularity theory for the area functional (in geometric measure theory)**
Author: Camillo De Lellis
Venue/year: International Congress of Mathematicians 2022, Vol. 2, Plenary lectures, EMS Press, pp. 872-913, 2023 publication of proceedings
DOI/arXiv/URL: DOI 10.4171/ICM2022/122; arXiv:2110.11324; <https://ems.press/content/book-chapter-files/33147>
Relevance: This is the apparent source of the problem. It explicitly states this exact two-plane tangent-cone uniqueness question and says it is still open, even in the basic case $m = 3$ in $\mathbb{R}^5$. It also explains that the issue is tied to whether Simon-type cylindrical tangent-cone results can be used without the no-hole condition.
2. **Cylindrical tangent cones and the singular set of minimal submanifolds**
Author: Leon Simon
Venue/year: Journal of Differential Geometry 38(3), 585-652, 1993
DOI/arXiv/URL: DOI 10.4310/jdg/1214454484; <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-38/issue-3/Cylindrical-tangent-cones-and-the-singular-set-of-minimal-submanifolds/10.4310/jdg/1214454484.full>
Relevance: Gives uniqueness for cylindrical tangent cones under structural hypotheses, including integrability and a no-hole condition. A two-plane cone with a common spine is a natural cylindrical-cone case, but the statement does not assume the no-hole condition, so this is only conditional progress.
3. **On the radial behavior of minimal surfaces and the uniqueness of their tangent cones**
Authors: William K. Allard and Frederick J. Almgren, Jr.
Venue/year: Annals of Mathematics 113(2), 215-265, 1981
DOI/arXiv/URL: DOI 10.2307/2006984; <https://annals.math.princeton.edu/198/113-2/p01>
Relevance: Foundational uniqueness theorem for regular/smooth-link tangent-cone settings. It does not cover the singular-link two-plane cone in the general common-spine case posed here.
4. **Analysis of singularities of area minimizing currents: planar frequency, branch points of rapid decay, and weak locally uniform approximation**
Authors: Brian Krummel and Neshan Wickramasekera
Venue/year: arXiv preprint, 2023
DOI/arXiv/URL: arXiv:2304.10653; DOI 10.48550/arXiv.2304.10653; <https://arxiv.org/abs/2304.10653>
Relevance: Develops the planar-frequency and decomposition framework used in the later uniqueness/rectifiability results. It provides structural alternatives but not the pointwise uniqueness asserted in the problem statement.
5. **Analysis of singularities of area minimizing currents: a uniform height bound, estimates away from branch points of rapid decay, and uniqueness of tangent cones**
Authors: Brian Krummel and Neshan Wickramasekera
Venue/year: arXiv preprint, 2023
DOI/arXiv/URL: arXiv:2304.10272; DOI 10.48550/arXiv.2304.10272; <https://arxiv.org/abs/2304.10272>
Relevance: Proves, together with the companion work, that locally area-minimizing currents of codimension at least two have unique tangent cones at $\mathcal{H}^{n-2}$ -almost every point, and proves rectifiability of the relevant singular set. This is substantial partial progress,

but it is an a.e. statement and does not force uniqueness at the arbitrary point p in the problem statement.

6. The Fine Structure of the Singular Set of Area-Minimizing Integral Currents III: Frequency 1 Flat Singular Points and $\mathcal{H}^{m-2}$ -a.e. Uniqueness of Tangent Cones

Authors: Camillo De Lellis, Paul Minter, Anna Skorobogatova

Venue/year: arXiv preprint, 2023; revised 2024

DOI/arXiv/URL: arXiv:2304.11553; DOI 10.48550/arXiv.2304.11553; <https://arxiv.org/abs/2304.11553>

Relevance: Independently proves that an area-minimizing integral current in codimension greater than one has a unique tangent cone, a superposition of planes, at $\mathcal{H}^{m-2}$ -almost every point, and contributes to countable $(m-2)$ -rectifiability of the singular set. Again, this is not a pointwise theorem for every singular point with a two-plane tangent cone.

7. Singularities and tangent cones for area-minimizing and semicalibrated currents

Author: Anna Skorobogatova

Venue/year: Séminaire Laurent Schwartz – EDP et applications (2024-2025), Talk no. 5, published online 2025

DOI/arXiv/URL: DOI 10.5802/slsedp.175; <https://proceedings.centre-mersenne.org/articles/10.5802/slsedp.175/>

Relevance: Survey of the recent breakthroughs. It states the current higher-dimensional result as $\mathcal{H}^{m-2}$ -a.e. uniqueness and rectifiability, confirming that the available broad theorem is almost-everywhere rather than pointwise for every singular point.

4.18.3.4. What remains unresolved

The missing theorem would need to prove that, for every interior singular point p satisfying the stated two-plane tangent-cone hypothesis, all blow-up sequences converge to the same cone, without adding a no-hole condition, integrability condition, genericity/a.e. restriction, boundary hypothesis, or low-dimensional assumption. I found no published or arXiv result proving that full statement.

4.18.4. Remaining open problems recorded by the database

4.18.4.1. Open problem 1

Let T be a three-dimensional area-minimizing integral current in $\mathbb{R}^5$, and let p be an interior singular point. Suppose that one tangent cone to T at p is the union of two distinct three-dimensional linear planes, each counted with multiplicity one. Must this cone be the unique tangent cone to T at p ?

Reference supplied by the database. Camillo De Lellis, “The regularity theory for the area functional (in geometric measure theory),” in *Proceedings of the International Congress of Mathematicians 2022*, Vol. 2, Plenary Lectures, EMS Press, Berlin, 2023, pp. 872–913, §16.7 (“Uniqueness of tangent cones”), the explicitly identified special case $m = 3$, $n = 2$ of the final open question. DOI: [10.4171/ICM2022/122](https://doi.org/10.4171/ICM2022/122); arXiv: [2110.11324](https://arxiv.org/abs/2110.11324).

4.18.5. Primary-source pointer

The originating ICM record is *The regularity theory for the area functional (in geometric measure theory)* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>.

4.19. 100. Emergent complex geometry [2022-ag-215]

Record: 2022-ag-215

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Emergent complex geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>

4.19.1. Problem statement

Let X be an n -dimensional Fano manifold. For each positive integer k , set

$$N_k := \dim_{\mathbb{C}} H^0(X, -kK_X).$$

Choose a basis $s_1^{(k)}, \dots, s_{N_k}^{(k)}$ of $H^0(X, -kK_X)$, and let $\det S^{(k)}$ be the Slater determinant section of $-kK_{X^{N_k}}$ given locally by

$$\det S^{(k)}(x_1, \dots, x_{N_k}) := \det(s_i^{(k)}(x_j)).$$

Let

$$D_{N_k} := \frac{1}{k} \operatorname{div}(\det S^{(k)})$$

be the anticanonical $\mathbb{Q}$ -divisor on X^{N_k} cut out by the multivalued section $(\det S^{(k)})^{1/k}$ of $-K_{X^{N_k}}$, and let $\operatorname{lct}(D_{N_k})$ denote its log canonical threshold.

Let $\mathcal{P}(X)$ be the space of probability measures on X . Fix a smooth Kähler metric on X , with volume form dV , and let E be the associated pluricomplex energy on $\mathcal{P}(X)$ in the anticanonical class: on smooth probability volume forms it is characterized, up to an additive constant, by the first variation $dE(\mu) = -u_\mu$, where u_μ solves

$$\frac{(\omega_0 + \frac{i}{2\pi} \partial \bar{\partial} u_\mu)^n}{\int_X \omega_0^n} = \mu$$

for the curvature form ω_0 of the fixed Hermitian metric on $-K_X$, and E is extended to $\mathcal{P}(X)$ as the greatest lower-semicontinuous extension. Let $\operatorname{Ent}_{dV}(\mu) = \int_X \log(d\mu/dV) d\mu$ when μ is absolutely continuous with respect to dV , and let $\operatorname{Ent}_{dV}(\mu) = +\infty$ otherwise. For $\beta < 0$, define

$$F_\beta(\mu) = \begin{cases} \beta E(\mu) + \operatorname{Ent}_{dV}(\mu), & E(\mu) < +\infty, \\ +\infty, & E(\mu) = +\infty. \end{cases}$$

Prove or disprove that, with N ranging through the integers N_k as $k \rightarrow \infty$,

$$\lim_{k \rightarrow \infty} \operatorname{lct}(D_{N_k}) = \Gamma(X), \quad \Gamma(X) := \sup_{\beta < 0} \left\{ -\beta : \inf_{\mu \in \mathcal{P}(X)} F_\beta(\mu) > -\infty \right\}.$$

4.19.2. Historical background, literature, and current progress

Verdict: partially resolved.

Precise problem checked: for every Fano manifold X , prove the full asymptotic identity

$$\lim_{k \rightarrow \infty} \text{lct}(D_{N_k}) = \Gamma(X),$$

where D_{N_k} is the N_k -point Slater-determinant anticanonical $\mathbb{Q}$ -divisor on X^{N_k} , and $\Gamma(X)$ is the macroscopic free-energy boundedness threshold defined from the pluricomplex energy plus entropy.

Current literature does not prove this identity in full generality.

Key literature and progress:

1. Title: “Emergent complex geometry” Authors: Robert J. Berman Venue/year: Proceedings of the International Congress of Mathematicians 2022, Vol. 4, EMS Press, 2022 DOI/arXiv/URL: DOI 10.4171/ICM2022/59; <https://ems.press/content/book-chapter-files/33209>; arXiv:2109.00307 Relevance: This is the source for this problem. Conjecture 2.7 states the same limiting identity between the Slater-determinant lct threshold and the free-energy threshold $\Gamma(X)$ for Fano manifolds. It is presented as a conjecture suggested by extending large-deviation/partition-function asymptotics to negative inverse temperature.
2. Title: “An invitation to Kähler-Einstein metrics and random point processes” Authors: Robert J. Berman Venue/year: arXiv preprint, 2020 DOI/arXiv/URL: arXiv:2003.11358; <https://arxiv.org/abs/2003.11358> Relevance: Section 3.2.2 formulates the same problem as Conjecture 3.10, $\gamma(X) = \Gamma(X)$, with $\gamma_k(X) = \text{lct}(X^{N_k}, D_k/k)$. The paper records one-sided and conditional progress and notes that the converse implication and convergence problem remain open in general.
3. Title: “On the K-stability of Fano varieties and anticanonical divisors” Authors: Kento Fujita and Yuji Odaka Venue/year: Tohoku Mathematical Journal (2) 70, no. 4, 511–521, 2018 DOI/arXiv/URL: DOI 10.2748/tmj/1546570823; arXiv:1602.01305 Relevance: Proves the algebro-geometric one-sided comparison underlying partial progress: the Berman-Gibbs/microscopic threshold is bounded above by the Fujita-Odaka delta thresholds, hence Berman-Gibbs stability implies K-stability. This does not give the missing reverse inequality needed for the full limit identity.
4. Title: “The probabilistic vs the quantization approach to Kähler-Einstein geometry” Authors: Robert J. Berman Venue/year: Mathematische Annalen 388, 4383–4404, 2024; published online 2023 DOI/arXiv/URL: DOI 10.1007/s00208-023-02627-5; arXiv:2109.06575 Relevance: Gives a quantitative bound on the partition functions and proves $\gamma(X) \leq$ the analytic/coercivity threshold; combined with known results this gives the direction “uniform Gibbs stability implies existence of a unique Kähler-Einstein metric.” The paper explicitly states that the missing lower bound on $\gamma(X)$, equivalently upgrading $\gamma(X) \leq \delta(X)$ to equality, remains to be established.
5. Title: “Kähler-Einstein metrics and Archimedean zeta functions” Authors: Robert J. Berman Venue/year: Proceedings of the 8th European Congress of Mathematics, EMS Press, 2023 DOI/arXiv/URL: <https://ems.press/content/book-chapter-files/28296>; arXiv:2112.04791 Relevance: Develops conditional/analytic approaches via Archimedean zeta functions and proves convergence/equivalence results for log Fano curves. It states that the converse to uniform Gibbs stability implying K-stability is widely open in general.

6. Title: “Microscopic stability thresholds and constant scalar curvature Kähler metrics” Authors: Takahiro Aoi Venue/year: arXiv preprint, 2024 DOI/arXiv/URL: arXiv:2410.22090; <https://arxiv.org/abs/2410.22090> Relevance: Extends related microscopic-threshold techniques to polarized manifolds and cscK metrics. It explicitly states that the limiting equality $\delta(L) = \gamma(L)$ is still open, even in the Fano case.
7. Title: “Gibbs polystability of Fano manifolds, stability thresholds and symmetry breaking” Authors: Rolf Andreasson, Robert J. Berman and Ludvig Svensson Venue/year: arXiv preprint, 2026 DOI/arXiv/URL: arXiv:2511.16173; <https://arxiv.org/abs/2511.16173> Relevance: Gives newer progress for log Fano manifolds with automorphisms. It proves the relevant microscopic/analytic threshold conjectures for log Fano curves and gives higher-dimensional examples and one-sided estimates, but it explicitly leaves the general log Fano/Fano conjectures for future work.

What remains unresolved:

The full reverse inequality/lower bound needed to prove

$$\Gamma(X) \leq \liminf_{k \rightarrow \infty} \text{lct}(D_{N_k})$$

for every Fano manifold is not known. Existing results prove one-sided bounds, implications for uniform Gibbs stability, conditional large-deviation/partition-function results, and complete curve/log-curve cases, but no citable theorem found covers all Fano manifolds and proves the exact equality in the problem statement.

4.19.3. Remaining open problems recorded by the database

4.19.3.1. Open problem 1

With $\gamma(X) := \liminf_{k \rightarrow \infty} \text{lct}(D_{N_k})$ and $\Gamma(X)$ the macroscopic stability threshold defined from F_β , prove the still-missing reverse inequality $\Gamma(X) \leq \gamma(X)$ for every Fano manifold X .

Reference supplied by the database. Robert J. Berman, “The probabilistic vs the quantization approach to Kähler–Einstein geometry,” *Mathematische Annalen* 388 (2024), 4383–4404, Conjecture 1.1 and Corollary 1.3 (which proves only $\gamma(X) \leq \Gamma(X)$, leaving the reverse implication). DOI: [10.1007/s00208-023-02627-5](https://doi.org/10.1007/s00208-023-02627-5).

4.19.3.2. Open problem 2

Let X be a uniformly Gibbs-stable Fano manifold. For the canonical N_k -particle Gibbs measures $\mu_{-1}^{(N_k)}$, let $\Gamma_{N_k, -1}$ be the law of the empirical measure and let Γ_{-1} be any weak limit point. Prove the “upper bound property of the mean energy”

$$\limsup_{k \rightarrow \infty} E^{(N_k)}(\mu_{-1}^{(N_k)}) \leq \int_{\mathcal{P}(X)} E(\mu) d\Gamma_{-1}(\mu).$$

Reference supplied by the database. Robert J. Berman, “An invitation to Kähler–Einstein metrics and random point processes,” *Surveys in Differential Geometry* 23 (2020), 35–87, equation (7.4), the discussion “Towards the missing Step 2,” and Theorem 7.6. arXiv: [2003.11358](https://arxiv.org/abs/2003.11358).

4.19.3.3. Open problem 3

For a uniformly Gibbs-stable Fano manifold X , prove that the meromorphic continuations in β of the canonical Archimedean partition functions $Z_{N_k, \beta}$ are holomorphic and nonvanishing on one complex neighborhood of the interval $[-1, 0]$ that is independent of k .

Reference supplied by the database. Robert J. Berman, “An invitation to Kähler–Einstein metrics and random point processes,” *Surveys in Differential Geometry* 23 (2020), 35–87, Section 7.1, especially the paragraph following Theorem 7.9 (the proposed uniform zero-free-neighborhood approach, explicitly described as completely open in general). arXiv: [2003.11358](https://arxiv.org/abs/2003.11358).

4.19.4. Primary-source pointer

The originating ICM record is *Emergent complex geometry* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>.

5. Symplectic and contact geometry

The symplectic and contact records are strongly shaped by Floer theory, Gromov–Witten theory, mirror symmetry, Hamiltonian dynamics, and contact topology. Many statements began as broad programs in ICM lectures rather than single yes–no conjectures. Consequently, the literature often proves one direction, a monotone or exact case, or a categorical enhancement while leaving foundational and general-case comparisons unresolved.

Coverage: 19 website records; **source years:** 1994–2014; **separately listed descendant open problems:** 39.

5.1. 101. Homological algebra of mirror symmetry [1994-ag-53]

Record: 1994–ag–53

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Homological algebra of mirror symmetry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=192>

5.1.1. Problem statement

Let (V, ω) be a closed symplectic manifold. For $\beta \in H_2(V, \mathbf{Z})$ and $n \geq 3$, let

$$I_{0,n;\beta} \in H_*\left(\overline{M}_{0,n}(\mathbf{C}) \times V^n; \mathbf{Q}\right)$$

denote the genus-zero Gromov–Witten homology class, where $\overline{M}_{0,n}$ is the Deligne–Mumford compactification of the moduli stack of stable genus-zero curves with n marked points. Let

$$\mathcal{H} := \bigoplus_k H^k(V, \mathbf{C})$$

be the total cohomology of V , regarded as a $\mathbf{Z}$ -graded super vector space. For an even element $\gamma \in \mathcal{H} \otimes \Lambda$, where Λ is an auxiliary supercommutative algebra, define the genus-zero Gromov–Witten pre-potential

$$\Phi(\gamma) := \sum_{\beta \in H_2(V, \mathbf{Z})} e^{-\int_{\beta} \omega} \sum_{n \geq 3} \frac{1}{n!} \int_{I_{0,n;\beta}} 1_{\overline{M}_{0,n}} \otimes \gamma \otimes \cdots \otimes \gamma,$$

where $1_{\overline{M}_{0,n}}$ is the identity cohomology class on $\overline{M}_{0,n}$.

Prove that if the symplectic cohomology class $[\omega] \in H^2(V, \mathbf{R})$ is sufficiently positive, in the sense that its pairings $\int_{\beta} \omega$ with the pseudo-holomorphic curve classes counted by the Gromov-Witten classes are taken large enough for the exponential weights $e^{-\int_{\beta} \omega}$ to decay, then the series defining Φ is absolutely convergent for γ in some neighborhood $\mathcal{U}$ of 0 in $\mathcal{H}$.

5.1.2. Historical background, literature, and current progress

5.1.3. Verdict: partially resolved

5.1.3.1. Precise problem

The problem asks for Kontsevich's convergence conjecture for the genus-zero Gromov-Witten prepotential: for an arbitrary closed symplectic manifold (V, ω) , does making the symplectic class sufficiently positive make

$$\sum_{\beta} \exp(-\int_{\beta} \omega) \sum_{n \geq 3} (1/n!) \langle I_{0,n;\beta}, 1 \times \gamma^n \rangle$$

absolutely convergent for γ in a neighborhood of zero in total cohomology?

This is a quantitative growth problem for genus-zero Gromov-Witten invariants as the curve class/energy increases. The full statement is not just formal GW theory over a Novikov ring; it asks for honest analytic convergence after choosing sufficiently decaying symplectic weights.

5.1.3.2. Original source

- **Title:** Homological Algebra of Mirror Symmetry
- **Author:** Maxim Kontsevich
- **Venue/year:** Proceedings of the International Congress of Mathematicians, Zurich 1994, Vol. I, Birkhauser, 1995, pp. 120-139
- **DOI/arXiv/URL:** DOI 10.1007/978-3-0348-9078-6_11; arXiv alg-geom/9411018; <https://arxiv.org/abs/alg-geom/9411018>
- **Relevance:** This is the source of the exact conjecture. Kontsevich states that the series Φ should be absolutely convergent near zero when $[\omega]$ is sufficiently positive, and immediately notes that without this conjecture one can instead work formally over a Novikov-type coefficient ring.

5.1.3.3. Literature giving partial progress

- **Title:** Convergence of quantum cohomology by quantum Lefschetz
- **Author:** Hiroshi Iritani
- **Venue/year:** Journal fuer die reine und angewandte Mathematik 610 (2007), 29-69
- **DOI/arXiv/URL:** DOI 10.1515/CRELLE.2007.067; arXiv math/0506236; <https://arxiv.org/abs/math/0506236>
- **Progress:** Proves convergence of twisted genus-zero theory under the assumption that the original genus-zero theory converges; as an application it proves convergence/semisimplicity results for projective toric manifolds. This is significant but

conditional or restricted to special algebraic targets, so it does not cover arbitrary closed symplectic manifolds.

- **Title:** On the Convergence of Gromov-Witten Potentials and Givental's Formula
- **Authors:** Tom Coates and Hiroshi Iritani
- **Venue/year:** Michigan Mathematical Journal 64 no. 3 (2015), 587-631
- **DOI/arXiv/URL:** DOI 10.1307/mmj/1441116660; arXiv 1203.4193; <https://arxiv.org/abs/1203.4193>
- **Progress:** Theorem 1.1 proves NF-convergence of the total descendant potential for smooth projective varieties satisfying Formal Semisimplicity, Genus-Zero Convergence, and Analytic Semisimplicity. Corollary 1.2 gives compact toric varieties and complete flag varieties; Corollary 1.3 gives certain total spaces of negative line bundles over compact toric varieties. This paper explicitly treats genus-zero convergence as an assumption in the main theorem and says convergence of quantum cohomology is not known in general, so it cannot resolve Kontsevich's full conjecture.
- **Title:** The genus 0 Gromov-Witten invariants of projective complete intersections
- **Author:** Aleksey Zinger
- **Venue/year:** Geometry & Topology 18 no. 2 (2014), 1035-1114
- **DOI/arXiv/URL:** DOI 10.2140/gt.2014.18.1035; arXiv 1106.1633; <http://arxiv.org/abs/1106.1633>
- **Progress:** Establishes mirror-formula structure and growth bounds for genus-zero Gromov-Witten invariants of projective complete intersections in the stated setting. These bounds imply positive-radius convergence for corresponding generating functions in these special cases, but only for projective complete intersections and specified insertions/classes, not the full closed symplectic case.
- **Title:** Some conjectures on the asymptotic behavior of Gromov-Witten invariants
- **Author:** Aleksey Zinger
- **Venue/year:** Handbook for Mirror Symmetry of Calabi-Yau and Fano Manifolds, Advanced Lectures in Mathematics 47, Higher Education Press/International Press, 2019, pp. 523-550; arXiv version 2016/2017
- **DOI/arXiv/URL:** arXiv 1610.02971; <https://arxiv.org/abs/1610.02971>
- **Progress:** Surveys and formulates energy-growth conjectures for GW invariants. It states that these asymptotic/energy-bound problems are outside current techniques in full generality and explains their connection to positive radii of convergence of GW generating functions. This is evidence that the general convergence problem remains unresolved.
- **Title:** Energy Bounds and Vanishing Results for the Gromov-Witten Invariants of the Projective Space
- **Author:** Aleksey Zinger
- **Venue/year:** Journal of Geometry and Physics 145 (2019), Article 103479
- **DOI/arXiv/URL:** DOI 10.1016/j.geomphys.2019.103479; arXiv 1801.07242; <https://arxiv.org/abs/1801.07242>
- **Progress:** Proves uniform energy bounds and vanishing results for arbitrary-genus GW

invariants of projective space, supporting the convergence philosophy in a very special family. It does not address arbitrary closed symplectic manifolds.

5.1.3.4. Why the full problem is not resolved

The exact problem is about every closed symplectic manifold after making $[\omega]$ sufficiently positive. The strongest convergence theorems found either:

- assume the target is a smooth projective variety and impose analytic/genus-zero convergence and semisimplicity hypotheses,
- prove convergence for special classes such as compact toric varieties, complete flag varieties, projective spaces, or projective complete intersections,
- or formulate the required energy-growth estimates as conjectures.

None of the citable literature found proves an unconditional analytic convergence theorem for the genus-zero Gromov-Witten prepotential for all closed symplectic manifolds. Therefore the correct status is **partially resolved**, not fully resolved.

5.1.4. Remaining open problems recorded by the database

5.1.4.1. Open problem 1

Let X be a smooth complex projective variety. Prove that its genus-zero primary Gromov–Witten potential has a nonempty domain of absolute convergence after the Novikov variables are specialized sufficiently far inside the ample cone.

Reference supplied by the database. David A. Cox and Sheldon Katz, *Mirror Symmetry and Algebraic Geometry*, Mathematical Surveys and Monographs **68**, American Mathematical Society, Providence, RI, 1999, Conjecture 8.1.2, DOI: [10.1090/surv/068](https://doi.org/10.1090/surv/068). See also Maxim Kontsevich and Yuri Manin, “Gromov-Witten Classes, Quantum Cohomology, and Enumerative Geometry,” *Communications in Mathematical Physics* **164** (1994), no. 3, 525–562, §4.3, DOI: [10.1007/BF02101490](https://doi.org/10.1007/BF02101490). Its still-open status is explicitly confirmed in Mark McLean, “Bounds for Genus Zero Gromov-Witten Invariants,” 2026, introduction following Theorem 1.1, arXiv: [2606.16997](https://arxiv.org/abs/2606.16997).

5.1.4.2. Open problem 2

Let (X, ω) be a compact Calabi–Yau threefold. Prove that there is a constant $C_X > 0$ such that every genus-zero, unmarked Gromov–Witten invariant satisfies

$$|\langle \gamma \rangle_{0, \beta}^X| \leq C_X^{\langle \omega, \beta \rangle} \quad \text{for all } \beta \in H_2(X; \mathbf{Z}).$$

Reference supplied by the database. Aleksey Zinger, “The Genus 0 Gromov–Witten Invariants of Projective Complete Intersections,” *Geometry & Topology* **18** (2014), no. 2, 1035–1114, Conjecture 1 and the Calabi–Yau threefold specialization discussed immediately afterward,

DOI: [10.2140/gt.2014.18.1035](https://doi.org/10.2140/gt.2014.18.1035), arXiv: [1106.1633](https://arxiv.org/abs/1106.1633). Restated as Conjecture 2.5 in Aleksey Zinger, “Some Conjectures on the Asymptotic Behavior of Gromov-Witten Invariants,” in *Handbook for Mirror Symmetry of Calabi-Yau and Fano Manifolds*, Advanced Lectures in Mathematics **47**, Higher Education Press and International Press, 2019, pp. 523–550, arXiv: [1610.02971](https://arxiv.org/abs/1610.02971). The exponential bound remains conjectural in Mark McLean, “Bounds for Genus Zero Gromov-Witten Invariants,” 2026, arXiv: [2606.16997](https://arxiv.org/abs/2606.16997).

5.1.5. Primary-source pointer

The originating ICM record is *Homological algebra of mirror symmetry* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=192>.

5.2. 102. Homological algebra of mirror symmetry [1994-ag-54]

Record: 1994-ag-54

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Homological algebra of mirror symmetry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocf.pdf#page=192>

5.2.1. Problem statement

Let (V, ω) be a closed $2n$ -dimensional symplectic manifold with $c_1(T_V) = 0$, and let W be a mirror-dual n -dimensional complex algebraic manifold. Let

$$LV = \{(x, L) : x \in V, L \subset T_x V \text{ is Lagrangian}\},$$

and fix the $\mathbb{Z}$ -covering $\overline{LV} \rightarrow LV$ that induces the universal cover on each Lagrangian-Grassmannian fiber. Let $F(V)$ denote Fukaya's A_∞ -category whose basic objects are Lagrangian submanifolds $\mathcal{L} \subset V$ equipped with a lift $\mathcal{L} \rightarrow \overline{LV}$ of the natural map $\mathcal{L} \rightarrow LV$; for transversely intersecting objects, the morphism space is the complex vector space generated by their intersection points, graded by the Maslov index, with differential and higher compositions defined by signed counts of pseudo-holomorphic discs weighted by $\exp(-\int \phi^* \omega)$. Allow, if necessary, a suitably enlarged Fukaya A_∞ -category, for example one whose objects include pairs $(\mathcal{L}, \mathcal{E})$ with $\mathcal{E}$ a unitary local system on $\mathcal{L}$. For an A_∞ -category C , let $\mathcal{D}^b(C)$ be the triangulated category $H(\text{Tw}(C))$ obtained from one-sided twisted complexes over C , and let $\text{Coh}(W)$ be the abelian category of coherent sheaves on W .

Prove or disprove that $\mathcal{D}^b(F(V))$, or the corresponding derived category of a suitable enlargement of $F(V)$, embeds as a full triangulated subcategory of $\mathcal{D}^b(\text{Coh}(W))$. In the strongest expected form, prove or disprove that this derived Fukaya category is equivalent to the bounded derived category of coherent sheaves on W .

5.2.2. Historical background, literature, and current progress

5.2.3. Verdict: partially resolved

5.2.3.1. Precise problem checked

The problem asks for Kontsevich's homological mirror symmetry conjecture in the original compact Calabi-Yau setting: for every closed symplectic manifold (V, ω) with $c_1(T_V) = 0$ and a mirror-dual complex algebraic manifold W , the derived Fukaya category, possibly after the standard enlargement by local systems/bounding data, should fully embed in $\mathcal{D}^b(\text{Coh}(W))$; in the strongest expected form it should be equivalent to $\mathcal{D}^b(\text{Coh}(W))$.

This is a universal statement over all such mirror pairs. A theorem for elliptic curves, K3 surfaces, torus fibrations, hypersurfaces, or Batyrev/Greene-Plesser families is therefore only par-

tial progress unless it covers all closed symplectic Calabi-Yau manifolds and all mirror-dual algebraic manifolds.

5.2.3.2. Status summary

The problem is not fully resolved. Live arXiv/web searches found many citable proofs for important special cases and broad families, but no theorem proving the statement for arbitrary closed symplectic Calabi-Yau manifolds with $c_1=0$ and arbitrary mirrors. Recent broad-family papers themselves describe the compact Calabi-Yau HMS theorem as known only in a limited list of cases before their new classes.

The correct status is therefore **partially resolved**, not fully open and not fully resolved.

5.2.3.3. Key literature and progress

1. Homological Algebra of Mirror Symmetry

- Authors: Maxim Kontsevich
- Venue/year: Proceedings of the International Congress of Mathematicians, Zurich 1994, Vol. I, Birkhauser, 1995, pp. 120-139
- arXiv/DOI/URL: arXiv:alg-geom/9411018, <https://arxiv.org/abs/alg-geom/9411018>, DOI <https://doi.org/10.48550/arXiv.alg-geom/9411018>
- Role: Original source of the homological mirror symmetry formulation. It formulates the conjecture; it does not prove the universal statement.

2. Categorical Mirror Symmetry: The Elliptic Curve

- Authors: Alexander Polishchuk, Eric Zaslow
- Venue/year: Advances in Theoretical and Mathematical Physics 2 (1998), 443-470
- arXiv/DOI/URL: arXiv:math/9801119, <https://arxiv.org/abs/math/9801119>, DOI <https://doi.org/10.4310/ATMP.1998.v2.n2.a9>
- Result/progress: Proves the HMS equivalence for elliptic curves, matching a suitable Fukaya category of the mirror two-torus with the derived category of coherent sheaves on the elliptic curve. This resolves the one-dimensional elliptic-curve case only.

3. Homological mirror symmetry for the quintic 3-fold

- Authors: Yuichi Nohara, Kazushi Ueda
- Venue/year: Geometry & Topology 16 (2012), 1967-2001
- arXiv/DOI/URL: arXiv:1103.4956, <https://arxiv.org/abs/1103.4956>, DOI <https://doi.org/10.2140/gt.2012.16.1967>
- Result/progress: Proves HMS for the quintic Calabi-Yau threefold. This is a major compact Calabi-Yau threefold example, but only one family/example, not the full conjecture.

4. Homological Mirror Symmetry for the Quartic Surface

- Authors: Paul Seidel
- Venue/year: Memoirs of the American Mathematical Society 236, no. 1116 (2015)
- arXiv/DOI/URL: arXiv:math/0310414, <https://arxiv.org/abs/math/0310414>, DOI <https://doi.org/10.1090/memo/1116>

- Result/progress: Proves Kontsevich’s mirror conjecture for the quartic surface on the symplectic side. This is a K3/quartic-surface case, not a general theorem for all Calabi-Yau symplectic manifolds.

5. Homological Mirror Symmetry for Calabi-Yau hypersurfaces in projective space

- Author: Nicholas Sheridan
- Venue/year: *Inventiones Mathematicae* 199 (2015), no. 1, 1-186
- arXiv/DOI/URL: arXiv:1111.0632, <https://arxiv.org/abs/1111.0632>, DOI <https://doi.org/10.1007/s00222-014-0507-2>
- Result/progress: Proves HMS for smooth d -dimensional Calabi-Yau hypersurfaces in projective space for $d > 2$ (including the quintic threefold). This is a broad projective-hypersurface family, but it is still far narrower than arbitrary closed symplectic Calabi-Yau manifolds.

6. Homological mirror symmetry without correction

- Author: Mohammed Abouzaid
- Venue/year: *Journal of the American Mathematical Society* 34 (2021), no. 4, 1059-1173
- arXiv/DOI/URL: arXiv:1703.07898, <https://arxiv.org/abs/1703.07898>, DOI <https://doi.org/10.1090/jams/973>
- Result/progress: For closed symplectic manifolds equipped with a Lagrangian torus fibration over a base $\mathbb{Q}$, proves a fully faithful embedding of the Fukaya category of tautologically unobstructed graded Lagrangians into the derived category of twisted coherent sheaves on a rigid analytic mirror, under the technical hypothesis $\pi_2(\mathbb{Q})=0$. This resembles the embedding form of the problem, but has strong extra hypotheses and does not cover arbitrary $c_1=0$ symplectic manifolds or all Fukaya objects.

7. Homological mirror symmetry for generalized Greene-Plesser mirrors

- Authors: Nick Sheridan, Ivan Smith
- Venue/year: *Inventiones Mathematicae* 224 (2021), 627-682
- arXiv/DOI/URL: arXiv:1709.08937, <https://arxiv.org/abs/1709.08937>, DOI <https://doi.org/10.1007/s00222-020-01018-w>
- Result/progress: Proves HMS for all Greene-Plesser mirror pairs and certain Batyrev-Borisov mirror pairs/Calabi-Yau complete intersections. These are important compact Calabi-Yau mirror families, but still special classes.

8. Homological mirror symmetry for Batyrev mirror pairs

- Authors: Sheel Ganatra, Andrew Hanlon, Jeff Hicks, Daniel Pomerleano, Nick Sheridan
- Venue/year: arXiv preprint, 2024, revised 2024
- arXiv/DOI/URL: arXiv:2406.05272, <https://arxiv.org/abs/2406.05272>, DOI <https://doi.org/10.48550/arXiv.2406.05272>
- Result/progress: Proves HMS for a large new class of Batyrev mirror pairs of Calabi-Yau hypersurfaces in toric varieties. The theorem has extra hypotheses, including dimension at least 3, characteristic conditions, smoothness/MPCS/connectedness conditions, and for the equivalence form assumes standard structural hypotheses on the relative Fukaya category. This is substantial progress but not a theorem for all closed symplectic Calabi-Yau manifolds.

9. Homological mirror symmetry for projective K3 surfaces

- Authors: Paul Hacking, Ailsa Keating
- Venue/year: arXiv preprint, 2025
- arXiv/DOI/URL: arXiv:2503.05680, <https://arxiv.org/abs/2503.05680>, DOI <https://doi.org/10.48550/arXiv.2503.05680>
- Result/progress: Proves HMS for projective K3 surfaces with integral Kahler class, with mirror a Picard-rank 19 K3 surface over $\mathbb{C}((q))$. This is a major K3 result, but it covers complex dimension two and integral Kahler classes, not the full arbitrary-dimensional statement.

10. A Gentle Introduction to Homological Mirror Symmetry

- Author: Raf Bocklandt
- Venue/year: London Mathematical Society Student Texts 99, Cambridge University Press, 2021
- DOI/URL: <https://doi.org/10.1017/9781108692458>
- Role: Survey/textbook background. It confirms HMS is a broad active program with many examples and variants, but it is not a resolving theorem for the universal compact Calabi-Yau statement.

5.2.3.4. Why no cited result fully resolves the problem statement

The problem is not restricted to elliptic curves, abelian varieties, K3 surfaces, hypersurfaces in projective or toric varieties, Batyrev/Greene-Plesser mirrors, or manifolds admitting a suitable torus fibration. The strongest recent broad-family results still impose geometric hypotheses and, in some cases, structural assumptions on the Fukaya category. Therefore their hypotheses do not imply the full statement for every closed symplectic $2n$ -manifold with $c_1(TV) = 0$ and every mirror-dual algebraic W .

What remains unresolved is a general construction/proof of the required full embedding or equivalence for arbitrary compact Calabi-Yau mirror pairs, including arbitrary closed symplectic Calabi-Yau manifolds, the correct universally enlarged Fukaya category, and the verification that it matches $D^b(\text{Coh}(W))$ without the special geometric hypotheses used in known cases.

5.2.4. Remaining open problems recorded by the database

5.2.4.1. Open problem 1

Let

$$V_\lambda = Z(Q_{1,\lambda}, Q_{2,\lambda}) \subset \mathbb{P}^5$$

be a smooth Calabi-Yau threefold given by the intersection of the two cubics

$$Q_{1,\lambda} = x_0^3 + x_1^3 + x_2^3 - 3\lambda x_3 x_4 x_5, \quad Q_{2,\lambda} = x_3^3 + x_4^3 + x_5^3 - 3\lambda x_0 x_1 x_2.$$

Prove homological mirror symmetry for V_λ : its split-closed derived Fukaya category is equivalent to the bounded derived category of coherent sheaves on either the Libgober-Teitelbaum mirror stack $\mathcal{U}_{LT,\lambda}$ or the derived-equivalent Batyrev-Borisov mirror stack $\mathcal{Z}_\lambda$.

Reference supplied by the database. Aimeric Malter, “A Derived Equivalence of the Libgober–Teitelbaum and the Batyrev–Borisov Mirror Constructions,” *International Mathematics Research Notices* **2024** (2024), no. 3, 2099–2137, introduction after Theorem 1.1 (explicit expectation that both mirror derived categories are equivalent to the Fukaya category of the intersection of two cubics), DOI: [10.1093/imrn/rnad081](https://doi.org/10.1093/imrn/rnad081).

5.2.5. Primary-source pointer

The originating ICM record is *Homological algebra of mirror symmetry* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=192>.

5.3. 103. Fibrations in Symplectic Topology [1998-geom-62]

Record: 1998-geom-62

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Fibrations in Symplectic Topology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=341>

5.3.1. Problem statement

Determine whether there is an intrinsic C^0 notion of closeness for symplectic structures ω and ω' , at least in dimension 4, not merely ordinary C^0 -closeness of cohomologous symplectic forms, such that sufficiently close symplectic structures are isotopic through symplectic structures.

In particular, determine whether the following type of intrinsic C^0 notion, or any analogous one, yields such a stability result: declare two symplectic structures ω and ω' to be ε -close on a compact domain K if

$$|w_G(U, \omega) - w_G(U, \omega')| \leq \varepsilon$$

for every open subset $U \subset K$, where $w_G(U, \omega)$ is the Gromov width of U with respect to ω , namely the supremum of the numbers πr^2 such that the standard ball $B^{2n}(r)$ symplectically embeds in (U, ω) .

5.3.2. Historical background, literature, and current progress

5.3.3. Status: partially resolved

5.3.3.1. Precise problem

The problem asks for a symplectic analogue of Colin's C^0 -stability theorem for contact structures: define an intrinsic C^0 notion of closeness for symplectic structures, preferably not just ordinary C^0 closeness of cohomologous closed 2-forms, such that sufficiently close symplectic structures on a fixed manifold are isotopic through symplectic structures. The question is asked at least in dimension 4. A proposed candidate is closeness measured by the Gromov widths of all open subsets in a compact domain:

$$|w_G(U, \omega) - w_G(U, \omega')| \leq \varepsilon$$

for every open $U \subset K$.

5.3.3.2. Verdict

I found no published or otherwise citable result that fully resolves this exact question. In particular, I found no theorem that defines an intrinsic C^0 topology on the space of symplectic

structures and proves the requested local isotopy stability, even in dimension 4, and no theorem proving that the all-open-subsets Gromov-width candidate implies such stability.

The problem is not completely untouched: there is substantial partial and adjacent progress in C^0 symplectic topology, capacity-based characterizations of symplectic maps, ordinary C^0 perturbations of symplectic forms in non-squeezing problems, and uniqueness/classification questions for symplectic forms. These results do not imply the full statement because their hypotheses concern maps rather than the topology on symplectic structures, use extrinsic C^0 norms on forms, add special deformation/nondegeneracy assumptions, or prove rigidity of capacities rather than isotopy of nearby structures.

5.3.3.3. Literature and progress

1. Dusa McDuff, “Fibrations in Symplectic Topology”, Documenta Mathematica, Extra Volume ICM 1998, Vol. I, 339-357, 1998. URL: <https://ems.press/content/book-chapter-files/27111>.

This is the original source of the question. In Question 3.2 McDuff asks for a symplectic analogue of Colin’s C^0 -stability result and explicitly says that the appropriate notion of C^0 -closeness is unclear. She also suggests the Gromov-width condition over all open subsets and states that the consequences of such a definition are not known.

2. Vincent Colin, “Stabilité topologique des structures de contact en dimension 3”, Duke Mathematical Journal 99(2), 329-351, 1999. DOI: 10.1215/S0012-7094-99-09912-X. URL: <https://projecteuclid.org/journals/duke-mathematical-journal/volume-99/issue-2/Stabilit%C3%A9-topologique-des-structures-de-contact-en-dimension-3/10.1215/S0012-7094-99-09912-X.full>.

This is the contact-stability theorem used as the model for McDuff’s question. It does not address symplectic structures.

3. Yong-Geun Oh and Stefan Muller, “The group of Hamiltonian homeomorphisms and C^0 -symplectic topology”, Journal of Symplectic Geometry 5(2), 167-219, 2007. arXiv:math/0402210. URL: <https://projecteuclid.org/journals/journal-of-symplectic-geometry/volume-5/issue-2/The-group-of-Hamiltonian-homeomorphisms-and-C0-symplectic-topology/jsg/1202004455.pdf>.

This develops C^0 symplectic topology for Hamiltonian homeomorphisms and continuous Hamiltonian flows. It gives a rigorous C^0 theory for maps on a fixed symplectic manifold, but it does not define an intrinsic C^0 topology on symplectic structures themselves or prove isotopy stability of nearby structures.

4. Vincent Humilière, Remi Leclercq, and Sobhan Seyfaddini, “Coisotropic rigidity and C^0 -symplectic geometry”, Duke Mathematical Journal 164(4), 767-799, 2015. DOI: 10.1215/00127094-2881701. arXiv:1305.1287. URL: <https://projecteuclid.org/journals/duke-mathematical-journal/volume-164/issue-4/----Custom-HTML----Coisotropic/10.1215/00127094-2881701.full>.

This proves C^0 -rigidity of coisotropic submanifolds and characteristic foliations under symplectic homeomorphisms, generalizing the Gromov-Eliashberg map-rigidity viewpoint. It is important C^0 rigidity, but its object is symplectic homeomorphisms/submanifolds, not closeness or isotopy of symplectic forms.

5. Lev Buhovsky and Emmanuel Opshtein, “Some quantitative results in C^0 symplectic geometry”, *Inventiones Mathematicae* 205(1), 1-56, 2016. DOI: 10.1007/s00222-015-0626-4. arXiv:1404.0875. URL: <https://hal.science/hal-03432689/>.

The paper studies behavior of symplectic homeomorphisms with respect to numerical invariants such as capacities and proves rigidity/flexibility results for submanifolds. This is relevant to the capacity side of the question, but it does not prove that Gromov-width closeness of symplectic structures forces isotopy.

6. Stefan Muller, “ C^0 -characterization of symplectic and contact embeddings and Lagrangian rigidity”, *International Journal of Mathematics* 30(9), 1950035, 48 pp., 2019. DOI: 10.1142/S0129167X19500356. arXiv:1607.03135. URL: <https://www.worldscientific.com/doi/10.1142/S0129167X19500356>.

This gives C^0 characterizations of symplectic embeddings/diffeomorphisms using shape and Lagrangian rigidity. It supports the idea that intrinsic symplectic data can characterize symplectic maps, but it is not a stability theorem for nearby symplectic structures.

7. Yasha Savelyev, “A remark on deformation of Gromov non-squeezing”, *Differential Geometry and its Applications*, 2025, article 102262. DOI: 10.1016/j.difgeo.2025.102262. arXiv:2302.13985. URL: <https://arxiv.org/abs/2302.13985>.

Savelyev proves that Gromov non-squeezing persists under certain ordinary C^0 perturbations of the symplectic form on the target, including special dimension-4 cases. This is the closest progress I found toward the Gromov-width/deformation direction. It still uses an extrinsic C^0 norm on forms and special non-squeezing hypotheses, and it does not produce an intrinsic topology with local isotopy stability.

8. Spencer Cattalani, “On Gromov Width under C^0 Deformations”, arXiv:2507.11466, 2025. DOI: 10.48550/arXiv.2507.11466. URL: <https://arxiv.org/abs/2507.11466>.

Cattalani constructs a bounded symplectic structure on $S^2 \times \mathbb{R}^4$ with infinite Gromov width, giving a higher-dimensional counterexample to a conjecture of Savelyev, and proves positive results for special classes. This clarifies the limits of ordinary C^0 deformation statements about Gromov width. It does not settle McDuff’s intrinsic C^0 stability problem, and the counterexample is not a theorem about the proposed all-open-subsets width topology on compact domains.

9. Dietmar Salamon, “Uniqueness of Symplectic Structures”, *Acta Mathematica Vietnamica* 38(1), 123-144, 2013. DOI: 10.1007/s40306-012-0004-x. arXiv:1211.2940. URL: <https://arxiv.org/abs/1211.2940>.

This survey records what is known about uniqueness, isotopy, homotopy, and deformation equivalence of symplectic forms in many examples. It gives useful background and many partial uniqueness results, especially in dimension 4, but not an intrinsic C^0 local stability theorem.

10. Shengzhen Ning, “Cohomologous symplectic forms with different Gromov widths”, arXiv:2505.09550, 2025. DOI: 10.48550/arXiv.2505.09550. URL: <https://arxiv.org/abs/2505.09550>.

This shows that in dimensions at least 6 there are closed manifolds with cohomologous symplectic forms of different Gromov widths. It is related to the role of Gromov width in distinguishing symplectic structures, but it is higher-dimensional and does not address the dimension-4 intrinsic C^0 stability question.

5.3.3.4. What remains unresolved

The full problem remains unresolved: find, or rule out, an intrinsic C^0 notion of closeness for symplectic structures, especially in dimension 4, for which sufficiently close symplectic structures are isotopic through symplectic structures. In particular, the literature found does not determine whether the proposed topology using $w_G(U, \omega)$ for all open $U \subset K$ has the desired stability property.

5.3.4. Remaining open problems recorded by the database

5.3.4.1. Open problem 1

Let ω_0 be the product symplectic form on $S^2 \times \mathbb{R}^2$, normalized so that $\int_{S^2} \omega_0 = 1$, and use its standard product metric to define the comass norm. If ω is any symplectic form with $\|\omega - \omega_0\| < \infty$, must its Gromov width satisfy $c_G(\omega) \leq 1 + \|\omega - \omega_0\|$?

Reference supplied by the database. Yasha Savelyev, “A Remark on Deformation of Gromov Non-squeezing,” *Differential Geometry and its Applications* **100** (2025), Article 102262, Conjecture 1 (specialized to the still-unresolved four-dimensional case), DOI: 10.1016/j.difgeo.2025.102262, arXiv:2302.13985, <https://doi.org/10.1016/j.difgeo.2025.102262>. Current status is recorded in Spencer Cattalani, “On Gromov Width under C^0 Deformations,” accepted in *Journal of Symplectic Geometry*, 2026 preprint, Conjecture 1, Theorem 2, and Remark 8 (counterexample in dimensions at least six, while the construction does not settle the four-dimensional case), arXiv:2507.11466, <https://arxiv.org/abs/2507.11466>.

5.3.5. Primary-source pointer

The originating ICM record is *Fibrations in Symplectic Topology* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=341>.

5.4. 104. Some Open Problems in Dynamical Systems [1998-geom-69]

Record: 1998-geom-69

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Some Open Problems in Dynamical Systems

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=799>

5.4.1. Problem statement

Let (M^{2n}, ω) be a compact C^∞ symplectic manifold with Euler-Poincaré characteristic $\chi(M^{2n}) = 0$, and suppose that M^{2n} does not admit a fibration over S^1 . Prove that every symplectic diffeomorphism $f : (M^{2n}, \omega) \rightarrow (M^{2n}, \omega)$ that is homotopic to the identity map has a fixed point. In fact, prove that the number of fixed points of f is at least

$$\min_X \#\{p \in M^{2n} : X(p) = 0\},$$

where the minimum is taken over all smooth symplectic vector fields X on (M^{2n}, ω) , equivalently over all vector fields X for which $i_X \omega = v$ for some closed 1-form v on M^{2n} .

5.4.2. Historical background, literature, and current progress

5.4.3. Verdict: partially resolved

5.4.3.1. Precise problem

The problem is Michael Herman's Conjecture 5.8 from "Some Open Problems in Dynamical Systems" (Documenta Mathematica, Extra Volume ICM 1998, Vol. II, 797-808). It asks whether a compact smooth symplectic manifold (M^{2n}, ω) with $\chi(M) = 0$ and no fibration over S^1 has the following fixed point property: every symplectic diffeomorphism $f : (M, \omega) \rightarrow (M, \omega)$ homotopic to the identity has a fixed point, and in fact

$$\#\text{Fix}(f) \geq \min_X \#\{p : X(p) = 0\},$$

where X ranges over symplectic vector fields, equivalently $i_X \omega$ ranges over closed 1-forms.

Tischler's theorem explains the fibration condition: a closed manifold carrying a nowhere-zero closed 1-form fibers over S^1 .

5.4.3.2. Current literature status

No published or otherwise citable result I found proves Herman's full statement in this generality. Existing results prove important special cases and weaker lower bounds, mainly for

symplectomorphisms lying in the identity component through symplectic isotopies, usually under nondegeneracy hypotheses, and with lower bounds given by Novikov homology rather than by the actual minimum number of zeros of closed 1-forms. This does not cover all symplectic diffeomorphisms merely homotopic to the identity, and it does not prove the stated zero-count lower bound in all cases.

5.4.3.3. Key sources and what they prove

1. Some Open Problems in Dynamical Systems

Michael R. Herman. Documenta Mathematica, Extra Volume ICM 1998, Vol. II, 797-808, 1998. URL: <https://ems.press/content/book-chapter-files/27206>

This is the original source. Section 5.8 states the conjecture exactly: if $\chi(M^{2n}) = 0$ and M^{2n} does not fiber over S^1 , then every symplectic diffeomorphism homotopic to the identity has a fixed point, with the stronger lower bound by the minimal number of zeros of a symplectic vector field. Herman also notes that the conjecture is easy for f C^1 -near the identity.

2. On fibering certain foliated manifolds over S^1

David Tischler. Topology 9, 153-154, 1970. DOI: 10.1016/0040-9383(70)90037-6.

Gives the standard implication used in Herman's formulation: a nowhere-zero closed 1-form on a compact manifold yields a fibration over S^1 . This identifies why the no-fibration hypothesis forces zeros for every closed 1-form, but it is not a fixed-point theorem for symplectomorphisms.

3. Symplectic fixed points, the Calabi invariant and Novikov homology

Hông Vân Lê and Kaoru Ono. Topology 34(1), 155-176, 1995. DOI: 10.1016/0040-9383(94)E0015-C. URL: https://archive.mpim-bonn.mpg.de/4017/1/preprint_1993_27.pdf

Proves a Floer-Novikov lower bound for nondegenerate symplectomorphisms isotopic to the identity through symplectomorphisms, under monotonicity-type assumptions. The lower bound is the sum of Novikov Betti numbers associated to the flux/Calabi invariant. This is substantial partial progress, but it has extra hypotheses and does not cover arbitrary homotopic-to-identity symplectomorphisms or Herman's stated minimum-zero bound.

4. Fixed points of non-Hamiltonian symplectomorphisms

Luisa D. Stelling. Journal of Geometric Analysis 11, 693-716, 2001. DOI: 10.1007/BF02930763.

URL: <https://link.springer.com/article/10.1007/BF02930763>

Studies an Arnold-type fixed point problem for non-Hamiltonian symplectic maps and obtains lower bounds for maps with sufficiently small Calabi invariant. This is a small-flux/small-Calabi special case, not the full Herman conjecture.

5. Floer-Novikov cohomology and symplectic fixed points

Kaoru Ono. Journal of Symplectic Geometry 3, 545-563, 2005. DOI: 10.4310/JSG.2005.v3.n4.a3.

Project Euclid URL: <http://projecteuclid.org/euclid.jsg/1154467629>

Extends Floer-Novikov methods and gives a lower bound for nondegenerate time-one maps of symplectic isotopies on compact symplectic manifolds in terms of minimal Novikov numbers. This improves the generality of the symplectic-isotopy case, but the lower bound can be weaker than the minimum number of zeros of closed 1-forms and still does not address all homotopic-to-identity symplectomorphisms.

6. **The Calabi invariant and the least number of periodic solutions of locally Hamiltonian equations**
 Hồng Vân Lê. arXiv:1511.00638, 2015. DOI: 10.48550/arXiv.1511.00638. URL: <https://arxiv.org/abs/1511.00638>
 Proves lower bounds for nondegenerate locally Hamiltonian equations on compact symplectic manifolds in terms of Novikov homology associated to the Calabi invariant, generalizing the homological Arnold conjecture. This is again a Floer-Novikov/Novikov-Betti bound, not Herman's full zero-count statement for all homotopic-to-identity symplectic diffeomorphisms.
7. **Floer-Novikov cohomology and symplectic fixed points, revisited**
 Kaoru Ono and Hồng Vân Lê. Proceedings of the International Geometry Center 13(4), 89-115, 2020. DOI: 10.15673/tmgc.v13i4.1809. URL: <https://journals.ontu.edu.ua/index.php/geometry/article/view/1809>
 States sharper modern versions: for a smooth symplectic isotopy $\{\phi_t\}$ from the identity with nondegenerate ϕ_1 , the number of fixed points is bounded below by Novikov-Betti numbers of the actual flux. The authors explicitly frame this as a result for symplectomorphisms symplectically isotopic to the identity. It is strong partial progress but does not cover maps merely homotopic to the identity.
8. **A Symplectic Fixed Point Theorem and Floer-Novikov Theory in the Presence of an Infinite Fundamental Group**
 Hồng Vân Lê and Kaoru Ono. Vietnam Journal of Mathematics 54, 441-450, 2026; published online 2025. DOI: 10.1007/s10013-025-00772-9. URL: <https://link.springer.com/article/10.1007/s10013-025-00772-9>
 Gives a newer Floer-Novikov existence result for 1-periodic orbits on certain product symplectic manifolds, including situations where a relevant Floer-Novikov homology group vanishes in a degree. This is additional partial progress, but it is explicitly for certain product manifolds and does not resolve Herman's all-manifolds/all-homotopic-symplectomorphisms statement.
9. **Arnold conjecture and Gromov-Witten invariant**
 Kenji Fukaya and Kaoru Ono. Topology 38(5), 933-1048, 1999. DOI: 10.1016/S0040-9383(98)00042-1.
 Together with the next item, this proves the homological Arnold conjecture for Hamiltonian diffeomorphisms on general compact symplectic manifolds. It covers the Hamiltonian subcase of Herman's problem, but Herman's conjecture is about all symplectic diffeomorphisms homotopic to the identity and uses closed 1-forms/symplectic vector fields, so this is only partial progress.
10. **Floer homology and Arnold conjecture**
 Gang Liu and Gang Tian. Journal of Differential Geometry 49(1), 1-74, 1998. DOI: 10.4310/jdg/1214460936. URL: <https://projecteuclid.org/euclid.jdg/1214460936>
 Another proof of the homological Arnold conjecture for Hamiltonian diffeomorphisms. It does not extend to arbitrary non-Hamiltonian symplectic diffeomorphisms homotopic to the identity.
11. **Closed 1-forms with at most one zero**
 Michael Farber and Dirk Schütz. Topology 45(3), 465-473, 2006. DOI: 10.1016/j.top.2005.06.006. URL: <https://maths.dur.ac.uk/users/dirk.schuetz/d10.pdf>
 Shows that every nonzero real cohomology class on a closed connected smooth manifold has a closed 1-form with at most one zero. This clarifies Herman's lower-bound term: when $H^1(M; \mathbb{R}) \neq 0$ and M does not fiber over S^1 , the minimum over nonzero closed

1-forms is often just one. This topological result does not supply the missing symplectic fixed point theorem.

5.4.3.4. What remains unresolved

The unresolved part is the full Herman conjecture: proving, for every compact symplectic (M, ω) with $\chi(M) = 0$ and no fibration over S^1 , that every symplectic diffeomorphism homotopic to the identity has a fixed point, and proving the stated lower bound by the minimum number of zeros of symplectic vector fields.

The main gaps in the existing literature are:

- results generally require f to be the time-one map of a symplectic isotopy from the identity, not merely homotopic to the identity;
- results generally assume nondegenerate fixed points or prove homological/Novikov-Betti lower bounds;
- Novikov-Betti bounds are weaker than the exact minimum-zero bound appearing in the problem statement;
- recent Floer-Novikov refinements still cover special classes or product situations rather than all compact symplectic manifolds in Herman's statement.

Therefore the correct status is **partially resolved**, not fully resolved.

5.4.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

5.4.5. Primary-source pointer

The originating ICM record is *Some Open Problems in Dynamical Systems* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=799>.

5.5. 105. Polterovich [1998-geom-73]

Record: 1998-geom-73

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Polterovich

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=403>

5.5.1. Problem statement

For every connected symplectic manifold (M, Ω) without boundary, let $\text{Ham}(M, \Omega)$ be the group of Hamiltonian diffeomorphisms. For a smooth Hamiltonian path $\{f_t\}_{t \in [0,1]}$ from ϕ to ψ , generated by a normalized Hamiltonian $F(x, t)$ (compactly supported in M if M is noncompact, and of zero mean with respect to the symplectic volume if M is closed), set

$$\text{length}\{f_t\} = \int_0^1 \left(\max_M F(\cdot, t) - \min_M F(\cdot, t) \right) dt,$$

and let $\rho(\phi, \psi)$ be the infimum of this length over all such paths. Prove that $(\text{Ham}(M, \Omega), \rho)$ has infinite diameter; equivalently, the distances $\rho(\text{id}, \phi)$ are unbounded as ϕ ranges over $\text{Ham}(M, \Omega)$.

5.5.2. Historical background, literature, and current progress

5.5.3. Verdict: partially resolved

5.5.3.1. Precise problem

The problem asks for the general Hofer diameter conjecture: for every connected symplectic manifold without boundary, with compactly supported Hamiltonians in the noncompact case and normalized Hamiltonians in the closed case, is the Hofer metric on $\text{Ham}(M, \Omega)$ unbounded, equivalently does $\text{Ham}(M, \Omega)$ have infinite Hofer diameter?

This matches Polterovich's ICM formulation:

- Title: Geometry on the Group of Hamiltonian Diffeomorphisms
- Author: Leonid Polterovich
- Venue/year: Documenta Mathematica, Extra Volume ICM 1998, Vol. II, pp. 401-410, 1998
- DOI/arXiv/URL: <https://ems.press/content/book-chapter-files/27169>
- Relevant item: Conjecture 6.A states that the diameter of $\text{Ham}(M, \Omega)$ with respect to Hofer's metric is infinite.
- Status relevance: This is the original all-manifolds problem, not a theorem.

5.5.3.2. Evidence that the full problem is not resolved

Recent literature still states the all-symplectic-manifolds assertion as open in general.

- Title: Hofer geometry of A3-configurations
- Author: Adrian Dawid
- Venue/year: arXiv preprint, 2025 version
- DOI/arXiv/URL: arXiv:2402.16773, <https://arxiv.org/abs/2402.16773>
- Relevant statement: The introduction says that the Hofer diameter conjecture asserts infinite Hofer diameter for all symplectic manifolds and that this question is still open in general.
- Applicability check: The paper proves quasi-flats for Liouville domains with an A3-configuration of exact Lagrangian spheres. This is a substantial partial result, but it has strong Liouville/exact/Lagrangian-configuration hypotheses and does not imply the problem statement for every connected boundaryless symplectic manifold.
- Title: Spectral diameter of negatively monotone manifolds
- Author: Yuhan Sun
- Venue/year: arXiv preprint 2024; listed by the author's page as accepted for Journal of Symplectic Geometry
- DOI/arXiv/URL: arXiv:2410.21416, <https://arxiv.org/abs/2410.21416>
- Relevant statement: The introduction describes the Hofer diameter conjecture as the question whether the Hofer norm is unbounded for any closed symplectic manifold, and proves quasi-isometric embeddings under negative-monotonicity and heavy-Lagrangian hypotheses.
- Applicability check: The theorem assumes a closed negatively monotone symplectic manifold of dimension at least six containing an incompressible heavy Lagrangian. These hypotheses are far narrower than the stated all connected symplectic manifolds.

5.5.3.3. Main partial progress

- Title: Hofer's diameter and Lagrangian intersections
- Author: Leonid Polterovich
- Venue/year: International Mathematics Research Notices, 1998, no. 4, pp. 217-223
- DOI/arXiv/URL: DOI 10.1155/S1073792898000178; arXiv:math/9801054, <https://arxiv.org/abs/math/9801054>
- Result: Proves that the Hamiltonian diffeomorphism group of the 2-sphere has infinite Hofer diameter.
- Applicability check: This resolves the special case $M = S^2$, not arbitrary symplectic manifolds.
- Title: Hofer's L-infinity-geometry: energy and stability of Hamiltonian flows, parts I and II
- Authors: Francois Lalonde and Dusa McDuff

- Venue/year: *Inventiones Mathematicae* 122 (1995), pp. 1-33 and 35-69; erratum *Inventiones Mathematicae* 123 (1996), p. 613
- DOI/arXiv/URL: DOI 10.1007/BF01231437 and 10.1007/BF01231438; arXiv:math/9503227 and arXiv:math/9503228
- Result: Early Hofer-geometry results used in Polterovich's ICM survey to establish infinite diameter for closed oriented surfaces of positive genus and related cases.
- Applicability check: Important surface and stability/geodesic results, but not a theorem for all symplectic manifolds.
- Title: On the action spectrum for closed symplectically aspherical manifolds
- Author: Matthias Schwarz
- Venue/year: *Pacific Journal of Mathematics* 193 (2000), no. 2, pp. 419-461
- DOI/arXiv/URL: <https://msp.org/pjm/2000/193-2/pjm-v193-n2-p10-p.pdf>
- Result: Develops spectral invariants for closed symplectically aspherical manifolds; Section 5.1 gives examples and methods yielding infinite Hofer diameter in symplectically aspherical settings.
- Applicability check: Requires symplectic asphericity or related hypotheses, so it does not cover rational/monotone/non-aspherical closed manifolds in general.
- Title: A comparison of Hofer's metrics on Hamiltonian diffeomorphisms and Lagrangian submanifolds
- Author: Yaron Ostrover
- Venue/year: *Communications in Contemporary Mathematics* 5 (2003), no. 5, pp. 803-811
- DOI/arXiv/URL: DOI 10.1142/S0219199703001154; arXiv:math/0207070, <https://arxiv.org/abs/math/0207070>
- Result: As a corollary, proves infinite Hofer diameter for closed symplectic manifolds with $\pi_2(M) = 0$.
- Applicability check: The hypothesis $\pi_2(M) = 0$ excludes many closed symplectic manifolds, so it is partial.
- Title: Monodromy in Hamiltonian Floer theory
- Author: Dusa McDuff
- Venue/year: *Commentarii Mathematici Helvetici* 85 (2010), no. 1, pp. 95-133
- DOI/arXiv/URL: DOI 10.4171/CMH/189; arXiv:0801.1328, <https://arxiv.org/abs/0801.1328>
- Result: Gives conditions under which asymptotic spectral invariants descend from the universal cover to $\text{Ham}(M, \omega)$; when they descend, $\text{Ham}(M, \omega)$ has infinite Hofer diameter. The paper states this establishes infinite diameter in many new cases.
- Applicability check: The result depends on extra conditions involving Chern class, quantum homology, minimal Chern number, or Gromov-Witten behavior; it is not a universal theorem.

- Title: Hofer’s metrics and boundary depth
- Author: Michael Usher
- Venue/year: Annales scientifiques de l’Ecole Normale Supérieure (4) 46 (2013), no. 1, pp. 57-129
- DOI/arXiv/URL: DOI 10.24033/asens.2185; arXiv:1107.4599, <https://arxiv.org/abs/1107.4599>
- Result: Theorem 1.1 proves infinite Hofer diameter, indeed infinite-dimensional quasi-isometric embeddings, for closed symplectic manifolds admitting a nonconstant autonomous Hamiltonian whose contractible closed orbits are all constant.
- Applicability check: This dynamical hypothesis covers many new examples but fails for other manifolds; Usher explicitly notes that it is not yet known whether Hofer’s metric is always unbounded.
- Title: Hofer geometry via toric degeneration
- Author: Yusuke Kawamoto
- Venue/year: Mathematische Annalen 390 (2024), pp. 721-762
- DOI/arXiv/URL: DOI 10.1007/s00208-023-02775-8; arXiv:2210.14726, <https://arxiv.org/abs/2210.14726>
- Result: Uses toric degeneration to produce homogeneous quasimorphisms and Hofer-geometry consequences for quadric hypersurfaces and del Pezzo surfaces.
- Applicability check: This gives further important closed examples, not all symplectic manifolds.

5.5.3.4. What remains unresolved

No citable theorem found proves the full statement for every connected symplectic manifold without boundary. The available literature proves infinite Hofer diameter in many classes, including closed surfaces, S^2 , closed symplectically aspherical manifolds, closed manifolds satisfying McDuff’s spectral-invariant descent criteria, manifolds satisfying Usher’s autonomous-Hamiltonian dynamical condition, and several more recent Liouville, monotone, negatively monotone, toric, and heavy-Lagrangian settings.

The gap is the absence of a theorem covering all closed symplectic manifolds and all non-compact connected symplectic manifolds in the exact formulation of the problem, without extra asphericity, monotonicity, Floer-theoretic descent, dynamical, exactness, or Lagrangian-configuration hypotheses. Therefore the correct status is partially resolved.

5.5.4. Remaining open problems recorded by the database

5.5.4.1. Open problem 1

Does the Hamiltonian diffeomorphism group of the monotone one-point blow-up of $\mathbb{C}P^2$ have infinite diameter with respect to Hofer’s metric?

Reference supplied by the database. Dusa McDuff, “Monodromy in Hamiltonian Floer theory,” *Commentarii Mathematici Helvetici* **85** (2010), no. 1, 95–133, discussion following Corollary 1.5 (the paper explicitly states that infinite Hofer diameter is unknown for the monotone one-point blow-up of $\mathbb{CP}^2$); DOI: [10.4171/CMH/189](https://doi.org/10.4171/CMH/189); arXiv: [0801.1328](https://arxiv.org/abs/0801.1328).

5.5.5. Primary-source pointer

The originating ICM record is *Polterovich* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=403>.

5.6. 106. Geometric Physics [1998-ag-74]

Record: 1998-ag-74

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Geometric Physics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=539>

5.6.1. Problem statement

Let M be a Calabi-Yau manifold of complex dimension d , let W be its mirror Calabi-Yau manifold, and let V be a holomorphic vector bundle on M with total Chern class $c(V) \in \bigoplus_p H^{p,p}(M)$. Represent the components of $c(V)$ by Poincaré-dual holomorphic cycles in M , equivalently by the corresponding collection of D-branes wrapped on those cycles.

Prove or disprove that the mirror of this collection representing V is a real d -dimensional submanifold $C \subset W$ such that holomorphicity of V corresponds to C being Lagrangian with respect to the Kähler form on W , and stability of V corresponds to C being a minimal-area Lagrangian cycle.

5.6.2. Historical background, literature, and current progress

5.6.3. Verdict: partially resolved

5.6.3.1. Precise problem

The problem asks for a general mirror-symmetry theorem for Calabi-Yau mirror pairs: given a Calabi-Yau manifold M of complex dimension d , its mirror W , and a stable holomorphic vector bundle V on M , does the D-brane charge represented by the Chern-class components of V correspond to a real d -dimensional submanifold $C \subset W$, with holomorphicity of V equivalent to C being Lagrangian and stability of V equivalent to C being minimal-area, more precisely a supersymmetric or special Lagrangian cycle?

This is stronger than ordinary homological mirror symmetry: it asks for an object-level geometric representative and for the stability/minimal-special-Lagrangian correspondence in all Calabi-Yau cases.

5.6.3.2. Original source

- Title: Extending Mirror Conjecture to Calabi-Yau with Bundles
- Author: Cumrun Vafa
- Venue: Communications in Contemporary Mathematics 1(1), 65-70
- Year: 1999

- DOI/arXiv/URL: DOI 10.1142/S0219199799000043; arXiv:hep-th/9804131; <https://arxiv.org/abs/hep-th/9804131>
- Relevance: This is the original conjectural source. It defines mirror symmetry for a Calabi-Yau with a stable bundle in terms of supersymmetric cycles on the mirror. It is a physics proposal, not a proof for arbitrary Calabi-Yau manifolds or arbitrary vector bundles.
- Title: Geometric Physics
- Author: Cumrun Vafa
- Venue: Proceedings of the International Congress of Mathematicians, Vol. I (Berlin, 1998), 537-556; Documenta Mathematica
- Year: 1998
- DOI/arXiv/URL: arXiv:hep-th/9810149; <https://arxiv.org/abs/hep-th/9810149>
- Relevance: This ICM survey contains the distinctive “extension to bundles” discussion underlying the problem statement wording and explicitly treats checks as ongoing rather than as a completed theorem.

5.6.3.3. Partial progress

- Title: Mirror Symmetry is T-Duality
- Authors: Andrew Strominger, Shing-Tung Yau, Eric Zaslow
- Venue: Nuclear Physics B 479, 243-259
- Year: 1996
- DOI/arXiv/URL: DOI 10.1016/0550-3213(96)00434-8; arXiv:hep-th/9606040; <https://arxiv.org/abs/hep-th/9606040>
- Progress: Gives the SYZ framework in which mirrors arise from dual special Lagrangian torus fibrations, with the mirror as a moduli space of such cycles with flat connections. This is foundational support for the statement, but it is not a proof for all Calabi-Yau manifolds or all stable bundles.
- Title: Categorical Mirror Symmetry: The Elliptic Curve
- Authors: Alexander Polishchuk, Eric Zaslow
- Venue: Advances in Theoretical and Mathematical Physics 2(2), 443-470
- Year: 1998
- DOI/arXiv/URL: DOI 10.4310/ATMP.1998.v2.n2.a9; arXiv:math/9801119; <https://arxiv.org/abs/math/9801119>
- Progress: Proves Kontsevich’s categorical mirror equivalence for elliptic curves and gives a detailed dictionary between coherent sheaves and Lagrangian objects. This is a genuine one-dimensional special case, not a general theorem for higher-dimensional Calabi-Yau manifolds or arbitrary stable bundles.
- Title: From Special Lagrangian to Hermitian-Yang-Mills via Fourier-Mukai Transform
- Authors: Naichung Conan Leung, Shing-Tung Yau, Eric Zaslow

- Venue: Advances in Theoretical and Mathematical Physics 4(6), 1319-1341; also AMS/IP Studies in Advanced Mathematics 23, 209-225
- Year: 2000/2001
- DOI/arXiv/URL: DOI 10.4310/ATMP.2000.v4.n6.a5; arXiv:math/0005118; <https://arxiv.org/abs/math/0005118>
- Progress: Verifies Vafa's mirror conjecture with bundles in a semi-flat special case. The hypotheses are restrictive: the mirror pair is assumed to be given by dual flat torus fibrations, singular fibers are avoided, and the main construction treats A-cycles that are sections, with multisections discussed separately. Therefore it does not cover the full statement.
- Title: Special Lagrangians, stable bundles and mean curvature flow
- Authors: R. P. Thomas, S.-T. Yau
- Venue: Communications in Analysis and Geometry 10(5), 1075-1113
- Year: 2002
- DOI/arXiv/URL: DOI 10.4310/CAG.2002.v10.n5.a8; arXiv:math/0104197; <https://arxiv.org/abs/math/0104197>
- Progress: Develops the Thomas-Yau stability and Lagrangian mean-curvature-flow conjectural framework. It proves uniqueness of special Lagrangians in Hamiltonian deformation classes under mild conditions and proves the conjecture in some symmetric examples, but it does not prove the general stability/existence correspondence.
- Title: Conjectures on Bridgeland stability for Fukaya categories of Calabi-Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow
- Author: Dominic Joyce
- Venue: EMS Surveys in Mathematical Sciences 2(1), 1-62
- Year: 2015
- DOI/arXiv/URL: DOI 10.4171/EMSS/8; arXiv:1401.4949; <https://arxiv.org/abs/1401.4949>
- Progress: Gives a modern conjectural formulation using Bridgeland stability on Fukaya categories and Lagrangian mean curvature flow with surgeries. Its role is to refine the conjecture; it is not a resolving theorem.
- Title: Moment Maps, Nonlinear PDE and Stability in Mirror Symmetry, I: Geodesics
- Authors: Tristan C. Collins, Shing-Tung Yau
- Venue: Annals of PDE 7, article 11
- Year: 2021
- DOI/arXiv/URL: DOI 10.1007/s40818-021-00100-7; arXiv:1811.04824; <https://arxiv.org/abs/1811.04824>
- Progress: Studies the deformed Hermitian-Yang-Mills equation, mirror to the special Lagrangian equation, from a variational and stability viewpoint. It proves geodesic and obstruction results in hypercritical/line-bundle settings and discusses toric Landau-Ginzburg mirrors. It also makes clear that the full Bridgeland-stability/PDE correspondence and higher-rank bundle case remain open.

5.6.3.4. Why this is not fully resolved

No located theorem proves the exact statement for arbitrary Calabi-Yau mirror pairs and arbitrary stable holomorphic vector bundles. The strongest direct verification, Leung-Yau-Zaslow, assumes a semi-flat dual torus-fibration model and mainly section-type A-cycles. The modern stability side is still formulated through Thomas-Yau/Joyce/Bridgeland conjectures and through the deformed Hermitian-Yang-Mills program, where general Bridgeland stability conditions in Calabi-Yau dimension greater than 2 and higher-rank non-abelian dHYM remain major unresolved issues.

Thus the literature contains substantial special cases and conceptual progress, but not a published or otherwise citable result covering the full statement/all cases asked in this problem.

5.6.4. Remaining open problems recorded by the database

5.6.4.1. Open problem 1

Let L be a graded Lagrangian submanifold of a Calabi-Yau manifold X . Normalize the phase of the holomorphic volume form so that the cohomological phase of L is zero. Suppose that either (i) for every graded Lagrangian connect-sum decomposition $L_1 \# L_2 \approx L$, the interval $[\phi(L_1), \phi(L_2)]$ is not contained in $(\inf_L \theta_L, \sup_L \theta_L)$, or (ii) for every such decomposition, the volume of L is at most the sum of the cohomological volumes of L_1 and L_2 . Prove that Lagrangian mean-curvature flow starting from L exists for all time and converges to the unique special Lagrangian in the Hamiltonian deformation class of L .

Reference supplied by the database. Richard P. Thomas and Shing-Tung Yau, “Special Lagrangians, stable bundles and mean curvature flow,” *Communications in Analysis and Geometry* **10** (2002), no. 5, 1075–1113, Conjecture 7.3 (conditions (7.1) and (7.2)). DOI: [10.4310/CAG.2002.v10.n5.a8](https://doi.org/10.4310/CAG.2002.v10.n5.a8). arXiv: [math/0104197](https://arxiv.org/abs/math/0104197).

5.6.4.2. Open problem 2

For a compact or suitably convex-at-infinity Calabi-Yau manifold (M, J, g, Ω) , construct a natural Bridgeland stability condition $(Z, \mathcal{P})$ on its derived Fukaya category such that the central charge is $Z([L, E, b]) = \int_L \Omega$ and every brane supported on a special Lagrangian of phase $e^{i\pi\phi}$ is semistable of phase ϕ .

Reference supplied by the database. Dominic Joyce, “Conjectures on Bridgeland stability for Fukaya categories of Calabi-Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow,” *EMS Surveys in Mathematical Sciences* **2** (2015), no. 1, 1–62, Conjecture 3.2(a)–(b). DOI: [10.4171/EMSS/8](https://doi.org/10.4171/EMSS/8). arXiv: [1401.4949](https://arxiv.org/abs/1401.4949).

5.6.4.3. Open problem 3

Let A_α be the category of objects of the derived Fukaya category represented by unobstructed graded Lagrangian branes whose phase functions take values in $(\pi\alpha, \pi(\alpha+1))$, and let $\overline{A}_\alpha$ be its isomorphism closure. Prove that $\overline{A}_\alpha$ is the heart of a bounded t -structure and that slope semistability in this heart, with central charge $Z([L, E, b]) = \int_L \Omega$, gives the expected Bridgeland stability condition.

Reference supplied by the database. Dominic Joyce, “Conjectures on Bridgeland stability for Fukaya categories of Calabi–Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow,” *EMS Surveys in Mathematical Sciences* **2** (2015), no. 1, 1–62, Conjecture 3.5. DOI: [10.4171/EMSS/8](https://doi.org/10.4171/EMSS/8). arXiv: [1401.4949](https://arxiv.org/abs/1401.4949).

5.6.4.4. Open problem 4

Prove Joyce’s neck-pinch conjecture for Lagrangian mean-curvature flow with surgeries in Calabi–Yau m -folds, $m \geq 2$: a Lawlor neck may shrink at a finite singular time to two transversely intersecting special-Lagrangian planes and continue as an immersed Lagrangian brane with the prescribed local system and bounding cochain, and such neck pinches are generic under sufficiently small Hamiltonian perturbations of the initial Lagrangian.

Reference supplied by the database. Dominic Joyce, “Conjectures on Bridgeland stability for Fukaya categories of Calabi–Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow,” *EMS Surveys in Mathematical Sciences* **2** (2015), no. 1, 1–62, Conjecture 3.16. DOI: [10.4171/EMSS/8](https://doi.org/10.4171/EMSS/8). arXiv: [1401.4949](https://arxiv.org/abs/1401.4949).

5.6.4.5. Open problem 5

For a generic unobstructed compact graded Lagrangian brane (L, E, b) in a compact or suitably convex-at-infinity Calabi–Yau manifold, construct the unique all-time Lagrangian mean-curvature flow with finitely many prescribed surgeries that stays in the isomorphism class of (L, E, b) in an enlarged derived Fukaya category and converges, as a graded Lagrangian integral current, to a finite ordered sum of special Lagrangians whose phases give the semistable factors of the brane.

Reference supplied by the database. Dominic Joyce, “Conjectures on Bridgeland stability for Fukaya categories of Calabi–Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow,” *EMS Surveys in Mathematical Sciences* **2** (2015), no. 1, 1–62, Conjecture 3.34. DOI: [10.4171/EMSS/8](https://doi.org/10.4171/EMSS/8). arXiv: [1401.4949](https://arxiv.org/abs/1401.4949).

5.6.5. Primary-source pointer

The originating ICM record is *Geometric Physics* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=539>.

5.7. 107. Geometric Physics [1998-ag-80]

Record: 1998-ag-80

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Geometric Physics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=539>

5.7.1. Problem statement

Let M be a Calabi-Yau manifold of complex dimension d , let $Q \in H_d(M, \mathbb{Z})$ be a middle-dimensional homology class, and let Ω denote a nowhere-zero holomorphic d -form on M . For a complex structure on M , define

$$V(Q, \Omega) = \frac{|\Omega(Q)|}{\sqrt{\int_M \Omega \wedge \bar{\Omega}}},$$

where $\Omega(Q)$ denotes the period of Ω over Q . Minimize $V(Q, \Omega)$ over the moduli space of complex structures of the Calabi-Yau, and suppose that the minimum exists and is attained at a non-degenerate Calabi-Yau; denote the minimum by $V_{\min}$.

Prove or disprove that the growth S of the cohomology of the moduli space of pairs (L, ∇) , where $L \subset M$ is a real d -dimensional minimal Lagrangian submanifold representing Q and ∇ is a flat connection on L , satisfies

$$S = \exp(c(d) V_{\min}^{d-1}),$$

where $c(d)$ is a universal constant depending only on d .

5.7.2. Historical background, literature, and current progress

5.7.3. Verdict: partially resolved

5.7.3.1. Precise problem

The problem asks for Vafa's proposed asymptotic count/entropy for A-branes on a Calabi-Yau d -fold: for a middle-dimensional class $Q \in H_d(M, \mathbb{Z})$, minimize the normalized period

$$V(Q, \Omega) = \frac{|\Omega(Q)|}{\sqrt{\int_M \Omega \wedge \bar{\Omega}}}$$

over complex structures, assume the minimum is attained at a nondegenerate Calabi-Yau, and prove that the growth of the cohomology of the moduli space of pairs (L, ∇) , with L a real d -dimensional minimal/special Lagrangian representative of Q and ∇ a flat connection on L , is

$$S = \exp(c(d) V_{\min}^{d-1})$$

with $c(d)$ depending only on d . This is a global enumerative/cohomological growth statement, not just a local deformation statement for special Lagrangians.

5.7.3.2. Source and related progress

- **Geometric Physics.** Cumrun Vafa. Proceedings of the International Congress of Mathematicians, Berlin 1998, Vol. I, Documenta Mathematica Extra Volume ICM 1998, pp. 537-556, 1998. DOI: <https://doi.org/10.4171/DMS/1-1/19>. arXiv: <https://arxiv.org/abs/hep-th/9810149>.
This is the exact source of the question. Section 6 frames the $d = 3$ case as a black-hole prediction and then conjectures the general d -dimensional formula $S = \exp(c(d)V_{\min}^{d-1})$. Vafa says the formula is vacuous for $d = 1$, verifiable for $d = 2$, predicted for $d = 3$, and asks whether it is true in all dimensions. The paper does not prove the full conjecture.
- **Minimal Cycles, Black Holes and QFT's.** Cumrun Vafa. Preprint, 1998. arXiv: <https://arxiv.org/abs/hep-th/9805213>. DOI: <https://doi.org/10.48550/arXiv.hep-th/9805213>.
Gives physical context for minimal cycles and black-hole entropy. It supports the motivation for the conjecture but does not provide a mathematical proof of the cohomology-growth formula for moduli of minimal Lagrangians.
- **Black Holes and Calabi-Yau Threefolds.** Cumrun Vafa. Advances in Theoretical and Mathematical Physics 2 (1998), 207-218. arXiv: <https://arxiv.org/abs/hep-th/9711067>. DOI: <https://doi.org/10.48550/arXiv.hep-th/9711067>.
Computes/compares black-hole entropy predictions in Calabi-Yau threefold settings and verifies a related surprising growth for cohomology of moduli of holomorphic curves in elliptic threefolds. This is related by the physics and mirror-symmetry motivation, but it is not the problem as stated, about all Calabi-Yau d -folds and moduli of minimal Lagrangian d -folds with flat connections.
- **Deformations of Calibrated Submanifolds.** Robert C. McLean. Communications in Analysis and Geometry 6 (1998), no. 4, 705-747. DOI: <https://doi.org/10.4310/CAG.1998.v6.n4.a4>. URL: <https://www.intlpress.com/site/pub/pages/journals/items/cag/content/vols/0006/0004/a004/>
Proves the fundamental local deformation theorem: the moduli space of nearby compact special Lagrangian deformations is smooth of dimension $b^1(L)$. This is necessary background for understanding the moduli spaces, but it is purely local and does not count global cohomology or prove Vafa's asymptotic formula.
- **The moduli space of special Lagrangian submanifolds.** Nigel J. Hitchin. Annali della Scuola Normale Superiore di Pisa, Classe di Scienze 25 (1997), no. 3-4, 503-515. arXiv: <https://arxiv.org/abs/dg-ga/9711002>. URL: <https://eudml.org/doc/84303>.
Develops the local geometry of the moduli space of deformations of a compact special Lagrangian, building on McLean. This is structural progress, not an enumerative theorem or an asymptotic cohomology-growth result.
- **On counting special Lagrangian homology 3-spheres.** Dominic Joyce. Contemporary Mathematics 314 (2002), 125-151. arXiv: <https://arxiv.org/abs/hep-th/9907013>. DOI: <https://doi.org/10.48550/arXiv.hep-th/9907013>.
Attempts to define an invariant of almost Calabi-Yau 3-folds by weighted counts of special Lagrangian rational homology 3-spheres. Joyce explicitly treats the singular behav-

ior needed for such invariants as not fully understood and formulates conjectural wall-crossing behavior. This is partial progress in dimension 3, but it is much narrower than Vafa's all- d , all-classes cohomology-growth formula.

- **Special Lagrangian submanifolds with isolated conical singularities. V. Survey and applications.** Dominic Joyce. *Journal of Differential Geometry* 63 (2003), 279-348. arXiv: <https://arxiv.org/abs/math/0303272>. DOI: <https://doi.org/10.48550/arXiv.math/0303272>.

Surveys and proves major results on singular special Lagrangians, desingularization, connected sums, and boundaries of moduli spaces. These results address important compactification/singularity issues but do not yield Vafa's global asymptotic cohomology count.

- **Sigma Models for Bundles on Calabi-Yau: A Proposal for Matrix String Compactifications.** C. Hofman and J.-S. Park. *Nuclear Physics B* 561 (1999), 125-156. DOI: [https://doi.org/10.1016/S0550-3213\(99\)00548-9](https://doi.org/10.1016/S0550-3213(99)00548-9). arXiv: <https://arxiv.org/abs/hep-th/9904150>.

Discusses Vafa's extension of mirror symmetry involving stable bundles on one side and minimal Lagrangian submanifolds on the mirror side. The authors propose sigma-model structures whose A-model and elliptic genus relate to moduli of stable bundles, but they do not compute the Vafa asymptotic growth formula.

- **Gauge Theory in Higher Dimensions, II.** Simon K. Donaldson and Ed Segal. *Surveys in Differential Geometry* 16, *Geometry of Special Holonomy and Related Topics*, International Press, 2011, pp. 1-41. arXiv: <https://arxiv.org/abs/0902.3239>.

Develops a formal Floer/gauge-theoretic framework connected to counting calibrated submanifolds and says that, for counting special Lagrangian submanifolds, a final theory is still lacking. This is strong evidence that the general enumerative foundation needed for Vafa's conjecture was not available.

- **Transversality for perturbed special Lagrangian submanifolds.** Emily Autumn Windes. *Annals of Global Analysis and Geometry* 68, article 6, 2025. DOI: <https://doi.org/10.1007/s10455-025-10006-3>. arXiv: <https://arxiv.org/abs/2406.17948>.

Proves that the moduli space of perturbed special Lagrangian submanifolds in the Donaldson-Segal setup is generically a set of isolated points. This is recent progress toward a counting theory in real dimension 3/Calabi-Yau dimension 3, but it does not define the full invariant, prove compactness/wall-crossing enough for Vafa's formula, or handle all dimensions.

- **Tropical Quantum Field Theory, Mirror Polyvector Fields, and Multiplicities of Tropical Curves.** Travis Mandel and Helge Ruddat. *International Mathematics Research Notices* 2023, no. 4, 3249-3304. DOI: <https://doi.org/10.1093/imrn/rnab332>. URL: <https://academic.oup.com/imrn/article/2023/4/3249/6454977>.

Relates tropical multiplicities to Joyce-type weights in settings where tropical curves produce Lagrangian submanifolds in torus fibrations. This is useful special-geometry progress, but it does not settle the global cohomology-growth formula in the problem statement.

5.7.3.3. What remains unresolved

No published or otherwise citable result found proves the full statement as written. The missing pieces include:

- a general, rigorous compactified moduli theory for minimal/special Lagrangian d -folds with flat connections in arbitrary compact Calabi-Yau d -folds;
- a theorem identifying the relevant cohomology growth of those moduli spaces for all classes Q , or even for all large multiples of Q ;
- a proof that the exponent is universally $c(d)V_{\min}^{d-1}$ with $c(d)$ independent of the Calabi-Yau and with exactly Vafa's period-minimization hypotheses;
- a verification that related stable-bundle or BPS/black-hole results imply the Lagrangian moduli statement without extra hypotheses. Current literature provides motivation, local deformation theory, proposed invariants, singularity/desingularization analysis, and some special-case or perturbed transversality results, but not a theorem covering the full problem.

Therefore the correct status is **partially resolved**, not fully resolved.

5.7.4. Remaining open problems recorded by the database

5.7.4.1. Open problem 1

For a generic almost Calabi-Yau threefold M , construct a rational-valued invariant $I : H_3(M, \mathbb{Z}) \rightarrow \mathbb{Q}$ by a finite weighted count of compact immersed, possibly singular, special Lagrangian rational homology 3-spheres in each class. For a nonsingular immersed N , the weight should be $|H_1(N, \mathbb{Z})|/|G(N)|$, and the invariant should be unchanged when the Kähler class varies while the period class of the holomorphic volume form is fixed up to nonzero complex scaling.

Reference supplied by the database. Dominic Joyce, “On counting special Lagrangian homology 3-spheres,” in *Topology and Geometry: Commemorating SISTAG*, Contemporary Mathematics **314**, American Mathematical Society, Providence, RI, 2002, 125–151, Conjecture 7.3(a)–(c). arXiv: [hep-th/9907013](https://arxiv.org/abs/hep-th/9907013). MR1941627.

5.7.4.2. Open problem 2

Prove the basic wall-crossing formula for Joyce's special-Lagrangian counting invariant. If primitive classes $\chi^+, \chi^- \in H_3(M, \mathbb{Z})$ align in phase and $I(\chi^+ + \chi^-)^\pm$ denote the values on the two sides of the corresponding wall, then

$$I(\chi^+ + \chi^-)^+ - I(\chi^+ + \chi^-)^- = (\chi^+ \cap \chi^-) I(\chi^+) I(\chi^-).$$

Reference supplied by the database. Dominic Joyce, “On counting special Lagrangian homology 3-spheres,” in *Topology and Geometry: Commemorating SISTAG*, Contemporary Mathematics **314**, American Mathematical Society, Providence, RI, 2002, 125–151, Conjecture 7.3(d), equation (7.9). arXiv: [hep-th/9907013](https://arxiv.org/abs/hep-th/9907013). MR1941627.

5.7.4.3. Open problem 3

Determine conditions under which the zero-dimensional moduli space of perturbed special Lagrangian 3-submanifolds in a fixed homology class and with fixed topology, for a generic tamed closed G_2 -pair on a closed 6-manifold, is compact; more generally, construct the anticipated compactification by rectifiable currents and describe the singular objects in its boundary well enough to define a count.

Reference supplied by the database. Emily Autumn Windes, “Transversality for perturbed special Lagrangian submanifolds,” *Annals of Global Analysis and Geometry* **68** (2025), article 6, Introduction, unnumbered compactness/compactification problem following the volume-bounds proposition. DOI: [10.1007/s10455-025-10006-3](https://doi.org/10.1007/s10455-025-10006-3). arXiv: [2406.17948](https://arxiv.org/abs/2406.17948).

5.7.4.4. Open problem 4

Construct a Floer theory for special Lagrangian 3-submanifolds of a Calabi–Yau threefold using the critical points and flow theory of the Donaldson–Segal Lagrange-multiplier functional, including the perturbation, compactness, and singularity analysis needed for the theory to be well defined.

Reference supplied by the database. Emily Autumn Windes, “Transversality for perturbed special Lagrangian submanifolds,” *Annals of Global Analysis and Geometry* **68** (2025), article 6, Introduction, unnumbered question on constructing a Floer theory for special Lagrangian submanifolds. DOI: [10.1007/s10455-025-10006-3](https://doi.org/10.1007/s10455-025-10006-3). arXiv: [2406.17948](https://arxiv.org/abs/2406.17948).

5.7.5. Primary-source pointer

The originating ICM record is *Geometric Physics* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=539>.

5.8. 108. Geometry of Symplectic Intersections [2002-geom-82]

Record: 2002-geom-82

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry of Symplectic Intersections

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=234>

5.8.1. Problem statement

Let $L \subset (\mathbb{C}^n, \omega_{\text{std}})$ be a closed Lagrangian submanifold, and let $\mu_L : \pi_2(\mathbb{C}^n, L) \rightarrow \mathbb{Z}$ denote its Maslov homomorphism, equivalently the Maslov class of L in $H^1(L; \mathbb{Z})$. Determine the possible classes μ_L for Lagrangian submanifolds of $\mathbb{C}^n$, subject to the Floer-theoretic vanishing principle that, since every compact subset of $\mathbb{C}^n$ can be Hamiltonianly displaced from itself, every such L whose self-Floer homology is well defined must satisfy $HF(L, L) = 0$.

5.8.2. Historical background, literature, and current progress

5.8.3. Verdict: partially resolved

5.8.3.1. Precise problem checked

The problem is broader than the original torus version of Audin's conjecture. In the cited ICM discussion, the question is not merely whether a Lagrangian torus has minimal Maslov number 2, but what restrictions Floer-theoretic vanishing imposes on the Maslov homomorphism/class of an arbitrary closed Lagrangian submanifold

$$L \subset (\mathbb{C}^n, \omega_{\text{std}}).$$

Since every compact subset of $\mathbb{C}^n$ is Hamiltonianly displaceable, any version of self-Floer homology that is defined for such an L must vanish. The intended problem is therefore the Maslov-class restriction/realization problem for closed Lagrangians in $\mathbb{C}^n$, with the original Audin torus conjecture as a distinguished special case.

5.8.3.2. Literature and current progress

1. Michele Audin, "Fibres normaux d'immersions en dimension double, points doubles d'immersions lagrangiennes et plongements totalement reels", *Commentarii Mathematici Helvetici* 63 (1988), 593-623. DOI: 10.1007/BF02566781. This is the original source traditionally cited for Audin's Maslov-class questions. Later papers identify the original concrete conjecture as the statement that every Lagrangian torus $T^n \subset \mathbb{C}^n$ has Maslov number 2, equivalently bounds a disk of Maslov index 2.

2. **Paul Biran**, “Geometry of Symplectic Intersections”, *Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. II*, Higher Education Press, 2002, 241-255; [arXiv:math/0304260](#). This is the source corresponding to the dataset entry. Section 4 states the displaceability/Floer-vanishing principle: for Lagrangians in manifolds with the non-intersection property, $HF(L, L) = 0$ whenever it is well-defined, and applying this to $\mathbb{C}^n$ gives restrictions on possible Maslov classes. Biran explicitly treats this as a source of restrictions, not as a complete classification.
3. **Yong-Geun Oh**, “Floer cohomology, spectral sequences, and the Maslov class of Lagrangian embeddings”, *International Mathematics Research Notices* 1996, no. 7, 305-346. DOI: [10.1155/S1073792896000219](#). Oh gives the Floer-theoretic spectral-sequence interpretation behind the Maslov-class restrictions. The results apply under hypotheses such as monotonicity/minimal Maslov-number assumptions and provide important restrictions, but not a complete answer for arbitrary closed Lagrangians in $\mathbb{C}^n$.
4. **Claude Viterbo**, “A new obstruction to embedding Lagrangian tori”, *Inventiones Mathematicae* 100 (1990), 301-320. DOI: [10.1007/BF01231188](#). Viterbo proved strong restrictions for Lagrangian tori, including early cases of the Audin torus problem. This is key partial progress but is restricted to tori.
5. **Leonid Polterovich**, “Monotone Lagrange submanifolds of linear spaces and the Maslov class in cotangent bundles”, *Mathematische Zeitschrift* 207 (1991), 217-222. DOI: [10.1007/BF02571385](#). Polterovich proved Maslov-class rigidity results for monotone Lagrangians and low-dimensional/special cases. These results are again partial: they do not classify possible Maslov classes for all closed Lagrangians in $\mathbb{C}^n$.
6. **Lev Buhovsky**, “The Maslov class of Lagrangian tori and quantum products in Floer cohomology”, *Journal of Topology and Analysis* 2 (2010), no. 1, 57-75. DOI: [10.1142/S1793525310000240](#); [arXiv:math/0608063](#). Buhovsky proved the monotone version of Audin’s torus conjecture in all dimensions: a monotone Lagrangian torus in $\mathbb{R}^{2n} \cong \mathbb{C}^n$ has minimal Maslov number 2. This resolves the monotone torus case only.
7. **Mihai Damian**, “Floer homology on the universal cover, Audin’s conjecture and other constraints on Lagrangian submanifolds”, *Commentarii Mathematici Helvetici* 87 (2012), 433-462/463. DOI: [10.4171/CMH/259](#); [arXiv:1006.3398](#). Damian proves generalized Audin-type statements for monotone displaceable Lagrangians, including that if L is aspherical, or more generally its universal cover has no odd homology, then its Maslov number is 2. The monotonicity and topological assumptions are extra hypotheses, so this is partial progress for the broader problem stated here.
8. **Kai Cieliebak and Klaus Mohnke**, “Punctured holomorphic curves and Lagrangian embeddings”, *Inventiones Mathematicae* 212 (2018), no. 1, 213-295. DOI: [10.1007/s00222-017-0767-8](#); [arXiv:1411.1870](#). Theorem 1.2(b) proves Audin’s original torus conjecture: every Lagrangian torus in $\mathbb{C}^n$ admits a disk of Maslov index 2. This fully resolves the torus special case, including non-monotone tori. It does not cover arbitrary closed Lagrangian submanifolds.
9. **Kei Irie**, “Chain level loop bracket and pseudo-holomorphic disks”, *Journal of Topology* 13 (2020), no. 2, 870-938. DOI: [10.1112/topo.12140](#); [arXiv:1801.04633](#). Irie’s work gives another proof of Audin’s conjecture in the torus case and proves/generalizes Fukaya’s string-topology approach for closed, oriented, spin $K(\pi, 1)$ Lagrangians, producing Maslov-index-2 holomorphic disks. These are important generalizations but still require strong topological hypotheses.
10. **Tobias Ekholm, Yakov Eliashberg, Emmy Murphy, and Ivan Smith**, “Constructing ex-

act Lagrangian immersions with few double points”, *Geometric and Functional Analysis* 23 (2013), 1772–1803. DOI: 10.1007/s00039-013-0243-6; arXiv:1303.0588. Their construction yields a Lagrangian embedding $S^1 \times S^2 \rightarrow \mathbb{R}_{\text{st}}^6 \cong \mathbb{C}^3$ with vanishing Maslov class. This rules out any unrestricted theorem saying that every closed Lagrangian in $\mathbb{C}^n$ must have nonzero Maslov class.

11. Yin Li, “Aspherical Lagrangian submanifolds, Audin’s conjecture and cyclic dilations”, arXiv:2308.05086, 2023; listed by the author as to appear in *Selecta Mathematica* (N.S.). This proves a general Audin-type Maslov-index-2 disk result for closed oriented $K(\pi, 1)$ Lagrangians in Liouville domains with finite first Gutt-Hutchings capacity, extending the $\mathbb{C}^n$ aspherical/ $K(\pi, 1)$ story. It remains conditional on asphericity-type assumptions and is not a classification for all closed Lagrangians.
12. Shuhao Li, “The topology of Lagrangian submanifolds via open-closed string topology”, arXiv:2604.08481, 2026 preprint. The paper proves a new partial Maslov-class rigidity result: for closed, oriented, spin $L \subset \mathbb{C}^n$, if $\pi_2(L) = 0$, then the Maslov class is nonzero. Its introduction also surveys the current state and explicitly notes the Ekholm-Eliashberg-Murphy-Smith example $S^2 \times S^1 \subset \mathbb{C}^3$ with vanishing Maslov class. Thus even the latest literature gives further partial restrictions, not a complete solution of the full problem.

5.8.3.3. Why this is not fully resolved

The original Audin torus conjecture is fully resolved by Cieliebak-Mohnke, with alternative/generalized proofs by Fukaya-Irie and later work. However, the dataset entry is stated for arbitrary closed Lagrangian submanifolds $L \subset \mathbb{C}^n$ and asks about possible Maslov classes/restrictions coming from Floer-theoretic vanishing. No cited theorem gives a complete characterization of all possible Maslov homomorphisms/classes for arbitrary closed Lagrangians in $\mathbb{C}^n$.

Moreover, the broad nonzero-Maslov formulation is not true without hypotheses: $S^1 \times S^2$ admits a Maslov-zero Lagrangian embedding in $\mathbb{C}^3$. Current positive results require extra assumptions such as torus, monotone, aspherical/ $K(\pi, 1)$, spin, $\pi_2(L) = 0$, nonpositive curvature, or vanishing H^2 . Therefore the correct status of the full problem is **partially resolved**, not fully resolved.

5.8.4. Remaining open problems recorded by the database

5.8.4.1. Open problem 1

Does the connected sum $T^3 \# T^3$ admit a Lagrangian embedding into $(\mathbb{C}^3, \omega_{\text{std}})$?

Reference supplied by the database. Shuhao Li, “The topology of Lagrangian submanifolds via open-closed string topology,” arXiv preprint (2026), Remark following Theorem 1.1, item (3), where this existence problem is explicitly identified as wide open; arXiv: 2604.08481, DOI: 10.48550/arXiv.2604.08481.

5.8.4.2. Open problem 2

If M_1 and M_2 are closed hyperbolic 3-manifolds, does their connected sum $M_1 \# M_2$ admit a Lagrangian embedding into $(\mathbb{C}^3, \omega_{\text{std}})$?

Reference supplied by the database. Shuhao Li, “The topology of Lagrangian submanifolds via open-closed string topology,” arXiv preprint (2026), Remark following Theorem 1.1, item (3), where the existence or nonexistence of Lagrangian embeddings of connected sums of hyperbolic 3-manifolds into $\mathbb{C}^3$ is explicitly identified as wide open; arXiv: [2604.08481](https://arxiv.org/abs/2604.08481), DOI: [10.48550/arXiv.2604.08481](https://doi.org/10.48550/arXiv.2604.08481).

5.8.5. Primary-source pointer

The originating ICM record is *Geometry of Symplectic Intersections* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=234>.

5.9. 109. Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures [2002-geom-83]

Record: 2002-geom-83

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=389>

5.9.1. Problem statement

Prouver ou réfuter que, sur toute variété de contact close de dimension trois (V, ξ) , tout flot de Reeb, c'est-à-dire le flot du champ de Reeb R_α associé à une forme de contact α telle que $\xi = \ker \alpha$ et caractérisé par $\alpha(R_\alpha) = 1$ et $i_{R_\alpha} d\alpha = 0$, admet une section de Poincaré-Birkhoff : une surface compacte qui rencontre toutes les orbites du flot, dont l'intérieur est transversal au flot et dont chaque composante du bord est une orbite périodique.

5.9.2. Historical background, literature, and current progress

5.9.3. Verdict: partially resolved

5.9.3.1. Precise problem

The problem asks whether every Reeb flow of every contact form α defining an arbitrary closed three-dimensional contact manifold (V, α) admits a Poincaré-Birkhoff/Birkhoff global surface of section: a compact surface meeting every orbit, with interior transverse to the Reeb flow, and with every boundary component a periodic Reeb orbit.

This is stronger than Giroux's ICM corollary, which asserts only that every closed contact 3-manifold admits some adapted Reeb flow with such a section.

5.9.3.2. Original source and status of the question

- Title: Geometrie de contact: de la dimension trois vers les dimensions superieures
- Author: Emmanuel Giroux
- Venue/year: Proceedings of the International Congress of Mathematicians, Vol. II, Beijing 2002, Higher Education Press, 2002, pp. 405-414
- arXiv/URL: arXiv:math/0305129, <https://arxiv.org/abs/math/0305129>
- Relevant point: Corollaire 6 states that on every closed contact 3-manifold there exists a Reeb flow admitting a Poincaré-Birkhoff section. Immediately afterward Giroux says that it is not excluded that every Reeb flow admits such a section, which would imply the Weinstein conjecture. Thus the exact all-Reeb-flows statement in this problem was not proved there.

5.9.3.3. Main partial progress

- Title: The dynamics on three-dimensional strictly convex energy surfaces
- Authors: Helmut Hofer, Krzysztof Wysocki, Eduard Zehnder
- Venue/year: Annals of Mathematics 148 (1998), no. 1, 197-289
- DOI/URL: 10.2307/120994, <https://annals.math.princeton.edu/articles/12987>
- Result: proves disk-like global surfaces of section for Hamiltonian/Reeb dynamics on strictly convex energy surfaces in $\mathbb{R}^4$. This is a foundational special case, but it assumes a convex/dynamically convex S^3 -type setting and does not cover arbitrary contact 3-manifolds or arbitrary Reeb flows.
- Title: On the existence of supporting broken book decompositions for contact forms in dimension 3
- Authors: Vincent Colin, Pierre Dehornoy, Ana Rechtman
- Venue/year: Inventiones mathematicae 231 (2023), no. 3, 1489-1539
- DOI/arXiv/URL: 10.1007/s00222-022-01160-7, arXiv:2001.01448, <https://arxiv.org/abs/2001.01448>
- Result: every nondegenerate contact form in dimension 3 is carried by a broken book decomposition. This gives a system of transverse surfaces and strong dynamical consequences, but a broken book is weaker than a Birkhoff/global surface of section. The paper formulates “Every Reeb vector field has a Birkhoff section” as a conjecture, so it does not resolve the full statement.
- Title: Existence of Birkhoff sections for Kupka-Smale Reeb flows of closed contact 3-manifolds
- Authors: Gonzalo Contreras, Marco Mazzucchelli
- Venue/year: Geometric and Functional Analysis 32 (2022), no. 5, 951-979
- DOI/arXiv/URL: 10.1007/s00039-022-00616-5, arXiv:2110.07491, <https://arxiv.org/abs/2110.07491>
- Result: any Reeb vector field on an orientable closed 3-manifold satisfying the Kupka-Smale condition admits a Birkhoff section. Consequently C^∞ -generic contact forms have Reeb flows with Birkhoff sections. This is strong generic progress, but the Kupka-Smale/nondegeneracy-transversality hypothesis is an extra assumption absent from the problem statement.
- Title: Generic properties of 3-dimensional Reeb flows: Birkhoff sections and entropy
- Authors: Vincent Colin, Pierre Dehornoy, Umberto Hryniewicz, Ana Rechtman
- Venue/year: Commentarii Mathematici Helvetici 99 (2024), no. 3, 557-611
- DOI/arXiv/URL: 10.4171/CMH/573, arXiv:2202.01506, <https://arxiv.org/abs/2202.01506>
- Result: if the Liouville measure of a nondegenerate contact form on a closed 3-manifold can be approximated by periodic Reeb orbits, then the Reeb flow admits a partial-strong Birkhoff section; using Irie’s equidistribution theorem this gives an open dense/generic existence result. This again does not cover arbitrary, possibly degenerate Reeb flows.

- Title: Homoclinic orbits, Reeb chords and nice Birkhoff sections for Reeb flows in 3D
- Authors: Vincent Colin, Umberto Hryniewicz, Ana Rechtman
- Venue/year: arXiv preprint, 2025
- arXiv/URL: arXiv:2501.11725, <https://arxiv.org/abs/2501.11725>
- Result: for C-infinity-generic contact forms defining a fixed co-oriented contact structure on a closed 3-manifold, the authors obtain embedded Birkhoff/global surfaces of section with additional prescribed features. This improves the generic picture but remains generic rather than universal.
- Title: From curve shortening to flat link stability and Birkhoff sections of geodesic flows
- Authors: Marcelo R. R. Alves, Marco Mazzucchelli
- Venue/year: Journal de l'Ecole polytechnique - Mathematiques 12 (2025), 801-851
- DOI/URL: 10.5802/jep.301, <https://jep.centre-mersenne.org/articles/10.5802/jep.301/>
- Result: proves Birkhoff sections for geodesic flows of any closed connected orientable Riemannian surface. These are Reeb flows on unit tangent bundles, so this is an important subclass, but not the full class of Reeb flows on arbitrary closed contact 3-manifolds.

5.9.3.4. What remains unresolved

I found no published or otherwise citable theorem proving that every Reeb flow, without genericity, nondegeneracy, Kupka-Smale, equidistribution, dynamical convexity, geodesic-flow, or other extra hypotheses, admits a compact Poincare-Birkhoff/global surface of section on every closed contact 3-manifold. the exact statement is therefore not fully resolved; the literature gives substantial partial progress only.

5.9.4. Remaining open problems recorded by the database

5.9.4.1. Open problem 1

Does every nondegenerate Reeb vector field on a closed contact three-manifold admit a Birkhoff section?

Reference supplied by the database. Vincent Colin, Pierre Dehornoy, and Ana Rechtman, “On the existence of supporting broken book decompositions for contact forms in dimension 3,” *Inventiones Mathematicae* 231, no. 3 (2023), 1489–1539, §4.6 (discussion preceding Conjecture 4.14). DOI: [10.1007/s00222-022-01160-7](https://doi.org/10.1007/s00222-022-01160-7). Preprint: [arXiv:2001.01448](https://arxiv.org/abs/2001.01448).

5.9.5. Primary-source pointer

The originating ICM record is *Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=389>.

5.10. 110. Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures [2002-geom-84]

Record: 2002-geom-84

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=389>

5.10.1. Problem statement

Prove that, for every closed cooriented contact manifold (V, ξ) and every contact form α defining ξ , the Reeb vector field R_α characterized by $\alpha(R_\alpha) = 1$ and $\iota_{R_\alpha} d\alpha = 0$ has at least one periodic orbit.

5.10.2. Historical background, literature, and current progress

5.10.3. Status check: conjecture de Weinstein

5.10.3.1. Verdict

Partially resolved.

5.10.3.2. Precise problem

The problem asks for the full modern Weinstein conjecture for contact manifolds: for every closed cooriented contact manifold (V, ξ) , in every odd dimension, and for every contact form α with $\ker \alpha = \xi$, the Reeb vector field R_α has at least one periodic orbit.

This is not just the three-dimensional conjecture, nor only the hypersurface-of-contact-type version in a specified symplectic manifold. It quantifies over all closed cooriented contact manifolds and all defining contact forms.

5.10.3.3. Current status

No published or otherwise citable result found resolves this exact statement in all dimensions and for all closed cooriented contact manifolds. The literature proves many important special cases, including all closed contact 3-manifolds, but in dimensions at least five the general case remains only partially understood.

5.10.3.4. Key literature and what it proves

1. Alan Weinstein, “On the hypotheses of Rabinowitz’ periodic orbit theorems”, *Journal of Differential Equations* 33(3), 353-358, 1979. DOI: 10.1016/0022-0396(79)90070-6.
 - Original source associated with the Weinstein conjecture. It reformulates periodic-orbit existence in contact-type terms, following Rabinowitz’s variational results, but does not prove the full arbitrary closed contact manifold statement.
2. Claude Viterbo, “A proof of Weinstein’s conjecture in $\mathbb{R}^{2n}$ ”, *Annales de l’Institut Henri Poincaré. C, Analyse non lineaire* 4(4), 337-356, 1987. URL: https://www.numdam.org/item/AIHPC_1987__4_4_337_0/
 - Proves the conjecture for contact-type hypersurfaces in $\mathbb{R}^{2n}$. This is a major high-dimensional result, but it requires the contact manifold to occur as such a hypersurface and therefore does not cover arbitrary closed cooriented contact manifolds.
3. Helmut Hofer, “Pseudoholomorphic curves in symplectizations with applications to the Weinstein conjecture in dimension three”, *Inventiones Mathematicae* 114, 515-563, 1993. DOI: 10.1007/BF01232679.
 - Establishes important three-dimensional cases using pseudoholomorphic curves, including overtwisted contact 3-manifolds and other cases, but not all dimensions and not all closed contact manifolds.
4. Clifford Henry Taubes, “The Seiberg-Witten equations and the Weinstein conjecture”, *Geometry & Topology* 11, 2117-2202, 2007. arXiv: math/0611007. DOI: 10.2140/gt.2007.11.2117.
 - Proves that if M is a compact oriented 3-manifold and α is a contact 1-form, then the Reeb vector field has a closed integral curve. This fully resolves the problem statement in dimension three only. The proof uses dimension-specific Seiberg-Witten/Floer input, so it does not imply the arbitrary-dimensional statement.
5. Michael Hutchings, “Taubes’s proof of the Weinstein conjecture in dimension three”, *Bulletin of the American Mathematical Society* 47(1), 73-125, 2010. arXiv: 0906.2444. DOI: 10.1090/S0273-0979-09-01282-8.
 - Survey of Taubes’s proof and surrounding literature. It states the 3-dimensional theorem and notes that in higher dimensions, although the conjecture is known for compact hypersurfaces of contact type in $\mathbb{R}^{2n}$ and other cases, the general problem is wide open. This confirms that Taubes’s theorem does not settle the full all-dimensional statement.
6. Peter Albers and Helmut Hofer, “On the Weinstein conjecture in higher dimensions”, *Commentarii Mathematici Helvetici* 84(2), 429-436, 2009. arXiv: 0705.3953.
 - Proves the Weinstein conjecture for closed PS-overtwisted contact manifolds, giving a contractible periodic orbit for every supporting contact form. This is a genuine higher-dimensional family but assumes the presence of a plastikstufe/PS-overtwisted structure, so it is only partial progress.
7. Peter Albers, Urs Fuchs, and Will J. Merry, “Orderability and the Weinstein conjecture”, *Compositio Mathematica* 151(12), 2251-2272, 2015. DOI: 10.1112/S0010437X15007642.
 - Proves the Weinstein conjecture for contact manifolds whose identity component of the contactomorphism group is non-orderable, linking orderability and hypertightness. This gives another significant class but does not cover arbitrary closed contact manifolds.

8. Hansjorg Geiges and Kai Zehmisch, “The Weinstein conjecture for connected sums”, *International Mathematics Research Notices* 2016(2), 325-342, 2016. DOI: 10.1093/imrn/rnv124. arXiv: 1407.4307.
 - Proves the conjecture for non-trivial contact connected sums under topological assumptions such as non-trivial fundamental group or torsion-free homology. These hypotheses exclude many contact manifolds and therefore do not resolve the full statement.
9. Matthew Strom Borman, Yakov Eliashberg, and Emmy Murphy, “Existence and classification of overtwisted contact structures in all dimensions”, *Acta Mathematica* 215(2), 281-361, 2015. DOI: 10.1007/s11511-016-0134-4. arXiv: 1404.6157.
 - Provides the modern all-dimensional theory of overtwisted contact structures. Together with results such as Albers-Hofer for PS-overtwisted/overtwisted-type structures, it enlarges the class of high-dimensional contact manifolds where Weinstein-type conclusions are known. It does not prove the conjecture for tight or arbitrary high-dimensional contact manifolds.
10. Yanqiao Ding, “A survey on the Weinstein conjecture”, *Frontiers of Mathematics in China* 20(1), 1-15, 2025. DOI: 10.3868/s140-DDD-025-0003-x.
 - Recent survey article on variational and pseudoholomorphic-curve approaches. It lists the major known special cases and later progress, but does not report a theorem resolving the arbitrary-dimensional closed contact manifold case.
11. Lina Deschamps, Levin Maier, and Tom Stalljohann, “On the contact type conjecture for exact magnetic systems”, arXiv: 2508.01113, 2025 preprint.
 - States the Weinstein conjecture for closed contact manifolds and explicitly notes that in dimension three it was resolved by Hofer and Taubes, while in higher dimensions it remains only partially understood. It then uses the conjecture only for classes where it is known, again confirming that no general higher-dimensional resolution is available.

5.10.3.5. Why the problem is not fully resolved

Taubes’s theorem proves the exact Reeb-periodic-orbit conclusion for every contact form on every closed oriented contact 3-manifold, so the conjecture is fully solved in dimension three. However, the statement has no dimension restriction. The higher-dimensional literature found in the search proves the result under extra hypotheses: contact-type hypersurfaces in special symplectic manifolds, fillability/cobordism assumptions, PS-overtwisted or overtwisted-type structures, non-orderability, connected-sum hypotheses, or other special geometric/topological conditions.

Thus the unresolved part is: closed cooriented contact manifolds of dimension at least five with arbitrary contact structure and arbitrary supporting contact form, without the additional hypotheses appearing in the known theorems.

5.10.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

5.10.5. Primary-source pointer

The originating ICM record is *Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=389>.

5.11. 111. Geometry and Nonlinear Analysis [2002-geom-85]

Record: 2002-geom-85

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>

5.11.1. Problem statement

Prove that if (M, g, ω) is a compact Kähler-Einstein complex surface with Kähler form ω , then every smooth embedded real surface $S_0 \subset M$ that is symplectic with respect to ω is isotopic in M to a smooth embedded real surface $S \subset M$ that is still symplectic with respect to ω and has zero mean curvature with respect to the Kähler-Einstein metric g .

5.11.2. Historical background, literature, and current progress

5.11.3. Verdict: partially resolved

5.11.3.1. Precise problem

The problem asks whether Tian's symplectic isotopy/minimal-surface problem holds in full:

Given any compact Kähler-Einstein complex surface (M, g, ω) and any smooth embedded real surface S_0 such that $\omega|_{S_0}$ is a positive area form, must S_0 be isotopic in M to another smooth embedded ω -symplectic surface S whose mean curvature vector with respect to g is zero?

Equivalently, every embedded symplectic surface in every compact Kähler-Einstein complex surface should have a symplectic minimal representative in its smooth isotopy class.

5.11.3.2. Original source

- Title: *Symplectic isotopy in four dimension*
- Author: Gang Tian
- Venue: *First International Congress of Chinese Mathematicians (Beijing, 1998)*, AMS/IP Studies in Advanced Mathematics 20, American Mathematical Society/International Press
- Year: 2001
- Pages: 143-147
- DOI/URL: <https://doi.org/10.1090/amsip/020/09>
- Role: This is the cited source for Tian's problem. Later papers quote it as the conjecture that every embedded orientable closed symplectic surface in a compact Kähler-Einstein surface is isotopic to a symplectic minimal surface.

5.11.3.3. Main partial progress

- Title: *Moving Symplectic Curves in Kähler-Einstein Surfaces*
- Authors: Jingyi Chen, Gang Tian
- Venue: *Acta Mathematica Sinica, English Series* 16(4), 541-548
- Year: 2000
- DOI/URL: <https://doi.org/10.1007/s101140000075>
- Result: Derives the parabolic equation for the Kähler angle along mean curvature flow in a Kähler-Einstein surface and proves that a symplectic curve remains symplectic as long as the smooth mean curvature flow exists.
- Limitation: This is preservation along an existing smooth flow, not global existence/convergence for all initial symplectic surfaces.
- Title: *Mean Curvature Flow of Surfaces in 4-Manifolds*
- Authors: Jingyi Chen, Jiayu Li
- Venue: *Advances in Mathematics* 163, 287-309
- Year: 2001
- DOI/URL: <https://doi.org/10.1006/aima.2001.2008>
- Result: Develops codimension-two mean-curvature-flow estimates used in the symplectic Kähler-Einstein setting, including preservation/monotonicity tools cited by later work.
- Limitation: Does not prove the full isotopy-to-minimal statement.
- Title: *Two-Dimensional Graphs Moving by Mean Curvature Flow*
- Authors: Jingyi Chen, Jiayu Li, Gang Tian
- Venue: *Acta Mathematica Sinica, English Series* 18(2), 209-224
- Year: 2002
- DOI/URL: <https://doi.org/10.1007/s101140200163>
- Result: Gives long-time existence and convergence in graphical cases.
- Limitation: the problem asks for arbitrary embedded symplectic surfaces, not only graphical initial data.
- Title: *Mean Curvature Flow of Surfaces in Einstein Four-Manifolds*
- Author: Mu-Tao Wang
- Venue: *Journal of Differential Geometry* 57, 301-338
- Year: 2001
- DOI/URL: <https://projecteuclid.org/euclid.jdg/1090348113>
- Result: Studies mean curvature flow of surfaces in Einstein four-manifolds; in the Kähler-Einstein/calibrated setting it is part of the standard evidence that the symplectic condition is preserved under the flow.
- Limitation: It does not establish global smooth flow and convergence for every symplectic surface.

- Title: *Singularities of Codimension Two Mean Curvature Flow of Symplectic Surfaces*
- Authors: Jingyi Chen, Jiayu Li
- Venue: arXiv preprint
- Year: 2002
- arXiv/URL: <https://arxiv.org/abs/math/0208227>
- Result: For a compact symplectic surface in a compact Kähler-Einstein surface, analyzes first singular-time tangent cones; the theorem gives tangent cones as finite unions of complex 2-planes and records no Type I singularities for symplectic mean curvature flow.
- Limitation: Type II singularities remain a central obstruction; tangent-cone control is not a proof that the flow always exists globally or reaches a smooth minimal representative.
- Title: *The Mean Curvature Flow Approach to the Symplectic Isotopy Problem*
- Authors: Xiaoli Han, Jiayu Li
- Venue: *International Mathematics Research Notices* 2005(26), 1611-1620
- Year: 2005
- DOI/arXiv/URL: <https://doi.org/10.1155/IMRN.2005.1611>; <https://arxiv.org/abs/math/0411284>
- Result: If the Kähler-Einstein surface has positive scalar curvature and the initial surface is sufficiently close to a holomorphic curve, the mean curvature flow exists globally and converges to a holomorphic curve.
- Limitation: Requires positive scalar curvature and a smallness/closeness hypothesis on the initial surface.
- Title: *Symplectic critical surfaces in Kähler surfaces*
- Authors: Xiaoli Han, Jiayu Li
- Venue: *Journal of the European Mathematical Society* 12(2), 505-527
- Year: 2010
- DOI/arXiv/URL: <https://doi.org/10.4171/JEMS/207>; <https://arxiv.org/abs/0711.2211>
- Result: Introduces a variational functional on symplectic surfaces, computes its Euler-Lagrange equation, proves ellipticity, and shows that compact symplectic critical surfaces satisfy strong minimal/constant-angle conclusions; for nonnegative scalar curvature the critical surfaces are holomorphic.
- Limitation: This is a variational framework and conditional structure theorem; it does not prove existence of such a critical/minimal surface in every isotopy class. The paper itself states the natural conjecture that each homotopy class of symplectic surfaces should contain a symplectic critical surface.
- Title: *Minimal surfaces and deformations of holomorphic curves in Kähler-Einstein manifolds*
- Author: Claudio Arezzo
- Venue: *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* 29(2), 473-481
- Year: 2000
- DOI/arXiv/URL: https://www.numdam.org/item/ASNSP_2000_4_29_

[2_473_0/](#)

- Result: Constructs stable nonholomorphic minimal surfaces in negative-scalar-curvature Kähler-Einstein four-manifolds.
- Limitation: Shows that in negative scalar curvature the target minimal surfaces need not be holomorphic, but does not prove existence in every symplectic isotopy class.
- Title: *Long time existence of the symplectic mean curvature flow*
- Authors: Xiaoli Han, Jiayu Li
- Venue: arXiv preprint
- Year: 2011
- arXiv/URL: <https://arxiv.org/abs/1107.0829>
- Result: In Kähler surfaces of constant positive holomorphic sectional curvature, proves long-time existence and convergence to a holomorphic curve under explicit curvature/Kähler-angle pinching hypotheses.
- Limitation: Applies only under strong pinching assumptions, not to all embedded symplectic surfaces.
- Title: *Symplectic mean curvature flow in CP²*
- Authors: Xiaoli Han, Jiayu Li, Liuqing Yang
- Venue: *Calculus of Variations and Partial Differential Equations* 48, 111-129
- Year: 2013
- DOI/URL: <https://doi.org/10.1007/s00526-012-0544-x>
- Result: Proves global existence and convergence for symplectic mean curvature flow in CP² under suitable curvature pinching conditions.
- Limitation: Covers a pinched subclass of initial surfaces in CP², not arbitrary symplectic surfaces.
- Title: *Gradient flow for beta-symplectic critical surfaces*
- Authors: Xiaoli Han, Jiayu Li, Jun Sun
- Venue: *Annales de l'Institut Henri Poincaré C, Analyse Non Linéaire* 41, 1083-1116
- Year: 2024
- DOI/URL: <https://doi.org/10.4171/AIHPC/100>
- Result: Studies a gradient flow for the beta-symplectic-critical-surface functional, proving preservation of the symplectic condition, a monotonicity formula, epsilon-regularity, and finite flat-plane tangent cones for the flow.
- Limitation: Still a regularity/flow framework; it does not prove existence of smooth minimal representatives for every isotopy class.
- Title: *Tangent Flows of Symplectic Mean Curvature Flows*
- Authors: Jingyi Chen, Xiaoli Han, Jiayu Li, Jun Sun
- Venue: *Journal of Mathematical Study* 59(1), 40-59
- Year: 2026
- DOI/URL: <https://doi.org/10.4208/jms.v59n1.26.03>

- Result: Recent tangent-flow progress: the tangent cone at the first blow-up time of mean curvature flow of a closed symplectic surface in a compact Kähler-Einstein surface is a finite union of planes in $\mathbb{R}^4$, and Type I* tangent cones are holomorphic cones.
- Limitation: This improves singularity analysis but still does not establish the required global smooth flow/convergence or an alternative all-class existence theorem.

5.11.3.4. Related symplectic-isotopy evidence

- Title: *On the holomorphicity of genus two Lefschetz fibrations*
- Authors: Bernd Siebert, Gang Tian
- Venue: *Annals of Mathematics* 161(2), 959-1020
- Year: 2005
- DOI/URL: <https://doi.org/10.4007/annals.2005.161.959>
- Result: Includes the strongest classical smooth $\mathbb{CP}^2$ symplectic-isotopy progress: smooth symplectic plane curves are isotopic to complex curves through degree 17.
- Limitation: This is only a low-degree result. Since $\mathbb{CP}^2$ with the Fubini-Study metric is Kähler-Einstein, the problem statement would in particular imply the unresolved high-degree smooth symplectic isotopy problem in $\mathbb{CP}^2$.
- Title: *A new approach to the symplectic isotopy problem*
- Author: Laura Starkston
- Venue: *Journal of Topology* 13(4), 1649-1689
- Year: 2020
- DOI/arXiv/URL: <https://doi.org/10.1112/topo.12158>; <https://arxiv.org/abs/1709.02544>
- Result: Recasts the smooth symplectic isotopy problem in $\mathbb{CP}^2$ in terms of Lagrangian disk data and explicitly notes that progress after Siebert-Tian's degree-17 result had stalled.
- Limitation: Provides an equivalent/reductive approach, not a full solution.

5.11.3.5. Why the full problem is not resolved

No citable result found proves the full statement for every compact Kähler-Einstein complex surface and every embedded symplectic surface. The literature proves preservation of symplecticity along smooth mean curvature flow, analyzes singularities, and proves convergence under additional assumptions such as graphical initial data, closeness to a holomorphic curve, positive scalar curvature, $\mathbb{CP}^2$, constant holomorphic sectional curvature, or curvature/Kähler-angle pinching.

The unrestricted case remains open. In particular, the Fubini-Study $\mathbb{CP}^2$ case is already a necessary subcase; for positive scalar curvature, known results relate symplectic minimal representatives to holomorphic curves, so a full proof would imply the still-open high-degree smooth symplectic isotopy problem in $\mathbb{CP}^2$. Thus the correct status is partially resolved, not fully resolved.

5.11.4. Remaining open problems recorded by the database

5.11.4.1. Open problem 1

Let $\Sigma \subset \mathbb{CP}^2$ be a smooth symplectically embedded surface representing $d[\mathbb{CP}^1]$. Must there exist $d(d-1)/2$ pairwise disjoint embedded Lagrangian disks whose interiors are disjoint from Σ and whose boundary curves form a d -splitting system on Σ , meaning that cutting Σ along them gives d genus-zero components, each with $d-1$ boundary components, with each boundary curve separating a distinct pair of components?

Reference supplied by the database. Laura Starkston, “A new approach to the symplectic isotopy problem,” *Journal of Symplectic Geometry* 18, no. 3 (2020), 939–960, Definition 1.1, Theorem 1.2, and the discussion immediately following Theorem 1.2. DOI: [10.4310/JSG.2020.v18.n3.a11](https://doi.org/10.4310/JSG.2020.v18.n3.a11). Preprint: [arXiv:1709.02544](https://arxiv.org/abs/1709.02544).

5.11.5. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

5.12. 112. Mirror Symmetry and Quantum Geometry (2) [2002-ag-94]

Record: 2002-ag-94

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Mirror Symmetry and Quantum Geometry (2)

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=422>

5.12.1. Problem statement

Prove or disprove that for any Calabi-Yau manifold X admitting a special Lagrangian torus fibration $\pi : X \rightarrow B$, the mirror of X is obtained, up to the quantum corrections required in the presence of singular fibers, by applying T-duality fiber by fiber: each smooth torus fiber $T^n = \pi^{-1}(b)$ is replaced by its dual torus $\widehat{T}^n \cong H^1(T^n, U(1))$, equivalently the moduli space of flat $U(1)$ connections, or holonomies, on T^n , and under this dualization a D-brane wrapped on a torus fiber is transformed into a D0-brane at the corresponding point of the dual fiber.

5.12.2. Historical background, literature, and current progress

5.12.3. Status check: Strominger-Yau-Zaslow (SYZ) conjecture

Verdict: partially resolved.

5.12.3.1. Precise problem checked

The problem asks for the full SYZ mirror construction: given a Calabi-Yau manifold X with a special Lagrangian torus fibration $\pi : X \rightarrow B$, the mirror should be obtained by dualizing each smooth torus fiber T^n to $\widehat{T}^n \cong H^1(T^n, U(1))$, with a mathematically precise treatment of singular fibers by quantum/instanton corrections, and with the corresponding brane transform identifying a fiber brane with a point/D0-brane on the dual fiber. This is stronger than proving only the existence of special Lagrangian tori on a large open set, or proving a topological/logarithmic analogue, or proving the construction for a particular class of examples.

5.12.3.2. Current status

No published or otherwise standard citable result proves the full statement in this generality. The literature gives major partial results: semi-flat/no-singular-fiber constructions, topological compactifications in special examples, the Gross-Siebert logarithmic/tropical reconstruction program, wall-crossing and open Gromov-Witten corrections for toric and related examples, family-Floer constructions in the absence of singular fibers and some local singular models, and recent metric SYZ results for maximal degenerations. These do not together imply the full

statement for arbitrary Calabi-Yau manifolds admitting special Lagrangian torus fibrations with singular fibers and quantum corrections.

5.12.3.3. Key literature and what it proves

- **Mirror Symmetry is T-Duality**; Andrew Strominger, Shing-Tung Yau, Eric Zaslow; *Nuclear Physics B* 479 (1996), 243-259; DOI: 10.1016/0550-3213(96)00434-8; arXiv: hep-th/9606040, <https://arxiv.org/abs/hep-th/9606040>. This is the original proposal. It argues that a mirror is the moduli space of special Lagrangian torus fibers with flat $U(1)$ -connections and that mirror symmetry is fiberwise T-duality. It is not a complete mathematical proof, especially around singular fibers.
- **Singularities of special Lagrangian fibrations and the SYZ Conjecture**; Dominic Joyce; *Communications in Analysis and Geometry* 11 (2003), no. 5, 859-907; arXiv: math/0011179, <https://arxiv.org/abs/math/0011179>. Joyce constructs and studies local examples of special Lagrangian fibrations and emphasizes that singularities are poorly understood; he argues that generic discriminants can be codimension one and that stronger/literal versions of SYZ are unlikely to be true as stated. This is evidence against treating the problem statement as fully resolved.
- **Topological Mirror Symmetry**; Mark Gross; *Inventiones Mathematicae* 144 (2001), 75-137; arXiv: math/9909015, <https://arxiv.org/abs/math/9909015>. Gross proves a topological version for certain T^3 -fibrations, including the quintic/mirror quintic at the topological level. This supports the SYZ picture but omits the special Lagrangian metric condition, analytic singular-fiber corrections, and full brane/Floer-theoretic mirror equivalence.
- **Mirror symmetry and the Strominger-Yau-Zaslow conjecture**; Mark Gross; *Current Developments in Mathematics 2012* (International Press, 2013), 133-191; DOI: 10.4310/CDM.2012.v2012.n1.a3; arXiv: 1212.4220, <https://arxiv.org/abs/1212.4220>. This survey records that the original differential-geometric conjecture had successes but that Joyce's work made the literal form unlikely; it describes the Gross-Siebert program as an algebro-geometric route inspired by SYZ, not as a proof of the full original statement.
- **From real affine geometry to complex geometry**; Mark Gross, Bernd Siebert; *Annals of Mathematics* 174 (2011), 1301-1428; DOI: 10.4007/annals.2011.174.3.1; arXiv: math/0703822, <https://arxiv.org/abs/math/0703822>. This gives a reconstruction theorem producing degenerations of Calabi-Yau manifolds from affine manifolds with singularities and scattering/tropical data. It is a major logarithmic/tropical realization of SYZ-style quantum corrections, but it does not prove that every special Lagrangian torus fibration produces the mirror by literal fiberwise T-duality across singular fibers.
- **Mirror symmetry and T-duality in the complement of an anticanonical divisor**; Denis Auroux; *Journal of Gokova Geometry Topology* 1 (2007), 51-91; arXiv: 0706.3207, <https://arxiv.org/abs/0706.3207>. Auroux studies concrete wall-crossing and quantum corrections in SYZ-type constructions. This is important special-case progress, not a general resolution.
- **SYZ mirror symmetry for toric Calabi-Yau manifolds**; Kwokwai Chan, Siu-Cheong Lau, Naichung Conan Leung; *Journal of Differential Geometry* 90 (2012), no. 2, 177-250; arXiv: 1006.3830, <https://arxiv.org/abs/1006.3830>. The authors construct SYZ mirrors of toric Calabi-Yau manifolds using T-duality corrected by open Gromov-Witten

invariants. This covers a rich special class, but not arbitrary Calabi-Yau fibrations or the full singular-fiber problem.

- **Family Floer cohomology and mirror symmetry**; Mohammed Abouzaid; arXiv preprint (2014), arXiv: 1404.2659, <https://arxiv.org/abs/1404.2659>. Abouzaid explains a family-Floer construction of the mirror from a Lagrangian torus fibration in the absence of singular fibers. The no-singular-fiber hypothesis is a substantial restriction relative to this problem.
- **Family Floer mirror space for local SYZ singularities**; Hang Yuan; *Forum of Mathematics, Sigma* 12 (2024), e119; DOI: 10.1017/fms.2024.107; arXiv: 2206.04652, <https://arxiv.org/abs/2206.04652>. Yuan gives a precise family-Floer/non-archimedean SYZ statement and proves it for toric Calabi-Yau manifolds with Gross Lagrangian fibrations, including local singular-fiber behavior. This is significant progress with singularities, but still special-case/local rather than the full conjecture.
- **Valuative independence and metric SYZ conjecture**; Yang Li; arXiv preprint (2026), arXiv: 2605.00516, <https://arxiv.org/abs/2605.00516>, together with **Valuative independence for Calabi-Yau varieties**; Harold Blum, Yuchen Liu; arXiv preprint (2026), arXiv: 2604.27890, <https://arxiv.org/abs/2604.27890>. These recent works address the metric SYZ conjecture for polarized maximal degenerations: roughly, the existence of a special Lagrangian T^n -fibration on an open subset of X_t of measure at least $1 - \delta$. Even if counted as resolving this metric formulation, it is not the same as the full mirror construction: it does not provide a global dual mirror with singular-fiber quantum corrections and the D-brane/D0-brane transform in all cases.

5.12.3.4. Claimed full-resolution result not counted

The search also found **The SYZ Conjecture through Computational Equivalence**; Logan Nye; *Advances in Pure Mathematics* 15 (2025), no. 2, 145-179; DOI: 10.4236/apm.2025.152007. Although it claims to prove SYZ, its conclusion is formulated through a quantum-complexity/computational-equivalence framework and does not supply a standard theorem in differential, symplectic, or algebraic geometry proving the full fiberwise dual torus construction with singular-fiber corrections. It therefore does not resolve the stated mathematical question.

5.12.3.5. What remains unresolved

The unresolved part is the general, precise, global SYZ theorem: starting from a special Lagrangian torus fibration with singular fibers on an arbitrary relevant Calabi-Yau manifold, construct the mirror by fiberwise dualization plus quantum corrections, prove that the corrected compactified dual is the mirror in the full mathematical sense intended by mirror symmetry, and establish the corresponding brane/Fourier-Mukai/family-Floer transform across singular fibers. Current results prove this package only in restricted settings or in weakened/reformulated versions.

5.12.4. Remaining open problems recorded by the database

5.12.4.1. Open problem 1

Prove that the two piecewise-smooth special Lagrangian fibrations F and F' constructed by Joyce are the generic local models for codimension-one singularities of special Lagrangian fibrations of almost Calabi–Yau threefolds: locally the fibrations should have the same topology, their fibres should approach those of F or F' , and the singular fibres should have conical singularities modeled on Joyce’s special Lagrangian T^2 -cone $C \subset \mathbb{C}^3$.

Reference supplied by the database. Dominic Joyce, “Singularities of Special Lagrangian Fibrations and the SYZ Conjecture,” *Communications in Analysis and Geometry* **11** (2003), no. 5, 859–907, Conjecture 5.5. arXiv: [math/0011179](https://arxiv.org/abs/math/0011179); journal PDF: [International Press](#).

5.12.4.2. Open problem 2

Prove that Joyce’s piecewise-smooth special Lagrangian fibration $\hat{F}$ is a generic local model for a codimension-two singularity of a special Lagrangian fibration of an almost Calabi–Yau threefold, including the transition where two multiplicity-one conical singular points merge into a multiplicity-two nonconical singularity.

Reference supplied by the database. Dominic Joyce, “Singularities of Special Lagrangian Fibrations and the SYZ Conjecture,” *Communications in Analysis and Geometry* **11** (2003), no. 5, 859–907, Conjecture 6.7. arXiv: [math/0011179](https://arxiv.org/abs/math/0011179); journal PDF: [International Press](#).

5.12.4.3. Open problem 3

Prove that a generic special Lagrangian fibration of an almost Calabi–Yau threefold which contains singular fibres is never a smooth map, but only a piecewise-smooth map; in particular, under a small generic deformation, the codimension-two graph-like discriminant of a smooth fibration should thicken into codimension-one ribbons.

Reference supplied by the database. Dominic Joyce, “Singularities of Special Lagrangian Fibrations and the SYZ Conjecture,” *Communications in Analysis and Geometry* **11** (2003), no. 5, 859–907, Conjecture 7.15 and the discussion immediately following it. arXiv: [math/0011179](https://arxiv.org/abs/math/0011179); journal PDF: [International Press](#).

5.12.4.4. Open problem 4

Prove or disprove that for generic mirror Calabi–Yau threefolds M and $\widehat{M}$, even when special Lagrangian fibrations $f : M \rightarrow B$ and $\widehat{f} : \widehat{M} \rightarrow \widehat{B}$ exist, in general there is no homeomorphism $B \cong \widehat{B}$ which simultaneously identifies their discriminant loci and makes their nonsingular 3-torus fibres have dual homology.

Reference supplied by the database. Dominic Joyce, “Singularities of Special Lagrangian Fibrations and the SYZ Conjecture,” *Communications in Analysis and Geometry* **11** (2003), no. 5, 859–907, Conjecture 8.1. arXiv: [math/0011179](https://arxiv.org/abs/math/0011179); journal PDF: [International Press](#).

5.12.4.5. Open problem 5

Construct the family-Floer mirror space for a higher-dimensional Lagrangian torus fibration whose singular fibres are not Lagrangian immersions, by defining and controlling the Floer theory of those singular fibres and gluing it compatibly to the dual-torus charts over the smooth locus.

Reference supplied by the database. Mohammed Abouzaid, “Family Floer Cohomology and Mirror Symmetry,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 813–836, especially Introduction and §1.2, item (2), which identifies nonimmersed higher-dimensional singular fibres as beyond existing family-Floer techniques. arXiv: [1404.2659](https://arxiv.org/abs/1404.2659).

5.12.4.6. Open problem 6

Let $\mathcal{X} \rightarrow D$ be a polarized toric degeneration with intersection complex $(\check{B}, \check{D})$, and let ω_t represent the first Chern class of the polarization on $\mathcal{X}_t$. Prove that there is an open set $\check{U} \subseteq \check{B}$, whose complement retracts onto the discriminant locus of $\check{B}$, such that every smooth fibre $\mathcal{X}_t$ is a symplectic compactification of the semi-flat torus bundle $\check{X}(\check{U})$.

Reference supplied by the database. Mark Gross, “Mirror Symmetry and the Strominger–Yau–Zaslow Conjecture,” in *Current Developments in Mathematics 2012*, International Press, 2013, pp. 133–191, conjecture in §4 following the construction of the intersection complex (the polarized-toric-degeneration symplectic compactification conjecture). DOI: [10.4310/CDM.2012.v2012.n1.a3](https://doi.org/10.4310/CDM.2012.v2012.n1.a3); arXiv: [1212.4220](https://arxiv.org/abs/1212.4220).

5.12.5. Primary-source pointer

The originating ICM record is *Mirror Symmetry and Quantum Geometry* (2) (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=422>.

5.13. 113. Symplectic field theory and its applications [2006-geom-102]

Record: 2006-geom-102

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Symplectic field theory and its applications

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=227>

5.13.1. Problem statement

Prove that for a Stein fillable contact manifold (V, ξ) , the cylindrical contact homology $CH_*(V, \xi)$ is well defined; that is, it is independent of the auxiliary choices of contact form λ , almost complex structure J , and augmentation used to form the linearized contact-homology complex. Determine whether the same independence holds more generally for symplectically fillable contact manifolds (V, ξ) .

5.13.2. Historical background, literature, and current progress

5.13.3. Verdict: partially resolved

5.13.3.1. Precise problem

The problem asks whether the cylindrical/linearized contact homology $CH_*(V, \xi)$ of a contact manifold is a well-defined contact invariant, independent of the contact form λ , compatible almost complex structure J , and the augmentation used to linearize the contact homology DGA. The asserted target range is at least all Stein fillable contact manifolds, and possibly all symplectically fillable contact manifolds.

This is stronger than the now-standard statement that the full contact homology algebra is well defined: it asks for a single cylindrical/linearized homology independent of augmentation, and for a theorem covering all Stein fillable or all symplectically fillable cases.

5.13.3.2. Original source

- Title: "Symplectic field theory and its applications"
- Author: Yakov Eliashberg
- Venue/year: International Congress of Mathematicians, Vol. I, European Mathematical Society, Zurich, 2007, pp. 217-246; ICM Madrid 2006 lecture
- DOI/URL: <https://doi.org/10.4171/022-1/10> ; PDF source found at <https://www.staff.science.uu.nl/~crain101/Floer/symp13.pdf>
- Relevant content: Eliashberg states that if the contact homology algebra has a unique augmentation, then the linearized homology is independent of λ, J, ϵ , and says it seems likely that cylindrical contact homology is well defined at least for all Stein fillable, per-

haps symplectically fillable, contact manifolds. The same passage lists only examples then known, including subcritical Stein-fillable manifolds with $c_1(\xi) = 0$.

5.13.3.3. Current progress

- Title: “Cylindrical contact homology of subcritical Stein-fillable contact manifolds”
- Author: Mei-Lin Yau
- Venue/year: Geometry & Topology 8 (2004), no. 3, 1243-1280
- DOI/arXiv/URL: <https://doi.org/10.2140/gt.2004.8.1243> ; arXiv:math/0409542
- Result/status: Computes and establishes cylindrical contact homology for subcritical Stein-fillable contact manifolds under the additional hypothesis $c_1(\xi) = 0$. This is a significant special case, but it does not cover all Stein fillable manifolds, nor symplectically fillable manifolds in general.
- Title: “ S^1 -equivariant symplectic homology and linearized contact homology”
- Authors: Frederic Bourgeois and Alexandru Oancea
- Venue/year: International Mathematics Research Notices 2017, no. 13, 3849-3937
- DOI/arXiv/URL: <https://doi.org/10.1093/imrn/rnw029> ; arXiv:1212.3731
- Result/status: Shows that, over rational coefficients, positive S^1 -equivariant symplectic homology is isomorphic to linearized contact homology when the latter is defined, and gives a rigorous S^1 -equivariant substitute. This supplies a powerful replacement and comparison theorem, but not a theorem that cylindrical/linearized contact homology is independent of augmentation for all Stein or symplectically fillable contact manifolds.
- Title: “Contact homology and virtual fundamental cycles”
- Author: John Pardon
- Venue/year: Journal of the American Mathematical Society 32 (2019), 825-919
- DOI/arXiv/URL: <https://doi.org/10.1090/jams/924> ; arXiv:1508.03873
- Result/status: Constructs contact homology in the Eliashberg-Givental-Hofer sense using coherent virtual fundamental cycles. This resolves the basic foundations of the full contact homology DGA/algebra as an invariant, but it does not by itself prove that the linearized homology for an augmentation is independent of the augmentation in the sense asked in the problem statement.
- Title: “Semi-global Kuranishi charts and the definition of contact homology”
- Authors: Erkao Bao and Ko Honda
- Venue/year: Advances in Mathematics 414 (2023), article 108864
- DOI/arXiv/URL: <https://doi.org/10.1016/j.aim.2023.108864> ; arXiv:1512.00580
- Result/status: Defines the contact homology algebra for any contact manifold via semi-global Kuranishi charts and proves invariance under auxiliary choices. Like Pardon, this addresses full contact homology, not the full augmentation-independence claim for cylindrical/linearized contact homology in all Stein/symplectically fillable cases.

- Title: “Contact Homology and Linearization Without DGA Homotopies”
- Author: Julian Chaidez
- Venue/year: International Mathematics Research Notices 2025, no. 21, article rnaf325
- DOI/arXiv/URL: <https://doi.org/10.1093/imrn/rnaf325> ; arXiv:2412.20676
- Result/status: Proves that the set of weak-equivalence classes of augmentations of a closed contact manifold is well defined up to canonical bijection, and that the linearized contact homology associated to an augmentation class is well defined up to non-canonical isomorphism. It also proves augmentation-independence for strongly asymptotically dynamically convex (SADC) contact manifolds, and notes that subcritical Stein-fillable contact manifolds with zero first Chern class are SADC. However, it explicitly does not supply a single augmentation-independent linearized contact homology for arbitrary Stein fillable or symplectically fillable contact manifolds.
- Title: “Contact manifolds with flexible fillings”
- Author: Oleg Lazarev
- Venue/year: Geometric and Functional Analysis 30 (2020), no. 1, 188-254
- DOI/arXiv/URL: <https://doi.org/10.1007/s00039-020-00524-6> ; arXiv:1610.04837
- Result/status: Develops asymptotically dynamically convex contact structures and proves filling-independence results for positive symplectic homology in that setting, including flexible-fillable cases under hypotheses. This is important partial progress related to the linearized contact-homology substitute, but it is not all Stein fillable or all symplectically fillable contact manifolds.
- Title: “Contact homology of contact manifolds and its applications”
- Author: Frederic Bourgeois
- Venue/year: arXiv preprint/survey arXiv:2504.16540, 2025; listed by the author as for a Celebratio Mathematica volume on Yakov Eliashberg
- DOI/arXiv/URL: <https://arxiv.org/abs/2504.16540>
- Result/status: A current survey explicitly distinguishes full contact homology, cylindrical contact homology, and linearized contact homology. It states that cylindrical contact homology is well defined under extra Reeb-index hypotheses excluding problematic rigid planes, and that linearized contact homology depends on the augmentation, while the collection over all augmentations is a contact invariant. This matches the conclusion that the full statement is not presently proved.

5.13.3.4. What remains unresolved

No citable theorem found proves the exact statement for all Stein fillable contact manifolds, let alone all symplectically fillable contact manifolds. The literature now gives rigorous full contact homology for general contact manifolds and rigorous invariance of the collection of linearized homologies over augmentation classes. It also gives augmentation uniqueness/independence for special classes such as SADC contact manifolds, including subcritical Stein-fillable cases with $c_1 = 0$, and related filling-independence results for positive symplectic homology under ADC/flexible hypotheses.

The missing part is the full augmentation-independence and cylindrical/linearized invariant claimed in the problem statement for every Stein fillable contact manifold, and a fortiori for every symplectically fillable contact manifold. Therefore the problem is partially resolved, not fully resolved.

5.13.4. Remaining open problems recorded by the database

5.13.4.1. Open problem 1

Is cylindrical contact homology well defined for every subcritical Stein-fillable contact manifold without assuming that the first Chern class $c_1(\xi)$ vanishes?

Reference supplied by the database. Yakov Eliashberg, “Symplectic field theory and its applications,” in *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Vol. I, European Mathematical Society, Zürich, 2007, pp. 217–246, §3.2, example (a) following formula (43). DOI: [10.4171/022-1/10](https://doi.org/10.4171/022-1/10).

5.13.5. Primary-source pointer

The originating ICM record is *Symplectic field theory and its applications* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=227>.

5.14. 114. Gromov-Witten invariants and moduli spaces of curves [2006-ag-108]

Record: 2006-ag-108

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Gromov-Witten invariants and moduli spaces of curves

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=801>

5.14.1. Problem statement

Let V^{2d} be a compact symplectic manifold with a compatible almost-complex structure such that $H^{\text{odd}}(V; \mathbb{C}) = 0$, and fix a basis $\{\gamma_1, \dots, \gamma_N\}$ of $H^*(V; \mathbb{C})$ with γ_1 the identity. For $g \geq 0$, let $F_g(t)$ be the genus- g Gromov-Witten potential, viewed as a formal power series in the big phase-space variables t_n^α with coefficients in the Novikov ring:

$$F_g(t) = \sum_{k \geq 0} \frac{1}{k!} \sum_{\substack{\alpha_1, \dots, \alpha_k \in \{1, \dots, N\} \\ n_1, \dots, n_k \geq 0}} t_{n_1}^{\alpha_1} \cdots t_{n_k}^{\alpha_k} \sum_{A \in H_2(V; \mathbb{Z})} q^A \langle \tau_{n_1}(\gamma_{\alpha_1}) \cdots \tau_{n_k}(\gamma_{\alpha_k}) \rangle_{g, A}.$$

Assume that V has semisimple quantum cohomology in the sense that, for the big quantum product

$$\omega_1 \circ \omega_2 = \llbracket \omega_1 \omega_2 \gamma^\alpha \rrbracket_0 \gamma_\alpha,$$

the endomorphism $W \mapsto \mathcal{X} \circ W$ of the span of the primary fields γ_α has distinct eigenvalues at generic points of the big phase space; here $\llbracket \cdots \rrbracket_0$ denotes genus-0 correlation functions, γ^α is dual to γ_α under the Poincaré pairing, and $\mathcal{X}$ is the Euler vector field determined by the grading and by cup product with $c_1(V)$.

Prove that, for every integer $g \geq 2$, F_g is explicitly recoverable from the universal equations, namely the partial differential equations for $F_0, \dots, F_g$ obtained by pulling tautological relations among powers of ψ -classes and boundary classes on the Deligne-Mumford spaces $\overline{M}_{h,k}$ of stable curves back to the stable-map moduli spaces $\overline{M}_{h,k}(V, A)$.

5.14.2. Historical background, literature, and current progress

5.14.3. Verdict: partially resolved

5.14.3.1. Precise problem

The problem asks whether, for a compact symplectic target V with no odd cohomology and generically semisimple big quantum cohomology, every higher-genus descendant potential F_g for $g \geq 2$ is explicitly recoverable from the universal differential equations obtained by pulling tautological/topological-recursion relations on $\overline{M}_{h,k}$ back to stable-map moduli spaces.

This is not just the broader Givental-Teleman semisimple reconstruction statement. The ex-

act issue is whether the universal equations coming from tautological relations are known to determine F_g in all genera.

5.14.3.2. Current status

No published theorem found proves the exact all-genus statement via universal equations/topological recursion relations. The literature gives substantial partial progress:

- genus 1 is known by Dubrovin-Zhang;
- genus 2 is known by Xiaobo Liu, using universal equations/topological recursion methods;
- Givental-Teleman proves a closely related but different all-genus semisimple reconstruction theorem for CohFT/Gromov-Witten theory;
- recent papers continue to produce new universal equations and still describe the all-universal-equations determination route as conjectural.

Therefore the correct status for this problem is partially resolved.

5.14.3.3. Key literature

1. Xiaobo Liu, “Gromov-Witten invariants and moduli spaces of curves”, in *Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. II*, European Mathematical Society, 2006, pp. 791-812. DOI: 10.4171/022-2/38. URL: <https://ems.press/books/standalone/21/422>.

This is the source matching the problem statement. Its abstract states the expectation that, for semisimple quantum cohomology, information from the geometry of moduli spaces of curves might determine all higher-genus Gromov-Witten invariants.

2. Boris Dubrovin and Youjin Zhang, “Bi-Hamiltonian hierarchies in 2D topological field theory at one-loop approximation”, *Communications in Mathematical Physics* 198(2), 311-361, 1998. DOI: 10.1007/s002200050480. arXiv/hep-th:9712232.

This is cited in later universal-equation literature as proving the genus-1 semisimple case.

3. Xiaobo Liu, “Genus-2 Gromov-Witten invariants for manifolds with semisimple quantum cohomology”, *American Journal of Mathematics* 129(2), 463-498, 2007. DOI: 10.1353/ajm.2007.0012. arXiv:math/0310410.

Gives a formula for the genus-2 generating function for semisimple quantum cohomology and proves the genus-2 Virasoro conjecture. This resolves the $g = 2$ case but not all $g \geq 2$.

4. Constantin Teleman, “The structure of 2D semi-simple field theories”, *Inventiones Mathematicae* 188, 525-588, 2012. DOI: 10.1007/s00222-011-0352-5. arXiv:0712.0160.

Teleman’s Theorem 8.15 says that Gromov-Witten classes of a compact symplectic manifold are uniquely determined by c_1 and the quantum multiplication law at a semisimple point. This confirms Givental’s higher-genus reconstruction conjecture, but it is not a proof that the tautological universal equations themselves explicitly determine every F_g .

5. Y.-P. Lee, “Invariance of tautological equations II: Gromov-Witten theory”, *Journal of the American Mathematical Society* 22(2), 331-352, 2009. DOI: 10.1090/S0894-0347-08-00616-4. arXiv:math/0605708.

Proves strong invariance properties of tautological equations under Givental loop-group actions. This supports the tautological-equation program but does not by itself supply an all-genus determination theorem for the F_g .

6. Emily Clader, Felix Janda, Xin Wang, and Dmitry Zakharov, “Topological recursion relations from Pixton’s formula”, *Michigan Mathematical Journal* 73(2), 227–241, 2023. DOI: 10.1307/mmj/20195795. URL: <https://projecteuclid.org/journals/michigan-mathematical-journal/volume-73/issue-2/Topological-Recursion-Relations-from-Pixtons-Formula/10.1307/mmj/20195795.full>.

The introduction explicitly states that, for semisimple targets, universal equations are known to determine all Gromov-Witten invariants in genus 1 or 2, but that the genus-3 analogue does not follow from the currently known universal equations.

7. Felix Janda and Xin Wang, “Universal Equations for Higher Genus Gromov-Witten Invariants from Hodge Integrals”, *Communications in Mathematical Physics* 405, Article 47, 2024. DOI: 10.1007/s00220-024-04931-w.

Establishes new all-genus universal equations, showing ongoing progress, but not a full reconstruction theorem from all such equations.

8. Xin Wang, “New Push-Forward Relation and Universal Equation for Higher Genus Gromov-Witten invariants”, *Communications in Mathematical Physics* 407, Article 118, 2026. DOI: 10.1007/s00220-026-05634-0.

Gives another new universal equation and push-forward relation. This is further partial progress rather than a complete solution.

9. Felix Janda and Xin Wang, “Structures in topological recursion relations”, arXiv:2601.20673, 2026.

This recent preprint still phrases the statement that the collection of all universal equations determines all higher-genus Gromov-Witten invariants from genus-zero data in the semisimple case as a conjecture, distinct from the Givental or Dubrovin-Zhang approaches.

5.14.3.4. What remains unresolved

What remains open is a citable all-genus theorem proving that, for every compact symplectic V satisfying the stated semisimplicity hypothesis and every $g \geq 2$, F_g is explicitly recoverable from the universal PDEs obtained from tautological relations on moduli spaces of curves.

Teleman’s theorem means the broader semisimple all-genus reconstruction problem is solved by CohFT classification/Givental reconstruction, but the exact universal-equations formulation in this problem is not fully resolved.

5.14.4. Remaining open problems recorded by the database

5.14.4.1. Open problem 1

For a target with semisimple quantum cohomology, prove that all genus-3 Gromov–Witten invariants are explicitly determined by genus-0 data through the universal equations obtained from tautological relations on moduli spaces of stable curves.

Reference supplied by the database. Emily Clader, Felix Janda, Xin Wang, and Dmitry Zakharov, “Topological recursion relations from Pixton’s formula,” *Michigan Mathematical Journal* **73** (2023), no. 2, 227–241, Introduction (explicitly observing that the genus-3 analogue of the known genus-1 and genus-2 reconstruction does not follow from the currently known universal equations). DOI: [10.1307/mmj/20195795](https://doi.org/10.1307/mmj/20195795); arXiv: [1704.02011](https://arxiv.org/abs/1704.02011).

5.14.4.2. Open problem 2

For every $g \geq 2$, prove that the universal pure-descendant Gromov–Witten equation in Janda and Wang’s Corollary 2 (their equation arising from the $r = 0$ extension of the Liu–Pandharipande boundary push-forward relation) is a consequence of the degree- g topological recursion relations of Clader–Janda–Wang–Zakharov.

Reference supplied by the database. Felix Janda and Xin Wang, “Universal equations for higher genus Gromov–Witten invariants from Hodge integrals,” *Communications in Mathematical Physics* **405** (2024), no. 2, Article 47, Conjecture 11 in §3.3 (verified there only for $g = 2$). DOI: [10.1007/s00220-024-04931-w](https://doi.org/10.1007/s00220-024-04931-w); arXiv: [2401.00899](https://arxiv.org/abs/2401.00899).

5.14.4.3. Open problem 3

Let a degree- g topological recursion relation contain a term $c(\tilde{\zeta}_{\Gamma_{0,g,[n]}})_*(1)$ for the bouquet graph with one genus-0 vertex, g loops, and all n markings. Assume every other graph has at least two vertices and that there are no κ -classes and no genus- h vertex decorated by a ψ -monomial of degree at least $h + \delta_h^0$. Prove that $c = 0$.

Reference supplied by the database. Felix Janda and Xin Wang, “Structures in topological recursion relations,” *Transactions of the American Mathematical Society, Series B* **12** (2025), 1223–1263, Conjecture 3. DOI: [10.1090/btran/238](https://doi.org/10.1090/btran/238); arXiv: [2601.20673](https://arxiv.org/abs/2601.20673).

5.14.5. Primary-source pointer

The originating ICM record is *Gromov–Witten invariants and moduli spaces of curves* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=801>.

5.15. 115. Gromov-Witten invariants and moduli spaces of curves [2006-ag-109]

Record: 2006-ag-109

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Gromov-Witten invariants and moduli spaces of curves

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=801>

5.15.1. Problem statement

Let V be a compact symplectic manifold, and let $F_g(t)$ be the genus- g Gromov-Witten descendant potential of V on the big phase space with coordinates t_n^α corresponding to descendants $\tau_n(\gamma_\alpha)$ of a homogeneous basis $\{\gamma_\alpha\}$ of $H^*(V; \mathbb{C})$. For vector fields $\mathcal{W}_i$ on the big phase space, write

$$\langle\langle \mathcal{W}_1 \cdots \mathcal{W}_k \rangle\rangle_g$$

for the k -th covariant derivative of F_g ; in particular $\langle\langle \mathcal{W} \rangle\rangle_g = \mathcal{W}(F_g)$.

Let $\mathcal{L}_n$, $n \geq -1$, be the Virasoro vector fields used in the vector-field formulation of the Gromov-Witten Virasoro constraints: they are defined recursively by $\mathcal{L}_n = -R^{n+1}(\mathcal{S})$, where $\mathcal{S} = -\sum_{m,\alpha} (t_m^\alpha - \delta_{m,1} \delta_{\alpha,1}) \tau_{m-1}(\gamma_\alpha)$ is the string vector field, with $\tau_{-1}(\gamma_\alpha) = 0$, and $R = GT + C$,

$$T(\mathcal{W}) = \tau_+(\mathcal{W}) - \langle\langle \mathcal{W} \gamma^\alpha \rangle\rangle_0 \gamma_\alpha.$$

Here τ_+ raises descendant level, $G(\tau_m(\gamma_\alpha)) = (m + b_\alpha) \tau_m(\gamma_\alpha)$ is the grading operator with $b_\alpha = \frac{1}{2}(\deg \gamma_\alpha - d + 1)$ when $\dim_{\mathbb{R}} V = 2d$ and, for a smooth projective d -fold with $\gamma_\alpha \in H^{p_\alpha, q_\alpha}(V)$, with the Virasoro weight $b_\alpha = p_\alpha - \frac{1}{2}(d - 1)$, C is induced by cup product with $c_1(V)$, and repeated Greek indices are summed using the Poincare pairing $\eta_{\alpha\beta} = \int_V \gamma_\alpha \cup \gamma_\beta$, so that $\gamma^\alpha = \eta^{\alpha\beta} \gamma_\beta$. Universal equations mean the partial differential equations for the potentials F_h obtained by translating tautological relations among ψ -classes and boundary strata on the Deligne-Mumford moduli spaces $\overline{\mathcal{M}}_{h,k}$ through the stabilization maps from moduli spaces of stable maps to V .

Prove that, for every compact symplectic manifold V , every $g \geq 1$, and every $n \geq 2$, the genus- g one-point Virasoro correlation $\langle\langle \mathcal{L}_n \rangle\rangle_g$ can be computed from $\langle\langle \mathcal{L}_1 \rangle\rangle_g$ together with lower-genus Gromov-Witten data, meaning the potentials F_h and their correlation functions for $h < g$. Prove also that, if V is a smooth projective variety, then the full Gromov-Witten Virasoro conjecture, namely the assertion that the total descendant potential $Z_V = \exp(\sum_{g \geq 0} \lambda^{2g-2} F_g)$ is annihilated by the half-Virasoro sequence of Eguchi-Hori-Xiong/Katz differential operators, follows from the universal equations together with the L_1 -constraint, i.e. the $n = 1$ Virasoro constraint.

5.15.2. Historical background, literature, and current progress

5.15.3. Verdict: partially resolved

5.15.3.1. Precise problem

The problem asks for an all-genus reduction statement in Gromov-Witten theory: for every compact symplectic target V , every $g \geq 1$, and every $n \geq 2$, the genus- g one-point Virasoro correlation $\langle\langle \mathcal{L}_n \rangle\rangle_g$ should be computable from $\langle\langle \mathcal{L}_1 \rangle\rangle_g$ and lower-genus data via universal equations coming from tautological relations on $\overline{M}_{g,k}$. For smooth projective V , this would imply that the full Eguchi-Hori-Xiong/Katz Virasoro conjecture follows from universal equations plus the L_1 -constraint.

5.15.3.2. Status

I found substantial low-genus and special-target progress, but no published or otherwise citable result proving the exact all-genus, all-compact-symplectic/smooth-projective statement in the problem statement. Recent literature still treats the corresponding all-genus determination by universal equations as conjectural, and describes the non-semisimple Virasoro problem as open. The correct status is therefore **partially resolved**.

5.15.3.3. Key literature and what it proves

1. Xiaobo Liu, "Gromov-Witten invariants and moduli spaces of curves," *Proceedings of the International Congress of Mathematicians, Vol. II*, European Mathematical Society, 2006, pp. 791-812; published 2007. DOI: 10.4171/022-2/38. URL: <https://ems.press/books/standalone/21/422>.

This is the source most closely matching the problem statement. Its abstract says the paper explains how much Gromov-Witten information is determined by moduli spaces of curves and says that, in the semisimple case, this information is believed to determine all higher genus invariants and help study the Virasoro conjecture. This is not a proof of the all-genus statement.

2. Xiaobo Liu, "Quantum product on the big phase space and the Virasoro conjecture," *Advances in Mathematics* 169(2), 313-375, 2002. DOI: 10.1006/aima.2001.2062; arXiv:math/0104030. URL: <https://arxiv.org/abs/math/0104030>.

This paper introduces the big-phase-space quantum product and the recursive Virasoro vector-field formalism used in the problem statement. It proves the genus-2 reduction: for any manifold satisfying the genus-1 L_1 -constraint, the genus-2 L_1 -constraint implies the genus-2 Virasoro conjecture, and for $n \geq 2$ the genus-2 L_n -constraint can be recursively computed from the genus-2 L_1 -constraint plus lower-genus terms. This is exactly analogous to this problem but only in genus 2, not all g .

3. Xiaobo Liu, "Elliptic Gromov-Witten invariants and Virasoro conjecture," *Communications in Mathematical Physics* 216, 705-728, 2001. DOI: 10.1007/s002200000357; arXiv:math/9907113. URL: <https://arxiv.org/abs/math/9907113>.

Gives necessary and sufficient conditions for the genus-1 Virasoro conjecture. This is low-genus progress and supports the $L_1/SL(2)$ -reduction pattern, but it does not prove the all-genus claim.

4. Xiaobo Liu and Gang Tian, “Virasoro constraints for quantum cohomology,” *Journal of Differential Geometry* 50(3), 537-590, 1998. DOI: 10.4310/jdg/1214424970; arXiv:math/9806028. URL: <https://arxiv.org/abs/math/9806028>.

Proves the genus-0 Virasoro constraints for quantum cohomology. This is foundational progress, but the problem concerns every $g \geq 1$.

5. Xiaobo Liu, “Genus-2 Gromov-Witten invariants for manifolds with semisimple quantum cohomology,” arXiv:math/0310410, 2004. URL: <https://arxiv.org/abs/math/0310410>.

Proves the genus-2 Virasoro conjecture for manifolds with semisimple quantum cohomology by giving a formula for F_2 . This is a special semisimple/low-genus result, not the general all-genus reduction in the problem statement.

6. Constantin Teleman, “The structure of 2D semi-simple field theories,” *Inventiones Mathematicae* 188, 525-588, 2012. DOI: 10.1007/s00222-011-0352-5. URL: <https://link.springer.com/article/10.1007/s00222-011-0352-5>.

Teleman’s classification confirms Givental’s higher-genus reconstruction conjecture and implies the Virasoro conjecture for targets with semisimple quantum cohomology. This is a powerful full-genus theorem under a semisimplicity hypothesis. It does not cover arbitrary smooth projective varieties or arbitrary compact symplectic manifolds, and it is not the claimed universal-equations-plus- L_1 reduction.

7. Andrei Okounkov and Rahul Pandharipande, “Virasoro constraints for target curves,” *Inventiones Mathematicae* 163, 47-108, 2006. DOI: 10.1007/s00222-005-0455-y; arXiv:math/0308097. URL: <https://arxiv.org/abs/math/0308097>.

Proves generalized Virasoro constraints for the relative Gromov-Witten theories of all nonsingular target curves. This is another major special-target result, not a proof for all smooth projective targets.

8. Tom Coates, Alexander Givental, and Hsian-Hua Tseng, “Virasoro constraints for toric bundles,” *Forum of Mathematics, Pi* 12, e4, 2024. DOI: 10.1017/fmp.2024.2. URL: <https://www.cambridge.org/core/journals/forum-of-mathematics-pi/article/virasoro-constraints-for-toric-bundles/8BE7221F9F98EBE37DEDAFB62F4ACBC5>.

Proves the Virasoro conjecture for toric bundles exactly when it holds for the base, with corollaries including bases with generically semisimple quantum cohomology, compact Riemann surfaces, K3 surfaces, and Calabi-Yau manifolds of dimension at least 3. This substantially enlarges the known cases, but remains a special-family theorem and does not imply the all-target statement.

9. Felix Janda and Xin Wang, “Universal Equations for Higher Genus Gromov-Witten Invariants from Hodge Integrals,” *Communications in Mathematical Physics* 405, article 47, 2024. DOI: 10.1007/s00220-024-04931-w. URL: <https://link.springer.com/article/10.1007/s00220-024-04931-w>.

Establishes new universal equations for higher-genus Gromov-Witten invariants from Hodge integrals. This is directly relevant progress on the universal-equation side, but it does not prove that universal equations plus L_1 imply all Virasoro constraints in every genus.

10. Xin Wang, “New Push-Forward Relation and Universal Equation for Higher Genus Gromov-Witten invariants,” *Communications in Mathematical Physics* 407, article 118, 2026. DOI: 10.1007/s00220-026-05634-0. URL: <https://link.springer.co>

[m/article/10.1007/s00220-026-05634-0](https://arxiv.org/abs/2601.20673).

Proves new push-forward tautological relations and obtains a new partial differential equation for higher-genus descendant Gromov-Witten invariants of any smooth projective variety. This is further progress, but still only supplies additional universal equations, not the full all-genus reduction asserted in the problem statement.

11. Felix Janda and Xin Wang, “Structures in topological recursion relations,” arXiv:2601.20673, 2026. URL: <https://arxiv.org/pdf/2601.20673>.

This preprint gives structural results and applications to Gromov-Witten universal equations. It explicitly states that it is still conjectured that all such equations determine all higher-genus Gromov-Witten invariants in terms of genus-0 invariants in the semisimple case. That language is inconsistent with a known proof of the stated all-genus claim.

12. Xin Wang, “Virasoro constraints for Hodge integrals,” arXiv:2506.10033, 2025. URL: <https://arxiv.org/pdf/2506.10033>.

A recent preprint on Virasoro constraints for Hodge integrals states that the Virasoro conjecture for targets with non-semisimple quantum cohomology is widely open. This supports the conclusion that the stated all-target smooth-projective implication is not known.

5.15.3.4. What remains unresolved

No source found proves that for every compact symplectic V , every $g \geq 1$, and every $n \geq 2$, $\langle\langle \mathcal{L}_n \rangle\rangle_g$ is computable from $\langle\langle \mathcal{L}_1 \rangle\rangle_g$ and lower-genus data via universal equations. Likewise, no source proves for all smooth projective varieties that universal equations together with the L_1 -constraint imply the full Eguchi-Hori-Xiong/Katz Virasoro constraints. The known results cover genus 0, genus 1/2 reductions, semisimple quantum cohomology, target curves, toric-bundle families, and many new universal equations, but not the full statement in this problem.

5.15.4. Remaining open problems recorded by the database

5.15.4.1. Open problem 1

Let X be an arbitrary smooth projective variety. Prove the genus-one L_1 Virasoro constraint for the descendant Gromov–Witten theory of X , equivalently $\Psi_{1,1;\emptyset} = 0$, without assuming that the quantum cohomology of X is semisimple.

Reference supplied by the database. Xin Wang, “Virasoro constraints for Hodge integrals,” *Communications in Number Theory and Physics* **19** (2025), no. 3, 631–682, Introduction (the vanishing of $\Psi_{1,n;\emptyset}$ for general targets is explicitly identified as open and reduced to $\Psi_{1,1;\emptyset}$); DOI: [10.4310/CNTP.250829234004](https://doi.org/10.4310/CNTP.250829234004); arXiv: [2506.10033](https://arxiv.org/abs/2506.10033).

5.15.4.2. Open problem 2

Let X and Y be smooth projective varieties. Prove that the Virasoro conjecture holds for the product $X \times Y$ if and only if it holds for both X and Y , without a semisimplicity hypothesis on either factor.

Reference supplied by the database. Hsian-Hua Tseng, “A note on Virasoro constraints for products,” preprint (2026), Problem 1.1 (with the case in which one factor has semisimple quantum cohomology proved as Theorem 1.2); arXiv: [2603.22486](https://arxiv.org/abs/2603.22486).

5.15.5. Primary-source pointer

The originating ICM record is *Gromov-Witten invariants and moduli spaces of curves* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=801>.

5.16. 116. Gauge theory and mirror symmetry [2014-geom-124]

Record: 2014-geom-124

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Gauge theory and mirror symmetry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1329>

5.16.1. Problem statement

Let T be a compact torus with Lie algebra $\mathfrak{t}$, let $T_{\mathbb{C}}^{\vee}$ be its complexified dual torus, and let X be a symplectic manifold with Hamiltonian T -action and moment map $\mu : X \rightarrow \mathfrak{t}^*$. Using the unitary mirror/spectral decomposition of the Fukaya category $\mathfrak{F}(X)$ over $T_{\mathbb{C}}^{\vee}$, write $\mathfrak{Vect}_{\tau}$ for the irreducible topological T -representation classified by $\tau \in T_{\mathbb{C}}^{\vee}$, i.e. the category of complex vector spaces with that topological T -action. If $q = \operatorname{Re} \log \tau \in \mathfrak{t}^*$ is a regular value of μ , prove that the multiplicity category, or τ -spectral component, of $\mathfrak{Vect}_{\tau}$ in $\mathfrak{F}(X)$ is equivalent to the Fukaya category of the symplectic quotient

$$X//_q T = \mu^{-1}(q)/T,$$

with imaginary curving, equivalently B -field, $\operatorname{Im} \log \tau$.

5.16.2. Historical background, literature, and current progress

5.16.3. Verdict: partially resolved

5.16.3.1. Problem identified

The problem asks for Teleman's torus symplectic-quotient conjecture: for every compact torus T , every Hamiltonian T -symplectic manifold X , and every $\tau \in T_{\mathbb{C}}^{\vee}$ whose real logarithm $q = \operatorname{Re} \log \tau$ is a regular moment-map value, the τ -spectral/multiplicity component of the Fukaya category $\mathfrak{F}(X)$ should be equivalent to the Fukaya category of the Marsden-Weinstein quotient $X//_q T = \mu^{-1}(q)/T$, with B -field $\operatorname{Im} \log \tau$.

This is Conjecture 4.2, "Torus symplectic quotients", in Teleman's ICM article.

5.16.3.2. Current literature status

The problem is not fully resolved in the exact generality of the problem. There is substantial partial progress: several papers prove Floer-cohomology-level, local mirror, exact/Liouville, or special-geometric versions of the expected relationship, and one recent preprint states a quasi-isomorphism under a detailed setup. However, I did not find a published or otherwise citable result proving the full spectral-component equivalence for arbitrary Hamiltonian compact-torus actions and arbitrary Fukaya categories with the stated B -field.

5.16.3.3. Key literature and applicability

Original source

- Title: “Gauge theory and mirror symmetry”
- Author: Constantin Teleman
- Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, Kyung Moon Sa, 2014, pp. 1309-1332
- arXiv/DOI/URL: arXiv:1404.6305, <https://arxiv.org/abs/1404.6305>, DOI <https://doi.org/10.48550/arXiv.1404.6305>
- Relevant result: Conjecture 4.2 states that the multiplicity of Vect_τ in $\mathfrak{F}(X)$ is the Fukaya category of the symplectic reduction at $\text{Re log } \tau$, with imaginary curving/B-field $\text{Im log } \tau$. Teleman explicitly frames it as a conjecture and notes singular values are not covered.

Regular-value and spectral-component progress

- Title: “Equivariant Fukaya categories at singular values”
- Authors: Yanki Lekili and Ed Segal
- Venue/year: Glasgow Mathematical Journal 68(2), May 2026, pp. 197-221; published online 2025
- DOI/arXiv/URL: DOI <https://doi.org/10.1017/S0017089525100487>; arXiv:2304.10969, <https://arxiv.org/abs/2304.10969>
- Relevant content: The paper states that Teleman and Fukaya proposed equivalence between Fukaya categories of symplectic quotients and equivariant/spectral Fukaya categories. It explains that for regular values the corresponding component should be the Fukaya category of the quotient, possibly with bulk deformation, and formulates extensions to singular values.
- Applicability check: This is progress and evidence, not a full proof of the problem. The paper repeatedly formulates conjectures, discusses examples and mirror heuristics, and points out possible sign ambiguities and bulk-deformation issues.
- Title: “Equivariant Lagrangian correspondence and a conjecture of Teleman”
- Authors: Siu-Cheong Lau, Nai-Chung Conan Leung, Yan-Lung Leon Li
- Venue/year: arXiv preprint, 2023; revised 2025
- DOI/arXiv/URL: arXiv:2312.13926, <https://arxiv.org/abs/2312.13926>, DOI <https://doi.org/10.48550/arXiv.2312.13926>
- Relevant content: The abstract says the equivariant Lagrangian-correspondence functor induces a quasi-isomorphism between the equivariant derived Fukaya category at a regular moment-map level and the derived Fukaya category of the corresponding quotient. The paper also proves precise relations between equivariant and reduced Lagrangian Floer cohomology and local mirror potentials.
- Applicability check: This is the closest result found, but it still does not fully settle the problem statement. Its main theorem is under Setup 4.14, assumes a free action on the relevant moment level and weak unobstructedness of L , L_π , and $\bar{L}$ (possibly after bulk deformation). Its Theorem 4.26 is explicitly a localized abelian mirror statement, with globalization deferred to forthcoming work. The problem asks for the entire τ -spectral Fukaya-category equivalence for all Hamiltonian compact-torus manifolds at all regular values, with the stated B -field and no extra unobstructedness/topological/toric/localization hy-

potheses.

Floer-homology-level and foundational partial results

- Title: “Equivariant Lagrangian Floer homology via cotangent bundles of EG_N ”
- Author: Guillem Cazassus
- Venue/year: Journal of Topology 17(1), 2024, article e12328
- DOI/arXiv/URL: DOI <https://doi.org/10.1112/topo.12328>; arXiv:2202.10097, <https://arxiv.org/abs/2202.10097>
- Relevant content: Constructs equivariant Lagrangian Floer homology for Hamiltonian compact-group actions and constructs Kirwan morphisms relating equivariant Floer complexes to Floer complexes on smooth reductions.
- Applicability check: This supplies important Floer-theoretic machinery, but it is not a Fukaya-category equivalence of spectral components.
- Title: “Equivariant Floer homology is isomorphic to reduced Floer homology”
- Author: Julio Sampietro Christ
- Venue/year: arXiv preprint, 2025
- DOI/arXiv/URL: arXiv:2505.02647, <https://arxiv.org/abs/2505.02647>, DOI <https://doi.org/10.48550/arXiv.2505.02647>
- Relevant content: Proves Cazassus’ conjecture that, under free action on the zero moment level, equivariant Lagrangian Floer homology agrees with Floer homology of the Marsden-Weinstein reduction.
- Applicability check: The theorem is for pairs of G -invariant Lagrangians contained in the zero level, with relative exactness and compactness/convexity assumptions, and gives a homology isomorphism rather than the full Fukaya-category spectral-component equivalence.
- Title: “Moment Lagrangian correspondences are unobstructed after bulk deformation”
- Author: Yao Xiao
- Venue/year: arXiv preprint, 2024
- DOI/arXiv/URL: arXiv:2405.11169, <https://arxiv.org/abs/2405.11169>, DOI <https://doi.org/10.48550/arXiv.2405.11169>
- Relevant content: Proves unobstructedness after bulk deformation for moment Lagrangian correspondences induced by symplectic reduction maps at free zero level sets, assuming equivariant Kuranishi structures and compatible perturbations.
- Applicability check: This addresses a key obstruction in the correspondence approach, but it is conditional on technical structures and does not itself prove the stated equivalence.

Exact/Liouville-sector and special-geometric results

- Title: “Equivariant Partially Wrapped Fukaya Categories on Liouville Sectors”
- Authors: Dongwook Choa, Jiawei Hu, Siu-Cheong Lau, Yan-Lung Leon Li

- Venue/year: arXiv preprint, 2025
- DOI/arXiv/URL: arXiv:2512.24382, <https://arxiv.org/abs/2512.24382>, DOI <https://doi.org/10.48550/arXiv.2512.24382>
- Relevant content: Develops equivariant wrapped/partially wrapped Fukaya categories for Liouville sectors and a correspondence theory. In the free-action case it produces a fully faithful A_∞ -functor from a full subcategory of the equivariant wrapped Fukaya category to the wrapped Fukaya category of the exact quotient.
- Applicability check: This is an exact/Liouville-sector result and is fully faithful from a full subcategory, not the full general equivalence for arbitrary Hamiltonian compact-torus symplectic manifolds and all spectral components.
- Title: “The Hamiltonian reduction of hypertoric mirror symmetry”
- Authors: Michael McBreen, Vivek Shende, Peng Zhou
- Venue/year: Advances in Mathematics 493, 2026, article 110881
- DOI/arXiv/URL: DOI <https://doi.org/10.1016/j.aim.2026.110881>; arXiv:2405.07955, <https://arxiv.org/abs/2405.07955>
- Relevant content: Proves a “Fukaya category commutes with reduction” theorem for the Hamiltonian torus action on a multiplicative hypertoric variety.
- Applicability check: This is a strong special-case theorem, but it is restricted to multiplicative hypertoric geometry and does not cover arbitrary Hamiltonian torus actions.

5.16.3.4. What remains unresolved

The unresolved part is the full, global categorical statement in Teleman’s Conjecture 4.2 exactly as in the problem statement: an equivalence between the τ -spectral/multiplicity category of $\mathfrak{F}(X)$ and the Fukaya category of $X//_q T$ with B -field $\text{Im } \log \tau$, for general compact torus T , general Hamiltonian T -symplectic manifolds X , and all regular values $q = \text{Re } \log \tau$, without imposing exact/Liouville, toric/hypertoric, relative exactness, weak unobstructedness, free-action-on-selected-Lagrangians, local mirror-chart, or bulk-deformation hypotheses beyond those in the problem statement.

Therefore the correct status is: partially resolved.

5.16.4. Remaining open problems recorded by the database

5.16.4.1. Open problem 1

Let

$$X = \{(x, y, z, w) \in \mathbb{C}^4 : xy + zw = 1\} \cong T^*S^3$$

and let S^1 act with weights $((1, -1, 1, -1))$. Prove that the spectral components of its S^1 -equivariant wrapped Fukaya category satisfy

$$\mathcal{W}^{S^1}(X)_1 \simeq \mathcal{W}(T^*S^2), \quad \mathcal{W}^{S^1}(X)_\lambda \simeq 0 \quad (\lambda \neq 1),$$

where the zero-level Hamiltonian reduction of (X) is T^*S^2 .

Reference supplied by the database. Yanki Lekili and Ed Segal, “Equivariant Fukaya Categories at Singular Values,” *Glasgow Mathematical Journal* **68** (2026), no. 2, 197–221, prediction in Example 2.8, especially equation (2.12), DOI: [10.1017/S0017089525100487](https://doi.org/10.1017/S0017089525100487), arXiv: [2304.10969](https://arxiv.org/abs/2304.10969).

5.16.5. Primary-source pointer

The originating ICM record is *Gauge theory and mirror symmetry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1329>.

5.17. 117. Quasi-morphisms and quasi-states in symplectic topology [2014-geom-132]

Record: 2014-geom-132

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Quasi-morphisms and quasi-states in symplectic topology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1167>

5.17.1. Problem statement

Let (M, ω) be a closed connected symplectic manifold, and let $C^\infty(M)$ and $C(M)$ denote the real-valued smooth and continuous functions on M . Let $\mathbb{A} \subset C^\infty(M)$ be a finite-dimensional Poisson-commutative vector subspace, meaning that $\{F, G\} = 0$ for every $F, G \in \mathbb{A}$. Define the moment map of $\mathbb{A}$ by

$$\Phi_{\mathbb{A}} : M \rightarrow \mathbb{A}^*, \quad \langle \Phi_{\mathbb{A}}(x), F \rangle = F(x).$$

Is it true that, for every such $\mathbb{A}$, there exist $p \in \mathbb{A}^*$ with $\Phi_{\mathbb{A}}^{-1}(p) \neq \emptyset$ and a symplectic quasi-state $\zeta : C(M) \rightarrow \mathbb{R}$ such that the fiber $\Phi_{\mathbb{A}}^{-1}(p)$ is heavy with respect to ζ ? Here a symplectic quasi-state is a normalized, monotone, homogeneous functional on $C(M)$ whose restriction to $C^\infty(M)$ is additive on Poisson-commuting functions, vanishes on functions supported in stably displaceable open subsets, and is invariant under $\text{Symp}_0(M)$, the identity component of the symplectomorphism group; a closed set $X \subset M$ is heavy with respect to ζ if $\zeta(H) \geq \inf_X H$ for every $H \in C(M)$. Equivalently, the desired fiber should satisfy

$$\zeta(H) \geq \inf_{\Phi_{\mathbb{A}}^{-1}(p)} H$$

for every $H \in C(M)$.

5.17.2. Historical background, literature, and current progress

5.17.3. Status: partially resolved

5.17.3.1. Problem identified

The problem asks whether every finite-dimensional Poisson-commutative subspace $\mathbb{A} \subset C^\infty(M)$ on an arbitrary closed connected symplectic manifold has a nonempty moment-map fiber $\Phi_{\mathbb{A}}^{-1}(p)$ that is heavy with respect to some genuine symplectic quasi-state $\zeta : C(M) \rightarrow \mathbb{R}$, where “genuine” is understood in the sense stated in the problem: normalized, monotone, homogeneous, additive on Poisson-commuting smooth functions, vanishing on stably displaceable supports, and $\text{Symp}_0(M)$ -invariant.

5.17.3.2. Current literature status

Original source

- Title: Quasi-morphisms and quasi-states in symplectic topology
- Author: Michael Entov
- Venue/year: Proceedings of the International Congress of Mathematicians 2014, Seoul; arXiv preprint 2014
- DOI/arXiv/URL: arXiv:1404.6408, <https://arxiv.org/abs/1404.6408>
- Relevant item: Question 4.9 asks whether any finite-dimensional Poisson-commutative subspace has a heavy fiber, at least with respect to some symplectic quasi-state. The preceding Corollary 4.8 gives only a non-displaceable fiber.
- Hypothesis/conclusion check: This is the source of the problem. The survey distinguishes genuine symplectic quasi-states from partial symplectic quasi-states; for arbitrary idempotents, including the fundamental class, one generally gets only the weaker partial version.

Earlier full result for non-displaceability, not heaviness

- Title: Quasi-states and symplectic intersections
- Authors: Michael Entov and Leonid Polterovich
- Venue/year: Commentarii Mathematici Helvetici 81 (2006), 75-99
- DOI/arXiv/URL: arXiv:math/0410338, <https://arxiv.org/abs/math/0410338>
- Relevant result: Theorem 2.1 proves that every finite-dimensional Poisson-commutative subspace of $C^\infty(M)$ has at least one non-displaceable fiber.
- Hypothesis/conclusion check: This covers all closed symplectic manifolds and all such A , but the conclusion is non-displaceability, not heaviness with respect to a genuine symplectic quasi-state.

Cases where genuine quasi-states are available

- Title: Symplectic quasi-states and semi-simplicity of quantum homology
- Authors: Michael Entov and Leonid Polterovich
- Venue/year: Toric Topology, Contemporary Mathematics 460, American Mathematical Society, 2008, 47-70
- DOI/arXiv/URL: DOI 10.1090/conm/460/09010; arXiv:0705.3735, <https://arxiv.org/abs/0705.3735>
- Relevant result: Constructs Calabi quasi-morphisms and genuine symplectic quasi-states for manifolds whose quantum homology has a suitable field direct summand.
- Hypothesis/conclusion check: Combined with later heavy-fiber theorems, this gives the literal conclusion for these field-split/field-factor cases. The field-factor hypothesis is extra and is not known for all closed symplectic manifolds.

Pseudoheavy fibers and simple partial quasi-states

- Title: Existence of pseudo-heavy fibers of moment maps
- Authors: Morimichi Kawasaki and Ryuma Orita
- Venue/year: Communications in Contemporary Mathematics 23 (2021), no. 5, 2050047
- DOI/arXiv/URL: DOI 10.1142/S0219199720500479; arXiv:1901.09395, <https://arxiv.org/abs/1901.09395>

- Relevant results: Theorem 1.7 proves that for any closed symplectic manifold, any finite-dimensional Poisson-commutative A , and any partial symplectic quasi-state, some fiber is pseudoheavy. Corollary 1.9 gives a heavy fiber when the partial symplectic quasi-state is simple.
- Hypothesis/conclusion check: Pseudoheaviness is weaker than heaviness, and simplicity is an extra hypothesis. The paper also gives examples of pseudoheavy fibers that are not heavy, so this does not resolve the problem statement as stated.

Strong recent heavy-fiber theorem in the spectral-invariant sense

- Title: Symplectic topology and ideal-valued measures
- Authors: Adi Dickstein, Yaniv Ganor, Leonid Polterovich, and Frol Zapolsky
- Venue/year: arXiv preprint, 2021; revised 2024
- DOI/arXiv/URL: arXiv:2107.10012, <https://arxiv.org/abs/2107.10012>
- Relevant contribution: Develops quantum-cohomology ideal-valued quasi-measures and SH-heavy sets, providing the framework used in later relative symplectic cohomology results.
- Hypothesis/conclusion check: This is major progress toward heavy-type rigidity and big-fiber theorems, but it does not itself supply a genuine scalar symplectic quasi-state for every closed symplectic manifold.
- Title: A characterization of heaviness in terms of relative symplectic cohomology
- Authors: Cheuk Yu Mak, Yuhan Sun, and Umut Varolgunes
- Venue/year: Journal of Topology 17 (2024), no. 1, e12327
- DOI/arXiv/URL: DOI 10.1112/topo.12327; arXiv:2301.12625, <https://arxiv.org/abs/2301.12625>
- Relevant results: Theorem 1.7 proves that a compact set is heavy if and only if its relative symplectic cohomology over the Novikov field is nonzero. Corollary 1.8 proves that the union of two Poisson-commuting non-heavy compact sets is non-heavy. Corollary 1.9 states that any involutive map $\Phi : M \rightarrow \mathbb{R}^N$ has at least one heavy fiber, and the paper explicitly says this gives a positive answer to Entov's Question 4.9.
- Hypothesis/conclusion check: Choosing a basis of A identifies Φ_A with an involutive map $M \rightarrow \mathbb{R}^N$, so Corollary 1.9 covers the same fibers. However, the heaviness in this paper is the standard spectral-invariant/idempotent notion, with the unit idempotent omitted in the introduction. This is the heavy-fiber conclusion for the homogenized spectral invariant/partial quasi-state framework. It implies the literal genuine-quasi-state version in cases where the chosen idempotent gives a genuine Entov-Polterovich symplectic quasi-state, for example field-factor cases. It does not by itself prove that every closed connected symplectic manifold admits a genuine symplectic quasi-state satisfying the stated axioms.

5.17.3.3. What remains unresolved

The original heavy-fiber question has very strong progress, and in the standard spectral-invariant/partial-quasi-state sense Mak-Sun-Varolgunes prove the expected heavy fiber for every involutive map. But the problem statement as written asks for a genuine symplectic

quasi-state on every closed connected symplectic manifold. I found no published or otherwise citable theorem proving that all closed connected symplectic manifolds admit such a genuine quasi-state, or that the Mak-Sun-Varolgunes heavy fiber can always be promoted to heaviness with respect to a genuine quasi-state satisfying the full additivity and homogeneity axioms.

Therefore the exact statement is partially resolved, not fully resolved.

5.17.4. Remaining open problems recorded by the database

5.17.4.1. Open problem 1

Construct a symplectic quasi-state on the standard symplectic four-torus $(T^4, \omega_{\text{std}})$, or prove that none exists.

Reference supplied by the database. Michael Entov, “Quasi-morphisms and quasi-states in symplectic topology,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, 1147–1171, §3 (explicitly identifies T^4 as the basic open case for existence of a symplectic quasi-state). [arXiv:1404.6408](https://arxiv.org/abs/1404.6408). Current evidence and status: Adi Dickstein and Frol Zapolsky, “Constraints on symplectic quasi-states,” *International Journal of Mathematics* **37**(7) (2026), DOI: [10.1142/S0129167X26500291](https://doi.org/10.1142/S0129167X26500291), [arXiv:2407.08014](https://arxiv.org/abs/2407.08014).

5.17.5. Primary-source pointer

The originating ICM record is *Quasi-morphisms and quasi-states in symplectic topology* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1167>.

5.18. 118. Gauge theory and mirror symmetry [2014-ag-152]

Record: 2014-ag-152

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Gauge theory and mirror symmetry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1329>

5.18.1. Problem statement

Let G be a compact Lie group, let $L \subset G$ be a Levi subgroup, and let $G^\vee$ and $L^\vee$ denote the Langlands dual groups. For a group H equal to $G^\vee$ or $L^\vee$, choose a Borel subgroup of $H_{\mathbb{C}}$ with unipotent radical N_H , Lie algebra $\mathfrak{n}_H$, and regular character $\chi_H \in \mathfrak{n}_H^*$. Define the Toda space

$$T(H) := (N_H, \chi_H) \backslash T^*H_{\mathbb{C}} // (N_H, \chi_H),$$

the holomorphic symplectic quotient of $T^*H_{\mathbb{C}}$ by the left and right N_H -actions, reduced at (χ_H, χ_H) . The Toda isomorphism identifies $T(H)$ with the BFM holomorphic symplectic manifold

$$BFM(H) = T_{\text{reg}}^*H_{\mathbb{C}} //_{\text{Ad}} H_{\mathbb{C}}.$$

Assume $N_{L^\vee} = N_{G^\vee} \cap L_{\mathbb{C}}^\vee$, let w_0 be a representative of the longest Weyl element of $G^\vee$, and let $w_{L^\vee}$ be a representative of the longest Weyl element of $L^\vee$. Let

$$N_{G^\vee} \times_{N_{L^\vee}} w_0 w_{L^\vee}^{-1} \cdot L_{\mathbb{C}}^\vee \times_{N_{L^\vee}} N_{G^\vee} \subset G_{\mathbb{C}}^\vee$$

be the inclusion that determines an open holomorphic symplectic embedding $T(L^\vee) \hookrightarrow T(G^\vee)$.

Prove or disprove that, via the Toda isomorphism, this embedding is the mirror of the symplectic induction functor SInd_L^G , which sends a categorical representation $\mathcal{C}$ of L to the Fukaya category of the flag variety G/L with coefficients in the associated local system of categories $G \times_L \mathcal{C} \rightarrow G/L$. Equivalently, prove or disprove that the flag variety G/L , viewed as the Hamiltonian G -manifold used for this induction and as a domain wall between pure three-dimensional L - and G -gauge theories, has as its mirror the above open embedding between the Toda/BFM spaces of $L^\vee$ and $G^\vee$.

5.18.2. Historical background, literature, and current progress

5.18.3. Verdict: partially resolved

5.18.3.1. Precise problem

The problem asks for the status of Teleman's Levi/Toda mirror-symmetry conjecture: for every compact Lie group G and Levi $L \subset G$, the open holomorphic symplectic embedding

$$T(L^\vee) \hookrightarrow T(G^\vee)$$

coming from the $N_{G^\vee} \times_{N_{L^\vee}} w_0 w_{L^\vee}^{-1} L_C^\vee \times_{N_{L^\vee}} N_{G^\vee}$ open piece should, after the Toda/BFM identification, be the mirror of the symplectic induction domain wall SInd_L^G implemented by the flag variety G/L , including the categorical Fukaya/KRS interpretation.

5.18.3.2. Status

This is not fully resolved. The holomorphic symplectic geometry on the BFM/Toda side is substantially established, and there is strong decategorified and special-case evidence from quantum cohomology and Rietsch mirrors. However, I found no citable theorem proving the full categorical statement in all compact Lie group/Levi cases. The exact mirror identification with symplectic induction still depends on foundations that the literature itself describes as conjectural, especially equivariant Fukaya categories with coefficients and the Kapustin-Rozansky-Saulina/Rozansky-Witten 2-category framework.

5.18.3.3. Key literature and progress

1. Gauge theory and mirror symmetry

Author: Constantin Teleman.

Venue/year: Proceedings of the International Congress of Mathematicians–Seoul 2014, Vol. II, Kyung Moon Sa, 1309–1332, 2014.

arXiv/URL: arXiv:1404.6305, <https://arxiv.org/abs/1404.6305>; PDF at <https://math.berkeley.edu/~teleman/math/icmtalk.pdf>.

Relevant results: This is the original source of the problem. Teleman proves the Toda/BFM holomorphic symplectic isomorphism in Theorem 6.3 and states that the Levi open cell inclusion determines an open embedding $T(L) \subset T(G)$. The exact mirror statement is Conjecture 6.5, not a theorem. Remark 6.7 explicitly says the conjecture is difficult to prove without precise definitions of equivariant Fukaya categories with coefficients and the KRS 2-category. Theorem 6.8 proves a smooth foliation of $\mathrm{BFM}(G^\vee)$ by leaves indexed by Levi/cuspidal data, but the categorical interpretation of those leaves is still conjectural.

2. Equivariant homology and K-theory of affine Grassmannians and Toda lattices

Authors: Roman Bezrukavnikov, Michael Finkelberg, Ivan Mirkovic.

Venue/year: Compositio Mathematica 141 no. 3, 746–768, 2005.

DOI/URL: <https://doi.org/10.1112/S0010437X04001228>; Cambridge page <https://www.cambridge.org/core/journals/compositio-mathematica/article/equivariant-homology-and-ktheory-of-affine-grassmannians-and-toda-lattices/EDBE1754F7F5B202B9998A90A9D3CD8D>.

Relevant results: Foundational BFM/Toda result identifying equivariant homology/K-theory of affine Grassmannians with universal centralizer/Toda structures. It supports the BFM side used by Teleman, but it does not prove the mirror-domain-wall assertion for symplectic induction.

3. A mirror symmetric construction of $qH_T^*(G/P)_{(q)}$

Author: Konstanze Rietsch.

Venue/year: Advances in Mathematics 217 no. 6, 2401–2442, 2008.

DOI/arXiv: <https://doi.org/10.1016/j.aim.2007.08.010>; arXiv:math/0511124.

Relevant results: Gives Lie-theoretic Landau-Ginzburg mirrors for G/P and recovers Peterson presentations of equivariant quantum cohomology after localizing quantum

parameters. This is key evidence for Teleman's final-section recovery of Rietsch mirrors, but it is a quantum-cohomology/mirror-family result, not a proof that the Toda open embedding is the KRS/Fukaya mirror of SInd_L^G .

4. Crystals and Mirror Constructions for Quotients

Author: George W. Melvin.

Venue/year: PhD thesis, University of California, Berkeley, 2017.

URL: <https://escholarship.org/uc/item/1xm099m7>; PDF also at https://personal.colby.edu/personal/g/gwmelvin/thesis_final.pdf.

Relevant results: Surveys Teleman's construction and explicitly distinguishes the proved foliation from the conjectural Fukaya-category statement. Melvin notes that the BFM foliation is proved in Teleman's Theorem 6.8, while the Fukaya-category interpretation depends on unproven homological mirror symmetry equivalences and the conjectural KRS 2-category. The thesis verifies a quantum-cohomology consequence in low-rank polygon-space examples, so it is partial progress only.

5. The role of Coulomb branches in 2D gauge theory

Author: Constantin Teleman.

Venue/year: Journal of the European Mathematical Society 23 no. 11, 3497–3520, 2021.

DOI/arXiv: <https://doi.org/10.4171/JEMS/1071>; arXiv:1801.10124.

Relevant results: Develops the Coulomb-branch/BFM interpretation and actions on equivariant quantum cohomology. This strengthens the rigorous BFM and quantum-cohomology framework, but it does not prove the full categorical Levi-domain-wall mirror statement.

6. Affine nil-Hecke algebras and quantum cohomology

Authors: Eduardo Gonzalez, Cheuk Yu Mak, Dan Pomerleano.

Venue/year: Advances in Mathematics 415, Article 108867, 2023.

DOI/arXiv: <https://doi.org/10.1016/j.aim.2023.108867>; arXiv:2202.05785.

Relevant results: Constructs nonabelian shift-operator actions and proves that $QH_G^*(M)$ defines a canonical holomorphic Lagrangian subvariety in $BFM(G_{\mathbb{C}}^V)$, confirming part of Teleman's expectation at the level of Lagrangian support. The authors still state that the relevant Rozansky-Witten 2-category is not yet rigorously defined, and their flag-variety example is described as conjectural pending comparison with the module structure. Thus it is important partial progress, not a full resolution.

7. Homological Mirror Symmetry for the universal centralizers

Author: Xin Jin.

Venue/year: arXiv preprint, 2022 revision.

arXiv/URL: arXiv:2206.09035, <https://arxiv.org/abs/2206.09035>. Earlier adjoint-case version: arXiv:2107.13395.

Relevant results: Proves homological mirror symmetry for universal centralizers/Toda spaces J_G , with a partially wrapped Fukaya category on J_G on the A-side and coherent sheaves on a quotient of a dual maximal torus on the B-side. This is substantial progress around Toda/BFM spaces, but it is not the same as Teleman's domain-wall statement: it does not identify the Levi open embedding $T(L^V) \hookrightarrow T(G^V)$ with the symplectic induction functor G/L between L - and G -gauge theories.

8. Equivariant Lagrangian correspondence and a conjecture of Teleman

Authors: Siu-Cheong Lau, Nai-Chung Conan Leung, Yan-Lung Leon Li.

Venue/year: arXiv preprint, 2023, revised later.

arXiv/URL: arXiv:2312.13926, <https://arxiv.org/abs/2312.13926>.

Relevant results: Addresses a Teleman conjecture about equivariant Lagrangian correspondences and the relation between disc potentials upstairs and downstairs under symplectic quotient. This is adjacent progress on Teleman’s quotient-mirror program, but it concerns a different quotient/potential statement and does not settle the Levi/Toda symplectic-induction domain wall in this problem.

5.18.3.4. What remains unresolved

The remaining missing theorem would have to give a rigorous all-cases categorical mirror equivalence showing that the open symplectic embedding $T(L^\vee) \hookrightarrow T(G^\vee)$, under the Toda/BFM isomorphism, represents the domain wall/functor SInd_L^G on categorical representations, with Fukaya categories of G/L with coefficient systems matching the appropriate KRS/Rozansky-Witten 2-categorical morphism. Existing results prove the geometric embedding, the Toda/BFM identification, related foliation statements, quantum-cohomology shadows, and some special or adjacent Teleman conjectures, but not this full categorical identification.

5.18.4. Remaining open problems recorded by the database

5.18.4.1. Open problem 1

Let G be a compact Lie group with maximal torus T , and let $q \in T^\vee$ specify a one-dimensional representation of T . Prove the torus-Levi case of symplectic-induction mirror symmetry: under the Toda isomorphism, the cotangent fibre at q maps to a Lagrangian leaf $\Lambda(q) \subset \mathrm{BFM}(G^\vee)$ representing the symplectically induced representation, equivalently the G -equivariant Fukaya category of G/T with quantum parameter q .

Reference supplied by the database. Constantin Teleman, “Gauge theory and mirror symmetry,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 1309–1332, Conjecture 6.5 and its torus-Levi specialization in Example 6.6. arXiv: [1404.6305](https://arxiv.org/abs/1404.6305); ICM proceedings PDF: [ICM 2014, Vol. II](#).

5.18.4.2. Open problem 2

Let $L \subset G$ be a Levi subgroup, let T be a maximal torus, and let ν be a regular value of the moment map for the Hamiltonian T -action on G/L . If $t \in Z(L_\mathbb{C})$ denotes the exponential of the symplectic structure on G/L and $\Lambda(t)$ is the Lagrangian support of its T -equivariant Fukaya category in $\mathrm{BFM}(T^\vee) = T^*T_\mathbb{C}^\vee$, prove that the Fukaya category of the symplectic reduction $(G/L)//_\nu T$ is the category $\mathrm{Hom}(S_\nu, \Lambda(t))$, where S_ν is the cotangent fibre over $\exp(\nu) \in T^\vee$.

Reference supplied by the database. George W. Melvin, *Crystals and Mirror Constructions for Quotients*, Ph.D. thesis, University of California, Berkeley, 2017, Conjecture 3.2.6 (attributed there to Constantin Teleman). eScholarship permalink: <https://escholarship.org/uc/item/1xm099m7>.

5.18.4.3. Open problem 3

Under the hypotheses of the preceding symplectic-reduction problem, assume that $(G/L)//_{\nu}T$ is Fano. Prove that its quantum cohomology is the Jacobian ring of the restriction of the T -equivariant superpotential to the fibre over $\exp(2\pi i\nu)$ of the equivariant-structure map on the mirror of G/L , with the quantum deformation obtained by varying $t \in Z(L_{\mathbb{C}})$; if G has finite nontrivial centre Z , account for every critical point with multiplicity $|Z|$.

Reference supplied by the database. George W. Melvin, *Crystals and Mirror Constructions for Quotients*, Ph.D. thesis, University of California, Berkeley, 2017, Conjecture 3.2.7. eScholarship permalink: <https://escholarship.org/uc/item/1xm099m7>.

5.18.5. Primary-source pointer

The originating ICM record is *Gauge theory and mirror symmetry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1329>.

5.19. 119. The structure of algebraic varieties [2014-ag-173]

Record: 2014-ag-173

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: The structure of algebraic varieties

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.1/ICM2014.1.pdf#page=411>

5.19.1. Problem statement

Prove that rational connectedness is determined by the underlying symplectic structure: if X and Y are smooth projective complex varieties whose underlying smooth manifolds, with their natural symplectic structures, are symplectomorphic, then X is rationally connected if and only if Y is rationally connected.

Here a projective variety Z is rationally connected if, for every finite set of points $z_1, \dots, z_r \in Z$, there exists a morphism $\phi : \mathbb{CP}^1 \rightarrow Z$ whose image contains $z_1, \dots, z_r$.

5.19.2. Historical background, literature, and current progress

5.19.3. Verdict: partially resolved

5.19.3.1. Precise problem

The problem asks whether rational connectedness is determined by the natural symplectic structure of a smooth projective complex variety: if smooth projective complex varieties X and Y are symplectomorphic as smooth manifolds with their natural symplectic forms, must X be rationally connected if and only if Y is rationally connected?

The literature usually formulates this as invariance under symplectic deformation equivalence. That is slightly broader than a single symplectomorphism of chosen natural symplectic forms, so positive results for symplectic deformation equivalence apply to the stated symplectomorphism formulation. No citable result found covers all smooth projective varieties in all dimensions.

5.19.3.2. Original source and restatement

- **Title:** Low degree polynomial equations: arithmetic, geometry and topology
Author: Janos Kollar
Venue/year: European Congress of Mathematics, Vol. I (Budapest, 1996), Progress in Mathematics 168, Birkhauser, 1998, pp. 255-288
arXiv/URL: arXiv:alg-geom/9607016, <https://arxiv.org/abs/alg-geom/9607016>
Relevant result: Conjecture 4.2.7 states that for smooth projective varieties over $\mathbb{C}$, rational connectedness should be invariant under symplectic deformation equivalence. The-

orem 4.2.10 proves the analogous invariance for uniruledness, and Kollar notes that the rational-connectedness conjecture follows in the case $\dim H^2(X, \mathbb{Q}) = 1$. This is evidence and a special case, not a full solution.

- **Title:** The structure of algebraic varieties
Author: Janos Kollar
Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. I, Kyung Moon Sa, 2014, pp. 395-419/420
arXiv/URL: arXiv:1407.7478, <https://arxiv.org/abs/1407.7478>
Relevant result: Conjecture 7.4 restates that being rationally connected is a property of the underlying symplectic structure and cites only partial results.

5.19.3.3. Partial progress

- **Title:** Rationally connected 3-folds and symplectic geometry
Author: Claire Voisin
Venue/year: Asterisque 322, 2008, pp. 1-21
arXiv/URL: arXiv:0801.1396, <https://arxiv.org/abs/0801.1396>; Numdam: https://www.numdam.org/item/AST_2008__322__1_0/
Relevant result: Theorem 0.8 proves the conjecture for compact Kahler threefolds when X is Fano or when X is rationally connected and $b_2(X) \leq 2$. The paper also observes the surface case. These are restricted cases only.
- **Title:** Symplectic geometry of rationally connected threefolds
Author: Zhiyu Tian
Venue/year: Duke Mathematical Journal 161, no. 5, 2012, pp. 803-843
DOI/arXiv: <https://doi.org/10.1215/00127094-1548398>; arXiv:1006.2486, <https://arxiv.org/abs/1006.2486>
Relevant result: Theorem 1.7 proves Kollar's conjecture in complex dimension 3. This fully resolves the threefold case, but not higher dimensions.
- **Title:** Symplectic geometry and rationally connected 4-folds
Author: Zhiyu Tian
Venue/year: Journal fur die reine und angewandte Mathematik 698, 2015, pp. 221-244
DOI/arXiv: <https://doi.org/10.1515/crelle-2013-0005>; arXiv:1208.4343, <https://arxiv.org/abs/1208.4343>
Relevant result: Theorem 1.2 proves rational connectedness for smooth projective fourfolds symplectic deformation equivalent to either a Fano fourfold of pseudo-index at least 2 or a rationally connected fourfold with second Betti number 2. This is partial fourfold progress and does not cover arbitrary fourfolds.
- **Title:** Symplectic invariance of rational surfaces on Kahler manifolds
Author: Jason Michael Starr
Venue/year: arXiv preprint, 2019 version
DOI/arXiv: <https://doi.org/10.48550/arXiv.1803.06412>; arXiv:1803.06412, <https://arxiv.org/abs/1803.06412>
Relevant result: The paper proves Kollar's conjecture for Kahler manifolds symplectically deformation equivalent to G/P and for certain low-degree complete intersections in such spaces. This is higher-dimensional evidence, but it has strong geometric hypotheses and does not imply the full all-varieties statement.
- **Title:** Towards the Symplectic Graber-Harris-Starr Theorems
Author: Zhiyu Tian

Venue/year: Acta Mathematica Sinica, English Series 42, 2026, pp. 791-805

DOI/arXiv: <https://doi.org/10.1007/s10114-026-5298-z>;
arXiv:1208.4340, <https://arxiv.org/abs/1208.4340>

Relevant result: This develops related questions about symplectic rational connectedness and Gromov-Witten invariants. It explicitly treats Kollar's conjecture as known for threefolds and then studies related inductive questions; it is not a proof of the full conjecture.

- **Title:** Gromov-Witten invariants and rationally connectedness

Author: Jianxun Hu

Venue/year: In *Gromov-Witten theory, gauge theory and dualities*, Proceedings of the Centre for Mathematics and its Applications, Australian National University, vol. 48, 2019, 23 pp.

URL: <https://maths.anu.edu.au/research/cma-proceedings/gromov-witten-theory-gauge-theory-and-dualities-anukioloa-6-16-january>

Relevant result: Survey-style account of symplectic rational connectedness and Gromov-Witten approaches. It records Tian's threefold theorem and Tian's fourfold partial results; it does not claim a full solution.

- **Title:** Cohomological splitting over rationally connected bases

Authors: Shaoyun Bai, Daniel Pomerleano, Guangbo Xu

Venue/year: arXiv preprint, 2024

DOI/arXiv: <https://doi.org/10.48550/arXiv.2406.00931>;
arXiv:2406.00931, <https://arxiv.org/abs/2406.00931>

Relevant result: Introduces and uses enumerative/symplectic rational connectedness for cohomological splitting of Hamiltonian fibrations. It cites Kollar's conjecture and the Voisin/Tian work as motivation, but its theorems concern Hamiltonian fibrations over enumeratively rationally connected bases, not invariance of algebraic rational connectedness for arbitrary symplectomorphic projective varieties.

5.19.3.4. What remains unresolved

The complete statement remains open in dimensions at least 4 for arbitrary smooth projective complex varieties. Existing results cover all dimensions up to 3 and several special fourfold or higher-dimensional families, but I found no published or otherwise citable theorem proving that every pair of symplectomorphic smooth projective complex varieties has the same rational-connectedness status.

5.19.4. Remaining open problems recorded by the database

5.19.4.1. Open problem 1

Is every smooth complex projective rationally connected variety symplectically rationally connected, in the sense that it has a nonzero genus-zero Gromov-Witten invariant with two point-class insertions (and possibly further insertions)?

Reference supplied by the database. Jianxun Hu, "Gromov-Witten Invariants and Rationally Connectedness," in Peter Bouwknegt, Brett Parker, and Bai-Ling Wang (eds.), *Gromov-Witten Theory, Gauge Theory and Dualities: Proceedings of the Centre for Mathematics and its Appli-*

cations, *Australian National University*, vol. 48, Australian National University, Canberra, 2019, pp. 74–96, Conjecture 3.4. ISBN 978-0-6481056-2-6. https://maths.anu.edu.au/files/4gromov-witten-invariants_and_rationally_connectedness.pdf

5.19.4.2. Open problem 2

For smooth complex projective fourfolds, is rational connectedness invariant under symplectic deformation equivalence (and hence under symplectomorphism)?

Reference supplied by the database. Zhiyu Tian, “Symplectic Geometry and Rationally Connected 4-Folds,” *Journal für die reine und angewandte Mathematik* 698 (2015), 221–244, Conjecture 1.1 and the fourfold problem studied in the paper. DOI: 10.1515/crelle-2013-0005. arXiv:1208.4343. <https://doi.org/10.1515/crelle-2013-0005>

5.19.4.3. Open problem 3

Let X be a smooth Fano fourfold of Picard number one, and let C be a general embedded very free rational curve obtained by the Kollár–Miyaoka–Mori quotient construction from a free rational curve of minimal anticanonical degree. Is the normal bundle $N_{C/X}$ balanced?

Reference supplied by the database. Zhiyu Tian, “Symplectic Geometry and Rationally Connected 4-Folds,” *Journal für die reine und angewandte Mathematik* 698 (2015), 221–244, Remark 1.5 (the explicit question on whether the normal bundle of a general such curve is balanced). DOI: 10.1515/crelle-2013-0005. arXiv:1208.4343. <https://doi.org/10.1515/crelle-2013-0005>

5.19.4.4. Open problem 4

Does every extremal ray of the Mori cone of a smooth complex projective fourfold contain a curve class that gives a nonzero genus-zero Gromov–Witten invariant, so that the ray remains effective under symplectic deformation?

Reference supplied by the database. Zhiyu Tian, “Symplectic Geometry and Rationally Connected 4-Folds,” *Journal für die reine und angewandte Mathematik* 698 (2015), 221–244, §3 (the explicitly identified problem of showing that all extremal rays support curve classes giving nonzero Gromov–Witten invariants). DOI: 10.1515/crelle-2013-0005. arXiv:1208.4343. <https://doi.org/10.1515/crelle-2013-0005>

5.19.5. Primary-source pointer

The originating ICM record is *The structure of algebraic varieties* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.1/ICM2014.1.pdf#page=411>.

6. Discrete and polyhedral geometry

This chapter ranges from face-number and realization problems to packing, covering, convex-position, and incidence questions. Exact extremal constants and dimension dependence are especially prominent. Computation and probabilistic methods frequently establish asymptotic bounds, whereas the original sharp statement remains open even when a great deal is known around it.

Coverage: 13 website records; **source years:** 1950–2022; **separately listed descendant open problems:** 12.

6.1. 120. Recent progress in the geometry of numbers [1950-geom-6]

Record: 1950-geom-6

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1950: Recent progress in the geometry of numbers

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1950.1/ICM1950.1.ocr.pdf#page=174>

6.1.1. Problem statement

For every positive integer n , every full-rank lattice $\Lambda \subset \mathbb{R}^n$ with determinant (covolume) 1, and every point $c = (c_1, \dots, c_n) \in \mathbb{R}^n$, there exists a lattice point $x = (x_1, \dots, x_n) \in \Lambda$ such that

$$\left| \prod_{i=1}^n (x_i - c_i) \right| \leq 2^{-n}.$$

6.1.2. Historical background, literature, and current progress

6.1.3. Verdict: partially resolved

6.1.3.1. Precise problem

The problem asks for the full inhomogeneous Minkowski conjecture on the product of linear forms: for every dimension n , every determinant-one full-rank lattice Λ in $\mathbb{R}^n$, and

every translate c in $\mathbb{R}^n$, there should be x in Λ with

$$\left| \prod_i (x_i - c_i) \right| \leq 2^{-n}.$$

This is equivalent to the usual formulation for n real non-homogeneous linear forms. If $\Lambda = A \mathbb{Z}^n$ with $|\det A| = 1$, then the coordinate functions of Az are n real linear forms $L_i(z)$, and the stated inequality becomes

$$\left| \prod_i (L_i(z) - c_i) \right| \leq |\det A| / 2^n.$$

Thus literature proving Minkowski's conjecture for products of n non-homogeneous linear forms in all dimensions would resolve the problem statement. Literature proving only fixed dimensions, estimates with constants worse than 2^{-n} , almost-everywhere results, function-field analogues, or results under extra lattice hypotheses is only partial progress.

6.1.3.2. Current status

No citable source found proves the full all-dimensional real Euclidean statement. The strongest exact published progress I found proves the conjecture through dimension $n \leq 10$. Therefore the problem is not fully resolved; the unresolved part is the exact statement for real determinant-one lattices in dimensions $n \geq 11$.

6.1.3.3. Key literature and applicability

1. Title: Ueber die Annaeherung an eine reelle Groesse durch rationale Zahlen
Authors: Hermann Minkowski
Venue/year: Mathematische Annalen 54 (1901), 91-124
DOI/URL: <https://doi.org/10.1007/BF01454200>; <https://eudml.org/doc/157996>
Role: Original source associated with Minkowski's conjecture. It is historical/original-source evidence, not a modern full resolution.
2. Title: On the inhomogeneous minimum of the product of n linear forms
Authors: B. J. Birch and H. P. F. Swinnerton-Dyer
Venue/year: Mathematika 3 (1956), 25-39
DOI/URL: <https://doi.org/10.1112/S0025579300000863>; <https://www.cambridge.org/core/journals/mathematika/article/on-the-inhomogeneous-minimum-of-the-product-of-n-linear-forms/1458B4BE234DD5013BC3BC14CAF79711>
Role: Gives the standard modern formulation and important reduction results. It does not prove the conjecture in all dimensions.
3. Title: Minkowski's conjecture, well-rounded lattices and topological dimension
Authors: Curtis T. McMullen
Venue/year: Journal of the American Mathematical Society 18 (2005), 711-734
DOI/URL: <https://doi.org/10.1090/S0894-0347-05-00483-2>; <https://people.math.harvard.edu/~ctm/papers/home/text/papers/mink/mink.pdf>
Role: Proves a major reduction via well-rounded lattices and proves Minkowski's conjecture for $n = 6$, assuming known covering-radius cases through dimension 6. Its Theorem

8.1 says covering-radius bounds for all well-rounded unimodular lattices in dimensions $k \leq n$ imply Minkowski's conjecture in those dimensions. This is a conditional reduction in general, not a full all-dimensional proof.

4. Title: On conjectures of Minkowski and Woods for $n = 7$
 Authors: R. J. Hans-Gill, Madhu Raka, Ranjeet Sehmi
 Venue/year: Journal of Number Theory 129(5), 1011-1033 (2009)
 DOI/URL: <https://doi.org/10.1016/j.jnt.2008.10.020>
 Role: Proves the relevant Woods/Minkowski route in dimension 7. This is a fixed-dimensional result only.
5. Title: On conjectures of Minkowski and Woods for $n = 8$
 Authors: R. J. Hans-Gill, Madhu Raka, Ranjeet Sehmi
 Venue/year: Acta Arithmetica 147(4), 337-385 (2011)
 DOI/URL: <https://doi.org/10.4064/aa147-4-3> ; <https://eudml.org/doc/279840>
 Role: Proves the relevant Woods/Minkowski route in dimension 8. This is a fixed-dimensional result only.
6. Title: On conjectures of Minkowski and Woods for $n = 9$
 Authors: Leetika Kathuria, Madhu Raka
 Venue/year: Proceedings - Mathematical Sciences 126(4), 501-548 (2016)
 DOI/arXiv/URL: <https://doi.org/10.1007/s12044-016-0295-2> ;
 arXiv:1410.5743 ; <https://arxiv.org/abs/1410.5743>
 Role: Proves Woods' conjecture for $n = 9$, from which Minkowski's conjecture follows in dimension 9 via McMullen's result. This does not cover $n \geq 10$.
7. Title: On conjectures of Minkowski and Woods for $n = 10$
 Authors: Leetika Kathuria, Madhu Raka
 Venue/year: Proceedings - Mathematical Sciences 132(2), article 45 (2022)
 DOI/arXiv/URL: <https://doi.org/10.1007/s12044-022-00679-2> ;
 arXiv:2009.09992 ; <https://arxiv.org/abs/2009.09992>
 Role: Proves Woods' conjecture and hence Minkowski's conjecture for $n = 10$. This is the largest exact-dimensional resolution I found. It proves the problem statement only when the positive integer n is 10, not for every n .
8. Title: Stable and well-rounded lattices in diagonal orbits
 Authors: Omri Nisan Solan
 Venue/year: Israel Journal of Mathematics 234, 501-519 (2019)
 DOI/URL: <https://doi.org/10.1007/s11856-019-1912-4>
 Role: Gives a newer diagonal-orbit/stable-lattice reduction and new proofs in dimensions up to 7 under finite covering-radius checks. It is a reduction/partial-progress paper, not an all-dimensional proof.
9. Title: Counterexamples to a conjecture of Woods
 Authors: Oded Regev, Uri Shapira, Barak Weiss
 Venue/year: Duke Mathematical Journal 166(13), 2443-2446 (2017)
 DOI/arXiv/URL: <https://doi.org/10.1215/00127094-2017-0018> ;
 arXiv:1604.07644
 Role: Disproves Woods' conjecture in high dimensions. This does not disprove or resolve Minkowski's conjecture; it only shows that one sufficient route through Woods cannot work in those dimensions.
10. Title: Counterexamples to the Woods Conjecture in dimensions $d \geq 24$
 Authors: Hao Chen, Liqing Xu

Venue/year: Journal de Theorie des Nombres de Bordeaux 31(3), 723-726 (2019)

DOI/URL: <https://doi.org/10.5802/jtnb.1105>; <https://jtnb.centre-mersenne.org/item/10.5802/jtnb.1105.pdf>

Role: Improves high-dimensional counterexamples to Woods' conjecture to $d \geq 24$. Again, this is not a counterexample to Minkowski's conjecture and does not settle the problem statement.

11. Title: Lecture Notes on Geometry of Numbers

Author: R. J. Hans-Gill

Venue/year: Springer Nature Singapore, 2024

DOI/URL: <https://doi.org/10.1007/978-981-99-9602-5>

Role: Recent survey/book source stating that Minkowski's conjecture is proved for $n \leq 10$ and discussing weaker estimates and open Woods dimensions beyond that. This supports the current-status assessment but does not contain a full all-dimensional proof.

12. Title: On covering radii in function fields

Author: Noy Soffer Aranov

Venue/year: Monatshefte fuer Mathematik 206, 705-734 (2025)

DOI/URL: <https://doi.org/10.1007/s00605-024-02049-8>

Role: Treats function-field analogues and cites the $n \leq 10$ real-dimensional literature. It is not applicable to this problem because the problem statement is about real lattices in $\mathbb{R}^n$.

6.1.3.4. What remains unresolved

The full statement asks for every positive integer n . Published exact results cover only fixed dimensions up to $n = 10$. For $n \geq 11$, I found estimates, reductions, almost-everywhere statements, and analogues in other settings, but no published or otherwise citable theorem whose hypotheses and conclusion imply

for every unimodular Λ in $\mathbb{R}^n$ and every c in $\mathbb{R}^n$,
there exists x in Λ with $|\prod_i (x_i - c_i)| \leq 2^{-n}$.

Therefore the literature only partially resolves the problem statement.

6.1.4. Remaining open problems recorded by the database

6.1.4.1. Open problem 1

Prove Minkowski's inhomogeneous product-of-linear-forms conjecture in the first unresolved dimension $n = 11$: for every unimodular lattice $\Lambda \subset \mathbb{R}^{11}$ and every $c \in \mathbb{R}^{11}$, there is an $x \in \Lambda$ such that $\left| \prod_{i=1}^{11} (x_i - c_i) \right| \leq 2^{-11}$.

Reference supplied by the database. R. J. Hans-Gill, Madhu Raka, and Ranjeet Sehmi, *Lecture Notes on Geometry of Numbers*, University Texts in the Mathematical Sciences, Springer Singapore, 2024, Chapter "Inhomogeneous Problems," pp. 161–193 (Minkowski's conjecture; recorded as proved only for $n \leq 10$), DOI: [10.1007/978-981-99-9602-5](https://doi.org/10.1007/978-981-99-9602-5). The endpoint $n = 10$ is proved in Leetika Kathuria and Madhu Raka, "On conjectures of Minkowski and Woods for

$n = 10$,” *Proceedings of the Indian Academy of Sciences (Mathematical Sciences)* **132** (2022), Article 45, DOI: [10.1007/s12044-022-00679-2](https://doi.org/10.1007/s12044-022-00679-2), arXiv:2009.09992.

6.1.4.2. Open problem 2

Prove Woods’s covering-radius conjecture in dimension 11: every well-rounded unimodular lattice in $\mathbb{R}^{11}$ has covering radius at most $\sqrt{11}/2$. This is an explicitly proposed intermediate problem that would imply the eleven-dimensional case of Minkowski’s conjecture.

Reference supplied by the database. Hao Chen and Liqing Xu, “Counterexamples to the Woods Conjecture in dimensions $d \geq 24$,” *Journal de Théorie des Nombres de Bordeaux* **31** (2019), no. 3, 723–726, introduction (explicitly identifying dimensions $10 \leq d \leq 23$ as open at that time), DOI: [10.5802/jtnb.1105](https://doi.org/10.5802/jtnb.1105). The dimension-10 case was subsequently settled by Leetika Kathuria and Madhu Raka, “On conjectures of Minkowski and Woods for $n = 10$,” *Proceedings of the Indian Academy of Sciences (Mathematical Sciences)* **132** (2022), Article 45, DOI: [10.1007/s12044-022-00679-2](https://doi.org/10.1007/s12044-022-00679-2), leaving $n = 11$ as the first open dimension; the implication to Minkowski’s conjecture is due to Curtis T. McMullen, “Minkowski’s conjecture, well-rounded lattices and topological dimension,” *Journal of the American Mathematical Society* **18** (2005), no. 3, 711–734, Theorem 8.1, DOI: [10.1090/S0894-0347-05-00483-2](https://doi.org/10.1090/S0894-0347-05-00483-2).

6.1.5. Primary-source pointer

The originating ICM record is *Recent progress in the geometry of numbers* (1950). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1950.1/ICM1950.1.ocr.pdf#page=174>.

6.2. 121. On the application of combinatorial analysis to number theory, geometry and analysis [1970-geom-16]

Record: 1970-geom-16

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1970: On the application of combinatorial analysis to number theory, geometry and analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.3/ICM1970.3.ocr.pdf#page=209>

6.2.1. Problem statement

For every integer $n \geq 3$, let $f(n)$ be the smallest integer with the following property: every set of $f(n)$ points in the plane, with no three of the points collinear, contains n points that are the vertices of a convex n -gon.

Prove or disprove that

$$f(n) = 2^{n-2} + 1$$

for every integer $n \geq 3$.

6.2.2. Historical background, literature, and current progress

6.2.3. Verdict: partially resolved

6.2.3.1. Precise problem

Let $f(n)$ be the least integer such that every set of $f(n)$ points in the plane in general position, meaning no three collinear, contains n points in convex position. The problem asks for the exact value of $f(n)$ for every $n \geq 3$, namely the Erdős-Szekeres happy-ending conjecture

$$f(n) = 2^{n-2} + 1.$$

The lower bound $f(n) \geq 2^{n-2} + 1$ is known, so a full resolution would require proving the matching upper bound $f(n) \leq 2^{n-2} + 1$ for every $n \geq 3$.

6.2.3.2. Status check

The problem is not fully resolved. Exact values are known through $n = 6$, but the first open exact case is $n = 7$, where the conjecture predicts $f(7) = 33$. Recent literature still describes the full conjecture as open. General upper bounds now have the correct exponential order up to a subexponential factor, but they do not imply the exact formula. Several newer papers prove relaxed or restricted versions, but their hypotheses or conclusions do not cover all point sets in the problem statement.

6.2.3.3. Key literature and progress

1. Title: "A combinatorial problem in geometry" Authors: Paul Erdős and George Szekeres Venue/year: *Compositio Mathematica* 2, 463-470, 1935 DOI/arXiv/URL: http://www.numdam.org/item?id=CM_1935__2__463_0 Result/relevance: Original Erdős-Szekeres paper proving finiteness of $f(n)$ and an early upper bound. This is the source of the convex-position problem, but it does not prove the exact formula.
2. Title: "On some extremum problems in elementary geometry" Authors: Paul Erdős and George Szekeres Venue/year: *Annales Universitatis Scientiarum Budapestinensis de Rolando Eötvös Nominatae, Sectio Mathematica* 3-4, 53-62, 1960-1961 DOI/arXiv/URL: no DOI located; standard citation appears in later publisher bibliographies Result/relevance: Gives the classical construction showing $f(n) \geq 2^{n-2} + 1$, the lower bound predicted by the conjecture. This is necessary but not sufficient for the stated equality.
3. Title: "On the application of combinatorial analysis to number theory, geometry and analysis" Authors: Paul Erdős Venue/year: *Actes du Congrès International des Mathématiciens*, Nice 1970, Tome 3, Gauthier-Villars, Paris, 201-210, 1971 DOI/arXiv/URL: https://www.renyi.hu/~p_erdos/1971-13.pdf Result/relevance: Records the problem in the same form: whether there is a smallest $f(n)$, Szekeres's conjecture $f(n) = 2^{n-2} + 1$, and the then-known exact cases. It is source evidence for the question, not a resolution.
4. Title: "Konvexe Fünfecke in ebenen Punktmengen" Authors: Heiko Harborth Venue/year: *Elemente der Mathematik* 33, 116-118, 1978 DOI/arXiv/URL: DOI 10.5169/seals-32945; <http://eudml.org/doc/141217> Result/relevance: Published proof of the $n = 5$ case, $f(5) = 9 = 2^3 + 1$. This is a special case only.
5. Title: "Computer solution to the 17-point Erdős-Szekeres problem" Authors: George Szekeres and Lindsay Peters Venue/year: *The ANZIAM Journal* 48(2), 151-164, 2006 DOI/arXiv/URL: DOI 10.1017/S144618110000300X; <https://doi.org/10.1017/S144618110000300X> Result/relevance: Computer-assisted proof that every 17-point general-position planar set contains a convex hexagon, so $f(6) = 17 = 2^4 + 1$. This verifies the conjecture through $n = 6$, but not beyond.
6. Title: "Fast Formal Proof of the Erdős-Szekeres Conjecture for Convex Polygons with at Most 6 Points" Authors: Filip Marić Venue/year: *Journal of Automated Reasoning* 62, 301-329, 2019 DOI/arXiv/URL: DOI 10.1007/s10817-017-9423-7; <https://doi.org/10.1007/s10817-017-9423-7> Result/relevance: Formal/SAT-based verification of the conjecture for polygons with at most 6 vertices. This strengthens confidence in the low-dimensional exact cases but still covers only $n \leq 6$.
7. Title: "The Erdős-Szekeres problem on points in convex position - a survey" Authors: Walter Morris and Valeriu Soltan Venue/year: *Bulletin of the American Mathematical Society* 37(4), 437-458, 2000 DOI/arXiv/URL: DOI 10.1090/S0273-0979-00-00877-6; <https://doi.org/10.1090/S0273-0979-00-00877-6> Result/relevance: Survey of known results and variants. Useful background confirming that the main exact problem remained far from solved as of 2000.
8. Title: "Forced Convex n-Gons in the Plane" Authors: Fan R. K. Chung and Ronald L. Graham Venue/year: *Discrete & Computational Geometry* 19, 367-371, 1998 DOI/arXiv/URL: DOI 10.1007/PL00009353; <https://doi.org/10.1007/PL00009353> Result/relevance: First small improvement over the Erdős-Szekeres upper bound, removing the final $+1$ for $n \geq 4$. It gives an upper bound much larger than

$$2^{n-2} + 1.$$

9. Title: “Finding Convex Sets Among Points in the Plane” Authors: Daniel J. Kleitman and Lior Pachter Venue/year: Discrete & Computational Geometry 19, 405-410, 1998 DOI/arXiv/URL: DOI 10.1007/PL00009358; <https://doi.org/10.1007/PL00009358> Result/relevance: Further improves the general upper bound, but still not to the conjectured exact value.
10. Title: “Note on the Erdős-Szekeres Theorem” Authors: Géza Tóth and Pavel Valtr Venue/year: Discrete & Computational Geometry 19, 457-459, 1998 DOI/arXiv/URL: DOI 10.1007/PL00009363; <https://doi.org/10.1007/PL00009363> Result/relevance: Improves the upper bound to $g(n) \leq \binom{2n-5}{n-2} + 2$. This applies to the same point-set problem but remains far above the conjectured exact threshold.
11. Title: “An Improved Upper Bound for the Erdős-Szekeres Conjecture” Authors: Hossein Nassajian Mojarrad and Georgios Vlachos Venue/year: Discrete & Computational Geometry 56, 165-180, 2016 DOI/arXiv/URL: DOI 10.1007/s00454-016-9791-5; arXiv:1510.06255 Result/relevance: Gives a stronger binomial-type upper bound. Its conclusion is still an upper bound larger than $2^{n-2} + 1$, so it does not resolve the exact formula.
12. Title: “On the Erdős-Szekeres convex polygon problem” Authors: Andrew Suk Venue/year: Journal of the American Mathematical Society 30(4), 1047-1053, 2017 DOI/arXiv/URL: DOI 10.1090/jams/869; arXiv:1604.08657 Result/relevance: Proves the asymptotic upper bound $f(n) = 2^{n+o(n)}$, more precisely an upper bound of the form $2^{n+O(n^{2/3} \log n)}$. This is a major near-asymptotic result for the same hypotheses, but it does not imply $f(n) = 2^{n-2} + 1$.
13. Title: “Two extensions of the Erdős-Szekeres problem” Authors: Andreas F. Holmsen, Hossein Nassajian Mojarrad, János Pach, and Gábor Tardos Venue/year: Journal of the European Mathematical Society 22(12), 3981-3995, 2020 DOI/arXiv/URL: DOI 10.4171/JEMS/1000; arXiv:1710.11415 Result/relevance: Improves the general upper-bound error term to $2^{n+O(\sqrt{n \log n})}$ and extends methods to related convexity structures. This still does not give the exact conjectured threshold.
14. Title: “A SAT attack on the Erdős-Szekeres conjecture” Authors: Martin Balko and Pavel Valtr Venue/year: European Journal of Combinatorics 66, 13-23, 2017 DOI/arXiv/URL: DOI 10.1016/j.ejc.2017.06.010 Result/relevance: Refutes a stronger Peters-Szekeres hypergraph-coloring conjecture and develops SAT-based approaches. Since it refutes a stronger auxiliary statement rather than proving the original point-set conjecture, it is partial/computational progress only.
15. Title: “The Erdős-Szekeres Conjecture Revisited” Authors: Jineon Baek and Martin Balko Venue/year: 41st International Symposium on Computational Geometry (SoCG 2025), LIPIcs 332, 13:1-13:15, 2025 DOI/arXiv/URL: DOI 10.4230/LIPIcs.SocG.2025.13; <https://doi.org/10.4230/LIPIcs.SocG.2025.13> Result/relevance: The abstract explicitly describes the Erdős-Szekeres conjecture as “still open.” The paper proves a tight relaxed version for split k -gons and proves the conjecture for decomposable sets, but split k -gons are weaker than k points in convex position and decomposability is an extra hypothesis. Therefore these results do not cover all point sets in the problem statement.
16. Title: “Notes on the 33-point Erdős-Szekeres problem” Authors: Dumitru Bogan Venue/year: preprint, December 2025 DOI/arXiv/URL: arXiv:2512.24061; <https://arxiv.org/abs/2512.24061> Result/relevance: Treats $ES(7)$, where the conjecture predicts $ES(7) = 33$, as the first open case. It gives a SAT encoding

and UNSAT certificates for some anchored subfamilies, but not a complete proof that every 33-point general-position set contains a convex heptagon.

6.2.3.4. What remains unresolved

The exact equality $f(n) = 2^{n-2} + 1$ remains unproved for all $n \geq 7$. In particular, the concrete first open case $f(7) = 33$ remains unresolved in citable literature found in this search. The available results either prove only low- n cases, provide weaker general upper bounds, or assume extra structure/replace convex k -gons by relaxed objects.

6.2.4. Remaining open problems recorded by the database

6.2.4.1. Open problem 1

Does every set of 33 points in the plane in general position contain seven points in convex position; equivalently, is $f(7) = 33$?

Reference supplied by the database. Bogdan Dumitru, “Notes on the 33-Point Erdős–Szekeres Problem,” arXiv:2512.24061 [math.CO], 2025, first open case stated in the abstract and §1, <https://doi.org/10.48550/arXiv.2512.24061>.

6.2.5. Primary-source pointer

The originating ICM record is *On the application of combinatorial analysis to number theory, geometry and analysis* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.3/ICM1970.3.ocr.pdf#page=209>.

6.3. 122. Combinatorics and convexity [1994-geom-52]

Record: 1994-geom-52

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

6.3.1. Problem statement

Prove or disprove that for every centrally symmetric convex d -dimensional polytope $P \subset \mathbb{R}^d$ with center at the origin (that is, $x \in P$ implies $-x \in P$), every combinatorial automorphism ϕ of P (equivalently, every automorphism of the face lattice of P , acting in particular on the vertices of P) has the following property: for every vertex v of P , $\phi(-v) = -\phi(v)$.

6.3.2. Historical background, literature, and current progress

6.3.3. Verdict: partially resolved / unresolved

6.3.3.1. Precise problem

The problem asks whether the antipodal vertex-pairing of an origin-centrally symmetric convex polytope is determined by its face lattice in the following strong sense: if $P = -P$ and ϕ is any combinatorial automorphism of P , must $\phi(-v) = -\phi(v)$ for every vertex v ? Equivalently, does the central symmetry $v \mapsto -v$ commute with the full combinatorial automorphism group of P ?

6.3.3.2. Status

I did not find a published or otherwise citable result that proves or disproves this statement for all centrally symmetric convex polytopes. The exact face-lattice automorphism question remains unresolved. The literature contains relevant partial and adjacent progress, but the hypotheses/conclusions do not cover the full statement.

6.3.3.3. Relevant literature and progress

1. Combinatorics and Convexity

Author: Gil Kalai

Venue/year: Proceedings of the International Congress of Mathematicians, Vol. 1, 2 (Zurich, 1994), Birkhauser, Basel, 1995, pp. 1363-1374

DOI/URL: https://doi.org/10.1007/978-3-0348-9078-6_131 ; author PDF: <https://www.ma.huji.ac.il/~kalai/co.pdf>

Relevance: Kalai records Perles' question in essentially the stated form: whether every combinatorial automorphism ϕ of a centrally symmetric d -polytope satisfies $\phi(-v) = -\phi(v)$. This is the original/open-problem source, not a resolution.

2. **Is combinatorial automorphism of symmetric convex polytope always antipodal?**

Author: Fedor Petrov, with comments by Frieder Ladisch and Richard Stanley

Venue/year: MathOverflow, asked 2015, modified 2016

DOI/arXiv/URL: <https://mathoverflow.net/questions/223642/is-combinatorial-automorphism-of-symmetric-convex-polytope-always-antipodal>

Relevance: This is a direct restatement of the same problem. The comments identify the equivalent formulation that the central symmetry should centralize all combinatorial automorphisms and point back to Kalai's ICM article. No answer resolving the problem is given.

3. **On combinatorial and affine automorphisms of polytopes**

Authors: Jurgen Bokowski, Gunter Ewald, Peter Kleinschmidt

Venue/year: Israel Journal of Mathematics 47 (1984), 123-130

DOI/URL: <https://doi.org/10.1007/BF02760511>

Relevance: The paper disproves the older conjecture that every combinatorial automorphism of a convex polytope is induced by an affine transformation. If a combinatorial automorphism of a centrally symmetric polytope is affine/geometric, then it preserves the center and hence the antipodal pairing; this paper shows that this affine-realizability route cannot settle Perles' question in general. It does not itself provide a centrally symmetric counterexample to the problem statement.

4. **Prescribing symmetries and automorphisms for polytopes**

Authors: Egon Schulte, Pablo Soberon, Gordon Ian Williams

Venue/year: In *Polytopes and Discrete Geometry*, Contemporary Mathematics 764, American Mathematical Society, Providence, RI, 2021, pp. 221-233

DOI/arXiv/URL: <https://doi.org/10.1090/conm/764/15359> ; arXiv:1902.05439

Relevance: Theorem 4.1 constructs, for every finite abelian group Γ of even order and involution σ , a centrally symmetric convex polytope with $G(P) = \Gamma(P) = \Gamma$ and σ realized as central symmetry. These are broad positive examples where the combinatorial automorphism group equals the geometric symmetry group, so central symmetry is central. This is only an existence/extra-hypothesis result and does not prove the property for arbitrary centrally symmetric polytopes.

5. **Rigidity, Tensegrity, and Reconstruction of Polytopes Under Metric Constraints**

Author: Martin Winter

Venue/year: International Mathematics Research Notices 2024, no. 9, 7721-7747; published online 2023

DOI/arXiv/URL: <https://doi.org/10.1093/imrn/rnad298> ; arXiv:2302.14194

Relevance: Winter proves several reconstruction/rigidity results and notes a subtle obstruction for centrally symmetric polytopes: David E. Speyer constructed a 12-vertex 4-polytope whose edge graph has an automorphism that does not preserve antipodality. This is strong negative evidence for graph-level variants, but it is explicitly an edge-graph automorphism statement, not a face-lattice automorphism statement. Therefore it does not disprove the problem.

6. **Does the edge-graph of a centrally symmetric polytope determine which vertices are antipodal?**

Author: David E. Speyer, answer to a MathOverflow question by Martin Winter

Venue/year: MathOverflow, 2023

DOI/arXiv/URL: <https://mathoverflow.net/questions/440534/does-the-edge-graph-of-a-centrally-symmetric-polytope-determine-which-vertices-a>

Relevance: Speyer gives the underlying 12-vertex 4-dimensional centrally symmetric polytope with an edge-graph symmetry interchanging non-antipodal vertices. This resolves a stronger-looking graph-only variant negatively, but it does not verify that the symmetry preserves all faces of the polytope. Thus it is not a counterexample to Perles' face-lattice automorphism question.

6.3.3.4. What remains unresolved

No located source proves that every face-lattice automorphism of every centrally symmetric convex polytope preserves antipodal vertex pairs, and no located source gives a centrally symmetric convex polytope with a genuine face-lattice automorphism violating antipodality. The full all-dimensions, all-polytopes statement in the problem remains open.

6.3.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

6.3.5. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>.

6.4. 123. Combinatorics and convexity [1994-geom-53]

Record: 1994-geom-53

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

6.4.1. Problem statement

For integers $d \geq 1$ and $n \geq d + 1$, let $C_d(n)$ denote the cyclic d -polytope with n vertices, i.e. the convex hull in $\mathbb{R}^d$ of n points on the moment curve $t \mapsto (t, t^2, \dots, t^d)$. Prove or disprove that, if K is any $(d-1)$ -dimensional simplicial complex with n vertices such that for every face $F \in K$ with r vertices the link $\text{lk}_K(F)$ has the same Euler characteristic as the sphere S^{d-r-1} , then for every k with $0 \leq k \leq d-1$, the number $f_k(K)$ of k -dimensional faces of K is at most $f_k(C_d(n))$, the number of k -dimensional faces of $C_d(n)$.

6.4.2. Historical background, literature, and current progress

6.4.3. Verdict: partially resolved

6.4.3.1. Precise problem

The problem asks for Klee's Eulerian-complex extension of Motzkin's Upper Bound Conjecture: if K is an Eulerian $(d-1)$ -dimensional simplicial complex on n vertices, meaning every face link has the Euler characteristic of the corresponding sphere, must

$$f_k(K) \leq f_k(C_d(n)) \quad 0 \leq k \leq d-1$$

hold for all d, n, k ? The $k = 0$ case is automatic because both sides have n vertices, so the real content is the componentwise upper bound for all higher face numbers.

6.4.3.2. Verdict

Partially resolved. The exact all-cases statement is not fully proved in the literature I found. Current survey literature states the conjecture in essentially this form and says that, in this generality, it remains wide open. However, many important subclasses are known.

6.4.3.3. Key literature and what it proves

1. Theodore S. Motzkin, "Comonotone curves and polyhedra, Abstract III", Bulletin of the American Mathematical Society 63 (1957), 35. DOI not found; cited in later literature, e.g. Cambridge Core references.

This is the original Motzkin upper-bound conjecture source for convex polytopes. It concerns polytopes, not arbitrary Eulerian complexes.

2. Victor Klee, “A combinatorial analogue of Poincaré’s duality theorem”, *Canadian Journal of Mathematics* 16 (1964), 517-531. DOI: 10.4153/CJM-1964-053-0. URL: <https://www.cambridge.org/core/journals/canadian-journal-of-mathematics/article/combinatorial-analogue-of-poincares-duality-theorem/0F30C9676FAA0E82E9266BC2D97E8805>
Klee introduced the Eulerian-complex framework and Dehn-Sommerville relations for this setting. This supplies structural background but not the full upper-bound theorem.
3. Victor Klee, “On the Number of Vertices of a Convex Polytope”, *Canadian Journal of Mathematics* 16 (1964), 701-720. DOI: 10.4153/CJM-1964-067-6. URL: <https://www.cambridge.org/core/journals/canadian-journal-of-mathematics/article/on-the-number-of-vertices-of-a-convex-polytope/62AF79696185093199301D13A02C5D4F>
Later surveys report that Klee proved the upper-bound assertion for Eulerian complexes with sufficiently many vertices and conjectured it for all Eulerian complexes. This is partial because the problem asks for every $n \geq d + 1$.
4. Peter McMullen, “The maximum numbers of faces of a convex polytope”, *Mathematika* 17(2) (1970), 179-184. DOI: 10.1112/S0025579300002850. URL: <https://www.cambridge.org/core/journals/mathematika/article/maximum-numbers-of-faces-of-a-convex-polytope/13483049ECEDF0F2D760C50BAE8D6F21>
Proves Motzkin’s upper-bound conjecture for convex polytopes. This covers boundary complexes of polytopes but not arbitrary Eulerian complexes.
5. Richard P. Stanley, “The Upper Bound Conjecture and Cohen-Macaulay Rings”, *Studies in Applied Mathematics* 54(2) (1975), 135-142. DOI: 10.1002/sapm1975542135. URL: <https://onlinelibrary.wiley.com/doi/10.1002/sapm1975542135>
Proves the upper-bound theorem via Cohen-Macaulay methods, in particular for simplicial spheres/homology spheres. This does not cover Eulerian complexes that are not Cohen-Macaulay.
6. Isabella Novik, “Upper bound theorems for homology manifolds”, *Israel Journal of Mathematics* 108 (1998), 45-82. DOI: 10.1007/BF02783042. URL: <https://link.springer.com/article/10.1007/BF02783042>
Proves the UBC for all odd-dimensional homology manifolds and for even-dimensional homology manifolds satisfying a Betti-number condition; the known consequences include Eulerian homology manifolds. This is still stronger local topology than an arbitrary Eulerian complex.
7. Patricia Hersh and Isabella Novik, “A short simplicial h -vector and the upper bound theorem”, *Discrete & Computational Geometry* 28(3) (2002), 283-289. DOI: 10.1007/s00454-002-0746-7. arXiv: math/0111302. URL: <https://arxiv.org/abs/math/0111302>
Theorem 1 proves the UBC for a class of odd-dimensional simplicial complexes, including odd-dimensional Eulerian complexes with isolated singularities. This leaves even-dimensional cases and more general singularities open.
8. Isabella Novik and Ed Swartz, “Face numbers of pseudomanifolds with isolated singularities”, *Mathematica Scandinavica* 110(2) (2012), 198-222. DOI: 10.7146/math.scand.a-15204. arXiv: 1004.5100. URL: <https://arxiv.org/abs/1004.5100>
Theorem 4.10 proves the h -number form of the UBC for Eulerian pseudomanifolds with

homologically isolated singularities, assuming $n \geq 3d - 4$. This is partial because it adds both a singularity condition and a vertex lower bound.

9. Steven Klee and Isabella Novik, “Face enumeration on simplicial complexes”, in *Recent Trends in Combinatorics*, IMA Volumes in Mathematics and its Applications 159, Springer, 2016, pp. 653-686. arXiv: 1505.06380. URL: <https://arxiv.org/abs/1505.06380>
Section 4.2 summarizes the state of the UBC for manifolds and Eulerian complexes: Klee proved sufficiently-many-vertices cases; McMullen, Stanley, Novik, Hersh-Novik, and Novik-Swartz prove the subclasses above; but Klee’s conjecture for all Eulerian complexes remains wide open.
10. Isabella Novik, “Face numbers: the upper bound side of the story”, *Proceedings of the International Congress of Mathematicians 2022*, Vol. 6, EMS Press, 2023, pp. 4622-4644. DOI: 10.4171/ICM2022/43. URL: <https://ems.press/content/book-chapter-files/33308>
Conjecture 4.1 states the same Eulerian-complex upper-bound assertion: for an Eulerian simplicial complex Δ of dimension $d - 1$ with n vertices, $f_i(\Delta) \leq f_i(C_d(n))$. The survey explicitly says that in this generality the conjecture remains wide open, while it is known for simplicial spheres, Eulerian manifolds, and some pseudomanifolds with mild singularities.

6.4.3.4. What remains unresolved

No source found proves the full statement for every Eulerian $(d - 1)$ -dimensional simplicial complex with arbitrary singularities and every $n \geq d + 1$. The published/citable results all impose extra hypotheses, such as being a polytope, a simplicial sphere or homology sphere, an Eulerian homology manifold, an odd-dimensional pseudomanifold with isolated singularities, a homologically isolated singularity condition, or sufficiently many vertices. Therefore the exact problem is not fully resolved, but it has substantial partial progress.

6.4.4. Remaining open problems recorded by the database

6.4.4.1. Open problem 1

Prove the Upper Bound Conjecture for even-dimensional Eulerian simplicial pseudomanifolds with isolated singularities: if Δ is such a complex of dimension $d - 1$ on n vertices, prove $f_i(\Delta) \leq f_i(C_d(n))$ for every i .

Reference supplied by the database. Isabella Novik and Ed Swartz, “Face numbers of pseudomanifolds with isolated singularities,” *Mathematica Scandinavica* **110** (2012), no. 2, 198–222, discussion immediately preceding Theorem 4.10 (explicitly identifying the even-dimensional isolated-singularity cases as open), DOI: [10.7146/math.scand.a-15204](https://doi.org/10.7146/math.scand.a-15204), arXiv:1004.5100.

6.4.5. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>.

6.5. 124. Combinatorics and convexity [1994-geom-54]

Record: 1994-geom-54

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

6.5.1. Problem statement

Prove or disprove that every centrally symmetric convex d -dimensional polytope $P \subset \mathbb{R}^d$ has at least 3^d nonempty faces in its face lattice, counting P itself as a face but not counting the empty face.

6.5.2. Historical background, literature, and current progress

6.5.3. Verdict: partially resolved

6.5.3.1. Precise problem

This problem is Kalai's 3^d conjecture for centrally symmetric polytopes:

Every centrally symmetric convex d -dimensional polytope P in $\mathbb{R}^d$ has at least 3^d nonempty faces, counting P itself and not counting the empty face.

Some papers use the equivalent convention “count the empty face but not P ”; since exactly one face is exchanged in either convention, the numerical bound is the same.

6.5.3.2. Why it is not fully resolved

I found no published or citable result proving the statement for all centrally symmetric d -polytopes in all dimensions. The literature consistently treats the general case as open beyond the known special cases. In particular, Novik's survey says that the 3^d conjecture is known for centrally symmetric simplicial polytopes, by duality for simple polytopes, and for all centrally symmetric polytopes of dimension at most 4, but is “wide open” in all other cases.

Thus the exact statement remains unresolved for arbitrary centrally symmetric d -polytopes in dimensions $d \geq 5$ that are not covered by the special classes below.

6.5.3.3. Progress in the literature

1. Title: The number of faces of centrally-symmetric polytopes Authors: Gil Kalai Venue: Graphs and Combinatorics 5(1), 389-391 Year: 1989 DOI/URL: <https://doi.org/10.1007/BF01788696> Role: Original source of the conjecture. It states the general all-dimensions problem for centrally symmetric convex polytopes.

2. Title: On the number of faces of centrally-symmetric simplicial polytopes Authors: Richard P. Stanley Venue: Graphs and Combinatorics 3, 55-66 Year: 1987 DOI/URL: <https://doi.org/10.1007/BF01788529> Result: Gives lower bounds for face numbers of centrally symmetric simplicial polytopes. These imply Kalai's 3^d bound for centrally symmetric simplicial polytopes; by polarity, the bound also holds for centrally symmetric simple polytopes. Limitation: This does not cover non-simple, non-simplicial centrally symmetric polytopes.
3. Title: Borsuk's theorem and the number of facets of centrally symmetric polytopes Authors: Imre Barany, Laszlo Lovasz Venue: Acta Mathematica Academiae Scientiarum Hungaricae 40, 323-329 Year: 1982 DOI/URL: <https://doi.org/10.1007/BF01903592> Result: Earlier facet-number progress for centrally symmetric polytopes and part of the background leading to the simplicial case. Limitation: It is not a proof of the total-face 3^d conjecture in full generality.
4. Title: On Kalai's Conjectures Concerning Centrally Symmetric Polytopes Authors: Raman Sanyal, Axel Werner, Guenter M. Ziegler Venue: Discrete & Computational Geometry 41, 183-198 Year: 2009 DOI/arXiv: <https://doi.org/10.1007/s00454-008-9104-8>; arXiv:0708.3661 Result: Proves Kalai's Conjecture A, the 3^d conjecture, for all centrally symmetric polytopes of dimension at most 4. It also disproves stronger conjectures B and C in higher dimensions. Limitation: The 3^d result is dimensional: it does not prove the conjecture for arbitrary dimensions $d \geq 5$.
5. Title: Face Numbers of Centrally Symmetric Polytopes Produced from Split Graphs Authors: Ragnar Freij, Matthias Henze, Moritz W. Schmitt, Guenter M. Ziegler Venue: Electronic Journal of Combinatorics 20(2), #P32 Year: 2013 DOI/arXiv: <https://doi.org/10.37236/3315>; arXiv:1201.5790 Result: Proves Kalai's 3^d conjecture for Hansen polytopes of split graphs, with formula $s(H(G)) = 3^d + p_G(C, S)$. Limitation: Hansen polytopes of split graphs are a special family, not all centrally symmetric polytopes.
6. Title: A note on Kalai's 3^d Conjecture Authors: Gregory R. Chambers, Elia Portnoy Venue: arXiv preprint Year: 2022, revised 2023 DOI/arXiv: <https://doi.org/10.48550/arXiv.2211.09215>; arXiv:2211.09215 Result: Proves the conjecture when the polytope is symmetric about d hyperplanes with orthogonal normal vectors, equivalently after an orthogonal change of coordinates, the coordinate-symmetric/unconditional case. Limitation: This is a symmetry-hypothesis special case.
7. Title: Kalai's 3^d conjecture for unconditional and locally anti-blocking polytopes Authors: Raman Sanyal, Martin Winter Venue: Proceedings of the American Mathematical Society 153(01), 279-290 Year: 2025 DOI/arXiv: <https://doi.org/10.1090/proc/16909>; arXiv:2308.02909 Result: Theorem 3 proves $s(P) \geq 3^d$ for unconditional polytopes, with equality exactly for Hanner polytopes. Theorem 4 proves the more general result for proper locally anti-blocking polytopes, with equality exactly for generalized Hanner polytopes. Limitation: Not every centrally symmetric polytope is unconditional or locally anti-blocking.
8. Title: A tale of centrally symmetric polytopes and spheres Author: Isabella Novik Venue: Recent Trends in Algebraic Combinatorics, Association for Women in Mathematics Series 16, Springer, 305-331 Year: 2019 arXiv/URL: arXiv:1711.09310; <https://arxiv.org/abs/1711.09310> Role: Survey source. It records the known cases and explicitly states that the 3^d conjecture is wide open outside centrally symmetric simplicial/simple polytopes and dimension at most 4.

6.5.3.4. Remaining unresolved part

The full statement for arbitrary centrally symmetric convex d -dimensional polytopes remains open in dimensions $d \geq 5$, except for the special classes listed above. Since the literature proves only special cases and no theorem found covers all centrally symmetric polytopes, the correct status is partially resolved.

6.5.4. Remaining open problems recorded by the database

6.5.4.1. Open problem 1

Prove that every centrally symmetric five-dimensional convex polytope has at least $3^5 = 243$ nonempty faces, counting the polytope itself and excluding the empty face.

Reference supplied by the database. Raman Sanyal, Axel Werner, and Günter M. Ziegler, “On Kalai’s conjectures concerning centrally symmetric polytopes,” *Discrete & Computational Geometry* **41** (2009), no. 2, 183–198, Conjecture A (the 3^d -conjecture) and the proof through dimension 4, DOI: [10.1007/s00454-008-9104-x](https://doi.org/10.1007/s00454-008-9104-x), [arXiv:0708.3661](https://arxiv.org/abs/0708.3661). Current open status is also recorded in Raman Sanyal and Martin Winter, “Kalai’s 3^d conjecture for unconditional and locally anti-blocking polytopes,” *Mathematische Zeitschrift* **307** (2024), article 46, DOI: [10.1007/s00209-024-03505-1](https://doi.org/10.1007/s00209-024-03505-1), [arXiv:2308.02909](https://arxiv.org/abs/2308.02909).

6.5.5. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>.

6.6. 125. Combinatorics and convexity [1994-geom-55]

Record: 1994-geom-55

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

6.6.1. Problem statement

Prove that for every simplicial $(d - 1)$ -dimensional sphere K , the facet-adjacency data determine the entire face structure of K up to combinatorial isomorphism. Equivalently, if one records, for each facet F of K , the d neighboring facets of F that share a codimension-one face with F , then this facet-neighbor puzzle has a unique solution.

6.6.2. Historical background, literature, and current progress

6.6.3. Status check

Verdict: partially resolved.

6.6.3.1. Precise problem

The problem asks whether every simplicial $(d-1)$ -sphere K is uniquely determined, up to combinatorial isomorphism, by its facet-ridge graph: the graph whose vertices are facets of K , with two facets adjacent exactly when they share a codimension-one face. Equivalently, if two simplicial spheres have isomorphic facet-ridge graphs, must the graph isomorphism extend to an isomorphism of simplicial complexes?

This is the standard reconstruction problem for simplicial spheres, formulated in recent papers as the Blind-Mani/Kalai conjecture. It is stronger than the classical polytopal case because it quantifies over all simplicial spheres, not only boundaries of simplicial polytopes and not only shellable spheres.

6.6.3.2. Literature and current progress

1. Title: Puzzles and polytope isomorphisms Authors: Roswitha Blind and Peter Mani-Levitska Venue: Aequationes Mathematicae 34(2-3), 287-297 Year: 1987 DOI/URL: <https://doi.org/10.1007/BF01830678> Relevance: Proves Perles' conjecture that the graph of a simple polytope determines its combinatorial structure. In dual form, this says the facet-ridge graph determines a simplicial polytope. This fully resolves the polytopal special case, but the problem asks about arbitrary simplicial spheres.

2. Title: A simple way to tell a simple polytope from its graph Author: Gil Kalai Venue: Journal of Combinatorial Theory, Series A 49(2), 381-383 Year: 1988 DOI/URL: [https://doi.org/10.1016/0097-3165\(88\)90064-7](https://doi.org/10.1016/0097-3165(88)90064-7) Relevance: Gives a short constructive proof of the Blind-Mani theorem. Again, this proves the dual statement for simplicial polytopes, not for all simplicial spheres.
3. Title: Graphs, skeleta and reconstruction of polytopes Author: Margaret M. Bayer Venue: Acta Mathematica Hungarica 155, 61-73 Year: 2018 DOI/arXiv/URL: <https://doi.org/10.1007/s10474-018-0804-0>, arXiv:1710.00118 Relevance: Survey of reconstruction results. It states the dual consequence of Blind-Mani/Kalai: the face lattice of a simplicial polytope can be reconstructed from its facet-ridge graph. This is background for why the problem is a generalization beyond polytopes.
4. Title: Subword Complexes and Kalai's Conjecture on Reconstruction of Spheres Authors: Cesar Ceballos and Joseph Doolittle Venue: Discrete & Computational Geometry 74, 23-48 Year: 2025 DOI/arXiv/URL: <https://doi.org/10.1007/s00454-025-00733-6>, arXiv:2206.15461 Relevance: States the conjecture in the exact form: for d -dimensional simplicial spheres P, Q , every facet-ridge graph isomorphism $FR(P) \rightarrow FR(Q)$ should uniquely extend to a simplicial isomorphism. Their Theorem 1.3 proves this only for finite-type spherical subword complexes. This is a special family of simplicial spheres, so it is partial progress only.
5. Title: Reconstructing a shellable sphere from its facet-ridge graph Author: Yirong Yang Venue: arXiv preprint Year: 2024, revised 2025 DOI/arXiv/URL: arXiv:2401.04220, <https://arxiv.org/abs/2401.04220> Relevance: Theorem 1.2 proves that the combinatorial structure of a shellable simplicial sphere is determined by its facet-ridge graph. This is the strongest positive result found. It covers all shellable simplicial spheres, hence all simplicial polytopes, but it does not cover arbitrary simplicial spheres. The paper explicitly remarks that it remains open whether the theorem holds for more general classes such as constructible spheres, partitionable spheres, or all simplicial spheres, and explains that the proof uses shellability to get acyclic good orientations.
6. Title: Unshellable triangulations of spheres Author: W. B. R. Lickorish Venue: European Journal of Combinatorics 12(6), 527-530 Year: 1991 DOI/URL: [https://doi.org/10.1016/S0195-6698\(13\)80103-5](https://doi.org/10.1016/S0195-6698(13)80103-5) Relevance: Shows that non-shellable triangulated spheres exist. This confirms that Yang's shellability hypothesis is a real restriction and not automatically satisfied by all simplicial spheres.
7. Title: A minimal counterexample to a strengthening of Perles' conjecture Author: Joseph Doolittle Venue: arXiv preprint Year: 2018 DOI/arXiv/URL: arXiv:1809.00662, <https://arxiv.org/abs/1809.00662> Relevance: Gives counterexamples to a stronger Perles-piece criterion and discusses barriers to extending polytopal reconstruction methods to arbitrary simplicial spheres. It does not disprove the exact reconstruction statement, but it shows that some natural stronger local-recognition approaches fail.

6.6.3.3. What remains unresolved

No published or otherwise citable result found proves that every simplicial sphere is determined by its facet-ridge graph. The literature resolves important subclasses: simplicial polytopes, finite-type spherical subword complexes, and shellable simplicial spheres. The exact all-cases statement in the problem remains open beyond these hypotheses.

Therefore the correct status is partially resolved, not fully resolved.

6.6.4. Remaining open problems recorded by the database

6.6.4.1. Open problem 1

Prove that every constructible simplicial sphere is uniquely determined, up to simplicial isomorphism, by its facet–ridge graph.

Reference supplied by the database. Yirong Yang, “Reconstructing a shellable sphere from its facet-ridge graph,” arXiv preprint (2024; revised 2025), open problem in the concluding paragraph asking whether the main reconstruction theorem extends to constructible spheres, [arXiv:2401.04220v2](https://arxiv.org/abs/2401.04220v2).

6.6.4.2. Open problem 2

Prove that every partitionable simplicial sphere is uniquely determined, up to simplicial isomorphism, by its facet–ridge graph.

Reference supplied by the database. Yirong Yang, “Reconstructing a shellable sphere from its facet-ridge graph,” arXiv preprint (2024; revised 2025), open problem in the concluding paragraph asking whether the main reconstruction theorem extends to partitionable spheres, [arXiv:2401.04220v2](https://arxiv.org/abs/2401.04220v2).

6.6.5. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>.

6.7. 126. Face Numbers of 4-Polytopes and 3-Spheres [2002-geom-77]

Record: 2002-geom-77

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Face Numbers of 4-Polytopes and 3-Spheres

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=605>

6.7.1. Problem statement

Prove that for every strongly regular cellulation P of the 3-sphere S^3 , meaning a regular cell decomposition of S^3 in which the intersection of any two cells is a face of both cells or is empty, if $f_i(P)$ denotes the number of i -dimensional faces for $i = 0, 1, 2, 3$, and $f_{03}(P)$ denotes the number of vertex-facet incidences, equivalently the number of chains consisting of a 0-dimensional face contained in a 3-dimensional face, then

$$f_{03}(P) - 3(f_0(P) + f_3(P)) \geq -10.$$

Equivalently, whenever the quotient is defined, the complexity

$$C(P) := \frac{f_{03}(P) - 20}{f_0(P) + f_3(P) - 10}$$

satisfies $C(P) \geq 3$.

6.7.2. Historical background, literature, and current progress

6.7.3. Verdict: partially resolved

6.7.3.1. Precise problem

The problem asks whether every strongly regular cellulation P of S^3 satisfies

$$f_{03}(P) - 3(f_0(P) + f_3(P)) \geq -10.$$

Equivalently, for non-simplex cases where the quotient is defined, it asks whether the normalized complexity

$$C(P) = \frac{f_{03}(P) - 20}{f_0(P) + f_3(P) - 10}$$

always satisfies $C(P) \geq 3$. This is the rank-5 / dimension-4 toric g_2 -nonnegativity inequality for the face lattice of a strongly regular CW 3-sphere.

6.7.3.2. Current status

I found no published or otherwise citable result proving this inequality for all strongly regular cellulations of S^3 . The problem is therefore not fully resolved. There is genuine partial progress: the inequality is known for all convex 4-polytope boundaries, hence for the polytopal subclass of strongly regular 3-spheres, and for several related/special classes. Those results do not cover arbitrary strongly regular CW 3-spheres.

6.7.3.3. Key literature and applicability

1. Face numbers of 4-Polytopes and 3-Spheres

Günter M. Ziegler, *Proceedings of the International Congress of Mathematicians, Vol. III (Beijing, 2002)*, Higher Education Press, 2002, pp. 625-634. arXiv:math/0208073, DOI: <http://doi.org/10.48550/arXiv.math/0208073>.

This is the original source for the fatness/complexity formulation. It states the exact inequality $f_{03} - 3(f_0 + f_3) \geq -10$, equivalently $C(P) \geq 3$, identifies it with $g_2^{\text{tor}}(P) \geq 0$, and says it was known for rational polytopes and all convex polytopes but not for strongly regular 3-spheres or Eulerian lattices. Thus it frames the exact unresolved extension.

2. Generalized h -vectors, intersection cohomology of toric varieties, and related results

Richard P. Stanley, in *Commutative Algebra and Combinatorics*, Advanced Studies in Pure Mathematics 11, Kinokuniya/North-Holland, 1987, pp. 187-213. URL: <https://math.mit.edu/~rstan/pubs/pubfiles/69.pdf>.

Stanley proves toric g -vector nonnegativity in the rational polytope setting. This implies the stated inequality for rational convex 4-polytopes, but not for arbitrary strongly regular 3-spheres.

3. Rigidity and the lower bound theorem. I

Gil Kalai, *Inventiones Mathematicae* 88 (1987), 125-151. DOI: <https://doi.org/10.1007/BF01405094>.

Kalai gives a rigidity-theoretic proof of the relevant toric g_2 inequality for general convex polytopes. This covers all 4-polytope boundary complexes, a proper subclass of the strongly regular 3-spheres in the problem statement.

4. Hard Lefschetz theorem for nonrational polytopes

Kalle Karu, *Inventiones Mathematicae* 157 (2004), 419-447. arXiv:math/0112087, DOI: <https://doi.org/10.1007/s00222-004-0358-3>.

Karu proves the Hard Lefschetz theorem for combinatorial intersection cohomology of general polytopes, giving another route to toric g -vector nonnegativity for all convex polytopes. This still does not apply to non-polytopal strongly regular CW 3-spheres.

5. Nonpolytopal Nonsimplicial Lattice Spheres with Nonnegative Toric g -Vector

Louis J. Billera and Eran Nevo, *Discrete & Computational Geometry* 48(4), 1048-1057, 2012. arXiv:1110.0739, DOI: <https://doi.org/10.1007/s00454-012-9449-x>.

This constructs many nonpolytopal nonsimplicial Gorenstein* meet semilattices with nonnegative toric g -vector, supporting Stanley's conjecture. It is partial progress, not a theorem for every strongly regular CW 3-sphere.

6. The flag f -vectors of Gorenstein* order complexes of dimension 3

Satoshi Murai and Eran Nevo, *Proceedings of the American Mathematical Society* 142(5), 1527-1538, 2014. arXiv:1108.0572, DOI: <https://doi.org/10.1090/S0002-9939-2014-11897-1>.

This characterizes rank-5 Gorenstein* poset cd-indices/flag vectors. It does not settle

the problem statement because strongly regular CW spheres impose the additional intersection-property/lattice condition, and the theorem is not a proof of $g_2^{\text{tor}} \geq 0$ for all such lattices.

7. A Lower Bound Theorem for Strongly Regular CW Spheres with up to $2d + 1$ Vertices

Lei Xue, *Discrete & Computational Geometry* 72, 1348-1368, 2024. arXiv:2207.13839, DOI: <https://doi.org/10.1007/s00454-023-00553-6>.

This gives sharp lower bounds for ordinary face numbers of strongly regular CW spheres/pseudomanifolds with few vertices. It is related lower-bound progress but does not prove the vertex-facet incidence inequality for all strongly regular cellulations of S^3 .

6.7.3.4. What remains unresolved

The missing case is the full non-polytopal strongly regular CW 3-sphere case: no searched paper proves that every such cellulation has $f_{03} - 3(f_0 + f_3) \geq -10$, and no searched paper gives a counterexample. The exact problem should therefore be classified as partially resolved, not fully resolved.

6.7.4. Remaining open problems recorded by the database

6.7.4.1. Open problem 1

Let S be a strongly regular 3-sphere that is 2-simple and 2-simplicial (every edge lies in exactly three facets and every 2-face is a triangle), and suppose that S is not the boundary of the 4-simplex. Is its fatness

$$F(S) = \frac{f_1(S) + f_2(S) - 20}{f_0(S) + f_3(S) - 10}$$

always at least 4?

Reference supplied by the database. Philip Brinkmann, *f-Vector Spaces of Polytopes, Spheres, and Eulerian Lattices*, doctoral dissertation, Department of Mathematics and Computer Science, Freie Universität Berlin, 2016, Chapter 1, Conjecture 1 (the text notes that this conjecture would imply $C \geq 3$ for 2-simple 2-simplicial 3-spheres); DOI: [10.17169/refubium-13151](https://doi.org/10.17169/refubium-13151); repository record: [Refubium](https://refubium.org/).

6.7.5. Primary-source pointer

The originating ICM record is *Face Numbers of 4-Polytopes and 3-Spheres* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=605>.

6.8. 127. Face Numbers of 4-Polytopes and 3-Spheres [2002-geom-78]

Record: 2002-geom-78

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Face Numbers of 4-Polytopes and 3-Spheres

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=605>

6.8.1. Problem statement

Prove that for every strongly regular CW 3-sphere P , meaning a regular cell decomposition of S^3 in which the intersection of any two cells is either empty or a common face of both cells, if $f_i(P)$ denotes the number of i -dimensional faces of P , then the fatness

$$F(P) := \frac{f_1(P) + f_2(P) - 20}{f_0(P) + f_3(P) - 10}$$

satisfies $F(P) \geq \frac{5}{2}$. Equivalently,

$$2(f_1(P) + f_2(P) - 20) \geq 5(f_0(P) + f_3(P) - 10).$$

6.8.2. Historical background, literature, and current progress

6.8.3. Verdict: partially resolved

6.8.3.1. Precise problem

The problem asks whether every strongly regular CW decomposition P of S^3 , i.e. every regular CW 3-sphere with the intersection property/face-poset lattice property, satisfies

$$F(P) = \frac{f_1(P) + f_2(P) - 20}{f_0(P) + f_3(P) - 10} \geq \frac{5}{2},$$

equivalently

$$5f_0(P) - 2f_1(P) - 2f_2(P) + 5f_3(P) \leq 10.$$

6.8.3.2. Status

I did not find any published or citable result proving this inequality for all strongly regular CW 3-spheres, nor any published counterexample. The exact all-cases problem remains unresolved. It is not fully open in the sense that several important subclasses and adjacent forms are known, so the correct verdict is partially resolved.

6.8.3.3. Literature and progress

1. **Face numbers of 4-Polytopes and 3-Spheres.** Günter M. Ziegler. Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. III, Higher Education Press, pp. 625-636, 2002/2003. arXiv:math/0208073, DOI/URL: <https://doi.org/10.48550/arXiv.math/0208073>.
This is the original source for the normalized fatness parameter used in the problem statement. Ziegler identifies the inequality $\varphi_0 + \varphi_3 \leq 2/5$, equivalently $F \geq 5/2$, as valid for 4-polytopes but explicitly says that its validity for strongly regular 3-spheres is open.
2. **The extended f-vectors of 4-polytopes.** Margaret M. Bayer. Journal of Combinatorial Theory, Series A 44(1), 141-151, 1987. DOI: [https://doi.org/10.1016/0097-3165\(87\)90066-5](https://doi.org/10.1016/0097-3165(87)90066-5).
Bayer's 4-polytope inequalities include the linear inequality $5f_0 - 2f_1 - 2f_2 + 5f_3 \leq 10$, i.e. $F \geq 5/2$, for convex 4-polytopes. This resolves the boundary-sphere-of-a-4-polytope subclass, but not arbitrary strongly regular CW 3-spheres.
3. **Lower-bound theorems for pseudomanifolds.** Tiong-Seng Tay. Discrete & Computational Geometry 13(2), 203-216, 1995. DOI: <https://doi.org/10.1007/BF02574038>.
The simplicial lower-bound theorem implies the fatness inequality for simplicial strongly regular 3-spheres, and by duality for simple cases. This is a proper special case, not the nonsimplicial strongly regular CW case in full.
4. **Fat 4-polytopes and fatter 3-spheres.** David Eppstein, Greg Kuperberg, and Günter M. Ziegler. In *Discrete Geometry: In Honor of W. Kuperberg's 60th Birthday*, Pure and Applied Mathematics 253, Marcel Dekker, pp. 239-265, 2003. arXiv:math/0204007, URL: <https://arxiv.org/abs/math/0204007>.
This constructs strongly regular CW 3-spheres of unbounded fatness. It resolves the upper/unbounded-fatness direction for strongly regular CW 3-spheres, but it gives no lower-bound theorem and no counterexample to $F \geq 5/2$.
5. **A flag vector of a 3-sphere that is not the flag vector of a 4-polytope.** Philip Brinkmann and Günter M. Ziegler. Mathematika 63(1), 260-271, 2017. arXiv:1506.08148, DOI: <https://doi.org/10.1112/S0025579316000267>.
The paper separates flag vectors of strongly regular 3-spheres from those of 4-polytopes, and states that although the lower bound for fatness is known for 4-polytopes, it is not known whether it holds for 3-spheres.
6. **Small f-vectors of 3-spheres and of 4-polytopes.** Philip Brinkmann and Günter M. Ziegler. Mathematics of Computation 87(314), 2955-2975, 2018. arXiv:1610.01028, DOI: <https://doi.org/10.1090/mcom/3300>.
This gives finite enumerations and examples showing that f-vectors of strongly regular 3-spheres differ from f-vectors of 4-polytopes. It explicitly lists "Is the fatness lower bound $F \geq 2.5$ valid for all 3-spheres?" as a key open problem; its enumeration does not cover all cases.
7. **The flag f-vectors of Gorenstein* order complexes of dimension 3.** Satoshi Murai and Eran Nevo. Proceedings of the American Mathematical Society 142(5), 1527-1538, 2014. arXiv:1108.0572, DOI: <https://doi.org/10.1090/S0002-9939-2014-11897-1>.
This characterizes cd-indices of rank-5 Gorenstein* posets/order complexes and a larger regular-CW setting. It does not resolve the problem because the problem concerns the original f-vector of strongly regular CW 3-spheres with the intersection/face-lattice prop-

erty.

8. **A Lower Bound Theorem for Strongly Regular CW Spheres with up to $2d + 1$ Vertices.** Lei Xue. Discrete & Computational Geometry 72(3), 1348-1368, 2024. arXiv:2207.13839, DOI: <https://doi.org/10.1007/s00454-023-00553-6>.

This proves sharp lower bounds on individual face numbers for strongly regular CW spheres in a bounded-vertex regime. It is substantial partial progress on lower-bound phenomena for strongly regular CW spheres, but it does not prove the stated fatness inequality for all strongly regular CW 3-spheres.

9. **Face numbers of shellable CW balls and spheres.** Joshua Hinman. Israel Journal of Mathematics, 2025. arXiv:2409.08427, DOI: <https://doi.org/10.1007/s11856-025-2887-y>.

This proves lower bounds for shellable strongly regular CW balls and spheres, and for CL-shellable plus dual CL-shellable face posets. The conclusions are not the exact global inequality $F \geq 5/2$ for all strongly regular CW 3-spheres.

10. **Fat Shellable Spheres.** Joshua Hinman. Preprint, 2025. arXiv:2509.20771, URL: <https://arxiv.org/abs/2509.20771>.

This constructs arbitrarily fat strongly regular CW 3-spheres that are shellable and dual shellable. It strengthens the upper/unbounded-fatness side, but it does not settle the lower-bound question.

6.8.3.4. What remains unresolved

The missing case is the full nonsimplicial, nonpolytopal class of strongly regular CW 3-spheres. Existing results prove the inequality for convex 4-polytope boundaries and simplicial/simple subclasses, and provide several related lower-bound theorems or enumerations, but no citable theorem found covers every strongly regular CW 3-sphere and implies

$$2(f_1 + f_2 - 20) \geq 5(f_0 + f_3 - 10).$$

6.8.4. Remaining open problems recorded by the database

6.8.4.1. Open problem 1

Let S be a strongly regular 3-sphere that is 2-simple and 2-simplicial (every edge lies in exactly three facets and every 2-face is a triangle), and suppose that S is not the boundary of the 4-simplex. Is its fatness

$$F(S) = \frac{f_1(S) + f_2(S) - 20}{f_0(S) + f_3(S) - 10}$$

always at least 4?

Reference supplied by the database. Philip Brinkmann, *f-Vector Spaces of Polytopes, Spheres, and Eulerian Lattices*, doctoral dissertation, Department of Mathematics and Computer Science, Freie Universität Berlin, 2016, Chapter 1, Conjecture 1; DOI: [10.17169/refubium-13151](https://doi.org/10.17169/refubium-13151); repository record: [Refubium](https://refubium.org/).

6.8.5. Primary-source pointer

The originating ICM record is *Face Numbers of 4-Polytopes and 3-Spheres* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=605>.

6.9. 128. Random Points, Convex Bodies, Lattices [2002-geom-87]

Record: 2002-geom-87

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Random Points, Convex Bodies, Lattices

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=512>

6.9.1. Problem statement

Let $K \subset \mathbb{R}^d$ be a rational polytope given by m linear inequalities with integral coefficients. Define the size of each inequality to be the number of bits needed to encode it as a binary string, and let ϕ be the sum of these sizes over the m defining inequalities. Let

$$I(K) = \text{conv}(\mathbb{Z}^d \cap K)$$

be the integer convex hull of K , and for a polytope P , let $f_i(P)$ denote the number of i -dimensional faces of P . Prove or disprove that, for every $i = 0, 1, \dots, d-1$,

$$f_i(I(K)) \ll \phi^{d-1},$$

where the implied constant in $\ll$ may depend on d and on the number m of defining inequalities.

6.9.2. Historical background, literature, and current progress

6.9.3. Verdict: partially resolved

6.9.3.1. Precise problem

The problem asks for the following all-face strengthening of the classical integer-hull vertex bound. If a rational polytope $K \subset \mathbb{R}^d$ is given by m integral linear inequalities of total binary encoding size ϕ , and

$$I(K) = \text{conv}(K \cap \mathbb{Z}^d),$$

then for every $i = 0, \dots, d-1$ one wants

$$f_i(I(K)) \leq C_{d,m} \phi^{d-1}.$$

The important point is that this is not just the vertex case $i = 0$; it asks for facets and all intermediate face numbers with the same exponent.

6.9.3.2. Original source

Title: “Random Points, Convex Bodies, Lattices”

Author: Imre Barany

Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. III, pp. 527-536; arXiv preprint, 2003

arXiv/URL: arXiv:math/0304462, <https://arxiv.org/abs/math/0304462>

Relevant passage: Section 4 defines $I(K)$, recalls the Cook-Hartmann-Kannan-McDiarmid bound for $f_0(I(K))$, and then states that “most likely” the same inequality holds for all $i = 0, \dots, d - 1$. This is exactly the question.

6.9.3.3. Progress found

Title: “On integer points in polyhedra”

Authors: William Cook, Martin Hartmann, Ravindran Kannan, Colin McDiarmid

Venue/year: Combinatorica 12, 27-37 (1992)

DOI/URL: <https://doi.org/10.1007/BF01191202>

Result: Gives an upper bound on the number of vertices of the integer hull of a rational polyhedron in terms of dimension, number of inequalities, and encoding size. In the usual row-size notation this is cited as

$$|\text{vert}(P^I)| \leq 2m^n (6n^2\phi)^{n-1}.$$

Since the maximum row size is at most the total size used in the problem statement, this proves the requested bound for $i = 0$. It does not prove the same exponent for facets or intermediate faces.

Title: “On integer points in polyhedra: A lower bound”

Authors: Imre Barany, Roger Howe, Laszlo Lovasz

Venue/year: Combinatorica 12, 135-142 (1992)

DOI/URL: <https://doi.org/10.1007/BF01204716>

Result: Constructs, for fixed d , rational simplices of size at most ϕ whose integer hulls have $\gg_d \phi^{d-1}$ vertices. This shows the vertex exponent $d - 1$ is sharp, and Barany notes that the same exponent would be best possible for all faces if the all-face statement were true.

Title: “On the number of vertices in integer linear programming problems”

Author: Nikolai Yu. Zolotykh

Venue/year: arXiv preprint, 2006

arXiv/DOI/URL: arXiv:math/0611356, <https://doi.org/10.48550/arXiv.math/0611356>

Result: Surveys and improves bounds for the number of vertices of integer hulls. Its facet discussion only derives weaker polynomial bounds from vertex bounds via McMullen’s general face-number theorem; it does not give the conjectured $C_{d,m}\phi^{d-1}$ bound for all faces.

Title: “New Bounds for the Vertices of the Integer Hull”

Authors: Sebastian Berndt, Klaus Jansen, Kim-Manuel Klein

Venue/year: Symposium on Simplicity in Algorithms (SOSA), SIAM, 2021, pp. 25-36; arXiv:2006.10847

DOI/URL: <https://doi.org/10.1137/1.9781611976496.3>, <https://arxiv.org/abs/2006.10847>

Result: Gives later improved bounds and enumeration results for vertices of integer hulls in canonical integer-programming settings. This confirms continuing progress on vertex complexity, but it does not resolve the all-face bound in the problem statement.

6.9.3.4. What is resolved

The vertex case $f_0(I(K)) \ll_{d,m} \phi^{d-1}$ is fully resolved and sharp. The full problem also follows in dimensions $d \leq 3$: for polygons, edges and vertices are comparable, and for 3-polytopes Euler/incidence relations bound edges and facets linearly in the number of vertices.

6.9.3.5. What remains unresolved

I found no published or otherwise citable theorem proving

$$f_i(I(K)) \ll_{d,m} \phi^{d-1}$$

for every i in arbitrary fixed dimension. In dimensions $d \geq 4$, the known vertex bound plus general polytope face-number estimates gives only a weaker polynomial upper bound with a larger exponent. Thus the exact all-face statement in the problem statement remains unresolved beyond the vertex and low-dimensional consequences.

6.9.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

6.9.5. Primary-source pointer

The originating ICM record is *Random Points, Convex Bodies, Lattices* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=512>.

6.10. 129. Random Points, Convex Bodies, Lattices [2002-geom-88]

Record: 2002-geom-88

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Random Points, Convex Bodies, Lattices

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=512>

6.10.1. Problem statement

Fix an integer $d \geq 2$ and a constant $D < \infty$. Let C^d denote the class of convex bodies in $\mathbb{R}^d$, and let $\mathbb{Z}^d$ be the standard integer lattice. For a convex body $L \subset \mathbb{R}^d$, let

$$I(L) = \text{conv}(\mathbb{Z}^d \cap L)$$

be its integer convex hull. For a polytope P , let $f_i(P)$ denote the number of i -dimensional faces of P .

Let G be the compact group of Euclidean isometries of $\mathbb{R}^d$ modulo translations by elements of $\mathbb{Z}^d$, equipped with its normalized Haar probability measure, and choose $g \in G$ according to this measure. For $K \in C^d$, define

$$u_K(x) = \text{Vol}(K \cap (2x - K)), \quad x \in K,$$

where $2x - K = \{2x - y : y \in K\}$ is the reflection of K about x , and write

$$K(u_K < t) = \{x \in K : u_K(x) < t\}.$$

Prove or disprove that, for every face dimension $i = 0, 1, \dots, d-1$, uniformly over all $K \in C^d$ for which the ratio of the radius of the smallest Euclidean ball containing K to the radius of the largest Euclidean ball contained in K is at most D , one has

$$\text{Vol } K(u_K < 1) \ll_{d,D} \mathbb{E}_g f_i(I(gK)) \ll_{d,D} \text{Vol } K(u_K < 1)$$

as $\text{Vol } K \rightarrow \infty$, where the Vinogradov constants implicit in $\ll_{d,D}$ depend only on d and D .

6.10.2. Historical background, literature, and current progress

6.10.3. Verdict: partially resolved

6.10.3.1. Problem checked

The problem asks for the conjectural extension of Barany's Theorem 4.3 from vertices to every face dimension: for fixed $d \geq 2$ and bounded roundness D , uniformly over all D -rounded

convex bodies $K \subset \mathbb{R}^d$ with $\text{Vol } K \rightarrow \infty$, prove

$$\text{Vol } K(u_K < 1) \ll_{d,D} \mathbb{E}_g f_i(I(gK)) \ll_{d,D} \text{Vol } K(u_K < 1)$$

for every $i = 0, \dots, d-1$, where $I(gK) = \text{conv}(\mathbb{Z}^d \cap gK)$ and g is a random Euclidean isometry modulo integer translations.

6.10.3.2. Literature and progress

1. Random points, convex bodies, lattices

Author: Imre Barany.

Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. III, 527-536; arXiv preprint 2003.

DOI/arXiv/URL: arXiv:math/0304462, <https://doi.org/10.48550/arXiv.math/0304462>.

Status contribution: This is the source of Theorem 4.3. It states the two-sided estimate only for $f_0(I(gK))$, the expected number of vertices. Immediately after the theorem it says that the same is probably true for i -dimensional faces, but that there is “no proof in sight.” Thus the source explicitly frames the stated all-face statement as a conjectural extension, not as a proved theorem.

2. The Randomized Integer Convex Hull

Authors: Imre Barany and Jiri Matousek.

Venue/year: Discrete & Computational Geometry 33, 3-25, 2005.

DOI/arXiv/URL: <https://doi.org/10.1007/s00454-003-0836-1>.

Status contribution: This is the main citable source for the randomized integer convex hull theorem. Its abstract and later survey formulations say it estimates the expected number of vertices of $I_L(K)$ for sufficiently round convex bodies of sufficiently large volume. It resolves the stated $i = 0$ case, but it does not prove the estimates for all higher-dimensional faces in general dimension.

3. Random points and lattice points in convex bodies

Author: Imre Barany.

Venue/year: Bulletin of the American Mathematical Society 45(3), 339-365, 2008.

DOI/arXiv/URL: <https://doi.org/10.1090/S0273-0979-08-01210-X>.

Status contribution: The survey restates the randomized integer hull theorem as Theorem 13.3:

$$\text{Vol } K(v < 1) \ll \mathbb{E} f_0(I_L(K)) \ll \text{Vol } K(v < 1)$$

for D -rounded K . It then says that most likely $\mathbb{E} f_j(I_L(K))$ behaves like $\mathbb{E} f_j(K_n)$ for all j , but that a proof looks hopeless at that time. This confirms that the all- j extension was not known in the standard survey literature.

4. The convex hull of the integer points in a large ball

Authors: Imre Barany and David G. Larman.

Venue/year: Mathematische Annalen 312, 167-181, 1998.

DOI/arXiv/URL: <https://doi.org/10.1007/s002080050217>.

Status contribution: This proves all face-number orders for the deterministic integer hull of a large Euclidean ball, $P_r = \text{conv}(\mathbb{Z}^d \cap rB^d)$. This is important related progress, but it is a special deterministic/smooth-body case and does not imply the uniform randomized statement for all D -rounded convex bodies.

5. Expected mean width of the randomized integer convex hull

Authors: Binh Hong Ngoc and Matthias Reitzner.

Venue/year: Mathematika 67(2), 422-433, 2021.

DOI/arXiv/URL: <https://doi.org/10.1112/mtk.12080>; arXiv:2003.06864.

Status contribution: This later paper studies the expected mean width difference and gives related intrinsic-volume upper bounds. It explicitly treats metric/intrinsic-volume quantities rather than f_i -face counts, and it does not resolve the all-face conjecture.

6.10.3.3. What remains unresolved

the full statement is not covered by the literature found. The known theorem proves the vertex case $i = 0$. In dimensions 2 and 3, elementary fixed-dimensional face-vector inequalities make the higher face counts comparable to the number of vertices once the integer hull is full-dimensional. However, for arbitrary fixed d , especially $d \geq 4$, no published or otherwise citable result found proves the required uniform two-sided estimate for every $i = 0, \dots, d - 1$ and every D -rounded convex body.

Therefore the problem is partially resolved, not fully resolved.

6.10.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

6.10.5. Primary-source pointer

The originating ICM record is *Random Points, Convex Bodies, Lattices* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=512>.

6.11. 130. Metric dimension reduction A snapshot of the Ribe program [2018-geom-149]

Record: 2018-geom-149

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

6.11.1. Problem statement

Determine the asymptotic order, up to universal constant factors, of the function $S(\delta)$ as $\delta \in (0, 1/2)$ varies, where $S(\delta)$ is the infimum of all constants $S > 0$ with the following property: for every $n \in \mathbb{N}$, there exist an integer $k \in \mathbb{N}$ with $k \leq S \log n$ and points $y_1, \dots, y_n \in \mathbb{R}^k$ such that, for every pair of distinct indices $i, j \in \{1, \dots, n\}$,

$$1 \leq \|y_i - y_j\|_2 \leq 1 + \delta,$$

where $\|\cdot\|_2$ denotes the Euclidean norm.

6.11.2. Historical background, literature, and current progress

6.11.3. Verdict: partially resolved

6.11.3.1. Precise problem

The problem asks for the asymptotic order, up to universal constant factors as $\delta \rightarrow 0$, of the least constant $S(\delta)$ such that every n admits a δ -almost equilateral n -point set in Euclidean dimension at most $S(\delta) \log n$; equivalently, for each δ , how large the exponential growth rate of the maximum size of a set in $\mathbb{R}^k$ with all distances in $[1, 1+\delta]$ can be.

6.11.3.2. Current progress

1. Metric dimension reduction: A snapshot of the Ribe program

- Authors: Assaf Naor
- Venue/year: Proceedings of the International Congress of Mathematicians–Rio de Janeiro 2018, Vol. I, World Scientific, pp. 759-837, 2018
- DOI/arXiv/URL: arXiv:1809.02376, <https://arxiv.org/abs/1809.02376>
- Relevance: This is the original source of the exact problem considered here, stated as Question 33, “dimension reduction for the vertices of the simplex.” Naor records the known bounds $S(\delta) \leq C/\delta^2$ and $S(\delta)$

$\geq c/(\delta^2 \log(1/\delta))$, and presents the logarithmic gap as unresolved.

2. Extensions of Lipschitz mappings into a Hilbert space

- Authors: William B. Johnson and Joram Lindenstrauss
- Venue/year: Conference in Modern Analysis and Probability, Contemporary Mathematics 26, American Mathematical Society, pp. 189-206, 1984
- DOI/arXiv/URL: available reprint <https://web.stanford.edu/class/cs114/readings/JL-Johnson.pdf>
- Relevance: The Johnson-Lindenstrauss lemma applied to the regular simplex gives the upper bound $S(\delta) \leq C/\delta^2$ after rescaling, since n equidistant points in Hilbert space can be embedded into $O(\delta^{-2} \log n)$ Euclidean dimensions with pairwise distances within factor $1+O(\delta)$.

3. Problems and results in extremal combinatorics-I

- Authors: Noga Alon
- Venue/year: Discrete Mathematics 273(1-3), pp. 31-53, 2003
- DOI/arXiv/URL: [https://doi.org/10.1016/S0012-365X\(03\)00227-9](https://doi.org/10.1016/S0012-365X(03)00227-9)
- Relevance: Gives the near-tight lower-bound method for dimension reduction of the simplex, cited by Naor for this problem. It yields the lower bound with the extra $\log(1/\delta)$ loss.

4. Perturbed Identity Matrices Have High Rank: Proof and Applications

- Authors: Noga Alon
- Venue/year: Combinatorics, Probability and Computing 18(1-2), pp. 3-15, 2009
- DOI/arXiv/URL: <https://doi.org/10.1017/S0963548307008917>; author PDF <https://web.math.princeton.edu/~nalon/PDFS/identity1.pdf>
- Relevance: Proposition 3.1 states that if $P_0, \dots, P_n$ are points in $\mathbb{R}^k$ whose mutual distances all lie in $[1, 1+\epsilon]$, with $1/\sqrt{n} \leq \epsilon \leq 1/10$, then $k \geq c \epsilon^{-2} \log n / \log(1/\epsilon)$. This directly implies $S(\delta) \geq c/(\delta^2 \log(1/\delta))$ for the problem. The paper explicitly identifies removal of the $\log(1/\epsilon)$ term as the main open problem for this rank method.

5. Optimal compression of approximate inner products and dimension reduction

- Authors: Noga Alon and Bo'az Klartag
- Venue/year: 58th IEEE Symposium on Foundations of Computer Science (FOCS 2017), pp. 639-650, 2017
- DOI/arXiv/URL: <https://doi.org/10.1109/FOCS.2017.65>; arXiv:1610.00239, <https://arxiv.org/abs/1610.00239>
- Relevance: This later work proves sharp results for approximate distance/inner-product sketches and lower bounds for general Euclidean dimension reduction in several regimes. It does not resolve the present problem because its lower-bound configurations and sketching/general-point-set conclusions do not prove an upper or lower bound matching the exact near-equilateral/simplex existence question.

6. The Johnson-Lindenstrauss Lemma Is Optimal for Linear Dimensionality Reduction

- Authors: Kasper Green Larsen and Jelani Nelson
- Venue/year: ICALP 2016, LIPIcs 55, Article 82, pp. 82:1-82:11, 2016

- DOI/arXiv/URL: <https://doi.org/10.4230/LIPICs.ICALP.2016.82>
- Relevance: Proves optimal $\Omega(\epsilon^{-2} \log n)$ lower bounds for linear maps on suitable point sets, and discusses why the simplex is not known to force this bound in the nonlinear/arbitrary-placement setting. Thus it is related evidence for JL optimality, but it does not imply the full answer to this problem.

6.11.3.3. What remains unresolved

The direct literature gives

$$c/(\delta^2 \log(1/\delta)) \leq S(\delta) \leq C/\delta^2.$$

I did not find a published or otherwise citable theorem proving the missing lower bound $S(\delta) \geq c/\delta^2$, nor a construction improving the upper bound below C/δ^2 for all n . Later sharp JL and sketching results concern general point sets, linear embeddings, additive sketches, or different spherical-code regimes, and their hypotheses/conclusions do not cover exactly the full near-equilateral/simplex problem stated here. The exact asymptotic order of $S(\delta)$ therefore remains unresolved.

6.11.4. Remaining open problems recorded by the database

6.11.4.1. Open problem 1

Is there a universal constant $c > 0$ such that, whenever $0 < \epsilon < 1/2$ and n is sufficiently large as a function of ϵ , every real $n \times n$ matrix $B = (b_{ij})$ with $b_{ii} = 1$ and $|b_{ij}| \leq \epsilon$ for $i \neq j$ satisfies $\text{rank}(B) \geq c\epsilon^{-2} \log n$?

Reference supplied by the database. Noga Alon, “Perturbed Identity Matrices Have High Rank: Proof and Applications,” *Combinatorics, Probability and Computing* 18 (2009), nos. 1–2, 3–15, Theorem 2.1 and §8 (“Concluding remarks”), main open problem (removal of the $\log(1/\epsilon)$ factor); Proposition 3.1 gives the application to almost-equilateral simplex vertices. DOI: 10.1017/S0963548307008917. <https://doi.org/10.1017/S0963548307008917>.

6.11.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

6.12. 131. Systems of points with Coulomb interactions [2018-geom-158]

Record: 2018-geom-158

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Systems of points with Coulomb interactions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=930>

6.12.1. Problem statement

For the Coulomb jellium crystallization problem in $\mathbb{R}^d$, let $C \subset \mathbb{R}^d$ be an infinite point configuration of density 1 in a uniform neutralizing background, and let $\mathbb{W}(C)$ denote the renormalized Coulomb energy per unit volume obtained from electric potentials H satisfying $-\Delta H = C_d(C - 1)$, after subtracting the divergent self-energy of the point charges.

Prove or disprove that there exists a dimension d_0 such that, for every $d \geq d_0$, no covolume-one lattice $\Lambda \subset \mathbb{R}^d$ minimizes $\mathbb{W}$ among all unit-density configurations.

6.12.2. Historical background, literature, and current progress

6.12.3. Verdict: partially resolved

6.12.3.1. Precise problem checked

The problem asks whether the Coulomb jellium renormalized energy problem has been resolved in high dimension: for unit-density infinite point configurations in a uniform neutralizing background in $\mathbb{R}^d$, with Coulomb renormalized energy $\mathbb{W}$, is it proved that there is a dimension d_0 such that for every $d \geq d_0$, no covolume-one lattice $\Lambda \subset \mathbb{R}^d$ is a global minimizer of $\mathbb{W}$ among all unit-density configurations?

6.12.3.2. Status

I did not find a theorem proving the high-dimensional assertion. In particular, I did not find a published or citable result giving a threshold d_0 , nor even a general theorem showing that all sufficiently high-dimensional Bravais lattices are beaten by non-lattice configurations for the Coulomb renormalized energy $\mathbb{W}$. The exact problem therefore remains unresolved.

The literature does contain substantial partial progress and related crystallization results, including proved lattice minimization in special dimensions. These do not imply the problem statement, because they concern fixed low/special dimensions or restricted classes rather than all sufficiently large d .

6.12.3.3. Relevant literature and progress

1. Title: “Systems of points with Coulomb interactions” Authors: Sylvia Serfaty Venue: Proceedings of the International Congress of Mathematicians 2018; arXiv preprint Year: 2018 arXiv/URL: arXiv:1712.04095, <https://arxiv.org/abs/1712.04095> Result/status contribution: This is the apparent source of the formulation. Section 4.3 states that determining $\min \mathbb{W}$ is open except for one-dimensional analogues; in dimension 2 the triangular lattice is known to minimize only among lattices, while its global optimality is conjectural. It also records that in $d \geq 3$ the minimization of $\mathbb{W}$ restricted to lattices was open except for known strict local minimizers in dimensions 4, 8, and 24. The text explicitly describes the high-dimensional non-lattice expectation as “folklore knowledge”, motivated by sphere packing. This source does not prove the high-dimensional assertion.
2. Title: “Crystallization for Coulomb and Riesz interactions as a consequence of the Cohn-Kumar conjecture” Authors: Mircea Petrache and Sylvia Serfaty Venue: Proceedings of the American Mathematical Society 148, 3047-3057 Year: 2020 DOI/arXiv/URL: DOI 10.1090/proc/15003; arXiv:1908.09714, <https://arxiv.org/abs/1908.09714> Result/status contribution: Theorem 2 proves that the Cohn-Kumar universal optimality conjecture for the relevant lattice implies minimization of the periodic renormalized energies and of $\mathbb{W}$ for Coulomb cases $d \geq 2$ and Riesz cases in the stated range. Corollary 3 applies this in dimensions 8 and 24, where the Cohn-Kumar conjecture is known, and shows that the E_8 lattice and the Leech lattice achieve the corresponding minima. This is genuine progress on Coulomb/Riesz jellium crystallization, but it proves lattice minimizers in dimensions 8 and 24 and does not prove that lattices fail to minimize in all sufficiently high dimensions.
3. Title: “Universal optimality of the E_8 and Leech lattices and interpolation formulas” Authors: Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko, and Maryna Viazovska Venue: Annals of Mathematics 196 Year: 2022 DOI/arXiv/URL: DOI 10.4007/annals.2022.196.3.1; arXiv:1902.05438, <https://arxiv.org/abs/1902.05438> Result/status contribution: This proves universal optimality of E_8 in dimension 8 and the Leech lattice in dimension 24 for completely monotone interaction potentials. Petrache-Serfaty use this result to obtain Coulomb/Riesz renormalized-energy crystallization in dimensions 8 and 24. It does not address arbitrary high dimensions or prove non-lattice optimality.
4. Title: “Coulomb and Riesz gases: The known and the unknown” Authors: Mathieu Lewin Venue: Journal of Mathematical Physics 63, 061101 Year: 2022 DOI/arXiv/URL: DOI 10.1063/5.0086835; arXiv:2202.09240, <https://arxiv.org/abs/2202.09240> Result/status contribution: This survey summarizes the current crystallization status. Theorem 36 states crystallization results for dimension 1 in specified ranges and for dimensions 8 and 24 for $s \in [d-2, \infty) \setminus \{d\}$, covering the Coulomb exponent $s = d-2$. The surrounding discussion emphasizes that many long-range crystallization questions remain open and that even the two-dimensional Cohn-Kumar conjecture is open. This supports that there is no known theorem resolving the stated high-dimensional non-lattice assertion.
5. Title: “Floating Wigner Crystal and Periodic Jellium Configurations” Author: Asbjorn Christian Drachmann Lauritsen Venue: Journal of Mathematical Physics 62, 063305 Year: 2021 DOI/arXiv/URL: DOI 10.1063/5.0049411; arXiv:2103.07975, <https://arxiv.org/abs/2103.07975> Result/status contribution: This paper develops periodic jellium configurations and related lower bounds, especially in dimension 2. It is relevant

to the renormalized-energy/jellium framework, but it does not prove high-dimensional non-lattice minimization.

6.12.3.4. What remains unresolved

The missing theorem is exactly the quantified high-dimensional statement: existence of d_0 such that, for every $d \geq d_0$, every covolume-one lattice in $\mathbb{R}^d$ fails to be a global minimizer of the Coulomb renormalized energy $\mathbb{W}$ among all unit-density configurations. Existing results prove special-dimensional lattice minimization, restricted lattice-class minimization, local optimality, or structural results for related periodic/jellium energies. None of these results gives the required high-dimensional non-lattice conclusion.

6.12.4. Remaining open problems recorded by the database

6.12.4.1. Open problem 1

Among all covolume-one lattices $L \subset \mathbb{R}^3$, does the body-centered-cubic lattice uniquely minimize the three-dimensional Coulomb jellium energy $\mathbb{W}(L)$ (equivalently, the analytically continued Epstein zeta energy $\zeta_L(1)$)?

Reference supplied by the database. Xavier Blanc and Mathieu Lewin, “The Crystallization Conjecture: A Review,” *EMS Surveys in Mathematical Sciences* **2** (2015), no. 2, 255–306, §2.5–§2.6 and Figure 7 (the explicit still-open prediction that BCC minimizes the three-dimensional Epstein zeta energy for $0 < s < 3/2$, which includes the Coulomb exponent $s = 1$), DOI: [10.4171/EMSS/13](https://doi.org/10.4171/EMSS/13), arXiv: [1504.01153](https://arxiv.org/abs/1504.01153). The equivalence with lattice jellium energy and the continuing absence of a three-dimensional global lattice-optimality theorem are recorded in Laurent Bétermin, “Three-Dimensional Lattice Ground States for Riesz and Lennard-Jones-Type Energies,” *Studies in Applied Mathematics* **150** (2023), no. 2, 354–415, §1.2 and §2, DOI: [10.1111/sapm.12533](https://doi.org/10.1111/sapm.12533).

6.12.5. Primary-source pointer

The originating ICM record is *Systems of points with Coulomb interactions* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=930>.

6.13. 132. the upper bound side of the story [2022-geom-175]

Record: 2022-geom-175

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: the upper bound side of the story

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=471>

6.13.1. Problem statement

Prove that for every integer $d \geq 4$, if $s(d, n)$ denotes the number of simplicial $(d - 1)$ -spheres with vertex set $[n] = \{1, \dots, n\}$, counted as labeled simplicial complexes up to equality on this fixed vertex set, and if $sn(d, n)$ denotes the number of those simplicial $(d - 1)$ -spheres that are $\lfloor d/2 \rfloor$ -neighborly, meaning that every subset of at most $\lfloor d/2 \rfloor$ vertices is a face, then

$$\lim_{n \rightarrow \infty} \frac{\log sn(d, n)}{\log s(d, n)} = 1.$$

6.13.2. Historical background, literature, and current progress

6.13.3. Verdict: partially resolved

6.13.3.1. Precise problem

For each fixed integer $d \geq 4$, let $s(d, n)$ be the number of labeled simplicial $(d-1)$ -spheres on the fixed vertex set $[n]$, and let $sn(d, n)$ be the number of those spheres that are $\lfloor d/2 \rfloor$ -neighborly. The question is whether

$$\lim_{n \rightarrow \infty} \frac{\log sn(d, n)}{\log s(d, n)} = 1$$

for every $d \geq 4$. Equivalently, it asks whether neighborly simplicial spheres account for the same logarithmic asymptotic growth as all simplicial spheres, in every dimension covered by the statement.

6.13.3.2. Current status

No citable result found proves this full limit for all $d \geq 4$. The literature gives substantial lower bounds for $sn(d, n)$, but the best known bounds do not imply the required log-ratio statement. Recent surveys explicitly still list the conjecture as out of reach and pose weaker lower-bound problems.

6.13.3.3. Key literature and progress**1. Many triangulated spheres**

Authors: Gil Kalai

Venue: *Discrete & Computational Geometry* 3, 1-14

Year: 1988

DOI/URL: <https://doi.org/10.1007/BF02187893>

Role: Original source for the asymptotic enumeration framework for simplicial spheres and for the speculation that there are very many neighborly spheres. Kalai's Section 6.3 phrases the log-ratio question for fixed even d ; later sources state the now-standard all- $d \geq 4$ formulation.

2. Many Neighborly Polytopes and Oriented Matroids

Author: Arnau Padrol

Venue: *Discrete & Computational Geometry* 50, 865-902

Year: 2013

DOI/arXiv/URL: <https://doi.org/10.1007/s00454-013-9544-7>,
arXiv:1202.2810, <https://arxiv.org/abs/1202.2810>

Role: Gives large families of neighborly polytopes and non-realizable neighborly oriented matroids. As summarized in Novik-Zheng and Novik's ICM survey, this implies only $sn(d, n) \geq 2^{\Omega(n \log n)}$ for the neighborly sphere count, for $d \geq 4$. This is partial progress, not the stated limit.

3. Many triangulated odd-dimensional spheres

Authors: Eran Nevo, Francisco Santos, Stedman Wilson

Venue: *Mathematische Annalen* 364, 737-762

Year: 2016

DOI/URL: <https://doi.org/10.1007/s00208-015-1232-x>

Role: Improves lower bounds for the total number $s(d, n)$ in odd sphere dimensions. Together with Kalai, Pfeifle-Ziegler, and Stanley-type upper bounds, current surveys summarize the best general bounds as $2^{\Omega(n^{\lfloor d/2 \rfloor})} \leq s(d, n) \leq 2^{O(n^{\lfloor d/2 \rfloor} \log n)}$ for fixed $d \geq 4$. These denominator bounds remain too coarse to settle the log-ratio conjecture.

4. Many neighborly spheres

Authors: Isabella Novik and Hailun Zheng

Venue: *Mathematische Annalen* 388, 969-984

Year: 2024 (published online 2022)

DOI/arXiv/URL: <https://doi.org/10.1007/s00208-022-02538-x>,
arXiv:2104.04476, <https://arxiv.org/abs/2104.04476>

Result: The abstract and Theorem 1.2 state that for every $d \geq 5$, there are at least $2^{\Omega(n^{\lfloor (d-1)/2 \rfloor})} \lfloor d/2 \rfloor$ -neighborly simplicial $(d-1)$ -spheres with n vertices. The paper explicitly says this improves Padrol's bound but is still far from resolving Kalai's conjecture. It also does not cover the full $d=4$ case in the problem statement.

5. Face numbers: the upper bound side of the story

Author: Isabella Novik

Venue: *Proceedings of the International Congress of Mathematicians 2022*, EMS Press, pp. 4622-4644

Year: 2023

DOI/URL: <https://doi.org/10.4171/ICM2022/43>, <https://ems.press/books/standalone/278/5571>

Role: Survey source for the exact all- $d \geq 4$ formulation as Conjecture 7.1. It states the current bounds on $s(d, n)$, summarizes Padrol and Novik-Zheng, and says that Conjecture 7.1 is likely out of reach; even proving $sn(d, n) \geq 2^{\Omega(n^{\lfloor d/2 \rfloor})}$ is presented as a more feasible open goal.

6. Neighborly spheres and transversal numbers

Authors: Isabella Novik and Hailun Zheng

Venue: *Open Problems in Algebraic Combinatorics*, Proceedings of Symposia in Pure Mathematics 110, American Mathematical Society, Providence, RI, pp. 113-128

Year: 2024

arXiv/URL: arXiv:2208.02071, <https://arxiv.org/abs/2208.02071>

Role: Open-problems survey. It restates Kalai's conjecture as Conjecture 4.1, identifies Novik-Zheng's $d \geq 5$ lower bound as the current best lower bound on $sn(d, n)$, and then poses the weaker Problem 4.2 asking whether $2^{\Omega(n^{\lfloor d/2 \rfloor})}$ neighborly spheres exist.

6.13.3.4. Why this is not fully resolved

The desired conclusion is a statement about the exact relative logarithmic growth of $sn(d, n)$ and $s(d, n)$ for every $d \geq 4$. The best lower bounds for $sn(d, n)$ are:

- $2^{\Omega(n \log n)}$ for all $d \geq 4$ from Padrol-type constructions;
- $2^{\Omega(n^{\lfloor (d-1)/2 \rfloor})}$ for all $d \geq 5$ from Novik-Zheng.

These do not imply

$$\log sn(d, n) / \log s(d, n) \rightarrow 1.$$

For even d , Novik-Zheng's exponent is one power of n below $\lfloor d/2 \rfloor$. For $d=4$, their theorem does not apply. For odd $d \geq 5$, the exponent matches $\lfloor d/2 \rfloor$, but the known upper bound for $s(d, n)$ still has an additional $\log n$ factor and no theorem identifies the logarithmic growth of $sn(d, n)$ with that of all spheres. Thus the literature gives meaningful partial progress, but not a full resolution of the problem statement.

Related recent work on centrally symmetric neighborly spheres, transversal numbers, missing faces, and small or special families of neighborly spheres concerns different problems or imposes extra structure; it does not prove the full all- $d \geq 4$ log-ratio limit for labeled simplicial spheres.

6.13.4. Remaining open problems recorded by the database

6.13.4.1. Open problem 1

For every fixed even integer $d \geq 4$, are there $2^{\Omega(n^{d/2})}$ distinct simplicial $(d-1)$ -spheres on n vertices that are $d/2$ -neighborly?

Reference supplied by the database. Isabella Novik and Hailun Zheng, "Neighborly spheres and transversal numbers," in *Open Problems in Algebraic Combinatorics*, Proceedings of Symposia in Pure Mathematics, vol. 110, American Mathematical Society, Providence, RI, 2024, pp. 113–128, Problem 4.2. [arXiv:2208.02071](https://arxiv.org/abs/2208.02071).

6.13.5. Primary-source pointer

The originating ICM record is *the upper bound side of the story* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=471>.

7. Applications of geometry / global analysis

These problems arise where geometric structures organize questions from mathematical physics, representation theory, topology, and nonlinear analysis. They include conjectural correspondences as well as concrete vanishing, rigidity, and enumeration statements. The research record therefore mixes theorem-driven progress with the gradual construction of a broader framework.

Coverage: 8 website records; **source years:** 1986–2018; **separately listed descendant open problems:** 15.

7.1. 133. E., Physics and geometry [1986-geom-38]

Record: 1986-geom-38

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: E., Physics and geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=369>

7.1.1. Problem statement

Determine the proper mathematical and logical setting underlying string theory, understood as a proposed framework for reconciling gravitation with quantum mechanics and known perturbatively through constructions involving Riemann surfaces, modular forms, and representations of infinite-dimensional Lie algebras, and formulate the fundamental generalization of the usual geometric concepts that in general relativity are expressed through Riemannian or pseudo-Riemannian geometry, metrics, connections, and curvature.

7.1.2. Historical background, literature, and current progress

7.1.3. Verdict: partially resolved

7.1.3.1. Precise problem identified

The problem is not a sharply formalized theorem. It is a broad programmatic question from Edward Witten's 1986 ICM address: find the proper mathematical and logical setting for string

theory as a framework reconciling gravitation and quantum mechanics, in which the usual Riemannian/pseudo-Riemannian language of spacetime, metrics, connections, and curvature is replaced or enlarged by a genuinely “stringy” geometry. The exact question to check is therefore whether the literature now supplies a complete mathematical framework for full string/M-theory, not only for special perturbative sectors or particular backgrounds.

7.1.3.2. Status assessment

The literature does not fully resolve this problem. There is extensive partial progress: perturbative string theory, two-dimensional conformal field theory, rational CFT, topological strings, mirror symmetry, D-branes, noncommutative geometry, derived/categorical geometry, string field theory, AdS/CFT, and matrix models all supply pieces of the new geometry Witten anticipated. However, recent surveys still treat the full definition of string theory/M-theory, especially a background-independent and generally non-perturbative formulation, as unknown. The best-developed frameworks are tied to sectors, backgrounds, extra hypotheses, or toy/simplified theories, so none covers the full statement.

7.1.3.3. Original source

- Title: “Physics and geometry”
- Author: Edward Witten
- Venue/year: Proceedings of the International Congress of Mathematicians, Berkeley 1986, Vol. 1, American Mathematical Society, 1987, pp. 267-303
- DOI/arXiv/URL: no DOI found; CERN scan: <https://cds.cern.ch/record/181783/files/er-000093203.pdf> ; IMU proceedings scan: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf>
- Relevance: This is the source of the problem statement. Witten describes string theory as involving Riemann surfaces, modular forms, and infinite-dimensional Lie algebra representation theory, and says the proper logical setting is not known but should involve a fundamental generalization of usual geometry.

7.1.3.4. Partial progress and why it is not a full resolution

Mathematical foundations of perturbative string theory and QFT

- Title: “Survey of mathematical foundations of QFT and perturbative string theory”
- Authors: Hisham Sati and Urs Schreiber
- Venue/year: arXiv preprint, 2011; introduction to the AMS volume below
- DOI/arXiv/URL: arXiv:1109.0955, <https://arxiv.org/abs/1109.0955> ; DOI: <https://doi.org/10.48550/arXiv.1109.0955>
- Progress: Surveys progress on mathematical formulations of quantum field theory and perturbative string theory.
- Limitation: The scope is explicitly perturbative. It does not define full non-perturbative, background-independent string/M-theory.
- Title: “Mathematical Foundations of Quantum Field Theory and Perturbative String Theory”

- Editors: Hisham Sati and Urs Schreiber
- Venue/year: Proceedings of Symposia in Pure Mathematics, Vol. 83, American Mathematical Society, 2011, 354 pp.
- DOI/arXiv/URL: AMS page <https://bookstore.ams.org/pspum-83/>
- Progress: Collects rigorous work on higher categories, quantization, defects, orientifolds, rational CFT, conformal nets, supersymmetric field theories, and generalized cohomology.
- Limitation: The title and contents are about QFT and perturbative string theory. This is foundational progress but not a complete setting for full string theory.

Current broad surveys

- Title: “A Panorama Of Physical Mathematics c. 2022”
- Authors: Ibrahima Bah, Daniel S. Freed, Gregory W. Moore, Nikita Nekrasov, Shlomo S. Razamat, Sakura Schafer-Nameki
- Venue/year: arXiv preprint, 2022; version 3 in 2024
- DOI/arXiv/URL: arXiv:2211.04467, <https://arxiv.org/abs/2211.04467> ; DOI: <https://doi.org/10.48550/arXiv.2211.04467>
- Progress: Gives an up-to-date survey of physical mathematics, including string theory/M-theory, topological strings, perturbation theory, anomalies, noncommutative spacetime, and holography.
- Limitation: Section 3.1 asks for the definition of string theory and M-theory and answers that it is not known. It explicitly treats matrix theory and AdS/CFT as profound but partial, background-dependent answers, not satisfactory definitions of full string/M-theory.
- Title: “Les Houches lectures on non-perturbative topological strings”
- Author: Marcos Marino
- Venue/year: arXiv preprint, 2024/2025 lecture notes for the Les Houches School on Quantum Geometry
- DOI/arXiv/URL: arXiv:2411.16211, <https://arxiv.org/abs/2411.16211> ; DOI: <https://doi.org/10.48550/arXiv.2411.16211>
- Progress: Reviews non-perturbative definitions for topological strings in important toric Calabi-Yau settings via spectral theory.
- Limitation: This is for topological string theory, a simplified subsector. The introduction emphasizes that non-perturbative definitions of string theories do not exist in general.
- Title: “String Theory and Math: Why This Marriage May Last. Mathematics and dualities of quantum physics”
- Author: Mina Aganagic
- Venue/year: Bulletin of the American Mathematical Society 53 (2016), no. 1, 93-115
- DOI/arXiv/URL: arXiv:1508.06642, <https://arxiv.org/abs/1508.06642> ; DOI: <https://doi.org/10.1090/bull/1517>
- Progress: Surveys how string dualities have generated major mathematical structures and conjectures.

- Limitation: It documents rich mathematical output from string theory, not a theorem giving the full mathematical/logical setting of string theory.

String field theory

- Title: “Non-commutative geometry and string field theory”
- Author: Edward Witten
- Venue/year: Nuclear Physics B 268 (1986), no. 2, 253-294
- DOI/arXiv/URL: DOI: [https://doi.org/10.1016/0550-3213\(86\)90155-0](https://doi.org/10.1016/0550-3213(86)90155-0)
- Progress: Early string field theory paper showing algebraic/noncommutative structures in string interactions.
- Limitation: It initiated an important framework but does not give a complete definition of all string/M-theory.
- Title: “String Field Theory: A Review”
- Authors: Ashoke Sen and Barton Zwiebach
- Venue/year: arXiv preprint, 2024; expanded review for the Handbook of Quantum Gravity
- DOI/arXiv/URL: arXiv:2405.19421, <https://arxiv.org/abs/2405.19421> ; DOI: <https://doi.org/10.48550/arXiv.2405.19421>
- Progress: Reviews consistent gauge-invariant string field theories for bosonic open/closed strings, open superstrings, heterotic strings, and type II strings, using A-infinity/L-infinity structures, moduli of Riemann surfaces, CFT/BCFT, and BV quantization.
- Limitation: This is major progress for string field theory, perturbation theory, and certain non-perturbative phenomena such as tachyon condensation, but it is not a generally accepted complete mathematical/logical foundation for full background-independent string/M-theory.

Mirror symmetry and stringy geometry

- Title: “A pair of Calabi-Yau manifolds as an exactly soluble superconformal theory”
- Authors: Philip Candelas, Xenia C. de la Ossa, Paul S. Green, Linda Parkes
- Venue/year: Nuclear Physics B 359 (1991), 21-74
- DOI/arXiv/URL: DOI: [https://doi.org/10.1016/0550-3213\(91\)90292-6](https://doi.org/10.1016/0550-3213(91)90292-6)
- Progress: Found landmark mirror-symmetry computations linking Calabi-Yau geometry, CFT, and enumerative predictions.
- Limitation: Resolves important special enumerative and compactification questions, not the full string-theoretic foundation.
- Title: “Homological Algebra of Mirror Symmetry”
- Author: Maxim Kontsevich

- Venue/year: Proceedings of the International Congress of Mathematicians, Zurich 1994, Birkhauser, 1995
- DOI/arXiv/URL: arXiv:alg-geom/9411018, <https://arxiv.org/abs/alg-geom/9411018>
- Progress: Formulates homological mirror symmetry via equivalences of Fukaya categories and derived categories, a major categorical form of stringy geometry.
- Limitation: It is a mirror-symmetry program/conjectural framework, proven in many special cases, not a complete foundation for all string theory.
- Title: “Mirror Symmetry is T-Duality”
- Authors: Andrew Strominger, Shing-Tung Yau, Eric Zaslow
- Venue/year: Nuclear Physics B 479 (1996), 243-259
- DOI/arXiv/URL: arXiv:hep-th/9606040, <https://arxiv.org/abs/hep-th/9606040> ; DOI: [https://doi.org/10.1016/0550-3213\(96\)00434-8](https://doi.org/10.1016/0550-3213(96)00434-8)
- Progress: Proposes the SYZ geometric mechanism for mirror symmetry via dual special Lagrangian torus fibrations.
- Limitation: It explains a central duality in favorable Calabi-Yau settings, but not the full mathematical setting of string theory.
- Title: “Open Strings and Extended Mirror Symmetry”
- Author: Johannes Walcher
- Venue/year: Fields Institute Communications 54, “Modular Forms and String Duality”, American Mathematical Society, 2008, pp. 279-297
- DOI/arXiv/URL: AMS page <https://pubs.ams.org/view?ProductCode=FIC%2F54> ; PDF mirror <https://inspirehep.net/files/524d5ed62a8b4c90a2ac42a3709b2c2b>
- Progress: Reviews open-string extensions of mirror symmetry and D-brane geometry.
- Limitation: It explicitly says there is no complete definition of string theory and no precise idea of the final generalized geometry, so the work is partial progress.

Noncommutative geometry and matrix theory

- Title: “Noncommutative Geometry and Matrix Theory: Compactification on Tori”
- Authors: Alain Connes, Michael R. Douglas, Albert Schwarz
- Venue/year: Journal of High Energy Physics 1998, no. 02, 003
- DOI/arXiv/URL: arXiv:hep-th/9711162, <https://arxiv.org/abs/hep-th/9711162> ; DOI: <https://doi.org/10.1088/1126-6708/1998/02/003>
- Progress: Applies noncommutative geometry to Matrix-theory compactifications and noncommutative tori.
- Limitation: Covers toroidal/noncommutative backgrounds in matrix theory, not a full universal geometry of string theory.
- Title: “String Theory and Noncommutative Geometry”
- Authors: Nathan Seiberg and Edward Witten

- Venue/year: Journal of High Energy Physics 1999, no. 09, 032
- DOI/arXiv/URL: arXiv:hep-th/9908142, <https://arxiv.org/abs/hep-th/9908142> ; DOI: <https://doi.org/10.1088/1126-6708/1999/09/032>
- Progress: Shows how noncommutative gauge theory arises from open strings in a background B-field and gives the Seiberg-Witten map.
- Limitation: It identifies a precise stringy noncommutative regime, not the full mathematical/logical setting of string theory.

Conformal field theory and categorical geometry

- Title: “Conformal field theory and a new geometry”
- Author: Liang Kong
- Venue/year: Proceedings of Symposia in Pure Mathematics 83, American Mathematical Society, 2011, pp. 199-242
- DOI/arXiv/URL: arXiv:1107.3649, <https://arxiv.org/abs/1107.3649> ; DOI: <https://doi.org/10.1090/pspum/083/2742430>
- Progress: Reviews open-closed rational conformal field theory via vertex operator algebras and proposes a geometry based on CFTs and D-branes.
- Limitation: The results concern rational/open-closed CFT and a proposed new geometry. They do not supply a complete formulation of arbitrary string backgrounds or non-perturbative string/M-theory.

7.1.3.5. Conclusion

the full question remains unresolved, but it is not “fully open” in the sense of having no progress. The correct status is partially resolved: many important sectors and mathematical structures have been developed and in some cases rigorously proved, but no citable theorem or framework fully covers the entire claimed setting for string theory in all cases.

7.1.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

7.1.5. Primary-source pointer

The originating ICM record is *E., Physics and geometry* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=369>.

7.2. 134. Mirror symmetry and moduli spaces of superconformal field theories [1994-geom-58]

Record: 1994-geom-58

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Mirror symmetry and moduli spaces of superconformal field theories

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=600>

7.2.1. Problem statement

Determine whether the following assertion holds: Let X be a compact manifold, let g be a Ricci-flat Kähler metric on X with respect to a fixed complex structure $\mathcal{J}$, let ω_g be its Kähler form, and let $B \in H^2(X, \mathbb{R}/\mathbb{Z})$ be the B -field in the corresponding $N = (2, 2)$ supersymmetric nonlinear σ -model. If the Kähler class $[\omega_g]$ lies sufficiently deep in the Kähler cone of the complex manifold $X_{\mathcal{J}}$, then the renormalization-flow trajectory starting from the sigma-model data (g, B) determines a unique conformally invariant theory, equivalently a unique $N = (2, 2)$ superconformal field theory fixed point in the same universality class.

7.2.2. Historical background, literature, and current progress

7.2.3. Status: partially resolved

7.2.3.1. Precise problem

The problem asks for a nonperturbative existence and uniqueness statement for Calabi-Yau nonlinear sigma models: for every compact manifold X , fixed complex structure $\mathcal{J}$, Ricci-flat Kahler metric g , and B -field $B \in H^2(X, \mathbb{R}/\mathbb{Z})$, with $[\omega_g]$ sufficiently deep in the Kahler cone, the renormalization-group trajectory of the $N = (2, 2)$ nonlinear sigma model should determine a unique conformally invariant theory, equivalently a unique $N = (2, 2)$ SCFT fixed point in the same universality class.

This is stronger than Yau's theorem on Ricci-flat Kahler metrics and stronger than perturbative beta-function cancellation. It requires a mathematically or physically citable result constructing the full IR SCFT and proving uniqueness of the RG-flow endpoint for the whole class of large-radius Ricci-flat Kahler sigma-model data.

7.2.3.2. Literature checked and progress

1. David R. Morrison, "Mirror symmetry and moduli spaces of superconformal field theories."

- Venue: Proceedings of the International Congress of Mathematicians, Zurich 1994, vol. 2, Birkhauser, 1995, pp. 1304-1314.
 - arXiv: [alg-geom/9411019](https://arxiv.org/abs/alg-geom/9411019).
 - DOI/URL: <https://doi.org/10.48550/arXiv.alg-geom/9411019> and https://doi.org/10.1007/978-3-0348-9078-6_125.
 - Relevance: This is the original source of the problem statement. Morrison explicitly formulates the assertion as an expectation: Ricci-flat Kahler metrics deep in the Kahler cone are expected to determine a unique conformally invariant theory in the same universality class. The article also warns that the one-loop semiclassical sigma-model moduli space is only an approximation to the true SCFT moduli space. It is not a proof of the full claim.
2. **Luis Alvarez-Gaume, Sidney R. Coleman, and Paul H. Ginsparg, “Finiteness of Ricci Flat $N = 2$ Supersymmetric σ -Models.”**
- Venue: Communications in Mathematical Physics 103 (1986), 423.
 - URL: <https://projecteuclid.org/journals/communication-s-in-mathematical-physics/volume-103/issue-3/Finiteness-of-Ricci-flat-N2-supersymmetric-sigma-models/cmp/1104114791.pdf>.
 - Relevance: Early perturbative finiteness result for Ricci-flat $N = 2$ supersymmetric sigma models. Later four-loop computations showed that naive Ricci-flat metrics are not finite without scheme/counterterm corrections, so this does not by itself prove the full conformal fixed-point statement.
3. **Marcus T. Grisaru, A. E. M. van de Ven, and Daniela Zanon, “Four Loop beta Function for the $N = 1$ and $N = 2$ Supersymmetric Nonlinear Sigma Model in Two-Dimensions.”**
- Venue: Physics Letters B 173 (1986), 423-428.
 - DOI: [https://doi.org/10.1016/0370-2693\(86\)90408-9](https://doi.org/10.1016/0370-2693(86)90408-9).
 - Relevance: Computes higher-loop beta-function contributions and shows why naive Ricci-flatness is not enough as an uncorrected all-order statement. This is important negative/diagnostic progress, not a resolution.
4. **Marcus T. Grisaru, A. E. M. van de Ven, and Daniela Zanon, “Two-Dimensional Supersymmetric Sigma Models on Ricci Flat Kahler Manifolds Are Not Finite.”**
- Venue: Nuclear Physics B 277 (1986), 388-408.
 - DOI: [https://doi.org/10.1016/0550-3213\(86\)90448-7](https://doi.org/10.1016/0550-3213(86)90448-7).
 - Relevance: Shows a nonzero four-loop obstruction for uncorrected Ricci-flat Kahler sigma models. This confirms that the full problem cannot be reduced to “Ricci-flat implies conformal” in a naive perturbative scheme.
5. **Dennis Nemeschansky and Ashoke Sen, “Conformal Invariance of Supersymmetric Sigma Models on Calabi-Yau Manifolds.”**
- Venue: Physics Letters B 178 (1986), 365-369.
 - DOI: [https://doi.org/10.1016/0370-2693\(86\)91394-8](https://doi.org/10.1016/0370-2693(86)91394-8).
 - Relevance: This is the main positive perturbative result: by using finite local counterterms or equivalently corrections to the Ricci-flat metric order by order, the sigma-model beta functions can be made to vanish to all orders in perturbation theory. This is substantial progress toward conformal invariance, but it is still an order-by-order perturbative statement. It does not prove convergence, construct the full nonperturbative SCFT, or prove uniqueness of the RG-flow endpoint for all large-radius data.

6. Ian T. Jardine and Callum Quigley, “Conformal Invariance of (0,2) Sigma Models on Calabi-Yau Manifolds.”

- Venue: Journal of High Energy Physics 03 (2018), 090.
- arXiv: 1801.04336.
- DOI: [https://doi.org/10.1007/JHEP03\(2018\)090](https://doi.org/10.1007/JHEP03(2018)090).
- Relevance: The abstract and introduction summarize the Nemeschansky-Sen result as showing that the Ricci-flat metric can be corrected order by order in perturbation theory to produce a conformally invariant (2,2) nonlinear sigma model, and extend analogous methods to (0,2) models under additional assumptions. This confirms the scope of the known result: all-order perturbative beta-function cancellation, not a full proof of RG-flow existence and uniqueness.

7. Philip Candelas, Xenia C. de la Ossa, Paul S. Green, and Linda Parkes, “A pair of Calabi-Yau manifolds as an exactly soluble superconformal theory.”

- Venue: Nuclear Physics B 359 (1991), 21-74.
- DOI: [https://doi.org/10.1016/0550-3213\(91\)90292-6](https://doi.org/10.1016/0550-3213(91)90292-6).
- Relevance: Provides exact and highly influential SCFT/mirror-symmetry evidence in special Calabi-Yau examples, especially the quintic mirror story and massless-sector computations. It does not cover arbitrary compact Ricci-flat Kahler targets or prove uniqueness of RG-flow endpoints from the stated sigma-model data.

8. Werner Nahm and Katrin Wendland, “A hiker’s guide to K3: Aspects of $N = (4,4)$ superconformal field theory with central charge $c = 6$.”

- Venue: Communications in Mathematical Physics 216 (2001), 85-138.
- DOI: <https://doi.org/10.1007/PL00005548>.
- Relevance: Gives a detailed description of the moduli space of $N = (4,4)$, $c = 6$ SCFTs related to K3 surfaces and locates many known K3 models. This is special to the hyperkahler/K3 setting and does not imply the full $N = (2,2)$ Calabi-Yau statement for all compact targets.

9. Yi-Zhi Huang, “Some open problems in mathematical two-dimensional conformal field theory.”

- Venue: Lie Algebras, Vertex Operator Algebras, and Related Topics, Contemporary Mathematics 695, American Mathematical Society, 2017, pp. 123-138.
- arXiv: 1606.04493.
- DOI: <https://doi.org/10.1090/conm/695/13999>.
- Relevance: Describes open problems in rigorous mathematical construction of two-dimensional CFTs. The discussion reinforces that mathematically constructing interacting nonlinear sigma-model CFTs remains difficult; it does not provide a theorem resolving the claim.

7.2.3.3. Why the full problem is not resolved

The strongest directly relevant results establish perturbative, order-by-order vanishing of beta functions after correcting the sigma-model couplings. They do not prove convergence of the perturbation series, existence of the full nonperturbative RG flow, existence of an actual SCFT fixed point for every large-radius Ricci-flat Kahler input, or uniqueness of that fixed point in the same universality class.

Known exact SCFTs, toroidal/Narain models, Gepner models, quintic mirror computations, and K3 moduli-space analyses are important special cases or evidence. They have extra structure or cover restricted families and therefore do not imply the full statement for arbitrary compact Ricci-flat Kahler manifolds with B -field deep in the Kahler cone.

7.2.3.4. Verdict

The problem is **partially resolved**. The literature gives substantial perturbative and special-case evidence, but I found no published or otherwise citable theorem proving the full existence-and-uniqueness statement in all cases covered by the problem statement.

7.2.4. Remaining open problems recorded by the database

7.2.4.1. Open problem 1

Construct a rigorous correspondence under which the Fermat quintic Calabi–Yau three-fold is assigned the ($N=2$) Gepner superconformal field theory built from five level-(3) ($N=2$) minimal models (with the usual projection/orbifold), and prove that this theory is the quantum nonlinear sigma model of the Fermat quintic at the Gepner point.

Reference supplied by the database. Yi-Zhi Huang, “Some Open Problems in Mathematical Two-Dimensional Conformal Field Theory,” in *Lie Algebras, Vertex Operator Algebras, and Related Topics*, Contemporary Mathematics **695**, American Mathematical Society, 2017, pp. 123–138, §4, Problem (CY-funct) (including the Fermat-quintic/Gepner-model requirement), DOI: 10.1090/conm/695/13999, arXiv:1606.04493, <https://arxiv.org/abs/1606.04493>. The special model was originally proposed in Doron Gepner, “Exactly Solvable String Compactifications on Manifolds of $SU(N)$ Holonomy,” *Physics Letters B* **199** (1987), no. 3, 380–388, DOI: 10.1016/0370-2693(87)90938-5.

7.2.5. Primary-source pointer

The originating ICM record is *Mirror symmetry and moduli spaces of superconformal field theories* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=600>.

7.3. 135. String Theory and Duality [1998-ag-82]

Record: 1998-ag-82

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: String Theory and Duality

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=231>

7.3.1. Problem statement

Let X and Y be Calabi-Yau threefolds used as the compact factors in ten-dimensional type II string compactifications to four-dimensional Minkowski space $M \cong \mathbb{R}^{3,1}$. For the type IIA compactification on X , suppose the local string-theoretic data consist of a Ricci-flat metric on X , a B -field in $H^2(X, \mathbb{R}/\mathbb{Z})$, a Ramond-Ramond field in $H^{\text{odd}}(X, \mathbb{R}/\mathbb{Z})$, and a dilaton-axion $\Phi \in \mathbb{C}$. For the type IIB compactification on Y , suppose the corresponding data consist of a Ricci-flat metric on Y , a B -field in $H^2(Y, \mathbb{R}/\mathbb{Z})$, a Ramond-Ramond field in $H^{\text{even}}(Y, \mathbb{R}/\mathbb{Z})$, and a dilaton-axion $\Phi \in \mathbb{C}$. Determine whether (X, Y) is a mirror pair, that is, whether these two compactified string theories are physically equivalent in the sense that the physics they determine in the uncompactified space M is isomorphic.

7.3.2. Historical background, literature, and current progress

7.3.3. Status check: mirror symmetry

Verdict: partially resolved.

7.3.3.1. Precise problem

The problem is not asking for a single enumerative mirror theorem or for Hodge-number symmetry. It uses the strongest physics formulation of Calabi-Yau threefold mirror symmetry: for Calabi-Yau threefolds X and Y , equipped with the type-II compactification data listed in the problem statement (Ricci-flat metric, B -field, Ramond-Ramond fields, and dilaton-axion), X and Y form a mirror pair exactly when type IIA string theory compactified on X and type IIB string theory compactified on Y give isomorphic four-dimensional physics on $M \cong \mathbb{R}^{3,1}$. Thus a full resolution would require a mathematically precise and proven equivalence of the full compactified string theories, including the quantum and nonperturbative data, for all cases covered by the statement.

7.3.3.2. Original/source formulation

- Title: "String Theory and Duality" Authors: Paul S. Aspinwall Venue/year: Proceedings of the International Congress of Mathematicians, Vol. II, Berlin 1998, pp. 229-238; arXiv

preprint, 1998 DOI/arXiv/URL: arXiv:math/9809004, <https://arxiv.org/abs/math/9809004> Relevance: This is the source matching the problem statement. It defines a mirror pair of Calabi-Yau threefolds as one for which type IIA on X is physically equivalent to type IIB on Y , and explicitly notes that the definition is mathematically unsatisfying because it depends on physics. It also treats the moduli-space identifications and curve-counting consequences as approximate or sectorial rather than a rigorous theorem proving full physical equivalence.

- Title: “Mirror Symmetry and the Type II String” Authors: David R. Morrison Venue/year: Nuclear Physics B Proceedings Supplements 46, 146-155, 1996 DOI/arXiv/URL: arXiv:hep-th/9512016, <https://arxiv.org/abs/hep-th/9512016> Relevance: Morrison formulates the same full-strength expectation: for a mirror pair, mirror symmetry should extend to an isomorphism between type IIA compactified on X and type IIB compactified on Y , with nonperturbative effects included. The paper studies implications for semiclassical moduli spaces; it is not a proof of the full equivalence.
- Title: “The geometry underlying mirror symmetry” Authors: David R. Morrison Venue/year: In New Trends in Algebraic Geometry, Cambridge University Press, 1999, pp. 283-310 DOI/arXiv/URL: DOI 10.1017/CBO9780511721540.011, <https://doi.org/10.1017/CBO9780511721540.011> Relevance: This survey explicitly says that precise mathematical formulations of string-theoretic mirror symmetry had been elusive, because the traditional formulation predicts equivalence of physical theories without specifying a geometric relationship. It supports treating the full physical-equivalence statement as unresolved, not as a theorem.

7.3.3.3. Partial progress and why it does not fully resolve the problem statement

- Title: “Duality in Calabi-Yau Moduli Space” Authors: Brian R. Greene and M. Ronen Plesser Venue/year: Nuclear Physics B 338, 15-37, 1990 DOI/arXiv/URL: DOI 10.1016/0550-3213(90)90622-K Progress: Gives foundational examples of mirror duality in Calabi-Yau/Gepner-orbifold moduli spaces and helped identify topologically distinct Calabi-Yau geometries yielding the same conformal field theory data. Limitation: This is a construction and analysis of important examples of world-sheet conformal field theories, not a theorem covering all Calabi-Yau threefold compactifications with the full type-II data in the problem statement.
- Title: “A Pair of Calabi-Yau Manifolds as an Exactly Soluble Superconformal Theory” Authors: Philip Candelas, Xenia C. de la Ossa, Paul S. Green, Linda Parkes Venue/year: Nuclear Physics B 359, 21-74, 1991 DOI/arXiv/URL: DOI 10.1016/0550-3213(91)90292-6 Progress: Produced the celebrated quintic mirror example and the mirror-symmetry predictions for rational curves, solving a key massless/topological sector of a particular mirror pair. Limitation: It concerns a specific family and the superconformal/topological data used for enumerative predictions. It does not prove full type IIA/type IIB equivalence for arbitrary Calabi-Yau threefolds, nor the RR/dilaton/nonperturbative parts of the stated compactification data.
- Title: “Dual Polyhedra and Mirror Symmetry for Calabi-Yau Hypersurfaces in Toric Varieties” Authors: Victor V. Batyrev Venue/year: Journal of Algebraic Geometry 3, no. 3, 493-535, 1994 DOI/arXiv/URL: arXiv:alg-geom/9310003, <https://arxiv.org/abs/alg-geom/9310003> Progress: Gives a broad mathematical construction of mirror candidate families for Calabi-Yau hypersurfaces in toric varieties via reflexive polytopes. Limitation: The result constructs mirror candidates and verifies Hodge-theoretic/combinatorial predictions in a large class. It does not prove the full physical

equivalence of type-II compactifications.

- Title: “Homological Algebra of Mirror Symmetry” Authors: Maxim Kontsevich Venue/year: Proceedings of the International Congress of Mathematicians, Zurich 1994, Birkhauser, 1995, pp. 120-139 DOI/arXiv/URL: DOI 10.1007/978-3-0348-9078-6_11; arXiv:alg-geom/9411018, <https://arxiv.org/abs/alg-geom/9411018> Progress: Formulates homological mirror symmetry as an equivalence between the derived category of coherent sheaves on one side and the Fukaya category on the mirror side. Limitation: This is a rigorous mathematical shadow of mirror symmetry for topological/open-string sectors. Even where proved, it is not the same as the full four-dimensional type-II physical equivalence with all compactification fields.
- Title: “A Mirror Theorem for Toric Complete Intersections” Authors: Alexander Givental Venue/year: Topological Field Theory, Primitive Forms and Related Topics, Progress in Mathematics 160, Birkhauser, 1998, pp. 141-175 DOI/arXiv/URL: arXiv:alg-geom/9701016, <https://arxiv.org/abs/alg-geom/9701016> Progress: Proves mirror formulas for genus-zero Gromov-Witten invariants of non-negative complete intersections in symplectic toric manifolds, a rigorous version of a major enumerative prediction of mirror symmetry. Limitation: The theorem covers a genus-zero A-model/B-model enumerative sector under toric complete-intersection hypotheses. It does not imply full equivalence of type IIA and type IIB compactified string theories.
- Title: “Mirror Principle I” and “The Candelas-de la Ossa-Green-Parkes Formula” Authors: Bong H. Lian, Kefeng Liu, Shing-Tung Yau Venue/year: Asian Journal of Mathematics 1, 729-763, 1997; Nuclear Physics B Proceedings Supplements 67, 106-114, 1998 DOI/arXiv/URL: arXiv:alg-geom/9712011, <https://arxiv.org/abs/alg-geom/9712011>; DOI 10.1016/S0920-5632(98)00126-1 Progress: Gives rigorous mirror-principle proofs including the Candelas-de la Ossa-Green-Parkes quintic formula. Limitation: These are rigorous enumerative mirror theorems, not full physical equivalence theorems for arbitrary Calabi-Yau compactifications.
- Title: “Mirror Symmetry is T-Duality” Authors: Andrew Strominger, Shing-Tung Yau, Eric Zaslow Venue/year: Nuclear Physics B 479, 243-259, 1996 DOI/arXiv/URL: DOI 10.1016/0550-3213(96)00434-8; arXiv:hep-th/9606040, <https://arxiv.org/abs/hep-th/9606040> Progress: Proposes the SYZ geometric mechanism: mirror symmetry should arise from dual special Lagrangian torus fibrations. Limitation: The paper argues for a geometric explanation and states a conjectural mechanism. The existence, compactification, singular-fiber behavior, and full connection to type-II physical equivalence are not proved in the all-case form needed by the problem statement.
- Title: “From Real Affine Geometry to Complex Geometry” Authors: Mark Gross and Bernd Siebert Venue/year: Annals of Mathematics 174, 1301-1428, 2011 DOI/arXiv/URL: DOI 10.4007/annals.2011.174.3.1; arXiv:math/0703822, <https://arxiv.org/abs/math/0703822> Progress: Constructs degenerations of Calabi-Yau manifolds from affine/tropical data and is a central result in the Gross-Siebert program, an algebro-geometric incarnation of SYZ ideas. Limitation: It constructs and controls important mirror families under degeneration/tropical hypotheses and supplies a program toward mirror symmetry. It does not prove full type-II string compactification equivalence in the stated sense.
- Title: “Dirichlet Branes and Mirror Symmetry” Authors: Paul S. Aspinwall, Tom Bridgeland, Alastair Craw, Michael R. Douglas, Mark Gross, Anton Kapustin, Gregory W. Moore, Graeme Segal, Balazs Szendroi, P. M. H. Wilson Venue/year: Clay Mathematics Monographs, Vol. 4, American Mathematical Society/Clay Mathematics Institute, 2009

DOI/arXiv/URL: URL <https://www.claymath.org/library/monographs/cmim04.pdf> Progress: Gives an extensive mathematical-physics account of D-branes and mirror symmetry. It states as Conjecture 1.3 that type IIA on a Calabi-Yau threefold X is dual to type IIB on a mirror Calabi-Yau threefold Y . Limitation: The relevant statement is explicitly presented as a conjecture. The book develops consequences and mathematical models of sectors of the conjecture rather than proving the full physical equivalence.

- Title: “Homological Mirror Symmetry for Calabi-Yau Hypersurfaces in Projective Space” Authors: Nick Sheridan Venue/year: *Inventiones Mathematicae* 199, no. 1, 1-186, 2015 DOI/arXiv/URL: DOI 10.1007/s00222-014-0507-2; arXiv:1111.0632, <https://arxiv.org/abs/1111.0632> Progress: Proves homological mirror symmetry for smooth d -dimensional Calabi-Yau hypersurfaces in projective space for $d \geq 3$, including the quintic threefold case. Limitation: This is a major categorical theorem for a specific class of mirror pairs and a specific mathematical formulation. It does not cover all Calabi-Yau threefolds and does not prove equivalence of the full type-II compactified physical theories including RR fields and dilaton-axion.
- Title: “Homological Mirror Symmetry for Generalized Greene-Plesser Mirrors” Authors: Nick Sheridan and Ivan Smith Venue/year: *Inventiones Mathematicae* 224, 627-682, 2021 DOI/arXiv/URL: DOI 10.1007/s00222-020-01018-w; arXiv:1709.08937, <https://arxiv.org/abs/1709.08937> Progress: Proves Kontsevich’s homological mirror symmetry conjecture for broad classes including Greene-Plesser mirror pairs and certain Batyrev-Borisov mirror constructions. Limitation: The hypotheses are restricted classes of mirror constructions, and the conclusion is a homological/categorical equivalence. It still does not prove the all-case, full string-theoretic physical equivalence.

7.3.3.4. Conclusion

No citable theorem found in the literature resolves exactly the full statement. The literature contains extensive and deep partial resolutions: explicit mirror constructions, rigorous enumerative mirror theorems, SYZ/Gross-Siebert geometric programs, and homological mirror symmetry proofs for important classes including the quintic and many Greene-Plesser/Batyrev-Borisov mirrors. However, these results either apply only to special classes, prove only topological/enumerative/categorical sectors, or require extra hypotheses. They do not establish a mathematically precise equivalence of the full type IIA and type IIB compactified string theories for all Calabi-Yau threefold mirror pairs with the complete field data listed in the problem statement. Therefore the status is partially resolved.

7.3.4. Remaining open problems recorded by the database

7.3.4.1. Open problem 1

For a mirror pair of Calabi-Yau threefolds $((X, Y))$, is the split-closed derived Fukaya category of (X) equivalent to the bounded derived category of coherent sheaves on (Y) ,

$$D^{\pi} \text{Fuk}(X) \simeq D^b \text{Coh}(Y),$$

with the analogous equivalence after exchanging (X) and (Y) ?

Reference supplied by the database. Maxim Kontsevich, “Homological Algebra of Mirror Symmetry,” in *Proceedings of the International Congress of Mathematicians, Zürich, 1994*, Vol. I, Birkhäuser, Basel, 1995, pp. 120–139, “Homological Mirror Conjecture,” arXiv: [alg-geom/9411018](https://arxiv.org/abs/alg-geom/9411018).

7.3.4.2. Open problem 2

Do mirror Calabi–Yau threefolds (X) and (Y) , near the appropriate large-complex-structure limits, admit special Lagrangian (3) -torus fibrations over a common base whose nonsingular fibers are fiberwise dual, so that the mirror is obtained as the moduli space of the fibers together with flat $(U(1))$ -connections (with the necessary corrections at singular fibers)?

Reference supplied by the database. Andrew Strominger, Shing-Tung Yau, and Eric Zaslow, “Mirror Symmetry Is T-Duality,” *Nuclear Physics B* 479, no. 1–2 (1996), 243–259, Strominger–Yau–Zaslow conjecture, DOI: [10.1016/0550-3213\(96\)00434-8](https://doi.org/10.1016/0550-3213(96)00434-8); arXiv: [hep-th/9606040](https://arxiv.org/abs/hep-th/9606040).

7.3.4.3. Open problem 3

For every positive integer (d) , does a general smooth quintic threefold in $\mathbb{P}^4$ contain only finitely many rational curves of degree (d) (and, in the strong form, are these curves smooth, isolated, and equipped with normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$)?

Reference supplied by the database. C. Herbert Clemens, “Some Results on Abel–Jacobi Mappings,” in Phillip A. Griffiths (ed.), *Topics in Transcendental Algebraic Geometry*, Annals of Mathematics Studies 106, Princeton University Press, Princeton, 1984, pp. 289–304, p. 300 (Clemens conjecture).

7.3.5. Primary-source pointer

The originating ICM record is *String Theory and Duality* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=231>.

7.4. 136. Mathematical Results Inspired by Physics [2002-ag-95]

Record: 2002-ag-95

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Mathematical Results Inspired by Physics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=446>

7.4.1. Problem statement

After fixing the underlying four-manifold and gauge data, let M_n denote the moduli space of self-dual connections in the n -th topological sector, and let $\chi(M_n)$ denote its Euler number. Prove that the Euler-number generating series

$$Z(\tau) = \sum_n \chi(M_n) q^n, \quad q = e^{2\pi i \tau},$$

is modular, in the sense that it transforms as a modular form in the modular parameter τ .

7.4.2. Historical background, literature, and current progress

7.4.3. Verdict: partially resolved

7.4.3.1. Precise problem

The problem asks for the general Vafa-Witten S-duality prediction: after fixing an oriented four-manifold and gauge/topological data, let M_n be the instanton/self-dual connection moduli space in the n -th sector. Is the Euler-number partition function

$$Z(\tau) = \sum_n \chi(M_n) q^n, \quad q = e^{2\pi i \tau},$$

modular in τ in the sense predicted by S-duality?

This is a broad all-cases question. It is not just the K3 case, the projective-plane case, rank 2, or the modern algebraic-geometry version for polarized projective surfaces.

7.4.3.2. Verdict

Partially resolved. There are rigorous results proving the predicted modularity or corrected modular/mock-modular behavior in important special cases, especially for K3 surfaces, rational/projective surfaces, and some algebraic Vafa-Witten invariants. I did not find a published or otherwise citable theorem that proves the full statement for arbitrary four-manifolds and all gauge data/topological sectors.

7.4.3.3. Key literature and status

1. Title: "A Strong Coupling Test of S-Duality"
 Authors: Cumrun Vafa and Edward Witten
 Venue: Nuclear Physics B 431, pp. 3-77
 Year: 1994
 DOI/arXiv/URL: DOI 10.1016/0550-3213(94)90097-3; arXiv:hep-th/9408074; <https://arxiv.org/abs/hep-th/9408074>
 Role: Original source of the S-duality prediction. It studies topologically twisted $N = 4$ supersymmetric Yang-Mills theory on four-manifolds and gives exact physical tests and formulas in examples. It is not a mathematical proof of the full all-four-manifold statement.
2. Title: "Euler Characteristics of $SU(2)$ Instanton Moduli Spaces on Rational Elliptic Surfaces"
 Author: Kota Yoshioka
 Venue: Communications in Mathematical Physics 205, pp. 501-517
 Year: 1999
 DOI/arXiv/URL: DOI 10.1007/s002200050687; arXiv:math/9805003; <https://arxiv.org/abs/math/9805003>
 Role: Computes Euler characteristics of $SU(2)$ instanton/Gieseker moduli spaces on rational elliptic surfaces and checks the modular property of $Z_{SU(2)}$ and $Z_{SO(3)}$ conjectured by Vafa and Witten. This is a genuine special-case verification, not a proof for all four-manifolds or all gauge groups.
3. Title: "On blowup formulae for the S-duality conjecture of Vafa and Witten"
 Authors: Wei-Ping Li and Zhenbo Qin
 Venue: Inventiones Mathematicae 136, pp. 451-482
 Year: 1999
 DOI/arXiv/URL: DOI 10.1007/s002220050316; arXiv:math/9805054; <https://arxiv.org/abs/math/9805054>
 Role: Proves blow-up formulae for virtual Hodge polynomials and Euler numbers of rank-2 sheaf/Uhlenbeck compactification spaces over algebraic surfaces, verifying the blow-up formulae predicted by the Vafa-Witten S-duality conjecture. This supports and propagates special cases but does not prove the base modularity statement in all cases.
4. Title: "From sheaves on P^2 to a generalization of the Rademacher expansion"
 Authors: Kathrin Bringmann and Jan Manschot
 Venue: American Journal of Mathematics 135(4), pp. 1039-1065
 Year: 2013
 DOI/arXiv/URL: DOI 10.1353/ajm.2013.0031; arXiv:1006.0915; <https://arxiv.org/abs/1006.0915>
 Role: Proves a Vafa-Witten conjecture for rank-2 sheaves on $\mathbb{P}^2$, expressing Euler-number generating functions as mixed mock modular forms. This is an important exact result, but it is for a specific surface and rank and also shows that the naive phrase "is a modular form" must be interpreted carefully in cases with modular anomaly/mock modularity.
5. Title: "Vafa-Witten Theory and Iterated Integrals of Modular Forms"
 Author: Jan Manschot
 Venue: Communications in Mathematical Physics 371, pp. 787-831
 Year: 2019
 DOI/arXiv/URL: DOI 10.1007/s00220-019-03389-5; arXiv:1709.10098; <https://arxiv.org/abs/1709.10098>

- Role: Analyzes modular anomaly for Vafa-Witten partition functions, including $SU(3)$ on $\mathbb{P}^2$, and identifies higher-depth mock modular behavior. This is strong progress for projective-plane/higher-rank cases, not a full all-manifold proof.
6. Title: “Vafa-Witten invariants for projective surfaces I: stable case”
 Authors: Yuuji Tanaka and Richard P. Thomas
 Venue: Journal of Algebraic Geometry 29, pp. 603-668
 Year: 2020
 DOI/arXiv/URL: DOI 10.1090/jag/738; arXiv:1702.08487; <https://arxiv.org/abs/1702.08487>
 Role: Gives a mathematical definition of Vafa-Witten invariants for polarized projective surfaces in the stable case via symmetric obstruction theories and virtual localization. When the Vafa-Witten vanishing theorem applies, the invariant is the signed Euler characteristic of the instanton moduli space; in general there are additional rational contributions. This is foundational but is limited to projective surfaces and does not prove the full statement.
 7. Title: “Vafa-Witten invariants for projective surfaces II: semistable case”
 Authors: Yuuji Tanaka and Richard P. Thomas
 Venue: Pure and Applied Mathematics Quarterly 13(3), pp. 517-562
 Year: 2017/2018
 DOI/arXiv/URL: DOI 10.4310/PAMQ.2017.v13.n3.a6; arXiv:1702.08488; <https://arxiv.org/abs/1702.08488>
 Role: Extends the definition to semistable Higgs pairs and calculates K3-surface invariants in terms of modular forms generalizing and proving Vafa-Witten predictions in that setting. This is a major special-case result for projective surfaces, not a proof for arbitrary four-manifolds and all gauge data.
 8. Title: “Sheaf counting on local K3 surfaces”
 Authors: Daves Maulik and Richard P. Thomas
 Venue: Pure and Applied Mathematics Quarterly 14(3-4), pp. 419-441
 Year: 2018/2019
 DOI/arXiv/URL: DOI 10.4310/PAMQ.2018.v14.n3.a1; arXiv:1806.02657; <https://arxiv.org/abs/1806.02657>
 Role: Proves a conjecture of Tanaka-Thomas used in the semistable K3-surface Vafa-Witten theory. This strengthens the K3 case but does not cover general four-manifolds.
 9. Title: “A proof of all ranks S-duality conjecture for K3 surfaces”
 Authors: Yunfeng Jiang and Hsian-Hua Tseng
 Venue: Asian Journal of Mathematics 28(2), pp. 317-344
 Year: 2024
 DOI/arXiv/URL: arXiv:2003.09562; <https://arxiv.org/abs/2003.09562>; journal listing <https://intlpress.com/JDetail/1829551722763767809>
 Role: Proves the all-ranks S-duality conjecture for K3 surfaces in the $SU(r)/Z_r$ Vafa-Witten framework, using Tanaka-Thomas $SU(r)$ invariants and twisted-sheaf counts. This is one of the strongest resolving-looking results, but its hypothesis is specifically K3 surfaces, so it does not imply the full statement.
 10. Title: “Virtual Refinements of the Vafa-Witten Formula”
 Authors: Lothar Goettsche and Martijn Kool
 Venue: Communications in Mathematical Physics 376, pp. 1-49
 Year: 2020
 DOI/arXiv/URL: DOI 10.1007/s00220-020-03748-7; arXiv:1703.07196; <https://arxiv.org/abs/1703.07196>

[v.org/abs/1703.07196](https://arxiv.org/abs/1703.07196)

Role: Formulates and checks refined conjectural formulas for virtual χ_y -genera and virtual Euler characteristics of rank-2 sheaf moduli spaces on surfaces with holomorphic 2-form, recovering part of Vafa-Witten's formula for minimal general type surfaces. It is explicitly conjectural in its broad form and thus is partial progress, not a full resolution.

11. Title: "SU(r) Vafa-Witten Invariants, Ramanujan's Continued Fractions, and Cosmic Strings"
 Authors: Lothar Goettsche, Martijn Kool, and Ties Laarakker
 Venue: Michigan Mathematical Journal 75(1), pp. 3-63
 Year: 2025
 DOI/arXiv/URL: DOI 10.1307/mmj/20226202; arXiv:2108.13413; <https://arxiv.org/abs/2108.13413>
 Role: Gives conjectural structure formulas for SU(r) Vafa-Witten partition functions for surfaces with holomorphic 2-form and proves/checks vertical parts in low virtual dimensions. This documents active progress but also confirms that broad higher-rank/projective-surface formulas remain conjectural beyond proved pieces.
12. Title: analytic work on Vafa-Witten equations on closed four-manifolds
 Authors/items: Yuuji Tanaka, "Some boundedness properties of solutions to the Vafa-Witten equations on closed four-manifolds," Q. J. Math. 68 (2017), pp. 1203-1225, DOI 10.1093/qmath/hax015, arXiv:1308.0862; Clifford H. Taubes, "The behavior of sequences of solutions to the Vafa-Witten equations," arXiv:1702.04610
 Role: Develops analytic compactness/boundedness behavior for the equations over closed four-manifolds. These papers are foundational for moduli-space theory, but they do not prove modularity of Euler-number generating series.

7.4.3.4. What remains unresolved

No cited theorem found proves that, for every fixed underlying four-manifold and all relevant gauge/topological data, the Euler-number generating series of self-dual connection/instanton moduli spaces transforms as the predicted modular form.

The strongest rigorous results have additional hypotheses: K3 surfaces; rational elliptic surfaces; $\mathbb{P}^2$; smooth projective/polarized algebraic surfaces; rank restrictions or specialized SU(r)/SU(r)/Z_r frameworks; stable or semistable sheaf/Higgs-pair compactifications; virtual rather than ordinary Euler characteristics; or modular completions/mock-modular forms rather than ordinary holomorphic modular forms.

Therefore the exact broad problem is not fully resolved. The current literature supports the prediction in many important cases and has a robust projective-surface theory, but the general four-manifold/gauge-data modularity statement remains open beyond those cases.

7.4.4. Remaining open problems recorded by the database

7.4.4.1. Open problem 1

Let S be a smooth projective surface with $b_1(S) = 0$, $p_g(S) > 0$, and with Seiberg–Witten basic classes 0 and $K_S \neq 0$. If $M_S^H(2, c_1, c_2)$ contains no strictly Gieseker H -semistable

sheaves, prove that its normalized virtual χ_y -genus is the coefficient of $x^{4c_2-c_1^2-3\chi(O_S)}$ in

$$8 \left(\frac{1}{2} \prod_{n \geq 1} \frac{1}{(1-x^{2n})^{10}(1-x^{2n}y)(1-x^{2n}y^{-1})} \right)^{\chi(O_S)} \left(\frac{2\bar{\eta}(x^4)^2}{\theta_3(x, y^{1/2})} \right)^{K_S^2}.$$

Reference supplied by the database. Lothar Göttsche and Martijn Kool, “Virtual Refinements of the Vafa–Witten Formula,” *Communications in Mathematical Physics* **376** (2020), 1–49, Conjecture 1.1. DOI: [10.1007/s00220-020-03748-7](https://doi.org/10.1007/s00220-020-03748-7); arXiv: [1703.07196](https://arxiv.org/abs/1703.07196).

7.4.4.2. Open problem 2

For a polarized projective surface and fixed charge $\alpha = (r, c_1, c_2)$, prove the Tanaka–Thomas universal wall-crossing formula expressing the virtual-localization invariants of stable Joyce–Song Higgs pairs $P_\alpha^\perp(N)$, for all sufficiently large auxiliary twists N , in terms of rational Vafa–Witten invariants VW_{α_i} . In particular, prove in full generality that the resulting Vafa–Witten invariant is independent of N ; when $H^{0,1}(S) \neq 0$ or $H^{0,2}(S) \neq 0$, prove the predicted one-term formula

$$P_{r,L,c_2}^\perp(N) = (-1)^{\chi(\alpha(N))-1} \chi(\alpha(N)) \text{VW}_{r,L,c_2}(S).$$

Reference supplied by the database. Yuuji Tanaka and Richard P. Thomas, “Vafa–Witten Invariants for Projective Surfaces II: Semistable Case,” *Pure and Applied Mathematics Quarterly* **13** (2017), no. 3, 517–562, Conjecture 1.2 (equivalently Conjecture 6.5 in the detailed development). DOI: [10.4310/PAMQ.2017.v13.n3.a6](https://doi.org/10.4310/PAMQ.2017.v13.n3.a6); arXiv: [1702.08488](https://arxiv.org/abs/1702.08488).

7.4.4.3. Open problem 3

Let (S, H) be a smooth polarized projective surface with $H_1(S, \mathbb{Z}) = 0$ and $p_g(S) > 0$, let r be prime, and let $c_1 \in H^2(S, \mathbb{Z})$ be algebraic. Prove that the mathematically defined $\text{SU}(r)$ and $\text{SU}(r)/\mathbb{Z}_r$ Vafa–Witten partition functions are meromorphic modular functions on the upper half-plane satisfying

$$Z_{c_1}^{\text{SU}(r)}(-1/\tau) = (-1)^{(r-1)\chi(O_S)} \left(\frac{r\tau}{i} \right)^{-e(S)/2} Z_{c_1}^{\text{SU}(r)/\mathbb{Z}_r}(\tau).$$

Reference supplied by the database. Yunfeng Jiang and Martijn Kool, “Twisted Sheaves and $\text{SU}(r)/\mathbb{Z}_r$ Vafa–Witten Theory,” *Mathematische Annalen* **382** (2022), nos. 1–2, 719–743, Conjecture 1.2. DOI: [10.1007/s00208-021-02303-6](https://doi.org/10.1007/s00208-021-02303-6); arXiv: [2006.10368](https://arxiv.org/abs/2006.10368).

7.4.4.4. Open problem 4

For every rank $r > 1$, prove the universal Seiberg–Witten structure formula for the virtual Euler characteristics of Gieseker–Maruyama moduli spaces $M_S^H(r, c_1, c_2)$ on smooth polarized surfaces with $b_1(S) = 0$ and $p_g(S) > 0$: namely, prove the existence of universal power series D_0 and D_{ij} , depending only on r , such that all coefficients $e^{\text{vir}}(M_S^H(r, c_1, c_2))$ are given by the formula of Göttsche–Kool–Laarakker involving only $\chi(O_S)$, K_S^2 , the Seiberg–Witten invariants of S , and the intersection products of their basic classes.

Reference supplied by the database. Lothar Göttsche, Martijn Kool, and Ties Laarakker, “SU(r) Vafa–Witten Invariants, Ramanujan’s Continued Fractions, and Cosmic Strings,” *Michigan Mathematical Journal* **75** (2025), no. 1, 3–63, Conjecture 1.10 (restated as Conjecture 4.1). DOI: [10.1307/mmj/20226202](https://doi.org/10.1307/mmj/20226202); arXiv: [2108.13413](https://arxiv.org/abs/2108.13413).

7.4.4.5. Open problem 5

For every rank $r > 1$, prove that the universal series C_0, C_{ij} governing the vertical Vafa–Witten contribution and the universal series D_0, D_{ij} governing the horizontal contribution are Fourier expansions of meromorphic functions on the upper half-plane and are exchanged by the modular S -transformation:

$$D_0(\tau) = C_0(-1/\tau), \quad D_{ij}(\tau) = C_{ij}(-1/\tau).$$

Reference supplied by the database. Lothar Göttsche, Martijn Kool, and Ties Laarakker, “SU(r) Vafa–Witten Invariants, Ramanujan’s Continued Fractions, and Cosmic Strings,” *Michigan Mathematical Journal* **75** (2025), no. 1, 3–63, Conjecture 1.13 (restated as Conjecture 4.2). DOI: [10.1307/mmj/20226202](https://doi.org/10.1307/mmj/20226202); arXiv: [2108.13413](https://arxiv.org/abs/2108.13413).

7.4.5. Primary-source pointer

The originating ICM record is *Mathematical Results Inspired by Physics* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=446>.

7.5. 137. Inverse problems for linear and non-linear hyperbolic equations [2018-geom-140]

Record: 2018-geom-140

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Inverse problems for linear and non-linear hyperbolic equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>

7.5.1. Problem statement

Let (M, g) be a 4-dimensional globally hyperbolic time-oriented Lorentzian manifold of signature $(-, +, +, +)$, written using the product representation $M = \mathbb{R} \times N$, and let $t_0 > 0$. Let $\mu \subset (0, t_0) \times N$ be a timelike path, let $V \subset M$ be an open neighborhood of μ , and let $p^-, p^+ \in \mu$ with $p^- \ll p^+$, where $p \ll q$ means that p and q can be joined by a future-pointing timelike curve. Write

$$I^+(p) = \{q \in M : p \ll q\}, \quad I^-(p) = \{q \in M : q \ll p\},$$

and define the chronological double cone

$$I(p^-, p^+) = I^+(p^-) \cap I^-(p^+).$$

For a nowhere-vanishing smooth coefficient a and a first-order differential operator $B(x, D)$ with smooth x -dependent coefficients, consider the nonlinear wave equation

$$\square_g u(x) + B(x, D)u(x) + a(x)u(x)^2 = f(x), \quad x \in (-\infty, t_0) \times N,$$

with $u = 0$ and $f = 0$ in $(-\infty, 0) \times N$. For sufficiently small C^k sources f supported in V , where k is sufficiently large, let u be the corresponding solution and let

$$L_V : f \mapsto u|_V$$

be the source-to-solution measurement operator on V . Given the timelike path μ , its neighborhood V , and the operator L_V , is it possible to construct the metric tensor g and the first-order operator $B(x, D)$ on $I(p^-, p^+)$, up to the natural local gauge ambiguity for this inverse problem, namely local changes of variables and the induced transformation of the differential operator that leave the observed source-to-solution data on V unchanged?

7.5.2. Historical background, literature, and current progress

7.5.3. Verdict: partially resolved

7.5.3.1. Precise problem checked

The problem asks whether the local active source-to-solution map near a timelike path for

$$\Box_g u + B(x, D)u + a(x)u^2 = f$$

on an arbitrary 4-dimensional globally hyperbolic Lorentzian manifold determines, on the chronological diamond $I(p^-, p^+)$, both the full Lorentzian metric tensor g and the arbitrary smooth first-order operator $B(x, D)$, up to the natural local gauge by changes of variables and induced operator transformations.

I did not find a published or citable theorem that proves this exact simultaneous recovery result for an arbitrary first-order term $B(x, D)$ in this local source-to-solution setting.

7.5.3.2. Main partial progress

1. Inverse problems for Lorentzian manifolds and non-linear hyperbolic equations

Authors: Yaroslav Kurylev, Matti Lassas, Gunther Uhlmann.

Venue/year: *Inventiones Mathematicae* 212 (2018), 781-857.

DOI/arXiv/URL: DOI 10.1007/s00222-017-0780-y; arXiv 1405.3386, <https://arxiv.org/abs/1405.3386>.

Result checked: Theorem 1.5 treats the local source-to-solution map for $\Box_g u + a(x)u^2 = f$ with nonvanishing a on a 3+1 dimensional globally hyperbolic Lorentzian manifold. It reconstructs the differentiable structure and the conformal class of g on $I(p^-, p^+)$, with the conformal factor only in additional cases such as Ricci-flat regions. It does not include an unknown first-order operator $B(x, D)$ and does not generally recover the full metric tensor.

2. Inverse problems for semilinear wave equations on Lorentzian manifolds

Authors: Matti Lassas, Gunther Uhlmann, Yiran Wang.

Venue/year: *Communications in Mathematical Physics* 360 (2018), 555-609.

DOI/arXiv/URL: DOI 10.1007/s00220-018-3133-7; arXiv 1606.06261, <https://arxiv.org/abs/1606.06261>.

Result checked: Theorems 1.2-1.5 and 6.1-6.2 extend the local source-to-solution method to general semilinear terms and scalar zeroth-order linear terms. The paper explicitly leaves determination of the cubic, quadratic, and linear Taylor terms to future work in the fixed-metric coefficient theorem, and its lower-order linear term is scalar $Q(x)u$, not an arbitrary first-order differential operator $B(x, D)u$.

3. Inverse problems for quadratic derivative nonlinear wave equations

Authors: Yiran Wang, Ting Zhou.

Venue/year: *Communications in Partial Differential Equations* 44 (2019), 1140-1178.

DOI/arXiv/URL: DOI 10.1080/03605302.2019.1607895; arXiv 1612.04437, <https://arxiv.org/abs/1612.04437>.

Result checked: Theorem 1.1 has nearly the same local timelike-path/source-to-solution geometry and recovers the conformal class of the metric, and in special cases the metric itself, for a quadratic derivative nonlinear equation. This is a different nonlinear model and does not recover a general first-order linear operator $B(x, D)$.

4. Inverse problems for non-linear hyperbolic equations with disjoint sources and receivers

Authors: Ali Feizmohammadi, Matti Lassas, Lauri Oksanen.

Venue/year: Forum of Mathematics, Pi 9 (2021), e10.

DOI/arXiv/URL: DOI 10.1017/fmp.2021.8; arXiv 2006.12158, <https://arxiv.org/abs/2006.12158>.

Result checked: Theorem 1.1 recovers the Lorentzian metric, not just its conformal class, for semilinear equations $\Box_{g,u} u + u^m = f$ with $m \geq 3$ except for the exceptional $(n, m) = (3, 3)$ case. It does not cover the quadratic nonlinearity $a(x)u^2$ in $3+1$ dimensions with arbitrary a , and it has no unknown first-order $B(x, D)$.

5. The Dirichlet-to-Neumann map for a semilinear wave equation on Lorentzian manifolds

Authors: Peter Hintz, Gunther Uhlmann, Jian Zhai.

Venue/year: Communications in Partial Differential Equations 47 (2022), 2363-2400.

DOI/arXiv/URL: DOI 10.1080/03605302.2022.2131866; arXiv 2103.08110, <https://arxiv.org/abs/2103.08110>.

Result checked: Theorem 1.3 recovers the Lorentzian metric and a quartic coefficient a from boundary Dirichlet-to-Neumann data up to diffeomorphism and a conformal obstruction. This is boundary data, assumes no conjugate points in the relevant region, uses a quartic nonlinearity, and does not contain an arbitrary first-order B .

6. An inverse boundary value problem for a semilinear wave equation on Lorentzian manifolds

Authors: Peter Hintz, Gunther Uhlmann, Jian Zhai.

Venue/year: International Mathematics Research Notices 2022, no. 17, 13181-13211.

DOI/arXiv/URL: DOI 10.1093/imrn/rnab088; arXiv 2005.10447, <https://arxiv.org/abs/2005.10447>.

Result checked: The main theorem recovers nonlinear Taylor coefficients from a Neumann-to-Dirichlet map when the metric is fixed and null geodesics have no cut points. It is coefficient recovery for a different boundary problem, not simultaneous recovery of g and B .

7. Detection of Hermitian connections in wave equations with cubic non-linearity

Authors: Xi Chen, Matti Lassas, Lauri Oksanen, Gabriel P. Paternain.

Venue/year: Journal of the European Mathematical Society 24 (2022), 2191-2232.

DOI/arXiv/URL: DOI 10.4171/JEMS/1136; <https://ems.press/journals/jems/articles/8965734>.

Result checked: Theorem 1 recovers a Hermitian connection, a first-order lower-order structure, from a source-to-solution map for a cubic connection wave equation in Minkowski space. The Lorentzian metric is fixed and flat, the equation is vector-valued with a connection structure, and it is not the arbitrary scalar $B(x, D)$ problem.

8. Inverse problem for connections in semi-linear wave equations

Authors: Lauri Oksanen, Ruochong Zhang.

Venue/year: arXiv preprint, 2025.

DOI/arXiv/URL: arXiv 2509.25971, <https://arxiv.org/abs/2509.25971>.

Result checked: Theorem 1.1 extends connection recovery from source-to-solution data to arbitrary globally hyperbolic Lorentzian manifolds, but the metric g is given and the operator is a Hermitian connection wave operator with cubic nonlinearity. It does not simultaneously reconstruct an unknown metric and a completely general first-order scalar operator.

9. Inverse problems for real principal type operators

Authors: Lauri Oksanen, Mikko Salo, Plamen Stefanov, Gunther Uhlmann.

Venue/year: American Journal of Mathematics 146 (2024), 161-240.

DOI/arXiv/URL: DOI 10.1353/ajm.2024.a922595; arXiv 2001.07599, <https://arxiv.org/abs/2001.07599>.

Result checked: The nonlinear theorem fixes the real-principal-type operator P and recovers the nonlinearity under analytic and cubic-or-higher assumptions; other theorems recover boundary/scattering data or lower-order ray transforms under separate hypotheses. It does not solve the local Lorentzian source-to-solution problem with unknown g and arbitrary first-order B .

7.5.3.3. Why the exact problem remains unresolved

The literature fully covers important pieces: local nonlinear interactions recover the conformal Lorentzian geometry for $\Box_g u + a u^2$; later work recovers the full metric in several different nonlinear or boundary settings; and recent work recovers connection-type first-order data when the metric and operator structure are fixed. However, I found no theorem whose hypotheses match the full setting and whose conclusion reconstructs both:

- the full metric tensor g on $I(p^-, p^+)$, and
- an arbitrary first-order differential operator $B(x, D)$,

from the same local source-to-solution operator for $\Box_g u + B(x, D)u + a(x)u^2 = f$ on arbitrary 3+1 dimensional globally hyperbolic Lorentzian manifolds, modulo only the natural local gauge.

Therefore the correct status is **partially resolved**, not fully resolved.

7.5.4. Remaining open problems recorded by the database

7.5.4.1. Open problem 1

Let (M, g) be a known globally hyperbolic Lorentzian manifold, let $\Omega \subset M$ be an open observation region, and set $\mathbb{D} = J^+(\Omega) \cap J^-(\Omega)$. For an unknown scalar potential V , let $\mathcal{L}_V f = u|_\Omega$, where $(\Box_g + V)u = f$ and u vanishes in the past. Does the linearized source-to-solution map $\mathcal{L}_V$ determine $V|_{\mathbb{D}}$? This is open even in the corresponding Minkowski-space formulation without additional geometric assumptions.

Reference supplied by the database. Ali Feizmohammadi and Lauri Oksanen, “Recovery of zeroth order coefficients in non-linear wave equations,” Journal of the Institute of Mathematics of Jussieu 21 (2022), no. 2, 367–393, explicit open question in Section 2 immediately after the definition of the linearized source-to-solution map $\mathcal{L}_V$, DOI: 10.1017/S1474748020000122, <https://doi.org/10.1017/S1474748020000122>; arXiv:1903.12636, <https://arxiv.org/abs/1903.12636>.

7.5.5. Primary-source pointer

The originating ICM record is *Inverse problems for linear and non-linear hyperbolic equations* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>.

7.6. 138. Inverse problems for linear and non-linear hyperbolic equations [2018-geom-141]

Record: 2018-geom-141

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Inverse problems for linear and non-linear hyperbolic equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>

7.6.1. Problem statement

Let (M, g) be a 4-dimensional globally hyperbolic time-oriented Lorentzian manifold of signature $(-, +, +, +)$. For $p, q \in M$, write $p \ll q$ if p and q can be joined by a future-pointing timelike curve, set

$$I^+(p) = \{q \in M : p \ll q\}, \quad I^-(p) = \{q \in M : q \ll p\},$$

and define the double cone

$$I(p, q) = I^+(p) \cap I^-(q).$$

Using the product representation $M = \mathbb{R} \times N$ of a globally hyperbolic spacetime, let $t_0 > 0$, let $\mu \subset (0, t_0) \times N$ be a timelike curve, and let $V \subset M$ be an open neighborhood of μ . For a nowhere-vanishing smooth coefficient a , consider the semilinear wave equation

$$\square_g u(x) + a(x)u(x)^2 = f(x), \quad x \in (-\infty, t_0) \times N,$$

with $u = 0$ and $f = 0$ on $(-\infty, 0) \times N$. For sufficiently small C^k sources f supported in V , let L_V be the active measurement operator

$$L_V : f \mapsto u|_V,$$

where u is the corresponding solution. Suppose that the given data consist of the timelike curve μ , its neighborhood $V \subset M$, an approximation to the measurement operator L_V with some error, and points $p^-, p^+ \in \mu$ with $p^- \ll p^+$. Is it possible to reconstruct the double cone $I(p^-, p^+)$ and the metric g on $I(p^-, p^+)$ with an error estimate controlled in terms of a priori geometric bounds for M and the error in the given data?

7.6.2. Historical background, literature, and current progress

7.6.3. Verdict: partially resolved

7.6.3.1. Precise problem

The problem asks for a quantitative stability/reconstruction theorem for the local active inverse problem for

$$\square_g u + a(x)u^2 = f$$

on a 4-dimensional globally hyperbolic Lorentzian spacetime. The requested result would start from approximate local source-to-solution data $L_V : f \mapsto u|_V$, a timelike observation curve μ , a neighborhood V , and points $p^- \ll p^+$ on μ , and would reconstruct both the causal diamond $I(p^-, p^+)$ and the Lorentzian metric g on it with an explicit error estimate controlled by a priori geometry and measurement error.

7.6.3.2. Partial progress found

1. Inverse problems for Lorentzian manifolds and non-linear hyperbolic equations

Authors: Yaroslav Kurylev, Matti Lassas, Gunther Uhlmann.

Venue/year: *Inventiones Mathematicae* 212(3), 781-857, 2018.

DOI/arXiv/URL: DOI 10.1007/s00222-017-0780-y; arXiv 1405.3386, <https://arxiv.org/abs/1405.3386>.

Relevant result: Theorem 1.5 treats the exact local active source-to-solution map for the quadratic equation with nowhere-vanishing a in dimension $1+3$. It determines $I(p^-, p^+)$ as a differentiable manifold and determines the conformal class of g there, from exact equality of the maps. This is the closest match to this problem, but it is not a stability theorem for approximate data and its general conclusion is conformal, not full metric recovery. Full metric recovery is only indicated in additional special situations.

2. Inverse problems for linear and non-linear hyperbolic equations

Author: Matti Lassas.

Venue/year: *Proceedings of the International Congress of Mathematicians 2018, Rio de Janeiro*, Vol. III, pp. 3739-3760, 2018.

DOI/URL: World Scientific chapter URL https://www.worldscientific.com/doi/10.1142/9789813272880_0199.

Relevant result: Survey/source discussing inverse problems for hyperbolic equations and the nonlinear Lorentzian active measurement method. It records the uniqueness technology but does not provide the requested quantitative stability/error estimate.

3. Inverse Problems for Semilinear Wave Equations on Lorentzian Manifolds

Authors: Matti Lassas, Gunther Uhlmann, Yiran Wang.

Venue/year: *Communications in Mathematical Physics* 360, 555-609, 2018.

DOI/arXiv/URL: DOI 10.1007/s00220-018-3135-7; arXiv 1606.06261, <https://arxiv.org/abs/1606.06261>.

Relevant result: Theorems 1.2 and 1.3 extend the exact-data source-to-solution uniqueness theory to more general semilinear nonlinearities $H(x, u)$. They again give conformal recovery in general, with full metric recovery only under extra assumptions such as H independent of x or Ricci-flat metrics. The paper explicitly uses exact equality of measurement maps and gives no error estimate for approximate active data.

4. Inverse problems for quadratic derivative nonlinear wave equations

Authors: Yiran Wang, Ting Zhou.

Venue/year: Communications in Partial Differential Equations 44(11), 1140-1158, 2019.

DOI/arXiv/URL: DOI 10.1080/03605302.2019.1612908; arXiv 1612.04437, <https://arxiv.org/abs/1612.04437>.

Relevant result: Gives exact-data recovery results for a different class of quadratic derivative nonlinear wave equations, sometimes recovering the metric up to diffeomorphism under structural hypotheses. It does not cover the scalar quadratic nonlinearity $a(x)u^2$ in the problem statement and gives no approximate-data stability theorem.

5. Reconstruction of Lorentzian Manifolds from Boundary Light Observation Sets

Authors: Peter Hintz, Gunther Uhlmann.

Venue/year: International Mathematics Research Notices 2019(22), 6949-6987, 2019.

DOI/arXiv/URL: DOI 10.1093/imrn/rnx320; arXiv 1705.01215, <https://arxiv.org/abs/1705.01215>.

Relevant result: Reconstructs topological, differentiable, and conformal structures from boundary light observation sets under boundary/convexity-type hypotheses. This is related to the geometric reconstruction stage, but it is not the local active semilinear measurement problem and is not a quantitative stability theorem for L_V .

6. The Dirichlet-to-Neumann map for a semilinear wave equation on Lorentzian manifolds

Authors: Peter Hintz, Gunther Uhlmann, Jian Zhai.

Venue/year: Communications in Partial Differential Equations 47(12), 2363-2400, 2022.

DOI/arXiv/URL: DOI 10.1080/03605302.2022.2122837; arXiv 2103.08110, <https://arxiv.org/abs/2103.08110>.

Relevant result: Recovers the metric and coefficient, up to natural obstructions, from a boundary Dirichlet-to-Neumann map for a quartic semilinear wave equation. This is a boundary inverse problem with exact data and a different nonlinearity/data type, so it does not resolve the local active approximate-data question in the problem statement.

7. Stability estimates for inverse problems for semi-linear wave equations on Lorentzian manifolds

Authors: Matti Lassas, Tony Liimatainen, Leyter Potenciano-Machado, Teemu Tyni.

Venue/year: Analysis & PDE 18, 1065-1118, 2025.

DOI/arXiv/URL: DOI 10.2140/apde.2025.18.1065; arXiv 2106.12257, <https://arxiv.org/abs/2106.12257>.

Relevant result: Proves a Hölder stability estimate for recovering a potential q in $\square_g u + qu^m = 0$, $m \geq 4$, from boundary Dirichlet-to-Neumann data on a fixed Lorentzian background. This is the closest stability result found, but it recovers a coefficient rather than the metric/double cone, uses boundary DN data rather than local active data, and does not cover the quadratic case $m = 2$.

7.6.3.3. What remains unresolved

No published or otherwise citable result found during the live search proves the full statement in the problem statement. In particular, the literature does not currently provide a theorem that, for arbitrary globally hyperbolic 4-dimensional (M, g) and nowhere-vanishing $a(x)$, reconstructs the double cone and the full Lorentzian metric from an approximate local active map L_V with an error estimate depending only on a priori geometric bounds and the data error.

The exact active-measurement problem is substantially understood at the level of conformal reconstruction, and several related boundary or coefficient-recovery problems have unique-

ness or stability results. The missing pieces for the problem are the quantitative stability of the nonlinear active reconstruction pipeline and full metric recovery in the general quadratic $a(x)u^2$ setting without extra hypotheses.

7.6.4. Remaining open problems recorded by the database

7.6.4.1. Open problem 1

Let (M, g) be a fixed, known globally hyperbolic Lorentzian manifold with timelike boundary, and consider $\square_g u + q(x)u^2 = 0$. Establish a quantitative (for example, Hölder) stability estimate for recovering the unknown coefficient q on the causally observable region from the nonlinear Dirichlet-to-Neumann map, without assuming boundary convexity or that two lightlike geodesics intersect at most once.

Reference supplied by the database. Matti Lassas, Tony Liimatainen, Leyter Potenciano-Machado, and Teemu Tyni, “Stability and Lorentzian geometry for an inverse problem of a semilinear wave equation,” *Analysis & PDE* 18 (2025), no. 5, 1065–1118, discussion immediately after Corollary 1.2 (the cases $m = 2$ and $m = 3$ are left for future work, with $m = 2$ requiring different techniques), DOI: 10.2140/apde.2025.18.1065, <https://doi.org/10.2140/apde.2025.18.1065>; arXiv:2106.12257, <https://arxiv.org/abs/2106.12257>.

7.6.5. Primary-source pointer

The originating ICM record is *Inverse problems for linear and non-linear hyperbolic equations* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>.

7.7. 139. Inverse problems for linear and non-linear hyperbolic equations [2018-geom-142]

Record: 2018-geom-142

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Inverse problems for linear and non-linear hyperbolic equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>

7.7.1. Problem statement

Let (M, g) be a four-dimensional globally hyperbolic time-oriented Lorentzian manifold, written as $M = \mathbb{R} \times N$, and let $t_0 > 0$. Let $\mu \subset (0, t_0) \times N$ be a timelike curve, let $V \subset M$ be an open neighborhood of μ , and let $p^-, p^+ \in \mu$ with $p^- \ll p^+$. For a sufficiently small smooth source f supported in V , let u be the solution of the semilinear wave equation

$$\square_g u(x) + a(x)u(x)^2 = f(x), \quad x \in (-\infty, t_0) \times N,$$

with $u = 0$ and $f = 0$ in $(-\infty, 0) \times N$, where $a \in C^\infty(M)$ is nowhere vanishing, and let

$$L_V f := u|_V$$

be the observed response in V . Can one construct a single such source f so that the data consisting of the set V and the single measurement $L_V f$ uniquely determine the chronological diamond

$$I(p^-, p^+) = I^+(p^-) \cap I^-(p^+),$$

where $I^+(p)$ and $I^-(p)$ are the chronological future and past of p , and uniquely determine the metric g on $I(p^-, p^+)$?

7.7.2. Historical background, literature, and current progress

Verdict: partially resolved.

7.7.2.1. Precise problem checked

The problem asks whether, for the four-dimensional globally hyperbolic Lorentzian semilinear wave equation

$$\square_g u + a(x)u^2 = f, \quad a \in C^\infty(M), a \neq 0,$$

one can choose a single sufficiently small smooth source f supported in an observer neighborhood V of a timelike curve so that the single observed field $u|_V = L_V f$, together with V ,

uniquely determines both the chronological diamond

$$I(p^-, p^+) = I^+(p^-) \cap I^-(p^+)$$

and the Lorentzian metric g on that diamond. The distinctive point is the one-source/one-measurement requirement; the data are not the full local source-to-solution map for all small sources.

7.7.2.2. Literature and progress

1. Original source / survey context.

- Title: “Inverse problems for linear and non-linear hyperbolic equations”
- Author: Matti Lassas
- Venue/year: Proceedings of the International Congress of Mathematicians, Rio de Janeiro 2018; World Scientific, 2018/2019
- DOI/URL: [doi:10.1142/9789813272880_0199](https://doi.org/10.1142/9789813272880_0199)
- Relevance: This survey discusses the local inverse problems for nonlinear hyperbolic equations that underlie the question. It does not give a theorem proving the one-fixed-source statement.

2. Main full-map Lorentzian nonlinear result.

- Title: “Inverse problems for Lorentzian manifolds and non-linear hyperbolic equations”
- Authors: Yaroslav Kurylev, Matti Lassas, Gunther Uhlmann
- Venue/year: *Inventiones Mathematicae* 212, 781-857, 2018
- DOI/arXiv/URL: [doi:10.1007/s00222-017-0780-y](https://doi.org/10.1007/s00222-017-0780-y); arXiv:1405.3386; <https://arxiv.org/abs/1405.3386>
- Result checked: The paper proves recovery results for a four-dimensional Lorentzian manifold from local source-to-solution data for nonlinear wave equations, using nonlinear interaction of waves. The data are a local source-to-solution map for many small sources supported in an observation region, not a single fixed source. The recovery is formulated in terms of the causal geometry/conformal structure, with metric recovery requiring additional information or hypotheses in the surrounding theory. Thus this is major partial progress but does not resolve the exact question.

3. General semilinear source-to-solution map result.

- Title: “Inverse Problems for Semilinear Wave Equations on Lorentzian Manifolds”
- Authors: Matti Lassas, Gunther Uhlmann, Yiran Wang
- Venue/year: *Communications in Mathematical Physics* 360, 555-609, 2018
- DOI/arXiv/URL: [doi:10.1007/s00220-018-3135-7](https://doi.org/10.1007/s00220-018-3135-7); arXiv:1606.06261; <https://arxiv.org/abs/1606.06261>
- Result checked: The theorem determines the conformal class of the Lorentzian metric from the full source-to-solution map for a semilinear wave equation, and obtains metric determination only under extra hypotheses such as special nonlinearity/geometry assumptions. It does not treat data consisting of one single source response $L_V f$, and its hypotheses/conclusions do not imply the full one-source metric recovery asked in the problem statement.

4. Disjoint source/receiver full-map progress.

- Title: “Inverse problems for nonlinear hyperbolic equations with disjoint sources and receivers”
- Authors: Ali Feizmohammadi, Matti Lassas, Lauri Oksanen
- Venue/year: Forum of Mathematics, Pi 9:e10, 2021
- DOI/arXiv/URL: [doi:10.1017/fmp.2021.11](https://doi.org/10.1017/fmp.2021.11); arXiv:2006.12158; <https://arxiv.org/abs/2006.12158>
- Result checked: This paper proves metric recovery in a causal diamond from full source-to-solution data in a disjoint source/receiver setting for equations of the form $\square_g u + u^m = f$, under restrictions including $m \geq 3$ and excluding the conformal exceptional case $(n, m) = (3, 3)$. The stated equation is quadratic, $m = 2$, with a general nowhere-vanishing coefficient $a(x)$, and the problem asks for one fixed source rather than a full map. Hence this is partial progress only.

5. Single-observer but still full-map progress.

- Title: “Determining Lorentzian manifold from non-linear wave observation at a single point”
- Authors: Medet Nursultanov, Lauri Oksanen, Leo Tzou
- Venue/year: arXiv preprint, 2023; Journal of Differential Equations 444, Article 113563, 2025
- DOI/arXiv/URL: arXiv:2310.06925; [doi:10.1016/j.jde.2025.113563](https://doi.org/10.1016/j.jde.2025.113563); <https://arxiv.org/abs/2310.06925>
- Result checked: This work reduces the observation geometry to a timelike curve/single-point-type observation for a nonlinear wave model, but it still uses a source-to-solution map for all suitable sources and concerns a cubic nonlinearity with conformal recovery. It does not prove that a single smooth source response determines the metric for the quadratic problem stated here.

6. Related one-measurement inverse-problem ideas.

- Title: “An inverse problem for the wave equation with one measurement and the pseudorandom source”
- Authors: Tapio Helin, Matti Lassas, Lauri Oksanen
- Venue/year: Analysis & PDE 5(5), 887-912, 2012
- DOI/URL: [doi:10.2140/apde.2012.5.887](https://doi.org/10.2140/apde.2012.5.887)
- Relevance: This shows that one carefully chosen pseudorandom source can determine boundary spectral data in a linear wave inverse problem. It is conceptually relevant to the one-measurement theme, but it is not the Lorentzian semilinear metric-recovery problem in the problem statement.
- Title: “Global recovery of a time-dependent coefficient for the wave equation from a single measurement”
- Authors: Ali Feizmohammadi, Yavar Kian
- Venue/year: Asymptotic Analysis 131(3-4), 513-539, 2023
- DOI/arXiv/URL: [doi:10.3233/ASY-221779](https://doi.org/10.3233/ASY-221779); arXiv:2002.11598; <https://arxiv.org/abs/2002.11598>
- Relevance: This is a genuine single-measurement theorem, but it recovers a coefficient for a wave equation on a known background geometry, not an unknown Lorentzian metric in the nonlinear four-dimensional setting of the problem.

7.7.2.3. What remains unresolved

the exact question remains unresolved in the literature I found. Existing nonlinear Lorentzian recovery theorems require the full source-to-solution map, or recover only

conformal/geometric structure without the full metric unless extra assumptions are imposed. Later results reduce receiver geometry or prove one-measurement recovery for different unknowns, but they do not cover all hypotheses and the full conclusion of the problem: a single smooth source $f \subset V$ for $\square_g u + a(x)u^2 = f$ in four spacetime dimensions determining both $I(p^-, p^+)$ and the metric g on that diamond.

7.7.3. Remaining open problems recorded by the database

7.7.3.1. Open problem 1

Let (M, g) be a compact non-trapping Riemannian manifold with nonempty boundary. Prove that its full scattering relation (including the travel time of every boundary-to-boundary geodesic) determines (M, g) up to a boundary-fixing isometry.

Reference supplied by the database. Günther Uhlmann, “The Cauchy Data and the Scattering Relation,” in *Geometric Methods in Inverse Problems and PDE Control*, IMA Volumes in Mathematics and its Applications **137**, Springer, 2004, pp. 263–287, Conjecture 4.1, <https://sites.math.washington.edu/~gunther/Papers/imafinal.pdf>. The non-trapping formulation and its explicit role as the remaining metric-recovery step after one pseudorandom-source measurement are stated in Tapio Helin, Matti Lassas, and Lauri Oksanen, “An Inverse Problem for the Wave Equation with One Measurement and the Pseudorandom Source,” *Analysis & PDE* **5** (2012), no. 5, 887–912, introduction after Corollary 1, <https://doi.org/10.2140/apde.2012.5.887>; arXiv:1011.2527.

7.7.4. Primary-source pointer

The originating ICM record is *Inverse problems for linear and non-linear hyperbolic equations* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>.

7.8. 140. Liouville quantum gravity as a metric space and a scaling limit [2018-geom-143]

Record: 2018-geom-143

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Liouville quantum gravity as a metric space and a scaling limit

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=226>

7.8.1. Problem statement

Let Q_n be a uniformly random planar quadrangulation with n faces. Regard Q_n as a topological surface by replacing each quadrilateral face with a Euclidean unit square $[0, 1]^2$ and gluing square edges according to the adjacency relations of Q_n , identifying matching boundary lengths. Choose a conformal map from this random surface to the Riemann sphere $\mathbb{S}^2$, and in a local conformal coordinate $z = x + iy$ write the induced metric as

$$e^{\lambda_n(z)}(dx^2 + dy^2),$$

thereby defining the random conformal factor λ_n .

Prove or disprove that, as $n \rightarrow \infty$, the fields λ_n converge in law as random distributions on $\mathbb{S}^2$ to $\sqrt{8/3}h$, where h is a Gaussian free field on $\mathbb{S}^2$, i.e. a Gaussian random distribution whose covariance is locally given by the Green's function of the Laplacian. Equivalently, prove or disprove that the associated area measures

$$e^{\lambda_n(z)} dx dy$$

converge in law weakly as random measures to the $\sqrt{8/3}$ -Liouville quantum gravity area measure associated with h , namely the Gaussian multiplicative chaos measure

$$\mu_h^{\sqrt{8/3}} = \lim_{\varepsilon \rightarrow 0} \varepsilon^{(\sqrt{8/3})^2/2} e^{\sqrt{8/3} h_\varepsilon(z)} dx dy = \lim_{\varepsilon \rightarrow 0} \varepsilon^{4/3} e^{\sqrt{8/3} h_\varepsilon(z)} dx dy,$$

where $h_\varepsilon(z)$ denotes the circle-average regularization of h .

7.8.2. Historical background, literature, and current progress

7.8.3. Verdict: partially resolved

7.8.3.1. Precise problem checked

The problem asks for a theorem proving that if Q_n is a uniformly random spherical planar quadrangulation with n faces, made into a polyhedral Riemann surface by replacing each face by a Euclidean unit square and then conformally uniformizing to $\mathbb{S}^2$, then the resulting conformal factor λ_n , or equivalently the pushed-forward Riemannian area measure $e^{\lambda_n(z)} dx dy$,

converges to the $\gamma = \sqrt{8/3}$ Liouville quantum gravity sphere measure associated with a Gaussian free field.

I found no published or otherwise citable theorem that proves this exact finite-sphere, uniform-quadrangulation, unit-square/Riemann-uniformization statement. The literature proves the metric-measure scaling limit of uniform quadrangulations and identifies the Brownian map with the $\sqrt{8/3}$ -LQG metric sphere, and it proves conformal-embedding convergence for some related random planar map models/embeddings, but these results do not imply convergence of the specific conformal factors or area measures in the problem statement.

7.8.3.2. Relevant literature and current progress

1. **Grégory Miermont**, “The Brownian map is the scaling limit of uniform random plane quadrangulations”, *Acta Mathematica* 210 (2013), 319-401. DOI: 10.1007/s11511-013-0096-8. arXiv:1104.1606.

This proves that uniform random plane quadrangulations, with graph distance rescaled by $n^{-1/4}$, converge to the Brownian map. This is a central result for the metric scaling limit of the same discrete objects appearing in the problem statement. It does not prove convergence of the Riemann uniformization of the unit-square surface, nor convergence of the conformal factor λ_n or of the pushed-forward Euclidean area measure in a fixed conformal coordinate.

2. **Jean-François Le Gall**, “Uniqueness and universality of the Brownian map”, *Annals of Probability* 41 (2013), 2880-2960. DOI: 10.1214/12-AOP792. arXiv:1105.4842.

This gives Brownian-map universality results for broad classes of random planar maps. It supports the metric-measure side of the $\sqrt{8/3}$ -LQG picture but, like Miermont’s theorem, is not a theorem about the conformal embedding obtained by gluing unit squares and uniformizing to the sphere.

3. **Bertrand Duplantier and Scott Sheffield**, “Liouville quantum gravity and KPZ”, *Inventiones Mathematicae* 185 (2011), 333-393. DOI: 10.1007/s00222-010-0308-1. arXiv:0808.1560.

This constructs and analyzes Liouville quantum gravity measures via Gaussian multiplicative chaos and explains the $\gamma = \sqrt{8/3}$ connection with random planar maps through KPZ. It supplies the rigorous LQG-measure framework used in the statement, but it does not prove the convergence of uniform quadrangulations under the unit-square conformal uniformization.

4. **François David, Antti Kupiainen, Rémi Rhodes, and Vincent Vargas**, “Liouville quantum gravity on the Riemann sphere”, *Communications in Mathematical Physics* 342 (2016), 869-907. DOI: 10.1007/s00220-016-2579-6. arXiv:1410.7318.

This gives a rigorous construction of Liouville quantum gravity on the sphere. It establishes the continuum target object, not the convergence of the discrete quadrangulation surfaces in the problem statement.

5. **Jason Miller and Scott Sheffield**, “Liouville quantum gravity and the Brownian map I: The QLE(8/3,0) metric”, *Inventiones Mathematicae* 219 (2020), 75-152. DOI: 10.1007/s00222-019-00905-1. arXiv:1507.00719; “II: Geodesics and continuity of the embedding”, *Annals of Probability* 49 (2021), 2732-2829. DOI: 10.1214/21-AOP1506. arXiv:1605.03563; “III: The conformal structure is determined”, *Probability Theory and Related Fields* 179 (2021), 1183-1211. DOI: 10.1007/s00440-021-01026-8.

arXiv:1608.05391.

This trilogy identifies the Brownian map with the $\sqrt{8/3}$ -Liouville quantum gravity sphere equipped with its metric and shows that the conformal structure is determined in the continuum. Together with the quadrangulation-to-Brownian-map convergence, it is very strong evidence for the assertion. However, the results do not state or imply, without additional work, that the conformal maps obtained by uniformizing the discrete unit-square quadrangulation surfaces have conformal factors or pushed-forward area measures converging to the LQG measure.

6. **Nicolas Curien, “A glimpse of the conformal structure of random planar maps”, *Communications in Mathematical Physics* 333 (2015), 1417-1463. DOI: 10.1007/s00220-014-2196-5. arXiv:1308.1807.**

This studies conformal structure for random planar maps using circle packing and Riemann uniformization in a related setting. It gives partial evidence and techniques for conformal convergence questions, but it is not a proof of the finite uniform quadrangulation/unit-square sphere statement in this problem.

7. **Ewain Gwynne, Nina Holden, and Xin Sun, “Convergence of uniform triangulations under the Cardy embedding”, *Acta Mathematica* 230 (2023), 93-203. DOI: 10.4310/ACTA.2023.v230.n1.a2. arXiv:1905.13207.**

This proves convergence to $\sqrt{8/3}$ -LQG for uniform triangulations under the Cardy embedding, a major conformal-embedding result for a closely related random planar map model. It is not the same model and not the same embedding: the problem concerns uniform quadrangulations made into unit-square Riemann surfaces and then uniformized to the sphere.

8. **Federico Bertacco, Ewain Gwynne, and Scott Sheffield, “Scaling limits of planar maps under the Smith embedding”, *Annals of Probability* 53 (2025). DOI: 10.1214/24-AOP1731. arXiv:2306.02988.**

This proves γ -LQG scaling limits for mated-CRT maps under the Smith embedding. It is another strong conformal-embedding theorem, but its hypotheses and embedding are different from the uniform quadrangulation/unit-square Riemann uniformization in the problem statement.

9. **Nina Holden and Pu Yu, “Circle packing and Riemann uniformization of random planar maps in an ergodic scale-free environment”, preprint (2026). arXiv:2603.06528.**

This recent preprint studies circle packing and Riemann uniformization for infinite random planar maps in a scale-free environment. It gives relevant progress on Riemann-uniformization methods but does not prove the specific convergence of finite uniform quadrangulations with unit-square faces to the $\sqrt{8/3}$ -LQG sphere measure.

10. **Jason Miller, “Liouville quantum gravity as a metric space and a scaling limit”, *Proceedings of the International Congress of Mathematicians 2018, Vol. IV, World Scientific* (2018), 2945-2971. arXiv:1712.01571.**

This survey explains the Brownian-map/ $\sqrt{8/3}$ -LQG metric equivalence and its role as the scaling limit of random planar maps. It is useful background and likely close to the source of the formulation, but it is a survey of metric scaling-limit results and does not contain a theorem proving the exact unit-square conformal-factor convergence asserted here.

7.8.3.3. What remains unresolved

The missing theorem is a direct convergence result for the conformal uniformization of the discrete polyhedral surfaces obtained from uniform spherical quadrangulations by gluing Euclidean unit squares. In particular, the literature I found does not prove that the random measures $e^{\lambda_n(z)} dx dy$, after a specified normalization of the Möbius freedom, converge to the $\sqrt{8/3}$ -LQG sphere area measure, nor that the fields λ_n converge as distributions to $\sqrt{8/3}$ times a Gaussian free field. Existing Brownian-map convergence and Brownian-map/LQG identification resolve the metric-measure scaling-limit side, while Cardy/Smith/circle-packing/Riemann-uniformization results resolve related models or embeddings, so the correct status is **partially resolved**, not fully resolved.

7.8.4. Remaining open problems recorded by the database

7.8.4.1. Open problem 1

Let $T_{\infty,\infty}$ be the uniform infinite half-plane triangulation, conformally uniformized by $\mathbb{H}$, and explore it by an independent chordal SLE_6 . If $\mathcal{H}^+(i)$ is the right horodistance after the i th peeling step and $\eta_i^+ := \mathcal{H}^+(i+1) - \mathcal{H}^+(i + \frac{1}{2})$ is the discrepancy between the next peeling edge and the half-step prediction, prove Curien's condition (*):

$$\sup_{t \in [0,1]} \frac{\eta_1^+ + \cdots + \eta_{[tn]}^+}{n^{2/3}} \xrightarrow[n \rightarrow \infty]{\mathbb{P}} 0.$$

Reference supplied by the database. Nicolas Curien, "A Glimpse of the Conformal Structure of Random Planar Maps," *Communications in Mathematical Physics* **333** (2015), no. 3, 1417–1463, Section 2.5, condition (*), <https://doi.org/10.1007/s00220-014-2196-5>; arXiv:1308.1807.

7.8.4.2. Open problem 2

Uniformize the uniform infinite half-plane triangulation in $\mathbb{H}$ so that the root-edge endpoints map to $-\frac{1}{2}$ and $\frac{1}{2}$ and the infinite end maps to ∞ . If $\mathcal{X}_n$ is the image of the n th boundary vertex to the right of the root, prove

$$\frac{\log \mathcal{X}_n}{\log n} \xrightarrow[n \rightarrow \infty]{\mathbb{P}} 3,$$

and determine whether the family $(n^{-3} \mathcal{X}_n)_{n \geq 1}$ is tight.

Reference supplied by the database. Nicolas Curien, "A Glimpse of the Conformal Structure of Random Planar Maps," *Communications in Mathematical Physics* **333** (2015), no. 3, 1417–1463, Section 5, first question ("location of the n th vertex"), <https://doi.org/10.1007/s00220-014-2196-5>; arXiv:1308.1807.

7.8.4.3. Open problem 3

Let ν be a random locally finite measure on $\mathbb{R}$, independent of a chordal SLE_6 curve γ in $\mathbb{H}$ from 0 to ∞ , and set

$$\Gamma_\nu^+ = \{(t, \nu[0, \gamma_t]) : \gamma_t \in \mathbb{R}_+\}, \quad \Gamma_\nu^- = \{(t, \nu[\gamma_t, 0]) : \gamma_t \in \mathbb{R}_-\}.$$

Let S^+ and S^- be independent spectrally negative 3/2-stable Lévy processes and let

$$\mathcal{M}^\pm = \{(t, -S_t^\pm) : S_t^\pm = \inf_{0 \leq s \leq t} S_s^\pm\}.$$

Does the identity in law $(\Gamma_\nu^+, \Gamma_\nu^-) \stackrel{d}{=} (\mathcal{M}^+, \mathcal{M}^-)$, up to a reparametrization of the first coordinate, uniquely characterize the law of ν ?

Reference supplied by the database. Nicolas Curien, “A Glimpse of the Conformal Structure of Random Planar Maps,” *Communications in Mathematical Physics* **333** (2015), no. 3, 1417–1463, Section 5.2, Question “Characterization,” <https://doi.org/10.1007/s00220-014-2196-5>; arXiv:1308.1807.

7.8.5. Primary-source pointer

The originating ICM record is *Liouville quantum gravity as a metric space and a scaling limit* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=226>.

8. Geometric flows and geometric PDE

The seven records in this chapter concern the use of evolution equations and nonlinear PDE to construct or classify geometric structures. Ricci-type flows, elliptic equations, and canonical-metric problems illustrate the characteristic division between short-time or conditional theory and global existence, convergence, and singularity control.

Coverage: 7 website records; **source years:** 1998–2014; **separately listed descendant open problems:** 9.

8.1. 141. Machedon (2) [1998-geom-71]

Record: 1998-geom-71

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Machedon (2)

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=51>

8.1.1. Problem statement

Let n be the number of spatial variables, let M be a Riemannian manifold, and let $\phi : \mathbb{R}^{1+n} \rightarrow M$ be a wave map satisfying, in local coordinates,

$$\square \phi^i + \Gamma_{jk}^i(\phi) \partial_\alpha \phi^j \partial^\alpha \phi^k = 0,$$

where indices are raised with the Minkowski metric $(-1, 1, \dots, 1)$.

Prove that the Cauchy problem for this equation is locally well posed in time for initial data

$$(\phi(0, \cdot), \partial_t \phi(0, \cdot)) = (\phi_0, \phi_1)$$

of Sobolev regularity

$$H^{n/2}(\mathbb{R}^n) \times H^{n/2-1}(\mathbb{R}^n),$$

using both the null-form structure

$$Q_0(u, v) = \partial_\alpha u \partial^\alpha v$$

occurring in the nonlinearity and the geometric information coming from the target manifold. In the model case $M = S^{k-1} \subset \mathbb{R}^k$, this geometric structure is expressed by the equivalent constrained equation

$$\square \phi + \phi(\partial_\alpha \phi \cdot \partial^\alpha \phi) = 0, \quad |\phi| = 1.$$

8.1.2. Historical background, literature, and current progress

8.1.3. Verdict: partially resolved

8.1.3.1. Precise problem checked

The problem asks for the sharp endpoint local well-posedness theory for wave maps

$$\square \phi^i + \Gamma_{jk}^i(\phi) \partial_\alpha \phi^j \partial^\alpha \phi^k = 0$$

from $\mathbb{R}^{1+n}$ to a Riemannian target M , with arbitrary initial data in the scaling-critical Sobolev class

$$H^{n/2}(\mathbb{R}^n) \times H^{n/2-1}(\mathbb{R}^n).$$

The issue is not merely the scalar null form Q_0 ; the endpoint theory must also use the target geometry, e.g. for $M = S^{k-1}$ the constrained equation

$$\square \phi + \phi(\partial_\alpha \phi \cdot \partial^\alpha \phi) = 0, \quad |\phi| = 1.$$

No published or otherwise standard citable result was found that proves this full endpoint local well-posedness statement for arbitrary large critical Sobolev data, all spatial dimensions, and all Riemannian targets. The problem is therefore not fully resolved.

8.1.3.2. Main partial progress

1. Local and global results for wave maps. I

Daniel Tataru. *Communications in Partial Differential Equations* 23 (1998), no. 9-10, 1781-1793. DOI: 10.1080/03605309808821400. URL: <https://math.berkeley.edu/~tataru/papers/wm.pdf>

This established scale-invariant small-data theory in critical Besov-type spaces in high dimensions. It was a critical-regularity result, but the data space is smaller than

$H^{n/2} \times H^{n/2-1}$, and the result is small-data, not arbitrary large endpoint-data local well-posedness.

2. On global existence and scattering for the wave maps equation

Daniel Tataru. *American Journal of Mathematics* 123 (2001), no. 1, 37-77. DOI: 10.1353/ajm.2001.0005.

This extended the critical small-data wave map theory, including low-dimensional aspects, in spaces adapted to the null form. It does not settle arbitrary critical Sobolev data for general targets.

3. Global regularity of wave maps I. Small critical Sobolev norm in high dimension

Terence Tao. *International Mathematics Research Notices* 2001, no. 6, 299-328. URL: <https://doi.org/10.1155/S1073792801000128>

Tao proved small-data global regularity at the critical Sobolev norm in high dimensions for sphere targets, using a microlocal renormalization/gauge mechanism. This is genuine $\dot{H}^{n/2} \times \dot{H}^{n/2-1}$ progress, but only for small data and special targets/dimensions.

4. Global regularity of wave maps II. Small energy in two dimensions

Terence Tao. *Communications in Mathematical Physics* 224 (2001), no. 2, 443-544. DOI: 10.1007/PL00005588.

This gives small-energy regularity in the two-dimensional critical case. It is important endpoint progress, but again small-data and not a general large-data local theory for all targets.

5. On the global regularity of wave maps in the critical Sobolev norm

Sergiu Klainerman and Igor Rodnianski. *International Mathematics Research Notices* 2001, no. 13, 655-677.

This is another critical Sobolev small-data result, using geometric/gauge structure. Its hypotheses do not cover arbitrary large critical data for arbitrary Riemannian targets.

6. On the well-posedness of the wave map problem in high dimensions

Andrea Nahmod, Atanas Stefanov, and Karen Uhlenbeck. *Communications in Analysis and Geometry* 11 (2003), no. 1, 49-83. URL: <https://www.intlpress.com/site/pub/pages/journals/items/cag/content/vols/0011/0001/a004/>

This treats high-dimensional wave maps into compact Lie groups and Riemannian symmetric spaces, obtaining small critical-data well-posedness in that geometric setting. It is special-target and small-data progress.

7. Rough solutions for the wave maps equation

Daniel Tataru. *American Journal of Mathematics* 127 (2005), no. 2, 293-377. DOI: 10.1353/ajm.2005.0014. PDF: <https://math.berkeley.edu/~tataru/papers/wavemaps.pdf>

This is the strongest general-target critical Sobolev result found. Tataru proves global well-posedness for initial data small in the critical homogeneous Sobolev space for wave maps into a uniformly isometrically embedded Riemannian manifold. The paper explicitly says the analogous large-data problem remains largely open. Thus it does not imply the full arbitrary-data local well-posedness statement.

8. Energy Dispersed Large Data Wave Maps in 2 + 1 Dimensions

Jacob Sterbenz and Daniel Tataru. *Communications in Mathematical Physics* 298 (2010), no. 1, 139-230. DOI: 10.1007/s00220-010-1061-4.

This develops a large-data theory in the energy-critical 2 + 1-dimensional case under energy-dispersion assumptions. It is a major large-data endpoint result, but it is dimension-specific and assumes additional dispersion/threshold-type hypotheses.

9. Regularity of Wave-Maps in Dimension $2 + 1$

Jacob Sterbenz and Daniel Tataru. *Communications in Mathematical Physics* 298 (2010), no. 1, 231-264. DOI: 10.1007/s00220-010-1062-3.

For compact targets in $2+1$ dimensions, this proves that regularity and dispersive bounds persist unless a soliton-like concentration occurs; together with the companion paper it yields threshold-type global regularity below harmonic-map obstructions. This is not the full arbitrary-target/all-dimensions endpoint local theory asked in the problem statement.

10. Global regularity of wave maps V. Large data local wellposedness and perturbation theory in the energy class

Terence Tao. arXiv preprint, 2008. arXiv: 0808.0368. URL: <https://arxiv.org/abs/0808.0368>

This gives large-data local well-posedness in the energy class for wave maps from $\mathbb{R}^{1+2}$ to hyperbolic space $\mathbb{H}^m$. It is strong evidence and a major special case, but it is restricted to dimension 2 and hyperbolic targets, and it remains an unpublished preprint.

11. Unconditional well-posedness for wave maps

Nader Masmoudi and Fabrice Planchon. *Journal of Hyperbolic Differential Equations* 9 (2012), no. 2, 223-237. DOI: 10.1142/S0219891612500075. arXiv: 1111.4374. URL: <https://arxiv.org/abs/1111.4374>

This proves uniqueness in the natural critical class $C([0, T]; \dot{H}^{d/2}) \times C^1([0, T]; \dot{H}^{d/2-1})$ for $d \geq 4$, under bounded-curvature hypotheses. It is a uniqueness theorem, not a complete existence and continuous-dependence theorem for arbitrary endpoint data.

12. Ill-posedness for one-dimensional wave maps at the critical regularity

Terence Tao. *American Journal of Mathematics* 122 (2000), no. 3, 451-463. arXiv: math/9811169. URL: <https://arxiv.org/abs/math/9811169>

This shows that in $\mathbb{R}^{1+1}$ the wave map equation is in general ill-posed in the critical space $\dot{H}^{1/2}$. Therefore, if the problem statement is read literally as including $n = 1$, the endpoint well-posedness assertion is not true in full generality. Most positive critical wave-map theory is consequently interpreted for $n \geq 2$ or with additional structure.

8.1.3.3. What remains unresolved

the exact statement would require a citable theorem giving local well-posedness for arbitrary data in $H^{n/2} \times H^{n/2-1}$, using both null form and geometry, for general Riemannian targets and all intended spatial dimensions. Existing literature proves:

- small-data global well-posedness at critical Sobolev regularity for broad classes of targets, culminating in Tataru's general-target small-data theorem;
- sharper special-target results such as spheres, compact Lie groups, symmetric spaces, and hyperbolic spaces;
- substantial $2 + 1$ -dimensional energy-critical large-data/threshold theory;
- uniqueness in critical classes in some higher-dimensional settings.

These do not cover arbitrary large endpoint Sobolev data for all general Riemannian targets. The current status of the problem is therefore **partially resolved**, not fully resolved.

8.1.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

8.1.5. Primary-source pointer

The originating ICM record is *Machedon* (2) (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=51>.

8.2. 142. On non-Kähler Calabi-Yau Threefolds with Balanced Metrics [2010-ag-139]

Record: 2010-ag-139

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: On non-Kähler Calabi-Yau Threefolds with Balanced Metrics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=711>

8.2.1. Problem statement

For an integer $k \geq 2$, let X be a non-Kähler Calabi-Yau threefold obtained by the Clemens-Friedman construction whose underlying smooth manifold is the connected sum $\#_k(S^3 \times S^3)$ of k copies of $S^3 \times S^3$, and let Ω be a nowhere-vanishing holomorphic three-form on X . Does there exist a Hermitian metric ω on X and a Hermitian metric h on the holomorphic tangent bundle $T'X$ such that, writing F_h for the curvature of h , R_ω for the curvature of the Hermitian connection determined by ω , and α' for the string parameter, the Strominger system

$$d(\|\Omega\|_\omega \omega^2) = 0, \quad F_h^{2,0} = F_h^{0,2} = 0, \quad F_h \wedge \omega^2 = 0,$$

$$\sqrt{-1} \partial \bar{\partial} \omega = \frac{\alpha'}{4} (\text{tr}(R_\omega \wedge R_\omega) - \text{tr}(F_h \wedge F_h))$$

is satisfied?

8.2.2. Historical background, literature, and current progress

8.2.3. Verdict: partially resolved

8.2.3.1. Precise problem

For each integer $k \geq 2$, and for a non-Kähler Calabi-Yau threefold X obtained by the Clemens-Friedman conifold-transition construction with underlying smooth manifold $\#_k(S^3 \times S^3)$, decide whether there are Hermitian metrics ω on X and h on $T^{\{1,0\}}X$ solving the full Strominger/Hull-Strominger system:

- the conformally balanced equation $d(\|\Omega\|_\omega \omega^2) = 0$;
- the Hermitian-Yang-Mills equations for the Chern curvature F_h on $T^{\{1,0\}}X$;
- the anomaly cancellation/Bianchi identity $i \partial \bar{\partial} \omega = (\alpha'/4) (\text{tr}(R_\omega \wedge R_\omega) - \text{tr}(F_h \wedge F_h))$.

8.2.3.2. Verdict

The problem is partially resolved, not fully resolved. The literature proves the conformally balanced part and the Hermitian-Yang-Mills tangent-bundle part for the relevant conifold-transition/Clemens setting, including the $\#_k(S^3 \times S^3)$ examples. I found no citable result proving the final anomaly cancellation equation for these Clemens-Friedman threefolds, and recent survey/lecture literature explicitly states that this additional equation has not yet been solved through conifold transitions.

8.2.3.3. Progress and relevant literature

1. C. Herbert Clemens, “Double solids”, *Advances in Mathematics* 47 no. 2 (1983), 107-230. DOI: 10.1016/0001-8708(83)90025-7. This is part of the original conifold-transition/Clemens background.
2. Robert Friedman, “Simultaneous resolution of threefold double points”, *Mathematische Annalen* 274 (1986), 671-689. DOI: 10.1007/BF01458602; and “On threefolds with trivial canonical bundle”, in *Complex Geometry and Lie Theory*, Proc. Sympos. Pure Math. 53, AMS, 1991, 103-134. DOI: 10.1090/pspum/053/1141199. These give the Friedman side of the Clemens-Friedman construction and smoothing criteria for nodal Calabi-Yau threefolds.
3. P. Lu and G. Tian, “The complex structure on a connected sum of $S^3 \times S^3$ with trivial canonical bundle”, *Mathematische Annalen* 298 (1994), 761-764. DOI: 10.1007/BF01459760; and “Complex structures on connected sums of $S^3 \times S^3$ ”, in *Manifolds and Geometry*, Symposia Mathematica XXXVI, Cambridge Univ. Press, 1996, 284-293. These are standard sources for the non-Kähler Calabi-Yau complex structures on connected sums of copies of $S^3 \times S^3$.
4. Jixiang Fu, “On non-Kähler Calabi-Yau threefolds with balanced metrics”, *Proceedings of the International Congress of Mathematicians 2010*, Vol. II, Hindustan Book Agency, 2010, 705-716. DOI/URL: https://doi.org/10.1142/9789814324359_0070. This ICM article identifies the Strominger system as the intended canonical-geometry problem for these balanced non-Kähler Calabi-Yau threefolds.
5. Jixiang Fu, Jun Li, and Shing-Tung Yau, “Balanced metrics on non-Kähler Calabi-Yau threefolds”, *Journal of Differential Geometry* 90 no. 1 (2012), 81-129. DOI: 10.4310/jdg/1335209490; arXiv: 0809.4748. The main theorem constructs balanced metrics on non-Kähler Calabi-Yau threefolds obtained by smoothing after contracting $(-1, -1)$ curves. Corollary 1.3 applies to the complex structures on $\#_k(S^3 \times S^3)$ for any $k \geq 2$. The paper also explains that this is meant as a step toward the Strominger system; it does not solve the anomaly cancellation equation.
6. Yuri D. Bozhkov, “The specific Hermitian geometry of certain three-folds”, *Rivista di Matematica della Università di Parma* (5) 4 (1995), 61-68. MR1395324; Zbl 0865.53059. This proves stability properties of the tangent bundle for the $\#_k(S^3 \times S^3)$ complex threefolds. Via Li-Yau’s non-Kähler Hermitian-Einstein theorem, this gives Hermitian-Yang-Mills metrics in the relevant stable setting, but it does not address the anomaly equation.
7. Tristan C. Collins, Sebastien Picard, and Shing-Tung Yau, “Stability of the tangent bundle through conifold transitions”, *Communications on Pure and Applied Mathematics* 77 (2024), 284-371. DOI: 10.1002/cpa.22135; arXiv: 2102.11170. Their main theorem proves that for small smoothings in a conifold transition, $T^{\{1, 0\}}X_t$ ad-

mits a Hermitian-Yang-Mills metric with respect to the Fu-Li-Yau conformally balanced metrics, and gives quantitative behavior near vanishing cycles. This supplies the second equation in the stated system for the tangent bundle, but again not the anomaly equation.

8. Sebastien Picard, “Calabi-Yau threefolds across quadratic singularities”, in 2024 MATRIX Annals, Part II, MATRIX Book Series 8, Springer, 2026. DOI: 10.1007/978-3-032-16206-9_17; arXiv: 2501.19313. Theorem 3.2 summarizes the current conifold-transition results: for small t , the non-Kähler smoothing has (Ω_t, g_t, h_t) satisfying $d(|\Omega_t|_{\Omega_t} \Omega_t^2) = 0$ and $F_{h_t} \wedge \Omega_t^2 = 0$ on the tangent bundle. Immediately afterward, the text says that the heterotic Bianchi identity/anomaly cancellation equation is an additional equation and “has not yet been solved through conifold transitions”; it is stated as Yau’s conjecture.
9. Anna Fino, Gueo Grantcharov, and Luigi Vezzoni, “Solutions to the Hull-Strominger System with Torus Symmetry”, Communications in Mathematical Physics 388 (2021), 947-967. DOI: 10.1007/s00220-021-04223-7; arXiv: 1901.10322. This gives full Hull-Strominger solutions on other non-Kähler Calabi-Yau threefolds, including certain manifolds with $S^3 \times S^3$ summands, but the topology and construction are different from the pure Clemens-Friedman $\#_k(S^3 \times S^3)$ problem considered here. It is related progress, not a resolution.

8.2.3.4. What remains unresolved

No located paper proves that, for every $k \geq 2$ and every Clemens-Friedman non-Kähler Calabi-Yau threefold diffeomorphic to $\#_k(S^3 \times S^3)$, one can choose the conformally balanced metric ω and tangent-bundle Hermitian-Yang-Mills metric h so that the coupled anomaly cancellation equation also holds. The missing step is not just a technical detail: recent accounts isolate it as the remaining Bianchi/anomaly equation in Yau’s conjectural Strominger-system picture through conifold transitions.

8.2.4. Remaining open problems recorded by the database

8.2.4.1. Open problem 1

Given a Clemens-Friedman complex threefold $X \cong \#_{k \geq 2}(S^3 \times S^3)$ and a holomorphic vector bundle $V \rightarrow X$, find a positive Hermitian form ω and a Hermitian metric h on V satisfying the Bianchi identity

$$dd^c \omega = -\operatorname{tr}(F_h \wedge F_h).$$

Reference supplied by the database. Mario García-Fernández, “Pluriclosed flow and the Hull-Strominger system,” Herts Mathematical Physics Seminar, University of Hertfordshire, 30 October 2024, slide 43, Problem, <https://hertsmathphys.github.io/2024-10-30-Garc%C3%ADa.pdf>.

8.2.5. Primary-source pointer

The originating ICM record is *On non-Kähler Calabi-Yau Threefolds with Balanced Metrics* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=711>.

8.3. 143. Extremal Kähler metrics [2014-geom-123]

Record: 2014-geom-123

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Extremal Kähler metrics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1035>

8.3.1. Problem statement

Let M be a Fano manifold, and let $m > 0$ be such that there is a smooth divisor $D \in |mK_M^{-1}|$. Consider the conical continuity method with initial equation

$$\text{Ric}(\omega_t) = t\omega_t + \frac{1-t}{m}[D],$$

where $[D]$ denotes the current of integration over D .

Determine whether the following assertion holds. If, as t approaches a singular time $T < 1$, the solutions converge to a pair (W, Δ) , where W is a $\mathbf{Q}$ -Fano variety and Δ is the limiting Weil divisor corresponding to D , then the process can be continued on the current pair by solving the weak conical Kähler-Ricci soliton equation

$$\text{Ric}(\omega_t) + L_{X_t}\omega_t = t\omega_t + \frac{1-t}{m}[\Delta],$$

where X_t is a holomorphic vector field preserving Δ and $[\Delta]$ is the current of integration over Δ . Moreover, after passing through only finitely many such singular times and replacing the pair (W, Δ) by the successive limiting pairs $(W_1, \Delta_1), \dots, (W_k, \Delta_k)$, one can solve this equation all the way up to $t = 1$. At $t = 1$, the final variety W_k is a $\mathbf{Q}$ -Fano variety admitting a weak Kähler-Ricci soliton, and there is a test configuration, equivalently a $\mathbf{C}^*$ -equivariant flat degeneration, for the polarized Fano manifold (M, K_M^{-1}) whose central fiber is W_k .

8.3.2. Historical background, literature, and current progress

8.3.3. Verdict: partially resolved

8.3.3.1. Precise problem checked

The statement is Székelyhidi's proposed continuation picture for the conical continuity method on a Fano manifold. Starting with a smooth anticanonical multiple divisor $D \in |mK_M^{-1}|$, one follows

$$\text{Ric}(\omega_t) = t\omega_t + \frac{1-t}{m}[D].$$

If the path reaches a singular time $T < 1$ and converges to a $\mathbf{Q}$ -Fano pair (W, Δ) , the conjectural claim is that one can continue on the limiting current pair by solving a weak conical Kähler-

Ricci soliton equation with a holomorphic vector field preserving Δ . After only finitely many such jumps, the process should reach $t = 1$, producing a $\mathbf{Q}$ -Fano variety W_k with a weak Kähler-Ricci soliton and a $\mathbf{C}^*$ -equivariant flat degeneration/test configuration of $(M, -K_M)$ with central fiber W_k .

The exact problem is therefore not merely the existence of some Kähler-Ricci soliton degeneration of M , but the existence and finite termination of this specific conical-continuity continuation through singular pairs.

8.3.3.2. Original source

- Title: *Extremal Kähler metrics*
- Author: Gábor Székelyhidi
- Venue: Proceedings of the International Congress of Mathematicians, Seoul 2014; arXiv preprint
- Year: 2014
- DOI/arXiv/URL: arXiv:1405.4836, <https://arxiv.org/abs/1405.4836>
- Relevance: Conjecture 11 in this article is the source of the problem. It explicitly describes the conical continuity method with divisor D , continuation through limiting pairs by weak conical Kähler-Ricci soliton equations, finite singular times, and the final test-configuration degeneration. This is stated as a conjectural picture, not as a theorem.

8.3.3.3. Later progress

- Title: *Kähler-Ricci flow, Kähler-Einstein metric, and K-stability*
- Authors: Xiuxiong Chen, Song Sun, Bing Wang
- Venue: Geometry & Topology
- Year: 2018
- DOI/arXiv/URL: DOI 10.2140/gt.2018.22.3145; arXiv:1508.04397, <https://arxiv.org/abs/1508.04397>
- Relevance: This proves major compactness and degeneration results for the Kähler-Ricci flow on Fano manifolds, including algebraic degeneration phenomena tied to K-stability/Kähler-Einstein existence. It supports the broader expectation that analytic limits are $\mathbf{Q}$ -Fano degenerations, but it is about the Kähler-Ricci flow/Kähler-Einstein setting and does not prove the finite conical-continuity soliton-jump procedure in the problem statement.
- Title: *The Kähler-Ricci flow and optimal degenerations*
- Authors: Ruadhaí Dervan, Gábor Székelyhidi
- Venue: Journal of Differential Geometry
- Year: 2020
- DOI/arXiv/URL: DOI 10.4310/jdg/1594266881; arXiv:1612.07299, <https://arxiv.org/abs/1612.07299>
- Relevance: This paper proves that, when the Kähler-Ricci flow does not converge to a Kähler-Einstein metric, the limiting behavior is related to an optimal degeneration and a Kähler-Ricci soliton type central fiber. This is close in spirit to the final degeneration predicted by the conjecture. However, it does not establish Székelyhidi's conical conti-

nunity path with divisors, continuation across singular times, or finite termination of that process.

- Title: *Algebraic uniqueness of Kähler-Ricci flow limits and optimal degenerations of Fano varieties*
- Author: Han Li
- Venue: Geometry & Topology
- Year: 2023
- DOI/arXiv/URL: DOI 10.2140/gt.2023.27.1465; arXiv:2009.01098, <https://arxiv.org/abs/2009.01098>
- Relevance: This proves algebraic uniqueness results for Kähler-Ricci flow limits and optimal degenerations of Fano varieties. It strengthens the algebraic understanding of the limiting soliton/degeneration picture, but it is still not a proof of the finite conical-continuity continuation through the pairs (W_i, Δ_i) .
- Title: *The existence of the Kähler-Ricci soliton degeneration*
- Authors: Chenyang Xu, Ziquan Zhuang
- Venue: Forum of Mathematics, Pi
- Year: 2024
- DOI/arXiv/URL: DOI 10.1017/fmp.2024.11; arXiv:2406.04590, <https://arxiv.org/abs/2406.04590>
- Relevance: The authors prove existence of a Kähler-Ricci soliton degeneration for any log Fano pair. In particular, the theorem gives an algebraic degeneration whose central fiber admits the relevant soliton structure in a broad log Fano setting. This resolves a major algebraic/existence part of the picture predicted by Székelyhidi. It still does not prove that the particular conical continuity method described in the problem can be continued through singular limiting pairs, nor that only finitely many such analytic singular times occur.
- Title: *Twisted conical Kähler-Ricci solitons on Fano manifolds*
- Authors: Zhiwei Wang, Xiaohua Zhu
- Venue: Science China Mathematics
- Year: 2023
- DOI/arXiv/URL: DOI 10.1007/s11425-022-2125-3; <https://link.springer.com/article/10.1007/s11425-022-2125-3>
- Relevance: This studies existence of twisted conical Kähler-Ricci solitons under stability/properness hypotheses. It is relevant to the weak conical soliton equation appearing in the conjectural continuation, but it supplies conditional existence results rather than the global finite-step continuation asserted in the problem statement.

8.3.3.4. Why the problem is not fully resolved

The literature now contains strong results proving existence and uniqueness properties of Kähler-Ricci soliton degenerations, especially via Kähler-Ricci flow and non-Archimedean/algebraic methods. These results cover important pieces of the predicted final output: a $\mathbf{Q}$ -Fano/log Fano central fiber, soliton structure, and test-configuration style

degeneration.

However, I found no published or otherwise citable theorem proving the full statement as written: starting from the conical continuity method with a smooth divisor $D \in |mK_M^{-1}|$, every singular time $T < 1$ can be crossed by solving the weak conical Kähler-Ricci soliton equation on the limiting pair, only finitely many such crossings occur, and this analytic procedure reaches $t = 1$. The closest later theorems establish related flow or algebraic degeneration results, often without the divisor path and without the finite sequence of conical-continuity jumps.

Thus the correct current status of the problem is **partially resolved**: major algebraic and flow-theoretic components of the expected endpoint are known, but the full conical-continuity finite-continuation conjecture remains unresolved.

8.3.4. Remaining open problems recorded by the database

8.3.4.1. Open problem 1

Prove that the continuation of the conical continuity method through limiting pairs (W_i, Δ_i) cannot encounter infinitely many singular parameters: only finitely many changes of the pair are required before the parameter reaches $(t=1)$.

Reference supplied by the database. Gábor Székelyhidi, “Extremal Kähler Metrics,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 1017–1032, Conjecture 11, arXiv: [1405.4836](https://arxiv.org/abs/1405.4836).

8.3.5. Primary-source pointer

The originating ICM record is *Extremal Kähler metrics* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1035>.

8.4. 144. Geometric approaches to semilinear elliptic equations [2014-geom-125]

Record: 2014-geom-125

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Geometric approaches to semilinear elliptic equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=957>

8.4.1. Problem statement

Let $u : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a bounded entire solution of the Allen-Cahn equation

$$\Delta u + u - u^3 = 0 \quad \text{in } \mathbb{R}^3.$$

Define the Morse index $m(u)$ to be the maximal dimension of a vector space $E \subset C_c^\infty(\mathbb{R}^3)$ such that

$$\mathcal{Q}(\psi, \psi) := \int_{\mathbb{R}^3} (|\nabla \psi|^2 + (3u^2 - 1)\psi^2) < 0$$

for every nonzero $\psi \in E$. Prove that if $m(u) = 1$ and $\nabla u \neq 0$ outside some bounded subset of $\mathbb{R}^3$, then u is axially symmetric: equivalently, there is a line in $\mathbb{R}^3$ such that u is invariant under all rotations about that line, or, after a rigid motion, $u(x_1, x_2, x_3) = U(\sqrt{x_1^2 + x_2^2}, x_3)$ for some function U .

8.4.2. Historical background, literature, and current progress

8.4.3. Status check

Verdict: partially resolved.

8.4.3.1. Precise problem

The problem asks whether every bounded entire solution

$$\Delta u + u - u^3 = 0 \quad \text{in } \mathbb{R}^3$$

with Morse index $m(u) = 1$ and with $\nabla u \neq 0$ outside a compact set must be axially symmetric, i.e. invariant under rotations about some line.

8.4.3.2. Current literature status

I found no published or citable preprint theorem that proves this exact implication. The closest results either assume axial symmetry from the start, assume detailed two-end asymptotics and

extra rate conditions, work only in dimension two, or prove a finite-end/asymptotic structure under an added bounded-energy-density hypothesis without proving axial symmetry.

8.4.3.3. Key literature and hypothesis checks

1. Manuel del Pino, Michal Kowalczyk, Juncheng Wei, “Entire solutions of the Allen-Cahn equation and complete embedded minimal surfaces of finite total curvature in $\mathbb{R}^3$ ”, *Journal of Differential Geometry* 93 (2013), 67-131. arXiv:0902.2047, URL: <https://arxiv.org/abs/0902.2047>.

This is the original source for the classification question. In Section 12, the authors state as an open expected classification that a bounded solution in $\mathbb{R}^3$ with finite Morse index and $\nabla u \neq 0$ outside a bounded set should have finitely many planar/catenoidal ends, and separately that if the Morse index is one then the solution should be axially symmetric with two catenoidal ends. The paper constructs Allen-Cahn solutions from nondegenerate finite-total-curvature minimal surfaces and proves Morse-index correspondence for those constructed examples, but it does not classify arbitrary Morse-index-one solutions.

2. Juncheng Wei, “Geometric approaches to semilinear elliptic equations”, in *Proceedings of the International Congress of Mathematicians, Seoul 2014*, Vol. III, Kyung Moon SA, 2014, pp. 941-964. URL: <https://www.mathunion.org/icm/proceedings>.

This survey/source context records the geometric Allen-Cahn program and points back to the finite-index and axial-symmetry classification problems of del Pino-Kowalczyk-Wei. It is not a resolving theorem.

3. Changfeng Gui, Yong Liu, Juncheng Wei, “Two-end solutions to the Allen-Cahn equation in $\mathbb{R}^3$ ”, *Advances in Mathematics* 320 (2017), 926-992. DOI: 10.1016/j.aim.2017.09.013. arXiv:1502.05963, URL: <https://arxiv.org/abs/1502.05963>.

This paper constructs and studies a complete family of axially symmetric two-end solutions with logarithmic growth rates. It is important progress toward the catenoid-type picture, but the solutions are axially symmetric by construction/definition in the analysis. It does not prove that an arbitrary Morse-index-one solution satisfying the problem statement hypotheses is axially symmetric.

4. Kelei Wang, Juncheng Wei, “Finite Morse index implies finite ends”, *Communications on Pure and Applied Mathematics* 72 (2019), 1044-1119. DOI: 10.1002/cpa.21812. arXiv:1705.06831, URL: <https://arxiv.org/abs/1705.06831>.

This proves finite Morse index implies finitely many ends and linear energy growth for Allen-Cahn solutions in $\mathbb{R}^2$, and in particular that Morse-index-one solutions in $\mathbb{R}^2$ have four ends. It is strong lower-dimensional progress, but it does not apply to the stated $\mathbb{R}^3$ problem.

5. Changfeng Gui, Kelei Wang, Juncheng Wei, “Axially symmetric solutions of Allen-Cahn equation with finite Morse index”, *Transactions of the American Mathematical Society* 373 (2020), 3649-3668. DOI: 10.1090/tran/8035. arXiv:1901.11263, URL: <https://arxiv.org/abs/1901.11263>.

The main result says that axially symmetric finite-index solutions in dimension three have finitely many ends, and that an axially symmetric solution with Morse index one has exactly two ends. This is close but conditional on the desired symmetry. Since axial symmetry is a hypothesis rather than a conclusion, the theorem does not resolve the problem statement.

6. Weizhao Liang, Jianmin Yang, “Qualitative properties of two-end solutions to the Allen-Cahn equation in $\mathbb{R}^3$ ”, *Advanced Nonlinear Studies* 25 (2025), no. 3, 871-901. DOI: 10.1515/ans-2023-0186. URL: <https://doi.org/10.1515/ans-2023-0186>.

This paper proves axial symmetry and monotonicity for two-end solutions whose nodal set has prescribed logarithmic asymptotics $z = k_i \log |x'| + c_i$ and whose rates satisfy $k_1 - k_2 > 2\sqrt{2}$, plus nonexistence in another rate range. This is genuine partial progress toward axial symmetry of two-end solutions, but the statement does not assume two-end asymptotics or the rate gap, and current literature does not derive those hypotheses from $m(u) = 1$ and $\nabla u \neq 0$ outside a compact set.

7. Enric Florit-Simon, “Phase transitions with bounded index: Parallels to De Giorgi’s conjecture”, arXiv preprint, submitted 2026. arXiv:2602.03136, DOI: 10.48550/arXiv.2602.03136, URL: <https://arxiv.org/abs/2602.03136>.

Theorem 1.4 proves a strong three-dimensional structure theorem for finite-index Allen-Cahn solutions with bounded energy density/quadratic energy growth: such solutions have finitely many parallel ends, planar or catenoidal. This confirms a substantial part of the del Pino-Kowalczyk-Wei finite-end picture under an extra energy-growth hypothesis. However, the paper explicitly says it does not address the related question of proving that solutions in $\mathbb{R}^3$ with two regular ends, or with index one, are axially symmetric. Thus it does not resolve the problem statement.

8.4.3.4. What remains unresolved

The missing step is a theorem that starts only from the stated hypotheses—bounded entire solution in $\mathbb{R}^3$, Morse index one, and $\nabla u \neq 0$ outside a compact set—and proves axial symmetry. Existing results either assume the axial symmetry, assume the two-end asymptotic model plus extra rate restrictions, work in $\mathbb{R}^2$, or add bounded energy density while still leaving the index-one axial-symmetry question open. Therefore the exact problem is not fully resolved.

8.4.4. Remaining open problems recorded by the database

8.4.4.1. Open problem 1

Prove that every bounded entire two-ended solution of the Allen–Cahn equation

$$\Delta u + u - u^3 = 0 \quad \text{in } \mathbb{R}^3$$

is axially symmetric.

Reference supplied by the database. Changfeng Gui, Kelei Wang, and Juncheng Wei, “Axially symmetric solutions of the Allen–Cahn equation with finite Morse index,” *Transactions of the American Mathematical Society* 373(5) (2020), 3649–3668. Explicit open conjecture stated in the introduction following Theorem 1.6 (“all two-ended solutions ... must be axially symmetric”). DOI: [10.1090/tran/8035](https://doi.org/10.1090/tran/8035). Preprint: [arXiv:1901.05186](https://arxiv.org/abs/1901.05186).

8.4.5. Primary-source pointer

The originating ICM record is *Geometric approaches to semilinear elliptic equations* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=957>.

8.5. 145. Ricci flows with unbounded curvature [2014-geom-133]

Record: 2014-geom-133

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Ricci flows with unbounded curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1051>

8.5.1. Problem statement

Let (M^3, g_0) be a smooth, complete three-dimensional Riemannian manifold with non-negative Ricci curvature, $\text{Ric}_{g_0} \geq 0$, with no assumption that the full curvature tensor of g_0 is bounded.

Prove that there exist a time $T > 0$ and a smooth Ricci flow $g(t)$ on M , defined for $t \in [0, T)$, such that

$$\frac{\partial}{\partial t} g(t) = -2 \text{Ric}_{g(t)}, \quad g(0) = g_0.$$

Moreover, prove that this flow preserves nonnegative Ricci curvature, $\text{Ric}_{g(t)} \geq 0$ for all $t \in [0, T)$, and is unique among smooth Ricci flows starting from g_0 and satisfying these properties.

8.5.2. Historical background, literature, and current progress

8.5.3. Verdict: partially resolved

8.5.3.1. Precise problem checked

The problem asks whether every smooth complete three-dimensional Riemannian manifold (M^3, g_0) with $\text{Ric}_{g_0} \geq 0$, allowing unbounded full curvature tensor, admits a short-time smooth Ricci flow $g(t)$ with $g(0) = g_0$, preserving nonnegative Ricci curvature, and unique among smooth flows with these properties. The surrounding literature usually states the source conjecture in the stronger form that the resulting flow should also be complete for positive times.

8.5.3.2. Current progress

1. Peter M. Topping, “Ricci flows with unbounded curvature”, Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, 1033-1048, Kyung Moon Sa, 2014. arXiv:1408.6866; DOI: 10.48550/arXiv.1408.6866. This is the original survey/source context. It explains why unbounded curvature creates nonuniqueness and completeness issues, and formulates the relevant program of starting Ricci flow without bounded curvature assumptions.

2. Yi Lai, “Producing 3D Ricci flows with non-negative Ricci curvature via singular Ricci flows”, *Geometry & Topology* 25 (2021), no. 7, 3629-3690. DOI: 10.2140/gt.2021.25.3629; arXiv:2004.05291. Theorem 1.3 proves that if (M, g) is a complete 3-dimensional Riemannian manifold with $\text{Ric} \geq 0$, then there is $T > 0$ and a smooth Ricci flow $(M, g(t))$ on $[0, T)$ with $g(0) = g$ and $\text{Ric}(g(t)) \geq 0$. The paper explicitly says that completeness of this flow is not guaranteed. It also does not prove the uniqueness asserted in the problem statement.
3. Bing-Long Chen, Guoyi Xu, and Zhuhong Zhang, “Local pinching estimates in 3-dim Ricci flow”, *Mathematical Research Letters* 20 (2013), no. 5, 845-855. DOI: 10.4310/MRL.2013.v20.n5.a3; arXiv:1206.1814. This proves preservation results for curvature cones in complete 3-dimensional Ricci flows without assuming bounded curvature, including nonnegative Ricci curvature. It supports the preservation part once an appropriate complete flow exists, but it is not an existence or uniqueness theorem for the full initial-data problem.
4. Raphael Hochard, “Short-time existence of the Ricci flow on complete, non-collapsed 3-manifolds with Ricci curvature bounded from below”, preprint, 2016. arXiv:1603.08726. This gives short-time complete existence under a Ricci lower bound plus a uniform positive lower bound on volumes of unit balls. Combined with preservation results, it covers a noncollapsed special case, but the statement allows collapse at infinity.
5. Esther Cabezas-Rivas and Burkhard Wilking, “How to produce a Ricci flow via Cheeger-Gromoll exhaustion”, *Journal of the European Mathematical Society* 17 (2015), no. 12, 3153-3194. DOI: 10.4171/JEMS/578. This constructs complete Ricci flows under non-negative complex sectional curvature. In dimension three this is nonnegative sectional curvature, which is strictly stronger than nonnegative Ricci curvature.
6. Andrew D. McLeod and Peter M. Topping, “Pyramid Ricci flow in higher dimensions”, *Mathematische Zeitschrift* 296 (2020), 511-523. DOI: 10.1007/s00209-020-02472-1. Conjecture 1.1 states the WPIC1 version; in dimension three this is the complete nonnegative-Ricci Ricci-flow existence problem. The paper constructs pyramid/local-in-time-space flows, not a single complete flow on $M \times [0, T)$.
7. Albert Chau and Adam Martens, “Pseudolocality and completeness for nonnegative Ricci curvature limits of 3D singular Ricci flows”, *Mathematische Zeitschrift* 307 (2024), no. 4, Paper No. 63. DOI: 10.1007/s00209-024-03524-6; arXiv:2307.08088. Theorem 1.1 proves completeness of Lai’s flow provided the initial metric satisfies a volume-ratio lower bound tending to zero at infinity; Theorem 1.2 also covers compactly supported perturbations of complete nonnegative-sectional-curvature metrics. These are significant partial results but add hypotheses not present in the problem statement.
8. Man-Chun Lee and Peter M. Topping, “Three-manifolds with non-negatively pinched Ricci curvature”, preprint, 2022; arXiv:2204.00504. It proves complete Ricci-flow existence for complete noncompact 3-manifolds with the stronger pinching condition $\text{Ric} \geq \epsilon R g \geq 0$, and is used in work on Hamilton’s pinching conjecture. This does not cover arbitrary nonnegative Ricci curvature.
9. Man-Chun Lee, “Uniqueness of Ricci flow with scaling invariant estimates”, preprint, submitted 2025, revised 2026; arXiv:2503.20292; DOI: 10.48550/arXiv.2503.20292. Theorem 1.1 proves uniqueness for complete noncompact Ricci flows satisfying $|Rm| \leq \alpha/t$; Corollary 1.1 gives uniqueness in a uniformly noncollapsed 3D $\text{Ric} \geq 0$ setting. This does not prove uniqueness for arbitrary smooth flows from arbitrary complete $\text{Ric} \geq 0$ initial data, nor for Lai’s general flow without an a priori completeness/curvature-decay hypothesis.

8.5.3.3. What remains unresolved

No published or otherwise citable result found covers the full problem exactly. Lai resolves the general existence and Ricci-preservation part as stated without positive-time completeness, but not uniqueness. The stronger source conjecture asking for a complete Ricci flow remains open in full generality: Chau-Martens, Hochard, Cabezas-Rivas-Wilking, and Lee-Topping cover important cases with extra volume, noncollapsing, sectional/complex-sectional curvature, or pinching assumptions. Existing uniqueness theorems also impose extra hypotheses such as completeness plus scaling-invariant curvature bounds or noncollapsing. Therefore the correct status is partially resolved.

8.5.4. Remaining open problems recorded by the database

8.5.4.1. Open problem 1

Let (M^3, g_0) be smooth and complete with $\text{Ric}_{g_0} \geq 0$, allowing unbounded curvature and collapse. Prove the short-time existence of a smooth complete Ricci flow $g(t)$ with $g(0) = g_0$, without requiring uniqueness.

Reference supplied by the database. Peter M. Topping, “Ricci flows with unbounded curvature,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, 1033–1048, §4 (explicitly states that short-time existence from complete three-manifolds with nonnegative Ricci curvature and unbounded curvature remains open). [arXiv:1408.6866](https://arxiv.org/abs/1408.6866).

8.5.4.2. Open problem 2

Let (M^3, g_0) be a smooth complete, possibly collapsed manifold with nonnegative sectional curvature and no upper curvature bound. Prove that the Cabezas-Rivas–Wilking Ricci flow from g_0 is unique among smooth complete Ricci flows preserving nonnegative sectional curvature and having initial metric g_0 .

Reference supplied by the database. Peter M. Topping, “Ricci flows with unbounded curvature,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, 1033–1048, Question following Theorem 4.1 (explicitly asks for uniqueness of the Cabezas-Rivas–Wilking solution), [arXiv:1408.6866](https://arxiv.org/abs/1408.6866). Man-Chun Lee, “Uniqueness of Ricci flow with scaling invariant estimates,” preprint (2025), proves the three-dimensional case under uniform noncollapse, leaving the collapsed case above open, [arXiv:2503.20292](https://arxiv.org/abs/2503.20292).

8.5.5. Primary-source pointer

The originating ICM record is *Ricci flows with unbounded curvature* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1051>.

8.6. 146. Ricci flows with unbounded curvature [2014-geom-134]

Record: 2014-geom-134

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Ricci flows with unbounded curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1051>

8.6.1. Problem statement

Let (M, g_0) be a complete Riemannian manifold with nonnegative complex sectional curvature. Suppose that $g(t)$ is the short-time Ricci flow on $[0, \varepsilon)$ constructed by Cabezas-Rivas and Wilking, with $g(0) = g_0$ and with nonnegative complex sectional curvature for every $t \in [0, \varepsilon)$. Determine whether this Ricci flow is unique: that is, whether every Ricci flow $\tilde{g}(t)$ with $\tilde{g}(0) = g_0$ agrees with $g(t)$ on their common time interval.

8.6.2. Historical background, literature, and current progress

8.6.3. Verdict: partially resolved

8.6.3.1. Precise problem

The problem asks whether the short-time Ricci flow constructed by Cabezas-Rivas and Wilking from any complete Riemannian manifold with nonnegative complex sectional curvature is unique. Equivalently: if another Ricci flow starts from the same initial metric, must it agree with the Cabezas-Rivas–Wilking flow on the common time interval? The question is meant in the unbounded-curvature, complete noncompact setting and without adding a global curvature bound such as $|\text{Rm}| \leq C/t$ to every competing flow.

8.6.3.2. Literature status

1. Peter M. Topping, “Ricci flows with unbounded curvature”, *Proceedings of the International Congress of Mathematicians, Seoul 2014*, pp. 1033–1048, arXiv:1408.6866, DOI 10.48550/arXiv.1408.6866. Topping states this exact question after quoting the Cabezas-Rivas–Wilking existence theorem: proving uniqueness of the solution is an open question, and a priori there could be many branches with the same initial metric. This is the source formulation of the problem, not a resolution.
2. Esther Cabezas-Rivas and Burkhard Wilking, “How to produce a Ricci flow via Cheeger-Gromoll exhaustion”, *Journal of the European Mathematical Society* 17 (2015), no. 12, 3153–3194, DOI 10.4171/JEMS/582, arXiv:1107.0606. Theorem 1.1 proves short-time existence for complete open manifolds with nonnegative complex sectional curvature, with no upper curvature bound. Corollary 1.3 gives a $c(n, v_0)/t$ curvature bound under a uniform lower volume bound on unit balls. Theorem 1.4

constructs complete flows with nonnegative/positive curvature conditions whose curvature is unbounded for all times or becomes unbounded later. This paper supplies the flow and shows why bounded-curvature uniqueness is not enough, but it does not prove uniqueness of the constructed flow in full generality.

3. **Bing-Long Chen and Xi-Ping Zhu**, “Uniqueness of the Ricci flow on complete noncompact manifolds”, *Journal of Differential Geometry* 74 (2006), no. 1, 119-154, DOI 10.4310/jdg/1175266184, arXiv:math/0505447. This proves uniqueness for complete noncompact Ricci flows with bounded curvature. It applies to the classical Hamilton-Shi setting and to any positive-time interval on which the relevant flows have uniformly bounded curvature, but it does not cover the unbounded-curvature Cabezas-Rivas–Wilking cases.
4. **Brett Kotschwar**, “An energy approach to the problem of uniqueness for the Ricci flow”, *Communications in Analysis and Geometry* 22 (2014), no. 1, 149-176, DOI 10.4310/CAG.2014.v22.n1.a3, arXiv:1206.3225. Kotschwar gives energy-method uniqueness results, including extensions beyond uniformly bounded curvature under additional growth/integrability hypotheses. These results are important partial progress but do not give unconditional uniqueness for every complete flow from an arbitrary complete initial metric with nonnegative complex sectional curvature.
5. **Li Sheng and Xiaojie Wang**, “On Uniqueness of Complete Ricci Flow Solution with Curvature Bounded from Below”, preprint, 2013, arXiv:1310.1611, DOI 10.48550/arXiv.1310.1611. The abstract states uniqueness for complete noncompact noncollapsing initial data whose complex sectional curvature is bounded below and scalar curvature is bounded above, in the class of complete metrics with complex sectional curvature bounded below. This has extra hypotheses: noncollapsing, scalar curvature upper bound, and a restricted competitor class. It therefore does not settle the all-cases question.
6. **Man-Chun Lee**, “On the uniqueness of Ricci flow”, preprint, revised 2018, arXiv:1706.06848, DOI 10.48550/arXiv.1706.06848. This studies complete noncompact flows with curvature bounded above by C/t and proves uniqueness under further hypotheses such as polynomial growth of the initial curvature and a smallness condition on the flow’s Ricci curvature. It is partial progress only.
7. **Man-Chun Lee**, “Uniqueness of Ricci flow with scaling invariant estimates”, preprint, arXiv:2503.20292v2, 2026, DOI 10.48550/arXiv.2503.20292. Theorem 1.1 proves that if two complete noncompact Ricci flows start from the same complete metric and both satisfy $|Rm(g(t))| + |Rm(\tilde{g}(t))| \leq \alpha/t$ on $(0, T]$, then they are equal. Corollary 1.2 gives a strong uniqueness consequence in dimension three for uniformly noncollapsed initial manifolds with $Ric \geq 0$. This is the closest later result found. It still assumes the scaling-invariant curvature bound for both flows, or derives it only in special settings such as the three-dimensional uniformly noncollapsed case. Cabezas-Rivas–Wilking explicitly exhibit higher-dimensional/collapsed phenomena with unbounded curvature not controlled by a global C/t bound, so this theorem does not imply uniqueness for all initial metrics in the problem statement.
8. **Richard H. Bamler, Esther Cabezas-Rivas and Burkhard Wilking**, “The Ricci flow under almost non-negative curvature conditions”, *Inventiones Mathematicae* 217 (2019), 95-126, DOI 10.1007/s00222-019-00864-7, arXiv:1707.03002. This broadens existence and smoothing results under almost nonnegative curvature hypotheses, including short-time existence without upper curvature bounds in related settings. It does not provide a full uniqueness theorem for the Cabezas-Rivas–Wilking flow.

8.6.3.3. Conclusion

No published or citable result found proves uniqueness for the full Cabezas-Rivas–Willing short-time Ricci flow from every complete initial metric with nonnegative complex sectional curvature against all competing Ricci flows from the same metric. The literature resolves several special classes, especially bounded-curvature flows, flows with $|Rm| \leq C/t$, and some noncollapsed low-dimensional settings. The collapsed/unbounded-curvature general case remains unresolved.

8.6.4. Remaining open problems recorded by the database

8.6.4.1. Open problem 1

Prove uniqueness of the Cabezas-Rivas–Willing Ricci flow when the initial manifold is a complete, possibly collapsed three-manifold of nonnegative sectional curvature and unbounded curvature.

Reference supplied by the database. Peter M. Topping, “Ricci flows with unbounded curvature,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, 1033–1048, explicit open question following Theorem 4.1. [arXiv:1408.6866](https://arxiv.org/abs/1408.6866). Man-Chun Lee, “Uniqueness of Ricci flow with scaling invariant estimates,” preprint (2025), Theorem 1.1 and Corollary 1.1, establishes uniqueness under a C/t curvature bound and, in dimension three, under uniform noncollapse; it does not cover the collapsed case stated above. [arXiv:2503.20292](https://arxiv.org/abs/2503.20292).

8.6.5. Primary-source pointer

The originating ICM record is *Ricci flows with unbounded curvature* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1051>.

8.7. 147. Geometric approaches to semilinear elliptic equations [2014-ag-153]

Record: 2014-ag-153

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Geometric approaches to semilinear elliptic equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=957>

8.7.1. Problem statement

Let $w : \mathbb{R} \rightarrow (-1, 1)$ be the one-dimensional Allen-Cahn heteroclinic profile, defined by

$$w'' + w - w^3 = 0, \quad w(0) = 0, \quad \lim_{\zeta \rightarrow \pm\infty} w(\zeta) = \pm 1,$$

equivalently $w(\zeta) = \tanh(\zeta/\sqrt{2})$. In $\mathbb{R}^3$, write $r = \sqrt{x_1^2 + x_2^2}$ and $z = x_3$, and consider bounded axially symmetric solutions $u = u(r, z)$ of the Allen-Cahn equation

$$u_{zz} + u_{rr} + r^{-1}u_r + u - u^3 = 0, \quad r \in [0, \infty), z \in \mathbb{R},$$

which are regular on the axis and satisfy

$$u(r, z) = u(r, -z), \quad u_r(0, z) = 0, \quad u_z > 0 \text{ for } z > 0, \quad u_r < 0 \text{ for } r > 0.$$

Prove or disprove that for each $\kappa \in (\sqrt{2}, \infty)$, there is a unique solution u_κ in this class for which there exists a constant c_κ , depending on κ , such that in the region $z > 0$,

$$u_\kappa(r, z) = w(z - \kappa \ln r + c_\kappa) + o(1) \quad \text{as } r \rightarrow \infty,$$

and that for $\kappa \in (0, \sqrt{2}]$, no solution in this class has such an asymptotic behavior.

8.7.2. Historical background, literature, and current progress

8.7.3. Verdict: partially resolved

8.7.3.1. Precise problem checked

The problem asks for the full classification of bounded axially symmetric Allen-Cahn solutions in $\mathbb{R}^3$, even in z , regular on the axis, satisfying $u_z > 0$ for $z > 0$ and $u_r < 0$ for $r > 0$, whose upper interface has logarithmic asymptotic

$$u(r, z) = \tanh\left(\frac{z - \kappa \log r + c_\kappa}{\sqrt{2}}\right) + o(1) \quad (r \rightarrow \infty, z > 0).$$

The asserted full answer is: for every $\kappa > \sqrt{2}$, exactly one such solution exists; for $0 < \kappa \leq \sqrt{2}$, no such solution exists.

8.7.3.2. Literature found

1. Two-end solutions to the Allen-Cahn equation in $\mathbb{R}^3$

Authors: Changfeng Gui, Yong Liu, Juncheng Wei.

Venue/year: *Advances in Mathematics* 320 (2017), 926-992.

DOI/arXiv/URL: DOI 10.1016/j.aim.2017.09.013; arXiv 1502.05963, <https://arxiv.org/abs/1502.05963>.

Relevant result: The paper defines the same axial two-end growth rate by

$$\|u(r, \cdot) - H(\cdot - k \log r + c)\|_{L^\infty(0, \infty)} \rightarrow 0$$

with $H(t) = \tanh(t/\sqrt{2})$. Its Theorem 1 proves existence for every $k \in (\sqrt{2}, \infty)$. It also proves monotonicity of two-end solutions with $k > \sqrt{2}$, and Theorem 2 proves nonexistence only for $k \in (0, \sqrt{2}/2]$. The introduction explicitly says that uniqueness for $k > \sqrt{2}$ and nonexistence for $k \in (\sqrt{2}/2, \sqrt{2}]$ are expected, not proved there.

2. Qualitative properties of two-end solutions to the Allen-Cahn equation in $\mathbb{R}^3$

Authors: Weizhao Liang, Jianmin Yang.

Venue/year: *Advanced Nonlinear Studies* 25(3) (2025), 871-901.

DOI/URL: DOI 10.1515/ans-2023-0186, <https://doi.org/10.1515/ans-2023-0186>.

Relevant result: The paper treats general two-end solutions with ends $z = k_i \log |x'| + c_i$. It proves axial symmetry and monotonicity when $k_1 - k_2 > 2\sqrt{2}$. In the symmetric situation of this problem, $k_1 = \kappa$, $k_2 = -\kappa$, so this condition is exactly $\kappa > \sqrt{2}$. It also proves nonexistence for $-\sqrt{2}/2 < k_2 < k_1 < \sqrt{2}/2$ and for $k_1 = -k_2 = \sqrt{2}/2$, which in the symmetric case covers $\kappa \leq \sqrt{2}/2$, but not $\sqrt{2}/2 < \kappa \leq \sqrt{2}$. It does not prove uniqueness for fixed $\kappa > \sqrt{2}$.

3. Solutions with multiple catenoidal ends to the Allen-Cahn equation in $\mathbb{R}^3$

Authors: Oscar Agudelo, Manuel del Pino, Juncheng Wei.

Venue/year: *Journal de Mathématiques Pures et Appliquées* 103(1) (2015), 142-218.

DOI: 10.1016/j.matpur.2014.03.010.

Relevance: This is earlier Lyapunov-Schmidt progress constructing axially symmetric Allen-Cahn solutions with catenoidal/logarithmic ends in special regimes. It is cited by Gui-Liu-Wei as partial prior progress and does not by itself give the full one-parameter classification or the missing nonexistence range.

4. Entire solutions of the Allen-Cahn equation and complete embedded minimal surfaces of finite total curvature in $\mathbb{R}^3$

Authors: Manuel del Pino, Michał Kowalczyk, Juncheng Wei.

Venue/year: *Journal of Differential Geometry* 93(1) (2013), 67-131.

URL: <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-93/issue-1/Entire-solutions-of-the-Allen-Cahn-equation-and-complete-embedded-minimal/10.4310/jdg/1361800878.full>.

Relevance: Earlier gluing construction connecting Allen-Cahn interfaces with large dilations of finite-total-curvature minimal surfaces, including catenoidal behavior. It supplies special large-growth examples, not the full threshold/uniqueness statement.

8.7.3.3. What remains unresolved

the exact statement is not fully resolved. The literature proves existence for all $\kappa > \sqrt{2}$, but I found no citable theorem proving uniqueness for each fixed $\kappa > \sqrt{2}$ in the stated class. The nonexistence side is also incomplete: current theorems rule out the symmetric small-rate range only up to $\kappa \leq \sqrt{2}/2$, while the interval

$$\sqrt{2}/2 < \kappa \leq \sqrt{2}$$

remains not covered by the cited nonexistence results. Therefore the status is **partially resolved**, not fully resolved.

8.7.4. Remaining open problems recorded by the database

8.7.4.1. Open problem 1

Prove that for every fixed growth rate $\kappa > \sqrt{2}$ there is, up to the normalizations imposed in the definition of an axially symmetric two-end solution, a unique two-end solution of the Allen–Cahn equation in $\mathbb{R}^3$ with asymptotic profile $H(z - \kappa \log r + c_\kappa)$.

Reference supplied by the database. Changfeng Gui, Yong Liu, and Juncheng Wei, “Two-end solutions to the Allen–Cahn equation in $\mathbb{R}^3$,” *Advances in Mathematics* **320** (2017), 926–992, unnumbered expected uniqueness statement immediately following Theorem 2. DOI: [10.1016/j.aim.2017.09.013](https://doi.org/10.1016/j.aim.2017.09.013); arXiv: [1502.05963](https://arxiv.org/abs/1502.05963).

8.7.4.2. Open problem 2

Prove that no axially symmetric two-end solution of the Allen–Cahn equation in $\mathbb{R}^3$ has growth rate

$$\frac{\sqrt{2}}{2} < \kappa \leq \sqrt{2}.$$

Reference supplied by the database. Changfeng Gui, Yong Liu, and Juncheng Wei, “Two-end solutions to the Allen–Cahn equation in $\mathbb{R}^3$,” *Advances in Mathematics* **320** (2017), 926–992, unnumbered expected nonexistence statement immediately following Theorem 2. DOI: [10.1016/j.aim.2017.09.013](https://doi.org/10.1016/j.aim.2017.09.013); arXiv: [1502.05963](https://arxiv.org/abs/1502.05963).

8.7.4.3. Open problem 3

Prove that every two-end solution of the Allen–Cahn equation in $\mathbb{R}^3$ has Morse index one.

Reference supplied by the database. Changfeng Gui, Yong Liu, and Juncheng Wei, “Two-end solutions to the Allen–Cahn equation in $\mathbb{R}^3$,” *Advances in Mathematics* **320** (2017), 926–992, unnumbered open expectation in the introduction following Theorem 2. DOI: [10.1016/j.aim.2017.09.013](https://doi.org/10.1016/j.aim.2017.09.013); arXiv: [1502.05963](https://arxiv.org/abs/1502.05963).

8.7.5. Primary-source pointer

The originating ICM record is *Geometric approaches to semilinear elliptic equations* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=957>.

9. Incidence, axiomatic, and classical geometry

These five records return to foundational geometric configurations and incidence structures. Their statements are often elementary to formulate, but their modern treatments draw on combinatorics, topology, algebra, and computational search. The surviving problems tend to be sharp universal claims for which many structured or low-dimensional cases are known.

Coverage: 5 website records; **source years:** 1893–2022; **separately listed descendant open problems:** 8.

9.1. 148. CONSIDÉRATIONS GÉNÉRALES SUR LA MESURE DE LA SIMPLICITÉ DANS LES SCIENCES MATHÉMATIQUES ET... [1893-geom-1]

Record: 1893-geom-1

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1893: CONSIDÉRATIONS GÉNÉRALES SUR LA MESURE DE LA SIMPLICITÉ DANS LES SCIENCES MATHÉMATIQUES ET APPLICATION L'ÉVALUATION THÉORIQUE DE LA SIMPLICITÉ DES TRACÉS GÉOMÉTRIQUES, OU GÉOMÉTROGRAPHIE

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893/ICM1893.ocr.pdf#page=159>

9.1.1. Problem statement

In straightedge-and-compass geometry, where the simplicity of a construction is the total number of elementary operations R_1, R_2, C_1, C_2, C_3 , with R_1 = passing the edge of the straightedge through a given point, R_2 = drawing a line, C_1 = placing one point of the compass at a given point, C_2 = placing one point of the compass at an undetermined point of a drawn line, and C_3 = drawing a circle, determine, for a given length AB and a given positive integer n , a construction of a length equal to $n \cdot AB$ that minimizes this total number of elementary operations.

9.1.2. Historical background, literature, and current progress

9.1.3. Verdict: partially resolved

9.1.3.1. Precise problem checked

The problem asks for the exact geometrographic optimization problem posed by Emile Lemoine: in ruler-and-compass geometry, with simplicity equal to the total number of elementary operations $R1, R2, C1, C2, C3$, determine, for every given segment AB and every positive integer n , a construction of a segment of length $n \cdot AB$ whose total operation count is minimal.

This is not merely the constructibility of $n \cdot AB$; repeated transfer of AB gives a construction. The unresolved part is proving optimality, uniformly in n , in Lemoine's specific operation-count model.

9.1.3.2. Literature checked

Original source Title: "Considerations generales sur la mesure de la simplicité dans les sciences mathématiques et application à l'évaluation théorique de la simplicité des traces géométriques, ou géométrographie"

Author: Emile Lemoine

Venue: Mathematical Papers Read at the International Mathematical Congress Held in Connection with the World's Columbian Exposition, Chicago, 1893, ed. E. H. Moore, O. Bolza, H. Maschke, H. S. White; Macmillan, New York

Year: 1896, pp. 143-154

DOI/arXiv/URL: https://archive.org/download/mathematicalpap00socigoog/mathematicalpap00socigoog_djvu.txt

Lemoine defines the five elementary operations and the "simplicity" coefficient as their total count. He then states the exact problem: given a length AB , find as simply as possible a length equal to n times AB . He explicitly says he does not know a general solution, and that the problem is of the same kind as, but more complicated than, Dellac's minimum-multiplication problem for computing A^n .

Geometrography surveys and later work Title: "Two Approaches to Geometrography"

Authors: Jorma K. Merikoski and Timo Tossavainen

Venue: Journal of Geometry and Graphics 14(1), pp. 15-28

Year: 2010

DOI/arXiv/URL: <https://www.heldermann-verlag.de/jgg/jgg14/j14h1toss.pdf>

This paper surveys Lemoine's operation-count model, gives the five basic operations, and describes later statistical approaches to construction accuracy. It studies example constructions such as perpendiculars and regular pentagons. It does not give a theorem determining the optimal construction of $n \cdot AB$ for all n , nor does it give lower bounds proving optimality in Lemoine's model.

Title: Computational Geometry: An Introduction

Authors: Franco P. Preparata and Michael Ian Shamos

Venue: Springer-Verlag, Texts and Monographs in Computer Science

Year: 1985

DOI/arXiv/URL: <https://www.cs.kent.edu/~dragan/CG/CG-Book.pdf>

The historical discussion treats Lemoine's geometrography as an early time-complexity measure for Euclidean constructions, but emphasizes that Lemoine did not prove necessary lower bounds for operation counts. This supports the conclusion that known geometrographic constructions are not, by themselves, resolutions of the optimality problem.

Title: "The Complexity of Geometric Constructions"

Author: Francois Labelle

Venue: Web project, McGill University

Year: 1997; updated 2014

DOI/arXiv/URL: <https://www.cs.mcgill.ca/~sqrt/cons/constructions.html>

Labelle's page collects efficient or "best known" constructions under a related line/circle drawing-count model and states that, once constructibility is known, little is available on the minimum number of required steps. This is useful evidence about the state of the construction-complexity literature, but it uses a different complexity measure and does not resolve Lemoine's all- n problem.

Related number-theoretic problem: addition chains Title: "Efficient Generation of Minimal Length Addition Chains"

Author: Edward G. Thurber

Venue: SIAM Journal on Computing 28(4), pp. 1247-1263

Year: 1999

DOI/arXiv/URL: <https://doi.org/10.1137/S0097539795295663> ; PDF: https://wwwhomes.uni-bielefeld.de/achim/siam_thurber.pdf

Thurber identifies Dellac's 1894 question on the minimum number of multiplications needed to compute A^m as the source of addition chains, defines the minimal addition-chain length $l(n)$, and gives exact branch-and-bound methods for generating minimal chains. This is substantial progress on the analogous arithmetic problem Lemoine mentioned. It does not imply an optimal ruler-and-compass construction for $n \leq AB$, because Lemoine's geometric problem has additional operation costs and geometric reuse issues.

Title: "Calculating Optimal Addition Chains"

Author: Neill Michael Clift

Venue: Computing 91, pp. 265-284

Year: 2011

DOI/arXiv/URL: <https://doi.org/10.1007/s00607-010-0118-8> ; article page: <https://link.springer.com/article/10.1007/s00607-010-0118-8>

Clift gives faster exact computation of optimal addition chains, including computation of all optimal addition-chain lengths for $n \leq 2^{32}$. This is finite-range and arithmetic-chain progress, not a proof resolving the geometrographic optimum for all positive integers n .

Title: "Computing Sequences with Addition Chains"

Authors: Peter Downey, Benton Leong, and Ravi Sethi

Venue: SIAM Journal on Computing 10(3), pp. 638-646

Year: 1981

DOI/arXiv/URL: <https://doi.org/10.1137/0210047>

This paper proves complexity results for generalized addition-chain sequence problems. These results concern arithmetic addition chains, not Lemoine's straightedge-and-compass operation

model, and therefore do not resolve the exact geometric problem.

9.1.3.3. Applicability check

the exact problem requires both:

1. for every positive integer n , an explicit construction of a segment of length $n \cdot AB$ in Lemoine's $R1$, $R2$, $C1$, $C2$, $C3$ model; and
2. a proof that no construction with a smaller total operation count exists in that same model.

The addition-chain literature addresses an analogous arithmetic optimization: minimizing multiplications for A^n , equivalently minimizing additions in a chain for n . Even a shortest addition chain does not automatically minimize Lemoine's geometric cost, because a geometric realization must pay for drawing/placing lines and circles, reusing already drawn loci, transferring lengths, and choosing starting points. Lemoine explicitly described his geometric problem as more complicated than the Dellac/addition-chain problem.

The geometrography literature gives definitions, examples, comparisons, best-known constructions, and statistical accuracy models. It does not provide a complete lower-bound theory or a theorem whose hypotheses and conclusion imply optimal $n \cdot AB$ constructions for all n .

9.1.3.4. What remains unresolved

No citable theorem was found that determines, for every positive integer n , the minimum possible total number of Lemoine elementary operations needed to construct a segment of length $n \cdot AB$, together with an optimal construction attaining that number. The problem is therefore not fully resolved. The literature gives related and partial progress through addition-chain theory and through studies of geometrographic complexity for selected constructions, so the appropriate verdict is partially resolved rather than fully resolved.

9.1.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

9.1.5. Primary-source pointer

The originating ICM record is *CONSIDÉRATIONS GÉNÉRALES SUR LA MESURE DE LA SIMPLICITÉ DANS LES SCIENCES MATHÉMATIQUES ET APPLICATION L'ÉVALUATION THÉORIQUE DE LA SIMPLICITÉ DES TRACÉS GÉOMÉTRIQUES, OU GÉOMÉTROGRAPHIE* (1893). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893/ICM1893.ocr.pdf#page=159>.

9.2. 149. V. Theory of Functions and Geometry [1893-geom-2]

Record: 1893-geom-2

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1893: V. Theory of Functions and Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893.2/ICM1893.2.ocr.pdf#page=43>

9.2.1. Problem statement

Determine the possible forms of the circular or spherical quadrangles and, more generally, polygons that arise in the conformal-geometric treatment of functions defined by second-order linear differential equations with more than three singular points, generalizing the hypergeometric case in which three singular points correspond to the vertices of a circular or spherical triangle.

9.2.2. Historical background, literature, and current progress

9.2.3. Verdict: partially resolved

9.2.3.1. Precise problem identified

The problem is not asking for a single numerical theorem about one equation. It is Klein's Schwarz-map program for second-order linear differential equations with more than three regular singular points. In the hypergeometric case, the ratio of two independent solutions maps the upper half-plane (or a disk) to a circular/spherical triangle. For equations with four or more singular points, the analogous Schwarz images are circular or spherical quadrilaterals and, in general, circular/spherical polygons. The problem is to obtain a general classification/overview of the possible forms of these polygons, equivalently to understand the corresponding Schwarzian/Fuchsian equations and accessory parameters. In the modern positive-curvature real/unitary formulation, this becomes the classification of spherical n -gons or of second-order Fuchsian equations with real parameters and unitary monodromy; for $n=4$ this is the Heun-equation case.

9.2.3.2. Original source

- Title: The Evanston Colloquium: Lectures on Mathematics
- Author: Felix Klein
- Venue/year: Macmillan, New York, 1894; lectures delivered at Northwestern University after the 1893 Chicago Congress of Mathematics; AMS reprints later issued
- DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893.2/ICM1893.2.ocr.pdf>
- Relevance: Lecture V says that for differential equations with more than three singular points, triangles must be replaced by quadrangles and other polygons, and that the dif-

ficuity is the geometrical problem of obtaining a general view of the possible polygon forms. This is exactly the problem represented by the problem statement.

9.2.3.3. Progress found

- Title: Ueber Kreisbogenpolygone. Erste Abhandlung
- Author: A. Schoenflies
- Venue/year: Mathematische Annalen 42 (1893), 377-408
- DOI/arXiv/URL: <https://doi.org/10.1007/BF01444164>
- Progress: Early work on circular-arc polygons, directly cited in Klein's discussion. It gives important constructive/reduction analysis for circular polygons, but later literature treats it as only a partial extension of Klein's triangle classification.
- Title: Ueber Kreisbogendreiecke und Kreisbogenvierecke
- Author: A. Schoenflies
- Venue/year: Mathematische Annalen 44 (1894), 105-124
- DOI/arXiv/URL: <https://doi.org/10.1007/BF01446976>
- Progress: Early special treatment of circular triangles and quadrilaterals. It does not give a modern complete all-cases classification of the polygons arising from arbitrary higher-singularity second-order equations.
- Title: Ueber die gestaltlichen Verhaeltnisse der Kreisbogenvierecke; Ueber die geometrischen Eigenschaften der Kreisbogenvierecke
- Author: W. Ihlenburg
- Venue/year: Goettinger Nachrichten (1908), 225-230; Nova Acta Leopoldina 91 (1909), 1-79 plus tables
- DOI/arXiv/URL: no DOI found; bibliographic data cited in Eremenko-Gabrielov-Tarasov, Conformal Geometry and Dynamics 20 (2016), 128-175
- Progress: Further analysis of circular quadrilaterals. Modern authors state that the Schoenflies-Ihlenburg algorithm constructs all circular quadrilaterals, but still "falls short of a complete classification": prescribed angles and sides need not determine a unique quadrilateral, and the geodesic/spherical cases are hard to isolate.
- Title: Metrics with conic singularities and spherical polygons
- Authors: Alexandre Eremenko, Andrei Gabrielov, Vitaly Tarasov
- Venue/year: Illinois Journal of Mathematics 58(3) (2014), 739-755
- DOI/arXiv/URL: <https://doi.org/10.1215/ijm/1441790388> ; arXiv:1405.1738, <https://arxiv.org/abs/1405.1738>
- Progress: Gives a modern formulation of spherical n-gons and states that the problem is equivalent to classification of some second-order linear differential equations with regular singularities, real parameters, and unitary monodromy. It enumerates/classifies the case in which exactly two corner angles are not multiples of π , with the remaining angles integral. This is substantial progress for arbitrary n , but it is a restricted case.
- Title: Metrics with four conic singularities and spherical quadrilaterals

- Authors: Alexandre Eremenko, Andrei Gabrielov, Vitaly Tarasov
- Venue/year: Conformal Geometry and Dynamics 20 (2016), 128-175
- DOI/arXiv/URL: <https://doi.org/10.1090/ecgd/295> ; arXiv:1409.1529, <https://arxiv.org/abs/1409.1529>
- Progress: Classifies spherical quadrilaterals up to isometry in cases with integer-angle hypotheses and identifies the equivalent problem as classification of Heun equations with real parameters and unitary monodromy. This resolves important quadrilateral subcases, but not all quadrilaterals/polygons asked for by Klein's general problem.
- Title: Spherical quadrilaterals with three non-integer angles
- Authors: Alexandre Eremenko, Andrei Gabrielov, Vitaly Tarasov
- Venue/year: Journal of Mathematical Physics, Analysis, Geometry 12 (2016), 134-167
- DOI/arXiv/URL: <https://doi.org/10.15407/mag12.02.134> ; arXiv:1504.02928, <https://arxiv.org/abs/1504.02928>
- Progress: Treats the quadrilateral case in which one angle is an integer multiple of π and the other three are not. It is another major Heun/unitary-monodromy subcase, but still assumes a special angle pattern.
- Title: Circular Pentagons and Real Solutions of Painleve VI Equations
- Authors: Alexandre Eremenko, Andrei Gabrielov
- Venue/year: Communications in Mathematical Physics 355 (2017), 51-95
- DOI/arXiv/URL: <https://doi.org/10.1007/s00220-017-2921-y> ; arXiv:1611.01356, <https://arxiv.org/abs/1611.01356>
- Progress: Associates a class of real Painleve VI solutions to one-parameter families of circular pentagons and gives an algorithm for special-point counts and order. This shows active progress beyond quadrilaterals, but it is a special family of pentagons, not a full classification of all possible polygon forms.
- Title: Fuchsian Equations with Three Non-Apparent Singularities
- Authors: Alexandre Eremenko, Vitaly Tarasov
- Venue/year: SIGMA 14 (2018), article 058, 12 pages
- DOI/arXiv/URL: <https://doi.org/10.3842/SIGMA.2018.058> ; arXiv:1801.08529, <https://arxiv.org/abs/1801.08529>
- Progress: Proves that a second-order Fuchsian equation with n singularities of which $n-3$ are apparent is related to a hypergeometric equation by a polynomial-coefficient differential operator, generically surjectively, and applies this to curvature-1 metrics with all but three angles integral. This is a strong special case, but it assumes $n-3$ apparent singularities/all but three integer angles.
- Title: Metrics of constant positive curvature with four conic singularities on the sphere
- Author: Alexandre Eremenko
- Venue/year: Proceedings of the American Mathematical Society 148(9) (2020), 3957-3965
- DOI/arXiv/URL: arXiv:1905.02537, <https://arxiv.org/abs/1905.02537>
- Progress: Proves finiteness of metrics for four prescribed points and prescribed non- 2π -

multiple angles. Finiteness is important, but it is not a classification or “general overview” of the possible quadrilateral forms.

- Title: Classification of Generic Spherical Quadrilaterals
- Author: Andrei Gabrielov
- Venue/year: Arnold Mathematical Journal 9 (2023), 151-203
- DOI/arXiv/URL: <https://doi.org/10.1007/s40598-022-00204-3> ; arXiv:2008.07999, <https://arxiv.org/abs/2008.07999>
- Progress: Classifies generic spherical quadrilaterals up to isometry, where “generic” means the side images lie on four distinct circles with no triple intersections. This is a modern classification theorem, but the genericity hypothesis excludes non-generic four-circle configurations and does not address higher polygons.
- Title: Metrics of constant positive curvature with conic singularities. A survey
- Author: Alexandre Eremenko
- Venue/year: survey/preprint, 2021; updated online PDF lists later 2023 references
- DOI/arXiv/URL: arXiv:2103.13364, <https://arxiv.org/abs/2103.13364> ; updated PDF: <https://www.math.purdue.edu/~eremenko/dvi/survey1.pdf>
- Progress/status: The survey states that for curvature $\kappa \leq 0$ the conic-singularity metric problem has a complete solution, but for $\kappa > 0$ the problem is wide open. It also states that classification of metrics is equivalent to classification of developing maps, and lists many partial results, including quadrilateral and polygon special cases. This is strong evidence against a fully resolved verdict.

9.2.3.4. Why this is not fully resolved

The literature gives complete answers for several important subfamilies: classical triangles, all-integer-angle/rational-function cases, n -gons with exactly two non-integer angles, quadrilaterals with one or more integer angles, generic quadrilaterals, and some apparent-singularity/Fuchsian reductions. These results all impose extra hypotheses on angles, monodromy, genericity, number of sides, or apparent singularities.

the problem asks for the general overview/classification of the possible quadrangles and more general polygons that replace Schwarz triangles for second-order equations with more than three singular points. No citable source found proves a theorem covering all such polygons or all associated Fuchsian equations/accessory parameters. Current surveys still describe the positive-curvature/spherical-polygon version as wide open in general.

Therefore the correct status is: partially resolved.

9.2.4. Remaining open problems recorded by the database

9.2.4.1. Open problem 1

Characterize all quadruples $(\alpha_0, \alpha_1, \alpha_2, \alpha_3) \in \mathbb{R}_{>0}^4$ for which there exists a spherical quadrilateral with geodesic sides and interior angles $\pi\alpha_0, \pi\alpha_1, \pi\alpha_2, \pi\alpha_3$. In particular, determine the additional necessary and sufficient conditions beyond the Gauss–Bonnet inequality $\sum_{j=0}^3 \alpha_j > 2$ when angles larger than π are allowed.

Reference supplied by the database. Alexandre Eremenko, “Possible shapes of spherical quadrilaterals,” problem note in the author’s *Unsolved Problems in Function Theory*, originally dated February 16, 2014, revised July 21, 2025, no problem number and no journal publication; the opening problem explicitly asks which prescribed angle quadruples are realizable. [PDF](#) and [problem-list page](#).

9.2.5. Primary-source pointer

The originating ICM record is *V. Theory of Functions and Geometry* (1893). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893.2/ICM1893.2.ocr.pdf#page=43>.

9.3. 150. Combinatorial theory [1970-geom-11]

Record: 1970-geom-11

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1970: Combinatorial theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.3/ICM1970.3.ocr.pdf#page=237>

9.3.1. Problem statement

Let $G = G(S)$ be a finite combinatorial geometry on a finite set S , with matroid closure operation, rank function r , geometric lattice $L(G)$ of flats, and characteristic polynomial p_G . Let $G^\perp(S)$ be the orthogonal geometry whose bases are exactly the complements in S of bases of G . Fix a finite field $GF(q)$, and define the critical exponent of G to be the least integer c such that $p_G(q^c) > 0$.

Prove or disprove Rota's form of Hadwiger's conjecture: the critical exponent of G is characterized by the absence of the corresponding forbidden segments, equivalently obstruction intervals $[A, B]$ of flats of length $r(B) - r(A)$, in $L(G)$ or in $L(G^\perp)$.

9.3.2. Historical background, literature, and current progress

9.3.3. Verdict: partially resolved

9.3.3.1. Precise problem checked

The problem is not about Hadwiger's theorem in convex geometry. It is the graph-minor Hadwiger conjecture expressed in Rota/Crapo critical-problem language for finite combinatorial geometries/matroids. In ordinary graph language the underlying conjecture is:

For every integer $t \geq 1$, every graph with no K_t minor is $(t - 1)$ -colorable; equivalently, every graph with chromatic number at least t has a K_t minor.

Rota's formulation says that for a finite combinatorial geometry $G(S)$, with flats forming the geometric lattice $L(G)$, dual/orthogonal geometry $G^\perp(S)$, and critical exponent defined by characteristic-polynomial values $p_G(q^c)$, Hadwiger's conjecture should be viewed as an obstruction criterion: critical exponent should be controlled by the absence of certain forbidden segments/obstruction intervals in $L(G)$ or $L(G^\perp)$.

9.3.3.2. Status

The full problem is not resolved. It is partially resolved: the exact Hadwiger conjecture is proved only in small cases, while the first exact open case is $t = 7$. Since the Rota critical-problem statement is a reformulation of the same graph-minor conjecture, the unresolved $t = 7$ graph case prevents a full resolution of the question.

9.3.3.3. Key literature

1. Hugo Hadwiger, “Über eine Klassifikation der Streckenkomplexe”, *Vierteljahrsschrift der Naturforschenden Gesellschaft in Zürich* 88 (1943), 133-142/143. URL: https://ngzh.ch/wp-content/uploads/2024/08/88_17.pdf

This is the original graph-theoretic source. The modern form is that every graph with chromatic number at least t contains a K_t minor.

2. Gian-Carlo Rota, “Combinatorial theory, old and new”, in *Actes du Congrès International des Mathématiciens* (Nice, 1970), Tome 3, Gauthier-Villars, Paris, 1971, pp. 229-233. URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.3/ICM1970.3.ocr.pdf>

This is the source of the stated language. Rota defines the critical exponent via $p_G(q^c) > 0$, describes the critical problem for combinatorial geometries, and states that the critical exponent is conjecturally related to absence of “forbidden segments” or “obstructions” in the geometric lattice of $G(S)$ or $G^\perp(S)$; he then says Hadwiger’s conjecture can be restated in these terms. This is a reformulation/program, not a proof.

3. Geoff Whittle, “Dowling group geometries and the critical problem”, *Journal of Combinatorial Theory, Series B* 47(1) (1989), 80-92. DOI: 10.1016/0095-8956(89)90067-1. URL: https://homepages.ecs.vuw.ac.nz/~whittle/pubs/dowling_geometries_and_critical_problem.pdf

Whittle extends the critical-problem framework using Dowling group geometries. He proves that for suitable matroids embeddable as minors of $Q_r(A)$, a critical exponent over a finite group A is well-defined and determined by characteristic-polynomial evaluation. He explicitly observes that this lets every case of Hadwiger’s conjecture be stated as a critical problem. This verifies the reformulation but does not resolve Hadwiger’s conjecture.

4. G. A. Dirac, “A Property of 4-Chromatic Graphs and some Remarks on Critical Graphs”, *Journal of the London Mathematical Society* s1-27(1) (1952), 85-92. DOI: 10.1112/jlms/s1-27.1.85.

Together with Hadwiger’s original work, this covers the small cases up to $t \leq 4$.

5. Klaus Wagner, “Über eine Eigenschaft der ebenen Komplexe”, *Mathematische Annalen* 114 (1937), 570-590. DOI: 10.1007/BF01594196. URL: <https://eudml.org/doc/159935>

Wagner proved the relevant K_5 -minor-free structure/equivalence with the Four Color Theorem. In combination with the Four Color Theorem, this gives Hadwiger’s conjecture for $t = 5$.

6. Kenneth Appel and Wolfgang Haken, “Every planar map is four colorable. Part I: Discharging”, *Illinois Journal of Mathematics* 21(3) (1977), 429-490. DOI: 10.1215/ijm/1256049011; Kenneth Appel, Wolfgang Haken, and John Koch, “Every planar map is four colorable. Part II: Reducibility”, *Illinois Journal of Mathematics* 21(3) (1977), 491-567. DOI: 10.1215/ijm/1256049012.

These prove the Four Color Theorem, which, through Wagner’s equivalence, supplies the $t = 5$ case.

7. Neil Robertson, Paul Seymour, and Robin Thomas, “Hadwiger’s conjecture for K_6 -free graphs”, *Combinatorica* 13(3) (1993), 279-361. DOI: 10.1007/BF01202354.

This proves the $t = 6$ case: every graph with no K_6 minor is 5-colorable. It is the strongest fully settled exact case.

8. Paul Seymour, “Hadwiger’s conjecture”, in *Open Problems in Mathematics*, Springer, 2016, pp. 417-437. DOI: 10.1007/978-3-319-32162-2_13.

Seymour’s survey records the conjecture as a major open problem and identifies K_7 -minor-free 6-colorability as the first unsettled exact case.

9. Sergey Norin, “Recent progress towards Hadwiger’s conjecture”, *Proceedings of the International Congress of Mathematicians 2022*, EMS Press, 2023, pp. 4606-4620. URL: <https://ems.press/content/book-chapter-files/33307>

Norin states Hadwiger’s conjecture, summarizes that it is known for $t \leq 6$, and says settling the exact conjecture for $t \geq 7$ remains extremely challenging. The paper surveys weakenings and asymptotic progress, not a complete solution.

10. Michelle Delcourt and Luke Postle, “Reducing linear Hadwiger’s conjecture to coloring small graphs”, *Journal of the American Mathematical Society* 38(2) (2025), 481-507. DOI: 10.1090/jams/1047; arXiv:2108.01633.

This gives the current type of asymptotic progress: every K_t -minor-free graph is $O(t \log \log t)$ -colorable and linear Hadwiger reduces to coloring small graphs. This is far weaker than the exact $(t - 1)$ -colorability required by Hadwiger’s conjecture.

11. Sergey Norin and Agnes Totschnig, “Every graph with no $K_7^{\vee}$ -minor is 6-colorable”, arXiv:2507.03244, 2025. URL: <https://arxiv.org/abs/2507.03244>

This recent preprint works near the $t = 7$ frontier and explicitly states that the next open case is $t = 7$, that every K_7 -minor-free graph is known to be 8-colorable, and that even 7-colorability for K_7 -minor-free graphs is open. Its theorem excludes the weaker graph $K_7^{\vee}$, not K_7 itself, so it is partial progress only.

9.3.3.4. Why no full-resolution theorem applies

The strongest exact published theorem is the $t = 6$ case of Robertson-Seymour-Thomas. The modern surveys and 2025 frontier work still identify $t = 7$ as open. Results such as $O(t\sqrt{\log t})$, $O(t(\log t)^{\beta})$, and $O(t \log \log t)$ -colorability are asymptotic weakenings and do not imply $(t - 1)$ -colorability. Likewise, results excluding $K_7^{\vee}$, K_7^- , or extra minors have stronger hypotheses than the full obstruction criterion and do not cover all K_7 -minor-free graphs.

Therefore the exact problem is not fully resolved, but it is not fully open either because the cases $t \leq 6$ and many weakenings/special cases are resolved.

9.3.4. Remaining open problems recorded by the database

9.3.4.1. Open problem 1

Prove the first unresolved exact case of Hadwiger’s conjecture: every graph with no K_7 minor is 6-colorable.

Reference supplied by the database. Paul Seymour, “Hadwiger’s conjecture,” in Michael Th. Rassias (ed.), *Open Problems in Mathematics*, Springer, Cham, 2016, pp. 417–437 (the first open case $t = 7$), DOI: [10.1007/978-3-319-32162-2_13](https://doi.org/10.1007/978-3-319-32162-2_13). Current-status confirmation: Sergey Norin and Agnes Totschnig, “Every graph with no $K_7^{\vee}$ -minor is 6-colorable,” arXiv:2507.03244 (2025), [arXiv:2507.03244](https://arxiv.org/abs/2507.03244); their theorem concerns the strictly stronger exclusion of $K_7^{\vee}$, not exclusion of K_7 alone.

9.3.4.2. Open problem 2

Prove the still-open weakening of the $t = 7$ case: every graph with no K_7 minor is 7-colorable.

Reference supplied by the database. Sergey Norin and Agnes Totschnig, “Every graph with no $K_7^{\vee}$ -minor is 6-colorable,” arXiv:2507.03244 (2025), introduction (identifying even 7-colorability of all K_7 -minor-free graphs as open), [arXiv:2507.03244](https://arxiv.org/abs/2507.03244). See also Sergey Norin, “Recent progress towards Hadwiger’s conjecture,” in *Proceedings of the International Congress of Mathematicians 2022*, Vol. 6, EMS Press, Berlin, 2023, pp. 4606–4620, DOI: [10.4171/ICM2022/143](https://doi.org/10.4171/ICM2022/143).

9.3.4.3. Open problem 3

Prove Hadwiger’s conjecture for graphs of independence number at most two: every n -vertex graph G with $\alpha(G) \leq 2$ contains a $K_{\lfloor n/2 \rfloor}$ minor.

Reference supplied by the database. Zoltán Füredi, András Gyárfás, and Gábor Simonyi, “Connected matchings and Hadwiger’s conjecture,” *Combinatorics, Probability and Computing* **14** (2005), no. 3, 435–438, Conjecture 1.1 and its $\alpha(G) = 2$ specialization, DOI: [10.1017/S0963548305006759](https://doi.org/10.1017/S0963548305006759). Current-status confirmation: Sergey Norin and Paul Seymour, “Dense minors of graphs with independence number two,” *Journal of Combinatorial Theory, Series B* **176** (2026), 101–110, introduction, DOI: [10.1016/j.jctb.2025.08.005](https://doi.org/10.1016/j.jctb.2025.08.005).

9.3.4.4. Open problem 4

Find a constant-factor improvement over the Duchet–Meyniel bound for graphs of independence number two: prove that there is an absolute $\varepsilon > 0$ such that every n -vertex graph G with $\alpha(G) = 2$ contains a complete minor of order at least $(1/3 + \varepsilon)n$.

Reference supplied by the database. Paul Seymour, “Hadwiger’s conjecture,” in Michael Th. Rassias (ed.), *Open Problems in Mathematics*, Springer, Cham, 2016, pp. 417–437, Question 4.8, DOI: [10.1007/978-3-319-32162-2_13](https://doi.org/10.1007/978-3-319-32162-2_13). Current-status confirmation: Sergey Norin and Paul Seymour, “Dense minors of graphs with independence number two,” *Journal of Combinatorial Theory, Series B* **176** (2026), 101–110, introduction (calling any constant-factor improvement a well-known open problem), DOI: [10.1016/j.jctb.2025.08.005](https://doi.org/10.1016/j.jctb.2025.08.005).

9.3.4.5. Open problem 5

Prove the Füredi–Gyárfás–Simonyi connected-matching conjecture: for every positive integer t , every graph G on $4t - 1$ vertices with $\alpha(G) = 2$ contains a connected matching of size at least t .

Reference supplied by the database. Zoltán Füredi, András Gyárfás, and Gábor Simonyi, “Connected matchings and Hadwiger’s conjecture,” *Combinatorics, Probability and Computing* **14** (2005), no. 3, 435–438, connected-matching conjecture, DOI: [10.1017/S0963548305006759](https://doi.org/10.1017/S0963548305006759). Current-status confirmation: Rong Chen and Zijian Deng, “Connected matching in graphs with independence number two,” arXiv:2409.05920 (2024), proving the conjecture only for $t \leq 22$, [arXiv:2409.05920](https://arxiv.org/abs/2409.05920).

9.3.5. Primary-source pointer

The originating ICM record is *Combinatorial theory* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.3/ICM1970.3.ocr.pdf#page=237>.

9.4. 151. Intersection Theorems for Finite Sets and Geometric Applications [1986-geom-39]

Record: 1986-geom-39

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: Intersection Theorems for Finite Sets and Geometric Applications

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.2/ICM1986.2.ocr.pdf#page=555>

9.4.1. Problem statement

Let X be an n -element set, and for an integer r write

$$\binom{X}{r} = \{A \subseteq X : |A| = r\}.$$

Let k and l be integers with $k > l \geq 0$ and $k \leq 2l + 1$. Prove that if $\mathcal{F} \subseteq \binom{X}{k}$ is a family of k -element subsets of X such that no two distinct members $F, F' \in \mathcal{F}$ satisfy $|F \cap F'| = l$, then

$$|\mathcal{F}| \leq \frac{\binom{2k-l-1}{k} \binom{n}{l}}{\binom{2k-l-1}{l}}.$$

Moreover, prove that equality holds if and only if there is an $(l, 2k - l - 1, n)$ -Steiner system $\mathcal{S} \subseteq \binom{X}{2k-l-1}$, meaning that every l -element subset of X is contained in exactly one member of $\mathcal{S}$, such that

$$\mathcal{F} = \Delta_k(\mathcal{S}),$$

where

$$\Delta_k(\mathcal{S}) = \{H \in \binom{X}{k} : H \subseteq S \text{ for some } S \in \mathcal{S}\}$$

is the k -shadow of $\mathcal{S}$.

9.4.2. Historical background, literature, and current progress

9.4.3. Verdict: partially resolved

9.4.3.1. Precise problem checked

The problem asks for the exact maximum size of a k -uniform family $\mathcal{F} \subseteq \binom{X}{k}$ with no two distinct members satisfying $|F \cap F'| = l$, in the range $k > l \geq 0$ and $k \leq 2l + 1$. The proposed answer is

$$|\mathcal{F}| \leq \frac{\binom{2k-l-1}{k} \binom{n}{l}}{\binom{2k-l-1}{l}},$$

with equality exactly when $\mathcal{F}$ is the k -shadow of an $(l, 2k - l - 1, n)$ -Steiner system.

Equivalently, setting $t = l + 1$, this is the $k < 2t$ Steiner-system regime of the Erdős-Sós forbidden intersection problem for $g(n, k, t)$, where one forbids intersections of size $t - 1$.

9.4.3.2. Literature status

Original/source context

- Title: *Intersection theorems for finite sets and geometric applications*
- Author: Peter Frankl
- Venue: Proceedings of the International Congress of Mathematicians, Vol. 1, 2, Berkeley, California, 1986, American Mathematical Society, Providence, RI, pp. 1419-1430
- Year: 1987
- DOI/arXiv/URL: no DOI found; bibliographic listing at <https://www.renyi.hu/~pfrankl/FPpubl.html>
- Relevance: This is the cited ICM source for the conjectural forbidden-intersection/Steiner-shadow extremal statement.

Early formulation and partial verification

- Title: *Erdős-Ko-Rado theorem—22 years later*
- Authors: Michel Deza and Peter Frankl
- Venue: SIAM Journal on Algebraic and Discrete Methods 4(4), 419-431
- Year: 1983
- DOI/arXiv/URL: <https://doi.org/10.1137/0604042>
- Relevance: Conjecture 3.8 states the same dichotomy for $m(n, k, \{0, \dots, k\} \setminus \{l\})$ for sufficiently large n . It records that the inequality in the $k \leq 2l + 1$ case was known when $k - l$ is a prime power, not in all cases.
- Title: *An extremal set theoretical characterization of some steiner systems*
- Author: Peter Frankl
- Venue: Combinatorica 3, 193-199
- Year: 1983
- DOI/arXiv/URL: <https://doi.org/10.1007/BF02579293>
- Relevance: This proves the exact upper bound and equality characterization under the extra arithmetic hypothesis that $k - l$ is prime. The Springer abstract states: if $k \leq 2t + 1$ and $k - t$ is prime, then the same bound holds and equality is characterized by the corresponding Steiner system. With $t = l$, this is precisely a prime-difference subcase of the problem, so it is only partial progress.
- Title: *Intersection theorems with geometric consequences*
- Authors: Peter Frankl and Richard M. Wilson
- Venue: Combinatorica 1, 357-368
- Year: 1981
- DOI/arXiv/URL: <https://doi.org/10.1007/BF02579457>
- Relevance: Frankl-Wilson polynomial-method bounds imply the inequality part in the

prime-power cases. Later literature explicitly summarizes this as covering $k - l$ a prime power. This still leaves composite non-prime-power values of $k - l$, and it does not by itself provide the full equality classification in every problem case.

- Title: *Forbidding just one intersection*
- Authors: Peter Frankl and Zoltán Füredi
- Venue: Journal of Combinatorial Theory, Series A 39(2), 160-176
- Year: 1985
- DOI/arXiv/URL: [https://doi.org/10.1016/0097-3165\(85\)90035-4](https://doi.org/10.1016/0097-3165(85)90035-4) ; PDF <https://www.renyi.hu/~pfrankl/j26.pdf>
- Relevance: This is a central paper on the Erdős-Sós forbidden-intersection problem. It solves the $k \geq 2t$ umvirate regime for large n and gives asymptotic/conditional progress in the $k < 2t$ regime, but it does not remove the arithmetic restrictions needed for the exact Steiner-system statement in this problem.

Current status source

- Title: *Stability for the complete intersection theorem, and the forbidden intersection problem of Erdős and Sós*
- Authors: David Ellis, Nathan Keller, and Noam Lifshitz
- Venue: Journal of the European Mathematical Society 26, 1611-1654
- Year: 2024
- DOI/arXiv/URL: <https://doi.org/10.4171/JEMS/1441> ; PDF <https://ems.press/content/serial-article-files/47528>
- Relevance: The introduction defines the forbidden intersection problem $g(n, k, t)$. It states that Frankl proved the $k < 2t$ Steiner-system bound when $k - t + 1$ is prime and that Frankl-Wilson proved the inequality when $k - t + 1$ is a prime power, but that removing the prime hypothesis remains open. Translating $t = l + 1$, the unresolved range $k < 2t$ is exactly the stated $k \leq 2l + 1$ range. Their main theorem resolves broad large- k ranges where the extremal families are Ahlswede-Khachatrian/umvirate-type families, not the full Steiner-shadow regime asked here.

9.4.3.3. Conclusion

The exact statement is not fully resolved in the literature. It is proved in important special cases, especially when $k - l$ is prime, and the inequality is known in prime-power cases. Recent literature still identifies the general $k < 2t$ case, equivalently $k \leq 2l + 1$, as a remaining open part of the Erdős-Sós forbidden intersection problem. What remains unresolved is the full bound and equality characterization for all integers $k > l \geq 0$ with $k \leq 2l + 1$, especially composite non-prime-power values of $k - l$.

9.4.4. Remaining open problems recorded by the database

9.4.4.1. Open problem 1

Prove the forbidden-intersection bound and its equality characterization in the remaining arithmetic regime where $k - l$ is not a prime power: if $k > l \geq 0$, $k \leq 2l + 1$, and $\mathcal{F} \subseteq \binom{[n]}{k}$ has $|A \cap B| \neq l$ for distinct $A, B \in \mathcal{F}$, then

$$|\mathcal{F}| \leq \frac{\binom{2k-l-1}{k} \binom{n}{l}}{\binom{2k-l-1}{l}},$$

with equality only for the k -shadow of an $(l, 2k - l - 1, n)$ -Steiner system.

Reference supplied by the database. David Ellis, Nathan Keller, and Noam Lifshitz, “Stability for the complete intersection theorem, and the forbidden intersection problem of Erdős and Sós,” *Journal of the European Mathematical Society* 26, no. 5 (2024), 1611–1654, Problem 1.3 and the discussion after Theorem 1.4 (Frankl’s conjecture; the prime-power cases follow from Frankl–Wilson, while removal of that hypothesis is stated to remain open), DOI: 10.4171/JEMS/1441. <https://doi.org/10.4171/JEMS/1441>

9.4.5. Primary-source pointer

The originating ICM record is *Intersection Theorems for Finite Sets and Geometric Applications* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.2/ICM1986.2.ocr.pdf#page=555>.

9.5. 152. Combinatorial geometry takes the lead [2022-geom-160]

Record: 2022-geom-160

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Combinatorial geometry takes the lead

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=433>

9.5.1. Problem statement

Prove that for every finite configuration of points $P_1, \dots, P_n$ in $\mathbb{R}^d$, if $\mathcal{F}_r$ denotes the set of distinct rank r flats determined by these points, where a rank r flat means an $(r - 1)$ -dimensional affine flat and a flat determined by the points means the minimal affine flat spanned by some subset of $\{P_1, \dots, P_n\}$, then the finite sequence $(|\mathcal{F}_r|)_r$, as r ranges over the possible ranks, is unimodal; that is, there exists an index m such that

$$|\mathcal{F}_1| \leq \dots \leq |\mathcal{F}_m| \geq |\mathcal{F}_{m+1}| \geq \dots.$$

9.5.2. Historical background, literature, and current progress

9.5.3. Status: partially resolved

9.5.3.1. Precise problem checked

The problem asks for Rota's unimodality conjecture for the Whitney numbers of the second kind of a real-realizable matroid: given a finite point configuration in $\mathbb{R}^d$, count the distinct affine flats of each rank determined by subsets of the points, and prove that the sequence of these counts is unimodal. Equivalently, after homogenizing the affine points to vectors $(P_i, 1)$, this asks whether the numbers of rank- r flats in every matroid realizable over $\mathbb{R}$ form a unimodal sequence.

9.5.3.2. Literature and progress

1. Gian-Carlo Rota, "Combinatorial theory, old and new," in *Actes du Congres International des Mathematiciens (Nice, 1970)*, Tome 3, Gauthier-Villars, Paris, 1971, pp. 229-233.
 - URL/source record: https://www.openproblemgarden.org/op/rotas_unimodal_conjecture
 - This is the original source cited in later papers for the conjecture that the rank-level sizes of a geometric lattice, equivalently Whitney numbers of the second kind of a simple matroid, are unimodal.
2. June Huh and Botong Wang, "Enumeration of points, lines, planes, etc.," *Acta Mathematica* 218 (2017), no. 2, 297-317.

- DOI: <https://doi.org/10.4310/ACTA.2017.v218.n2.a2>
 - arXiv: <https://arxiv.org/abs/1609.05484>
 - Theorem/result: for a finite subset E of the projectivization of an r -dimensional vector space over a field, with L_p the rank- p subspaces spanned by subsets of E , they prove top-heavy inequalities and lower-half monotonicity. In particular, for $p \leq q$ and $p + q \leq r$, $|L_p| \leq |L_{r-q}|$, with an order-preserving injection, and for $p < r/2$, $|L_p| \leq |L_{p+1}|$.
 - Applicability: this includes the real affine point configurations after homogenization/restricting to the affine span, so it gives major partial progress for exactly the stated setting.
 - Limitation: it does not prove the full unimodality statement. The paper explicitly states that Rota conjectured unimodality of the rank-level sizes and that “the unimodality for the upper half remains as an outstanding open problem.” Thus Huh-Wang proves the lower-half/top-heavy part, not the required eventual decrease for all upper ranks.
3. Tom Braden, June Huh, Jacob P. Matherne, Nicholas Proudfoot, and Botong Wang, “Singular Hodge theory for combinatorial geometries,” arXiv preprint, 2020; version 4, 2023.
- arXiv: <https://arxiv.org/abs/2010.06088>
 - DOI: <https://doi.org/10.48550/arXiv.2010.06088>
 - Theorem/result: Theorem 1.1 proves the Dowling-Wilson top-heavy conjecture for all matroids: for any $k \leq j \leq \text{rk } M - k$, $|L_k(M)| \leq |L_j(M)|$, with compatible injections.
 - Applicability: this strengthens Huh-Wang by removing representability hypotheses, so it also covers the real-realizable configurations in the problem statement.
 - Limitation: this still proves top-heaviness/order-matchings, not full unimodality of the entire sequence of Whitney numbers of the second kind.
4. Luis Ferroni and Benjamin Schroeter, “Face enumeration for split matroid polytopes,” *Combinatorics, Probability and Computing*, published online 2025.
- DOI: <https://doi.org/10.1017/S0963548325000021>
 - arXiv: <https://arxiv.org/abs/2310.05487>
 - Relevance: Section 3.2 summarizes the status after the top-heavy theorem and says that the vector counting flats of each rank is the Whitney-number sequence of the second kind; after noting that the Top-Heavy Conjecture was settled, it states that it remains open to prove or disprove unimodality of the Whitney numbers of the second kind.
5. Paul D. Seymour, “On the points-lines-planes conjecture,” *Journal of Combinatorial Theory, Series B* 33 (1982), no. 1, 17-26.
- DOI: [https://doi.org/10.1016/0095-8956\(82\)90054-5](https://doi.org/10.1016/0095-8956(82)90054-5)
 - Relevance: proves a special stronger log-concavity-type points-lines-planes inequality under additional hypotheses, such as no line containing five or more points. This is special-case progress and does not cover arbitrary real point configurations or the full rank sequence.

9.5.3.3. Verdict

The problem is **partially resolved**, not fully resolved. The exact problem’s setting is covered by the Huh-Wang representable-matroid theorem and by the later all-matroids top-heavy theorem, but those results prove only lower-half monotonicity/top-heaviness. They do not imply

the full unimodality sequence required by Rota’s conjecture, because they do not control the possible behavior within the upper half after the middle ranks. No citable result found in the current literature proves full unimodality for every real point configuration.

9.5.4. Remaining open problems recorded by the database

9.5.4.1. Open problem 1

Let M be a simple matroid representable over $\mathbb{R}$, of rank r , and let $W_k(M)$ be its number of rank- k flats. Is the upper-half sequence

$$W_{\lfloor r/2 \rfloor}(M), W_{\lfloor r/2 \rfloor + 1}(M), \dots, W_r(M)$$

unimodal? Equivalently, can the already proved lower-half monotonicity/top-heavy inequalities be completed by ruling out a decrease followed by a later increase among the upper rank levels?

Reference supplied by the database. June Huh and Botong Wang, “Enumeration of Points, Lines, Planes, Etc.,” *Acta Mathematica* **218** (2017), no. 2, 297–317, discussion immediately after Conjecture 1 explicitly identifying unimodality of the “upper half” as the outstanding open problem, DOI: [10.4310/ACTA.2017.v218.n2.a2](https://doi.org/10.4310/ACTA.2017.v218.n2.a2), arXiv: [1609.05484](https://arxiv.org/abs/1609.05484). The full second-kind Whitney-number unimodality problem remains open after the top-heavy theorem; see Luis Ferroni and Benjamin Schröter, “Valuative Invariants for Large Classes of Matroids,” *Journal of the London Mathematical Society* (2) **110** (2024), no. 3, article e12984, §8, DOI: [10.1112/jlms.12984](https://doi.org/10.1112/jlms.12984), arXiv: [2208.04893](https://arxiv.org/abs/2208.04893).

9.5.5. Primary-source pointer

The originating ICM record is *Combinatorial geometry takes the lead* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=433>.

10. Convex geometry

The three convex-geometric records focus on extremal approximation and dimension-dependent structure. They exemplify a persistent divide between existence theorems and effective or sharp versions: algorithms and asymptotic bounds may be available even when optimal constants or exact generality remain unknown.

Coverage: 3 website records; **source years:** 1990–2022; **separately listed descendant open problems:** 1.

10.1. 153. Geometric Algorithms and Algorithmic Geometry [1990-geom-45]

Record: 1990-geom-45

Subfield: Convex geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1990: Geometric Algorithms and Algorithmic Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=227>

10.1.1. Problem statement

Determine whether the following computational problem is NP-hard: given a convex polytope $K \subset \mathbb{R}^n$ as the solution set of an explicit finite system of rational linear inequalities, and given an error tolerance $\varepsilon > 0$, compute an ellipsoid that approximates, with error at most ε , the Löwner-John ellipsoid of K , namely the unique minimum-volume ellipsoid containing K .

10.1.2. Historical background, literature, and current progress

Verdict: partially resolved.

Precise problem checked: given a bounded convex polytope K in $\mathbb{R}^n$ by an explicit rational H-description $Ax \leq b$, and a tolerance $\varepsilon > 0$, compute an ellipsoid approximating the exact Loewner-John ellipsoid of K , i.e. the unique minimum-volume ellipsoid containing K , to arbitrary prescribed accuracy. This is the H-presentation outer/minimum-volume enclosing ellipsoid problem, not the V-presentation point-set MVEE problem and not the polar maximum-volume inscribed ellipsoid problem.

The literature does not appear to contain a theorem fully resolving the Khachiyan-Todd conjecture as stated, namely NP-hardness of arbitrary-accuracy approximation of the true minimum-volume enclosing ellipsoid for all rational H-polytopes. There is substantial partial progress and closely related hardness/relaxation work, but the available citable results either cover different input presentations, the polar inscribed problem, exact/containment hardness claims without a precise arbitrary-accuracy approximation theorem for this problem, or coarse/relaxation-based outer ellipsoids.

Key literature and status:

1. Title: Geometric algorithms and algorithmic geometry Authors: Laszlo Lovasz Venue/year: Proceedings of the International Congress of Mathematicians, Kyoto 1990, Vol. 1, Springer-Verlag, 1991, pp. 139-154 DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf> Relevance: Section 3.1 states the original presentation-dependent contrast: for a polytope given by vertices, the Loewner-John ellipsoid can be approximated to any epsilon in polynomial time; for a polytope given by inequalities, the polar maximum-volume inscribed ellipsoid can be approximated in polynomial time; Khachiyan and Todd conjectured that approximating the outer Loewner-John ellipsoid from a linear-inequality presentation is NP-hard.
2. Title: On the complexity of approximating the maximal inscribed ellipsoid for a polytope Authors: Leonid G. Khachiyan and Michael J. Todd Venue/year: Mathematical Programming 61, 137-159, 1993 DOI/arXiv/URL: <https://doi.org/10.1007/BF01582144> Relevance: This gives polynomial-time approximation results for the maximum-volume ellipsoid inscribed in an H-polytope. It resolves the polar/inscribed side mentioned by Lovasz, but it does not solve the minimum-volume circumscribed ellipsoid problem for H-polytopes.
3. Title: On Khachiyan's algorithm for the computation of minimum-volume enclosing ellipsoids Authors: Michael J. Todd and E. Alper Yildirim Venue/year: Discrete Applied Mathematics 155(13), 1731-1744, 2007 DOI/arXiv/URL: <https://doi.org/10.1016/j.dam.2007.02.013> Relevance: Provides polynomial approximation algorithms for the MVEE of a finite point set, equivalently a V-presented polytope. This is an important special case, but it does not cover an H-polytope with exponentially many vertices.
4. Title: Minimal Ellipsoid Circumscribing a Polytope Defined by a System of Linear Inequalities Authors: Jun-Ya Gotoh and Hiroshi Konno Venue/year: Journal of Global Optimization 34, 1-14, 2006 DOI/arXiv/URL: <https://doi.org/10.1007/s10898-005-3883-8> Relevance: Gives constraint-generation algorithms for the H-polytope circumscribed ellipsoid problem. The paper emphasizes that vertex-based MVEE algorithms may fail to be efficient when the inequality description has many vertices, and reports numerical success in low dimensions. It does not prove a polynomial-time arbitrary-accuracy algorithm or NP-hardness theorem resolving the conjecture.
5. Title: Sharpening geometric inequalities using computable symmetry measures Authors: Rene Brandenberg and Stefan Konig Venue/year: Mathematika 61(3), 559-580, 2015 DOI/arXiv/URL: <https://doi.org/10.1112/S0025579314000291> ; <https://arxiv.org/abs/1310.4368> Relevance: Explicitly records the state of the H-presented MVEE problem after the earlier work: it says arbitrary-accuracy approximation of the maximum-volume inscribed ellipsoid is polynomial-time, while it is not known whether the same is true for the minimum-volume enclosing ellipsoid; it cites the Khachiyan-Todd conjecture that the latter is NP-hard. It also notes

a coarse multiplicative outer approximation from the inscribed ellipsoid and John's theorem, at about $(1+\epsilon)^d$, with possible symmetry-based improvements. These are approximation-factor results, not arbitrary-accuracy computation of the true Loewner-John ellipsoid.

6. Title: On the complexity of four polyhedral set containment problems Authors: Robert M. Freund and James B. Orlin Venue/year: Mathematical Programming 33, 139-145, 1985 DOI/arXiv/URL: <https://doi.org/10.1007/BF01582241> Relevance: A foundational containment-complexity reference used in later MVEP papers to justify hardness of containment/feasibility subproblems. It supports why H-description outer ellipsoid constraints are algorithmically hard, but it is not itself a theorem proving arbitrary-accuracy hardness for computing the Loewner-John ellipsoid of every rational H-polytope.
7. Title: Finding Minimum Volume Circumscribing Ellipsoids Using Generalized Copositive Programming Authors: Areesh Mittal and Grani A. Hanasusanto Venue/year: Operations Research 70(5), 2867-2882, 2022; published online 2021 DOI/arXiv/URL: <https://doi.org/10.1287/opre.2021.2156>; <https://arxiv.org/abs/1807.07507> Relevance: Gives an exact generalized copositive reformulation of the minimum-volume ellipsoid problem for compact convex sets and SDP restrictions for sets described by affine and convex quadratic inequalities, including polytopes. The paper states that finding the MVEE of a polytope described by affine inequalities is NP-hard, citing Boyd-Vandenberghe and Freund-Orlin, but its main positive results are exact copositive reformulation and tractable conservative SDP approximations. These approximations return feasible outer ellipsoids and are compared to scaled MVIE methods, but they are not a full arbitrary-accuracy algorithm or a self-contained approximation-hardness theorem matching the problem statement.
8. Title: Robust Optimization for Models with Uncertain Second-Order Cone and Semidefinite Programming Constraints Authors: Jianzhe Zhen, Frans J. C. T. de Ruiter, Ernst Roos, and Dick den Hertog Venue/year: INFORMS Journal on Computing 34(1), 196-210, 2022 DOI/arXiv/URL: <https://doi.org/10.1287/ijoc.2020.1025> Relevance: Develops robust-optimization reformulations and approximation/lower-bound techniques, with numerical experiments including minimum-volume circumscribing ellipsoids of polytopes. This is further partial progress on tractable approximations and bounds, not a full resolution of the Khachiyan-Todd arbitrary-accuracy H-polytope question.

Conclusion: the exact question remains unresolved in the sense required here. The strongest current literature I found provides polynomial algorithms for V-presented MVEE and H-presented maximal inscribed ellipsoids, coarse John-type outer approximations, constraint-generation methods, exact copositive reformulations, and SDP/decision-rule approximations. I did not find a published or otherwise citable theorem whose hypotheses and conclusion fully prove NP-hardness of computing an epsilon-accurate Loewner-John ellipsoid for every rational H-polytope, nor a polynomial-time algorithm for that exact task.

10.1.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

10.1.4. Primary-source pointer

The originating ICM record is *Geometric Algorithms and Algorithmic Geometry* (1990). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=227>.

10.2. 154. Isomorphic and almost-isometric problems in high-dimensional convex geometry [2006-geom-96]

Record: 2006-geom-96

Subfield: Convex geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Isomorphic and almost-isometric problems in high-dimensional convex geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1557>

10.2.1. Problem statement

Fix a positive integer k . Determine whether there exist constants $c(k), c'(k) > 0$, depending only on k , such that for every $0 < \varepsilon < 1$, if

$$N = \left\lceil c'(k) \left(\frac{1}{\varepsilon} \right)^{c(k)} \right\rceil,$$

then for every centrally symmetric convex body $K \subset \mathbb{R}^N$ (that is, $K = -K$, with K compact, convex, and with nonempty interior), there exist $r > 0$ and a k -dimensional linear subspace $E \subset \mathbb{R}^N$ such that

$$(1 - \varepsilon)rD_E \subset K \cap E \subset (1 + \varepsilon)rD_E,$$

where $D_E = \{x \in E : |x| \leq 1\}$ is the unit ball of E for the standard Euclidean norm inherited from $\mathbb{R}^N$. Determine in particular whether one may take $c(k) = (k - 1)/2$.

10.2.2. Historical background, literature, and current progress

10.2.3. Verdict: partially resolved

10.2.3.1. Precise problem

This is the fixed-dimensional, almost-isometric Dvoretzky problem: for each fixed positive integer k , whether every sufficiently high-dimensional centrally symmetric convex body has a k -dimensional central section whose Banach-Mazur distance to the Euclidean ball is $1 + O(\varepsilon)$, with the ambient dimension N bounded by a polynomial in $1/\varepsilon$. The conjectured sharp exponent is $c(k) = (k - 1)/2$.

10.2.3.2. Status

The exact problem is not fully resolved. Current literature still treats the polynomial dependence on $1/\varepsilon$ for arbitrary centrally symmetric convex bodies as the conjectural almost-isometric variant of Dvoretzky's theorem. The full result is known in dimension $k = 2$, and

polynomial/near-optimal estimates are known under strong symmetry assumptions on the norm/body, but these do not cover arbitrary centrally symmetric convex bodies for all $k \geq 3$.

10.2.3.3. Key literature and progress

1. Title: Isomorphic and almost-isometric problems in high-dimensional convex geometry
 Authors: Bo'az Klartag
 Venue: Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. II, pp. 1547-1562
 Year: 2006
 DOI/arXiv/URL: <https://www.weizmann.ac.il/math/klartag/sites/math.klartag/files/uploads/congress.pdf>
 Role: Original source for this problem's Question 1.2. Klartag states the fixed- k polynomial question, records the conjectured exponent $(k - 1)/2$, notes the special $k = 2$ result, and says that the problem was unresolved even for $k = 3$ at that time.
2. Title: A weak version of the epsilon-Dvoretzky conjecture for normed spaces
 Authors: Bo'az Klartag and Tomer Novikov
 Venue: Israel Journal of Mathematics
 Year: 2026
 DOI/arXiv/URL: DOI 10.1007/s11856-026-2911-x; arXiv:2206.13273; <https://link.springer.com/article/10.1007/s11856-026-2911-x>
 Role: Current citable status update. It states the best general Dvoretzky-type bound as $n \geq \exp(Ck|\log \varepsilon|/\varepsilon)$, which is still exponential in $1/\varepsilon$. It explicitly describes polynomial dependence as the conjectural almost-isometric variant, known for $k = 2$ or for 1-symmetric norms. Its main theorem gives a polynomial-dimensional subspace whose norm is close to an unconditional norm, not necessarily close to a Euclidean norm, so it does not imply the conclusion.
3. Title: Dichotomies, structure, and concentration in normed spaces
 Authors: Grigoris Paouris and Petros Valettas
 Venue: Advances in Mathematics 332, pp. 438-464
 Year: 2018
 DOI/arXiv/URL: DOI 10.1016/j.aim.2018.05.022; arXiv:1708.05149; <https://arxiv.org/abs/1708.05149>
 Role: Gives the best general quantitative Dvoretzky estimate cited in the recent literature: subspace dimension at least $c\varepsilon \log n / \log(1/\varepsilon)$. Inverting this for fixed k gives n exponential, not polynomial, in $1/\varepsilon$.
4. Title: A few observations on the connections between local theory and some other fields
 Authors: Vitali D. Milman
 Venue: Geometric Aspects of Functional Analysis (1986/87), Lecture Notes in Mathematics 1317, Springer, pp. 283-289
 Year: 1988
 DOI/arXiv/URL: cited in Klartag-Novikov 2026 and Klartag 2006
 Role: Source cited for the conjectural polynomial dependence and for the $k = 2$ case, whose proof is attributed to Gromov. This is a genuine special case only; it does not settle $k \geq 3$.
5. Title: Almost Euclidean sections in spaces with a symmetric basis
 Authors: Jean Bourgain and Joram Lindenstrauss
 Venue: Geometric Aspects of Functional Analysis (1987-88), Lecture Notes in Mathematics 1376, Springer, pp. 278-288

Year: 1989

DOI/arXiv/URL: cited in Klartag 2006 and Klartag-Novikov 2026

Role: Establishes polynomial/near-optimal dependence for spaces with a symmetric basis, a strong extra symmetry assumption. This is partial progress only because arbitrary centrally symmetric convex bodies need not have such symmetries.

6. Title: Almost Euclidean sections in symmetric spaces and concentration of order statistics
 Authors: Konstantin E. Tikhomirov
 Venue: Journal of Functional Analysis 265, pp. 2074-2088
 Year: 2013
 DOI/arXiv/URL: cited at <https://link.springer.com/article/10.1007/s11856-026-2911-x>
 Role: Further progress in symmetric spaces, again under symmetry assumptions not present in the problem.
7. Title: Explicit Euclidean embeddings in permutation invariant normed spaces
 Authors: Daniel Fresen
 Venue: Advances in Mathematics 266, pp. 1-16
 Year: 2014
 DOI/arXiv/URL: DOI 10.1016/j.aim.2014.07.017; arXiv:1401.0203; <https://arxiv.org/abs/1401.0203>
 Role: Gives explicit Euclidean embeddings in permutation-invariant normed spaces with $n \lesssim \log N / \log(1/\varepsilon)$. This is another special-symmetry result and does not cover arbitrary centrally symmetric convex bodies.
8. Title: Topological aspects of the Dvoretzky theorem
 Authors: Dmitri Burago, Sergei Ivanov, and Serge Tabachnikov
 Venue: Journal of Topology and Analysis 2, pp. 453-467
 Year: 2010
 DOI/arXiv/URL: DOI 10.1142/S1793525310000410; arXiv:0907.5041; <https://arxiv.org/abs/0907.5041>
 Role: Studies topological approaches and their limitations. It does not prove the full polynomial Euclidean-section statement.

10.2.3.4. What remains unresolved

No located published or citable theorem proves that for every fixed $k \geq 3$ and every centrally symmetric convex body $K \subset \mathbb{R}^N$, a $(1 + \varepsilon)/(1 - \varepsilon)$ -Euclidean k -dimensional section exists once N is merely polynomial in $1/\varepsilon$. The known full arbitrary-body estimates remain exponential in $1/\varepsilon$, while the polynomial results require extra symmetry or prove a weaker non-Euclidean conclusion.

10.2.4. Remaining open problems recorded by the database

10.2.4.1. Open problem 1

Does the almost-isometric Dvoretzky conjecture hold already for $k = 3$? More precisely, do constants $C, c > 0$ exist such that, for every $0 < \varepsilon < 1$, every centrally symmetric convex body $K \subset \mathbb{R}^N$ with $N = \lfloor c\varepsilon^{-C} \rfloor$ has a three-dimensional subspace E and an $r > 0$

satisfying

$$(1 - \varepsilon)rD_E \subset K \cap E \subset (1 + \varepsilon)rD_E?$$

Can one in particular take the conjecturally sharp exponent $C = 1$?

Reference supplied by the database. Bo'az Klartag, "Isomorphic and almost-isometric problems in high-dimensional convex geometry," in *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Vol. II, European Mathematical Society, Zürich, 2006, pp. 1547–1562, discussion following Question 1.2. [EMS Press article](#).

10.2.5. Primary-source pointer

The originating ICM record is *Isomorphic and almost-isometric problems in high-dimensional convex geometry* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1557>.

10.3. 155. Some questions related to the reverse Minkowski theorem [2022-geom-169]

Record: 2022-geom-169

Subfield: Convex geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Some questions related to the reverse Minkowski theorem

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=747>

10.3.1. Problem statement

Determine whether there are universal constants $c, C > 0$ and a fixed polynomial P such that, for every integer $n \geq 1$ and every centrally symmetric convex body $K \subseteq \mathbb{R}^n$ (that is, $K = -K$) with induced Minkowski norm

$$\|x\|_K = \inf\{s \geq 0 : x \in sK\},$$

if $X \sim N(0, I_n)$ is a standard Gaussian vector in $\mathbb{R}^n$, then

$$c \leq \mathbb{E}[\|X\|_K] \min_{\substack{W \subseteq \mathbb{R}^n \text{ linear subspace} \\ 1 \leq d = \dim(W) \leq n}} \text{vol}_d(\pi_W(K))^{1/d} \leq CP(\log n),$$

where π_W is orthogonal projection onto W and vol_d is d -dimensional Lebesgue volume on W .

10.3.2. Historical background, literature, and current progress

10.3.3. Verdict: partially resolved

10.3.3.1. Precise problem

The problem asks whether there are universal constants $c, C > 0$ and a fixed polynomial P such that every centrally symmetric convex body $K \subseteq \mathbb{R}^n$ satisfies

$$c \leq \mathbb{E}\|X\|_K \min_{1 \leq d = \dim W \leq n} \text{vol}_d(\pi_W K)^{1/d} \leq CP(\log n),$$

where $X \sim N(0, I_n)$. Equivalently, a large Gaussian norm expectation should always be explained, up to a polylogarithmic factor, by some low-volume orthogonal projection of K .

10.3.3.2. Current literature status

Original source

- Title: Some questions related to the reverse Minkowski theorem
- Author: Oded Regev

- Venue: Proceedings of the International Congress of Mathematicians 2022, Vol. 6, EMS Press, pp. 4898-4912
- Year: 2022; EMS online publication 2023
- DOI/URL: <https://doi.org/10.4171/ICM2022/134>, <https://ems.press/books/standalone/278/5582>
- Relevance: Section 5 formulates exactly this projection question as equation (5.3). Regev states that the analogous section bound follows from Milman-Pisier, but that the projection bound is not known for arbitrary symmetric convex bodies. The paper proves the projection bound only for Voronoi cells of lattices, using the reverse Minkowski theorem.

Earlier formulation and partial progress

- Title: Towards Strong Reverse Minkowski-type Inequalities for Lattices
- Authors: Daniel Dadush and Oded Regev
- Venue: 57th IEEE Symposium on Foundations of Computer Science (FOCS 2016), pp. 447-456
- Year: 2016
- DOI/arXiv/URL: <https://doi.org/10.1109/FOCS.2016.55>, arXiv:1606.06913, <https://arxiv.org/abs/1606.06913>
- Relevance: Section 10 introduces the same continuous convex-geometric relaxation. It records that the lower bound is standard and that the upper bound with sections $K \cap W$ follows from Milman-Pisier, but explicitly says the corresponding upper bound with projections $\pi_W(K)$ is not known for all symmetric convex bodies.
- Title: Gaussian processes and mixed volumes
- Authors: V. D. Milman and G. Pisier
- Venue: Annals of Probability 15, no. 1, pp. 292-304
- Year: 1987
- DOI/URL: <https://doi.org/10.1214/aop/1176992271>, <https://projecteuclid.org/journals/annals-of-probability/volume-15/issue-1/Gaussian-Processes-and-Mixed-Volumes/10.1214/aop/1176992271.full>
- Relevance: The volume-number theorem gives the analogous upper bound, up to $O(\log^2(n+1))$, when $\text{vol}_d(\pi_W K)$ is replaced by $\text{vol}_d(K \cap W)$. This does not imply the stated projection statement because $\pi_W K$ can be larger than $K \cap W$.
- Title: A reverse Minkowski theorem
- Authors: Oded Regev and Noah Stephens-Davidowitz
- Venue: Annals of Mathematics (2) 199, no. 1, pp. 1-49
- Year: 2024
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2024.199.1.1>, arXiv:1611.05979
- Relevance: This proves Dadush's lattice reverse Minkowski conjecture for stable lattices and implies the desired projection-type estimate for Voronoi cells of lattices. A Voronoi cell is a special centrally symmetric convex body, so this is not a proof for all K .

- Title: The Subspace Flatness Conjecture and Faster Integer Programming
- Authors: Victor Reis and Thomas Rothvoss
- Venue: arXiv preprint; conference version in FOCS 2023
- Year: 2023; updated versions through 2026
- arXiv/URL: arXiv:2303.14605, <https://arxiv.org/abs/2303.14605>
- Relevance: This settles a related Kannan-Lovasz/subspace-flatness problem up to $O(\log^3 n)$, using the reverse Minkowski theorem. Its conclusion concerns the covering radius $\mu(\Lambda, K)$ and the best volume-based lattice lower bound $\mu_{KL}(\Lambda, K)$, not the Gaussian norm expectation times the minimum projection volume for every symmetric convex body. It therefore does not resolve this problem's exact question.

10.3.3.3. What remains unresolved

The exact upper bound

$$\mathbb{E}\|X\|_K \min_{1 \leq d = \dim W \leq n} \text{vol}_d(\pi_W K)^{1/d} \leq \text{polylog}(n)$$

for every centrally symmetric convex body $K \subseteq \mathbb{R}^n$ remains unresolved in the literature I found.

Known results cover:

- the universal lower bound;
- the section analogue with $K \cap W$, by Milman-Pisier;
- the projection statement for lattice Voronoi cells, via Regev-Stephens-Davidowitz;
- related lattice covering-radius/subspace-flatness problems.

These do not cover all centrally symmetric convex bodies and therefore do not fully resolve the problem.

10.3.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

10.3.5. Primary-source pointer

The originating ICM record is *Some questions related to the reverse Minkowski theorem* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=747>.

Part II.

Audit and Reference Apparatus

A. Master coverage table

This table is the completeness certificate for the volume. Every one of the 155 Geometry records in the frozen API payload appears exactly once in the numbered body.

No.	ID	Year	Subfield	Open	Status
1	1958-geom-7	1958	Global/Riemannian geometry	1	Partially resolved
2	1970-geom-12	1970	Global/Riemannian geometry	1	Open / open-ended
3	1970-geom-13	1970	Global/Riemannian geometry	4	Partially resolved
4	1974-geom-20	1974	Global/Riemannian geometry	1	Partially resolved
5	1974-geom-23	1974	Global/Riemannian geometry	1	Partially resolved
6	1978-geom-25	1978	Global/Riemannian geometry	2	Partially resolved
7	1978-geom-28	1978	Global/Riemannian geometry	1	Partially resolved
8	1978-geom-29	1978	Global/Riemannian geometry	1	Partially resolved
9	1983-geom-32	1983	Global/Riemannian geometry	2	Partially resolved
10	1983-geom-33	1983	Global/Riemannian geometry	1	Partially resolved
11	1983-geom-35	1983	Global/Riemannian geometry	0	Partially resolved
12	1994-geom-51	1994	Global/Riemannian geometry	1	Partially resolved
13	1994-geom-59	1994	Global/Riemannian geometry	0	Partially resolved
14	1998-geom-63	1998	Global/Riemannian geometry	1	Open / open-ended
15	1998-geom-64	1998	Global/Riemannian geometry	1	Partially resolved
16	1998-geom-74	1998	Global/Riemannian geometry	1	Partially resolved
17	2002-geom-76	2002	Global/Riemannian geometry	0	Partially resolved
18	2002-geom-81	2002	Global/Riemannian geometry	2	Partially resolved
19	2002-ag-91	2002	Global/Riemannian geometry	1	Partially resolved
20	2002-ag-92	2002	Global/Riemannian geometry	0	Partially resolved
21	2002-ag-93	2002	Global/Riemannian geometry	3	Partially resolved
22	2006-geom-91	2006	Global/Riemannian geometry	0	Open / open-ended
23	2010-geom-114	2010	Global/Riemannian geometry	0	Partially resolved
24	2010-geom-115	2010	Global/Riemannian geometry	1	Partially resolved
25	2010-geom-116	2010	Global/Riemannian geometry	1	Partially resolved
26	2010-geom-118	2010	Global/Riemannian geometry	1	Partially resolved
27	2010-geom-119	2010	Global/Riemannian geometry	0	Partially resolved
28	2014-geom-122	2014	Global/Riemannian geometry	0	Partially resolved
29	2014-geom-128	2014	Global/Riemannian geometry	1	Partially resolved
30	2014-geom-129	2014	Global/Riemannian geometry	1	Partially resolved
31	2014-geom-130	2014	Global/Riemannian geometry	0	Partially resolved
32	2014-geom-135	2014	Global/Riemannian geometry	1	Partially resolved
33	2014-geom-136	2014	Global/Riemannian geometry	1	Partially resolved
34	2014-geom-137	2014	Global/Riemannian geometry	1	Partially resolved
35	2018-geom-138	2018	Global/Riemannian geometry	1	Partially resolved
36	2022-geom-161	2022	Global/Riemannian geometry	2	Partially resolved
37	2022-geom-162	2022	Global/Riemannian geometry	1	Partially resolved
38	2022-geom-163	2022	Global/Riemannian geometry	3	Partially resolved
39	2022-geom-164	2022	Global/Riemannian geometry	1	Partially resolved
40	2022-geom-166	2022	Global/Riemannian geometry	2	Partially resolved
41	1958-geom-8	1958	Metric and topological geometry	1	Partially resolved
42	1994-geom-61	1994	Metric and topological geometry	1	Partially resolved
43	1998-geom-65	1998	Metric and topological geometry	0	Partially resolved
44	1998-geom-72	1998	Metric and topological geometry	2	Partially resolved

No.	ID	Year	Subfield	Open	Status
45	2002-geom-79	2002	Metric and topological geometry	1	Partially resolved
46	2002-geom-89	2002	Metric and topological geometry	1	Partially resolved
47	2002-geom-90	2002	Metric and topological geometry	1	Partially resolved
48	2006-geom-95	2006	Metric and topological geometry	2	Partially resolved
49	2006-geom-101	2006	Metric and topological geometry	0	Partially resolved
50	2010-geom-121	2010	Metric and topological geometry	1	Partially resolved
51	2014-geom-126	2014	Metric and topological geometry	1	Partially resolved
52	2018-geom-144	2018	Metric and topological geometry	1	Partially resolved
53	2018-geom-145	2018	Metric and topological geometry	1	Open / open-ended
54	2018-geom-146	2018	Metric and topological geometry	0	Partially resolved
55	2018-geom-147	2018	Metric and topological geometry	2	Partially resolved
56	2018-geom-148	2018	Metric and topological geometry	1	Partially resolved
57	2018-geom-150	2018	Metric and topological geometry	1	Partially resolved
58	2018-geom-151	2018	Metric and topological geometry	0	Open / open-ended
59	2018-geom-152	2018	Metric and topological geometry	0	Open / open-ended
60	2018-geom-153	2018	Metric and topological geometry	1	Partially resolved
61	2018-geom-154	2018	Metric and topological geometry	1	Partially resolved
62	1912-geom-4	1912	Classical and local differential geometry	0	Partially resolved
63	1958-geom-9	1958	Classical and local differential geometry	0	Partially resolved
64	1970-geom-14	1970	Classical and local differential geometry	5	Partially resolved
65	1978-geom-24	1978	Classical and local differential geometry	1	Partially resolved
66	1978-geom-26	1978	Classical and local differential geometry	1	Partially resolved
67	1986-geom-41	1986	Classical and local differential geometry	1	Partially resolved
68	1986-geom-42	1986	Classical and local differential geometry	1	Partially resolved
69	1986-geom-43	1986	Classical and local differential geometry	1	Partially resolved
70	1986-geom-44	1986	Classical and local differential geometry	0	Partially resolved
71	2006-ag-115	2006	Classical and local differential geometry	1	Partially resolved
72	2006-ag-116	2006	Classical and local differential geometry	1	Partially resolved
73	2010-geom-104	2010	Classical and local differential geometry	0	Partially resolved
74	2010-geom-105	2010	Classical and local differential geometry	1	Partially resolved
75	2010-geom-106	2010	Classical and local differential geometry	2	Partially resolved
76	2010-geom-107	2010	Classical and local differential geometry	3	Partially resolved
77	2010-geom-108	2010	Classical and local differential geometry	0	Partially resolved
78	2010-geom-109	2010	Classical and local differential geometry	1	Partially resolved
79	2010-geom-110	2010	Classical and local differential geometry	0	Partially resolved
80	2014-geom-127	2014	Classical and local differential geometry	1	Partially resolved
81	2022-ag-217	2022	Classical and local differential geometry	2	Partially resolved
82	1978-ag-18	1978	Manifolds and global analysis	1	Partially resolved
83	1978-geom-27	1978	Manifolds and global analysis	0	Partially resolved
84	1983-geom-34	1983	Manifolds and global analysis	0	Partially resolved
85	1990-geom-47	1990	Manifolds and global analysis	0	Partially resolved
86	1994-geom-60	1994	Manifolds and global analysis	1	Partially resolved
87	2002-ag-88	2002	Manifolds and global analysis	4	Partially resolved
88	2006-geom-97	2006	Manifolds and global analysis	2	Partially resolved
89	2006-geom-98	2006	Manifolds and global analysis	3	Partially resolved
90	2006-geom-99	2006	Manifolds and global analysis	2	Partially resolved
91	2010-geom-103	2010	Manifolds and global analysis	1	Partially resolved
92	2010-geom-111	2010	Manifolds and global analysis	1	Partially resolved
93	2014-ag-161	2014	Manifolds and global analysis	3	Partially resolved
94	2018-ag-177	2018	Manifolds and global analysis	0	Partially resolved
95	2022-geom-165	2022	Manifolds and global analysis	2	Partially resolved
96	2022-geom-168	2022	Manifolds and global analysis	0	Partially resolved
97	2022-geom-170	2022	Manifolds and global analysis	5	Partially resolved
98	2022-geom-172	2022	Manifolds and global analysis	5	Partially resolved
99	2022-geom-174	2022	Manifolds and global analysis	1	Partially resolved
100	2022-ag-215	2022	Manifolds and global analysis	3	Partially resolved

No.	ID	Year	Subfield	Open	Status
101	1994-ag-53	1994	Symplectic and contact geometry	2	Partially resolved
102	1994-ag-54	1994	Symplectic and contact geometry	1	Partially resolved
103	1998-geom-62	1998	Symplectic and contact geometry	1	Partially resolved
104	1998-geom-69	1998	Symplectic and contact geometry	0	Partially resolved
105	1998-geom-73	1998	Symplectic and contact geometry	1	Partially resolved
106	1998-ag-74	1998	Symplectic and contact geometry	5	Partially resolved
107	1998-ag-80	1998	Symplectic and contact geometry	4	Partially resolved
108	2002-geom-82	2002	Symplectic and contact geometry	2	Partially resolved
109	2002-geom-83	2002	Symplectic and contact geometry	1	Partially resolved
110	2002-geom-84	2002	Symplectic and contact geometry	0	Partially resolved
111	2002-geom-85	2002	Symplectic and contact geometry	1	Partially resolved
112	2002-ag-94	2002	Symplectic and contact geometry	6	Partially resolved
113	2006-geom-102	2006	Symplectic and contact geometry	1	Partially resolved
114	2006-ag-108	2006	Symplectic and contact geometry	3	Partially resolved
115	2006-ag-109	2006	Symplectic and contact geometry	2	Partially resolved
116	2014-geom-124	2014	Symplectic and contact geometry	1	Partially resolved
117	2014-geom-132	2014	Symplectic and contact geometry	1	Partially resolved
118	2014-ag-152	2014	Symplectic and contact geometry	3	Partially resolved
119	2014-ag-173	2014	Symplectic and contact geometry	4	Partially resolved
120	1950-geom-6	1950	Discrete and polyhedral geometry	2	Partially resolved
121	1970-geom-16	1970	Discrete and polyhedral geometry	1	Partially resolved
122	1994-geom-52	1994	Discrete and polyhedral geometry	0	Partially resolved
123	1994-geom-53	1994	Discrete and polyhedral geometry	1	Partially resolved
124	1994-geom-54	1994	Discrete and polyhedral geometry	1	Partially resolved
125	1994-geom-55	1994	Discrete and polyhedral geometry	2	Partially resolved
126	2002-geom-77	2002	Discrete and polyhedral geometry	1	Partially resolved
127	2002-geom-78	2002	Discrete and polyhedral geometry	1	Partially resolved
128	2002-geom-87	2002	Discrete and polyhedral geometry	0	Partially resolved
129	2002-geom-88	2002	Discrete and polyhedral geometry	0	Partially resolved
130	2018-geom-149	2018	Discrete and polyhedral geometry	1	Partially resolved
131	2018-geom-158	2018	Discrete and polyhedral geometry	1	Partially resolved
132	2022-geom-175	2022	Discrete and polyhedral geometry	1	Partially resolved
133	1986-geom-38	1986	Applications of geometry / global analysis	0	Partially resolved
134	1994-geom-58	1994	Applications of geometry / global analysis	1	Partially resolved
135	1998-ag-82	1998	Applications of geometry / global analysis	3	Partially resolved
136	2002-ag-95	2002	Applications of geometry / global analysis	5	Partially resolved
137	2018-geom-140	2018	Applications of geometry / global analysis	1	Partially resolved
138	2018-geom-141	2018	Applications of geometry / global analysis	1	Partially resolved
139	2018-geom-142	2018	Applications of geometry / global analysis	1	Partially resolved
140	2018-geom-143	2018	Applications of geometry / global analysis	3	Partially resolved
141	1998-geom-71	1998	Geometric flows and geometric PDE	0	Partially resolved
142	2010-ag-139	2010	Geometric flows and geometric PDE	1	Partially resolved
143	2014-geom-123	2014	Geometric flows and geometric PDE	1	Partially resolved
144	2014-geom-125	2014	Geometric flows and geometric PDE	1	Partially resolved
145	2014-geom-133	2014	Geometric flows and geometric PDE	2	Partially resolved
146	2014-geom-134	2014	Geometric flows and geometric PDE	1	Partially resolved
147	2014-ag-153	2014	Geometric flows and geometric PDE	3	Partially resolved
148	1893-geom-1	1893	Incidence, axiomatic, and classical geometry	0	Partially resolved
149	1893-geom-2	1893	Incidence, axiomatic, and classical geometry	1	Partially resolved
150	1970-geom-11	1970	Incidence, axiomatic, and classical geometry	5	Partially resolved
151	1986-geom-39	1986	Incidence, axiomatic, and classical geometry	1	Partially resolved
152	2022-geom-160	2022	Incidence, axiomatic, and classical geometry	1	Partially resolved
153	1990-geom-45	1990	Convex geometry	0	Partially resolved
154	2006-geom-96	2006	Convex geometry	1	Partially resolved
155	2022-geom-169	2022	Convex geometry	0	Partially resolved

B. Original ICM source catalogue

The entries below catalogue the original ICM sources associated with the 155 records. Later research literature is retained in the problem-by-problem narratives.

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 14. **1978.** *Some Problems in Complex Analytic Geometry with Growth Conditions.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=141> Associated records: 1978-geom-27.
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83. **2022.** *Emergent complex geometry.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99> Associated records: 2022-geom-161, 2022-geom-162, 2022-ag-215.
84. **2022.** *Evolution of form and shape.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=267> Associated records: 2022-ag-217.
85. **2022.** *Kähler manifolds with curvature bounded below.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219> Associated records: 2022-geom-163, 2022-geom-164.
86. **2022.** *Lattice subgroups acting on manifolds.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=33> Associated records: 2022-geom-165.
87. **2022.** *Mean curvature and variational theory.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=339> Associated records: 2022-geom-166.
88. **2022.** *Rigidity, lattices, and invariant measures beyond homogeneous dynamics.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=129> Associated records: 2022-geom-168.
89. **2022.** *Some questions related to the reverse Minkowski theorem.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=747> Associated records: 2022-geom-169.
90. **2022.** *The regularity theory for the area functional (in geometric measure theory).* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313> Associated records: 2022-geom-170, 2022-geom-172, 2022-geom-174.
91. **2022.** *the upper bound side of the story.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=471> Associated records: 2022-geom-175.

C. Index of separately recorded open problems

The website payload contains 202 separately enumerated descendant or residual questions in addition to the 155 main statements. Their full wording and supplied references appear in the relevant body sections; this appendix provides a compact locator.

No.	Record	Subfield	Count
1	1958-geom-7	Global/Riemannian geometry	1
2	1970-geom-12	Global/Riemannian geometry	1
3	1970-geom-13	Global/Riemannian geometry	4
4	1974-geom-20	Global/Riemannian geometry	1
5	1974-geom-23	Global/Riemannian geometry	1
6	1978-geom-25	Global/Riemannian geometry	2
7	1978-geom-28	Global/Riemannian geometry	1
8	1978-geom-29	Global/Riemannian geometry	1
9	1983-geom-32	Global/Riemannian geometry	2
10	1983-geom-33	Global/Riemannian geometry	1
12	1994-geom-51	Global/Riemannian geometry	1
14	1998-geom-63	Global/Riemannian geometry	1
15	1998-geom-64	Global/Riemannian geometry	1
16	1998-geom-74	Global/Riemannian geometry	1
18	2002-geom-81	Global/Riemannian geometry	2
19	2002-ag-91	Global/Riemannian geometry	1
21	2002-ag-93	Global/Riemannian geometry	3
24	2010-geom-115	Global/Riemannian geometry	1
25	2010-geom-116	Global/Riemannian geometry	1
26	2010-geom-118	Global/Riemannian geometry	1
29	2014-geom-128	Global/Riemannian geometry	1
30	2014-geom-129	Global/Riemannian geometry	1
32	2014-geom-135	Global/Riemannian geometry	1
33	2014-geom-136	Global/Riemannian geometry	1
34	2014-geom-137	Global/Riemannian geometry	1
35	2018-geom-138	Global/Riemannian geometry	1
36	2022-geom-161	Global/Riemannian geometry	2
37	2022-geom-162	Global/Riemannian geometry	1
38	2022-geom-163	Global/Riemannian geometry	3
39	2022-geom-164	Global/Riemannian geometry	1
40	2022-geom-166	Global/Riemannian geometry	2
41	1958-geom-8	Metric and topological geometry	1
42	1994-geom-61	Metric and topological geometry	1
44	1998-geom-72	Metric and topological geometry	2
45	2002-geom-79	Metric and topological geometry	1
46	2002-geom-89	Metric and topological geometry	1

No.	Record	Subfield	Count
47	2002-geom-90	Metric and topological geometry	1
48	2006-geom-95	Metric and topological geometry	2
50	2010-geom-121	Metric and topological geometry	1
51	2014-geom-126	Metric and topological geometry	1
52	2018-geom-144	Metric and topological geometry	1
53	2018-geom-145	Metric and topological geometry	1
55	2018-geom-147	Metric and topological geometry	2
56	2018-geom-148	Metric and topological geometry	1
57	2018-geom-150	Metric and topological geometry	1
60	2018-geom-153	Metric and topological geometry	1
61	2018-geom-154	Metric and topological geometry	1
64	1970-geom-14	Classical and local differential geometry	5
65	1978-geom-24	Classical and local differential geometry	1
66	1978-geom-26	Classical and local differential geometry	1
67	1986-geom-41	Classical and local differential geometry	1
68	1986-geom-42	Classical and local differential geometry	1
69	1986-geom-43	Classical and local differential geometry	1
71	2006-ag-115	Classical and local differential geometry	1
72	2006-ag-116	Classical and local differential geometry	1
74	2010-geom-105	Classical and local differential geometry	1
75	2010-geom-106	Classical and local differential geometry	2
76	2010-geom-107	Classical and local differential geometry	3
78	2010-geom-109	Classical and local differential geometry	1
80	2014-geom-127	Classical and local differential geometry	1
81	2022-ag-217	Classical and local differential geometry	2
82	1978-ag-18	Manifolds and global analysis	1
86	1994-geom-60	Manifolds and global analysis	1
87	2002-ag-88	Manifolds and global analysis	4
88	2006-geom-97	Manifolds and global analysis	2
89	2006-geom-98	Manifolds and global analysis	3
90	2006-geom-99	Manifolds and global analysis	2
91	2010-geom-103	Manifolds and global analysis	1
92	2010-geom-111	Manifolds and global analysis	1
93	2014-ag-161	Manifolds and global analysis	3
95	2022-geom-165	Manifolds and global analysis	2
97	2022-geom-170	Manifolds and global analysis	5
98	2022-geom-172	Manifolds and global analysis	5
99	2022-geom-174	Manifolds and global analysis	1
100	2022-ag-215	Manifolds and global analysis	3
101	1994-ag-53	Symplectic and contact geometry	2
102	1994-ag-54	Symplectic and contact geometry	1
103	1998-geom-62	Symplectic and contact geometry	1
105	1998-geom-73	Symplectic and contact geometry	1
106	1998-ag-74	Symplectic and contact geometry	5
107	1998-ag-80	Symplectic and contact geometry	4
108	2002-geom-82	Symplectic and contact geometry	2
109	2002-geom-83	Symplectic and contact geometry	1
111	2002-geom-85	Symplectic and contact geometry	1
112	2002-ag-94	Symplectic and contact geometry	6

No.	Record	Subfield	Count
113	2006-geom-102	Symplectic and contact geometry	1
114	2006-ag-108	Symplectic and contact geometry	3
115	2006-ag-109	Symplectic and contact geometry	2
116	2014-geom-124	Symplectic and contact geometry	1
117	2014-geom-132	Symplectic and contact geometry	1
118	2014-ag-152	Symplectic and contact geometry	3
119	2014-ag-173	Symplectic and contact geometry	4
120	1950-geom-6	Discrete and polyhedral geometry	2
121	1970-geom-16	Discrete and polyhedral geometry	1
123	1994-geom-53	Discrete and polyhedral geometry	1
124	1994-geom-54	Discrete and polyhedral geometry	1
125	1994-geom-55	Discrete and polyhedral geometry	2
126	2002-geom-77	Discrete and polyhedral geometry	1
127	2002-geom-78	Discrete and polyhedral geometry	1
130	2018-geom-149	Discrete and polyhedral geometry	1
131	2018-geom-158	Discrete and polyhedral geometry	1
132	2022-geom-175	Discrete and polyhedral geometry	1
134	1994-geom-58	Applications of geometry / global analysis	1
135	1998-ag-82	Applications of geometry / global analysis	3
136	2002-ag-95	Applications of geometry / global analysis	5
137	2018-geom-140	Applications of geometry / global analysis	1
138	2018-geom-141	Applications of geometry / global analysis	1
139	2018-geom-142	Applications of geometry / global analysis	1
140	2018-geom-143	Applications of geometry / global analysis	3
142	2010-ag-139	Geometric flows and geometric PDE	1
143	2014-geom-123	Geometric flows and geometric PDE	1
144	2014-geom-125	Geometric flows and geometric PDE	1
145	2014-geom-133	Geometric flows and geometric PDE	2
146	2014-geom-134	Geometric flows and geometric PDE	1
147	2014-ag-153	Geometric flows and geometric PDE	3
149	1893-geom-2	Incidence, axiomatic, and classical geometry	1
150	1970-geom-11	Incidence, axiomatic, and classical geometry	5
151	1986-geom-39	Incidence, axiomatic, and classical geometry	1
152	2022-geom-160	Incidence, axiomatic, and classical geometry	1
154	2006-geom-96	Convex geometry	1

D. Data provenance, reproducibility, and limitations

D.1. Included project files

data/geometry_155_snapshot.json

The exact Geometry-only records used to generate the body.

data/icm_records_full_2026-07-28.json

The complete API payload frozen at collection time.

data/coverage_155.csv

One audit row per body problem, including status, review flag, open-problem count, source metadata, and text lengths.

data/source_catalog.csv

Unique original sources and the record identifiers attached to them.

data/link_inventory.tsv

Every distinct URL extracted from statements, progress narratives, and remaining-problem references, with referring record identifiers.

scripts/generate_survey.py

The deterministic generator for chapters and audit appendices.

D.2. Rebuilding

From the project directory, run:

```
python3 scripts/generate_survey.py
latexmk -xelatex -interaction=nonstopmode -halt-on-error main.tex
```

This requires Python 3, Pandoc 3, and a TeX distribution containing XeLaTeX and `latexmk`. Regeneration reads the included frozen JSON snapshot and does not contact the live website.

The generator asserts that there are exactly 155 Geometry records and that all ten subfield counts agree with the website’s category metadata. A failed assertion stops generation rather than silently producing an incomplete volume.

D.3. Interpretive limitations

This edition records a rapidly changing research landscape at a specific timestamp. Four limitations should remain visible.

1. A website status is not a substitute for a referee’s verification of the cited theorem.

2. “No later resolution found” is evidence about a literature search, not a theorem of nonexistence.
3. Several ICM questions were originally programs rather than sharply delimited conjectures; later authors may formulate their surviving cores differently.
4. New preprints or accepted papers after the snapshot date can change a verdict without changing the historical value of the earlier record.

D.4. Recommended scholarly use

Use this survey as a map into the primary literature. Before quoting a status claim in new research, follow the supplied DOI or arXiv link, check the exact hypotheses and publication version, and search for work appearing after 28 July 2026. The explicit record identifier and frozen snapshot make corrections and future editions traceable.

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